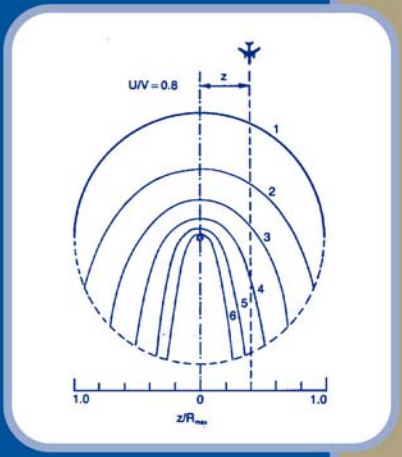
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J.S. PRZEMIENIECKI

MATHEMATICAL METHODS IN DEFENSE ANALYSES, THIRD EDITION

AIAA Education Series

J.S. PRZEMIENIECKI
EDITOR-IN-CHIEF

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Mathematical Methods in Defense Analyses

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Mathematical Methods in Defense Analyses

Third Edition

J. S. Przemieniecki, Ph.D., D.Sc.

Air Force Institute of Technology

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Dedicated to the freedom fighters throughout the world and those who made the ultimate sacrifice for freedom and human rights.

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Foreword

The publication of *Mathematical Methods in Defense Analyses* by John Przemieniecki—an important contribution in the Education Series of the American Institute of Aeronautics and Astronautics—marks a significant step in the coming of age of the application of mathematical and scientific methods in defense analyses. The initial major impulse for this movement goes back to the famous operations-research triumphs of World War II. Originally classified, many of the more relevant techniques have continued, for a variety of reasons, to be less accessible than is desirable or convenient, especially for the beginning student. This book will do much to fill the gap.

As both Senior Dean of the Air Force Institute of Technology, and the Editor-in-Chief of the AIAA Education Series of the American Institute of Aeronautics and Astronautics, Dr. Przemieniecki is exceptionally well qualified to translate the lore of his subject into a useful text, and to aid the student he has augmented the standard material with his own work. His chapters give techniques applicable to analyses of reliability and many different combat processes such as search, detection, allocation of effort, and attrition. To test its effectiveness, the text has been used in draft form by students at the Air Force Institute of Technology.

The reader will find it rewarding and especially important to reflect on the words of the final chapter on modeling. It forcefully reminds us that not all mathematical treatments of a subject are useful, and not all models are good representations of reality. Moreover, there are many important aspects of combat where our understanding is inadequate for a quantitative representation. Dr. Przemieniecki's final chapter thus brings into focus the current resurgence of interest in model verification and validation.

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Washington, DC, 1990

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Preface to 3rd Edition

"Si vis pacem, para bellum." If you want peace, prepare for war. This Latin adage has been the hallmark of rational thought about national defense since the dawn of time. It was reiterated by George Washington over 200 years ago in his famous statement, "To be prepared for war is one of the most effectual means of preserving the peace . . . ." This observation is still true today; to preserve peace, to preserve our cherished values, and to preserve our way of life requires a strong defense to deter any potential aggressor.

Strong defense requires an understanding of the defense capabilities as well as the capabilities of potential adversaries. Various analytical techniques and modeling of these capabilities contribute significantly to this understanding and provide a rational basis for deployment and employment of the available defense forces. *Mathematical Methods in Defense Analyses* is a collection of various mathematical methods used in military applications of operations research and analysis; it represents a rational development of many diverse analytical tools that can be applied to the solution of military modeling. This text is not intended to be a text on operations research: there are many such texts. Rather, its main purpose is to provide an introduction to the many techniques not normally discussed in a single volume on operations research. Thus, the present text should be useful as an introduction to the field for defense analysts in the Armed Services as well as for researchers in defense industry. The text has been used in graduate courses in the Strategic and Tactical Sciences program at the Air Force Institute of Technology. In most cases the materials in this book were derived from published sources; however, in many cases entirely new material was developed in the process of preparing the manuscript.

The second edition of this text contained additional material in several chapters. It included a new chapter on optimization methods with typical examples showing applications of linear programming, and also an expanded compendium of subroutines previously published in the AIAA Education Series as a separate volume. These subroutines can be used to compute numerical values for functions or tables derived in this text. Each subroutine is provided with a detailed reference to the equation(s) from which it was derived to ensure that its intended application is consistent with the assumptions used in the derivation. A solutions manual is available, and it can be requested directly from the author (e-mail jprzemie@msn.com).

In the third edition, a new Chapter 7 on "Theater Missile Defense" extends the concept of layered defense against intercontinental ballistic missiles to shorter range missiles or cruise missiles in a military theater of operations or in land/air battles. The layered defense is the basis for the U.S. National Ballistic Missile Defense (NBMD) and the Theater Missile Defense (TMD) systems under development

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designed to intercept long-range and medium-range to short-range missiles, respectively. The new chapter provides a detailed analysis of performance of these systems for three types of targeting assignments.

In the arduous task of completing the manuscript the author was assisted by many people. Particularly, special thanks must go to Dr. Albert H. Moore, Lt. Col. James N. Bexfield, and Clayton J. Thomas, who have read the manuscript and provided valuable comments. Lt. Gen. Glenn A. Kent and Brig. Gen. Frank S. Goodell provided valuable suggestions for improvements and Helena Czarnecki prepared graphics for the new Chapter 7. Their help is gratefully acknowledged.

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1
Scientific Methods in Military Operations

Mathematical methods of analysis play an increasingly significant role in planning military operations and determining optimal use of the available defense forces to ensure national security. These methods provide a rational approach for exploring different alternatives for force deployments and for determining the best strategies and tactics in actual combat engagements. This chapter describes in general some of the well-established mathematical techniques used in studies and analyses for defense.

1.1 Introduction

The most important attributes of military commanders are those that enable them to conduct successful military operations. These attributes were once equated to the military qualities of warriors—their operational prowess, wisdom, dedication, and courage—but in today's technologically intensive armed forces another important dimension must be included among those necessary qualities of military leaders. This new dimension is technological competence. It enables a commander to make decisions based not only on experience and judgment, but also on scientific methods of analysis and information processing for the planning and execution of military operations.

Technological competence will become increasingly essential during the new century and beyond, when many of the underlying concepts of strategy and tactics will undergo significant evolutionary changes as a result of the introduction of new weapon systems. Advanced weapons with autonomous capabilities, such as precision-guided munitions and strategic defense systems, will require a new generation of military officers and civilian planners capable of using the full potential of the emerging computerized force structure. Also, with limited defense budgets, not only must the systems themselves be cost-effective over the total life cycle, but their performance in the operational environment must be optimal.

The use of scientific methods to enhance the effectiveness and success of military operations is not new. These methods have been used to develop new weapon systems and to improve the effectiveness of their employment in combat. About 214 B.C., Archimedes helped break the Roman naval siege of Syracuse by designing machines of war, including catapults and missile throwers that terrified the Romans. One legend is that Archimedes constructed concave mirrors that burned the ships of Roman general Marcus Claudius Marcellus by concentrating the sun's rays on them.[1] This unusual defensive measure could perhaps be described as the first application of a directed energy weapon. Napoleon, during his Egyptian expedition of 1798, enlisted the services of the famous French mathematician Jean Baptiste Joseph Fourier.[2] In more recent times, World War I witnessed the first serious attempts to apply established scientific methods to the solution of military

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operational problems. In 1916, Frederick Lanchester developed a mathematical analysis of air combat,[3] which later served as a fundamental model in developing theories of combat and calculating attrition rates in military operations. Also, during World War I, Thomas Edison conducted investigations into submarine warfare for the U.S. Navy.[4] World War II provided still further impetus to such applications, and the discipline of operations research, in which scientific methods were applied to improve military operations, was born.[5]

This new discipline originated in the United Kingdom, where it was called "operational research." It resulted from the initiative of A. P. Rowe, Superintendent of the Bawdsey Research Station, who organized research teams to develop effective techniques for using the then newly developed radar to locate enemy aircraft. At the outbreak of war on 1 September 1939, Rowe sent a small group of scientists from Bawdsey to the Royal Air Force fighter command headquarters at Stanmore.[5] This group played a very important role in developing interception tactics for British fighter aircraft, tactics that played a decisive role in the Battle of Britain. By 1941 formal operational research groups had been in all three of Britain's military services.

As in Britain, the introduction of radar was responsible for stimulating scientific development in the U.S. Army Air Forces. In October 1942, General Henry "Hap" Arnold urged all commands to establish operations research groups, and, by the end of the war, 26 such groups were in existence.[6]

After World War II, operations research in both military and industrial applications began growing rapidly. In the United States, the Operations Research Society of America was formed in 1952, followed by the Military Operations Research Society (MORS) in 1966. Also, a number of private and governmental research organizations were established, such as the Institute of Defense Analyses in Washington, DC, and the Studies Analysis and Gaming Agency within the Department of Defense. In Germany, Industrieanlagen-Betriebesgesellschaft was established as the principal defense research institution for the German armed forces. In the United Kingdom, the Defense Operational Analysis Establishment was created to assist the British Ministry of Defense. Some of the other prominent centers of military operations research in Europe today are the Supreme Headquarters Allied Powers, Europe, (SHAPE) Technical Center in the Hague, the Norwegian Defense Research Establishment, and the Centre Interarmees de Recherches Operationelles in France.

The importance of operations research is emphasized at the highest echelons of the U.S. Army, Navy, and Air Force. The Army has an office of the Deputy Under Secretary (Operations Research); the Navy operates the Center for Naval Analyses; and the Air Force has its Studies and Analysis Agency.

In the former Soviet Union, applications of scientific methods to military operations were receiving much attention. Having achieved a substantial numerical superiority in weaponry, the Soviets were placing greater emphasis on combat readiness of their armed forces. Their force mobility, striking force, and firepower were constantly increasing, and the forces were showing greater concern for qualitative development.[7] In pursuing this goal the Soviets were striving for continuing introduction of the achievements of science and technology into military practice. Modern mathematical techniques were being applied to determine the optimum use of weapon systems, to achieve a better theoretical understanding

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of the conduct of large military operations, and to analyze the effectiveness of weapon systems.[8–10]

Many of the earlier applications of mathematical methods to the analysis of military operations in the United States are described at length in the U.S. Army Engineering Design Handbook on *Army Weapons Systems Analysis*.[11,12] The objective of this handbook was to present the necessary background material for the emerging field of military operations research and to record some of the methodology for evaluation of Army weapon systems and materiel. The handbook was published in two parts. Part One included the following topics: application of weapon systems analysis, management of weapon system resources, fundamentals of offense and defense, target damage assessment, factors affecting target selection, weapon delivery errors, single-shot hit probabilities, vulnerability and lethality, stochastic duels, fuzing, multiple round hit probabilities, target coverage and target damage, reliability and availability of systems, maintainability, mobility, maneuverability, agility, logistics planning, and weapon systems effectiveness evaluation. Part Two included the following topics: measures of effectiveness, target detection and search strategies, combat theories, weapon equivalence studies, human factors in weapon systems, cost analysis, weapon system life cycle cost, cost estimation techniques, survivability, countermeasures, war games, and evaluation analysis examples.

The scientific methods of analysis and optimization have now found acceptance at every level of military operations, starting with the planning of strategies, tactics, and doctrines through force deployment and employment in actual combat. The President of the United States and the National Command Authorities use the results of extensive analyses when various options for action are submitted by the military departments and the Department of Defense for the final decision. Also, at lower echelons the decision-makers rely heavily on the analytical studies that define measures of system effectiveness and present a process of determining appropriate criteria for choosing preferred alternatives. The role of analysis within the framework of military planning is shown in Fig. 1.1. Its importance in the decision-making process must not be underestimated; however, the preponderance of quantitative data now available through the use of computers must be properly interpreted in order to be of value to the military planner or commander, who must therefore have an expert understanding of the analytical process generating the data.

For military operations much of the data is generated in terms of probabilities of certain events. The probability theory, a branch of mathematical analysis, can be used for these operations to express the likelihood of an event's occurrence in terms of the probability function, whose value varies between 0 and 1 (or between 0 and 100%). Some of the typical applications include the analysis of gun or missile dispersion data to determine the circular error probable, the determination of the probabilities of kill, the generation of mathematical models for missile allocation, and the design of test procedures through which systems or equipment reliability can be established with a specified degree of confidence.

It should be understood that, because of the probabilistic nature of most military operations, one can only expect to calculate an estimate of the event probability or the level of confidence to achieve a specified equipment or system reliability. Thus, in the decision-making process it is unavoidable that some subjective judgment must be made; however, this judgment will be greatly strengthened if the

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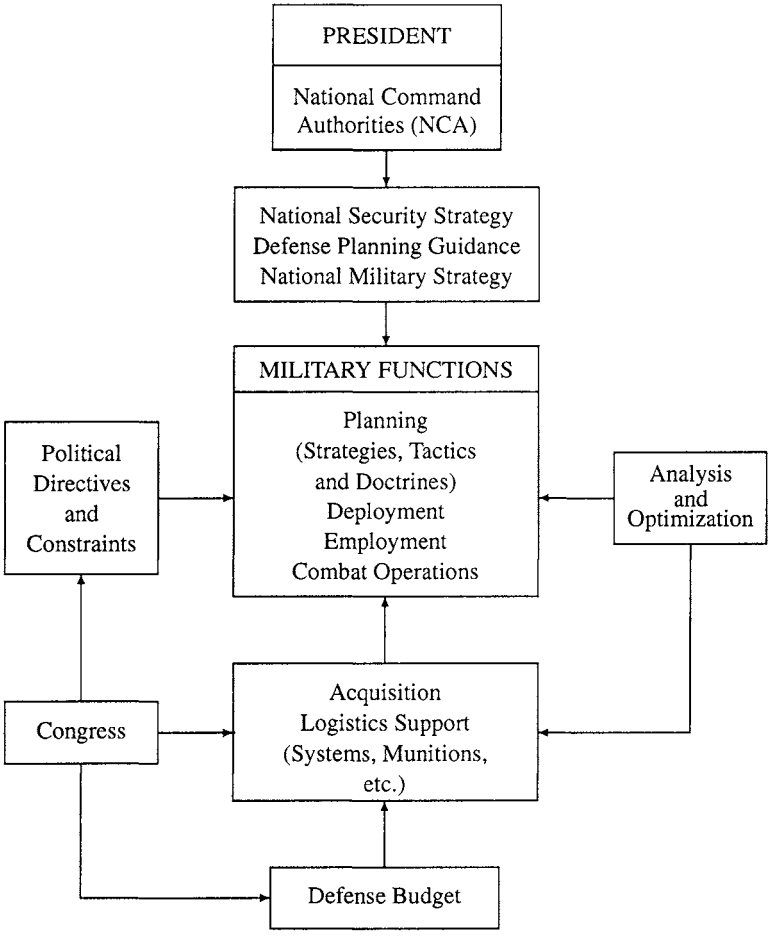


Fig. 1.1 Role of analysis within the framework of military planning.

decision-maker has a full understanding of the nature of the probabilistic process involved. This decision-making process for military operations is shown in Fig. 1.2. The decision-maker asks several basic questions: What is involved in accomplishing the mission? What are the probabilities of success of the mission? What is the level of risk? Is the level of risk acceptable? If the answer to the last question is yes, he proceeds with the mission. If the answer is no, there are a number of options available, such as eliminating or reducing the risk involved, modifying or using alternate tactics, or canceling the operation.

1.2 Mathematical Methods

To illustrate the scope of scientific methods and techniques in military operations, the rest of this chapter provides a cursory summary of the more important categories.

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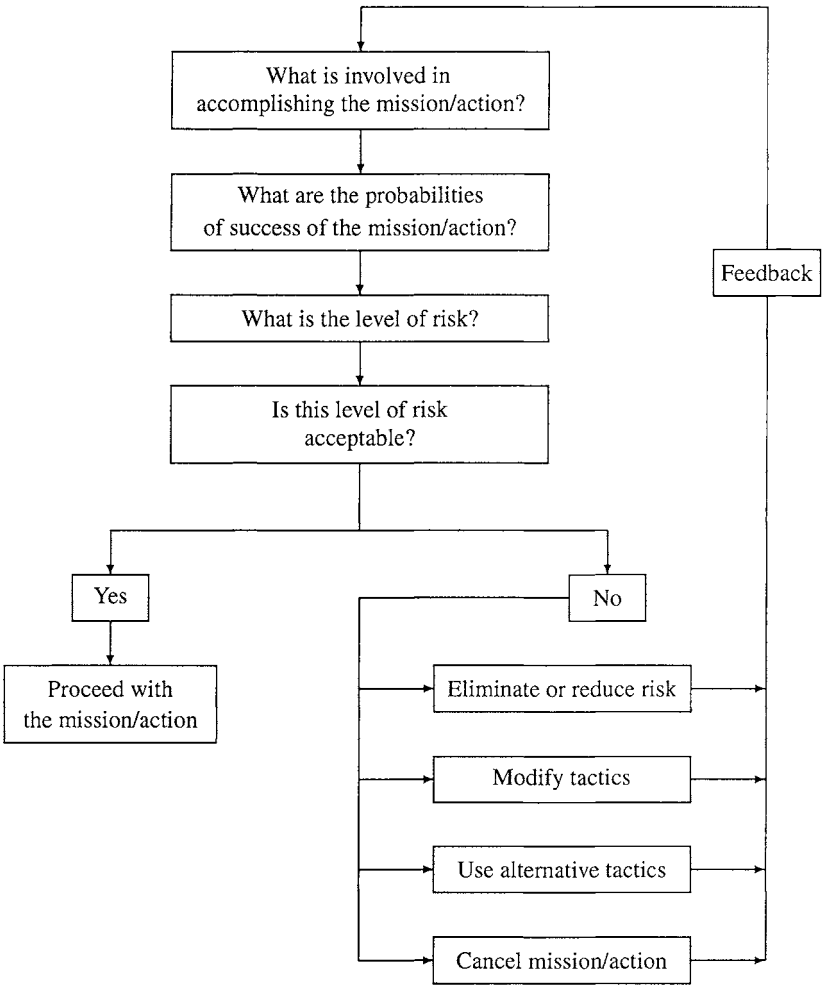


Fig. 1.2 Decision-making process.

Theory of Combat

As previously mentioned, in 1916 Lanchester[3] developed basic concepts for the theory of combat. He stated that if the number of Blue combatants (or weapons) is denoted by X, then the attrition rate of the Blue force dX_B/dt is equal to the product of the effective firing rate of the Red force α_R and the number of the Red combatants (or weapons) X_R. Mathematically this is expressed as

$$\frac{dX_B}{dt} = -\alpha_R X_R \tag{1.1}$$

where t denotes time and the negative sign signifies attrition, i.e., force reduction

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in combat. Similarly, for the Red combatants (or weapons),

$$\frac{dX_R}{dt} = -\alpha_B X_B \tag{1.2}$$

These equations or their modified forms, in which α_B and α_R are not constant but vary with time and depend on the size of forces involved, are still used today in various analytical studies to express attrition rates in combat. The analytical results obtained from the solutions of these equations can only be regarded as a gross approximation of the real situation. Nevertheless, they allow us to obtain a useful insight into the dynamics of combat and to identify the critical parameters affecting the attrition rates.

Decision Theory

Recently, the methods of statistical inference have been modified under the general description of decision theory to analyze decision-making in the face of uncertainty—a common situation in military operations. Decision theory deals with a set of possible states that can be modified by the acts of the decision-maker. If all possible consequences of the acts are known and can be assigned utility factors, then an optimal choice of acts can be determined that maximizes the utility of all consequences incurred. The available choices are best illustrated by drawing a decision tree, on which all possible decision paths are shown. If several consecutive decisions are needed, the selection process may call for the use of computers to find the optimum path.

Linear Programming

The analysis techniques known as linear programming have been developed to solve problems of allocation of resources subject to various constraining conditions. Such problems arise whenever a number of activities must be performed, but limitations on either the amount of resources available or the manner in which these resources can be allocated prevent the performance of each separate activity in the most effective way. The linear programming technique determines how these resources should be allocated in order to optimize the total effectiveness of the system. Problems that can be solved using this technique include the allocation of cargo among different types of aircraft for air logistics support, allocation of sorties to a target, allocations of bombs or missiles to a target system, and munitions storage. Linear programming is actually an optimization technique used to get a performance index (an objective function) to its maximum or minimum within the constraints of resources or technology limitations. Some specific applications of these techniques are described in Chapter 11.

Queuing Theory

Queuing theory deals with people or items in sequence and is used to minimize the cost of providing service or the waiting time of users. Many military maintenance problems can be treated as queuing problems, because items requiring service are like users of service. Moreover, inventory problems and sortie generation can be treated as queuing problems. Some military operations consist of a network

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of tasks to be carried out only once, some simultaneously, and some in sequence. In such processes coordination of starting times is necessary to optimize the final outcome. Queuing theory is useful here in determining optimal schedules.

Network Analysis

Network theory can be applied to the network routing problems that commonly arise in communication and airlift support. A typical problem is finding an optimum route (fastest, least costly, shortest, or most survivable) between two or more locations in relation to the total time, cost, distance, or survivability. Air Force applications of network theory are important in flight scheduling, logistics support, and determining the survivability of communication networks or strategic bombers.

Game Theory

Game theory is used to study conflict situations involving more than one decision-maker. Its mathematical form is, in many respects, similar to situations represented by common games of strategy, such as poker, tic-tac-toe, checkers, bridge, and chess. Conflict situations arise if the interests of two or more sides pursuing different goals clash. Without doubt, any situation in combat action can be classified as a conflict situation. The main purpose of using game theory is to draw up plans of action for the rational behavior of the players involved. The plans represent optimal strategies for each player. Game theory provides a number of theorems applicable to studies of choices of different military strategies.

Differential Games

Differential games are used to study conflict situations in which players may vary their strategies with time. Only a very limited number of problems can be solved. The most important class for which solutions are available is that of pursuit games in which two persons are involved, the pursuer and the evader. This class of problems has practical applications in the study of pursuit and evasion in air battles or the pursuit of a missile by an antimissile.

War-Gaming and Simulation

Through the process of modeling the real environment, military operations can be simulated by means of war-gaming or computer simulations. In war-gaming, which may involve map maneuvers, sand tables, or computer interactive games, people play roles to simulate the decision process, whereas simulations use computer algorithms to represent the decision-making process. Computer simulations usually employ random numbers to determine the outcome of random events and, in such cases, because of the probabilistic nature of the outcome, are called Monte Carlo simulations.

Simulated warfare provides a means of gaining experience, exploring the consequences of alternative strategies and tactics, identifying weaknesses, and improving skills without the necessity of "acting out" the situation in the real world. Gaming exercises stimulate the search for novel and more effective strategies and tactics and encourage innovation. But, more importantly, motivations aroused in

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war-gaming in peacetime may have carryover values that will pay off handsomely in the ultimate test of actual war.

Future Directions of Research

A recent report by the Committee on the Next Decade in Operations Research (CONDOR) describes several research areas in operations research likely to produce significant results over the next decade, some of which will clearly have applications in military operations analysis.[9] The following major areas have been identified in the report: 1) optimization; 2) stochastic processes; 3) operations research/artificial intelligence; 4) operational and modeling science; and 5) manufacturing and logistics. In particular, integration of operations research (OR) and artificial intelligence (AI) should lead to better methods and procedures to support problem-solving and decision-making for military operations. As described in the report,

> The two disciplines approach this common objective in fundamentally different but complementary ways: AI problem solving techniques tend to be inferential and to rely on expert knowledge and heuristics; OR uses algorithmic, mathematically based approaches. Artificial intelligence emphasizes qualitative aspects of problems; OR emphasizes the quantitative aspects. A careful integration of these two approaches to problem solving shows significant promise for improving the capability and, notably, the acceptability of problem solving systems.[13]

As an example, recent application of artificial intelligence in military logistics can be cited.[14] In this application, a neural network is used that learns very complex relationship from historical data between input factors (historical data on repairable items) and output variables (forecasted inventory needs). The system is based on a three-layer artificial neural network. When the initial learning process based on historical data is completed, the network can forecast its projected needs for the next period. It also measures its own error and periodically corrects itself. The system appears to be well suited for demand forecasting of repairable items in the Air Force inventory.

1.3 Quantitative Competence

The fast-growing requirement for technical managers of complex military operations has generated the need for operational commanders who possess quantitative skills at almost the same level as the engineer or professional analyst. The art of military operations has now been transformed into an "art-science" specialty, which evolves around modern weapon technologies and specific methods of analysis. Simply knowing the technology of weapon systems is not good enough; operational commanders must recognize strategic and tactical implications of the technologies involved and the available means of exploitation of these technologies in a possible military conflict.

This text is intended as a general introduction to the mathematical analysis of military operations. It should provide the necessary background knowledge for the military planners who use the results of more complex studies and analyses. For a

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more detailed treatment of some of the topics discussed in this text, the reader is referred to several other publications on modeling of military operations.[15–26]

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Problems

1.1. Describe briefly the origin of modern mathematical methods applied to military operations.

1.2. Discuss the concept of the Lanchester equations of combat.

1.3. List typical military applications of the following methods:

(a) Combat theory
(b) Decision theory
(c) Linear programming
(d) Queuing theory
(e) Network analysis
(f) Game theory
(g) Differential games
(h) War-gaming simulation

1.4. What are the future directions in operations research for military applications?

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2
Characteristic Properties of Weapons

Among the characteristic parameters of weapons, the most important ones are those that describe the weapon accuracy and its effectiveness against a specified target. These parameters can only be accurately assessed from the actual live tests of the system by collecting appropriate empirical data. This chapter addresses the accuracy of weapon systems based on the collected data from instrumented ranges. A specific case of a gun firing test is used to illustrate the computation of the various parameters describing the weapon accuracy. These parameters include the arithmetic mean, the median, the mean deviation, the standard deviation, and the coefficient of variation. The concept of the mean point of impact is also defined, and the sources of error causing deviation from the aiming point are briefly discussed.

2.1 Weapon Performance Data

Weapon performance is usually characterized in terms of the following parameters: 1) accuracy; 2) range; 3) rate of delivery (fire rate); 4) warhead or munitions round effectiveness against a given target; 5) time of target acquisition, pointing, and tracking; 6) time of flight to the target; 7) weight and size; 8) cost; and 9) reliability and maintainability. In any given weapon system, however, the most important characteristic parameters are its accuracy, effectiveness against a specified target, and reliability. These parameters are somewhat related, because, as the weapon's accuracy decreases, so does its effectiveness against the target.

Weapon accuracy is defined as the location of the point of impact for a given aim point on the target. The distance between the point of impact and the aim point is a random function. The precise magnitude of this distance cannot be predicted in advance; however, some general characteristics of this randomness can be estimated. There are many factors that cause this phenomenon, such as the fact that no two projectiles or missiles are identical (production tolerances), varying accuracy in aiming, maneuvering target, variable atmospheric conditions, errors caused by the human operator, etc. As a result of the combined influence of these factors, the actual trajectory of the projectile or missile never coincides precisely with the calculated trajectory, and the point of impact on the target or within the target area (or volume) inevitably deviates from the aim point.

In firing impact projectiles or missiles against a surface target (tank or ship), the random deviation of the impact point from the target aim point is measured by the coordinates x and y in the dispersion plane passing through the target. The dispersion plane can be actual, such as the surface of the Earth, or imagined, as is the case of an air combat as shown in Figs. 2.1 and 2.2. Likewise, for proximity fuse warheads the explosion point in space is measured by the coordinates x, y, and z in the dispersion volume as shown in Fig. 2.3 for the ground-to-air defense weapons [surface-to-air missiles (SAMs) and antiaircraft artillery (AAA)]. The random

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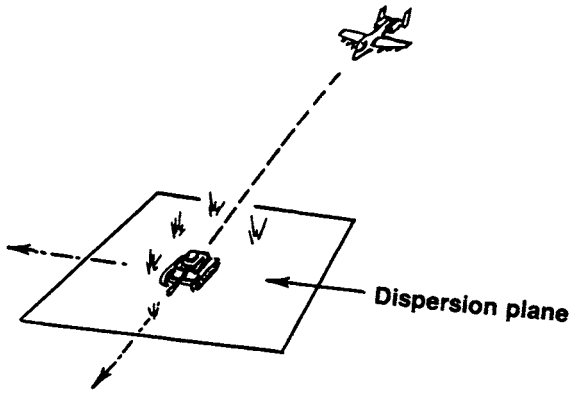


Fig. 2.1 Dispersion plane for air-to-ground attack.

nature of the dispersion, of course, does not allow the use of any deterministic measures of weapon performance. Consequently, special probabilistic parameters of the weapon system must be used to measure its accuracy.

To illustrate the process of obtaining the accuracy of a weapon system, the results of a gun firing test for a certain type of ammunition will be analyzed. A typical dispersion pattern of the impact points is illustrated in Fig. 2.4. To simplify the analysis, however, the cross-range or deflection dispersion will not be considered. Thus, the purpose of the test is to determine only the weapon range for a given aim point. The gun is fired on an instrumented range in which markers are placed at 20-m intervals and special instruments are set up to count the impacts between the markers. The aim point is located at 1450 m from the gun. During the test, the results shown in Table 2.1 were obtained.

The results of this test may be conveniently illustrated by a frequency histogram shown in Fig. 2.5, in which the frequency of occurrence (number of impacts, f) is plotted for each 20-m interval over the complete range. The width of each block in the histogram represents the group interval of 20-m; and the height of each block

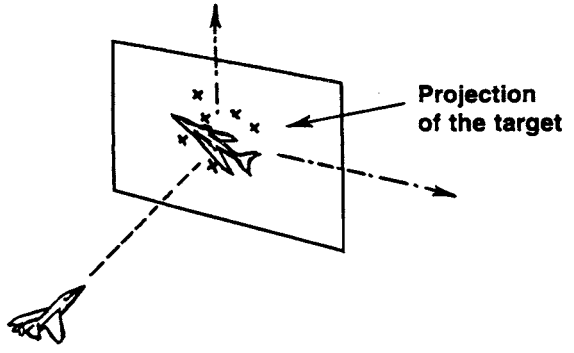


Fig. 2.2 Dispersion plane for air-to-air combat.

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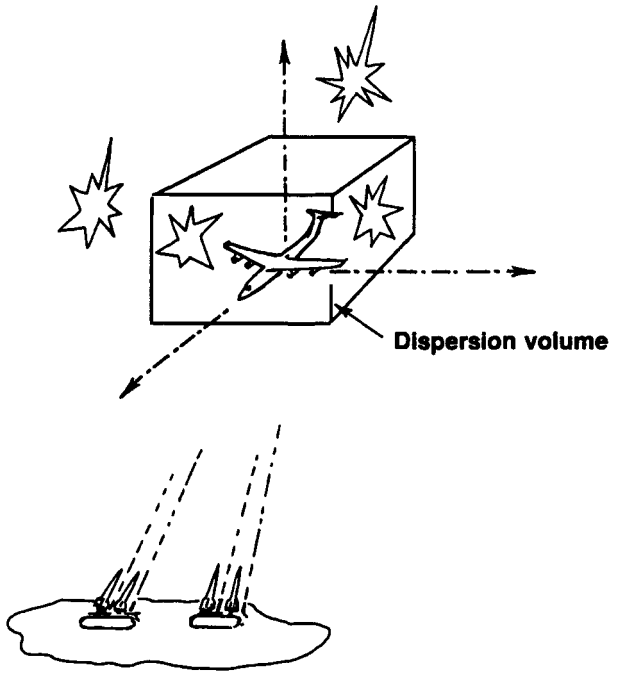


Fig. 2.3 Dispersion volume for ground-to-air defense.

represents the number of impacts in the corresponding group. If the midpoints on the tops of the frequency histogram are joined by straight lines, the resulting figure is a frequency polygon.

To compare distributions containing different total numbers of firings, the vertical scale can be replaced by the ratio of the number of impacts in the given group to the total number of firings (in this case 100). If the number of firings is large, the new vertical scale will represent the probability, p, of a single impact in a given range interval. The resulting histogram then becomes the probability histogram. Likewise, the frequency polygon becomes the probability polygon. If the width of each range interval in the probability histogram is assumed to be equal to unity,

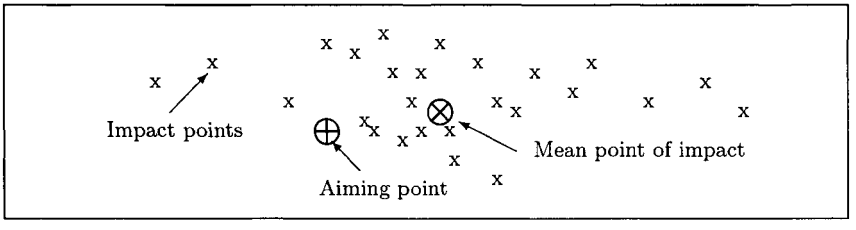


Fig. 2.4 Weapon firing dispersion.

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Table 2.1 Gun Firing Test

Range, m	1340 to 1360	1360 to 1380	1380 to 1400	1400 to 1420	1420 to 1440	1440 to 1460	1460 to 1480	1480 to 1500	1500 to 1520	1520 to 1540	1540 to 1560	1560 to 1580	1340 to 1580
Number of rounds	1	1	4	9	23	27	22	8	3	1	0	1	100

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then the area of the histogram will be unity. The probability histogram can also be used for ranges other than those specified by the range intervals. For example, the probability of an impact between 1420 and 1480 m on the range is obtained by the summation of the probabilities in the intervals 1420–1440, 1440–1460, and 1460–1480, i.e., summation of the probabilities of mutually exclusive events. Hence, using Fig. 2.5, the probability of impact is obtained as,

$$P(1420 < R < 1480) = 0.23 + 0.27 + 0.22 = 0.72$$

where R denotes the location of impact on the range.

If the number of firings is greatly increased and the size of the range intervals correspondingly decreased, then the probability polygon will tend toward a smooth curve. By increasing the number of firings indefinitely, it would be possible, in the limit, to analyze the whole infinite population of firings instead of a small number as in the example of Table 2.1. Thus, the smooth curve obtained by this limiting process represents the distribution of the population of infinite firings just as the probability polygon represents the distribution of the finite sample. This smooth curve is referred to as the probability distribution or probability density.

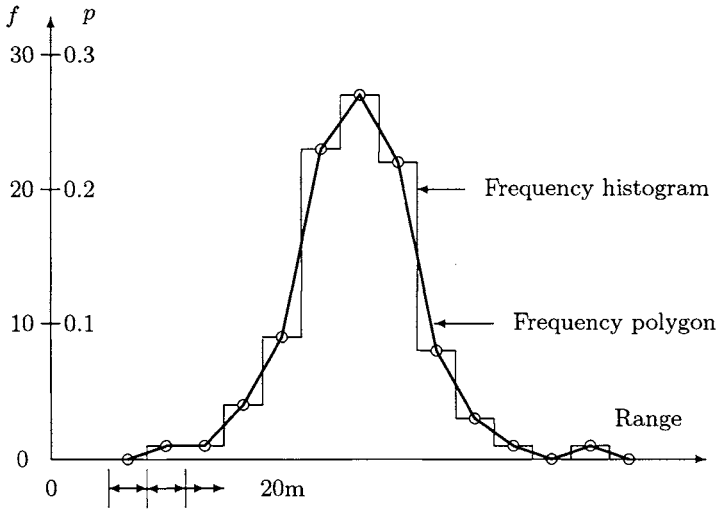


Fig. 2.5 Frequency histogram and polygon for a gun firing test.

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The area under the probability distribution curve between any two ordinates represents the probability that a single observation (impact), taken at random from the complete population, will fall within the corresponding range. This idea is most useful in practical applications because it applies to the whole population instead of to a single sample. This concept is explained more fully in this chapter.

Numerical data, such as those obtained from a gun firing test, can be analyzed further to obtain statistical data useful in evaluating the weapon. The tabular presentation, frequency histogram or polygon, and probability histogram or polygon are all particularly useful in presenting the numerical information; however, it is usually necessary to condense the data even further and to obtain certain characteristic parameters that describe the essential properties of the distribution of the sample. These parameters will then become the estimates for the corresponding parameters for the whole (infinite) population. There are two main parameters of this type. The first parameter provides a measure of central tendency of the data, and the second gives a measure of the scatter or dispersion. Another parameter sometimes used is the median of the finite sample data.

In describing the weapon system characteristics thus far, only the accuracy has been discussed as the main parameter. The other parameter is, of course, the firing rate of the system. This is measured in units of rounds or launched missiles per unit time. The firing rate can be combined with the probability of hit for a single round or a missile to determine the effective firing rate that causes damage or casualties. This concept is discussed in Chapter 4.

2.2 Empirical Averages

The calculation of empirical averages of a particular set of data will be considered next. There are several methods of measuring empirical averages in order to generate a single parameter describing the data. The most useful parameter is the arithmetic mean. The other parameter used in some applications is the median.

Arithmetic Mean (Average)

If there are N observations $X_1, X_2, \ldots, X_N$, then the arithmetic mean is defined as

$$\overline{X} = \frac{1}{N}(X_1 + X_2 + \cdots X_N) = \frac{1}{N}\sum_{k=1}^{N} X_k \tag{2.1}$$

If the observation X_k occurs f_k times, i.e., the observations are clustered as was the case of the gun firing test in Table 2.1, then the arithmetic mean is defined as

$$\overline{X} = \frac{f_1X_1 + f_2X_2 + \cdots + f_NX_N}{f_1 + f_2 + \cdots f_N} = \frac{\sum_{k=1}^{N} f_kX_k}{\sum_{k=1}^{N} f_k} \tag{2.2}$$

Thus, for the example in Table 2.1, using the midpoint observations for X_k,

$$\overline{X} = \frac{1 \times 1350 + 1 \times 1370 + 4 \times 1390 + \cdots}{1 + 1 + 4 + \cdots} = \frac{144{,}900}{100} = 1{,}449\,\text{m}$$

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Median

The median is defined as the middle observation when the observations are ranked in order of size. In the example of Table 2.1, the ordering would be done in terms of the range values. The formal definition of the median is

$$\begin{aligned} M &= \tfrac{1}{2}\left[\left(\tfrac{1}{2}N\right)\text{th observation} + \left(\tfrac{1}{2}N+1\right)\text{th observation}\right]; \quad N \text{ even} \\ M &= \left[\tfrac{1}{2}(N+1)\right]\text{th observation}; \qquad\qquad\qquad\qquad N \text{ odd} \end{aligned} \tag{2.3}$$

To find the median for the gun firing test considered here, it will be assumed that the impact points within the range interval containing the median were uniformly distributed. Since N is even, it is necessary to use the $(\frac{1}{2}N)$th and $(\frac{1}{2}N+1)$th observation, i.e., the 50th and 51st observation:

$$\begin{aligned} \text{50th observation} &= 1440 + 12 \times 20/27 = 1448.9 \\ \text{51st observation} &= 1440 + 13 \times 20/27 = 1449.6 \end{aligned}$$

Hence,

$$M = \tfrac{1}{2}(1448.9 + 1449.6) = 1449.3\,\text{m}$$

which, in this case, is very close to the arithmetic mean of 1449 m.

2.3 Scatter or Dispersion

The arithmetic mean by itself is not sufficient to give a complete picture of the distribution of statistical data. For example, the arithmetic mean of a set of three numbers, 9.99, 10.00, and 10.01, is 10.00, but so is the mean of 1.0, 10.0, and 19.0. Obviously, the mean number 10 is more representative of the first set than it is of the second set. The difference in the two sets of three numbers lies in the scatter or dispersion about the mean. This implies that another parameter is required to measure in some convenient way the amount of scatter or dispersion. This would indicate how closely the original set of observations is packed around the mean. Just as in the case of the empirical averages of data, there are several ways in which the scatter or dispersion can be measured. These are represented by such parameters as range, interquartile range, mean deviation, standard deviation, and coefficient of variation.

Range

This is simply taken as the total range in the sample from the smallest to the largest observation. Although easy to calculate, it is not very representative because it can vary considerably between different samples taken from the same population. Furthermore, it does not convey any information regarding how closely the observations are clustered around the mean.

Interquartile Range

When observations are ranked and the median is found, this process divides the observations into two equal parts. The medians of these two equal parts are then

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found, and these two new medians are referred to as the lower and upper quartile. The range between the two quartiles is the interquartile range. It contains 50% of all observations clustered on either side of the median. The interquartile range will also vary from sample to sample of the same population, but it is more stable in this respect than the range.

Mean Deviation

The mean deviation about the arithmetic mean is defined as the arithmetic mean of the absolute values of the deviations from the mean. Mathematically this is expressed as

$$S = \frac{1}{N}\sum_{k=1}^{N} |X_k - \overline{X}| \tag{2.4}$$

for single observations and

$$S = \frac{\sum_{k=1}^{N} f_k |X_k - \overline{X}|}{\sum_{k=1}^{N} f_k} \tag{2.5}$$

for clustered observations where, as before, f_k is the frequency of the X_k observation.

Similar expressions can be used for the mean deviation about the median by substituting M for $\overline{X}$ in Eqs. (2.4) and (2.5).

Standard Deviation

The standard deviation is defined as the root-mean-square of the deviation from the arithmetic mean and is usually denoted by σ. Thus, for separate observations,

$$\sigma = \pm\left[\frac{1}{N}\sum (X_k - \overline{X})^2\right]^{\frac{1}{2}} \tag{2.6}$$

where $\sum$ denotes summation from $k = 1$ to N. For clustered observations,

$$\sigma = \pm\left[\frac{\sum f_k (X_k - \overline{X})^2}{\sum f_k}\right]^{\frac{1}{2}} \tag{2.7}$$

The σ^2 is known as the variance. Equations (2.6) and (2.7) can be simplified as follows. From Eq. (2.6)

$$\begin{aligned}\sigma^2 &= \frac{1}{N}\left(\sum X_k^2 - 2\sum X_k \overline{X} + \sum \overline{X}^2\right) \\ &= \frac{1}{N}\left(\sum X_k^2 - 2\overline{X}\sum X_k + \overline{X}^2 N\right)\end{aligned} \tag{2.8}$$

Substitution of Eq. (2.1) into Eq. (2.8) leads to

$$\sigma^2 = \frac{1}{N}\left(\sum X_k^2 - \overline{X}^2 N\right)$$

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Table 2.2 Calculation of Standard Deviation

X_k	X_k^2	f_k	$f_k X_k^2$
1350	1.8225×10^6	1	1.8225×10^6
1370	1.8769×10^6	1	1.8769×10^6
1390	1.9321×10^6	4	7.7284×10^6
1410	1.9881×10^6	9	17.8929×10^6
1430	2.0449×10^6	23	47.0327×10^6
1450	2.1025×10^6	27	56.7675×10^6
1470	2.1609×10^6	22	47.5398×10^6
1490	2.2201×10^6	8	17.7608×10^6
1510	2.2801×10^6	3	6.8403×10^6
1530	2.3409×10^6	1	2.3409×10^6
1550	2.4025×10^6	0	0
1570	2.4649×10^6	1	2.4649×10^6
	Totals	100	210.0676×10^6

Hence,

$$\sigma = \pm \left(\sum \frac{X_k^2}{N} - \overline{X}^2 \right)^{\frac{1}{2}} \tag{2.9}$$

which is, of course, simpler to use than Eq. (2.6). For clustered observations, Eq. (2.7) can be simplified as

$$\sigma = \pm \left(\frac{\sum f_k X_k^2}{\sum f_k} - \overline{X}^2 \right)^{\frac{1}{2}} \tag{2.10}$$

The results of the gun firing tests from Table 2.1 will now be used to determine the standard deviation. The numerical calculations are shown in Table 2.2. Noting that the arithmetic mean $\overline{X} = 1449$, the standard deviation is calculated from Eq. (2.10) to give

$$\sigma = \pm \left(\frac{210.0676 \times 10^6}{100} - 2.0996 \times 10^6 \right)^{\frac{1}{2}}$$

$$= \pm 32.79 \text{ m}$$

Coefficient of Variation

It is sometimes useful to express the standard deviation as a percentage of the arithmetic mean. This is obtained from the formula

$$c = \frac{\sigma}{\overline{X}} \times 100\% \tag{2.11}$$

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Thus, for the example of the gun firing test, the coefficient of variation is

$$c = \frac{32.79}{1449} \times 100\% = 2.26\%$$

2.4 Mean Point of Impact: Systematic Error

The mean point of impact is defined as the arithmetic mean of all of the observations. Its location is equivalent to the position of the center of gravity of unit masses placed at the points of impact of all rounds.

If the dispersion of weapon rounds is analyzed, it is found that, in general, the mean point of impact does not coincide with the aiming point. This implies that the rounds were in effect aimed at the mean point of impact and not at the aiming point. The difference between the two points is referred to as the systematic error. Thus, if a particular round impact is selected from all of those shown in Fig. 2.4, the reason that it failed to hit the aiming point is twofold: 1) the usual randomness characteristics of the weapon and 2) the systematic error.

In statistics, systematic errors are defined as those that can be foreseen and therefore eliminated; however, in weapon applications this is not necessarily true. For example, in high-altitude bombing the systematic error might be partly due to an instrument error of the bombsight itself or a personal error of the bombsight operator, all of which can be compensated for, and partly due to errors caused by inadequate knowledge of the wind velocity, the aircraft maneuver while releasing the bombs, and the structural deformations of the bomb platform.

Problems

2.1. The point of impact of some 25 rounds of mortar fire from the same weapon resulted in the following data, measured in 10-m intervals along the range:

Point of impact, m	500–510	510–520	520–530	530–540	540–550
Number of impacts	2	6	11	5	1

2.2. Show that the location in range of the arithmetic mean point of impact is the same as the location of the center of gravity of unit masses placed at the points of impact of individual rounds.

2.3. Derive Eq. (2.10).

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3
Passive Targets

Starting with the concept of the probability of a single hit (and the companion concept of a single miss), this chapter develops three related models of the probability distributions: the binomial distribution, the normal (Gaussian) distribution, and the Poisson distribution. The normal distribution is then applied to calculate the probabilities of destruction of a single-point passive target placed 1) along the line of sight, 2) in a plane, and 3) in space for cases where the aiming point either coincides with the target or is placed at some specified offset distance from the target. These probabilities are derived for the "cookie cutter" damage functions; i.e., the target is destroyed if it is within a radius R from the weapon impact point and is saved if it is outside that radius. For a point target in a two-dimensional space, four different cases are considered: 1) zero aiming offset and equal variances; 2) offset aiming and equal variances; 3) zero aiming offset and unequal variances; and 4) offset aiming and unequal variances. The concepts of linear error probable, circular error probable, and spherical error probable are introduced as simple physical measures of weapon accuracy. In addition, the fractional damage of a uniform value circular target is discussed as a typical example of more complex calculations. Four different damage functions are also discussed with the objective of introducing a more accurate representation of the damage mechanism. The following damage functions are presented: Gaussian, exponential, inclined step (including a step function), log-normal, and shrapnel damage. Finally, a new concept of an effective weapon radius is introduced that can be used to replace the actual damage function with an equivalent cookie cutter function.

3.1 Binomial Distribution

In this section the probability of multiple hits on a stationary passive target will be considered when the probability of a single hit by an individual weapon is known. It will be assumed that the single-hit probability is p and the probability of miss is q, so that $p + q = 1$. Furthermore, it will also be assumed that these probabilities are the same for all of the weapons of the same kind. Therefore, the subsequent analysis will be equally applicable to cases of multiple shots either from several identical weapons or from a single weapon.

If n shots are expended, then the probability of zero hits is clearly

$$P_n(0) = q\,q\,q\cdots(n \text{ times}) = q^n \tag{3.1}$$

The probability that the first shot is a hit and the rest are misses is

$$p\,q\,q\cdots$$

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where q appears $(n-1)$ times. The probability that the second shot is a hit while the rest are misses is

$$q\ p\ q\ \cdots$$

where q also appears $(n-1)$ times. And finally, the probability of the nth shot being a hit while the rest are misses is

$$q\ q\ q\ \cdots q\ p$$

Thus, the probability of one hit in n shots is the probability associated with all of the individual outcomes, i.e.,

$$P_n(1) = npq^{n-1} \tag{3.2}$$

Using similar reasoning, the general case of the probability of r hits in n shots can now be considered. The total number of events representing r hits in n shots is equal to the number of ways r objects can be arranged in n spaces, which is given by the combinatorial formula

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

Hence, the probability of r hits in n shots is equal to

$$P_n(r) = \binom{n}{r} p^r q^{n-r} = \frac{n!}{r!\,(n-r)!} p^r q^{n-r} \tag{3.3}$$

which is valid for $r = 0, 1, 2, \ldots, n$. Equation (3.3) represents the probability distribution of hits in n shots. Coincidently, the formula for $P_n(r)$ is the same as the $(r+1)$th term in the binomial expansion of $(q+p)^n$; therefore, it is usually referred to as the binomial probability distribution. The summation of all probabilities $P_n(r)$ represents all possible events; i.e., it represents a certainty. Therefore,

$$\sum_{r=0}^{n} P_n(r) = 1 \tag{3.4a}$$

To verify the preceding statement, Eq. (3.3) is substituted into Eq. (3.4a). The resulting series is indeed the binomial expansion of $(p+q)^n$, i.e.,

$$\binom{n}{0} p^0 q^n + \binom{n}{1} p^1 q^{n-1} + \binom{n}{2} p^2 q^{n-2} + \cdots + \binom{n}{n} p^n q^0 = (p+q)^n = 1 \tag{3.4b}$$

In many actual applications one is interested not in the probability that there will be a specific number of r hits but in the probability that there will be *at least*

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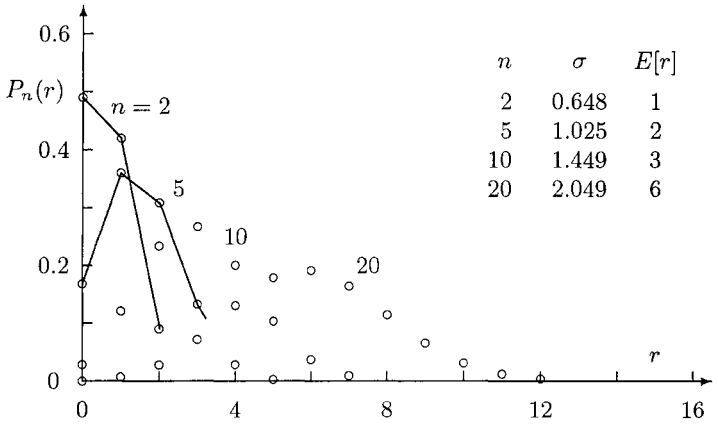


Fig. 3.1 Binomial probability distribution $P_n(r)$ for $p = 0.3$.

r hits. This is obtained from

$$P_n(r \geq r_{\min}) = \sum_{r=r_{\min}}^{n} P_n(r)$$

$$= \sum_{r=r_{\min}}^{n} \binom{n}{r} p^r (1-p)^{n-r} \tag{3.5}$$

Equation (3.5) is presented in a tabular form in Appendix A (Table A.1) for a range of values of p, n, and r. It should be noted here that, when $r_{\min}$ is small, it may be computationally more advantageous to combine Eqs. (3.4a) and (3.5) so that

$$P_n(r \geq r_{\min}) = 1 - \sum_{r=0}^{r_{\min}-1} P_n(r) \tag{3.6}$$

To illustrate the binomial probability distribution, Fig. 3.1 gives the variation of $P_n(r)$ with r for the values of $n = 2, 5, 10$, and 20 and $p = 0.3$. This demonstrates clearly that for a fixed value of p the graph of the binomial probability distribution spreads out as n increases, whereas its peak value decreases.

Keeping in mind the fact that probability represents a model for relative frequency of occurrence, the statistical average of the discrete random variable r may be defined by

$$E[r] = \sum_{r=0}^{n} r P_n(r) \tag{3.7}$$

This statistical average is often called the mean or the expectation (hence the symbol E) or the expected value. It should be noted that this value is equivalent to the location of the center of gravity of weights equal to the probability $P_n(r)$ placed at r with the total weight equal to unity [because the condition (3.4a) must be satisfied]. Equation (3.7) is analogous to Eq. (2.2) for the mean value $\overline{X}$ if r is substituted for X_k and $P_n(r)$ for f_k and $\sum f_k = 1$.

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To calculate the expected value of the binomial distribution, Eq. (3.3) is substituted into Eq. (3.7). This leads to

$$\begin{aligned} E[r] &= \sum_{r=0}^{n} \frac{n!\,r}{r!\,(n-r)!} p^r q^{n-r} \\ &= \sum_{r=1}^{n} \frac{n!\,r}{r!\,(n-r)!} p^r q^{n-r} \\ &= np \sum_{r=1}^{n} \frac{(n-1)!}{(r-1)!\,(n-r)!} p^{r-1} q^{n-r} \\ &= np \sum_{s=0}^{n-1} \frac{(n-1)!}{s!\,(n-1-s)!} p^s q^{n-1-s} \\ &= np(p+q)^{n-1} = np \end{aligned} \tag{3.8}$$

where a new variable $s = r-1$ was introduced. Thus, the expected value for the binomial probability distribution is $E[r] = np$. This occurs near a point where the probability reaches a maximum value. In general, the value $E[r] = np$ will not be an integer. Therefore, in military applications it will be necessary to approximate to an integer to represent the expected number of r hits in the total of n shots, as shown in Fig. 3.1.

To determine the variance of the binomial probability distribution, the following substitutions must be made: $X_k = r$, $f_k = P_n(r)$, and $\overline{X} = E[r]$ in Eq. (2.10). This leads to

$$\begin{aligned} \sigma^2 &= \sum_{r=0}^{n} r^2 P_n(r) - E^2[r] \\ &= \sum_{r=0}^{n} \frac{n!\,r^2}{r!\,(n-r)!} p^r q^{n-r} - n^2 p^2 \\ &= \sum_{r=1}^{n} \frac{n!\,r^2}{r!\,(n-r)!} p^r q^{n-r} - n^2 p^2 \\ &= np \sum_{r=1}^{n} \frac{(n-1)!\,r}{(r-1)!\,(n-r)!} p^{r-1} q^{n-r} - n^2 p^2 \\ &= np \sum_{s=0}^{n-1} \frac{(n-1)!\,(s+1)}{s!\,(n-1-s)!} p^s q^{n-1-s} - n^2 p^2 \\ &= np \sum_{s=0}^{n-1} \frac{(n-1)!\,s}{s!\,(n-1-s)!} p^s q^{n-1-s} + np \sum_{s=0}^{n-1} \frac{(n-1)!}{s!\,(n-1-s)!} p^s q^{n-1-s} - n^2 p^2 \\ &= np(n-1)p + np - n^2 p^2 = np(1-p) = npq \end{aligned} \tag{3.9}$$

and

$$\sigma = \pm[np(1-p)]^{\frac{1}{2}} \tag{3.10}$$

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3.2 Normal (Gaussian) Distribution

The binomial probability distribution becomes very awkward to use when n is very large. To simplify the practical application of the binomial distribution, this discrete distribution can be modified for large values of n into a continuously varying distribution. The derivation of this approximation makes use of the following double inequality[1]:

$$\sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n+1/(12n+1)} < n! < \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n+1/12n} \tag{3.11}$$

For large values of n when $1/(12n+1)$ and $1/12n$ can be neglected compared with n, this inequality leads to the Stirling's formula for $n!$:

$$n! \cong \sqrt{2\pi n}\, n^n e^{-n} \tag{3.12}$$

If Eq. (3.12) is substituted into Eq. (3.3), it can be shown that

$$P_n(r) = \left[\frac{n}{2\pi r(n-r)}\right]^{\frac{1}{2}} \left(\frac{np}{r}\right)^r \left(\frac{nq}{n-r}\right)^{n-r} \tag{3.13}$$

Next, a new variable defined as

$$\delta = r - np \tag{3.14}$$

is introduced that represents a distance measured from the mean value of the distribution $\bar{x} = np$. Equation (3.13) can now be rewritten in terms of this new variable so that

$$\begin{aligned} P_n(r) &= \left[\frac{n}{2\pi(np+\delta)(nq-\delta)}\right]^{\frac{1}{2}} \left(\frac{np}{np+\delta}\right)^{np+\delta} \left(\frac{nq}{nq-\delta}\right)^{nq-\delta} \\ &= \left[\frac{1}{2\pi npq(1+\delta/np)(1-\delta/nq)}\right]^{\frac{1}{2}} \left(\frac{1}{1+\delta/np}\right)^{np+\delta} \left(\frac{1}{1-\delta/nq}\right)^{nq-\delta} \end{aligned} \tag{3.15}$$

The denominator in the first factor in Eq. (3.15) for large n can be approximated for fixed values of δ as

$$2\pi npq(1+\delta/np)(1-\delta/nq) \cong 2\pi npq \tag{3.16}$$

unless p is close to zero or unity, causing either np or $nq = n(1-p)$ to be very small even for large values of n. To obtain an approximate expression for the second and third factor in Eq. (3.15), a new variable is introduced:

$$Y = \left(\frac{1}{1+\delta/np}\right)^{np+\delta} \left(\frac{1}{1-\delta/nq}\right)^{nq-\delta} \tag{3.17}$$

so that

$$\ln Y = (np+\delta)\ln\left(1+\frac{\delta}{np}\right) + (nq-\delta)\ln\left(1-\frac{\delta}{nq}\right)$$

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Noting that

$$\ell n(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

for $|x| < 1$, it follows that

$$\ell n\, Y = (np+\delta)\left(\frac{\delta}{np} - \frac{\delta^2}{2n^2p^2} + \cdots\right) + (nq-\delta)\left(-\frac{\delta}{nq} - \frac{\delta^2}{2n^2q^2} - \cdots\right)$$

$$= \delta + \frac{\delta^2}{np} - \frac{\delta^2}{2np} + \cdots - \delta + \frac{\delta^2}{nq} - \frac{\delta^2}{2nq} + \cdots$$

$$\cong \frac{\delta^2}{2np} + \frac{\delta^2}{2nq} = (p+q)\frac{\delta^2}{2npq} = \frac{\delta^2}{2npq}$$

Hence,

$$Y = \exp\left(-\frac{\delta^2}{2npq}\right) \tag{3.18}$$

Substitution of Eqs. (3.10) and (3.18) into Eq. (3.15) results in the approximate expression for $P_n(r)$ when n is large, i.e.,

$$P_n(r) = \frac{1}{\sqrt{2\pi npq}} \exp\left(-\frac{\delta^2}{2npq}\right) \tag{3.19}$$

Finally, the substitution of $\sigma^2 = npq$, $r = x$, $\overline{x} = np$, and $\delta^2 = (x-\overline{x})^2$ leads to

$$P_n(r) = p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left[-\frac{(x-\overline{x})^2}{2\sigma^2}\right] \tag{3.20}$$

which is described as the density of normal distribution of probability. It should be noted that, because of large n, x can be treated as a continuous variable instead of the discrete variable r. This formula can also be used as a reasonable approximation for the binomial distribution for large n ($n > 20$), provided $0.1 < p < 0.9$.

3.3 Poisson Distribution

The normal distribution becomes inaccurate as an approximation to the binomial distribution when n increases while p decreases in such a way that the product np remains essentially constant and $p < 0.1$. For these cases an alternative approximation is possible as described in the following.

For $r = 0$ it follows from Eq. (3.3) that

$$P_n(0) = (1-p)^n \tag{3.21}$$

Similarly, for $r = 1$,

$$P_n(1) = \lambda(1-p)^{n-1} \tag{3.22}$$

where the parameter $\lambda = np$ (const) is introduced. Now, from Eq. (3.21),

$$\ell n\, P_n(0) = n\ \ell n(1-p) = n\left(-p - \frac{p^2}{2} - \frac{p^3}{3} - \cdots\right)$$

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and, because p is small,

$$\ln P_n(0) = -np = -\lambda$$

Hence,

$$\lim_{n\to\infty} P_n(0) = P(0) = e^{-\lambda} \tag{3.23}$$

Similarly, from Eq. (3.22),

$$\ln P_n(1) = \ln \lambda + (n-1)\ln(1-p)$$
$$\cong \ln \lambda - np$$

Hence,

$$\ln\left[\frac{P_n(1)}{\lambda}\right] = -\lambda$$

and

$$\lim_{n\to\infty} P_n(1) = P(1) = \lambda e^{-\lambda} \tag{3.24}$$

Also, by the same reasoning,

$$P(2) = \frac{\lambda^2}{2!}e^{-\lambda} \tag{3.25}$$

and for a general term it follows by induction that

$$P(r) = \frac{\lambda^r}{r!}e^{-\lambda} = \frac{(np)^r}{r!}e^{-np} \tag{3.26}$$

This approximation to the binomial distribution when n is very large and p is very small is the Poisson distribution.

The Poisson distribution may be used to represent military operations in which events occur randomly in space or time at a constant average rate λ. Some of the applications include the following:

1) Analysis of queuing problems when items are arriving randomly in a queue at a constant rate, e.g., awaiting maintenance or battle damage repair.

2) Analysis of component reliability when failures of a particular component occur randomly at a constant rate.

3) Assessment of battle damage when weapon projectiles or fragments are distributed randomly over a specified area.

Just as in the binomial distribution, the Poisson distribution must satisfy

$$\sum_{r=0}^{\infty} P(r) = 1 \tag{3.27}$$

This can be demonstrated easily by substituting Eq. (3.26) into Eq. (3.27) so that

$$P(0) + P(1) + P(2) + \cdots = e^{-\lambda}\left(\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \cdots\right) = e^{-\lambda}e^{\lambda} = 1$$

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Another useful concept is the cumulative Poisson probability, i.e., the probability that at least a specified number of hits will occur. This is obtained from

$$P(r \geq r_{\min}) = \sum_{r=r_{\min}}^{\infty} P(r)$$

$$= \sum_{r=r_{\min}}^{\infty} \frac{\lambda^r}{r!} e^{-\lambda} = 1 - \sum_{r=0}^{r_{\min}-1} \frac{\lambda^r}{r!} e^{-\lambda} \tag{3.28}$$

For convenience, Eq. (3.28) is also tabulated in Appendix A (Table A.2) for a range of values of λ between 0.1 and 5 and r from 1 to 15.

As an example of the application of the Poisson distribution, consider the following case. At a certain location away from an exploding bomb, the average number of shrapnel fragments per square foot is 0.1. It is necessary to determine the probability that at least one fragment will strike a target area representing a frontal area of 2 ft^2. In this case,

$$\lambda = 0.1 \times 2 = 0.2 \text{ fragments/2 ft}^2 \text{ of area or } 0.2 \text{ fragments/target area}$$

The probability of zero fragments hitting the target area is

$$P(0) = \frac{(0.2)^0}{0!} e^{-0.2} = 0.8187$$

and the probability of at least one hit is therefore,

$$P(r) = 1 - P(0) = 1 - 0.8187 = 0.1813$$

3.4 Probability Density Distribution: Linear Error Probable (LEP)

As was demonstrated in Sec. 3.2, the normal probability distribution is expressed by

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left[-\frac{(x-\bar{x})^2}{\sigma^2}\right] \tag{3.20}$$

This distribution is symmetrical about the mean position (e.g., mean point of impact) and has the general bell shape as indicated in Fig. 3.2. Next it will be necessary to define some basic concepts pertaining to the calculation of probabilities from the normal probability density distribution curve.

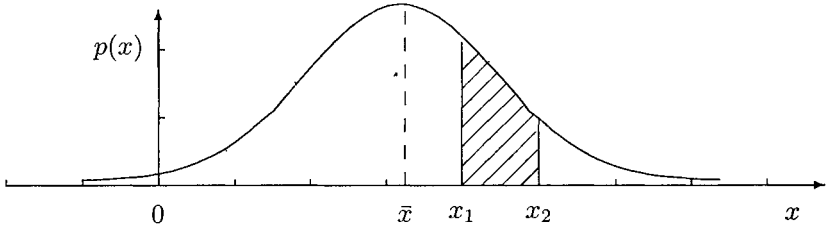


Fig. 3.2 Probability of x occurring between x_1 and x_2.

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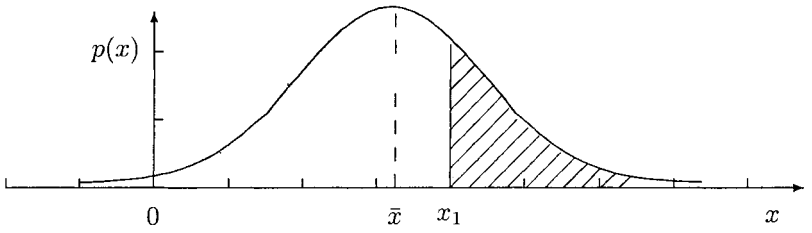


Fig. 3.3 Probability of x occurring between x_1 and infinity.

The area under the curve between x_1 and x_2, i.e.,

$$P(x_1 < x < x_2) = \int_{x_1}^{x_2} p(x)\,dx \tag{3.29}$$

represents the probability of x occurring between x_1 and x_2 (Fig. 3.2). The area under the curve between x_1 and $x = \infty$ (the tail area of distribution), i.e.,

$$P(x_1 < x < \infty) = \int_{x_1}^{\infty} p(x)\,dx \tag{3.30a}$$

represents the probability of x occurring between x_1 and ∞ (Fig. 3.3). The complete area under the curve represents the probability of x occurring between $-\infty$ and $+\infty$, which, of course, must be a certainty of occurrence represented by unity so that

$$P(-\infty < x < +\infty) = \int_{-\infty}^{+\infty} p(x)\,dx = 1 \tag{3.31}$$

The formal verification of the preceding equation will be shown later in this chapter when the circular error probable is discussed.

In practical applications Eq. (3.30a) is usually computed in terms of u standard deviations from the mean value $\bar{x}$ as defined by

$$x = \bar{x} + u\sigma \tag{3.32a}$$

or

$$u = \frac{x - \bar{x}}{\sigma} \tag{3.32b}$$

Substitution of Eq. (3.20) into Eq. (3.30a) and change in the variable x into u through Eq. (3.32b) with $dx = \sigma\,du$ finally leads to

$$P(u_1 < u < \infty) = \frac{1}{\sqrt{2\pi}} \int_{u_1}^{\infty} e^{-\frac{1}{2}u^2}\,du \tag{3.30b}$$

which is independent of the standard deviation σ. The preceding equation cannot be evaluated in terms of known functions. It has therefore been computed numerically and is available as special tables (see Appendix A, Table A.3).

In most statistical work and in the analysis of navigational errors, the convenient measure of error is the standard deviation, but in weapon effect analysis one generally uses the 50% error, or the so-called error probable. This implies that the tail areas on either side of the mean value $\bar{x}$ must be equal to 0.25 as shown in

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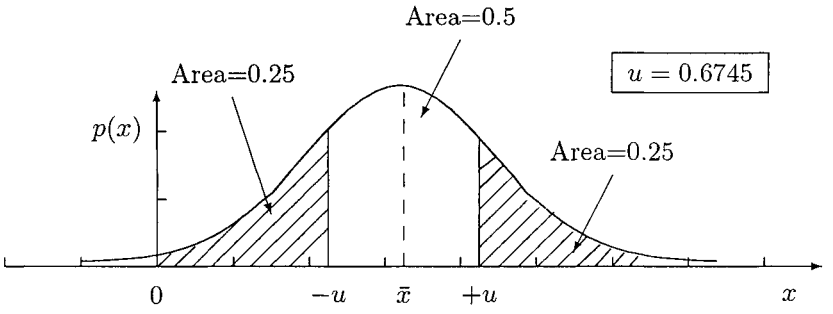


Fig. 3.4 Linear error probable or probability of 50% error.

Fig. 3.4. From Table A.3 it follows that for this case $u = \pm 0.6745$. And because the total area under the curve is equal to unity, the central area is therefore equal to 0.50 and defines the linear error probable (LEP). Hence, using the definition for u in Eq. (3.32b), the LEP in range is given by

$$\text{LEP}_{\text{range}} = 0.6745\sigma \tag{3.33}$$

The application of Eq. (3.33) will be illustrated by an example. For a particular tank gun the range error probable is 30 m. It is required to find the probability of hitting a target 20 m long when aiming at the center of the target. It will be assumed that there is no systematic error and the normal distribution is centered around the aiming point (see Fig. 3.5). In this case, $\bar{x} = 0$, $\sigma = 30/0.6745 = 44.48$ m, $x = \pm 10$ m, and $u = \pm 10/44.48 = \pm 0.225$.

From Table A.3,

$$P(u > 0.225) = 0.4110$$

Hence, the probability of hitting the target is

$$P(-0.225 < u < +0.225) = 1 - 2 \times 0.4110 = 0.178$$

This value represents the single-shot probability of hitting the target and is usually denoted by P_{SSH}. The probability of miss is therefore $(1 - P_{\text{SSH}})$. If the target is fired upon n times, then the probability of miss is $(1 - P_{\text{SSH}})^n$, and the probability

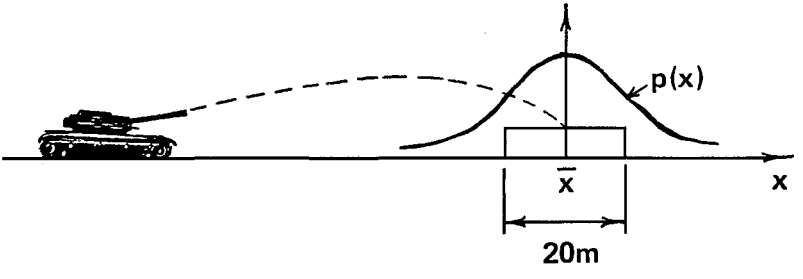


Fig. 3.5 Target within a normal probability distribution $p(x)$.

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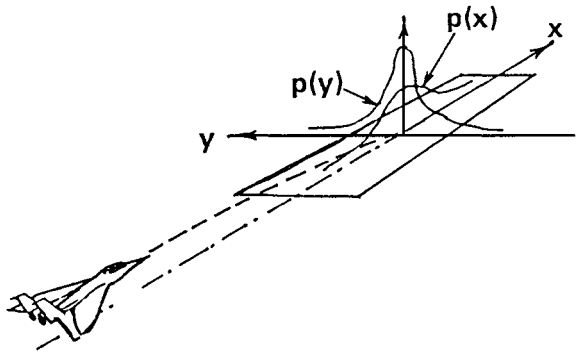


Fig. 3.6 Aircraft attacking a rectangular ground target.

of hit is

$$P_n = 1 - (1 - P_{\text{SSH}})^n \tag{3.34}$$

3.5 Single-Shot Probability of Hit for a Rectangular Target

In this section a target area represented by a rectangle will be considered. The sides of the rectangle will be assumed to be parallel to the range and cross-range directions for which the standard deviations σ_x and σ_y or the range error probable (REP) and the cross-range error probable (CREP) are known (see Fig. 3.6). Furthermore, to simplify the results the range and cross-range errors will be assumed to be independent so that the probability distribution for the target becomes simply the product of the range and cross-range probabilities, i.e.

$$p(x, y) = p(x)p(y) \tag{3.35}$$

where

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left[-\frac{x^2}{2\sigma_x^2}\right] \tag{3.36}$$

$$p(y) = \frac{1}{\sqrt{2\pi}\sigma_y} \exp\left[-\frac{y^2}{2\sigma_y^2}\right] \tag{3.37}$$

The expressions for $p(x)$ and $p(y)$ assume that the normal density distributions are centered around the aiming point. Hence, the probability of hitting a rectangular target of dimensions $2a \times 2b$ becomes

$$P = \iint_{\text{TargetArea}} p(x, y)\,dx\,dy = \int_{-b}^{b}\int_{-a}^{a} p(x)p(y)\,dx\,dy$$

$$= \int_{-a}^{a} p(x)\,dx \int_{-b}^{b} p(y)\,dy = P_{\text{range}} \times P_{\text{cross-range}}$$

$$= A \times B \tag{3.38}$$

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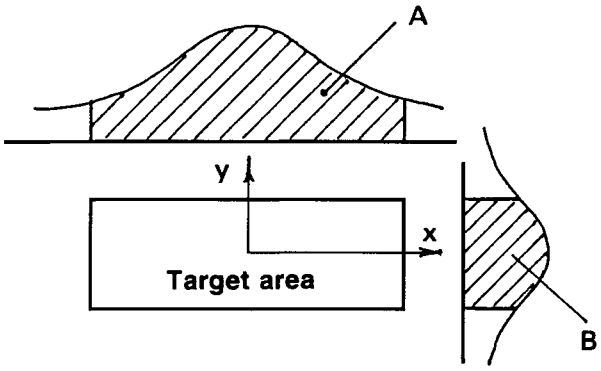


Fig. 3.7 Probability distribution in range and cross-range directions.

where the two probability density distributions are shown in Fig. 3.7 with the corresponding areas under the probability density curves for the range and cross range within the target designated by the shaded areas A and B.

To facilitate the calculation of probabilities, tables of the shaded areas in terms of the ratio of target dimensions to the error probables can be used. For convenience, such a table is reproduced in Appendix A (see Table A.5).

3.6 Probability of Target Kill or Hit for Multiple Shots

If the probability of kill given hit is $P(K|H)$, then the probability of survival after one hit is $1 - P(K|H)$, and for $z - 1$ hits it is $(1 - P(K|H))^{z-1}$. If on the zth hit the target is killed, then the probability of this event is

$$P_z(K) = \text{Probability of survival after } (z-1) \text{ hits} \times \text{Probability of kill after one hit}$$

$$= [1 - P(K|H)]^{z-1} P(K|H) \tag{3.39}$$

The preceding equation is plotted in Fig. 3.8 for a series of values of $P(K|H)$. It should also be noted that Eq. (3.39) represents a probability distribution for which

$$\sum_{z=1}^{\infty} P_z(K) = 1 \tag{3.40}$$

must be satisfied. To demonstrate this important condition, Eq. (3.39) is substituted into Eq. (3.40) so that

$$\sum_{z=1}^{\infty} [1 - P(K|H)]^{z-1} P(K|H)$$

$$= P(K|H) + P(K|H)[1 - P(K|H)] + P(K|H)[1 - P(K|H)]^2 + \cdots$$

$$= P(K|H)\{1 + [1 - P(K|H)] + [1 - P(K|H)]^2 + \cdots\}$$

$$= \frac{P(K|H)}{1 - [1 - P(K|H)]} = 1$$

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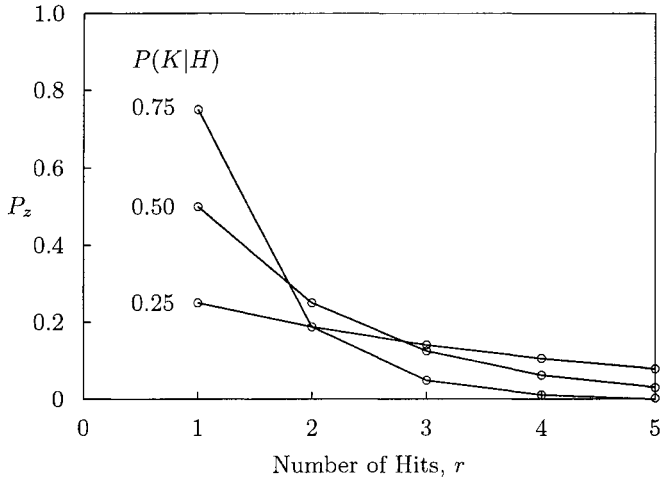


Fig. 3.8 Probability distribution for different values of $P(K|H)$.

where the third line represents a geometric series for which the summation to infinity is known.

By an analogous argument Eq. (3.39) can also be used to determine the probability of a hit on the zth attempt after $z-1$ misses. Thus,

$$P_z(H) = (1 - P_{SSH})^{z-1} P_{SSH} \tag{3.41}$$

where P_{SSH} represents the single-shot probability of hit. The occurrence of at least one hit in n shots is given by

$$P_z(H \geq 1) = \sum_{z=1}^{n} (1 - P_{SSH})^{z-1} P_{SSH} = 1 - (1 - P_{SSH})^n \tag{3.42}$$

To derive the last expression of Eq. (3.42), the summation term must be expanded, i.e.,

$$\begin{aligned} P_n(H \geq 1) &= \sum_{z=1}^{n} (1 - P_{SSH})^{z-1} P_{SSH} \\ &= P_{SSH} + P_{SSH}(1 - P_{SSH}) + P_{SSH}(1 - P_{SSH})^2 + \cdots \\ &\quad + P_{SSH}(1 - P_{SSH})^{n-1} \\ &= P_{SSH}[1 + (1 - P_{SSH}) + (1 - P_{SSH})^2 + \cdots + (1 - P_{SSH})^{n-1}] \end{aligned}$$

It can now be observed that the series expansion on the right is a sum of n terms of a geometric series so that

$$P_n(H \geq 1) = P_{SSH} \frac{1 - (1 - P_{SSH})^n}{1 - (1 - P_{SSH})} = 1 - (1 - P_{SSH})^n$$

The expression $1 - (1 - P_{SSH})^n$ can also be derived from the binomial distribution of probabilities. For the case of at least one hit the binomial probability from

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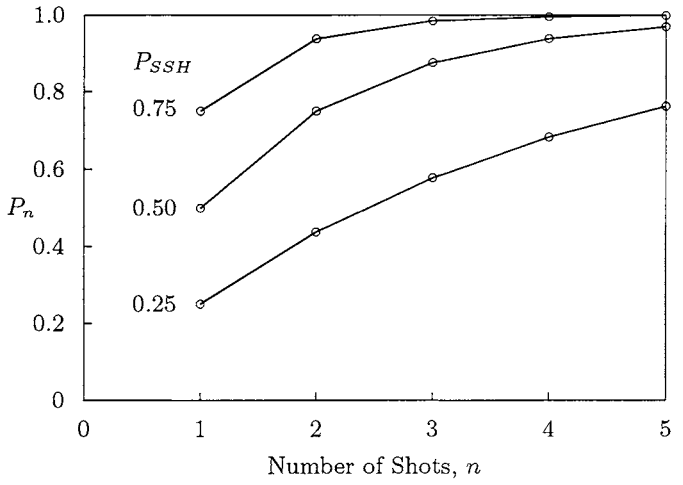


Fig. 3.9 Variation of $P_n(H \geq 1)$ with n, i.e., probability of at least one hit in n shots (P_{SSH} is the single-shot probability of hit).

Eqs. (3.5) and (3.3) is given by

$$P_n(r \geq 1) = \sum_{r=1}^{n} P_n(r) = 1 - \sum_{r=0}^{0} P_n(r) = 1 - P_n(0)$$

$$= 1 - \binom{n}{0} q^n = 1 - (1 - P_{SSH})^n$$

which agrees with the previous results (note that $p = P_{SSH}$).

The variation of $P_n(H \geq 1)$ with n is shown in Fig. 3.9 for $P_{SSH} = 0.25$, 0.5, and 0.75. This demonstrates that for low values of P_{SSH}, an increased number of shots increases significantly the overall probability of hit. On the other hand, for high values of P_{SSH} the improvement is only marginal. Thus for precision-guided munitions (PGM) only one shot per target would be needed.

3.7 Probability of Destruction of a Point Target: Circular Error Probable (CEP)

When the dimensions of the target are small compared with the effective radius of the weapon within which the target can be destroyed, the target itself can then be represented by a point in the mathematical model of the attack. In this section the single-shot probability of target destruction for such point targets will be determined. This probability is usually denoted by P_{SSK}, referring to the single-shot probability of kill. For convenience, however, in the subsequent analysis the subscript SSK will be dropped. If P denotes the probability that a single shot destroys the target, then the probability that the target is destroyed by at least one shot of n

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independently aimed shots is given by Eq. (3.42) with $P_{SSK} = P$, i.e.,

$$P(n \text{ shots}) = 1 - (1 - P)^n \tag{3.43}$$

It will now be assumed that the probability density distribution for the weapon aimed at a point target is given by the simple joint probability distribution

$$p(x, y) = p(x)p(y) \tag{3.35}$$

for which covariance $\sigma_{xy} = 0$ (for definition of covariance any standard textbook on probability can be consulted). Hence, from Eq. (3.20),

$$p(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left[-\frac{(x - x_0)^2}{2\sigma_x^2} - \frac{(y - y_0)^2}{2\sigma_y^2}\right] \tag{3.44}$$

where for convenience x and y coordinates are measured from the point target, (x_0, y_0) are the mean coordinates of the weapon impact distribution, and σ_x and σ_y are the mean standard deviations in the x and y directions, respectively. The probability of impact within a circle of radius R measured from the point target (i.e., the origin of the x and y coordinates) is

$$P = \iint_{(x^2+y^2)^{\frac{1}{2}} < R} p(x, y)\, dx\, dy \tag{3.45}$$

Examination of the expression for $p(x, y)$ in Eq. (3.44) reveals that the following four cases may be considered:

Case 1: $x_0 = y_0 = 0$; $\sigma_x^2 = \sigma_y^2 = \sigma^2$

Case 2: $x_0 \neq 0$, $y_0 \neq 0$; $\sigma_x^2 = \sigma_y^2 = \sigma^2$

Case 3: $x_0 = y_0 = 0$; $\sigma_x^2 \neq \sigma_y^2$

Case 4: $x_0 \neq 0$, $y_0 \neq 0$; $\sigma_x^2 \neq \sigma_y^2$

If the probability of target destruction when the weapon impact is at (x, y) is denoted by $d(x, y)$, then the unconditional probability of target destruction by a single weapon is expressed by the integral

$$P_d = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d(x, y)p(x, y)\, dx\, dy \tag{3.46}$$

The preceding expression can also be used if the target incurs only partial damage. In such cases $d(x, y)$ then represents the expected fraction of the target destroyed when the weapon impacts at (x, y), and $P - d$ then becomes the unconditional expected fraction of the target destroyed.

Most damage functions employed in numerical calculations are circularly symmetric, i.e., $d(x, y)$ may be expressed as a function of a single variable $r = (x^2 + y^2)^{\frac{1}{2}}$. The simplest form of this function is the so-called cookie cutter damage function for which $d(x, y) = 1$ when $r \leq R$ and $d(x, y) = 0$ elsewhere, as shown in Fig. 3.10. Other functions are discussed in Sec. 3.13.

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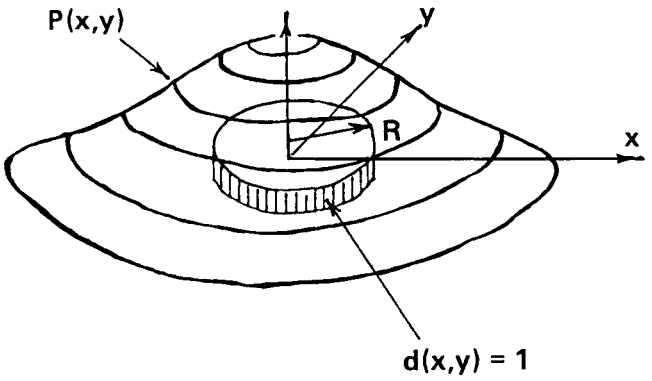


Fig. 3.10 Cookie cutter damage function $d(x, y)$.

It should be observed that *the probability of destruction of a point target for a cookie-cutter damage function with a lethal radius R is the same as the probability of hit within a circle of radius R, i.e.,* $P_d = P$. Thus, although the two probabilities are calculated from identical formulas they represent two different functions.

Case 1: Zero Offset and Equal Variances

For this case $x_0 = y_0 = 0$ and $\sigma_x^2 = \sigma_y^2 = \sigma^2$. Substitution of the cookie cutter function

$$d(x, y) = \begin{cases} 1; & r \leq R \\ 0; & r > R \end{cases} \tag{3.47}$$

and Eq. (3.44) into Eq. (3.46) and the subsequent use of polar coordinates (see Fig. 3.11) leads to

$$\begin{aligned} P_d\left(\frac{R}{\sigma}\right) &= \int_0^{2\pi} \int_0^R \frac{1}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) r\, dr\, d\theta \\ &= \int_0^R \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) dr = -\exp\left(-\frac{r^2}{2\sigma^2}\right)\Bigg|_0^R \\ &= 1 - \exp\left(-\frac{R^2}{2\sigma^2}\right) \end{aligned} \tag{3.48}$$

Similarly, it can be shown from Eqs. (3.44) and (3.45) that the probability of weapon impact within a radius r is given by

$$P(r) = 1 - \exp\left(-\frac{r^2}{2\sigma^2}\right) \tag{3.49}$$

which is identical in form to Eq. (3.48).

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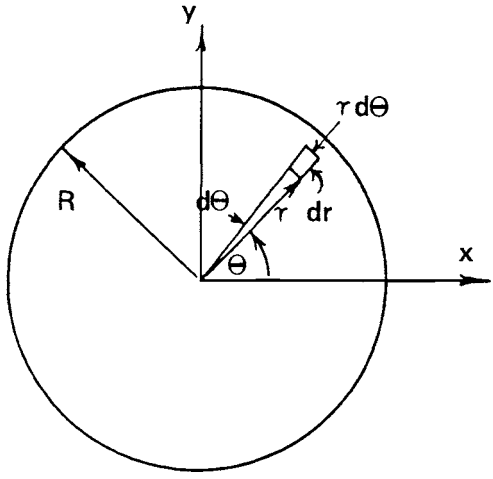


Fig. 3.11 Polar coordinates.

Next, a companion concept to the range or cross-range error probable or the 50% error will be introduced. If R_0 denotes the radius of a circle for which

$$P(R_0) = 1 - \exp\left(-\frac{R_0^2}{2\sigma^2}\right) = \frac{1}{2} \tag{3.50}$$

then the meaning of such a radius is that 50% of all the impact points for the probability distribution $p(x, y) = p(r)$ will fall within this radius $r \leq R_0$ and the remaining 50% will fall outside $r > R_0$. The radius R_0 is referred to as the circular error probable (CEP) and has a very important role in assessing the accuracy of any weapon system.

Taking the logarithm of Eq. (3.50),

$$-\frac{R_0^2}{2\sigma^2} = \ln \frac{1}{2} \tag{3.51}$$

Hence,

$$R_0^2 = -2\sigma^2 \ln \tfrac{1}{2} = 1.386294\,\sigma^2$$

and

$$R_0 = 1.1774\,\sigma \tag{3.52}$$

Equation (3.49) can also be expressed in a somewhat different form in terms of the circular error probable R_0. It will be assumed that $P(r)$ is expressible in an equivalent form as

$$P(r) = 1 - \exp\left(-\frac{r^2}{2\sigma^2}\right) = 1 - \left(\frac{1}{2}\right)^k \tag{3.53}$$

where k is an unknown parameter. From Eq. (3.53) it follows that

$$\frac{r^2}{2\sigma^2} = -k \ln \frac{1}{2}$$

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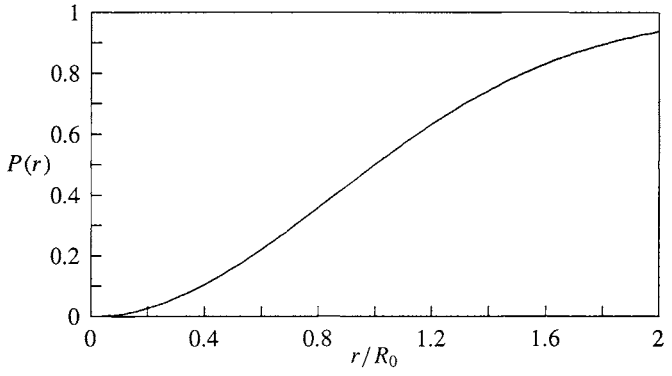


Fig. 3.12 Probability of weapon impact within radius r (R_0 is the circular error probable).

and

$$k = -\frac{r^2}{2\sigma^2}\left(\ell_n \frac{1}{2}\right)^{-1} \tag{3.54}$$

Subsequently, the substitution of $\ell_n(\frac{1}{2})$ from Eq. (3.51) into Eq. (3.54) leads to

$$k = \frac{r^2}{R_0^2} \tag{3.55}$$

Hence,

$$P(r) = 1 - \left(\tfrac{1}{2}\right)^{r^2/R_0^2} \tag{3.56}$$

Also, from Eqs. (3.49) and (3.52),

$$P(r) = 1 - \exp\left(-0.693147\frac{r^2}{R_0^2}\right) \tag{3.57}$$

Naturally, Eqs. (3.56) and (3.57) are equivalent. These equations are plotted in Fig. 3.12 as functions of the ratio r/R_0. It should be noted that Fig. 3.12 clearly underscores the importance of keeping the radius R_0 = CEP as small as possible in relation to the effective radius $r = R$ (lethal radius) to ensure a high probability of destruction of the target.

Case 2: Offset Distribution and Equal Variances

As in the previous case $\sigma_x^2 = \sigma_y^2 = \sigma^2$. The offset of the mean point of impact will be defined as

$$r_0 = \left(x_0^2 + y_0^2\right)^{\frac{1}{2}} \tag{3.58}$$

The probability of target destruction for this case, when the cookie cutter damage

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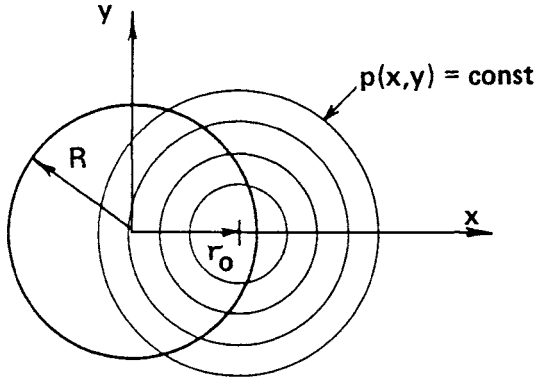


Fig. 3.13 Offset distribution and equal variances.

function is assumed, is calculated from Eqs. (3.46) and (3.44). Hence,

$$P_d = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d(x, y)p(x, y)\,dx\,dy$$
$$= \iint_{(x^2+y^2)^{\frac{1}{2}} \le R} \frac{1}{2\pi\sigma^2} \exp\left[-\frac{(x-x_0)^2}{2\sigma^2} - \frac{(y-y_0)^2}{2\sigma^2}\right] dx\,dy \qquad (3.59)$$

where R is the radius of the damage function $d(x, y)$. To simplify subsequent calculations the coordinate axes are rotated so that r_0 lies along the positive x axis as indicated in Fig. 3.13. Hence,

$$P_d = \iint_{(x^2+y^2)^{\frac{1}{2}} \le R} \frac{1}{2\pi\sigma^2} \exp\left[-\frac{(x-r_0)^2}{2\sigma^2} - \frac{y^2}{2\sigma^2}\right] dx\,dy \qquad (3.60)$$

while subsequent transformation to polar coordinates yields

$$P_d\left(\frac{R}{\sigma}, \frac{r_0}{\sigma}\right) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{r_0^2}{2\sigma^2}\right) \int_0^R \int_0^{2\pi} r \exp\left(-\frac{r^2}{2\sigma^2} + \frac{r r_0 \cos\theta}{\sigma^2}\right) d\theta\,dr \qquad (3.61)$$

where $P_d = P_d(R/\sigma, r_0/\sigma)$ is used to indicate the functional dependence on the two ratios R/σ and r_0/σ. To integrate Eq. (3.61), the following expression for the modified Bessel function of the first kind of order zero in Poisson's form is used[2,3]:

$$I_0(z) = \frac{1}{\pi}\int_0^{\pi} \cosh(z\cos\theta)\,d\theta = \frac{1}{\pi}\int_0^{\pi} \exp(z\cos\theta)\,d\theta \qquad (3.62)$$

Because $z\cos\theta$ is symmetrical about $\theta = \pi$,

$$2I_0(z) = \frac{1}{\pi}\int_0^{2\pi} \exp(z\cos\theta)\,d\theta \qquad (3.63)$$

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Hence, substituting Eq. (3.63) into Eq. (3.61),

$$P_d\left(\frac{R}{\sigma}, \frac{r_0}{\sigma}\right) = \frac{1}{\sigma^2} \exp\left(-\frac{r_0^2}{2\sigma^2}\right) \int_0^R r \exp\left(-\frac{r^2}{2\sigma^2}\right) I_0\left(\frac{r r_0}{\sigma^2}\right) dr \tag{3.64}$$

Now introducing a new variable u defined by

$$r = Ru^{\frac{1}{2}} \tag{3.65}$$

and

$$dr = \frac{R}{2u^{\frac{1}{2}}} du \tag{3.66}$$

the limits of integration 0 to R in Eq. (3.64) become 0 to 1, which is more convenient for subsequent computations. Substituting Eqs. (3.65) and (3.66) into Eq. (3.64) with the new limits of integration,

$$P_d\left(\frac{R}{\sigma}, \frac{r_0}{\sigma}\right) = \frac{R^2}{2\sigma^2} \exp\left(-\frac{r_0^2}{2\sigma^2}\right) \int_0^1 \exp\left(-\frac{R^2 u}{2\sigma^2}\right) I_0\left(\frac{R r_0 u^{\frac{1}{2}}}{\sigma^2}\right) du \tag{3.67}$$

The integral in Eq. (3.67) can be evaluated by replacing the Bessel function $I_0(z)$ by its series expansion[3]

$$I_0(z) = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{2}z\right)^{2n}}{(n!)^2}$$

so that

$$I_0\left(\frac{R r_0 u^{\frac{1}{2}}}{\sigma^2}\right) = \sum_{n=0}^{\infty} \left(\frac{R^2}{2\sigma^2}\right)^n \left(\frac{r_0^2}{2\sigma^2}\right)^n \frac{u^n}{(n!)^2} \tag{3.68}$$

and

$$\begin{aligned} P_d\left(\frac{R}{\sigma}, \frac{r_0}{\sigma}\right) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{r_0^2}{2\sigma^2}\right)^n \exp\left(-\frac{r_0^2}{2\sigma^2}\right) \\ &\quad \times \frac{1}{n!} \left(\frac{R^2}{2\sigma^2}\right)^{n+1} \int_0^1 u^n \exp\left(\frac{-R^2 u}{2\sigma^2}\right) du \\ &= \sum_{n=0}^{\infty} f_n g_n \end{aligned} \tag{3.69}$$

where

$$\begin{aligned} f_n &= \frac{1}{n!} \left(\frac{r_0^2}{2\sigma^2}\right)^n \exp\left(-\frac{r_0^2}{2\sigma^2}\right) \\ &= \frac{1}{n} \left(\frac{r_0^2}{2\sigma^2}\right) f_{n-1} \end{aligned} \tag{3.70}$$

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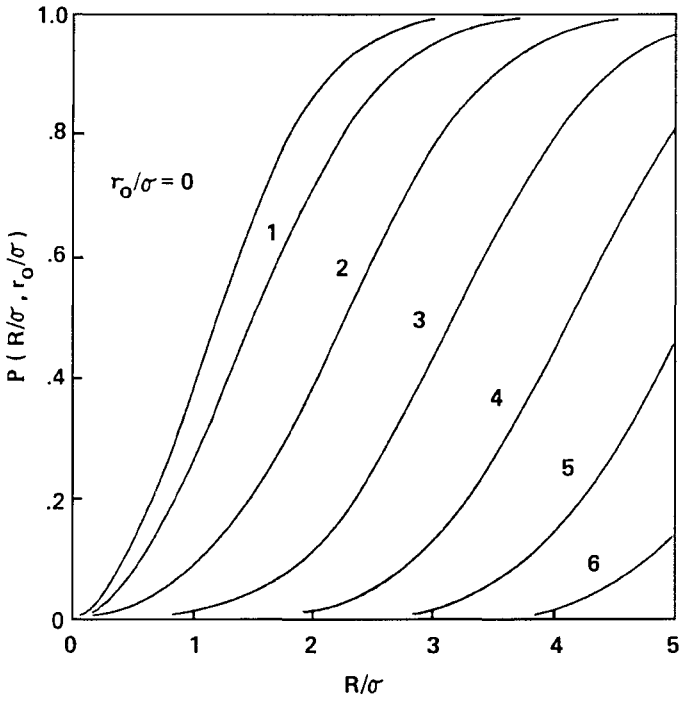


Fig. 3.14 Variation of the probability of destruction of a point target on surface, $P_d(R/\sigma, r_0/\sigma)$, as a function of R/σ and r_0/σ where R = lethal radius, r_0 = offset of the mean point of impact, and σ = standard deviation.

and

$$g_n = \frac{1}{n!}\left(\frac{R^2}{2\sigma^2}\right)^{n+1}\int_0^1 u^n \exp\left(-\frac{R^2 u}{2\sigma^2}\right) du$$

$$= -\frac{1}{n!}\left(\frac{R^2}{2\sigma^2}\right)^{n}\exp\left(-\frac{R^2}{2\sigma^2}\right) + g_{n-1} \tag{3.71}$$

The recursive expression in Eq. (3.71) is obtained through integration by parts.

Equation (3.69) is plotted in Fig. 3.14 for a series of values of the offset parameter r_0/σ. It should be noted that when $r_0/\sigma = 0$, Eq. (3.69) reduces to the special case of zero offset and equal variance (Case 1). Tables of the probability function (3.69) have been computed numerically in the past using different computational schemes.[4–6]

Case 3: Zero Offset and Unequal Variances

For this case $x_0 = y_0 = 0$ and $\sigma_x^2 \neq \sigma_y^2$ (see Fig. 3.15) and the probability of target damage for the cookie cutter damage function can be expressed in terms of two parameters $R/\sigma_{\max}$ and $c = \sigma_y^2/\sigma_x^2$, where without any loss of generality it is

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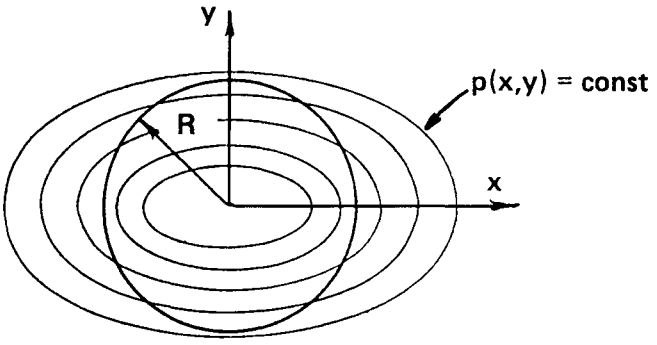


Fig. 3.15 Zero offset and unequal variances.

assumed that $\sigma_x^2 > \sigma_y^2$. The probability function $P_d(R/\sigma_{\max}, c)$ has been computed numerically by several authors,[7,8] and is shown in Fig. 3.16 as a function of R/σ_x and c. When $c = 1.0$ the probability function represents the case of zero offset and equal variances (Case 1).

Case 4: Offset Distribution and Unequal Variances

For this case $x_0 \neq 0$, $y_0 \neq 0$ and $\sigma_x^2 \neq \sigma_y^2$ as illustrated in Fig. 3.17. The probability function is usually expressed in terms of four parameters so that

$$P_d = P_d\left(\frac{R}{\sigma_y}, \frac{1}{c}, \frac{x_0}{\sigma_x}, \frac{y_0}{\sigma_y}\right) \tag{3.72}$$

where, as before, $c = \sigma_y^2/\sigma_x^2$. This function can only be computed numerically.[9]

3.8 Probability of Hit of a Rectangular Target: The Polya–Williams Approximation

The probability of hit of a rectangular target of dimensions $2a \times 2b$ is obtained from Eqs. (3.36), (3.37), and (3.38) as

$$P_{\text{rectangle}} = \frac{1}{\sqrt{2\pi}\sigma_x}\int_{-a}^{a}\exp\left(-\frac{x^2}{2\sigma_x^2}\right)dx\frac{1}{\sqrt{2\pi}\sigma_y}\int_{-b}^{b}\exp\left(-\frac{y^2}{2\sigma_y^2}\right)dy \tag{3.73}$$

Case 1: Equal Variances: Square Target

For a square target of dimensions $2a \times 2a$, when $\sigma_x = \sigma_y = \sigma$, Eq. (3.73) simplifies to

$$\begin{aligned} P_{\text{square}} &= \frac{1}{\sqrt{2\pi}\sigma}\int_{-a}^{a}\exp\left(-\frac{x^2}{2\sigma^2}\right)dx\frac{1}{\sqrt{2\pi}\sigma}\int_{-a}^{a}\exp\left(-\frac{y^2}{2\sigma^2}\right)dy \\ &= \frac{1}{2\pi\sigma^2}\int_{-a}^{a}\int_{-a}^{a}\exp\left[-\frac{(x^2+y^2)}{2\sigma^2}\right]dx\,dy \end{aligned} \tag{3.74}$$

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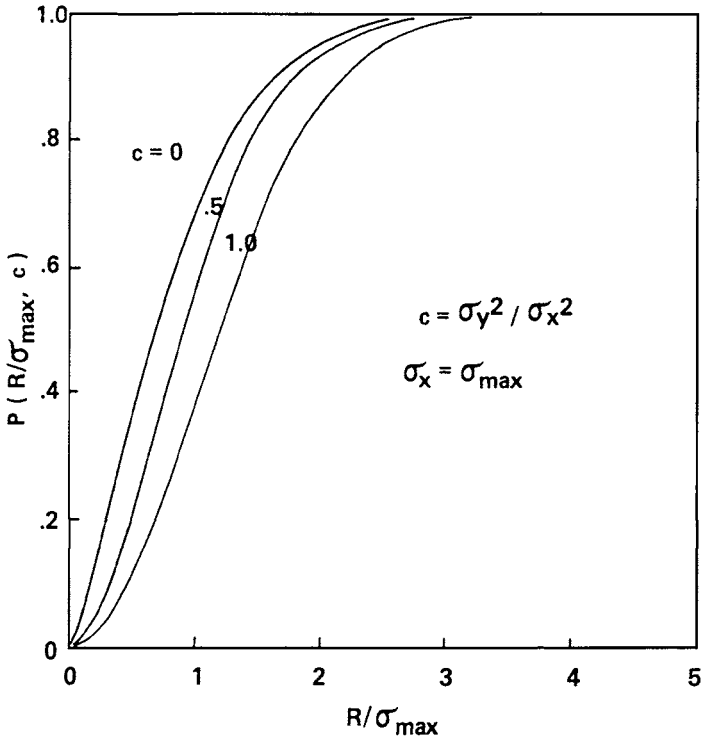


Fig. 3.16 Probability of destruction of a point target on surface, $P_d(R/\sigma, c)$, for the case of zero offset and unequal variances where $c = \sigma_y^2/\sigma_x^2$ and R is the lethal radius.

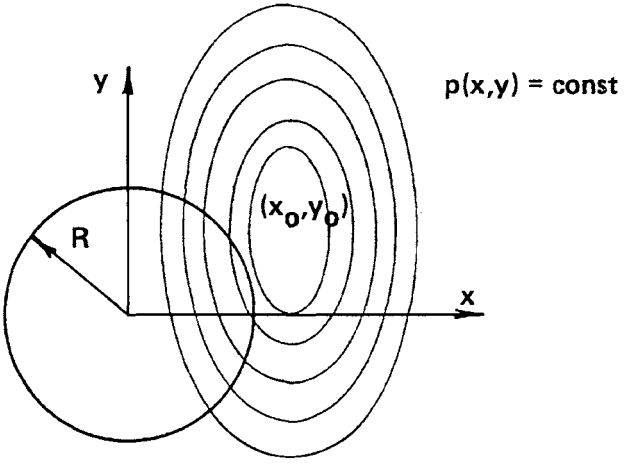


Fig. 3.17 Offset distribution and unequal variances.

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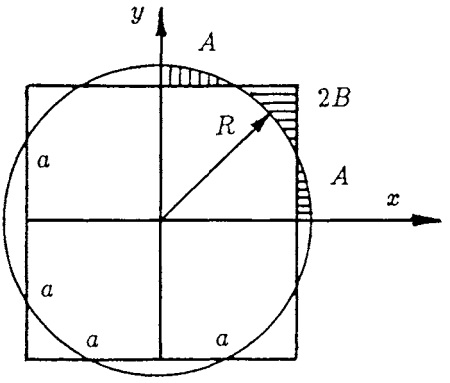


Fig. 3.18 Equivalent circular area for a square target ($A = B$).

According to Polya–Williams approximation[10,11] if, instead of performing integration over the square target, an equivalent circular area is used then the relative error in P_{square} is never more than about 1.42 percent. The equivalent radius R is obtained from $\pi R^2 = 4a^2$ so that $R = 2a/\sqrt{\pi}$. Observing from Fig. 3.18 that the integrand in Eq. (3.74) over the area B is smaller than over A, it follows that

$$P_{\text{square}} = \frac{1}{2\pi\sigma^2}\int_{-a}^{a}\int_{-a}^{a}\exp\left[-\frac{(x^2+y^2)}{2\sigma^2}\right]dx\,dy$$
$$\leq \iint_{(x^2+y^2)^{1/2}\leq R}\exp\left[-\frac{(x^2+y^2)}{2\sigma^2}\right]dx\,dy \tag{3.75}$$

Hence, changing to polar coordinates it follows that

$$P_{\text{square}} \approx \frac{1}{2\pi\sigma^2}\int_0^{2\pi}\int_0^{R}\exp\left(-\frac{r^2}{2\sigma^2}\right)r\,dr\,d\theta$$
$$= -\exp\left(-\frac{r^2}{2\sigma^2}\right)\Big|_0^R = 1-\exp\left(-\frac{R^2}{2\sigma^2}\right)$$
$$= 1-\exp\left(-\frac{2a^2}{\pi\sigma^2}\right) \tag{3.76}$$

A comparison of the Polya–Williams approximation with the exact values for a square target is shown in Table 3.1.

Case 2: Nonequal Variances: Rectangular Target

Observing from Eqs. (3.73) and (3.76) that

$$\frac{1}{\sqrt{2\pi}\sigma}\int_{-a}^{a}\exp\left(-\frac{x^2}{2\sigma^2}\right)dx \approx \left[1-\exp\left(-\frac{2a^2}{\pi\sigma^2}\right)\right]^{\frac{1}{2}} \tag{3.77}$$

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Table 3.1 Error in the Polya–Wiliams (P-W) approximation for the probability of hit of a square target (REP = range error probable)

a/REP	P–W Approximation	Exact Value of P_{square}	Percentage Error
0	0.0000	0.0000	0.0000
0.2	0.0115	0.0115	0.027
0.4	0.0453	0.0452	0.107
0.6	0.0990	0.0988	0.232
0.8	0.1692	0.1685	0.395
1.0	0.2515	0.2500	0.581
1.2	0.3410	0.3384	0.778
1.4	0.4331	0.4290	0.969
1.6	0.5236	0.5177	1.139
1.8	0.6087	0.6011	1.276
2.0	0.6860	0.6768	1.371
2.2	0.7538	0.7433	1.415
2.4	0.8114	0.8001	1.410
2.6	0.8588	0.8473	1.356
2.8	0.8968	0.8856	1.262
3.0	0.9262	0.9158	1.137

Eq. (3.73) can be approximated as

$$P_{\text{rectangle}} = \frac{1}{\sqrt{2\pi}\sigma_x}\int_{-a}^{a}\exp\left[-\frac{x^2}{2\sigma_x^2}\right]dx\,\frac{1}{\sqrt{2\pi}\sigma_y}\int_{-b}^{b}\exp\left[-\frac{y^2}{2\sigma_y^2}\right]dy$$

$$\approx\left\{\left[1-\exp\left(-\frac{2a^2}{\pi\sigma_x^2}\right)\right]\left[1-\exp\left(-\frac{2b^2}{\pi\sigma_y^2}\right)\right]\right\}^{\frac{1}{2}} \tag{3.78}$$

3.9 Probability of Hit of an Elliptic Target with Unequal Variances

The Polya–Williams approximation can also be used for elliptical targets. This involves replacing the ellipse by a rectangle of the same area, i.e.

$$\pi cd = 2a2b \tag{3.79}$$

where c and d are the semiaxes of the ellipse. From Fig. 3.19 it follows that a and b are less than c and d, respectively. Hence, by choosing

$$a = c\sqrt{\pi}/2 \tag{3.80}$$

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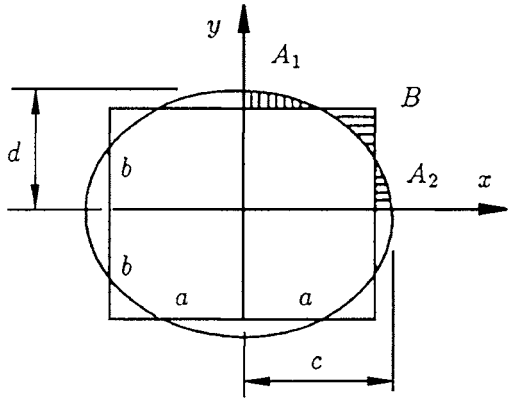


Fig. 3.19 Equivalent circular area for an elliptical target ($B = A_1 + A_2$).

and

$$b = d\sqrt{\pi}/2 \tag{3.81}$$

Eq. (3.79) is satisfied. Substituting Eq. (3.80) and (3.81) into Eq. (3.78) it follows that

$$P_{\text{ellipse}} \approx \left\{\left[1 - \exp\left(-\frac{c^2}{2\sigma_x^2}\right)\right]\left[1 - \exp\left(-\frac{d^2}{2\sigma_y^2}\right)\right]\right\}^{\frac{1}{2}} \tag{3.82}$$

For a circular target $c = d = R$ and the Polya–Williams approximation becomes

$$P_{\text{circle}} \approx \left\{\left[1 - \exp\left(-\frac{R^2}{2\sigma_x^2}\right)\right]\left[1 - \exp\left(-\frac{R^2}{2\sigma_y^2}\right)\right]\right\}^{\frac{1}{2}} \tag{3.83}$$

3.10 Probability of Destruction of a Point Target in Space

As an extension of the two-dimensional case a point target in space will be considered. The point target is small in relation to the lethal radius R of a space weapon that is equally effective in all directions. The aiming offset point is at a distance r_0 from the target as shown in Fig. 3.20. Also, this point will be assumed to lie, without any loss of generality, on the z axis; thus, the three-dimensional probability density function based on equal variances can be expressed as

$$\begin{aligned} p(x, y, z) &= \frac{1}{2^{\frac{3}{2}}\pi^{\frac{3}{2}}\sigma^3} \exp\left[-\frac{x^2}{2\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{(z - r_0)^2}{2\sigma^2}\right] \\ &= \frac{1}{2^{\frac{3}{2}}\pi^{\frac{3}{2}}\sigma^3} \exp\left(-\frac{r_0^2}{2\sigma^2}\right) \exp\left(-\frac{r^2}{2\sigma^2} + \frac{r r_0 \cos\varphi}{\sigma^2}\right) \end{aligned} \tag{3.84}$$

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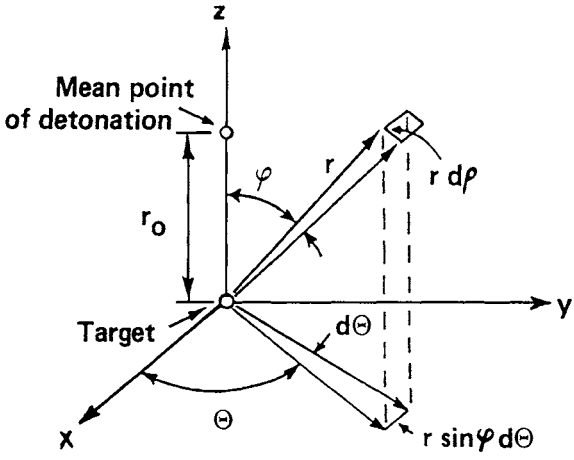


Fig. 3.20 Point target in space.

The probability of destruction of a point target in space is calculated next from the integral

$$P_d = \int_0^R \int_0^\pi \int_0^{2\pi} p(x, y, z) r^2 \sin\varphi \, dr \, d\varphi \, d\theta$$

$$= \frac{1}{2^{\frac{3}{2}} \pi^{\frac{3}{2}} \sigma^3} \exp\left(-\frac{r_0^2}{2\sigma^2}\right)$$

$$\times \int_0^R \int_0^\pi \int_0^{2\pi} \exp\left(-\frac{r^2 - 2r r_0 \cos\varphi}{2\sigma^2}\right) r^2 \sin\varphi \, dr \, d\varphi \, d\theta$$

$$= \frac{1}{2^{\frac{1}{2}} \pi^{\frac{1}{2}} \sigma^3} \exp\left(-\frac{r_0^2}{2\sigma^2}\right) \int_0^R \int_0^\pi \exp\left(-\frac{r^2}{2\sigma^2}\right) \exp\left(\frac{r r_0 \cos\varphi}{\sigma^2}\right) r^2 \sin\varphi \, dr \, d\varphi$$

$$= \frac{1}{2^{\frac{1}{2}} \pi^{\frac{1}{2}} \sigma r_0} \int_0^R r \left\{ \exp\left[-\frac{(r - r_0)^2}{2\sigma^2}\right] - \exp\left[-\frac{(r + r_0)^2}{2\sigma^2}\right] \right\} dr \qquad (3.85)$$

Subsequent integration is possible with substitution of the following variables:

$$\rho = \frac{r}{\sqrt{2}\sigma}, \quad \text{or}$$

$$= \frac{R}{\sqrt{2}\sigma}, \quad \text{when used as an integral limit or a parameter} \qquad (3.86)$$

$$\rho_0 = \frac{r_0}{\sqrt{2}\sigma} \qquad (3.87)$$

$$u = \rho - \rho_0 \qquad (3.88)$$

$$v = \rho + \rho_0 \qquad (3.89)$$

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Substitution of the preceding variables into Eq. (3.85) leads to

$$P_d = \frac{1}{\sqrt{\pi}\rho_0}\int_{-\rho_0}^{\rho-\rho_0}(u+\rho_0)e^{-u^2}\,du - \frac{1}{\sqrt{\pi}\rho_0}\int_{\rho_0}^{\rho+\rho_0}(v-\rho_0)e^{-v^2}\,dv \quad (3.90)$$

which can now be integrated. This results in an expression in terms of exponential and error functions given by

$$P_d = P_d\left(\frac{R}{\sigma}, \frac{r_0}{\sigma}\right)$$

$$= \frac{1}{2\sqrt{\pi}\rho_0}\left\{e^{-(\rho+\rho_0)^2} - e^{-(\rho-\rho_0)^2} + \sqrt{\pi}\rho_0[\mathrm{erf}(\rho+\rho_0) + \mathrm{erf}(\rho-\rho_0)]\right\} \quad (3.91)$$

where the error function erf(z) is defined as (see Appendix A, Table A.4)

$$\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-z^2}\,dz \quad (3.92)$$

Equation (3.91) is plotted in Fig. 3.21 for a series of values of the offset parameter r_0/σ. When $r_0/\sigma = 0$ (i.e., $\rho_0 = 0$), Eq. (3.91) reduces to a special case of zero offset. Since ρ_0 appears in the denominator of Eq. (3.91), L'Hospital's rule can be

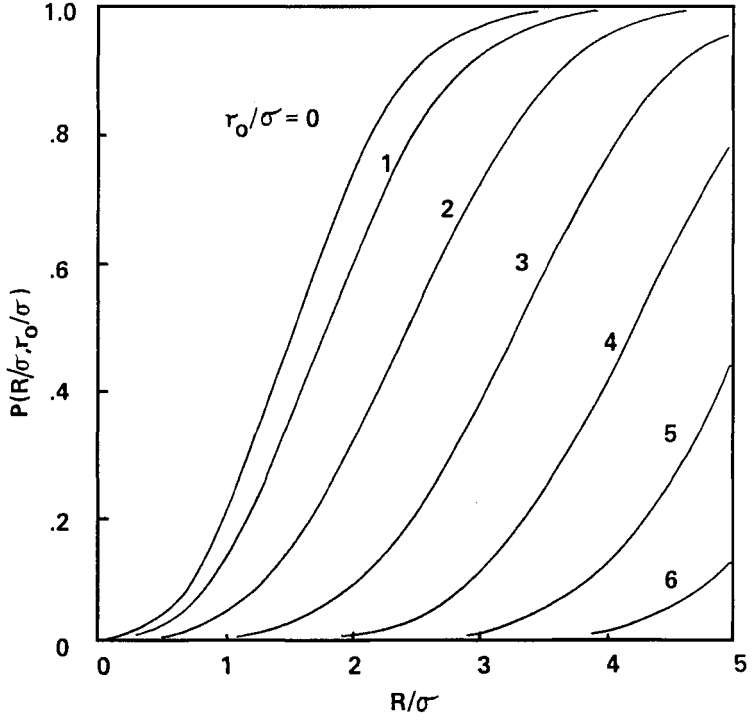


Fig. 3.21 Variation of the probability of destruction of a point target in space, $P_d(R/\sigma, r_0/\sigma)$, as a function of R/σ and r_0/σ where R is the lethal radius, r_0 the offset of the mean point of detonation, and σ the standard deviation.

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applied to obtain the limiting value of P_d when $r_0 = 0$. This leads to

$$P_d\left(\frac{R}{\sigma}, 0\right) = -\frac{2}{\sqrt{\pi}}\rho e^{-\rho^2} + \text{erf}(\rho) \tag{3.93}$$

3.11 Linear, Circular, and Spherical Error Probables

In Sec. 3.7 the CEP was defined as the radius for which 50% of impact points fall within the radius, and the remaining 50% fall outside this radius. This definition was used when $r_0 = 0$; however, it can also be extended to nonzero values of r_0. Thus, equating Eq. (3.69) to the probability of 0.5, i.e.,

$$\sum_{n=0}^{\infty} f_n g_n = \frac{1}{2} \tag{3.94}$$

results in a relationship between R/σ and r_0/σ for the CEP. Similarly, from Eq. (3.91),

$$\frac{1}{2\sqrt{\pi}\rho_0}\left\{e^{-(\rho+\rho_0)^2} - e^{-(\rho-\rho_0)^2} + \sqrt{\pi}\rho_0[\text{erf}(\rho+\rho_0) + \text{erf}(\rho-\rho_0)]\right\} = \frac{1}{2} \tag{3.95}$$

gives the corresponding relationship between R/σ and r_0/σ for the spherical error probable (SEP).

For completeness, a similar expression can also be derived for the linear case; however, its usefulness is somewhat limited because of the simplifying assumption of the dispersion in one direction only. This simple model is represented in Fig. 3.22, where as before R is the lethal radius (range in this case). The aiming offset is represented by x_0 and the standard deviation by σ. It can be easily demonstrated that the probability of destruction of a point target in one-dimensional space (range) is given by

$$P_d\left(\frac{R}{\sigma}, \frac{x_0}{\sigma}\right) = \frac{1}{2}\left[\text{erf}\left(\frac{R-x_0}{\sqrt{2}\sigma}\right) + \text{erf}\left(\frac{R+x_0}{\sqrt{2}\sigma}\right)\right] \tag{3.96}$$

which is illustrated in Fig. 3.23 as a function of R/σ for a series of values of x_0/σ. Hence, the linear error probable LEP is obtained from

$$P_d\left(\frac{R}{\sigma}, \frac{x_0}{\sigma}\right) = \frac{1}{2} \tag{3.97}$$

The special case for $x_0 = 0$ was previously discussed in Sec. 3.4.

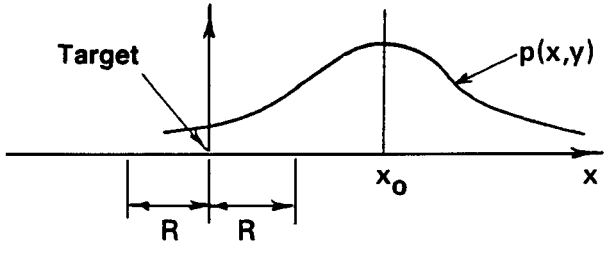


Fig. 3.22 One-dimensional model for the linear error probable.

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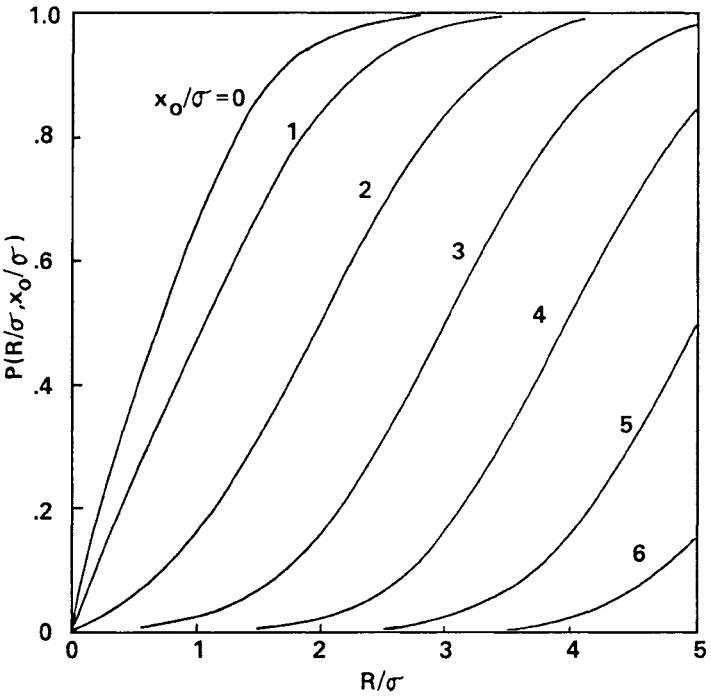


Fig. 3.23 Variation of the probability of destruction of a point target placed along the line of sight, $P_d(R/\sigma, x_0/\sigma)$, as a function of R/σ and x_0/σ where R is the lethal range (radius), x_0 the offset of the mean point of impact, and σ the standard deviation.

Equations (3.94), (3.95), and (3.97) are plotted in Fig. 3.24, which shows clearly how the lethal radius R must be increased with an increase in the offset parameters x_0/σ or r_0/σ to achieve the 0.5 (i.e., 50%) probability of target destruction. The linear error probable (LEP) is also referred to as the range error probable for the range probability distribution. For cross-range computations, the corresponding designation is the cross-range error probable (CREP). The important relationships between the error probables and the standard deviation σ for zero offset can also be obtained from Eqs. (3.94), (3.95), and (3.97). This leads to

$$\text{LEP, REP, or CREP} = 0.6745\sigma \tag{3.98}$$

$$\text{CEP} = 1.1774\sigma \tag{3.99}$$

$$\text{SEP} = 1.5382\sigma \tag{3.100}$$

3.12 Expected Fractional Damage of a Uniform-Valued Target

In Sec. 3.7 the size of the target was small in relation to the lethal radius of the damage function; therefore, it could be approximated as a point in the x-y plane. In many tactical and strategic applications targets are spread over a large area and

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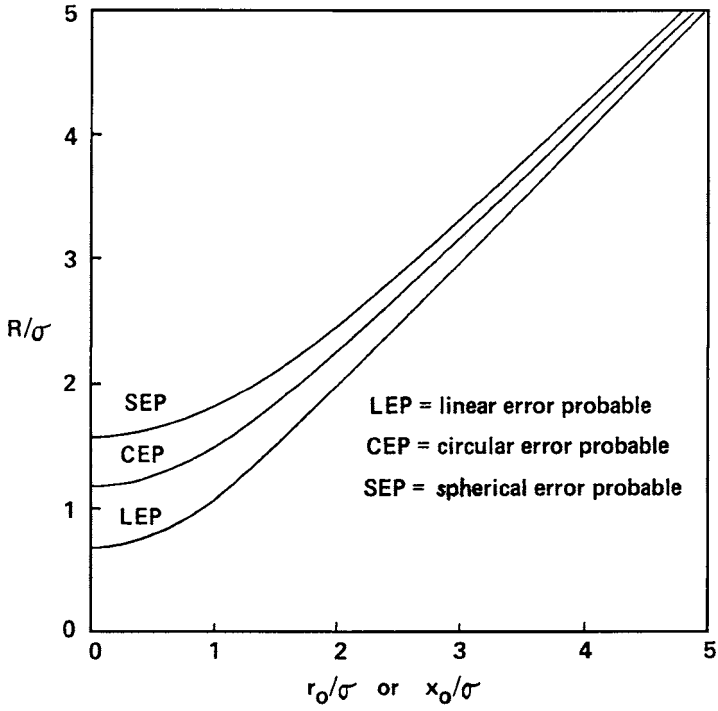


Fig. 3.24 Relationship between the lethal radius (or range) for the 50% error probable, i.e., 50% probability of destruction of a point target in one, two, and three dimensions.

cannot be represented as point targets. For such targets only a fraction of the target may be destroyed when a missile or warhead hits the target area. In this section a method of determining the expected fraction of the target destroyed by a single weapon will be presented.

It will be assumed that the target has a uniform value everywhere within the radius K in the x-y plane and zero value outside this radius. As before, the joint probability density function of the impact point of the weapon is $p(x, y)$ and the lethal radius of the weapon is R. It will also be assumed that an area $dx_t\, dy_t$ at a point (x_t, y_t) within the target area is destroyed with the probability $d(x-x_t, y-y_t)$ per unit target area by a weapon impacting at the point (x, y). Thus, the probability that the point (x_t, y_t) within the target is destroyed is a function of the distance from the point of impact (see Fig. 3.25). If one assumes that $d(x-x_t, y-y_t)\, dx_t\, dy_t$ also represents the expected fractional damage at the point (x_t, y_t), then the expected fraction of the target destroyed by a single weapon is obtained from[12]

$$E = \iint_{(x_t^2+y_t^2)^{\frac{1}{2}} \le K} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d(x-x_t, y-y_t)p(x, y)\, dx\, dy\, dx_t\, dy_t \qquad (3.101)$$

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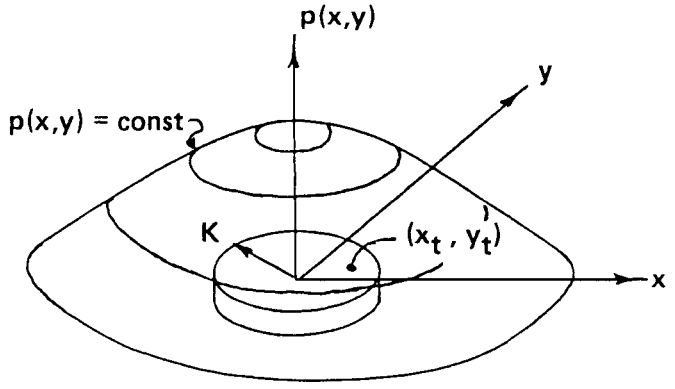


Fig. 3.25 Uniform value target on a surface.

It should be noted that the integral

$$\iint_{(x_t^2+y_t^2)^{\frac{1}{2}} \le K} d(x - x_t, y - y_t)\, dx_t\, dy_t$$

in Eq. (3.101) represents the fractional damage of the target if the weapon impacts at the point (x, y). This integral is combined with the joint probability density function $p(x, y)$ and integrated over the lethal area of the weapon to obtain the expected fraction of the target destroyed by firing a single weapon.

If a cookie cutter damage function $d(x - x_t, y - y_t)$ per unit target area is assumed to be distributed uniformly so that

$$d(x - x_t, y - y_t) = \begin{cases} \dfrac{1}{\pi K^2} & \text{for } \left(x_t^2 + y_t^2\right)^{\frac{1}{2}} \le K \\ 0 & \text{for } \left(x_t^2 + y_t^2\right)^{\frac{1}{2}} > K \end{cases} \tag{3.102}$$

then Eq. (3.101) simplifies to

$$E = \frac{1}{\pi K^2} \iint_A \iint_B p(x, y)\, dx\, dy\, dx_t\, dy_t \tag{3.103}$$

where A represents the region $(x_t^2 + y_t^2)^{\frac{1}{2}} \le K$ and B represents the region $[(x - x_t)^2 + (y - y_t)^2]^{\frac{1}{2}} \le R$. Because of the mathematical difficulties with Eq. (3.103), the joint probability density function $p(x, y)$ is usually assumed as the bivariate Gaussian probability density function

$$p(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \tag{3.104}$$

The advantage of this function is that it is circularly symmetric and centered on

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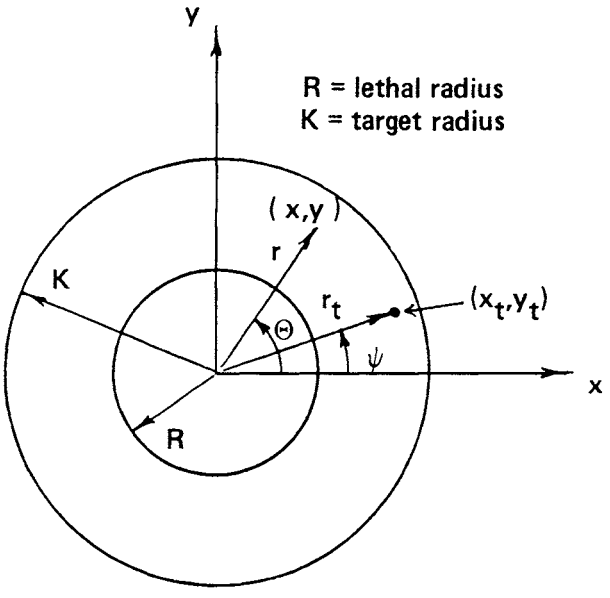


Fig. 3.26 Definition of variables r, θ, r_t, and ψ.

the target. To evaluate the integrals in Eq. (3.103), a new function Q is defined as

$$Q(x_t, y_t) = \iint_B p(x, y)\, dx\, dy \tag{3.105a}$$

where $p(x, y)$ is the circularly symmetric Gaussian density function given by Eq. (3.104). Hence,

$$Q(x_t, y_t) = \frac{1}{2\pi\sigma^2} \iint_B \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) dx\, dy \tag{3.105b}$$

To proceed further with the integration, the variables x_t and y_t are replaced with a set of auxiliary variables ξ and η given by (see Fig. 3.26)

$$\xi = x - x_t; \qquad \eta = y - y_t \tag{3.106}$$

which, when substituted into Eq. (3.105b), lead to

$$Q(x_t, y_t) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x_t^2 + y_t^2}{2\sigma^2}\right) \times \iint_{(\xi^2+\eta^2)^{\frac{1}{2}} \le R} \exp\left(-\frac{\xi^2 + \eta^2 + 2\xi x_t + 2\eta y_t}{2\sigma^2}\right) d\xi\, d\eta \tag{3.105c}$$

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Now the introduction of an additional set of variables r, r_t, θ, ψ, and α defined by (see Fig. 3.26)

$$\xi = r\cos\theta; \qquad \eta = r\sin\theta \tag{3.107}$$

$$\xi^2 + \eta^2 = r^2 \tag{3.108}$$

$$x_t = r_t\cos\psi; \qquad y_t = r_t\sin\psi \tag{3.109}$$

$$\alpha = \theta - \psi \tag{3.110}$$

results in the following expression for $Q(x_t, y_t)$:

$$\begin{aligned} Q(x_t, y_t) &= \frac{1}{2\pi\sigma^2}\exp\left(-\frac{r_t^2}{2\sigma^2}\right) \\ &\quad \times \int_0^R \int_0^{2\pi} \exp\left(-\frac{r^2}{2\sigma^2}\right)\exp\left[\frac{(x_t\cos\theta + y_t\sin\theta)r}{\sigma^2}\right] r\,d\theta\,dr \\ &= \frac{1}{2\pi\sigma^2}\exp\left(-\frac{r_t^2}{2\sigma^2}\right) \\ &\quad \times \int_0^R \int_0^{2\pi} \exp\left(-\frac{r^2}{2\sigma^2}\right)\exp\left[\frac{rr_t\cos(\theta-\psi)}{\sigma^2}\right] r\,d\theta\,dr \\ &= \frac{1}{2\pi\sigma^2}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\int_0^R\int_0^{2\pi}\exp\left(-\frac{r^2}{2\sigma^2}\right)\exp\left(\frac{rr_t\cos\alpha}{\sigma^2}\right) r\,d\alpha\,dr \\ &= \frac{1}{\sigma^2}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\int_0^R \exp\left(-\frac{r^2}{2\sigma^2}\right) I_0\left(\frac{rr_t}{\sigma^2}\right) r\,dr \\ &= P_d\left(\frac{R}{\sigma}, \frac{r_t}{\sigma}\right) \end{aligned} \tag{3.111}$$

where the integration with respect to α was accomplished by using Eq. (3.63). Comparing the previous equation with Eq. (3.64), it is clear that $P_d(R/\sigma, r_t/\sigma)$ represents the probability of destruction of a point target for a given standard deviation σ when the lethal radius is R and the aiming offset is r_t.

Returning now to the equation for the damage expectation E, Eq. (3.103) can be rewritten as

$$\begin{aligned} E\left(\frac{R}{\sigma}, \frac{K}{\sigma}\right) &= \frac{1}{K^2}\iint_A P_d\left(\frac{R}{\sigma}, \frac{r_t}{\sigma}\right) dx_t\,dy_t \\ &= \frac{1}{\pi K^2}\int_0^K \int_0^{2\pi} P_d\left(\frac{R}{\sigma}, \frac{r_t}{\sigma}\right) r_t\,dr_t\,d\psi \\ &= \frac{2}{K^2}\int_0^K P_d\left(\frac{R}{\sigma}, \frac{r_t}{\sigma}\right) r_t\,dr_t \end{aligned}$$

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Integrating the preceding expression by parts,

$$E\left(\frac{R}{\sigma},\frac{K}{\sigma}\right)=P_d\left(\frac{R}{\sigma},\frac{r_t}{\sigma}\right)\frac{r_t^2}{K^2}\bigg|_{r_t=0}^{r_t=K}-\frac{1}{K^2}\int_0^K\frac{d}{dr_t}P_d\left(\frac{R}{\sigma},\frac{r_t}{\sigma}\right)r_t^2\,dr_t$$

$$=P_d\left(\frac{R}{\sigma},\frac{K}{\sigma}\right)-\frac{1}{K^2}\int_0^K\frac{d}{dr_t}P_d\left(\frac{R}{\sigma},\frac{r_t}{\sigma}\right)r_t^2\,dr_t \qquad (3.112)$$

Equation (3.111) is then used to evaluate the derivative under the integral sign. Hence,

$$\frac{d}{dr_t}P_d\left(\frac{R}{\sigma},\frac{r_t}{\sigma}\right)=-\frac{r_t}{\sigma^4}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\int_0^R\exp\left(-\frac{r^2}{2\sigma^2}\right)I_0\left(\frac{rr_t}{\sigma^2}\right)r\,dr$$

$$+\frac{1}{\sigma^2}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\int_0^R\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}I_0'\left(\frac{rr_t}{\sigma^2}\right)r\,dr$$

$$=-\frac{r_t}{\sigma^2}P_d\left(\frac{R}{\sigma},\frac{r_t}{\sigma}\right)$$

$$+\frac{1}{\sigma^4}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\int_0^R\exp\left(-\frac{r^2}{2\sigma^2}\right)I_1\left(\frac{rr_t}{\sigma^2}\right)r^2\,dr$$

The last term of the preceding expression can be integrated by parts to give

$$\frac{1}{\sigma^4}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\int_0^R\exp\left(-\frac{r^2}{2\sigma^2}\right)I_1\left(\frac{rr_t}{\sigma^2}\right)r^2\,dr$$

$$=-\frac{1}{\sigma^4}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\sigma^2\exp\left(-\frac{r^2}{2\sigma^2}\right)r\;I_1\left(\frac{rr_t}{\sigma^2}\right)\bigg|_{r=0}^{r=R}$$

$$+\frac{1}{\sigma^4}\exp\left(-\frac{r_t^2}{2\sigma^2}\right)\int_0^R\sigma^2\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{d}{dr}\left[rI_1\left(\frac{rr_t}{\sigma^2}\right)\right]dr$$

$$=-\frac{1}{\sigma^2}\exp\left(-\frac{r_t^2+R^2}{2\sigma^2}\right)RI_1\left(\frac{Rr_t}{\sigma^2}\right)+\frac{r_t}{\sigma^2}P_d\left(\frac{R}{\sigma},\frac{r_t}{\sigma}\right)$$

Noting that

$$\frac{d}{dz}[zI_1(z)]=zI_0(z)$$

it follows that

$$\frac{d}{dr_t}P_d\left(\frac{R}{\sigma},\frac{r_t}{\sigma}\right)=-\exp\left(-\frac{r_t^2+R^2}{2\sigma^2}\right)\frac{R}{\sigma^2}I_1\left(\frac{Rr_t}{\sigma^2}\right) \qquad (3.113)$$

Finally, the substitution of Eq. (3.113) into (3.112) results in the following expression:

$$E\left(\frac{R}{\sigma},\frac{K}{\sigma}\right)=P_d\left(\frac{R}{\sigma},\frac{K}{\sigma}\right)+\frac{1}{K^2}\int_0^K\exp\left(-\frac{r_t^2+R^2}{2\sigma^2}\right)\frac{R}{\sigma^2}I_1\left(\frac{Rr_t}{\sigma^2}\right)r_t^2\,dr_t$$

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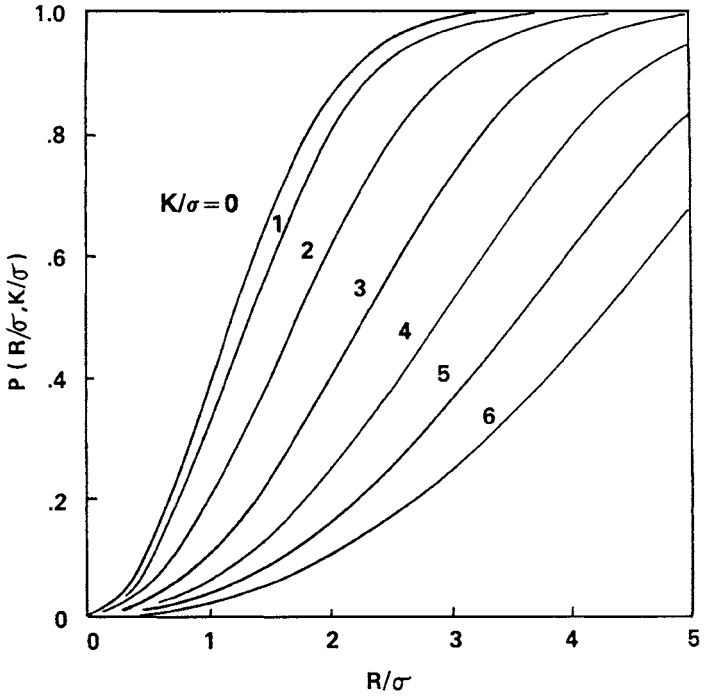


Fig. 3.27 Expected fraction of damage for a circular target of radius K and weapon lethal radius R (σ is the standard deviation).

Here again the second term on the right can be integrated by parts to give

$$E\left(\frac{R}{\sigma}, \frac{K}{\sigma}\right) = P_d\left(\frac{R}{\sigma}, \frac{K}{\sigma}\right) - \frac{R}{K}\exp\left(-\frac{K^2+R^2}{2\sigma^2}\right) I_1\left(\frac{RK}{\sigma^2}\right) + \frac{R^2}{K^2} P_d\left(\frac{K}{\sigma}, \frac{R}{\sigma}\right) \tag{3.114}$$

It should be noted that $P_d(R/\sigma, K/\sigma) \neq P_d(K/\sigma, R/\sigma)$, because the expression for P_d is not symmetric with respect to R and K. For a given set of values of R/σ and K/σ, the probabilities $P_d(R/\sigma, K/\sigma)$ and $P_d(K/\sigma, R/\sigma)$ can be obtained from Eq. (3.67). The damage expectation function $E(R/\sigma, K/\sigma)$ is plotted in Fig. 3.27 as a function of the lethal radius parameter R/σ for a series of values of the target radius parameter K/σ.

3.13 Damage Functions for a Point Target in a Plane

Section 3.7 introduced the concept of the cookie cutter damage function $d(x, y)$ given by

$$d(x, y) = \begin{cases} 1; & r \leq R \\ 0; & r > R \end{cases} \tag{3.45}$$

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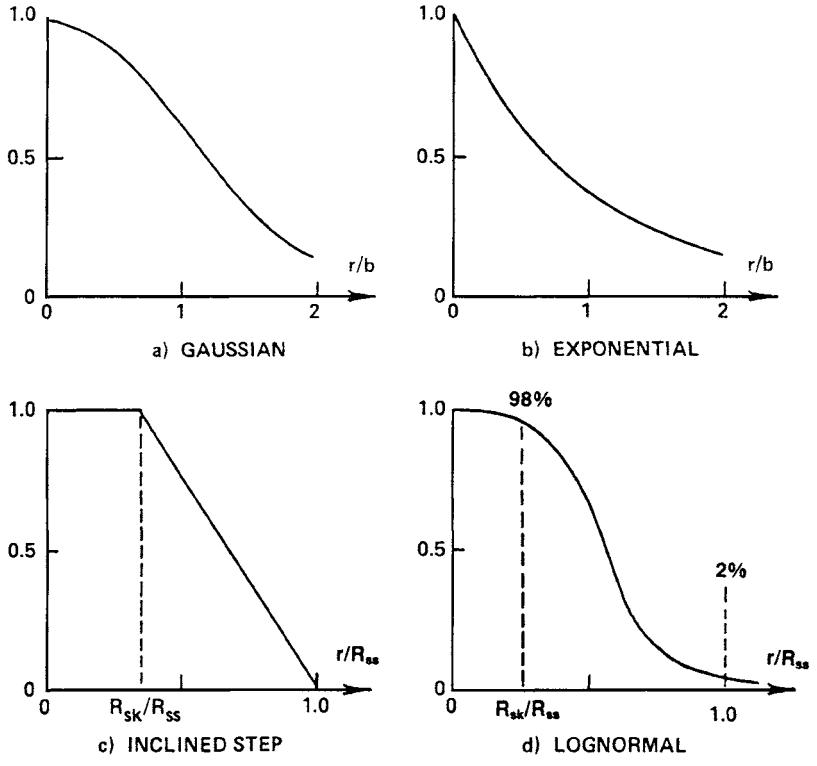


Fig. 3.28 Typical damage functions $d(r)$.

This implied that a point target outside the lethal radius R escaped any damage, whereas the target inside that radius was completely destroyed. A more realistic representation of the damage function would be to select a function with a continuous variable r such that this function would be decreasing as the distance r from the warhead burst increases. Several such functions will be considered to calculate a more accurate probability of destruction of a point target with zero offset of the aiming point; i.e., the weapon will be assumed to be aimed at the target point. The cases that involve aiming offset cannot be evaluated in terms of known functions, and numerical integration must be employed. In the subsequent section, however, the concept of an equivalent weapon radius will be introduced that, in turn, can be used with the simple cookie cutter damage function for the cases of zero aiming offset. The four typical damage functions considered here are illustrated in Fig. 3.28.

Gaussian Damage Function

The damage function (the expected fraction of the target destroyed when the warhead impacts at a distance r from the target) for the Gaussian distribution is of the form

$$d(r) = \exp\left(-\frac{r^2}{2b^2}\right) \tag{3.115}$$

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where b is a constant. The variation of $d(r)$ with r/b is shown in Fig. 3.28a. Substitution of Eq. (3.99) into Eq. (3.44) results in an expression for the probability of target destruction

$$\begin{aligned} P_d &= \iint d(x, y)p(x, y)\,dx\,dy \\ &= \int_0^{\infty}\int_0^{2\pi} \frac{1}{2\pi\sigma^2}\exp\left(-\frac{r^2}{2b^2}\right)\exp\left(-\frac{r^2}{2\sigma^2}\right) r\,dr\,d\theta \\ &= \int_0^{\infty} \frac{r}{\sigma^2}\exp\left[-\frac{r^2(\sigma^2+b^2)}{2\sigma^2 b^2}\right] dr = \frac{b^2}{b^2+\sigma^2} \end{aligned} \tag{3.116}$$

Exponential Damage Function

A damage function similar to the Gaussian function is of the form

$$d(r) = \exp\left(-\frac{r}{b}\right) \tag{3.117}$$

which is shown in Fig. 3.28b. For this case, after integration with respect to θ,

$$\begin{aligned} P_d &= \int_0^{\infty} \exp\left(-\frac{r}{b}\right)\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr \\ &= \exp\left(\frac{\sigma^2}{2b^2}\right)\int_0^{\infty}\exp\left[-\frac{(rb+\sigma^2)^2}{2\sigma^2 b^2}\right]\frac{r}{\sigma^2}\,dr \end{aligned} \tag{3.118}$$

which can be integrated further with the substitution of

$$z = \frac{rb+\sigma^2}{\sqrt{2}\sigma b} \tag{3.119}$$

so that finally

$$P_d = 1 - \frac{\sqrt{2\pi}\sigma}{2b}\exp\left(\frac{\sigma^2}{2b^2}\right)\left[1 - \operatorname{erf}\left(\frac{\sqrt{2}\sigma}{2b}\right)\right] \tag{3.120}$$

Inclined Step Damage Function

The inclined step damage function can be expressed in terms of a sure kill radius R_{SK} and a sure save radius R_{SS} by the following expression:

$$d(r) = \begin{cases} 1, & 0 \le r < R_{SK} \\ \dfrac{R_{SS}-r}{R_{SS}-R_{SK}}, & R_{SK} \le r < R_{SS} \\ 0, & r \ge R_{SS} \end{cases} \tag{3.121}$$

as shown in Fig. 3.28c. The probability of destruction in this case becomes

$$\begin{aligned} P_d = &\int_0^{R_{SK}} 1 \times \exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr + \int_{R_{SK}}^{R_{SS}} \frac{R_{SS}-r}{R_{SS}-R_{SK}}\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr \\ &+ \int_{R_{SS}}^{\infty} 0 \times \exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr \end{aligned} \tag{3.122}$$

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To express the preceding integral in terms of known functions, it is necessary to substitute a new variable $z = r/\sqrt{2}\sigma$, and after some relatively simple manipulations, including integration by parts, it can be shown that

$$P_d = 1 - \frac{\sqrt{2\pi}\sigma}{2(R_{SS} - R_{SK})}\left[\text{erf}\left(\frac{\sqrt{2}R_{SS}}{2\sigma}\right) - \text{erf}\left(\frac{\sqrt{2}R_{SK}}{2\sigma}\right)\right] \tag{3.123}$$

Log-Normal Damage Function

Probably the most accurate representation of the damage function is the log-normal distribution introduced by Bridgman.[13] This function can be represented in a slightly modified form as (see Fig. 3.28d)

$$d(r) = 1 - \int_0^r \frac{1}{\sqrt{2\pi}\beta r} \exp\left[-\frac{\ell n^2(r/\alpha)}{2\beta^2}\right] dr \tag{3.124}$$

Bridgman assumed that the two constants α and β can be determined from a sure kill radius R_{SK} and a sure save radius R_{SS}, which he specified to correspond to 98% and 2% of the damage to the target, respectively. Naturally, the specified percentages could be defined differently, but the general principle would remain the same for all cases. The log-normal damage function can be simplified further by noting that

$$\frac{1}{\sqrt{2\pi}}\int_0^r \frac{1}{\beta r}\exp\left[-\frac{\ell n^2(r/\alpha)}{2\beta^2}\right] dr = \frac{1}{2}\text{erf}\left[\frac{\ell n(r/\alpha)}{\sqrt{2}\beta}\right]\Bigg|_{r=0}^{r}$$

$$= \frac{1}{2}\left\{\text{erf}\left[\frac{\ell n(r/\alpha)}{\sqrt{2}\beta}\right] - \text{erf}(-\infty)\right\} = \frac{1}{2}\left\{\text{erf}\left[\frac{\ell n(r/\alpha)}{\sqrt{2}\beta}\right] + 1\right\} \tag{3.125}$$

Hence, substituting Eq. (3.125) into Eq. (3.124) the log-normal damage function becomes

$$d(r) = \frac{1}{2}\left\{1 - \text{erf}\left[\frac{\ell n(r/\alpha)}{\sqrt{2}\beta}\right]\right\} \tag{3.126}$$

The two constants α and β for the pair of damage probabilities assumed by Bridgman[13] are calculated from

$$0.98 = d(R_{SK}) \tag{3.127}$$

$$0.02 = d(R_{SS}) \tag{3.128}$$

Hence, from Eqs. (3.126) and (3.127),

$$0.98 = \frac{1}{2}\left\{1 - \text{erf}\left[\frac{\ell n(R_{SK}/\alpha)}{\sqrt{2}\beta}\right]\right\}$$

or

$$0.96 = -\text{erf}\left[\frac{\ell n(R_{SK}/\alpha)}{\sqrt{2}\beta}\right] = -\text{erf}(z_{SK}) \tag{3.129}$$

which is satisfied with

$$z_{SK} = -1.45222 \tag{3.130}$$

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obtained from Table A.3. Similarly, from Eqs. (3.126) and (3.128),

$$0.02 = \frac{1}{2}\left\{1 - \text{erf}\left[\frac{\ell_n(R_{SS}/\alpha)}{\sqrt{2}\beta}\right]\right\}$$

or

$$0.96 = \text{erf}\left[\frac{\ell_n(R_{SS}/\alpha)}{\sqrt{2}\beta}\right] = \text{erf}(z_{SS}) \tag{3.131}$$

which is satisfied by

$$z_{SS} = 1.45222 \tag{3.132}$$

Substituting the values of z_{SK} and z_{SS} from Eqs. (3.130) and (3.132) into Eqs. (3.129) and (3.131),

$$-1.45222\sqrt{2}\beta = \ell_n(R_{SK}/\alpha) = \ell_n(R_{SK}) - \ell_n(\alpha) \tag{3.133}$$

$$1.45222\sqrt{2}\beta = \ell_n(R_{SS}/\alpha) = \ell_n(R_{SS}) - \ell_n(\alpha) \tag{3.134}$$

Now solving Eqs. (3.133) and (3.134) for α and β results in the following expressions:

$$\alpha = (R_{SK}R_{SS})^{\frac{1}{2}} \tag{3.135}$$

$$\beta = \frac{1}{2 \times 1.45222\sqrt{2}}\ell_n\left(\frac{R_{SS}}{R_{SK}}\right) = \frac{1}{2\sqrt{2}z_{SS}}\ell_n\left(\frac{R_{SS}}{R_{SK}}\right) \tag{3.136}$$

The variation of the log-normal damage function for different values of ratio R_{SK}/R_{SS} is shown in Fig. 3.29 with R_{SK} and R_{SS} corresponding to 98% and 2% damage, respectively.

The probability of damage of a point target for which the damage function $d(r)$ is represented by Eq. (3.124) can now be obtained from

$$\begin{aligned} P_d &= \int_0^\infty d(r)\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr \\ &= \frac{1}{2}\int_0^\infty \left\{1 - \text{erf}\left[\frac{\ell_n(r/\alpha)}{\sqrt{2}\beta}\right]\right\}\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr \\ &= -\frac{1}{2}\exp\left(-\frac{r^2}{2\sigma^2}\right)\Big|_{r=0}^{\infty} - \frac{1}{2}\int_0^\infty \text{erf}\left[\frac{\ell_n(r/\alpha)}{\sqrt{2}\beta}\right]\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr \\ &= \frac{1}{2}\left\{1 - \int_0^\infty \text{erf}\left[\frac{\ell_n(r/\alpha)}{\sqrt{2}\beta}\right]\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr\right\} \end{aligned} \tag{3.137}$$

The integral in the preceding expression can be evaluated numerically for a given set of values α, β, and σ.

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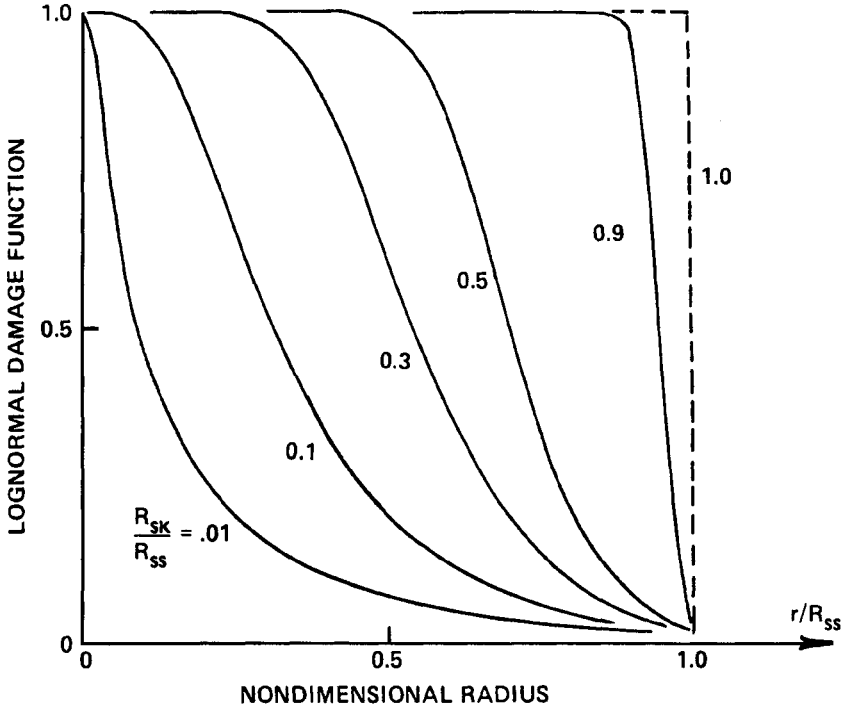


Fig. 3.29 Log-normal damage function for various values of R_{SK}/R_{SS} with R_{SK} and R_{SS} at 98% and 2%, respectively.

3.14 Shrapnel Damage Functions

In this section damage functions are derived for a warhead with exploding shrapnels *uniformly distributed* around the point of detonation. Both the surface and space target damage functions are considered.

Surface Targets

Introducing three new parameters

s_a = vulnerable segment length of the surface target
N = number of shrapnels in the radial direction
and r = distance from the detonation point

it follows that, for a single shrapnel from the warhead to hit the target at the radius r_a, the following condition must be satisfied:

$$s_a N = 2\pi r_a$$

or

$$r_a = \frac{s_a N}{2\pi} \tag{3.138}$$

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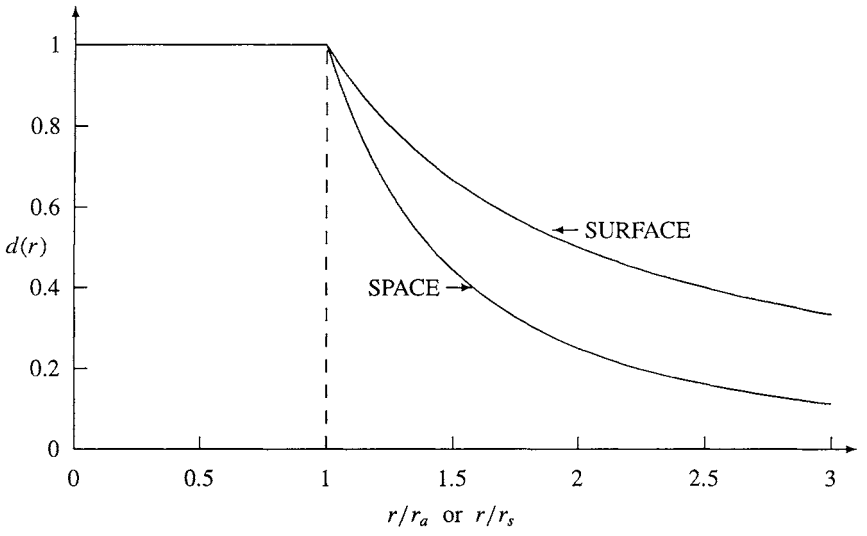


Fig. 3.30 Shrapnel damage functions for surface and space targets.

If it is assumed that a single hit will destroy the target, then for $0 \leq r \leq r_a$ there is certainty that the target is destroyed. On the other hand, for $r > r_a$ single shrapnels will be impacting segments of length s obtained from

$$sN = 2\pi r \tag{3.139}$$

and the damage function $d(r)$ at a distance r from the detonation point is given by the ratio s_a/s. Hence, the damage function can be represented by

$$d(r) = \begin{cases} 1; & 0 \leq r \leq r_a \\ s_a/s = r_a/r; & r > r_a \end{cases} \tag{3.140}$$

This shrapnel damage function for surface targets is shown in Fig. 3.30 as a function of r_a/r.

Substitution of Eq. (3.140) into Eq. (3.46) results in an expression for the probability of destruction of a surface target with the shrapnel damage function. Hence,

$$\begin{aligned} P_d &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d(x, y)p(x, y)\,dx\,dy \\ &= \int_0^{\infty}\int_0^{2\pi} \frac{d(r)}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) r\,dr\,d\theta \\ &= \int_0^{\infty} \frac{d(r)}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) r\,dr \\ &= \frac{1}{\sigma^2}\int_0^{r_a} \exp\left(-\frac{r^2}{2\sigma^2}\right) r\,dr + \frac{r_a}{\sigma^2}\int_{r_a}^{\infty} \exp\left(-\frac{r^2}{2\sigma^2}\right) dr \end{aligned} \tag{3.141}$$

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Introducing a new variable

$$z = r/\sqrt{2}\sigma \tag{3.142}$$

Equation (3.141) becomes

$$P_d = 2\int_0^{r_a/\sqrt{2}\sigma} z\exp(-z^2)\,dz + \frac{\sqrt{2}r_a}{\sigma}\int_0^{\infty}\exp(-z^2)\,dz - \frac{\sqrt{2}r_a}{\sigma}\int_0^{r_a/\sqrt{2}\sigma}\exp(-z^2)dz$$

$$= 1 - \exp\left(-\frac{r_a^2}{2\sigma^2}\right) + \frac{\sqrt{2\pi}\,r_a}{2\sigma}\left[1 - \operatorname{erf}\left(\frac{r_a}{\sqrt{2}\sigma}\right)\right] \tag{3.143}$$

Space Targets

For a space target the following parameter is introduced:

A_s = vulnerable surface area of the space target.

Assuming that N shrapnels are distributed radially in all directions, it follows that, for a single shrapnel from the warhead to hit the target vulnerable area A_s at radius r_s, the following relationship must be satisfied

$$A_s N = \pi r_s^2$$

or

$$r_s^2 = \frac{A_s N}{\pi} \tag{3.144}$$

Also at radius r

$$r^2 = \frac{AN}{\pi} \tag{3.145}$$

where A is an area impacted by a single shrapnel. It is obvious that, in this case, the damage function $d(r)$ for $r > r_s$ is given by the ratio A_s/A, while $d(r) = 1$ for $0 \le r \le r_s$. Hence, the damage function $d(r)$ for a space target can be represented by

$$d(r) = \begin{cases} 1; & 0 \le r \le r_s \\ A_s/A = r_s^2/r^2; & r > r_s \end{cases} \tag{3.146}$$

This shrapnel damage function for space targets is shown in Fig. 3.30 as a function of r_s/r.

It should be noted that as shrapnels move through the air their velocity decreases due to friction and their kinetic energy decreases to a point where they become ineffective on the target. Of course, this is not so for shrapnels traveling outside the Earth's atmosphere.

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Substitution of Eq. (3.146) into Eq. (3.46) expressed in spherical coordinates results in an expression for the probability of space target destruction given by

$$P_d = \frac{1}{2^{3/2}\pi^{3/2}\sigma^3}\int_0^{\infty}\int_0^{\pi}\int_0^{2\pi} d(r)\exp\left(-\frac{r^2}{2\sigma^2}\right) r^2 \sin\varphi\, dr\, d\varphi\, d\theta$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma^3}\int_0^{\infty}\int_0^{\pi} d(r)\exp\left(-\frac{r^2}{2\sigma^2}\right) r^2 \sin\varphi\, dr\, d\varphi$$

$$= \frac{2}{\sqrt{2\pi}\,\sigma^3}\int_0^{\infty} d(r)\exp\left(-\frac{r^2}{2\sigma^2}\right) r^2\, dr$$

$$= \frac{2}{\sqrt{2\pi}\,\sigma^3}\int_0^{r_s} \exp\left(-\frac{r^2}{2\sigma^2}\right) r^2\, dr + \frac{2r_s^2}{\sqrt{2\pi}\,\sigma^3}\int_0^{\infty}\exp\left(-\frac{r^2}{2\sigma^2}\right) dr$$

$$- \frac{2r_s^2}{\sqrt{2\pi}\,\sigma^3}\int_0^{r_s}\exp\left(-\frac{r^2}{2\sigma^2}\right) dr \tag{3.147}$$

Introducing again the variable $z = r/\sqrt{2}\sigma$, Eq. (3.147) transforms into

$$P_d = \frac{4}{\sqrt{\pi}}\int_0^{\frac{r_s}{\sqrt{2}\sigma}} \exp(-z^2)z^2\, dz + \frac{2r_s^2}{\sqrt{\pi}\sigma^2}\int_0^{\infty}\exp(-z^2)\, dz$$

$$- \frac{2r_s^2}{\sqrt{\pi}\,\sigma^2}\int_0^{\frac{r_s}{\sqrt{2}\sigma}}\exp(-z^2)\, dz$$

which when integrated by parts becomes

$$P_d = -\frac{2}{\sqrt{\pi}}\exp(-z^2)z\Big|_0^{\frac{r_s}{\sqrt{2}\sigma}} + \frac{2}{\sqrt{\pi}}\int_0^{\frac{r_s}{\sqrt{2}\sigma}}\exp(-z^2)\, dz$$

$$+ \frac{2r_s^2}{\sqrt{\pi}\,\sigma^2}\int_0^{\infty}\exp(z^2)\, dz - \frac{2r_s^2}{\sqrt{\pi}\,\sigma^2}\int_0^{\frac{r_s}{\sqrt{2}\sigma}}\exp(-z^2)\, dz$$

$$= -\frac{2r_s}{\sqrt{2\pi}\,\sigma}\exp\left(-\frac{r_s^2}{2\sigma^2}\right) + \operatorname{erf}\left(\frac{r_s}{\sqrt{2}\sigma}\right) + \frac{r_s^2}{\sigma^2}\left[1 - \operatorname{erf}\left(\frac{r_s}{\sqrt{2}\sigma}\right)\right] \tag{3.148}$$

3.15 Weapon Effective Radius (R_w) for Surface Targets

The simple concept of the cookie cutter damage function can be applied to the calculation of the probability of destruction of a point target for zero aiming offset by introducing the concept of an effective weapon radius R_w defined by the following equation:

$$P_d = \iint_{(x^2+y^2)^{\frac{1}{2}} \le R_w} p(x, y)\, dx\, dy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d(x, y)p(x, y)\, dx\, dy \tag{3.149}$$

The preceding equation ensures that the cookie cutter calculation based on the

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effective radius R_w results in the same probability of damage as the exact computation using the actual damage function $d(x, y)$. For the Gaussian probability distribution $p(x, y)$, the basic expressions in Eq. (3.149) can be transformed into polar coordinates and integrated between $\theta = 0$ and 2π. Hence,

$$P_d = \int_0^{R_w} \exp\left(-\frac{r^2}{2\sigma^2}\right) \frac{r}{\sigma^2}\, dr = \int_0^{\infty} d(r) \exp\left(-\frac{r^2}{2\sigma^2}\right) \frac{r}{\sigma^2}\, dr \qquad (3.150)$$

which upon integration of the first integral leads to the fundamental expression from which R_w can be obtained:

$$1 - \exp\left(-\frac{R_w^2}{2\sigma^2}\right) = \int_0^{\infty} d(r) \exp\left(-\frac{r^2}{2\sigma^2}\right) \frac{r}{\sigma^2}\, dr \qquad (3.151)$$

Equation (3.151) will now be solved for R_w for the damage functions discussed in the previous sections.

The concept of the weapon effective radius R_w allows the analyst to replace the actual damage function with the cookie cutter function of radius R_w. This means that the probability of destruction of a point target can be obtained from

$$P_d = \iint_{(x^2+y^2)^{\frac{1}{2}} \le R_w} p(x, y)\, dx\, dy \qquad (3.45)$$

where the integration is performed only over the region ascribed by the radius R_w. The equivalent cookie cutter radius is shown for the case of an exponential damage function in Fig. 3.31, where the continuously diminishing damage function $d(r/b)$ from $r/b = 0$ to $r/b = \infty$ is replaced with $d(r/b) = 1$ between $r/b = 0$ and $r/b = 0.9188$.

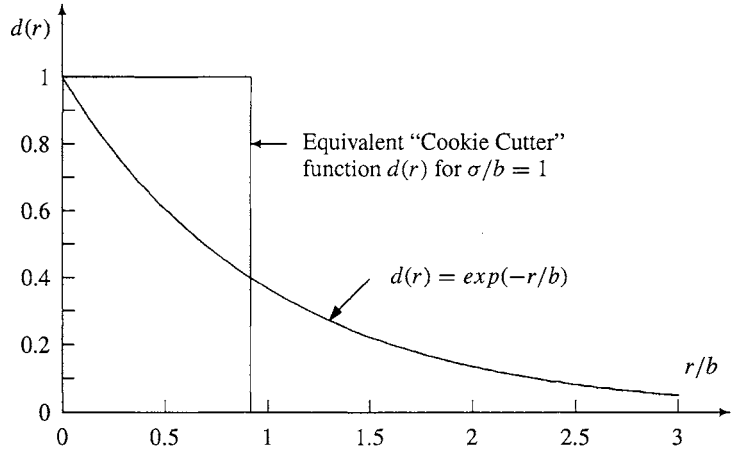


Fig. 3.31 Comparison of the actual exponential damage function with its equivalent cookie cutter function for $\sigma/b = 1.0$.

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If the aiming offset is not zero then Eq. (3.149) for the calculation of the effective radius R_w must be modified as

$$P_d = \iint_{(x^2+y^2)^{\frac{1}{2}} < R_w} p(x - x_0, y - y_0)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d(x, y)p(x - x_0, y - y_0)\,dx\,dy \tag{3.152}$$

where (x_0, y_0) are the coordinates of the aiming point and, as before, the target is placed at the origin.

Gaussian Damage Function

Combining Eq. (3.116) with Eq. (3.151) results in the following equation:

$$1 - \exp\left(-\frac{R_w^2}{2\sigma^2}\right) = \frac{b^2}{b^2 + \sigma^2}$$

or

$$\frac{R_w}{\sigma} = \left[\ell_n\left(1 + \frac{b^2}{\sigma^2}\right)^2\right]^{\frac{1}{2}} \tag{3.153}$$

Exponential Damage Function

Combining Eq. (3.120) with Eq. (3.151) leads to

$$\exp\left(-\frac{R_w^2}{2\sigma^2}\right) = \frac{\sqrt{2\pi}\sigma}{2b}\exp\left(\frac{\sigma^2}{2b^2}\right)\left[1 - \text{erf}\left(\frac{\sqrt{2}\sigma}{2b}\right)\right]$$

which can be rearranged as

$$\frac{R_w}{\sigma} = \left\{-2\,\ell_n\left(\frac{\sqrt{2\pi}\sigma}{2b}\right) - \frac{\sigma^2}{b^2} - 2\,\ell_n\left[1 - \text{erf}\left(\frac{\sqrt{2}\sigma}{2b}\right)\right]\right\}^{\frac{1}{2}} \tag{3.154}$$

Inclined Step Damage Function

Combining Eq. (3.123) with Eq. (3.151), it can be shown that for this case,

$$\frac{R_w}{\sigma} = \left(\ell_n\left\{\left[\frac{\sqrt{2\pi}\sigma}{2(R_{SS} - R_{SK})}\right]\left[\text{erf}\left(\frac{\sqrt{2}R_{SS}}{2\sigma}\right) - \text{erf}\left(\frac{\sqrt{2}R_{SK}}{2\sigma}\right)\right]\right\}^{-2}\right)^{\frac{1}{2}} \tag{3.155}$$

Log-Normal Damage Function

Combining Eq. (3.137) with Eq. (3.151), it can be shown that, for this case, the effective radius R_w can be calculated from

$$\frac{R_w}{\sigma} = \left\{\ell_n\left[\frac{1}{2}\left(1 + \int_0^{\infty}\text{erf}\left[\frac{\ell_n(r/\alpha)}{\sqrt{2}\beta}\right]\exp\left(-\frac{r^2}{2\sigma^2}\right)\frac{r}{\sigma^2}\,dr\right)\right]^{-2}\right\}^{\frac{1}{2}} \tag{3.156}$$

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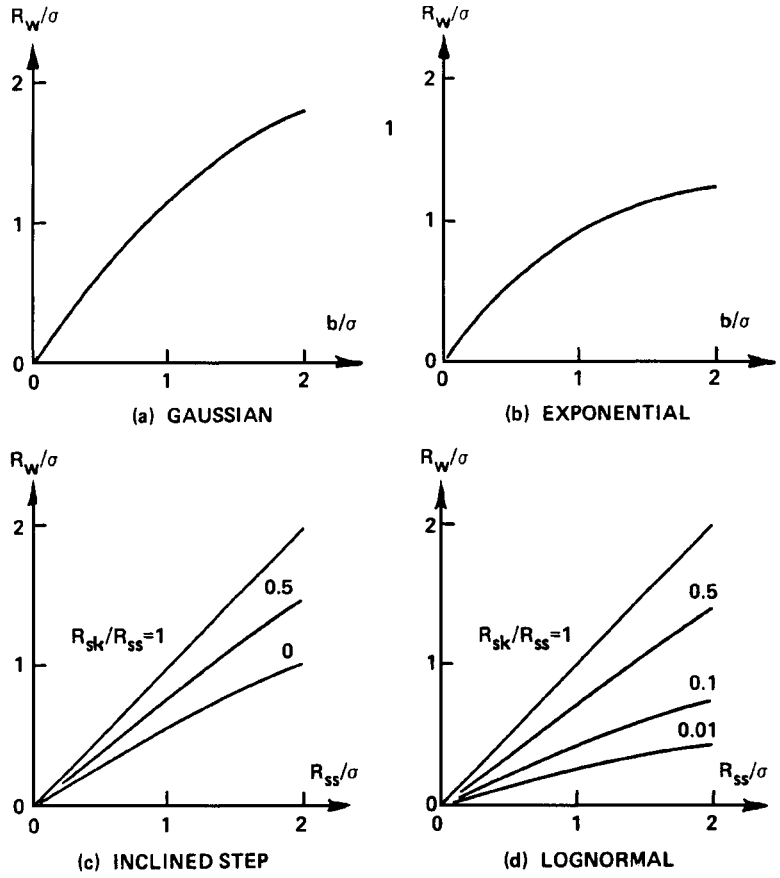


Fig. 3.32 Weapon effective radius R_w for typical damage functions.

The variation of the weapon effective radius with b/σ for the Gaussian and exponential damage functions and with R_{SS}/σ for the inclined step and log-normal functions are shown in Figs. 3.32a–3.32d. It is interesting to note the similarities between the inclined step and log-normal cases. In fact, when $R_{SK} = R_{SS}$, i.e., the case of a step function, the two damage functions give identical results for the weapon radius:

$$R_w = R_{SS} \tag{3.157}$$

as would be expected, because the two functions degenerate into the cookie cutter functions.

Shrapnel Damage Function

Combining Eq. (3.143) with Eq. (3.151) leads to

$$1 - \exp\left(-\frac{R_w^2}{2\sigma^2}\right) = 1 - \exp\left(-\frac{r_a^2}{2\sigma^2}\right) + \frac{\sqrt{2\pi}\, r_a}{2\sigma}\left[1 - \operatorname{erf}\left(\frac{r_a}{\sqrt{2}\sigma}\right)\right] \tag{3.158}$$

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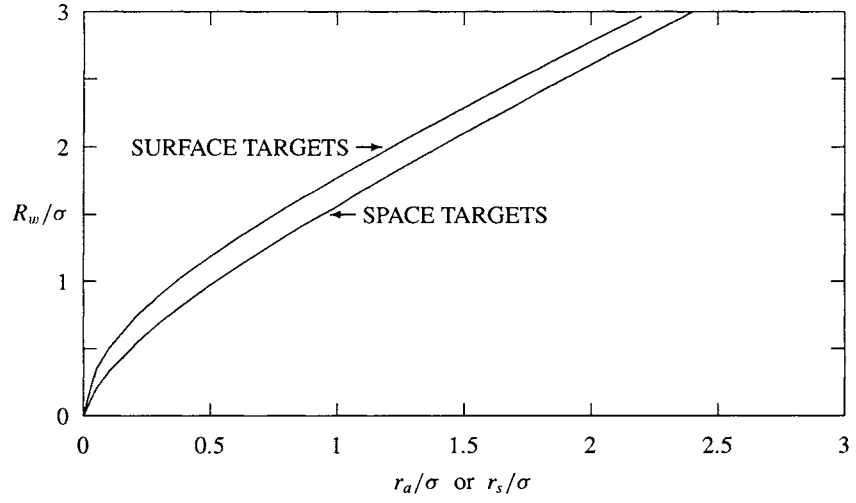


Fig. 3.33 Weapon effective radius R_w for surface and space targets with shrapnel damage functions.

Hence, it can be shown that

$$\frac{R_w}{\sigma} = \left\{-2\,\ell_n\left(\exp\left(-\frac{r_a^2}{2\sigma^2}\right) - \frac{\sqrt{2\pi}\,r_a}{2\sigma}\left[1 - \text{erf}\left(\frac{r_a}{\sqrt{2}\,\sigma}\right)\right]\right)\right\}^{\frac{1}{2}} \tag{3.159}$$

The variation of R_w/σ with r_a/σ from the preceding equation is shown in Fig. 3.33.

3.16 Weapon Effective Radius (R_w) for Space Targets

The probability of destruction of a point target with zero offset and a spherical cookie cutter damage function with the radius R_w is obtained from Eq. (3.85), with $r_0 = 0$, i.e.

$$P_d = \frac{1}{\sqrt{2\pi}\,\sigma^3}\int_0^{R_w}\int_0^{\pi} \exp\left(-\frac{r^2}{2\sigma^2}\right) r^2 \sin\varphi\, dr\, d\varphi$$

which upon integration by parts leads to

$$P_d = -\frac{2}{\sqrt{\pi}}\exp\left(-\frac{R_w^2}{2\sigma^2}\right)\frac{R_w}{\sqrt{2}\sigma} + \text{erf}\left(\frac{R_w}{\sqrt{2}\sigma}\right) \tag{3.160}$$

The weapon effective radius R_w for a shrapnel damage function Eq. (3.146) is obtained by equating Eq. (3.159) to Eq. (3.148) that leads to

$$\begin{aligned} P_d &= -\frac{2}{\sqrt{\pi}}\exp\left(-\frac{R_w^2}{2\sigma^2}\right)\frac{R_w}{\sqrt{2}\sigma} + \text{erf}\left(\frac{R_w}{\sqrt{2}\sigma}\right) \\ &= -\frac{2r_s}{\sqrt{2\pi}\,\sigma}\exp\left(-\frac{r_s^2}{2\sigma^2}\right) + \text{erf}\left(\frac{r_s}{\sqrt{2}\sigma}\right) + \frac{r_s^2}{\sigma^2}\left[1 - \text{erf}\left(\frac{r_s}{\sqrt{2}\sigma}\right)\right] \end{aligned} \tag{3.161}$$

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The preceding equation can only be solved numerically for R_w/σ as a function of r_s/σ. This is shown in Fig. 3.33.

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Problems

3.1. Determine the probability of a single hit on the rectangular target 5 m × 3 m with a missile having

Range Error Probable (REP) = Cross-Range Error Probable (CREP) = 3 m

using (a) Table A.5 and (b) Table A.3.

3.2. Show that the expected value of the normal distribution

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-\overline{x})^2}{2\sigma^2}\right]$$

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is $\bar{x}$, i.e.,

$$E[x] = \int_{-\infty}^{\infty} xp(x)\,dx = \bar{x}$$

3.3. Determine the probability of destruction of a point target in a plane when the aiming offset $r_0 = \sigma$ and the lethal radius $R = 2\sigma$ where σ is the standard deviation. What is the probability of target destruction for four shots?

Hint: use Fig. 3.14 for the single-shot probability.

3.4. Show that the probability of destruction of a point target in space with zero aiming offset is given by

$$P\left(\frac{R}{\sigma}, 0\right) = -\frac{2}{\sqrt{\pi}} \exp\left(-\frac{R^2}{2\sigma^2}\right) \frac{R}{\sqrt{2}\sigma} + \operatorname{erf}\left(\frac{R}{\sqrt{2}\sigma}\right)$$

3.5. Derive Eq. (3.83).

3.6. Derive Eq. (3.105).

3.7. Derive Eq. (3.107).

3.8. Prove that $R_w = R_{SS}$ for the inclined step damage function when $R_{SK} = R_{SS}$ where R_w is the weapon effective radius, R_{SK} the sure kill radius, and R_{SS} the sure save radius.

Hint: in the general expression for R_w expand the term "erf" as a power series in R_{SK} and R_{SS}.

3.9. Prove that $R_w = R_{SS}$ for the log-normal damage function for $R_{SK} = R_{SS}$ where R_w, R_{SK}, and R_{SS} are defined in Problem 3.8.

3.10. Calculate R_w/σ for a Gaussian damage function for $\sigma/b = 0.5$.

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4
Deterministic Combat Models

The previous chapter dealt with passive targets, i.e., targets without any active defenses. The passive targets represent special cases of military engagements. More typical cases involve active defenses, such as combat engagements, in which troops or weapons interact with each other. This chapter describes the modeling of homogeneous combat engagements based on the deterministic approach, which leads to a set of differential equations from which the strength of each side of the engaging forces can be determined at any given time. The other approach to the modeling of combat engagements is to treat them probabilistically and determine the probability of occurrence of different strength levels as the battle progresses. Such probability combat models are described in Chapter 5. This chapter begins by discussing the fundamental concepts of the attrition rate coefficients for use in various deterministic models. The combat models include the Lanchester model for directed fire, the Lanchester model for area fire, a guerrilla warfare model, and several others. A mobile attack model is also discussed and the Lanchester model for directed fire is applied to the Iwo Jima battle in World War II to illustrate the practical applications of one of the models. This chapter deals with the mathematical formulation of modeling of combat engagements for which analytical solutions are possible in terms of known tabulated functions; it also provides the theoretical foundation for the numerical solutions of the more general cases, discussed in Chapter 7, where the attrition coefficients may vary with time or the fire on enemy troops or weapons may be reallocated during the battle.

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4.1 Effective Firing Rate: Attrition Rate Coefficients

The basic concept in mathematical modeling of combat engagements centers around the notion of the attrition rates of combat troops, weapons, vehicles, military or economic resources, or some other specified targets. The attrition rates are expressed as numbers of individual targets (combatants, weapons, etc.) destroyed in a unit of time or as an area effectively destroyed in a unit of time. In reality these rates vary with time and location, and they may also depend on the troop or weapon concentration; availability of adequate logistics supply; adequate command, control, and communications; intelligence information; prior training; and, perhaps most importantly, the resolve of the combatants to engage in the battle to a maximum level possible constrained only by physical limitations.

The attrition rate can be related to the total time T required to destroy one combatant or weapon. The simplest relationship is

$$T = t_{a_{RB}} + \frac{1}{a_B P_{\mathrm{SSK}_{RB}}} \tag{4.1}$$

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where

$t_{a_{RB}}$ = time for Blue combatant or weapon (B) acquire a Red target (R)
a_B = firing rate of a single Blue combatant while engaging an acquired target
$P_{\mathrm{SSK}_{RB}}$ = single-shot probability of kill ($0 \leq P_{\mathrm{SSK}_{RB}} \leq 1$)

The first subscript refers to the target that suffers the attrition, and the second one refers to the firer. The firing rate a_B depends not only on the capability itself but also on the skills of the weapon operator. The product $a_B P_{\mathrm{SSK}_{RB}}$ is the rate at which acquired targets are destroyed by a single firer. Noting that the reciprocal of the rate is the time in which a single target is destroyed, it follows from Eq. (4.1) that the effective firing rate for the combatants is represented by

$$a_{RB} = \frac{1}{T} = \frac{1}{t_{a_{RB}} + \frac{1}{a_B P_{\mathrm{SSK}_{RB}}}} \tag{4.2}$$

while the corresponding effective firing rate for the Red combatants is

$$a_{BR} = \frac{1}{t_{a_{BR}} + \frac{1}{a_R P_{\mathrm{SSK}_{RB}}}} \tag{4.3}$$

4.2 Markovian Attrition Rates: Probabilistic Rates[1–4]

Note: *Because of the mathematical complexities, this section can be omitted at the first reading. It is not required for any subsequent derivations; it is included here only to demonstrate that the attrition rates derived on the basis of purely physical considerations may also be derived from a probabilistic analysis.*

Weapon systems may be classified as Markov chain fire weapons for which the outcome of firing of a round by the weapon depends on the outcome of the immediately preceding round; if the target is destroyed, the firing stops whereas if the target is undamaged, the firing continues, but its outcome is independent of the preceding rounds. For such systems the probabilistic attrition rates can be determined from the sequence of events from target acquisition to destruction. The formal mathematical derivation is rather complicated, and the reader can go directly to Eq. (4.30) for the final results. The full analysis is shown here only to demonstrate that the intuitive physical approach to the problem of computing the attrition coefficients is in fact in agreement with the probabilistic analysis. Therefore, to analyze the firing sequence of events, the following parameters are introduced:

t_a = time to acquire the target
t_1 = time to fire the first round
t_h = time to fire a round given the preceding round was a hit
t_m = time to fire a round given the preceding round was a miss
t_f = time of flight of the projectile from weapon to target
z = number of hits required to destroy the target (each hit does not necessarily destroy the target)

The sequence of firings begins with the target acquisition lasting t_a seconds. The first round is then fired and reaches the target some $(t_1 + t_f)$ seconds later. If the first round is a miss, the next round will arrive at the target $(t_m + t_f)$ seconds later.

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If, on the other hand, the first round hits the target without destroying it ($z > 1$), the next round will arrive at the target ($t_h + t_f$) seconds later. The sequence of firings with hits and misses is continued until the final hit destroys the target. The total number of rounds fired, both hits and misses, is denoted by N, and the last hit destroys the target. For such a sequence of firings, the time to destroy a target may be written as

$$\begin{aligned} T &= t_a + (t_1 + t_f) + (t_h + t_f)(z - 1) + (t_m + t_f)(N - z) \\ &= c_1 + c_{2N} \end{aligned} \tag{4.4}$$

where

$$t_1 = t_a + t_1 - t_h + (t_h - t_m)z \tag{4.5}$$

and

$$c_2 = t_m + t_f \tag{4.6}$$

The weapon lethality characteristics against a particular target are described as the conditional probability of a kill given a hit. This probability is usually expressed as $P(K \mid H)$. It depends on the extent of the mechanical damage caused by striking a target and on the resulting loss of combat effectiveness. Consequently, the target may require more than one hit to be completely destroyed. Thus, the probability of destroying the target after z hits is obtained directly from the conditional probability (of two independent events) as

$$P_z(K) = [1 - P(K \mid H)]^{z-1} P(K \mid H) \tag{4.7}$$

where $1 - P(K \mid H)$ denotes the probability of survival after one hit, and $[1 - P(K \mid H)]^{z-1}$ denotes the probability of survival after $z - 1$ hits. The function $P_z(K)$ can be treated as the probability density function for the variable z [see Eq. (3.37)].

Next, the weapon's accuracy capabilities will be considered in terms of the number of rounds fired (hits and misses) to destroy the target (note that in general more than one hit is required to destroy the target). To derive the probability distribution of the number of rounds N, the following probabilities will be defined:

P_1 = first round probability of a hit
u = conditional probability of a hit given the preceding round fired hit the target
v = conditional probability of a hit given the preceding round fired missed the target

If the target is destroyed after z hits, it will be necessary to examine the sequence of firings that end with a hit. This sequence can occur in two mutually exclusive and collectively exhaustive ways. Thus, the probability of occurrence of the sequence can be expressed as

$$f_z(N) = f_z(N \mid H, H) + f_z(N \mid M, H) \tag{4.8}$$

where the first term is the probability of getting z hits in N firings in which the first and last rounds of the sequence are hits, and the second term is the probability of getting z hits in N firings in which the first round is a miss and the last one is a hit.

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Table 4.1 Hit/miss firing sequence

Event	Result
First r_1 firings	Event hit occurs every time
Next s_1 firings	Event miss occurs every time
Next r_2 firings	Event hit occurs every time
Next s_2 firings	Event miss occurs every time
...	...
Next s_{k-1} firings	Event miss occurs every time
Next r_k firings where $k \geq 1$	Event hit occurs every time

To determine $f_z(N \mid H, H)$, the typical sequence of events given in Table 4.1 must be considered. The joint occurrence of these events has the probability

$$P_1 u^{r_1-1}[(1-u)(1-v^{s_1-1})](vu^{r_2-1})[(1-u)(1-v^{s_2-1})]\cdots(vu^{r_k-1})$$
$$= P_1 u^{r_1+r_2+\cdots+r_k-k}(1-u)^{k-1}(1-v)^{s_1+s_2+\cdots+s_k-(k-1)}v^{k-1} \tag{4.9}$$

Noting that the total number of hits is z and the total number of misses is $N-z$, the following simple relations can be used:

$$\sum_{i=1}^{k} r_i = z \tag{4.10}$$

and

$$\sum_{i=1}^{k-1} s_i = N - z \tag{4.11}$$

Substitution of Eqs. (4.10) and (4.11) into Eq. (4.9) leads to

$$P_1 u^{z-k}(1-u)^{k-1}(1-v)^{N-z-k+1}v^{k-1} \tag{4.12}$$

The probability of occurrence as expressed by Eq. (4.12) can also occur in many different sequences arranged in accordance with basic sequence $r_1, s_1, r_2, s_2, \ldots, s_{k-1}, r_k$. The number of hits z can be expressed as a sum of k positive integers r_i arranged in

$$\binom{z-1}{k-1}$$

combinations, and the number of misses $N-z$ can be expressed as a sum of $k-1$ positive integers s_i arranged in

$$\binom{N-z-1}{k-2}$$

combinations (for the derivation of the combinatorial formulas see Ref. 5, i.e., the problem of occupancy, the number of distinguishable distributions into different

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cells in which no cell remains empty). Therefore, for the case of N firings to obtain z hits with the first and last firing being hits with probabilities of P_1 and 1.0, respectively, and the hits and misses occurring in k and $k-1$ groups, respectively, the total number of combinations is given by the product

$$\binom{z-1}{k-1} \times \binom{N-z-1}{k-2}$$

and the probability of occurrence of any of the mutually exclusive sequences is obtained by the addition of the individual probabilities [Eq. (4.12)]

$$P_1\binom{z-1}{k-1}\binom{N-z-1}{k-2}u^{z-k}(1-u)^{k-1}(1-v)^{N-z-k+1}v^{k-1} \tag{4.13}$$

where $z \leq N$ and $1 \leq k \leq z$. Because the sequences for different values of k are also mutually exclusive events, the first term in Eq. (4.8) can be computed as the summation of the probabilities given by Eq. (4.13) over all values of k, i.e.,

$$f_z(N \mid H, H) = \begin{cases} P_1 u^{z-1} & \text{for } N = z \\ P_1 \displaystyle\sum_{k=2}^{z} \binom{z-1}{k-1}\binom{N-z-1}{k-2} u^{z-k} & \\ \quad \times (1-u)^{k-1} v^{k-1} (1-v)^{N-z-k+1} & \text{for } N > z \end{cases} \tag{4.14}$$

The term for $k = 1$ is not included in Eq. (4.14), because

$$\binom{N-z-1}{k-2} = 0 \quad \text{for} \quad k = 1 \quad \text{and} \quad N > z$$

To determine $f_z(N \mid M, H)$, the typical sequence of events given in Table 4.2 must be considered. The probability of the joint occurrence of these events is

Table 4.2 Miss/hit firing sequence

Event	Result
First s_1 firings	Event miss occurs every time
Next r_1 firings	Event hit occurs every time
Next s_2 firings	Event miss occurs every time
Next r_2 firings	Event hit occurs every time
...	...
Next s_k firings	Event miss occurs every time
Next r_k firings	Event hit occurs every time

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given by

$$[(1-P_1)(1-v)^{s_1-1}](vu^{r_1-1})[(1-u)(1-v)^{s_2-1}] \times (vu^{r_2-1})\cdots[(1-u)(1-v)^{s_k-1}](vu^{r_k-1}) = (1-P_1)(1-v)^{s_1+s_2+\cdots+s_k-k}(1-u)^{k-1}v^k u^{r_1+r_2+\cdots+r_k-1} \quad (4.15)$$

Also, for this case,

$$\sum_{i=1}^{k} s_i = N - z \quad (4.16)$$

and

$$\sum_{i=1}^{k} r_i = z \quad (4.17)$$

Hence, the probability of a typical sequence becomes

$$(1-P_1)(1-v)^{N-z-k}(1-u)^{k-1}v^k u^{z-k} \quad (4.18)$$

Also, the number of misses $N-z$ can occur in

$$\binom{N-z-1}{k-1}$$

combinations, whereas the number of hits z occurs in

$$\binom{z-1}{k-1}$$

combinations. Hence, it is clear that

$$f_z(N \mid M, H) = (1-P_1)\sum_{k=1}^{z}\binom{z-1}{k-1}\binom{N-z-1}{k-1}(1-v)^{N-z-k}(1-u)^{k-1}v^{z-k} \quad (4.19)$$

Substitution of Eqs. (4.14) and (4.19) into Eq. (4.8) leads to

$$f_z(N) = \begin{cases} P_1 u^{z-1} & \text{for } N = z \\ P_1\sum_{k=2}^{z}\binom{z-1}{k-1}\binom{N-z-1}{k-2} \times u^{z-k}(1-u)^{k-1}v^{k-1}(1-v)^{N-z-k+1} \\ \quad +(1-P_1)\sum_{k=1}^{z}\binom{z-1}{k-1}\binom{N-z-1}{k-1} \times u^{z-k}(1-u)^{k-1}v^k(1-v)^{N-z-k} & \text{for } N > z \end{cases} \quad (4.20)$$

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To proceed further with the analysis, it is necessary to introduce the characteristic function of Eq. (4.20), which is defined as[6]

$$\Phi_N(s) = \sum_{N=0}^{\infty} e^{isN} f_z(N) \tag{4.21}$$

where s is a dummy variable and $i^2 = -1$. It can be shown that substitution of Eq. (4.20) into Eq. (4.21) after some considerable algebraic manipulation leads to (see Appendix B)

$$\Phi_N(s) = e^{isz}\left[P_1 + (1-P_1)\frac{v}{1-q\,e^{is}}\right]\left[u + (1-u)\frac{v\,e^{is}}{1-q\,e^{is}}\right]^{z-1} \tag{4.22}$$

where

$$q = 1 - v \tag{4.23}$$

To demonstrate that $f_z(N)$ is a probability function, it is necessary to show that

$$\Phi_N(0) = \sum_{N=0}^{\infty} f_z(N) = 1 \tag{4.24}$$

which is clearly satisfied by Eq. (4.22) when $s = 0$.

The expected value of N is obtained from

$$E(N) = \frac{1}{i}\left.\frac{d\Phi_N(s)}{ds}\right|_{s=0}$$

$$= z + \frac{1-P_1}{v} + \frac{(1-u)(z-1)}{v} \tag{4.25}$$

It is interesting to observe here that, if all rounds fired are independent of each other, i.e., $P_1 = u = v$, then Eq. (4.25) reduces to

$$E(N) = \frac{z}{v} \tag{4.26}$$

which is the mean of the negative binomial distribution.[7]

From Eq. (4.4) it follows that the conditional expected time to defeat a target for a given number of hits z is given by

$$E(T \mid z) = c_1 + c_2 E(N) \tag{4.27}$$

The unconditional expected value of T, on the other hand, is given by

$$E(T) = \sum_{z=1}^{\infty} E(T \mid z) P_z(K) \tag{4.28}$$

where $P_z(K)$ given by Eq. (4.7) is the probability density distribution for the variable z. Hence, it can be shown from Eqs. (4.5–4.7), (4.25), (4.27), and (4.28) that

$$E(T) = t_a + t_1 - t_h + \frac{t_h + t_f}{P(K \mid H)} + \frac{t_m + t_f}{v}\left[\frac{1-u}{P(K \mid H)} + u - P_1\right] \tag{4.29}$$

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If necessary, several simplifying assumptions can be introduced into Eq. (4.29), such as the following:

1) Negligible target acquisition time, i.e., $t_a = 0$;

2) Uniform rate of fire, i.e., $t_1 = t_h = t_m = 1/a$, where a represents the firing rate;

3) Statistical independence among firing outcomes, i.e., $u = v = P_1 = P(K \mid H)$;

4) Negligible time of flight, i.e., $t_f = 0$.

When all of the preceding assumptions are used, Eq. (4.29) simplifies to

$$E(T) = \frac{1}{aP(K \mid H)} \tag{4.30}$$

which agrees with Eq. (4.1). The reason for the derivation of Eqs. (4.29) and (4.30) through the probabilistic approach is to demonstrate formally that the simplified Markovian attrition rates $1/E(T)$ are in fact equal to the attrition rates derived on the basis of purely physical considerations of the problem. Equation (4.30) plays a central role in determining the attrition rates in any combat engagements depicted through mathematical modeling. These attrition rates will be used extensively in the remainder of this chapter.

4.3 Lanchester Model for Directed Fire: Square Law

In 1916, Frederic Lanchester developed a mathematical analysis of air combat,[8,9] and his analysis has since served as the fundamental model for developing theories of combat and for calculating attrition of forces in military engagements. The overall concept is based on a simple mathematical expression equating the attrition rate to the product of the opposing force effective firing rate of a single force unit or weapon per unit of time and the total number of opposing units or weapons engaged in combat. Thus, if the total number of Blue forces (either people or weapons) is denoted by m_B, its effective firing rate is a_B and the corresponding quantities for the Red forces are m_R and a_R, then the governing differential equations can be written as

$$\frac{dm_B}{dt} = -a_R m_R \tag{4.31}$$

and

$$\frac{dm_R}{dt} = -a_B m_B \tag{4.32}$$

where t denotes the time. These equations also imply that the following assumptions must be met during the combat engagement:

1) Both forces are homogeneous and are continually engaged in combat.

2) Each unit or individual weapon is within the maximum weapon range of all of the opposing units.

3) Collateral damage within the target area is negligible.

4) The attrition coefficients a_B and a_R also include the probabilities of the target being destroyed when hit.

5) The effective firing rates are independent of the opposing force level.

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6) Each unit is aware of the location and condition of all opposing units so that its fire is directed only to live units or functioning weapons. Thus, when a target is destroyed, search begins immediately for a new target.

7) Fire is uniformly distributed over surviving units.

Because of assumption 6, the conceptual model represented by Eqs. (4.31) and (4.32) is also referred to as the Lanchester model with transfer of fire or directed fire model. To simplify the subsequent analysis, the effective firing rates a_B and a_R will be taken as constants; however, in more refined computations these rates may be assumed as known functions of time.

Differentiation of Eq. (4.31) with respect to t and substitution of Eq. (4.32) leads to the second-order differential equation

$$\frac{d^2 m_B}{dt^2} - a_B a_R m_B = 0 \tag{4.33}$$

The appropriate initial conditions for this equation at $t = 0$ are

$$m_B(0) = M_B \tag{4.34}$$

the initial strength of the Blue force, and

$$m_R(0) = M_R \tag{4.35}$$

the initial strength of the Red force.

From Eqs. (4.33) and (4.31) it can be shown that for the initial conditions given previously,

$$m_B(t) = M_B \cosh \sqrt{a_B a_R}\, t - M_R \sqrt{\frac{a_R}{a_B}} \sinh \sqrt{a_B a_R}\, t \tag{4.36}$$

$$m_R(t) = M_R \cosh \sqrt{a_B a_R}\, t - M_B \sqrt{\frac{a_B}{a_R}} \sinh \sqrt{a_B a_R}\, t \tag{4.37}$$

The preceding equations can be rewritten in the nondimensional form as

$$\mu_B = \cosh \tau - \Phi_0^{-1} \sinh \tau \tag{4.38}$$

$$\mu_R = \cosh \tau - \Phi_0 \sinh \tau \tag{4.39}$$

where

$$\mu_B = \frac{m_B(t)}{M_B} \tag{4.40}$$

is the nondimensional force strength (Blue),

$$\mu_R = \frac{m_R(t)}{M_R} \tag{4.41}$$

is the nondimensional force strength (Red),

$$\tau = \sqrt{a_B a_R}\, t \tag{4.42a}$$

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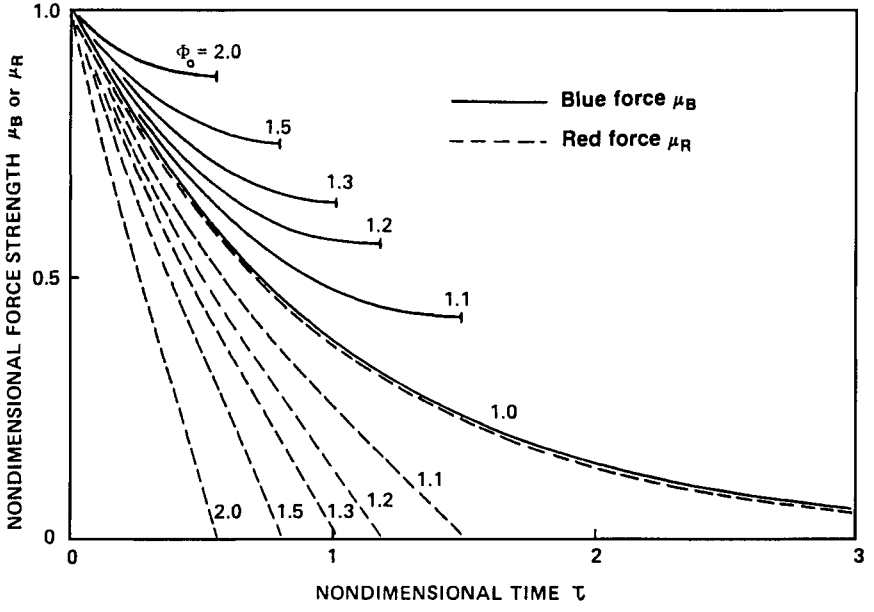


Fig. 4.1 Variation of the nondimensional force strength μ_B (Blue forces) or μ_R (Red forces) with the nondimensional time τ for $\Phi_0 \geq 1$ (directed fire combat).

is the nondimensional time, and

$$\Phi_0 = \frac{M_B}{M_R}\sqrt{\frac{a_B}{a_R}} \tag{4.43a}$$

is the superiority parameter.

The nondimensional representation of μ_B and μ_R allows for a better understanding of the mechanics of combat and the significant parameters controlling its outcome. It is interesting to note that the parameter $\sqrt{a_B a_R}$ in the nondimensional time τ represents in effect combat intensity of the battle engagement measured in terms of a geometric mean of effective firing rates of the two combatant forces. The variation of μ_B and μ_R with τ is shown in Figs. 4.1 and 4.2 for various values of Φ_0. The curves for μ_B (Blue force) are presented by solid lines, and the curves for μ_R (Red force) are represented by dashed lines. The curves for μ_B and μ_R merge for $\Phi_0 = 1$. If $\Phi_0 > 1$, the Blues are stronger than the Reds, and the combat engagement will end after a certain time in victory for the Blues; if $\Phi_0 < 1$, the opposite is true; and if $\Phi_0 = 1$, neither side has an advantage. In the special case when $\Phi_0 = 1$, the two opposing forces are equal in effective strengths, and Eqs. (4.38) and (4.39) reduce to

$$\mu_B = \mu_R = \cosh\tau - \sinh\tau = \exp(-\tau) \tag{4.44}$$

The time τ_0 needed to annihilate the Reds when $\Phi_0 > 1$ is determined from

$$\mu_R = \cosh\tau_0 - \Phi_0 \sinh\tau_0 = 0 \tag{4.45a}$$

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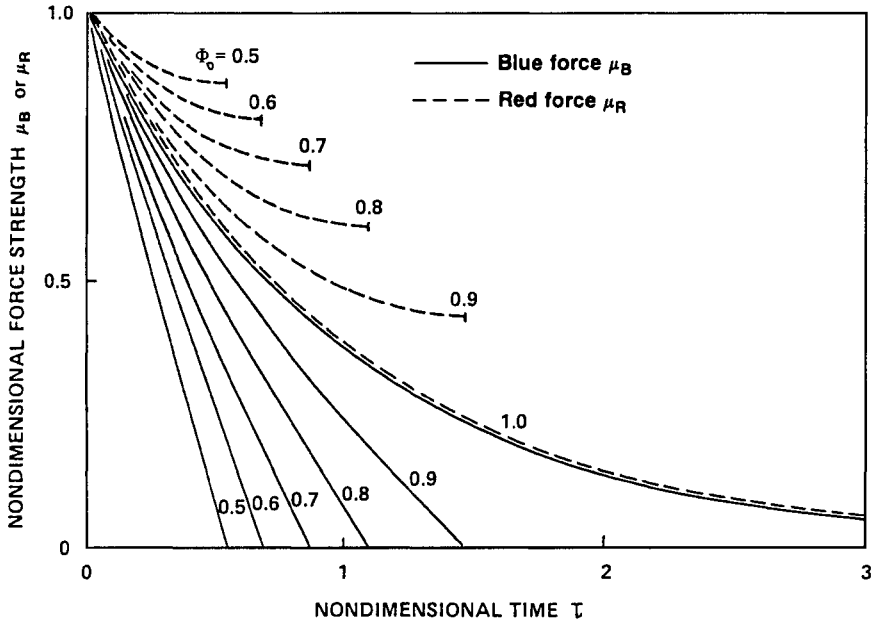


Fig. 4.2 Variation of the nondimensional force strength μ_B (Blue forces) or μ_R (Red forces) with the nondimensional time τ for $\Phi_0 \leq 1$ (directed fire combat).

Hence,

$$\tau_0 = \tanh^{-1}\left(\frac{1}{\Phi_0}\right) = \frac{1}{2}\,\ell n\frac{\Phi_0 + 1}{\Phi_0 - 1} \tag{4.46a}$$

and

$$t_0 = \frac{1}{2\sqrt{a_B a_R}}\,\ell n\frac{\Phi_0 + 1}{\Phi_0 - 1} \tag{4.46b}$$

Similarly, the time τ_0 needed to annihilate the Blues when $\Phi_0 < 1$ is determined from

$$\mu_B = \cosh\tau_0 - \Phi_0^{-1}\sinh\tau_0 = 0 \tag{4.45b}$$

Hence,

$$\tau_0 = \tanh^{-1}(\Phi_0) = \frac{1}{2}\,\ell n\frac{1 + \Phi_0}{1 - \Phi_0} \tag{4.46c}$$

and

$$t_0 = \frac{1}{2\sqrt{a_B a_R}}\,\ell n\frac{1 + \Phi_0}{1 - \Phi_0} \tag{4.46d}$$

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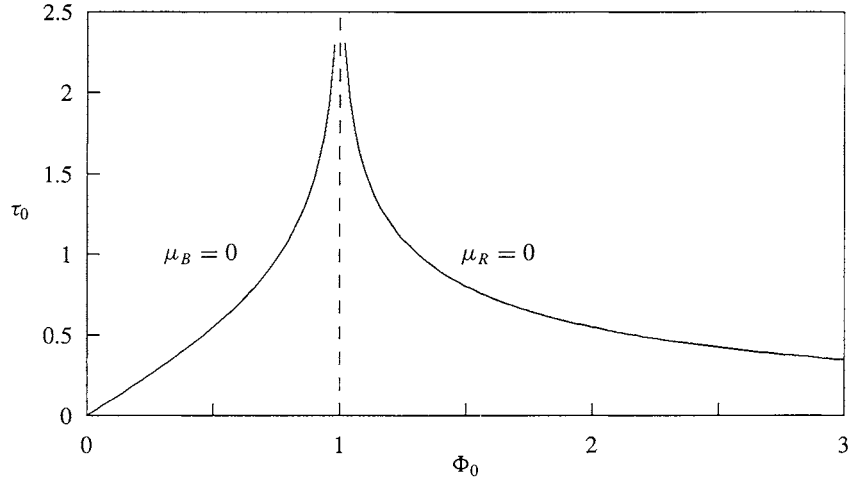


Fig. 4.3 Time to annihilate either the Red or Blue forces as a function of the superiority parameter Φ_0.

Equations (4.46a) and (4.46c) are illustrated in Fig. 4.3 for $\mu_R = 0$ and μ_B, respectively. This figure clearly demonstrates the dominant effect of the superiority parameter Φ_0 on the time needed to annihilate either the Red or Blue forces.

To obtain a functional relationship between μ_B and μ_R described by Eqs. (4.38) and (4.39), the two equations are squared, the resulting first equation is multiplied by Φ_0^2, and then the second equation is subtracted from the first. This leads to the following relationship:

$$\Phi_0^2 \mu_B^2 - \mu_R^2 = \Phi_0^2 - 1 \tag{4.47}$$

which is usually described as the Lanchester square law. This law provides a relationship between the opposing force strengths (μ_B and μ_R) at any given time.

The superiority parameter Φ_0 depends on two separate ratios: the quantitative ratio M_B/M_R and the qualitative ratio a_B/a_R. According to Eq. (4.43), an increase of M_B by a factor of two doubles the parameter Φ_0, whereas doubling of the effective firing rate a_B increases Φ_0 by a factor of the square root of two. This means that doubling of the quantitative strength is equivalent to a fourfold increase of the qualitative strength.

4.4 Lanchester Model for Area Fire: Linear Law

The Lanchester model for directed fire described an operation in which the transfer of fire from a target that has been hit to one that is not yet hit is realized immediately. This model assumed that accurate information about which targets have been hit is available to each side, and that the time necessary to acquire this information and to transfer the fire to targets that have not yet been hit can be accounted for in calculating the effective firing rate, as indicated by Eqs. (4.1) and (4.4). Consequently, the model represented a highly organized combat with complete and immediate information on the conditions of the opposing force.

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The next model will represent the case of a poorly organized combat where information on the enemy's condition is not available. The firing is simply directed into the general area where the enemy's combat units or weapons are located. This model will also be based on the first five assumptions used in the directed fire model,[10,11] i.e.,

1) Both forces are homogeneous and are continually engaged in combat.

2) Each unit or individual weapon is within the maximum range of all the opposing forces.

3) Collateral damage within the target area is negligible.

4) The attrition coefficients a_B and a_R also include the probabilities of the target being destroyed when hit.

5) The effective firing rates are independent of the enemy's force level.

The remaining assumptions, however, are different. They include the following:

6) Each unit is aware of only the general area in which enemy forces are located and directs its fire into this area without receiving any feedback information about the damage inflicted.

7) Fire is uniformly distributed over the area in which enemy forces are located.

8) All units are uniformly distributed over the area.

In the derivation of the governing equations for the area fire model, the same notation as for the directed fire will be used. To determine the decrease in the number of Blue combat units (or weapons) during the time Δt, however, a different approach must be taken because of the probabilistic nature of the area fire. This decrease is equal (with negative sign) to the number of successful hits from the Reds; however, because the firing is now into the general area where the Blue units are located, only a fraction of the total Red force will hit the Blue units. The probability that a successful shot from the Reds will fall on a combat unit of the Blue forces that has survived is equal to the ratio m_B/M_B. Consequently,

$$\Delta m_B = -\frac{a_R \Delta t\, m_R m_B}{M_B} \tag{4.48}$$

Hence, in the limit as Δt tends to zero,

$$\frac{dm_B}{dt} = -\frac{a_R m_R m_B}{M_B} \tag{4.49}$$

Similar reasoning for the Red forces leads to

$$\frac{dm_R}{dt} = -\frac{a_B m_B m_R}{M_R} \tag{4.50}$$

Using Eqs. (4.40)–(4.43), the preceding equations can be written in the nondimensional form as

$$\dot{\mu}_B = -\Phi_0^{-1} \mu_B \mu_R \tag{4.51}$$

$$\dot{\mu}_R = -\Phi_0 \mu_B \mu_R \tag{4.52}$$

where the overdots denote differentiation with respect to the nondimensional time τ. It should be noted that some publications use a_B/M_R and a_R/M_B as attrition

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coefficients for the area fire model instead of a_B and a_R used in this text. Therefore, when comparing the present analysis with other solutions, the meaning of a_B and a_R may be different.

Now further differentiation of Eq. (4.51) and substitution of Eq. (4.52) leads to the following nonlinear differential equation:

$$\ddot{\mu}_B \mu_B = \dot{\mu}_B^2 - \Phi_0 \dot{\mu}_B \mu_B^2 \tag{4.53}$$

which is of the form[12]

$$\ddot{y}y = \dot{y}^2 + (f + gy^2)\dot{y} - y(\dot{f} - \dot{g}y^2) \tag{4.54}$$

for which the first derivative is given by

$$\dot{y} = y(C_1 + gy) - f \tag{4.55}$$

where C_1 is a constant and f and g are functions of the independent variable (time). For the problem at hand, $y = \mu_B(\tau)$, $f = 0$, $g = -\Phi_0$, $\dot{f} = 0$, and $\dot{g} = 0$. Therefore, Eq. (4.55) reduces to

$$\dot{y} = C_1 y - \Phi_0 y^2 \tag{4.56}$$

The preceding differential equation is a special case of the Bernoulli equation

$$\dot{y} = F(\tau)y + G(\tau)y^k \tag{4.57}$$

where k is an integer and $F(\tau)$ and $G(\tau)$ are arbitrary functions of τ. The solution of Eq. (4.57) is given by[13]

$$y^{1-k} = Y_1 + Y_2 \tag{4.58}$$

where

$$Y_1 = C_2 e^{\psi} \tag{4.59}$$

$$Y_2 = (1-k)e^{\psi} \int e^{-\psi} G(\tau)\, d\tau \tag{4.60}$$

and

$$\psi = (1-k) \int F(\tau)\, d\tau \tag{4.61}$$

For the present case $k = 2$, $F(\tau) = C_1$, and $G(\tau) = -\Phi_0$. Therefore,

$$\psi = -\int C_1\, d\tau = -C_1\tau \tag{4.62}$$

$$Y_1 = C_2 e^{-C_1\tau} \tag{4.63}$$

$$Y_2 = -e^{-C_1\tau} \int e^{C_1\tau}(-\Phi_0)\, d\tau = \frac{\Phi_0}{C_1} \tag{4.64}$$

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provided $C_1 \neq 0$. Hence,

$$\mu_B = y = \frac{1}{Y_1 + Y_2} = \frac{C_1}{C_1 C_2 e^{-C_1 \tau} + \Phi_0} \tag{4.65}$$

and

$$\mu_R = \frac{-\Phi_0 \dot{\mu}_B}{\mu_B} = \frac{-C_1^2 C_2 \Phi_0}{\Phi_0 e^{C_1 \tau} + C_1 C_2} \tag{4.66}$$

The two unknown constants C_1 and C_2 can now be determined from the initial conditions $\mu_B = \mu_R = 1$ at $\tau = 0$. This leads to

$$\mu_B = \frac{\Phi_0^2 - 1}{\Phi_0^2 - e^{-(\Phi_0 - 1/\Phi_0)\tau}}; \qquad \Phi_0 \neq 1 \tag{4.67}$$

and

$$\mu_R = \frac{\Phi_0^2 - 1}{\Phi_0^2 e^{(\Phi_0 - 1/\Phi_0)\tau} - 1}; \qquad \Phi_0 \neq 1 \tag{4.68}$$

When $\Phi_0 = 1$ and $C_1 = 0$, Eqs. (4.63) and (4.64) become

$$Y_1 = C_2 \tag{4.69}$$

$$Y_2 = -\int (-1)\, d\tau = \tau \tag{4.70}$$

Hence, it can be shown that

$$\mu_B = \mu_R = \frac{1}{1 + \tau} \tag{4.71}$$

The variation of μ_B and μ_R with τ is shown on Figs. 4.4 and 4.5. If $\Phi_0 > 1$, then μ_B and μ_R decrease asymptotically to the values $\mu_B(\infty) = 1 - 1/\Phi_0^2$ and $\mu_R(\infty) = 0$. If $\Phi_0 < 1$, then μ_B and μ_R decrease asymptotically to $\mu_B(\infty) = 0$ and $\mu_R(\infty) = 1 - \Phi_0^2$. For $\Phi_0 = 1$ neither side has an advantage and $\mu_B(\infty) = \mu_R(\infty) = 0$. The rates of decrease in μ_B and μ_R are significantly less than the corresponding rates for the previous model representing the directed fire combat. It is interesting to compare the force strengths μ_B and μ_R at different times during the directed fire and area fire battles. Such a comparison is made in Table 4.3 for the case where the Blues have a 2 : 1 superiority, i.e., $\Phi_0 = 2$. This table clearly demonstrates the reduced efficiency of the area fire compared with the directed fire. The area fire has a more protracted, sluggish development; the advantage of one side over the other is expressed less strongly, and the decrease in the fraction of forces surviving the engagement is slower.

Elimination of τ from Eqs. (4.67) and (4.68) leads to a functional relationship between μ_B and μ_R given by

$$\Phi_0^2 \mu_B - \mu_R = \Phi_0^2 - 1 \tag{4.72}$$

which is true even for $\Phi_0 = 1$, because for this case $\mu_B = \mu_R$. Because of its linear form, the preceding relationship is usually referred to as the Lanchester linear law.

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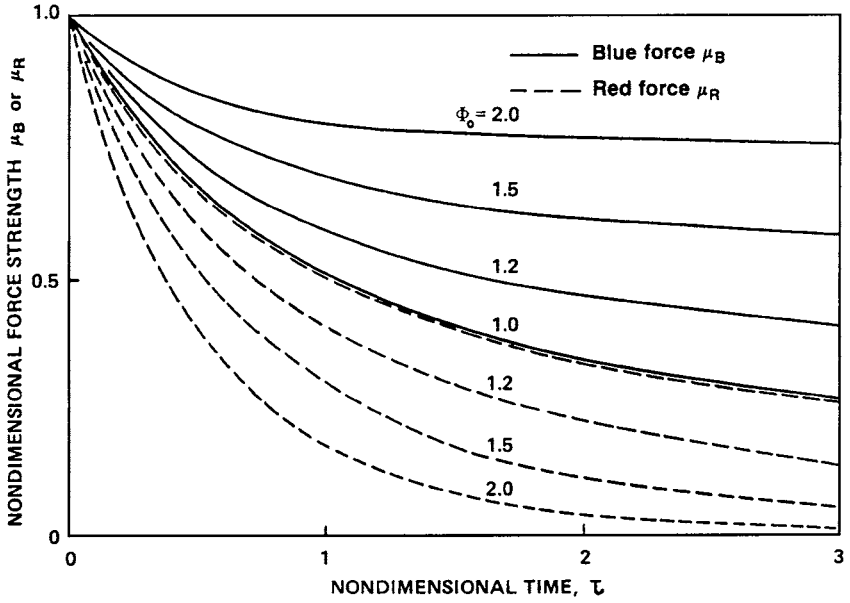


Fig. 4.4 Variation of the nondimensional force strength μ_B (Blue forces) or μ_R (Red forces) with the nondimensional time τ for $\Phi_0 \geq 1$ (area fire combat).

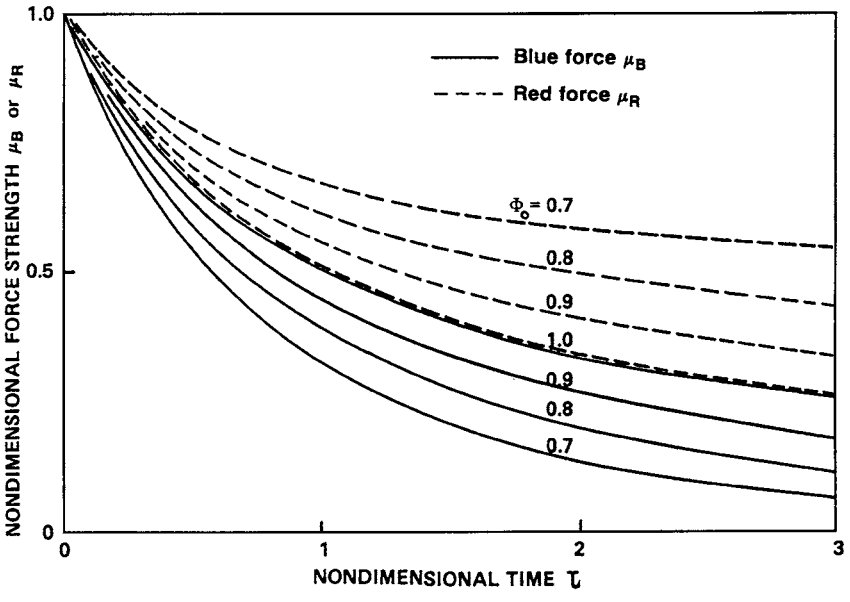


Fig. 4.5 Variation of the nondimensional force strength μ_B (Blue forces) or μ_R (Red forces) with the nondimensional time τ for $\Phi_0 \leq 1$ (area fire combat).

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Table 4.3 Comparison of force strengths for $\Phi_0 = 2$

	Directed Fire		Area Fire	
Time τ	μ_B	μ_R	μ_B	μ_R
0	1.0000	1.0000	1.0000	1.0000
0.1	0.9549	0.8047	0.9556	0.8225
0.2	0.9194	0.6174	0.9205	0.6819
0.3	0.8931	0.4363	0.8922	0.5689
0.4	0.8757	0.2596	0.8693	0.4771
0.5	0.8671	0.0854	0.8504	0.4017
0.5493	0.8660	0.0000	0.8424	0.3695

4.5 Guerrilla Warfare Model: Mixed Law

In a combat situation when the Blue forces are caught in an ambush, at least for the initial stages of the ensuing battle, the Blue forces will be exposed to a directed fire by the Reds but will only be able to reciprocate in the form of an area fire because the precise location of the Red forces will be largely unknown. Under these circumstances the governing differential equations can be written as[14,15]

$$\frac{dm_B}{dt} = -a_R m_R \tag{4.73}$$

and

$$\frac{dm_R}{dt} = -\frac{a_B m_B m_R}{M_R} \tag{4.74}$$

Equations (4.73) and (4.74) can also be written in the nondimensional form as

$$\dot{\mu}_B = -\frac{\mu_R}{\Phi_0} \tag{4.75}$$

and

$$\dot{\mu}_R = -\Phi_0 \mu_B \mu_R \tag{4.76}$$

Elimination of μ_R from Eqs. (4.75) and (4.76) leads to

$$\ddot{\mu}_B = -\Phi_0 \mu_B \dot{\mu}_B \tag{4.77}$$

which can be integrated directly so that

$$\dot{\mu}_B = -\tfrac{1}{2}\Phi_0 \mu_B^2 - C_1 \tag{4.78}$$

where C_1 is a constant of integration, and the negative sign with C_1 is used for convenience. Subsequent integration of Eq. (4.78) leads to

$$-\int \frac{d\mu_B}{C_1 + \frac{1}{2}\Phi_0 \mu_B^2} = \int d\tau = \tau + C_2 \tag{4.79}$$

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where C_2 is the second constant of integration. The left side of Eq. (4.79) is of the standard form, which results in the expression

$$-\frac{1}{\sqrt{\frac{1}{2}C_1\Phi_0}}\,\mathrm{Tan}^{-1}\,\frac{\mu_B}{\sqrt{2C_1/\Phi_0}} = \tau + C_2 \tag{4.80}$$

where Tan represents the tangent function with an argument in the range between 0 and $\frac{1}{2}\pi$. Rearranging the preceding equation,

$$\mu_B = -\sqrt{\frac{2C_1}{\Phi_0}}\,\mathrm{Tan}\left[\sqrt{\frac{1}{2}C_1\Phi_0}\,(\tau + C_2)\right] \tag{4.81}$$

which can then be substituted into Eq. (4.75) to give

$$\mu_R = C_1\Phi_0\left\{1 + \mathrm{Tan}^2\left[\sqrt{\frac{1}{2}C_1\Phi_0}\,(\tau + C_2)\right]\right\} \tag{4.82}$$

The unknown constant C_1 determined from the initial conditions $\mu_B = \mu_R = 1$ at $\tau = 0$ is given by

$$C_1 = \frac{2 - \Phi_0^2}{2\Phi_0} \tag{4.83}$$

To determine the remaining constant C_2, however, three separate cases must be considered: 1) $\Phi_0 < \sqrt{2}$, 2) $\Phi_0 > \sqrt{2}$, and 3) $\Phi_0 = \sqrt{2}$, corresponding to $C_1 > 0$, $C_1 < 0$, and $C_1 = 0$, respectively.

Case 1: $\Phi_0 < \sqrt{2}$, $C_1 > 0$

$$\mu_B = \frac{1 - \frac{\sqrt{2-\Phi_0^2}}{\Phi_0}\,\mathrm{Tan}\left(\frac{1}{2}\tau\sqrt{2-\Phi_0^2}\right)}{1 + \frac{\Phi_0}{\sqrt{2-\Phi_0^2}}\,\mathrm{Tan}\left(\frac{1}{2}\tau\sqrt{2-\Phi_0^2}\right)} \tag{4.84}$$

$$\mu_R = \frac{2 - \Phi_0^2}{2}\left\{1 + \frac{\Phi_0^2}{2 - \Phi_0^2}\left[\frac{1 - \frac{\sqrt{2-\Phi_0^2}}{\Phi_0}\,\mathrm{Tan}\left(\frac{1}{2}\tau\sqrt{2-\Phi_0^2}\right)}{1 + \frac{\Phi_0}{\sqrt{2-\Phi_0^2}}\,\mathrm{Tan}\left(\frac{1}{2}\tau\sqrt{2-\Phi_0^2}\right)}\right]^2\right\} \tag{4.85}$$

$$2\mu_R - \Phi_0^2\mu_B^2 = 2 - \Phi_0^2 \tag{4.86}$$

The Red forces win when $\mu_B = 0$ at the time given by

$$\tau_f = \frac{2}{\sqrt{2 - \Phi_0^2}}\,\mathrm{Tan}^{-1}\,\frac{\Phi_0}{\sqrt{2 - \Phi_0^2}} \tag{4.87}$$

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while their residual strength remains as

$$\mu_R(\tau_f) = \frac{2 - \Phi_0^2}{2} \tag{4.88}$$

It should be noted that for $0 < \tau < \tau_f$ the condition $2 - \frac{1}{2}\Phi_0^2 < \frac{1}{2}\pi$ required for the Tan function is always satisfied.

Case 2: $\Phi_0 > \sqrt{2}$, $C_1 < 0$

$$\mu_B = \frac{1 + \frac{\sqrt{\Phi_0^2 - 2}}{\Phi_0} \tanh\left(\frac{1}{2}\tau\sqrt{\Phi_0^2 - 2}\right)}{1 + \frac{\Phi_0}{\sqrt{\Phi_0^2 - 2}} \tanh\left(\frac{1}{2}\tau\sqrt{\Phi_0^2 - 2}\right)} \tag{4.89}$$

$$\mu_R = \frac{\Phi_0^2}{2}\left[\frac{1 + \frac{\sqrt{\Phi_0^2 - 2}}{\Phi_0} \tanh\left(\frac{1}{2}\tau\sqrt{\Phi_0^2 - 2}\right)}{1 + \frac{\Phi_0}{\sqrt{\Phi_0^2 - 2}} \tanh\left(\frac{1}{2}\tau\sqrt{\Phi_0^2 - 2}\right)}\right]^2 - \frac{\Phi_0^2 - 2}{2} \tag{4.90}$$

$$\Phi_0^2\mu_B^2 - 2\mu_R = \Phi_0^2 - 2 \tag{4.91}$$

which is the same as Eq. (4.86). In this case the Blue forces will win at $\tau = \infty$, and their residual strength will be

$$\mu_B(\infty) = \frac{\Phi_0^2 - 2}{\Phi_0^2} \tag{4.92}$$

Case 3: $\Phi_0 = \sqrt{2}$, $C_1 = 0$

$$\mu_B = \frac{1}{1 + \frac{1}{2}\sqrt{2}\,\tau} \tag{4.93}$$

$$\mu_R = \frac{1}{\left(1 + \frac{1}{2}\sqrt{2}\,\tau\right)^2} \tag{4.94}$$

and the relationship between μ_B and μ_R reduces to

$$\mu_B^2 - \mu_R = 0 \tag{4.95}$$

In this case there is no win for either the Reds or the Blues, except that the Red strength is diminishing much more rapidly than that of the Blues.

The variation of μ_B and μ_R with τ for the guerrilla warfare model, in which the Blues are attacked by the Red guerrilla forces, is shown in Fig. 4.6. When $1 < \Phi_0 < \sqrt{2}$, μ_B and μ_R intersect at times less than τ_f given by Eq. (4.87). This means that for this range there is an initial advantage for the Blues because the Reds are suffering higher proportional losses, but eventually this situation is reversed and the Blues suffer higher losses at later times. The intersection of μ_B and μ_R

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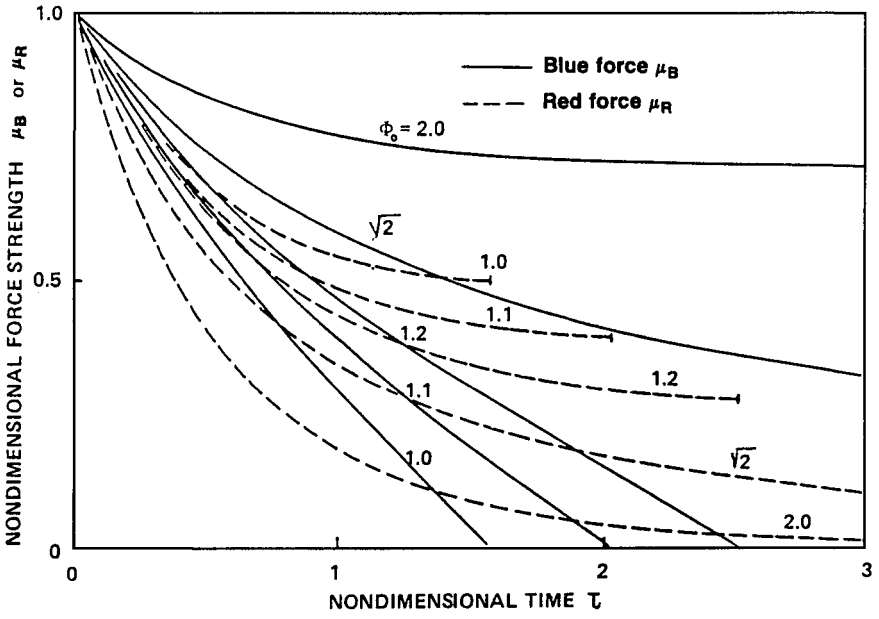


Fig. 4.6 Variation of the nondimensional force strength μ_B (Blue forces) or μ_R (Red forces) with the nondimensional time τ; guerrilla warfare (Blue forces ambushed by Red forces).

can be easily seen in Fig. 4.6. This can also be demonstrated from Eq. (4.86). If $\mu_B = \mu_R = \mu$, then Eq. (4.86) becomes

$$2\mu - \Phi_0^2\mu^2 = 2 - \Phi_0^2; \qquad \Phi_0 < \sqrt{2} \tag{4.96}$$

Hence,

$$\mu = \frac{1 \pm \left(\Phi_0^2 - 1\right)}{\Phi_0^2}; \qquad \Phi_0 < \sqrt{2} \tag{4.97}$$

Also, the condition

$$\mu < 1 \tag{4.98}$$

must be satisfied, which, when combined with Eq. (4.97) and the restriction that $\Phi_0 < \sqrt{2}$, leads to

$$\mu = \frac{2 - \Phi_0^2}{\Phi_0^2}; \qquad 1 < \Phi_0 < \sqrt{2} \tag{4.99}$$

4.6 Autonomous Fire Model: Logarithmic Law

In another addition to the list of possible combat models, Peterson[16,17] hypothesized that in some combat engagements the attrition rates may be proportional to

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the number of troops exposed to the enemy's fire. This may occur in the early stages of a small unit engagement in which vulnerability of a force dominates its ability to acquire enemy targets. This may also occur in situations where autonomous weapons are used that are triggered automatically as the opposing forces appear within their ranges. For these situations the governing attrition equations become

$$\frac{dm_B}{dt} = -b_R m_B \tag{4.100}$$

and

$$\frac{dm_R}{dt} = -b_B m_R \tag{4.101}$$

where b_B and b_R denote the attrition coefficients for the autonomous weapons. The solutions to these equations are in this case independent from each other and are given by

$$m_B = M_B e^{-b_R t} \tag{4.102}$$

and

$$m_R = M_R e^{-b_B t} \tag{4.103}$$

where, as before, M_B and M_R represent force strengths at $t = 0$ of the Blue and Red combatants, respectively. The nondimensional representation of Eqs. (3.102) and (3.103) is simply

$$\mu_B = \exp\left(-\sqrt{\frac{b_R}{b_B}}\,\tau\right) \tag{4.104}$$

$$\mu_R = \exp\left(-\sqrt{\frac{b_B}{b_R}}\,\tau\right) \tag{4.105}$$

Thus, for the autonomous fire model the governing equations are not dependent on the quantitative superiority. It should also be noted that the implicit assumption used here in this model is that there are always enough autonomous weapons to engage all forces coming within the effective range of weapons.

As for the previous models, the functional relationship between μ_B and μ_R can be obtained for Eqs. (4.104) and (4.105). This leads to

$$\ln \mu_B = \frac{b_R}{b_B} \ln \mu_R \tag{4.106}$$

which is described as the logarithmic law.

4.7 Geometric Mean Model: Linear Law

In this model it is assumed that the attrition rates for both forces are proportional to the geometric mean of the opposing forces. The governing differential equations are represented by

$$\frac{dm_B}{dt} = -a_R \sqrt{m_B m_R} \tag{4.107}$$

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and

$$\frac{dm_R}{dt} = -a_B\sqrt{m_B m_R} \tag{4.108}$$

which can be linearized through the substitution of

$$x = \sqrt{m_B} \tag{4.109}$$

and

$$y = \sqrt{m_R} \tag{4.110}$$

Hence,

$$\frac{dx}{dt} = -\frac{1}{2}a_R y \tag{4.111}$$

and

$$\frac{dy}{dt} = -\frac{1}{2}a_B x \tag{4.112}$$

Because the preceding equations are of the same form as Eqs. (4.31) and (4.32) (Lanchester model for directed fire), it follows that

$$x = \sqrt{M_B}\cosh\left(\frac{1}{2}\sqrt{a_B a_R}\,t\right) - \sqrt{\frac{M_R a_R}{a_B}}\sinh\left(\frac{1}{2}\sqrt{a_B a_R}\,t\right) \tag{4.107}$$

$$y = \sqrt{M_R}\cosh\left(\frac{1}{2}\sqrt{a_B a_R}\,t\right) - \sqrt{\frac{M_B a_B}{a_R}}\sinh\left(\frac{1}{2}\sqrt{a_B a_R}\,t\right) \tag{4.108}$$

that can be arranged in the standard nondimensional form as

$$\mu_B = \left(\cosh\tfrac{1}{2}\tau - \Psi_0^{-\frac{1}{2}}\sinh\tfrac{1}{2}\tau\right)^2 \tag{4.115}$$

and

$$\mu_R = \left(\cosh\tfrac{1}{2}\tau - \Psi_0^{\frac{1}{2}}\sinh\tfrac{1}{2}\tau\right)^2 \tag{4.116}$$

where Ψ is a new force superiority parameter defined as

$$\Psi_0 = \frac{M_B a_B}{M_R a_R} \tag{4.117a}$$

Elimination of τ from Eqs. (4.115) and (4.116) leads to the following functional relationship:

$$\Psi_0\mu_B - \mu_R = \Psi_0 - 1 \tag{4.118a}$$

Although the preceding relationship represents a linear law, its form is somewhat different from the Lanchester model for area fire. It should be noted that in both cases the force superiority parameters are different.

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For the special case when $\Psi_0 = 1$, the two opposing forces are equal in strength and

$$\mu_B = \mu_R = \left(\cosh \tfrac{1}{2}\tau - \sinh \tfrac{1}{2}\tau\right)^2 = e^{-\tau} \tag{4.119}$$

which is the same as the case of the directed fire model when $\Phi_0 = 1$.

The time τ_0 needed to annihilate the Reds is determined from

$$\mu_R = \left(\cosh \tfrac{1}{2}\tau_0 - \Psi_0^{\frac{1}{2}} \sinh \tfrac{1}{2}\tau_0\right)^2 = 0 \tag{4.120}$$

Hence,

$$\tau_0 = 2 \tanh^{-1} \Psi_0^{-\frac{1}{2}} = \ell_n \frac{\Psi_0^{\frac{1}{2}} + 1}{\Psi_0^{\frac{1}{2}} - 1} \tag{4.121a}$$

and

$$t_0 = \frac{1}{\sqrt{a_B a_R}} \ell_n \frac{\Psi_0^{\frac{1}{2}} + 1}{\Psi_0^{\frac{1}{2}} - 1} \tag{4.121b}$$

4.8 Helmbold Models: Size Effects[18]

When two forces of grossly unequal sizes are engaged in a combat battle, inefficiencies of scale should be accounted for in the mathematical models of such engagements. It is clear that, when a large force engages a much smaller enemy unit, not all elements of the larger force can be brought into the combat simultaneously because of the physical constraints of space and timing. For such situations Helmbold[18] proposed a modification of Lanchester equations for directed fire in which the effective firing rate is modified by a factor depending on only the force ratio. In his modification, the basic equations are given by

$$\frac{dm_B}{dt} = -a_R E_R \left(\frac{m_B}{m_R}\right) m_R \tag{4.122}$$

and

$$\frac{dm_R}{dt} = -a_B E_B \left(\frac{m_R}{m_B}\right) m_B \tag{4.123}$$

where E_B and E_R denote the effective firing modification factors that model the inefficiencies of scale. The functional form of these factors can be deduced from the following general observations. First, both E_B and E_R should be increasing functions of their argument. That is, when $m_B/m_R > 1$, the Red forces have more targets to choose from and the Blue force attrition rate dm_B/dt should increase above the rate for which $m_B/m_R = 1$. Similarly, when $m_R/m_B > 1$ the Red force attrition rate dm_R/dt should increase. Second, the inefficiencies of scale should be the same for both sides. And third, because E_B and E_R are modifying

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factors about the quantitative parity state, it follows that $E_B(1)$ and $E_R(1)$ should be equal to unity. Mathematically, this can be expressed as

$$E_B(u) = E_R(u) = E(u)$$

$$E(1) = 1$$

$$E(u) = \text{ increasing function of } u$$

where u is the argument of functions E_B or E_R.

Helmbold[18] considered a special case for which $E(u)$ is a power function, i.e.,

$$E(u) = u^c; \qquad c > 0 \tag{4.124}$$

Substituting Eq. (4.124) into Eqs. (4.122) and (4.123),

$$\frac{dm_B}{dt} = -a_R \left(\frac{m_B}{m_R}\right)^{1-W} m_R \tag{4.125}$$

$$\frac{dm_R}{dt} = -a_B \left(\frac{m_R}{m_B}\right)^{1-W} m_B \tag{4.126}$$

where

$$W = 1 - c \tag{4.127}$$

is usually referred to as the Weiss parameter.

When the exponent c in Eq. (4.124) takes on the values of 0, $\frac{1}{2}$, and 1, Eqs. (4.125) and (4.126) reduce to the previously discussed models. These three cases are represented in Table 4.4. Thus, as the exponent c is changed from 0 to 1, the basic equations for the directed fire model are modified to the geometric mean model and then into the autonomous fire model, in which the attrition rates on each side are proportional to the number of troops exposed to the enemy fire.

Table 4.4 Helmbold models

$c = 1 - W$	W	Governing equations	Type of model
0	1	$dm_B/dt = -a_R m_R$ $dm_R/dt = -a_B m_B$	Directed fire
$\frac{1}{2}$	$\frac{1}{2}$	$dm_B/dt = -a_R(m_B m_R)^{\frac{1}{2}}$ $dm_R/dt = -a_B(m_B m_R)^{\frac{1}{2}}$	Geometric mean
1	0	$dm_B/dt = -b_R m_B$ $dm_R/dt = -b_B m_R$	Autonomous fire

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4.9 Autonomous/Directed Fire Model

Another type of combat model assumes autonomous fire by the Blue forces and directed fire by the Red forces. This situation is represented by the following governing equations of combat:

$$\frac{dm_B}{dt} = -a_R m_R \tag{4.128}$$

$$\frac{dm_R}{dt} = -b_B m_R \tag{4.129}$$

The second equation can be integrated directly so that

$$m_R = M_R e^{-b_B t} \tag{4.130}$$

which, when substituted into the first equation, leads to

$$m_B = \frac{a_R}{b_B} M_R e^{-b_B t} + M_B - \frac{a_R}{b_B} M_R \tag{4.131}$$

Rearranging Eqs. (4.130) and (4.131) into nondimensional form,

$$\mu_B = \Psi_0^{-1} \exp\left(-\sqrt{\frac{b_B}{a_R}}\,\tau\right) + 1 - \Psi_0^{-1} \tag{4.132}$$

and

$$\mu_R = \exp\left(-\sqrt{\frac{b_B}{a_R}}\,\tau\right) \tag{4.133}$$

with

$$\tau = \sqrt{b_B a_R}\, t \tag{4.42b}$$

and where the parameter Ψ_0 is of the same form as in the geometric mean model and is defined as

$$\Psi_0 = \frac{b_B M_B}{a_R M_R} \tag{4.117b}$$

Also, it can be demonstrated easily from Eqs. (4.132) and (4.133) that the functional relationship between μ_B and μ_R for this model is of the same form as the geometric mean model relationship:

$$\Psi_0 \mu_B - \mu_R = \Psi_0 - 1 \tag{4.118b}$$

4.10 Force Parity

A convenient way of defining force parity in a combat engagement is to assume that for this to occur the proportional losses on both sides must be equal; i.e., both

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Table 4.5 Force parity conditions

Model type	Parity condition[a]
Directed fire	$\Phi_0 = 1$
Area fire	$\Phi_0 = 1$
Autonomous fire	$b_R/b_B = 1$
Geometric mean fire	$\Psi_0 = 1$
Autonomous/directed fire	$\Psi_0 = 1$

[a] $\Phi_0 = (M_B/M_R)(a_B/a_R)^{\frac{1}{2}}$
$\Psi_0 = a_B M_B/a_R M_R$ (for geometric mean model)
$= b_B M_B/a_R M_R$ (for autonomous/directed fire)

sides continue to diminish their strengths proportionately. Mathematically this is expressed as

$$\mu_B = \mu_R \tag{4.134}$$

The condition of force parity depends on the model representation used to simulate a particular combat situation. The necessary conditions to satisfy Eq. (4.134) for different models considered in this chapter are listed in Table 4.5. These conditions are obtained simply by inspection of the general solutions for μ_B and μ_R for each case.

4.11 Battle Disengagement: Force Breakpoints[19]

In most battle engagements the losing side will not continue to the point at which its forces are completely annihilated, and it will break off the engagement and stop further casualties. Military planners usually assume that a combat unit will try to disengage from the enemy when it has suffered a certain percentage of casualties. Of course, the casualty percentage will vary depending on many factors, the most important of which is the unit's determination to continue the engagement. Some planners assume that this will occur at a 30% level for a company-sized unit in an offensive action and at a 50% level in a defensive action. It may therefore be assumed that in a combat engagement the first side to be reduced to a specified force level (force breakpoint) will automatically break off the engagement and "lose" the battle. The condition of force breakpoint for the Red force (i.e., when the Blue force wins) can be expressed simply for the case of the directed fire model as

$$\mu_B > \mu_{BB} \tag{4.135}$$

and

$$\mu_R = \mu_{RB} \tag{4.136}$$

where μ_{BB} and μ_{RB} are the breakpoint levels of the Blue and Red forces, respectively. When Eqs. (4.38) and (4.39) are substituted into Eqs. (4.135) and (4.136) the following relationships are obtained:

$$\cosh \tau - \Phi_0^{-1} \sinh \tau > \mu_{BB} \tag{4.137}$$

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Table 4.6 Superiority conditions for Blue forces in terms of breakpoint levels

Type of model	Condition for the Red forces to disengage first
Directed fire	$\Phi_0 > \left[\left(1-\mu_{RB}^2\right)/\left(1-\mu_{BB}^2\right)\right]^{\frac{1}{2}}$
Area fire	$\Phi_0^2 > (1-\mu_{RB})/(1-\mu_{BB})$
Guerrilla warfare	$\Phi_0^2 > 2(1-\mu_{RB})/(1-\mu_{BB}^2)$; $\Phi_0^2 \neq 2$
	or $\mu_{RB} > \mu_{BB}^2$; $\Phi_0^2 = 2$
Autonomous fire	$b_B/b_R > \ln \mu_{RB}/\ln \mu_{BB}$
Geometric mean fire	$\Psi_0 > (1-\mu_{RB})/(1-\mu_{BB})$
Autonomous/directed fire	$\Psi_0 > (1-\mu_{RB})/(1-\mu_{BB})$

and

$$\cosh \tau - \Phi_0 \sinh \tau = \mu_{RB} \tag{4.138}$$

Because both sides of the preceding inequality are positive, both equations can be squared and the nondimensional time τ eliminated. This leads to the inequality

$$\Phi_0 > \sqrt{\frac{1-\mu_{RB}^2}{1-\mu_{BB}^2}} \tag{4.139}$$

which represents the superiority condition for the Blue force to compel the Red force to disengage before the Blue force reaches it own breakpoint μ_{BB}.

Superiority conditions similar to Eq. (4.139) can also be determined for other combat models discussed in this chapter. These conditions are summarized in Table 4.6.

4.12 Variable Attrition Coefficients: Mobile Attack Model

Although variable attrition coefficient models may be used in any of the previously discussed models, this section will discuss only the directed fire model for which the governing differential equations can be written as

$$\frac{dm_B}{dt} = -a_R(t)m_R \tag{4.140}$$

and

$$\frac{dm_R}{dt} = -a_B(t)m_B \tag{4.141}$$

where $a_B(t)$ and $a_R(t)$ denote time-dependent attrition coefficients. As before, these coefficients depend on many different variables, such as force separation, firing rates, rates of target acquisition, etc. Eliminating m_R from Eqs. (4.140) and (4.141), the following second-order differential equation is obtained:

$$\frac{d^2m_B}{dt^2} - \frac{1}{a_R}\frac{da_R}{dt}\frac{dm_B}{dt} - a_B a_R m_B = 0 \tag{4.142}$$

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It can be demonstrated that the solution to this equation is given by[20]

$$m_B(t) = C_1 m_{1B}(t) + C_2 m_{2B}(t) \tag{4.143}$$

where C_1 and C_2 are constants determined by the initial conditions, and $m_{1B}(t)$ and $m_{2B}(t)$ are the complementary functions of Eq. (4.142) described as the hyperbolic-like general Lanchester functions. Once these functions are known, the solution for $m_R(t)$ is obtained from Eq. (4.140). Analytical solutions have been derived for cases where the attrition coefficients are expressed as

$$a_B(t) = k_B(t + C)^{\mu} \text{ and } a_R(t) = k_R(t + C)^{\nu} \text{ with } C \geq 0 \tag{4.144}$$

or

$$a_B(t) = k_B(t + C + A)^{\mu} \text{ and } a_R(t) = k_R(t + C)^{\nu} \text{ with } A, C \geq 0 \tag{4.145}$$

where A, C, k_B, k_R, μ, and ν are constants. The results are rather complex and are expressible in terms of infinite series solutions; however, a relatively simple solution can be obtained for the special case when

$$a_B(t) = k_B h(t) \tag{4.146}$$

and

$$a_R(t) = k_R h(t) \tag{4.147}$$

where $h(t)$ denotes the common time-varying factor in the two attrition coefficients. For this case the differential Eq. (4.142) reduces to the form

$$\frac{d^2x}{dt^2} - \frac{1}{h(t)}\frac{dh(t)}{dt}\frac{dx}{dt} - k_B k_R h^2(t)x = 0 \tag{4.148}$$

where

$$x = m_B \tag{4.149}$$

Equation (4.148) can be solved with the substitution of a new variable:

$$\theta = \sqrt{k_B k_R}\int_0^t h(s)\,ds \tag{4.150}$$

Observing that

$$\frac{d\theta}{dt} = \sqrt{k_B k_R}\,h(t)$$

$$\frac{d^2\theta}{dt^2} = \sqrt{k_B k_R}\,\frac{dh(t)}{dt}$$

$$\frac{dx}{dt} = \frac{dx}{d\theta}\frac{d\theta}{dt} = \frac{dx}{d\theta}\sqrt{k_B k_R}\,h(t)$$

$$\frac{d^2x}{dt^2} = \frac{dx}{d\theta}\sqrt{k_B k_R}\,\frac{dh(t)}{dt} + \frac{d^2x}{d\theta^2}k_B k_R h^2(t)$$

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and then substituting the preceding expressions into Eq. (4.148) the following result is obtained:

$$\frac{d^2x}{d\theta^2} - x = 0 \tag{4.151}$$

The exact solution of this equation is

$$x = C_1 \sinh\theta + C_2 \cosh\theta \tag{4.152}$$

where C_1 and C_2 are constants. Now, with Eq. (4.152) as the basic solution, it can be easily demonstrated that

$$\mu_B = \cosh\theta - \Phi_0^{-1} \sinh\theta \tag{4.153}$$

and

$$\mu_R = \cosh\theta - \Phi_0 \sinh\theta \tag{4.154}$$

where, as before, $\mu_B = m_B/M_B$ and $\mu_R = m_R/M_R$, and Φ_0 is defined as

$$\Phi_0 = \frac{M_B}{M_R}\sqrt{\frac{k_B}{k_R}} \tag{4.43b}$$

The variable attrition rates also occur in mobile attacks. The simplest model for these cases is one of constant relative speed of the mobile forces in battlefield engagement. Mathematically such models can be represented by

$$\frac{dm_B}{dt} = -\alpha_R(r)m_R \tag{4.155}$$

and

$$\frac{dm_R}{dt} = -\alpha_B(r)m_B \tag{4.156}$$

where r denotes the separation range between the opposing forces, and $\alpha_R(r)$ and $\alpha_B(r)$ are the range-dependent attrition coefficients. The range r for a constant relative speed is expressed by

$$r(t) = r_0 - vt \tag{4.157}$$

with

$$v = v_B + v_R \tag{4.158}$$

where r_0 denotes the opening range, v is the relative closing speed of the attacking forces, and v_B and v_R are the constant velocities of advancement of the Blue and Red forces, respectively (see Fig. 4.7).

As the force separation r changes with time, the attrition coefficients $\alpha_B(r)$ and $\alpha_R(r)$ will also vary. One possible model is to use an expression for α_B or α_R based on the maximum effective range of the weapon systems used in the engagement. For example, Bonder and Farrell[2] used the following:

$$\alpha_B(r) = \begin{cases} a_B(1 - r/r_B)^{\mu} & \text{for } 0 \le r < r_B \\ 0 & \text{for } r \ge r_B \end{cases} \tag{4.159}$$

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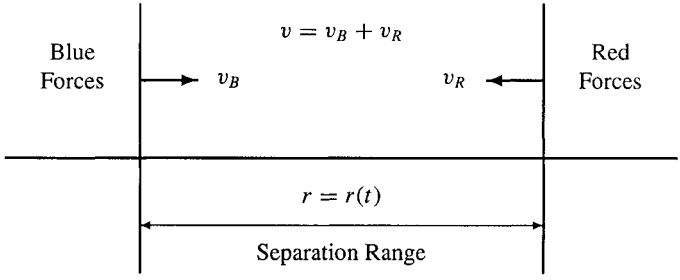


Fig. 4.7 Mobile attack model.

and

$$\alpha_R(r) = \begin{cases} a_R(1 - r/r_R)^{\nu} & \text{for } 0 \le r < r_R \\ 0 & \text{for } r \ge r_R \end{cases} \tag{4.160}$$

where r_B and r_R denote the maximum effective ranges for the Blue and Red weapon systems, respectively. Here the exponents μ and ν, such that $\mu, \nu \ge 0$, are used to model the range dependency of the attrition coefficients. The variation of $\alpha_B(r)$ and $\alpha_R(r)$ with the range parameters r/r_B or r/r_R, as expressed by Eqs. (4.159) or (4.160), is shown in Fig. 4.8 for a series of values of μ or ν.

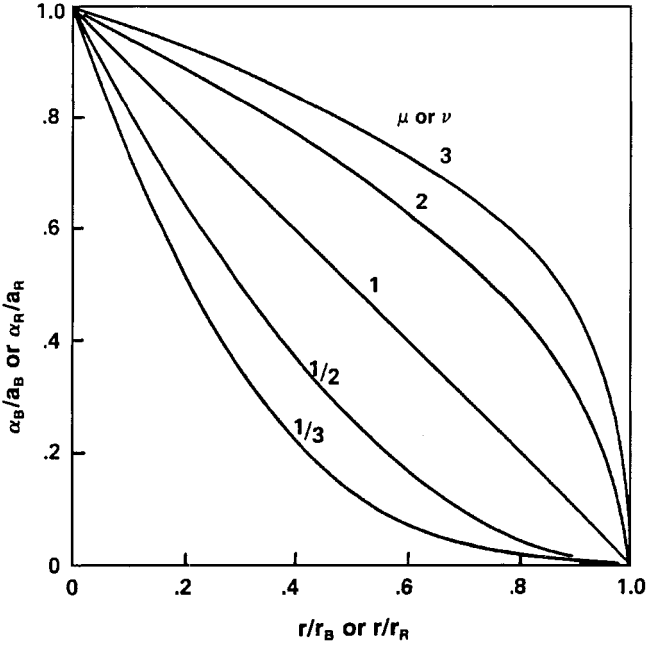


Fig. 4.8 Variation of the attrition coefficients α_B/a_B or α_R/a_R with the range parameters r/r_B or r/r_R.

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Naturally, the values of $\mu, \nu = 0$ represent constant attrition coefficients. Higher exponents μ or ν depict a poor weapon performance for wide separations that improves rapidly as the separating distance between the opposing forces diminishes.

Substitution of Eq. (4.157) into Eqs. (4.159) and (4.160) transforms the range-dependent attrition coefficients into time-dependent variables:

$$\alpha_B(t) = \begin{cases} a_B(v/r_B)^{\mu}(r_B/v - r_0/v + t)^{\mu} & \text{for } 0 \leq r < r_B \\ 0 & \text{for } r \geq r_B \end{cases} \tag{4.161}$$

and

$$\alpha_R(t) = \begin{cases} a_R(v/r_R)^{\nu}(r_R/v - r_0/v + t)^{\nu} & \text{for } 0 \leq r < r_R \\ 0 & \text{for } r \geq r_R \end{cases} \tag{4.162}$$

It should be noted that these coefficients are exactly of the same form as those given in by Eqs. (4.144) and (4.145) with

$$C = \frac{r_R - r_0}{v} \tag{4.163}$$

and

$$A = \frac{r_B - r_R}{v} \tag{4.164}$$

Hence, the starting parameter C relates the maximum range of the Red weapons to the opening range of the battle. If $C > 0$, the battle begins within the maximum effective range of the Red weapons. The offset parameter A expresses the difference in the maximum effective ranges of the Blue and Red forces. If $A > 0$, the maximum effective range of the Blue forces is greater than that of the Red weapons.

4.13 Force Reinforcements in Combat

If the reinforcements are added at the rate R_B and R_R for the Blue and Red forces, respectively, the governing equations for the directed fire combat become

$$\frac{dm_B}{dt} = -a_R m_R + R_B \tag{4.165}$$

$$\frac{dm_R}{dt} = -a_B m_B + R_R \tag{4.166}$$

Differentiation of Eq. (4.165) with respect to time and substitution of Eq. (4.166) leads to the second-order differential equation

$$\ddot{m}_B - a_B a_R m_B = -a_R R_R + \dot{R}_B = P(t) \tag{4.167}$$

where

$$P(t) = -a_R R_R + \dot{R}_B \tag{4.168}$$

and overdots denote differentiation with respect to time. The solution of Eq. (4.167)

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when $P(t) = P_0 = \text{const}$ is given by

$$m_B = \dot{m}_B(0)\frac{1}{\sqrt{a_B a_R}}\sinh\left(\sqrt{a_B a_R}\,t\right) + m_B(0)\cosh\left(\sqrt{a_B a_R}\,t\right)$$

$$+\frac{P_0}{a_B a_R}\left[\cosh\left(\sqrt{a_B a_R}\,t\right) - 1\right] \tag{4.169}$$

where the first two terms represent the complementary function, and the third term is the particular integral. When the initial force strength $m_B(0)$ and the attrition rate $\dot{m}_B(0)$ are both equal to 0 and $P_0 = 1$, then

$$m_B(t) = \Lambda(t) = \frac{\cosh\left(\sqrt{a_B a_R}\,t\right) - 1}{a_B a_R} \tag{4.170}$$

By analogy with structural dynamics problems, the function $\Lambda(t)$ may be defined as *indicial admittance* for the system. It represents the force growth due to a constant reinforcement rate described by the unit step function $P_0 = 1$ for $0 < t < \infty$.

Arbitrary Variation of P(t)[21]

An arbitrary variation of $P(t)$ may assumed to consist of a series of step increases ΔP as shown in Fig. 4.9. Because the system under consideration is linear, the growth (or decrease) in response to $P(t)$ may be evaluated as the cumulative action

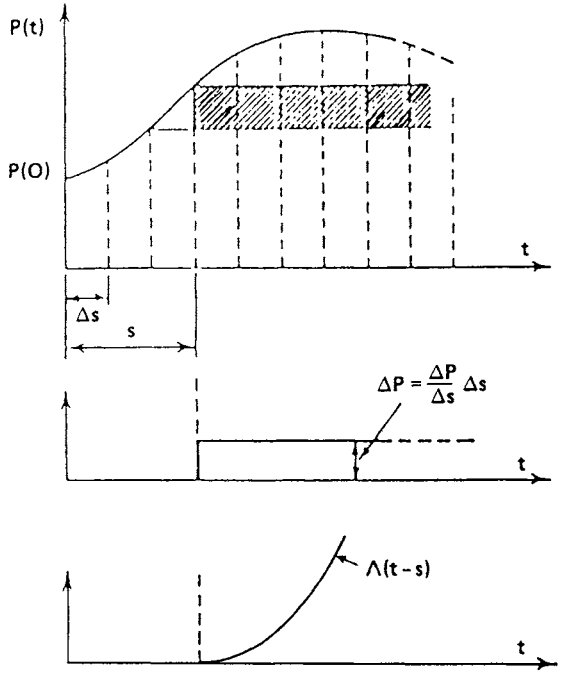


Fig. 4.9 Arbitrary variation of the function $P(t)$.

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of the individual step increases. Thus, using the results for a unit step increase, the particular integral for arbitrary variation of $P(t)$ can be obtained from

$$[m_B(t)]_{PI} = P(0)\Lambda(t) + \sum \Delta P \Lambda(t-s)$$
$$= P(0)\Lambda(t) + \sum \frac{\Delta P}{\Delta s}\Lambda(t-s)\,\Delta s$$

Hence, in the limit

$$[m_B(t)]_{PI} = P(0)\Lambda(t) + \int_0^t \frac{dP}{ds}\Lambda(t-s)\,ds \tag{4.171}$$

An equivalent expression may be obtained by integrating by parts:

$$[m_B(t)]_{PI} = P(0)\Lambda(t) + P(s)\Lambda(t-s)\Big|_{s=0}^{s=t} - \int_0^t P(s)\Lambda'(t-s)\,ds$$
$$= P(0)\Lambda(t) + P(t)\Lambda(0) - P(0)\Lambda(t) - \int_0^t P(s)\Lambda'(t-s)\,ds$$
$$= P(t)\Lambda(0) - \int_0^t P(s)\Lambda'(t-s)\,ds$$

where primes denote differentiation with respect to s. From Eq. (4.170) it follows that $\Lambda(0) = 0$; hence, the particular integral becomes

$$[m_B(t)]_{PI} = -\int_0^t P(s)\Lambda'(t-s)\,ds$$
$$= \frac{1}{\sqrt{a_B a_R}}\int_0^t P(s)\sinh\left[\sqrt{a_B a_R}\,(t-s)\right]ds \tag{4.172}$$

Substituting Eq. (4.168) for $P(s)$ in Eq. (4.172) and integrating by parts,

$$[m_B(t)]_{PI} = \frac{1}{\sqrt{a_B a_R}}\int_0^t (-a_R R_R + \dot{R}_B)\sinh\left[\sqrt{a_B a_R}\,(t-s)\right]ds$$
$$= -\frac{1}{\sqrt{a_B a_R}}\int_0^t a_R R_R \sinh\left[\sqrt{a_B a_R}\,(t-s)\right]ds$$
$$+ \frac{1}{\sqrt{a_B a_R}} R_B \sinh\left[\sqrt{a_B a_R}\,(t-s)\right]\Big|_{s=0}^{s=t}$$
$$+ \int_0^t R_B \cosh\left[\sqrt{a_B a_R}\,(t-s)\right]ds$$
$$= -\frac{1}{\sqrt{a_B a_R}} R_B(0)\sinh(\sqrt{a_B a_R}\,t)$$
$$- \sqrt{a_B/a_R}\int_0^t R_R \sinh\left[\sqrt{a_B a_R}\,(t-s)\right]ds$$
$$+ \int_0^t R_B \cosh\left[\sqrt{a_B a_R}\,(t-s)\right]ds \tag{4.173}$$

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Noting that $m_B(0) = M_B$ and $m_R(0) = M_R$, the complete solution for $m_B(t)$ can be expressed as

$$\begin{aligned} m_B(t) &= \frac{\dot{m}_B(0)}{\sqrt{a_B a_R}} \sinh\left(\sqrt{a_B a_R}\, t\right) + m_B(0) \cosh\left(\sqrt{a_B a_R}\, t\right) + [m_B]_{PI} \\ &= M_B \cosh\left(\sqrt{a_B a_R}\, t\right) - \sqrt{\frac{a_R}{a_B}}\, M_R \sinh\left(\sqrt{a_B a_R}\, t\right) \\ &\quad + \int_0^t R_B \cosh\left[\sqrt{a_B a_R}\,(t-s)\right] ds \\ &\quad - \sqrt{\frac{a_B}{a_R}} \int_0^t R_R \sinh\left[\sqrt{a_B a_R}\,(t-s)\right] ds \end{aligned} \tag{4.174}$$

Because of the symmetry of Eqs. (4.165) and (4.166), the solution for $m_R(t)$ can be obtained by interchanging appropriate coefficients. Hence,

$$\begin{aligned} m_R(t) &= M_R \cosh\left(\sqrt{a_B a_R}\, t\right) - \sqrt{\frac{a_B}{a_R}}\, M_B \sinh\left(\sqrt{a_B a_R}\, t\right) \\ &\quad + \int_0^t R_R \cosh\left[\sqrt{a_B a_R}\,(t-s)\right] ds - \sqrt{\frac{a_R}{a_B}} \int_0^t R_B \sinh\left[\sqrt{a_B a_R}\,(t-s)\right] ds \end{aligned} \tag{4.175}$$

When both R_B and R_R are constants, Eqs. (4.174) and (4.175) reduce to

$$\begin{aligned} m_B(t) &= \left(M_B - \frac{R_R}{a_B}\right) \cosh\left(\sqrt{a_B a_R}\, t\right) \\ &\quad - \left(\sqrt{\frac{a_R}{a_B}}\, M_R - \frac{R_B}{\sqrt{a_B a_R}}\right) \sinh\left(\sqrt{a_B a_R}\, t\right) + \frac{R_R}{a_B} \end{aligned} \tag{4.176}$$

and

$$\begin{aligned} m_R(t) &= \left(M_R - \frac{R_B}{a_R}\right) \cosh\left(\sqrt{a_B a_R}\, t\right) \\ &\quad - \left(\sqrt{\frac{a_B}{a_R}}\, M_B - \frac{R_R}{\sqrt{a_B a_R}}\right) \sinh\left(\sqrt{a_B a_R}\, t\right) + \frac{R_B}{a_R} \end{aligned} \tag{4.177}$$

4.14 Mixed Models

Combination of the directed and autonomous fire models leads to governing equations of combat of the following form[22]:

$$\dot{m} = -an - cm + P \tag{4.178}$$

$$\dot{n} = -bm - dn + Q \tag{4.179}$$

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where

m = Blue forces
n = Red forces
a, b = attrition coefficients (directed fire)
c, d = attrition coefficients (autonomous fire)
P, Q = reinforcement rates

The coefficients c and d can also be interpreted as a natural attrition caused by troops or weapons becoming inoperative due to stresses and strains of the battle engagement. Differential equations of the type represented by Eqs. (4.178) and (4.179) also occur in the analysis of the struggle between animal species and in chemical reactions.

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4.15 Iwo Jima Battle[23,24]

The battle of Iwo Jima in 1945, for which detailed statistics are available, can be used as a verification of the Lanchester directed fire model developed in Sec. 4.3.[25] This battle, fought on the island of Iwo Jima, 660 miles south of Tokyo, was one of the fiercest battles of World War II. The island was an important strategic point in the Pacific. Situated halfway across the distance of 1,500 miles between Tokyo and the U.S. bases on Saipan in the Marianas, Iwo Jima was developed by the Japanese as a base from which their fighter aircraft could easily intercept U.S. bombers on the way to Japan. For the Americans the island was most desirable as a potential base for their own operations close to the Japanese mainland.

The American invasion of Iwo Jima began on 19 February 1945. The Japanese had orders to fight to the last man, and that is exactly what they did. The American forces suffered heavy casualties during the 36 days of fighting. Iwo Jima cost the lives of 6,000 Marines, but, in the next five months prior to the end of World War II in the Pacific, more than 2,000 B-29 bombers were able to land on the island.

During the invasion of Iwo Jima the U.S. forces landed 54,000 troops on the first day, none on the second, 6,000 on the third, none on the fourth and fifth, 13,000 on the sixth, and none thereafter.[26] Hence, for this battle,

$$R_B(t) = \begin{cases} 54{,}000 & 0 < t \leq 1 \\ 0 & 1 < t \leq 2 \\ 6{,}000 & 2 < t \leq 3 \\ 0 & 3 < t \leq 4 \\ 0 & 4 < t \leq 5 \\ 3{,}000 & 5 < t \leq 6 \\ 0 & 6 < t \leq 36 \end{cases} \quad \text{troops/day} \tag{4.180}$$

The preceding figures included the personnel of all units that were actually put ashore. This included combat as well as supporting units. There were no American troops on Iwo Jima prior to the start of the battle. The daily count of all American combat losses was kept by Captain Clifford Morehouse,[27] but unfortunately no such records were available for the Japanese forces. Based on the figure of 20,000 Japanese troops killed and 1,083 prisoners, Newcomb arrived at the figure

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of 21,000 for the Japanese forces at the beginning of the battle. However, this figure is low because he did not include some of the living and dead found in caves on the island after the initial count. Thus, the figure of 21,500 is probably a better estimate. There were no Japanese reinforcements during the course of the battle. The battle effectively ceased on the 28th day, and the island was declared "secured" by the American forces, but some sporadic combat continued until the 36th day.

The Iwo Jima battle can therefore be described by Eqs. (4.165) and (4.166) with $R_R = 0$, i.e.,

$$\dot{m}_B = -a_R m_R + R_B \tag{4.181}$$

$$\dot{m}_R = -a_B m_B \tag{4.182}$$

where the subscripts B and R refer to the American and Japanese forces, respectively. Integrating Eq. (4.182) between $t = 0$ and 36, it follows that

$$m_R(36) - m_R(0) = -a_B \int_0^{36} m_B(t)\,dt \tag{4.183}$$

The integral on the right hand side of Eq. (4.183) can be approximated by the Riemann sum[28]

$$\int_0^{36} m_B(t)\,dt = \sum_{i=1}^{36} m_B(i) \quad \text{man} - \text{days} \tag{4.184}$$

where $m_B(i)$ is the number of effective American troops on the ith day of the battle. Hence, using the data available from Morehouse,[27]

$$a_B = \frac{m_R(0) - m_R(36)}{\sum_{i=1}^{36} m_B(i)} = \frac{21{,}500 - 0}{2{,}037{,}000} = 0.0106 \tag{4.185}$$

Next, if Eq. (4.181) is integrated,

$$\begin{aligned} m_B(36) &= -a_R \int_0^{36} m_R(t)\,dt + \int_0^{36} R_B\,dt \\ &= -a_R \int_0^{36} m_R(t)\,dt + 73{,}000 \end{aligned} \tag{4.186}$$

There were 52,000 effective American troops on the 36th day of the battle. Hence,

$$a_R = \frac{73{,}000 - 52{,}000}{\int_0^{36} m_R(t)\,dt} = \frac{21{,}000}{\int_0^{36} m_R(t)\,dt} \tag{4.187}$$

As before, the integral in Eq. (4.187) may be approximated by the Riemann sum

$$\int_0^{36} m_R(t)\,dt = \sum_{j=1}^{36} m_R(j) \tag{4.188}$$

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where $m_R(j)$ is obtained from Eq. (4.183) as

$$m_R(j) = m_R(0) - a_B \int_0^j m_B(t)\,dt$$

$$= m_R(0) - a_B \sum_{i=1}^{j} m_B(i) \tag{4.189}$$

Now, if the number of the American troops on the ith day of the battle is substituted in Eq. (4.189), it follows then from Eq. (4.187) that

$$a_R = \frac{21{,}000}{387{,}600} = 0.0543 \tag{4.190}$$

(Note that Ref. 28 gives $\sum m_R(j)$ for the first 28 days as 372,500 man-days.) The additional contribution to Eq. (4.188) for the remaining 8 days (i.e., days 29–36) was computed using Eq. (4.189) to be 15,100 man-days.

The comparison of the actual American troop strength with the values predicted by Eqs. (4.181) and (4.182) is shown in Fig. 4.10. The agreement is excellent. It should be noted that the Japanese troops had a clear qualitative superiority, because $a_R/a_B = 0.0544/0.0106 = 5.1$. This occurred mainly because of the well-established defense posture, such as extensive Japanese fortifications, tunnels, and natural caves. The American forces, on the other hand, had a clear quantitative superiority. For example, on the eighth day of the battle, $m_B(8)/m_R(8) = 64{,}500/16{,}500 = 3.9$.

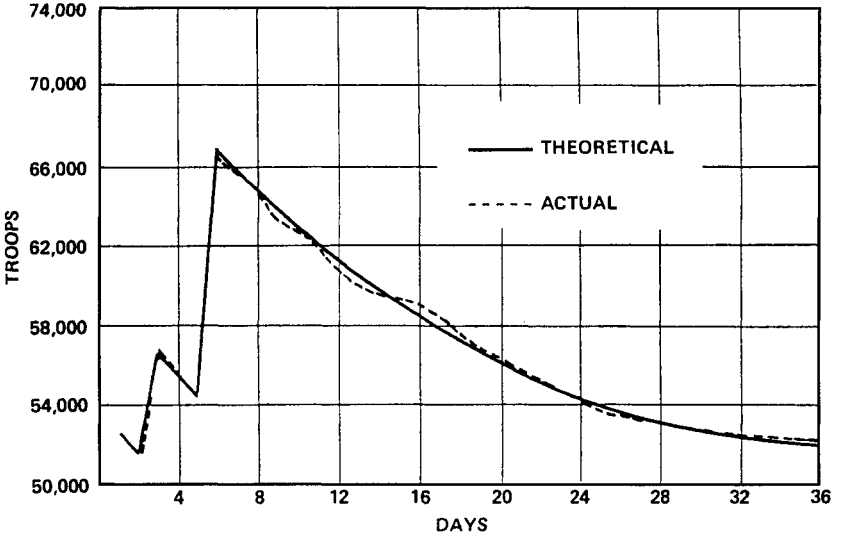


Fig. 4.10 Number of the American troops in action at the battle of Iwo Jima.

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Because there were no reinforcements after the eighth day of the battle, the results of Sec. 4.3 can be applied to compute the force superiority parameter for that day:

$$\Phi = \Phi(8) = \frac{m_B(8)}{m_R(8)}\sqrt{\frac{a_B}{a_R}} = \frac{64{,}500}{16{,}500}\sqrt{\frac{0.0106}{0.0543}} = 1.72 \tag{4.191}$$

Also, using Eq. (4.46b) the time to annihilate the Japanese forces can be obtained from

$$t_0 = \frac{1}{2\sqrt{a_B a_R}}\ell_n\frac{\Phi+1}{\Phi-1} = \frac{1}{2\sqrt{0.0106\times 0.0543}}\ell_n\frac{2.72}{0.72} = 28\text{ days} \tag{4.192}$$

which agrees with the time for the end of the battle on the 36th day, i.e., 28 days after the 8th day of the battle.

4.16 Casualty Exchange Ratio

A convenient way of measuring the effectiveness of tactical engagements represented by Lanchester models is to introduce the casualty exchange ratio Ψ defined by

$$\Psi = \frac{\text{Number of Blue force casualties from } t = 0 \text{ to } t = t_0}{\text{Number of Red force casualties from } t = 0 \text{ to } t = t_0} \tag{4.193}$$

where t_0 is the time taken to defeat either *all* Red forces or *all* Blue forces.[29] Therefore, this ratio represents the average number of Blue casualties per single casualty suffered by the Red force. The ratio depends on the type of engagement specified by the Lanchester model of combat. In this section, only the cases of the directed fire and area fire models will be considered for the purpose of deriving formulas for the casualty exchange ratio Ψ.

Directed Fire

From Eq. (4.193), it follows that

$$\Psi = \frac{M_B - m_B(t_0)}{M_R - m_R(t_0)} \tag{4.193a}$$

Depending which side is going to win, Eq. (193a) takes on two different forms. If the Blue forces win at $t = t_0$, then $m_R(t_0) = 0$ and the exchange ratio becomes

$$\Psi_B = \frac{M_B - m_B(t_0)}{M_R} = \frac{M_B}{M_R}\left(1 - \frac{m_B(t_0)}{M_B}\right)$$

$$= \frac{M_B}{M_R}(1 - \mu_B(\tau_0)) \tag{4.194}$$

Also, from the Lanchester square law, Eq. (4.47), it follows that when $\mu_R(\tau_0) = 0$ (i.e., when all Red forces are defeated)

$$\mu_B^2 = \frac{\Phi_0^2 - 1}{\Phi_0^2}; \quad \Phi_0 \geq 1 \tag{4.195}$$

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Substituting Eq. (4.195) into Eq. (4.194) and using $\Phi_0 = (M_B/M_R)\sqrt{a_B/a_R}$

$$\Psi_B = \frac{M_B}{M_R}\left(1 - \sqrt{1 - \frac{M_R^2}{M_B^2}\frac{a_R}{a_B}}\right) \tag{4.196}$$

Likewise, if the Red forces win at $t = t_0$, then $m_B(t_0) = 0$ and the exchange ratio becomes

$$\Psi_R = \frac{M_B}{M_R - m_R(t_0)} = \frac{M_B}{M_R(1 - m_R(t_0)/M_R)}$$

$$= \frac{M_B/M_R}{(1 - \mu_R(\tau_0))} \tag{4.197}$$

Using again the Lanchester square law, Eq. (4.47), it follows that when $\mu_B(\tau_0) = 0$

$$\mu_R(\tau_0) = \sqrt{1 - \Phi_0^2} \tag{4.198}$$

Hence,

$$\Psi_R = \frac{M_B/M_R}{\left(1 - \sqrt{1 - (M_B^2/M_R^2)(a_B/a_R)}\right)} \tag{4.199}$$

The exchange ratios given by Eqs. (4.196) and (4.199) are plotted in Fig 4.11 as functions of (M_B/M_R) for different values of (a_B/a_R); the exchange ratio Ψ_B is shown below the line for $\Phi_0 = 1$ and Ψ_R is shown above the line. The figure

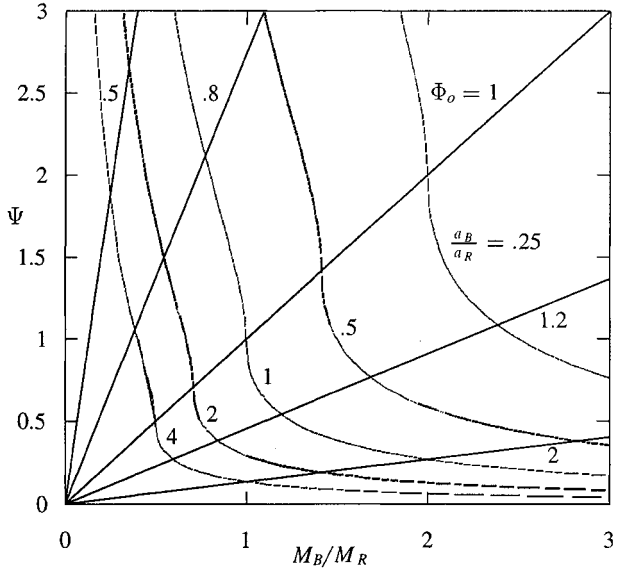


Fig. 4.11 Variation of the exchange ratio Ψ with (M_B/M_R) for different values of (a_B/a_R) (Directed fire model).

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illustrates that the Blue force casualty losses diminish with both the increases in the quantitative ratio M_B/M_R and the qualitative ratio a_B/a_R.

Area Fire

Starting again with the case when all Red forces are defeated (i.e., when $\mu_R = 0$), it follows from the Lanchester linear law (Eq. (4.72)) that

$$\mu_B = 1 - \Phi_0^{-2}; \quad \Phi_0 \geq 1 \tag{4.200}$$

Hence, substituting Eq. (4.200) into Eq. (4.194) and using $\Phi_0 = \frac{M_B}{M_R}\sqrt{\frac{a_B}{a_R}}$

$$\Psi_B = \frac{M_B}{M_R}\left(1 - 1 + \Phi_0^{-2}\right) = \left(\frac{M_B}{M_R}\right)^{-1}\left(\frac{a_B}{a_R}\right)^{-1} \tag{4.201}$$

Similarly, it can be demonstrated that for the case when all Blue forces are defeated (i.e., when $\mu_B = 0$)

$$\mu_R = 1 - \Phi_0^2 \tag{4.202}$$

and

$$\Psi_R = \frac{M_B}{M_R - m_R(t_0)} = \frac{M_B}{M_R}\frac{1}{(1 - \mu_R(\tau_0))}$$
$$= \frac{M_B}{M_R}\Phi_0^{-2} = \left(\frac{M_B}{M_R}\right)^{-1}\left(\frac{a_B}{a_R}\right)^{-1} \tag{4.203}$$

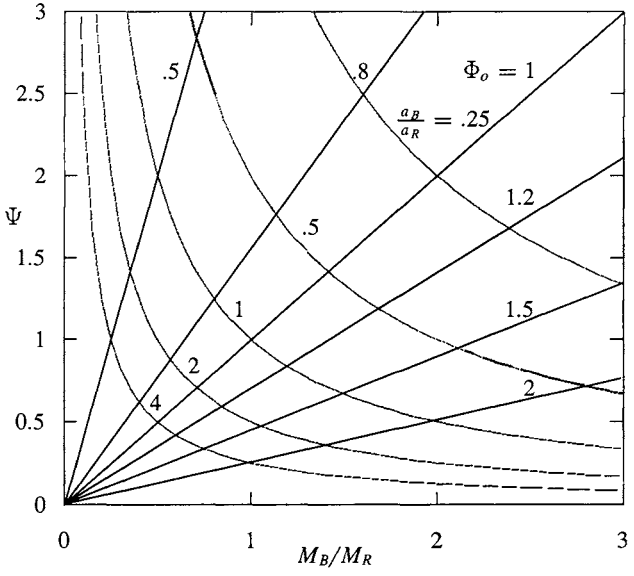


Fig. 4.12 Variation of the exchange ratio Ψ with (M_B/M_R) for different values of (a_B/a_R) (Area fire model).

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which is the same as Eq. (4.201). The exchange ratios Ψ_B or Ψ_R for the area fire model are plotted in Fig. 4.12 as functions of M_B/M_R for different values of the ratio a_B/a_R. It should also be noted that the exchange ratios Ψ can be defined differently. For example, the ratio can be specified as a casualty ratio when μ_R or μ_B are reduced to a certain level, e.g. 25%.

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Problems

4.1. Show that Eq. (4.7) is the probability density distribution for the variable z denoting the number of hits required to destroy the target.

4.2. Show that for the Lanchester model of directed fire for $\Phi_0 > 1$, the Blue force strength at the time of the complete annihilation of the Reds is given by

$$\mu_B = \sqrt{\frac{\Phi_0^2 - 1}{\Phi_0^2}}$$

4.3. Derive Eq. (4.72).

4.4. Derive Eq. (4.106).

4.5. Derive the conditions given in Table 4.6 for the Red forces to disengage first when the breakpoint levels for the Blue and Red forces are μ_{BB} and μ_{RB}, respectively, for the following combat models:

(a) Area fire model
(b) Guerrilla warfare
(c) Autonomous fire
(d) Geometric mean
(e) Autonomous/directed fire

4.6. Derive the exchange ratios Ψ for the case of a guerrilla warfare tactical engagement, i.e., when the Blue force is ambushed by the Red force.

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5
Probabilistic Combat Models

The previous chapter described the deterministic modeling of combat engagements that led to a set of differential equations. In this chapter, instead of determining the strength levels of the combatants or weapons, the probabilities of strength levels at a particular time will be determined for several types of combat engagements. As an introduction to the topic, the chapter starts with a sequential combat duel (time-independent combat), followed by a continuous combat duel (time-dependent combat) that clearly demonstrates the principle of surprise in any tactical engagement. Next, this chapter describes the complete probabilistic analysis for the case of a one-on-one combat engagement for the directed fire and area fire models. The results of the probabilistic models are compared with the deterministic solutions discussed in Chapter 4. Finally, the probability of survival of targets attacked by aircraft is derived for the three basic types of target allocation assignments: the uniform allocation assignment, the random allocation assignment, and the "shoot-look-shoot" allocation assignment (perfect assignment).

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5.1 Sequential Combat Duel: Time-Independent Combat

This section considers a sequential combat duel of two weapons or units firing at each other, i.e., one-on-one duel. Naturally this is a very simplified engagement model that will only be used to illustrate the mathematical principles involved in obtaining probabilities of one weapon or unit defeating the other.

The probability that a particular round will destroy the opposing weapon will be assumed to be equal to p for the Blue weapon and q for the Red weapon. The probabilities of the weapons surviving the rounds fired by the opposing side will therefore be given by $1 - p$ and $1 - q$ for the Red and Blue weapons, respectively. Also, a word of caution must be added here. The probabilities p and q used in this section should not be confused with the probabilities p and q in Chapter 2, where $q = 1 - p$.

In the sequential combat duel it will be assumed initially that the Blue weapon is fired first, then the Red weapon, then the Blue weapon, etc. It will also be assumed that the result of the previous firing is known before the next round is fired, and this assumption will allow one to formulate the individual probabilities for each event in the firing sequence of the duel. Table 5.1 summarizes the event probabilities for the first three rounds fired by each of the opposing weapons.

The probability of the Red weapon being destroyed in the sequential combat duel given in Table 5.1, in which rounds are fired in the sequence 1st Blue, 1st Red, 2nd Blue, 2nd Red, 3rd Blue, 3rd Red, . . . , etc., is obtained as the summation of individual probabilities (i.e., addition of probabilities of mutually exclusive

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Table 5.1 Sequential combat duel

Event	Event description	Event probability
(1)	Red is destroyed by 1st Blue round	p
(2)	Red survives 1st Blue round	$1-p$
(3)	Blue is destroyed by 1st Red round	$(1-p)q$
(4)	Blue survives 1st Red round	$(1-p)(1-q)$
(5)	Red is destroyed by 2nd Blue round	$(1-p)(1-q)p$
(6)	Red survives 2nd Blue round	$(1-p)^2(1-q)$
(7)	Blue is destroyed by 2nd Red round	$(1-p)^2(1-q)q$
(8)	Blue survives Red round	$(1-p)^2(1-q)^2$
(9)	Red is destroyed by 3rd Blue round	$(1-p)^2(1-q)^2p$
(10)	Red survives 3rd Blue round	$(1-p)^3(1-q)^2$
(11)	Blue is destroyed by 3rd Red round	$(1-p)^3(1-q)^2q$
(12)	Blue survives 3rd Red round	$(1-p)^3(1-q)^3$
etc.	etc.	etc.

events), which leads to

$$P_R = (1) + (5) + (9) + \cdots$$
$$= p + (1-p)(1-q)p + (1-p)^2(1-q)^2p + \cdots \tag{5.1}$$

Observing that Eq. (5.1) is the sum of a geometric series, a simplified expression for a finite number of fire exchanges can be obtained:

$$P_R(n) = p\,\frac{1-(1-p)^n(1-q)^n}{1-(1-p)(1-q)} \tag{5.2}$$

where n denotes the number of rounds fired by the Blue weapon. For a sufficiently large number of rounds in Eq. (5.2), n may be taken as $n = \infty$. Noting that

$$\lim_{n\to\infty}(1-p)^n(1-q)^n = 0$$

it follows that

$$P_R(n=\infty) = \frac{p}{1-(1-p)(1-q)} \tag{5.3}$$

Similarly, the probability of the Blue weapon being destroyed is found from

$$P_B = (3) + (7) + (11) + \cdots$$
$$= (1-p)q + (1-p)^2(1-q)q + (1-p)^3(1-q)^2q + \cdots \tag{5.4}$$

$$P_B(n) = (1-p)q\,\frac{1-(1-p)^n(1-q)^n}{1-(1-p)(1-q)} \tag{5.5}$$

$$P_B(n=\infty) = \frac{(1-p)q}{1-(1-p)(1-q)} \tag{5.6}$$

where n denotes the number of rounds fired by the Red weapon.

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The preceding expressions for probabilities were obtained on the assumption that the Blue round was fired first followed by alternating rounds from the opposing forces to form a sequence 1st Blue, 1st Red, 2nd Blue, 2nd Red, etc. Another order in sequential firings is for the Red weapon to fire first followed by alternating rounds in a sequence 1st Red, 1st Blue, 2nd Red, 2nd Blue, etc. In this case the probabilities become

$$P_B^*(n) = q\,\frac{1-(1-p)^n(1-q)^n}{1-(1-p)(1-q)} \tag{5.7}$$

$$P_B^*(n=\infty) = \frac{q}{1-(1-p)(1-q)} \tag{5.8}$$

$$P_R^*(n) = p(1-q)\frac{1-(1-p)^n(1-q)^n}{1-(1-p)(1-q)} \tag{5.9}$$

$$P_R^*(n=\infty) = \frac{p(1-q)}{1-(1-p)(1-q)} \tag{5.10}$$

where an asterisk is used to indicate that these probabilities refer to the sequence starting with the Red weapon fired first.

The variation of $P_R(n=\infty)$ and $P_R^*(n=\infty)$ with p and q, i.e., Eqs. (5.3) and (5.10), are plotted in Fig. 5.1. In this graphic representation it is evidently clear that when the Blue weapon is fired first, the Blue side will have advantage in that the probabilities of destroying Red weapons are higher than in the case where the Red weapon commences firing first. This occurs because $P_R(n) \geq P_R^*(n)$. Similarly, the variations of $P_B(n=\infty)$ and $P_B^*(n=\infty)$ with p and q, i.e., Eqs. (5.6) and (5.8), are plotted in Fig. 5.2. Here again it is evident that the probability of survival of the Blue weapon is enhanced if the Blue weapon is fired first because $1-P_B(n) > 1-P_B^*(n)$. The reader can verify that $P_B(n)+P_R(n)=1$ and $P_B^*(n)+P_R^*(n)=1$.

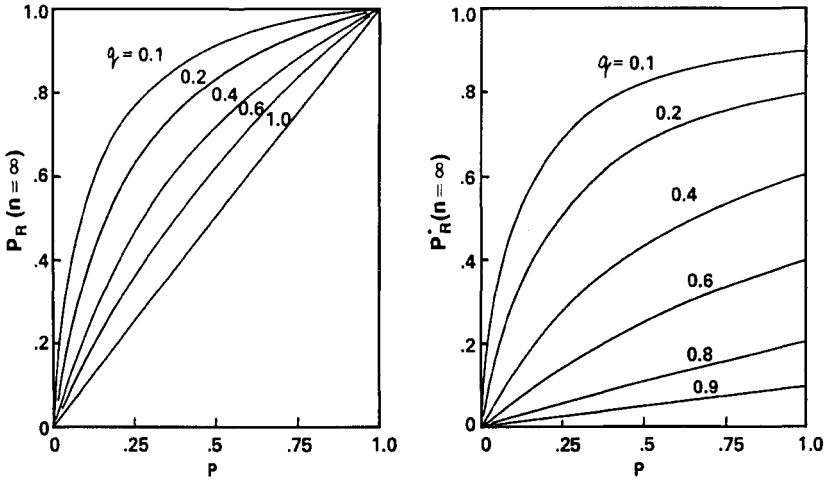


Fig. 5.1 Variation of $P_R(n=\infty)$ and $P_R^*(n=\infty)$ with p and q.

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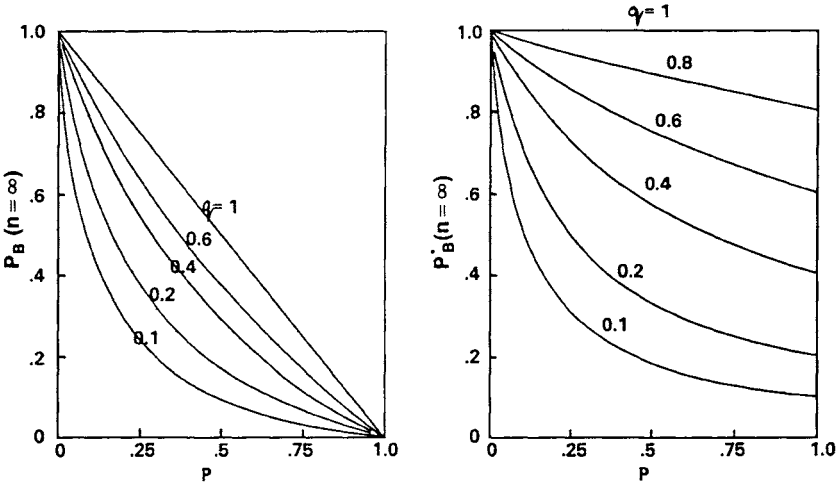


Fig. 5.2 Variation of $P_B(n = \infty)$ and $P_B^*(n = \infty)$ with p and q.

5.2 Continuous Combat Duel: Time-Dependent Combat

The sequential combat duel is a special case of two weapons firing at each other. A more realistic representation of this one-on-one duel would be to assume that each weapon fires continuously at a constant rate of specified number of rounds per unit time. The constant rate can also be interpreted as the mean rate over a period of time. To analyze the continuous combat duel, it will be assumed that the duel begins at $t = 0$, and from that instant on both sides will fire simultaneously at each other with the rates α_B and α_R for the Blue and Red forces, respectively. $P_B(t)$ and $P_R(t)$ will denote the probabilities that the Blue weapons and Red weapons, respectively, are destroyed in time t. These probabilities can be compared with $P_B(n)$ and $P_R(n)$ of the previous section, where the probabilities depended only on the number of rounds fired by each side. If at any instant $t \geq 0$, the duel still goes on, i.e., if both weapons have survived the previous fire exchange, then in a sufficiently small time interval Δt the following events can occur as described in Table 5.2. As before, p and q denote the single round probabilities that the opposing side target will be destroyed.

Table 5.2 describes a complete set of mutually exclusive and exhaustive events. Also, the time interval is assumed to be sufficiently small so that both $\alpha_B \Delta t$ and $\alpha_R \Delta t$ are less than or equal to unity (note that $\alpha_B \Delta t = 1$ implies that only one round has been fired in Δt). Consequently, these products can be treated as the probabilities of single rounds being fired by the Blue and Red weapons, respectively. The terms $O(\Delta t^2)$ indicate the higher-order terms that reflect the continuous process of the firing action; they can be ignored in subsequent analysis. The first event in Table 5.2 is simply the product of the probability of a single round destroying the Red weapon and the probability of a single round from the Blue weapon occurring during Δt. If $\alpha_B \Delta t$ was equal to unity, this would have represented the certainty that the Blue weapon was fired. In order to use the product $\alpha_B \Delta t$, it follows that $\alpha_B \Delta t < 1$. All other event probabilities in Table 5.2 are self-explanatory and can be derived from the basic principles.

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Table 5.2 Continuous combat duel

Event	Event description	Event probability
(1)	Blue weapon fires and destroys Red weapon	$p\alpha_B\,\Delta t + O(\Delta t^2)$
(2)	Red weapon fires and destroys Blue weapon	$q\alpha_R\,\Delta t + O(\Delta t^2)$
(3)	Blue weapon fires and does not destroy its target	$(1-p)\alpha_B\,\Delta t + O(\Delta t^2)$
(4)	Red weapon fires and does not destroy its target	$(1-q)\alpha_R\,\Delta t + O(\Delta t^2)$
(5)	Neither Blue nor Red fire their weapons	$1 - \alpha_B\,\Delta t - \alpha_R\,\Delta t + O(\Delta t^2)$
(6)	Both weapons fire in time Δt and do not destroy their targets	$1 - p\alpha_B\,\Delta t - q\alpha_R\,\Delta t + O(\Delta t^2)$
(7)	Both weapons fire simultaneously	$O(\Delta t^2)$

The probability $P_B(t)$ that the Blue weapon is destroyed in time t can be related to the probability $P_B(t+\Delta t)$. From Table 5.2 it is clear that

$$P_B(t+\Delta t) = (2) + (6)P_B(t)$$

Hence,

$$P_B(t+\Delta t) = q\alpha_R\Delta t + (1 - p\alpha_B\Delta t - q\alpha_R\Delta t)P_B(t)$$

$$\frac{P_B(t+\Delta t) - P_B(t)}{\Delta t} = q\alpha_R - (p\alpha_B + q\alpha_R)P_B(t)$$

Now in the limit, as Δt tends to 0, the preceding expression changes into a differential equation

$$\frac{dP_B(t)}{dt} = q\alpha_R - (p\alpha_B + q\alpha_R)P_B(t) \tag{5.11}$$

By similar reasoning the following differential equation for the probability $P_R(t)$ is obtained

$$\frac{dP_R(t)}{dt} = p\alpha_B - (p\alpha_B + q\alpha_R)P_R(t) \tag{5.12}$$

The solutions of Eqs. (5.11) and (5.12) are

$$P_B(t) = C_1 e^{-(p\alpha_B + q\alpha_R)t} + \frac{q\alpha_R}{p\alpha_B + q\alpha_R} \tag{5.13}$$

and

$$P_R(t) = C_2 e^{-(p\alpha_B + q\alpha_R)t} + \frac{p\alpha_B}{p\alpha_B + q\alpha_R} \tag{5.14}$$

where C_1 and C_2 are arbitrary constants that can be determined from the initial conditions $P_B(0) = P_R(0) = 0$. This leads to

$$P_B(t) = \frac{q\alpha_R}{p\alpha_B + q\alpha_R}\left[1 - e^{-(p\alpha_B + q\alpha_R)t}\right] \tag{5.15}$$

$$P_R(t) = \frac{p\alpha_B}{p\alpha_B + q\alpha_R}\left[1 - e^{-(p\alpha_B + q\alpha_R)t}\right] \tag{5.16}$$

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or, expressing the preceding equations in terms of nondimensional parameters,

$$P_B(\tau) = \frac{1}{1+\beta}\left[1 - \exp\left(-\frac{\beta+1}{\sqrt{\beta}}\,\tau\right)\right] \tag{5.17}$$

$$P_R(\tau) = \frac{\beta}{1+\beta}\left[1 - \exp\left(-\frac{\beta+1}{\sqrt{\beta}}\,\tau\right)\right] \tag{5.18}$$

where

$$\beta = \frac{p\alpha_B}{q\alpha_R} \quad \text{(fire superiority parameter)} \tag{5.19}$$

$$\tau = \sqrt{p\alpha_B q\alpha_R}\,t \quad \text{(nondimensional time)} \tag{5.20}$$

The variations of P_B and P_R with the nondimensional time τ for a series of values of the fire superiority parameter β are shown in Fig. 5.3. This figure clearly depicts the significance of $\beta = p\alpha_B/q\alpha_R$.

As an illustration of the application of Eqs. (5.17) and (5.18) to a typical scenario, consider a close air support (CAS) fire engagement between an A-10 aircraft and a tank with antiaircraft artillery (AAA). Assuming, for the example, that $p = q$ and $\alpha_B/\alpha_R = 2$, it follows from Eqs. (5.17) and (5.18) that, for large times ($\tau = \infty$),

$$P_R = 0.667$$

$$P_B = 0.333$$

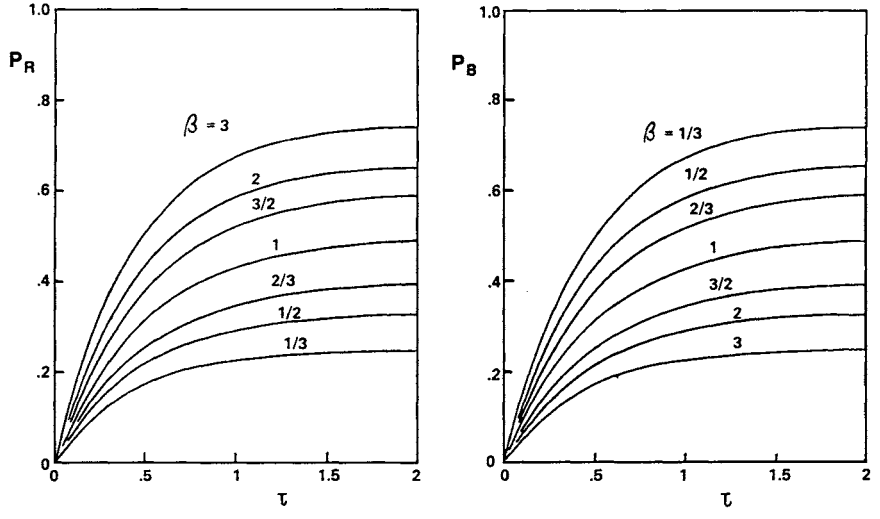


Fig. 5.3 Variation of $P_R(\tau)$ and $P_B(\tau)$ with τ for a series of values of the fire superiority parameter β.

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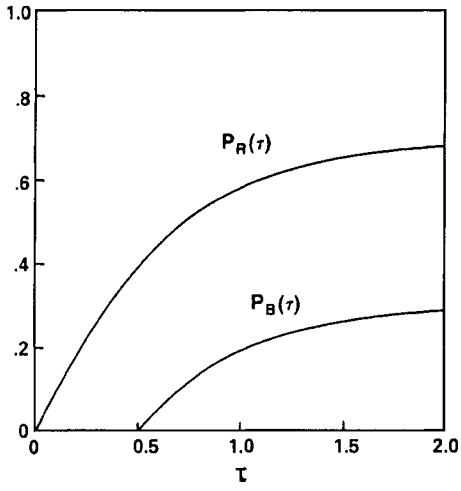


Fig. 5.4 Variation of $P_R(\tau)$ and $P_B(\tau)$ with nondimensional time τ for $\beta = 2$ and the delay of Red response of $\tau_0 = 0.5$.

Also, for any value of the nondimensional time τ,

$$\frac{P_R(\tau)}{P_B(\tau)} = \frac{\beta/(1+\beta)}{1/(1+\beta)} = \beta = 2$$

Thus, the probability of destroying the Red tank by the A-10 aircraft is two times greater than the probability of destroying the A-10 aircraft by the Red tank.

The probability of the Red weapon being destroyed can be greatly increased if the Blue weapon can achieve an element of surprise by commencing firing for t_0 units of time before the Red weapon has a chance to respond. This situation is depicted in Fig. 5.4. For the initial phase $0 \leq t \leq t_0$, the probabilities $P_B(t)$ and $P_R(t)$ can be computed from Eqs. (5.15) and (5.16) by substituting $\alpha_R = 0$, i.e., zero firing rate for the Red weapon. Hence,

$$P_B(t) = 0; \qquad 0 \leq t \leq t_0 \tag{5.21}$$

and

$$P_R(t) = 1 - e^{-p\alpha_B t}; \qquad 0 \leq t \leq t_0 \tag{5.22}$$

which can be modified using the nondimensional parameters

$$\tau = \sqrt{p\alpha_B q\alpha_R}\, t \quad \text{and} \quad \beta = p\alpha_B/q\alpha_R,$$

where α_R is the Red weapon firing rate for $t > t_0$. This leads to the expressions

$$P_B(\tau) = 0; \qquad 0 \leq \tau \leq \tau_0 \tag{5.23}$$

and

$$P_R(\tau) = 1 - e^{-\sqrt{\beta}\tau}; \qquad 0 \leq \tau \leq \tau_0 \tag{5.24}$$

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where

$$\tau_0 = \sqrt{p\alpha_B q\alpha_R}\, t_0 \tag{5.25}$$

To calculate the probabilities P_B and P_R for $t > t_0$, it should be observed that, at $t = t_0$,

$$P(\text{probability of Red weapon survival at } t = t_0) = 1 - P_R(t_0) = e^{-p\alpha_B t_0} \tag{5.26}$$

This result can then be used to determine P_B and P_R from

$$P_B = P(\text{probability of Blue weapon being destroyed given Red weapon survival}) \times P(\text{probability of Red weapon survival at } t = t_0) \tag{5.27}$$

$$P_R = P(\text{probability of Red weapon being destroyed by the time } t_0) + P(\text{probability of Red weapon being destroyed during the time } t - t_0 \text{ given Red weapon survival}) \times P(\text{probability of Red weapon survival at } t = t_0) \tag{5.28}$$

Hence,

$$P_B = \frac{q\alpha_R}{p\alpha_B + q\alpha_R}\left[1 - e^{-(p\alpha_B + q\alpha_R)(t-t_0)}\right] e^{-p\alpha_B t_0} \quad \text{for} \quad t \geq t_0 \tag{5.29}$$

$$P_R = 1 - e^{-p\alpha_B t_0} + \frac{p\alpha_B}{p\alpha_B + q\alpha_R}\left[1 - e^{-(p\alpha_B + q\alpha_R)(t-t_0)}\right] e^{-p\alpha_B t_0} \quad \text{for} \quad t \geq t_0 \tag{5.30}$$

or, using the previously defined nondimensional parameters β and α,

$$P_B(\tau) = \frac{1}{1+\beta}\left\{1 - \exp\left[-\frac{\beta+1}{\sqrt{\beta}}(\tau - \tau_0)\right]\right\}\exp\left(-\sqrt{\beta}\,\tau_0\right) \quad \text{for} \quad \tau \geq \tau_0 \tag{5.31}$$

$$P_R(\tau) = 1 - \exp\left(-\sqrt{\beta}\,\tau_0\right) + \frac{\beta}{1+\beta}\left\{1 - \exp\left[-\frac{\beta+1}{\sqrt{\beta}}(\tau - \tau_0)\right]\right\}\exp\left(-\sqrt{\beta}\,\tau_0\right) \quad \text{for} \quad \tau \geq \tau_0 \tag{5.32}$$

When $\tau = \infty$, Eq. (5.32) becomes

$$P_R(\infty) = 1 - \frac{1}{1+\beta}\exp\left(-\sqrt{\beta}\,\tau_0\right) \tag{5.33}$$

This equation is plotted in Fig. 5.5 for a range of values of the superiority parameter β to illustrate effect of delayed response by the Red weapon after time τ_0 has elapsed. For example, even if the Blues have an inferior rate of fire, e.g., $\beta = 2/3$, P_R can be increased from $P_R = 0.4$ for $\tau_0 = 0$ to $P_R = 0.6011$ for $\tau_0 = 0.5$. This confirms the important principle of surprise in any tactical engagement.

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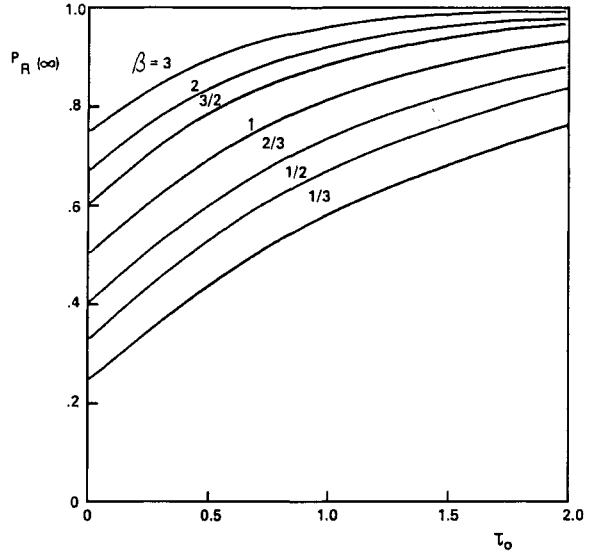


Fig. 5.5 Variation of $P_R(\infty)$ with τ_0 for a series of values of the superiority parameter β.

5.3 Continuous Combat States: Directed Fire Model[1]

The previous sections considered the cases of sequential and continuous duels between two weapons or two combat units, where a unit was treated as a single entity. This section will consider the case of tactical combat of two homogeneous forces engaging each other with some specified effective attrition rates. The engagement will be assumed to be governed by the overall assumptions used for the Lanchester model for the directed fire used in Sec. 4.3. Starting with the initial strengths m_0 and n_0 for the Blue and Red forces, respectively, the subsequent strengths m and n will be determined probabilistically. Thus, at the time t from the beginning of the battle engagement, the probability of the occurrence of any particular set of values (m, n) will be determined as a function of the initial values m_0 and n_0 and the effective attrition rates a_R and a_B for the Blue and Red forces, respectively. For simplicity, a new notation will be introduced here:

$$a = a_B \tag{5.34}$$

$$b = a_R \tag{5.35}$$

Thus, in the deterministic model of directed fire, the governing equations can be written, using this notation, as

$$\frac{dm}{dt} = -bn \tag{5.36}$$

and

$$\frac{dn}{dt} = -am \tag{5.37}$$

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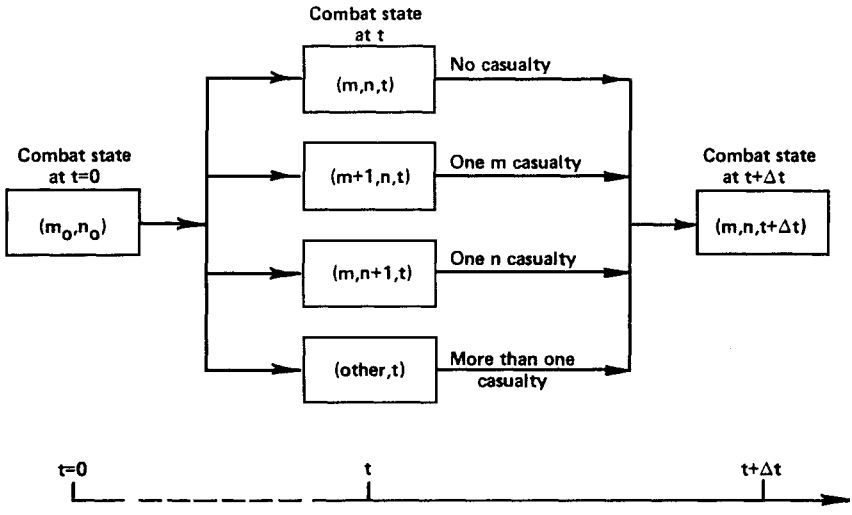


Fig. 5.6 Possible combat states.

Next, both m and n will be treated as random variables such that $m = 0, 1, 2, \ldots, m_0$ and $n = 0, 1, 2, \ldots, n_0$ and the casualties will be assumed to occur randomly over time. Thus, the number of live combatants (or functioning weapons) will be a random variable. The solution for m and n will be obtained in terms of the probabilities for a given combat state described by a pair of numbers (m, n).

Unlike the deterministic model, where there is a discontinuity in the solution when the strength of one of the combatants is reduced to 0 at the time $t = t_0$, the probabilistic model does not have this limitation because its solution is also valid for $t > t_0$. Of course, the practical validity of the probabilistic model for such cases when $m \gg n$ or $n \gg m$ is clearly questionable.

A particular combat state (m, n) at time $t + \Delta t$ can be reached from a number of other combat states as shown in Fig. 5.6. It should be noted that the future state at $t + \Delta t$ depends only on the present state at time t and independent of the past history, i.e., independent of the path from (m_0, n_0) to the particular state at time t. Such probabilistic situations are described as Markov chains and can be expressed by the forward Kolmogorov equations.

Consider next the product $bn\ \Delta t = -\Delta m$ obtained from Eq. (5.36). If $bn\ \Delta t = 1$, then one casualty occurs for the Blue force (m). If, on the other hand, $bn\ \Delta t < 1$ then this product can be considered to be the probability of occurrence of one casualty in m in time Δt. Likewise, $am\ \Delta t < 1$ can be considered to be the probability of one casualty in n. These probabilities can then be used to determine the probability of the combat state (m, n) at time t, which will be denoted as $P(m, n, t)$.

The initial conditions for $P(m, n, t)$ are simply

$$P(m, n, 0) = \begin{cases} 1 & \text{for } m = m_0 \text{ and } n = n_0 \\ 0 & \text{otherwise} \end{cases} \tag{5.38}$$

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From the definition of conditional probability it follows immediately that

$$P(m, n, t + \Delta t) = P(m, n, t)\, P(\text{no casualty occurs on both sides in } \Delta t)$$
$$+ P(m + 1, n, t)\, P(\text{one } m \text{ casualty in } \Delta t)$$
$$+ P(m, n + 1, t)\, P(\text{one } n \text{ casualty in } \Delta t)$$
$$+ P(\ldots, \ldots, t)\, P(\text{more than one casualty in } \Delta t) \tag{5.39}$$

where $P(\ldots, \ldots, \ldots)$ is the probability of an event described in parentheses, i.e., $P(m + i, n + j, t)$ where $i, j = 0, 1, 2, \ldots$. It can now be observed from the definition of the products $bn\Delta t$ and $am\Delta t$ that

$$P(\text{no casualties on both sides in } \Delta t) = 1 - bn\, \Delta t - am\, \Delta t \tag{5.40}$$
$$P(\text{one } m \text{ casualty in } \Delta t) = bn\, \Delta t \tag{5.41}$$
$$P(\text{one } n \text{ casualty in } \Delta t) = am\, \Delta t \tag{5.42}$$
$$P(\text{more than one casualty in } \Delta t) = 0 \tag{5.43}$$

The substitution of Eqs. (5.40)–(5.43) into Eq. (5.39) results in the following governing equation:

$$P(m, n, t + \Delta t) = P(m, n, t)(1 - bn\, \Delta t - am\, \Delta t)$$
$$+ P(m + 1, n, t)bn\, \Delta t + P(m, n + 1, t)am\, \Delta t \tag{5.44}$$

The preceding equation can be transformed into nondimensional form with

$$\Phi_0 = \sqrt{\frac{a}{b}}\, \frac{m_0}{n_0} \tag{5.45}$$
$$\tau = \sqrt{ab}\, t \tag{5.46}$$

Hence,

$$P(m, n, \tau + \Delta\tau) = P(m, n, \tau)\left(1 - \Phi_0^{-1} \frac{nm_0}{n_0}\, \Delta\tau - \Phi_0 \frac{mn_0}{m_0}\, \Delta\tau\right)$$
$$+ P(m + 1, n, \tau)\Phi_0^{-1} \frac{nm_0}{n_0}\, \Delta\tau + P(m, n + 1, \tau)\Phi_0 \frac{mn_0}{m_0}\, \Delta\tau \tag{5.47}$$

Equation (5.47) can be rewritten as

$$\frac{P(m, n, \tau + \Delta\tau) - P(m, n, \tau)}{\Delta\tau} = -P(m, n, \tau)\left(\Phi_0^{-1} \frac{nm_0}{n_0} + \Phi_0 \frac{mn_0}{m_0}\right)$$
$$+ P(m + 1, n, \tau)\Phi_0^{-1} \frac{nm_0}{n_0} + P(m, n + 1, \tau)\Phi_0 \frac{mn_0}{m_0} \tag{5.48a}$$

which in the limit as $\Delta\tau \to 0$ becomes

$$\frac{dP(m, n, \tau)}{d\tau} = -P(m, n, \tau)\left(\Phi_0^{-1} \frac{nm_0}{n_0} + \Phi_0 \frac{mn_0}{m_0}\right)$$
$$+ P(m + 1, n, \tau)\Phi_0^{-1} \frac{nm_0}{n_0} + P(m, n + 1, \tau)\Phi_0 \frac{mn_0}{m_0} \tag{5.48b}$$

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The preceding equation is the Kolmogorov differential equation for the probability of combat states (m, n) for the directed fire model.

Solutions to Eq. (5.48b) can be most conveniently obtained from its finite-difference representation by Eq. (5.47) for a specified time increment $\Delta\tau$ subject to the restrictions $bn\,\Delta t < 1$ and $am\,\Delta t < 1$, which can rewritten for the limiting cases when $m = m_0$ and $n = n_0$ as

$$\Delta\tau < \frac{\Phi_0}{m_0} \tag{5.49}$$

and

$$\Delta\tau < \frac{\Phi_0^{-1}}{n_0} \tag{5.50}$$

Equation (5.47) is valid only for the combat states other than those involving the boundary values $m = 0$ and $n = 0$. The boundary conditions, other than those for $m = 0$ and $n = 0$, can be obtained directly from Eq. (5.47) by eliminating redundant terms. This results in the following equations:

For $m = m_0$ and $n = n_0$,

$$P(m_0 + 1, n_0, \tau) = P(m_0, n_0 + 1, \tau) = 0$$

and

$$P(m_0, n_0, \tau + \Delta\tau) = P(m_0, n_0, \tau)\left(1 - \Phi_0^{-1} m_0\,\Delta\tau - \Phi_o n_0\,\Delta\tau\right) \tag{5.51}$$

For $0 < m < m_0$ and $n = n_0$,

$$P(m, n_0, \tau + \Delta\tau) = P(m, n_0, \tau)\left(1 - \Phi_0^{-1} m_0\,\Delta\tau - \Phi_0 \frac{m n_0}{m_0}\,\Delta\tau\right)$$
$$+ P(m + 1, n_0, \tau)\Phi_0^{-1} m_0\,\Delta\tau \tag{5.52}$$

For $m = m_0$ and $0 < n < n_0$,

$$P(m_0, n, \tau + \Delta\tau) = P(m_0, n, \tau)\left(1 - \Phi_0^{-1} \frac{n m_0}{n_0}\,\Delta\tau - \Phi_0 n_0\,\Delta\tau\right)$$
$$+ P(m_0, n + 1, \tau)\Phi_0 n_0\,\Delta\tau \tag{5.53}$$

On the boundaries $m = 0$ and $n = 0$ the fundamental Eq. (5.39) must be modified. For example, for $m = 0$ and $0 < n \leq n_0$,

$$P(0, n, \tau + \Delta\tau) = P(0, n, \tau)\,P(\text{no casualty occurs on both sides in time } \Delta\tau)$$
$$+ P(1, n, \tau)\,P(\text{one } m \text{ casualty in time } \Delta\tau)$$
$$+ P(0, n + 1, \tau)\,P(\text{one } n \text{ casualty in time } \Delta\tau)$$

It is obvious that for $m = 0$ the probabilities $P(\text{no casualty occurs on both sides in time } \Delta\tau) = 1$ and $P(\text{one } n \text{ casualty in time } \Delta\tau) = 0$. Hence,

$$P(0, n, \tau + \Delta\tau) = P(0, n, \tau) \times 1 + P(1, n, \tau)\Phi_0^{-1} \frac{n m_0}{n_0}\,\Delta\tau \tag{5.54}$$

Similarly, it can be shown that, for $n = 0$ and $0 < m \leq m_0$,

$$P(m, 0, \tau + \Delta\tau) = P(m, 0, \tau) + P(m, 1, \tau)\Phi_0 \frac{m n_0}{m_0}\,\Delta\tau \tag{5.55}$$

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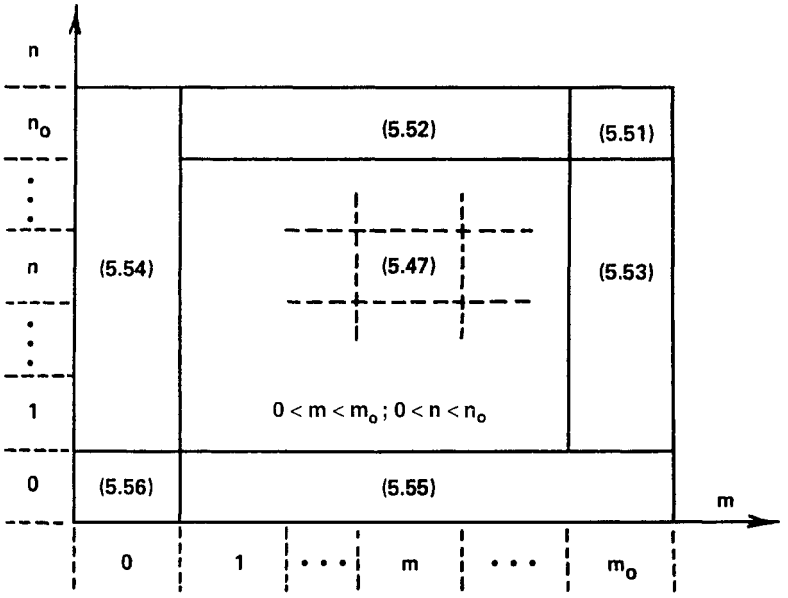


Fig. 5.7 Combat state space; regions of validity of probability equations (directed fire model).

Finally, for $m = 0$ and $n = 0$,

$$P(0, 0, \tau) = 0 \tag{5.56}$$

because it is impossible to reach the combat state (0, 0).

The regions of validity of the preceding equations are shown in Fig. 5.7. The solution to these equations for any given set of values of Φ_0, m_0, n_0, and $\Delta\tau$ can be obtained numerically. The solution progresses in a series of steps $\Delta\tau$ with new values of P computed from the previous step. The results of such computations for $\Phi_0 = 1.5$, $m_0 = n_0 = 20$, and $\Delta\tau = 0.01$ are shown in Figs. 5.8–5.11 for $\tau = 0.05, 0.25, 0.5$, and 0.75.

As a check of the numerical values obtained from the computations of $P(m, n, t)$,

$$\sum_{m=0}^{m_0} \sum_{n=0}^{n_0} P(m, n, t) = 1 \tag{5.57}$$

can be used because $P(m, n, t)$ represents the probability distribution for all states, which, of course, must sum up to unity (i.e., certainty). In addition, the expected values of m and n can be obtained from

$$E[m] = \sum_{m=0}^{m_0} \sum_{n=0}^{n_0} m P(m, n, t) \tag{5.58}$$

and

$$E[n] = \sum_{m=0}^{m_0} \sum_{n=0}^{n_0} n P(m, n, t) \tag{5.59}$$

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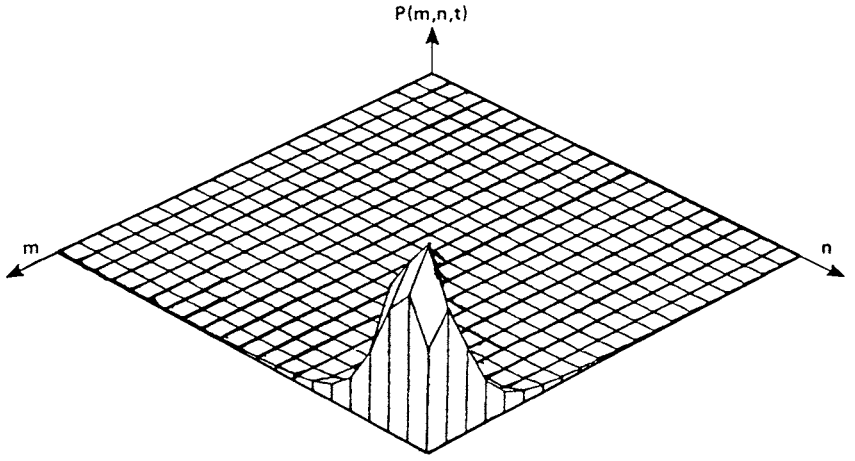


Fig. 5.8 Probability distribution for the directed fire model for $\Phi_0 = 1.5$ and $\tau = 0.05$.

It is interesting to compare the results of the expected values $E[m]$ and $E[n]$ with the results of the deterministic combat model. Such a comparison is made in Table 5.3 for the example used in Figs. 5.8–5.11, i.e., for $\Phi_0 = 1.5$.

Although both $E[m]$ and $m(\tau)$ and $E[n]$ and $n(\tau)$ show a very good agreement for small values of the nondimensional time τ, for larger values of τ the difference becomes more pronounced. This is not surprising because of the differences in the fundamental nature of the solution. $E[m]$ and $E[n]$ decrease asymptotically with time, and $n(\tau)$ decreases to 0 at the time $\tau = \tau_0$. Thus, in general, deterministic results should not be equated to their corresponding probabilistic expected values.

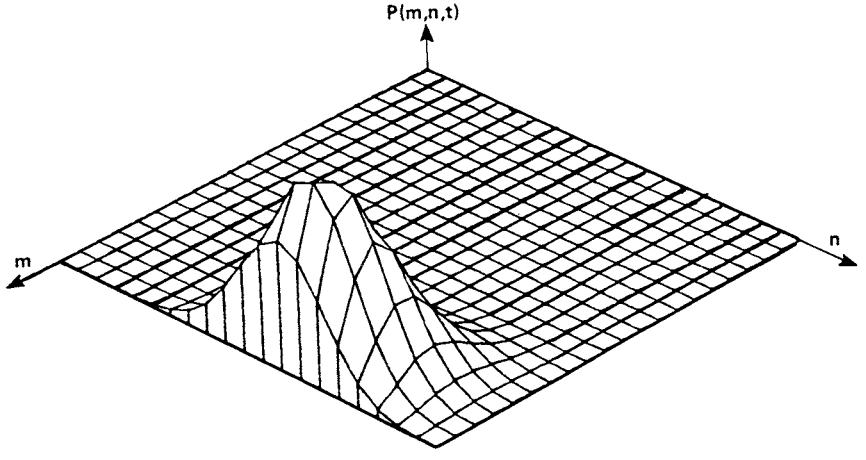


Fig. 5.9 Probability distribution for the directed fire model for $\Phi_0 = 1.5$ and $\tau = 0.25$.

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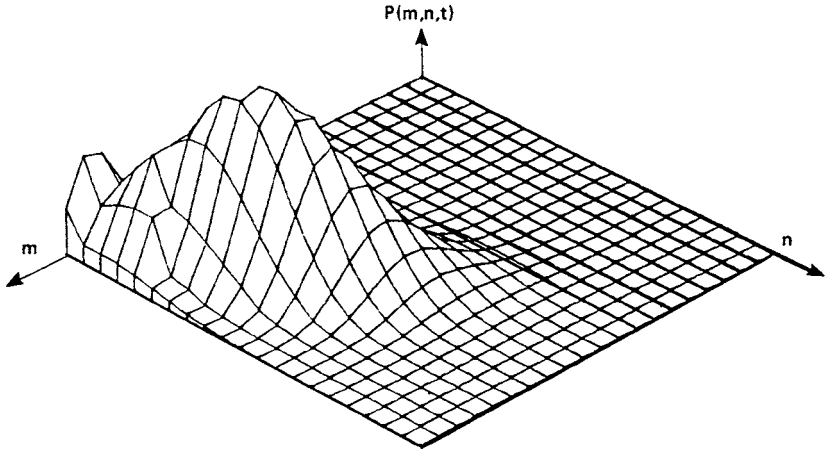


Fig. 5.10 Probability distribution for the directed fire model for $\Phi_0 = 1.5$ and $\tau = 0.50$.

5.4 Continuous Combat States: Area Fire Model

The governing equations for the deterministic model of area fire combat can be written as

$$\frac{dm}{dt} = -b\frac{mn}{m_0} \tag{5.60}$$

$$\frac{dn}{dt} = -a\frac{mn}{n_0} \tag{5.61}$$

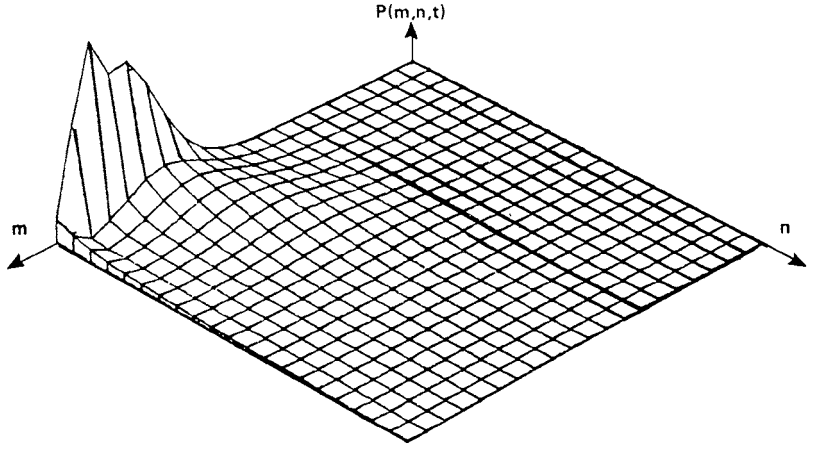


Fig. 5.11 Probability distribution for the directed fire model for $\Phi_0 = 1.5$ and $\tau = 0.75$.

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Table 5.3 Comparison of the expected values $E[m]$ and $E[n]$ with the deterministic values $m(\tau)$ and $n(\tau)$ for $\Phi_0 = 1.5$, $m_0 = 20$, and $n_0 = 20$ (directed fire model)

τ	$E[m]$	$m(\tau)$	$E[n]$	$n(\tau)$
0.0000	20.00	20.00	20.00	20.00
0.0500	19.35	19.36	18.52	18.52
0.2500	17.24	17.26	13.03	13.05
0.5000	15.56	15.60	6.94	6.92
0.7500	14.80	14.93	2.62	1.22
0.8047	14.72	14.91	2.04	0.00

Rewriting the preceding equations in the finite-difference form,

$$\Delta m = -b\frac{mn}{m_0}\Delta t = -\Phi_0^{-1}\frac{mn}{n_0}\Delta\tau \tag{5.62}$$

$$\Delta n = -a\frac{mn}{n_0}\Delta t = -\Phi_0\frac{mn}{m_0}\Delta\tau \tag{5.63}$$

which for sufficiently small Δt or $\Delta\tau$ such that $\Delta m \leq 1$ and $\Delta n \leq 1$ can be regarded as the probabilities of occurrence of one casualty in m or n.

As in the previous case of directed fire the initial conditions for $P(m, n, t)$ are

$$P(m, n, 0) = \begin{cases} 1 & \text{for } m = m_0 \text{ and } n = n_0 \\ 0 & \text{otherwise} \end{cases} \tag{5.38}$$

Likewise,

$$\begin{aligned} P(m, n, t + \Delta t) = {} & P(m, n, t)\, P(\text{no casualty occurs on both sides in } \Delta t) \\ & + P(m + 1, n, t)\, P(\text{one } m \text{ casualty in } \Delta t) \\ & + P(m, n + 1, t)\, P(\text{one } n \text{ casualty in } \Delta t) \\ & + P(\ldots, \ldots)\, P(\text{more than one casualty in } \Delta t) \end{aligned} \tag{5.39}$$

However, the component probabilities are different in this case, i.e.,

$$P(\text{no casualties on both sides in } \Delta t) = 1 - b\frac{nm}{m_0}\Delta t - a\frac{mn}{n_0}\Delta t \tag{5.64}$$

$$P(\text{one } m \text{ casualty in } \Delta t) = b\frac{nm}{m_0}\Delta t \tag{5.65}$$

$$P(\text{one } n \text{ casualty in } \Delta t) = a\frac{mn}{n_0}\Delta t \tag{5.66}$$

$$P(\text{more than one casualty in } \Delta t) = 0 \tag{5.67}$$

The substitution of Eqs. (5.64)–(5.67) into Eq. (5.39) leads to

$$\begin{aligned} P(m, n, t + \Delta t) = {} & P(m, n, t)\left(1 - b\frac{nm}{m_0}\Delta t - a\frac{mn}{n_0}\Delta t\right) \\ & + P(m + 1, n, t)\, b\frac{nm}{m_0}\Delta t + P(m, n + 1, t)\, a\frac{mn}{n_0}\Delta t \end{aligned} \tag{5.68}$$

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Here again the use of the nondimensional parameters Φ_0 and $\Delta\tau$ defined by Eqs. (5.45) and (5.46) is useful. Hence,

$$P(m,n,\tau+\Delta\tau) = P(m,n,\tau)\left(1-\Phi_0^{-1}\frac{mn}{n_0}\Delta\tau-\Phi_0\frac{mn}{m_0}\Delta\tau\right) + P(m+1,n,\tau)\Phi_0^{-1}\frac{mn}{n_0}\Delta\tau + P(m,n+1,\tau)\Phi_0\frac{mn}{m_0}\Delta\tau \tag{5.69}$$

Equation (5.69) can be rewritten as

$$\frac{P(m,n,\tau+\Delta\tau)-P(m,n,\tau)}{\Delta\tau} = -P(m,n,\tau)\left(\frac{\Phi_0^{-1}}{n_0}+\frac{\Phi_0}{m_0}\right)mn + P(m+1,n,\tau)\Phi_0^{-1}\frac{mn}{n_0} + P(m,n+1,\tau)\Phi_0\frac{mn}{m_0} \tag{5.70}$$

which in the limit becomes

$$\frac{dP(m,n,\tau)}{d\tau} = -P(m,n,\tau)\left(\frac{\Phi_0^{-1}}{n_0}+\frac{\Phi_0}{m_0}\right)mn + P(m+1,n,\tau)\Phi_0^{-1}\frac{mn}{n_0} + P(m,n+1,\tau)\Phi_0\frac{mn}{m_0} \tag{5.71}$$

The preceding equation is the Kolmogorov differential equation for the probability of combat states (m,n) for the area fire combat model.

The restrictions imposed on the size of $\Delta\tau$ are the same as those for the directed fire model, i.e., Eqs. (5.49) and (5.50). Equation (5.69) is valid only for the combat states other than those involving the boundary values m_0 and n_0 for which, as before, separate equations must be generated.

For $m = m_0$ and $n = n_0$,

$$P(m_0+1,n_0,\tau) = P(m_0,n_0+1,\tau) = 0$$

and

$$P(m_0,n_0,\tau+\Delta\tau) = P(m_0,n_0,\tau)\left(1-\Phi_0^{-1}m_0\Delta\tau-\Phi_0 n_0\Delta\tau\right) \tag{5.72}$$

For $0 < m < m_0$ and $n = n_0$,

$$P(m,n_0,\tau+\Delta\tau) = P(m,n_0,\tau)\left(1-\Phi_0^{-1}m\Delta\tau-\Phi_0\frac{mn_0}{m_0}\Delta\tau\right) + P(m+1,n_0,\tau)\Phi_0^{-1}(m+1)\Delta\tau \tag{5.73}$$

For $m = m_0$ and $0 < n < n_0$,

$$P(m_0,n,\tau+\Delta\tau) = P(m_0,n,\tau)\left(1-\Phi_0^{-1}\frac{m_0 n}{n_0}\Delta\tau-\Phi_0 n\Delta\tau\right) + P(m_0,n+1,\tau)\Phi_0(n+1)\Delta\tau \tag{5.74}$$

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On the boundaries $m = 0$ and $n = 0$ the fundamental Eq. (5.39) must be modified. For example, for $m = 0$ and $0 < n \leq n_0$,

$$P(0, n, \tau + \Delta\tau) = P(0, n, \tau)\, P(\text{no casualty occurs on both sides in time } \Delta\tau)$$
$$+ P(1, n, \tau)\, P(\text{one } m \text{ casualty in time } \Delta\tau)$$
$$+ P(0, n+1, \tau)\, P(\text{one } n \text{ casualty in time } \Delta\tau)$$

It is obvious that for $m = 0$ the probabilities P(no casualty occurs on both sides in time $\Delta\tau$) $= 1$ and P(one n casualty in time $\Delta\tau$) $= 0$. Hence,

$$P(0, n, \tau + \Delta\tau) = P(0, n, \tau) \times 1 + P(1, n, \tau)\Phi_0^{-1} \frac{n}{n_0} \Delta\tau \tag{5.75}$$

Similarly, it can be shown that, for $n = 0$ and $0 < m \leq m_0$,

$$P(m, 0, \tau + \Delta\tau) = P(m, 0, \tau) \times 1 + P(m, 1, \tau)\Phi_0 \frac{m}{m_0} \Delta\tau \tag{5.76}$$

Finally, for $m = 0$ and $n = 0$,

$$P(0, 0, \tau) = 0 \tag{5.77}$$

because it is impossible to reach the combat state (0, 0).

The regions of validity of the preceding equations are shown in Fig. 5.12. The solution to these equations for any given set of values of Φ_0, m_0, n_0, and $\Delta\tau$ can be obtained numerically. As in the previous case the solution progresses in a series

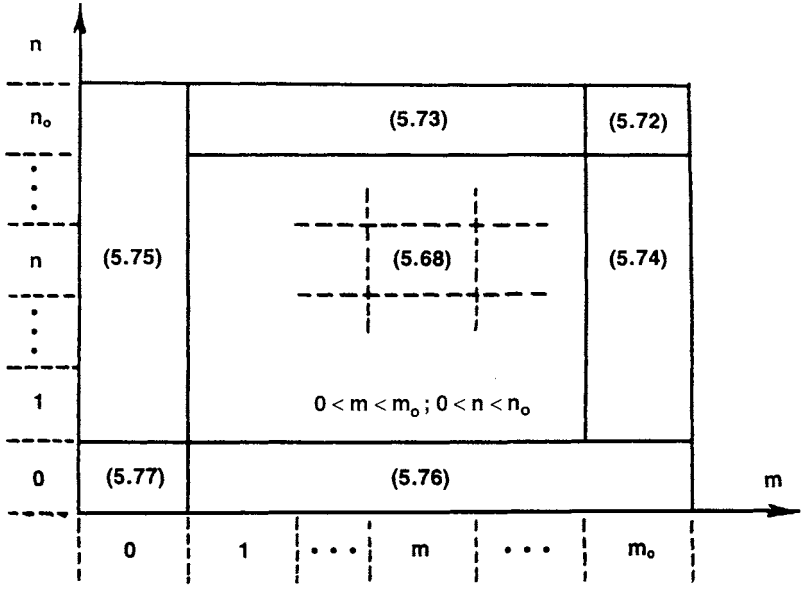


Fig. 5.12 Combat state space; regions of validity of probability equations (area fire model).

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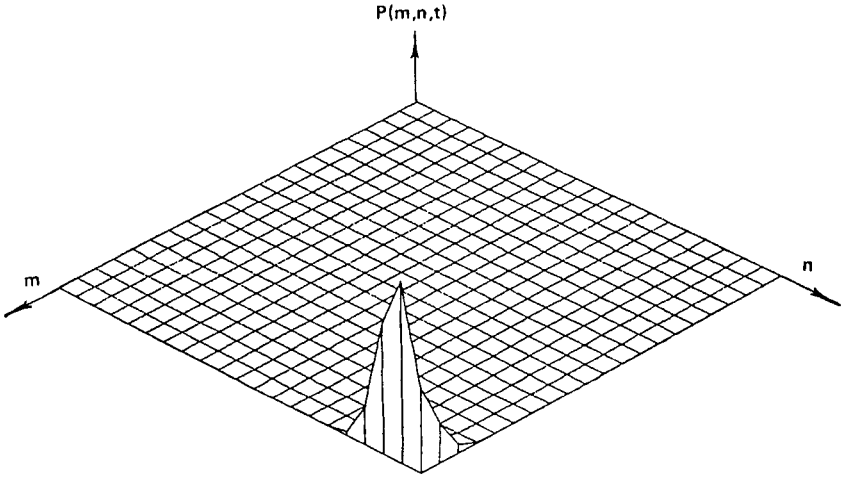


Fig. 5.13 Probability distribution for the area fire model for $\Phi_0 = 1.5$ and $\tau = 0.05$.

of steps $\Delta\tau$ with new values of P computed from the previous step. The results of such computations for $\Phi_0 = 1.5$, $m_0 = n_0 = 20$, and $\Delta\tau = 0.01$ are shown in Figs. 5.13–5.16 for $\tau = 0.05, 0.25, 0.5$, and 0.75. The differences between the two models—the directed and area fire models—are illustrated more effectively by topological plots of the probability distributions as shown in Figs. 5.17 and 5.18. These plots demonstrate clearly the slow, protracted nature of an area fire engagement. As it would be expected, area fire is not very effective compared with the directed fire engagement because the attrition coefficients in such cases are dependent on the ratios m/m_0 and n/n_0, which are decreasing as the engagement progresses.

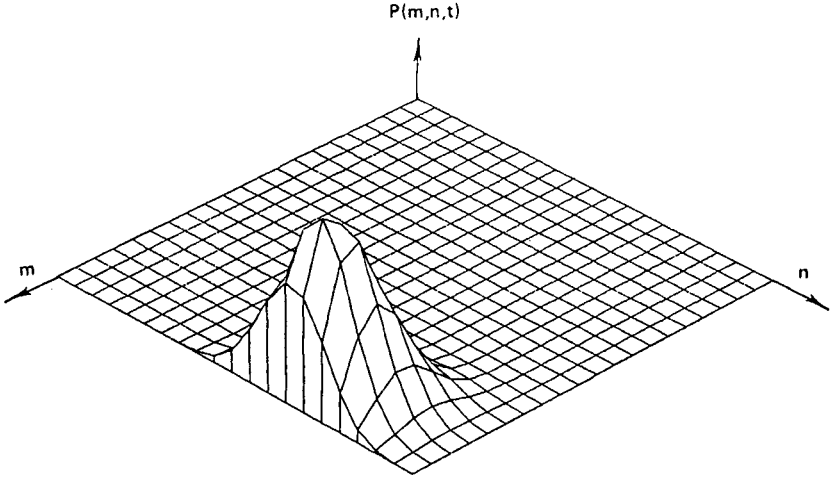


Fig. 5.14 Probability distribution for the area fire model for $\Phi_0 = 1.5$ and $\tau = 0.25$.

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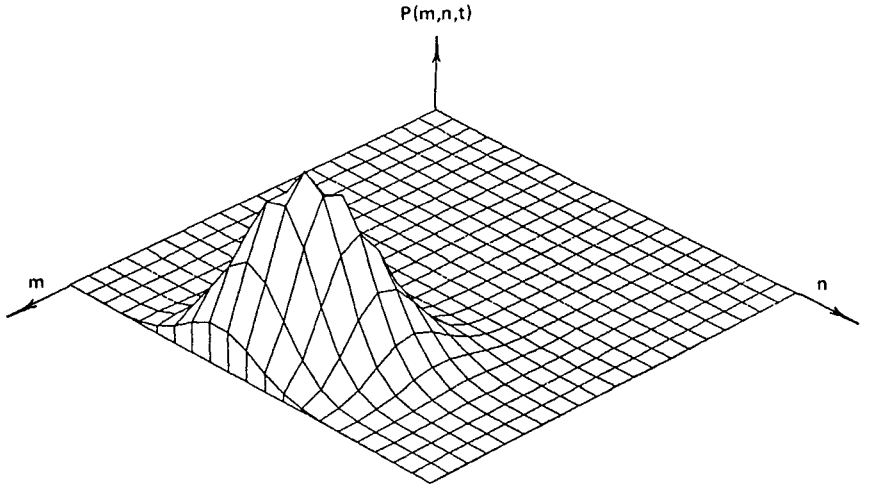


Fig. 5.15 Probability distribution for the area fire model for $\Phi_0 = 1.5$ and $\tau = 0.50$.

The expected values $E[m]$ and $E[n]$ can be obtained from Eqs. (5.58) and (5.59), with $P(m, n, t)$ denoting the probabilities for the area fire engagement. As before, Eq. (5.57) can also be used as a check on the accuracy of the solution. A comparison of the expected values $E[m]$ and $E[n]$ with the corresponding deterministic values $m(\tau)$ and $n(\tau)$ for an area fire model is shown in Table 5.4. Here, unlike in the case of the directed fire model, the probabilistic and deterministic solutions show excellent agreement because the two representations have a common feature. Both tend to 0 for large values of time.

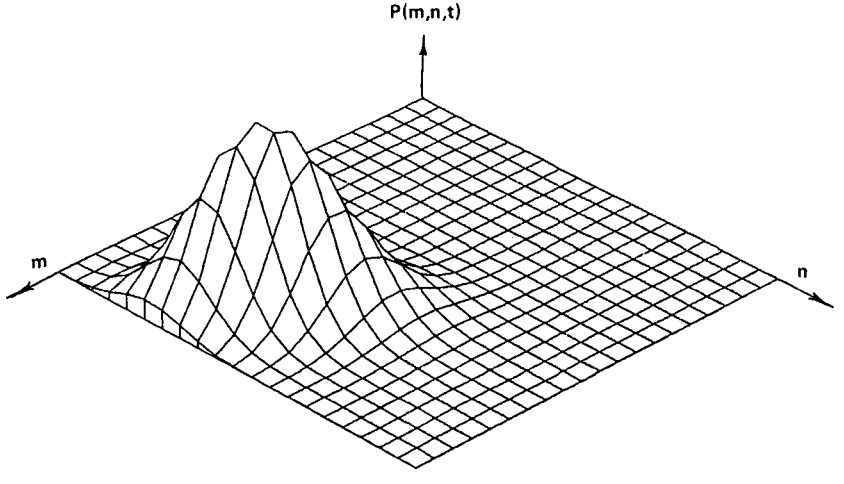


Fig. 5.16 Probability distribution for the area fire model for $\Phi_0 = 1.5$ and $\tau = 0.75$.

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Table 5.4 Comparison of the expected values $E[m]$ and $E[n]$ with the deterministic values $m(\tau)$ and $n(\tau)$ for $\Phi_0 = 1.5$, $m_0 = 20$, and $n_0 = 20$ (area fire model)

τ	$E[m]$	$m(\tau)$	$E[n]$	$n(\tau)$
0.00	20.00	20.00	20.00	20.00
0.05	19.36	19.37	18.56	18.58
0.25	17.37	17.38	14.08	14.12
0.50	15.70	15.72	10.33	10.36
0.75	14.58	14.58	7.81	7.80
1.00	13.79	13.77	6.02	5.98

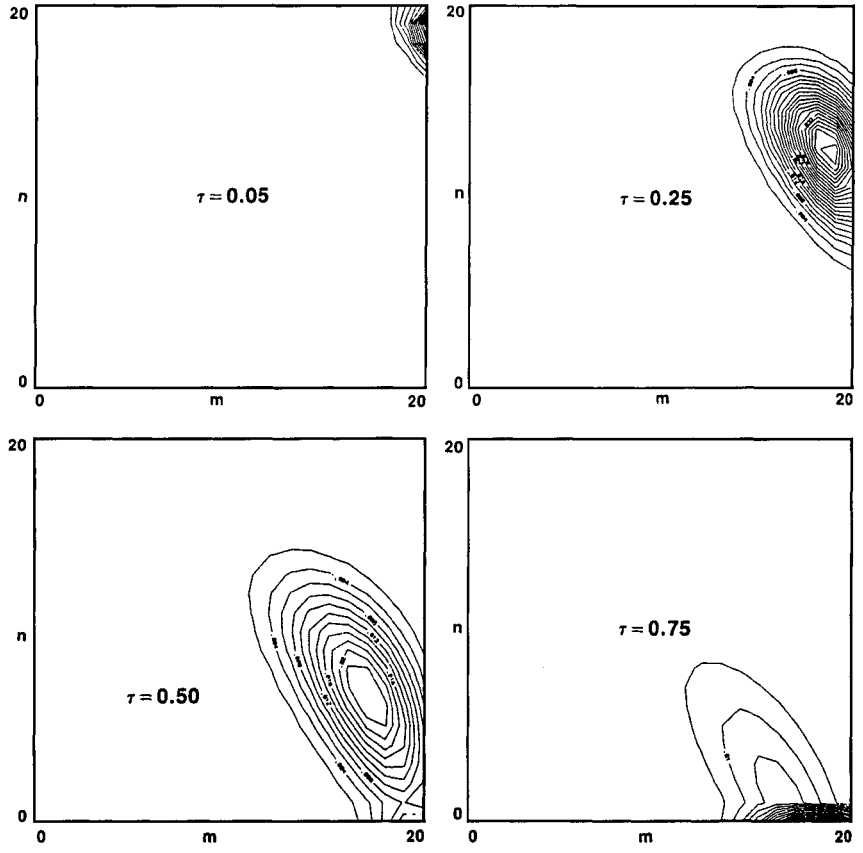


Fig. 5.17 Topographical representation of the probability distribution for the directed fire model ($\Phi_0 = 1.5$).

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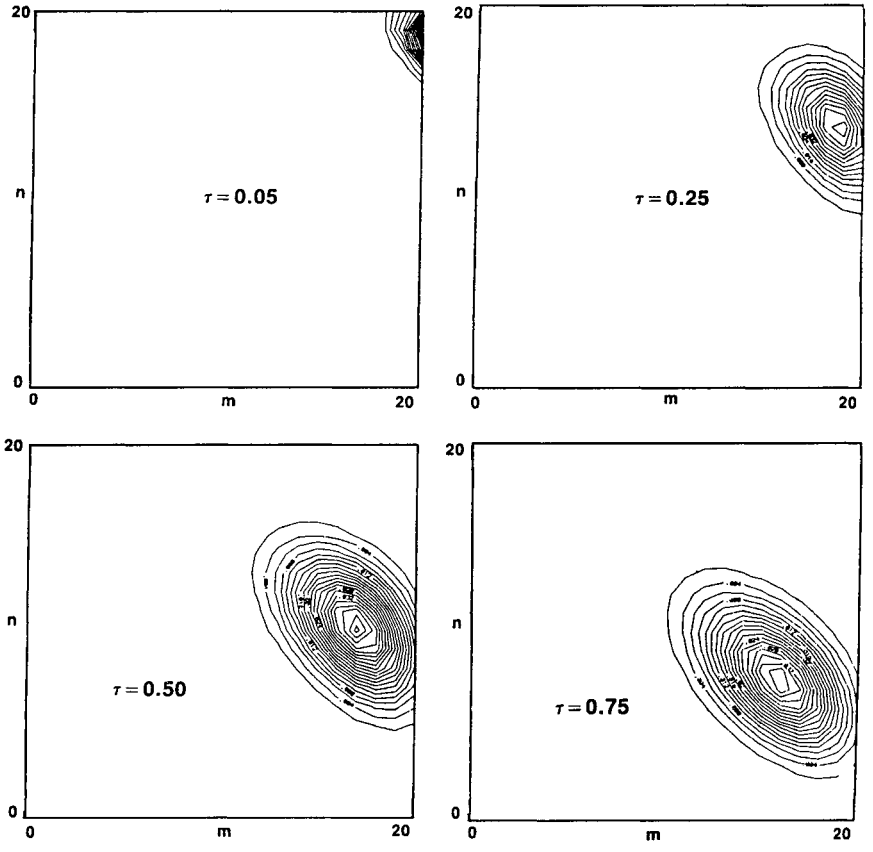


Fig. 5.18 Topographical representation of the probability distribution for the area fire model ($\Phi_0 = 1.5$).

5.5 Many-on-Many Engagements[2]

In Sec. 5.1 a sequential combat duel model was described. It involved two combatants engaging each other sequentially; therefore, the outcome of the duel depended only on the sequence of events. A similar situation arises when several attackers are engaging several targets. This is often described as the "many-on-many" engagement as opposed to the "one-on-one" engagement, wherein a single attacker engages a single target. The mathematical model for the probabilities of target survival or the probabilities of target kill can be developed for the case where the targets do not shoot back at the attackers; however, the roles of attackers and targets could be interchanged at some specified intervals, in terms of some specified numbers of shots, and the results can be suitably combined to produce a more general solution for an interactive engagement. In fact, the sequential combat duel in Sec. 5.1 is a special case of an interactive one-on-one engagement with only a single shot expended by the attacker, followed by a single shot expended by the defender (target), etc.

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The mathematical expressions developed for the many-on-many engagements will be derived from the known one-on-one effectiveness of the attack. Because several targets will be involved in the engagement, the method of allocating attackers to the various available targets will determine the overall effectiveness of the engagement. The allocation method used in combat is influenced by the situational awareness and the effectiveness of the available command, control, and communications (C^3). In this section, three basic types of allocation of targets will be considered: uniform allocation assignment, random allocation assignment, and shoot-look-shoot allocation assignment (perfect assignment). It will be demonstrated that the shoot-look-shoot assignment of targets is the most effective of the three types, but its implementation, in general, may not be easy because it requires a very efficient command and control and complete information about the status of destroyed targets.

Next, some basic relationships will be described for an engagement that consists of M attackers, each with J shots (missiles, bullets, laser intercepts, etc.) attacking a target. If the total number of shots expended is N, then

$$N = MJ \tag{5.78}$$

The probability of the ith shot destroying the target will be denoted by p_i, where $i = 1, 2, \ldots, N$. Hence, the probability that the target is not destroyed by the ith shot is $1 - p_i$, and the probability that the target survives the first k shots is

$$S(k) = (1 - p_1)(1 - p_2)\cdots(1 - p_k) = \prod_{i=1}^{k}(1 - p_i) \tag{5.79}$$

If all of the weapons used by the M attackers are identical, then $p_i = p$ for all i, and the probability of target survival for k shots is given by

$$S(k) = (1 - p)^k \tag{5.80}$$

For such cases, if the total number of targets is T, then the expected number of targets surviving the attack can be obtained from

$$E_S = TS(k) \tag{5.81}$$

Uniform Assignment of Targets

For the simplest case when all weapons are identical, one of the effective ways is to allocate the shots (or missiles) as uniformly as possible, as shown in Fig. 5.19a. If N is an integral multiple of T, then each target will be assigned exactly

$$L = \frac{N}{T} \tag{5.82}$$

shots, and the probability of survival of each target will be given by

$$S_I(L) = (1 - p)^L = (1 - p)^{N/T} \tag{5.83}$$

If, on the other hand, L is not an integer, then the ratio $L = N/T$ can be expressed as

$$L = I + F \tag{5.84}$$

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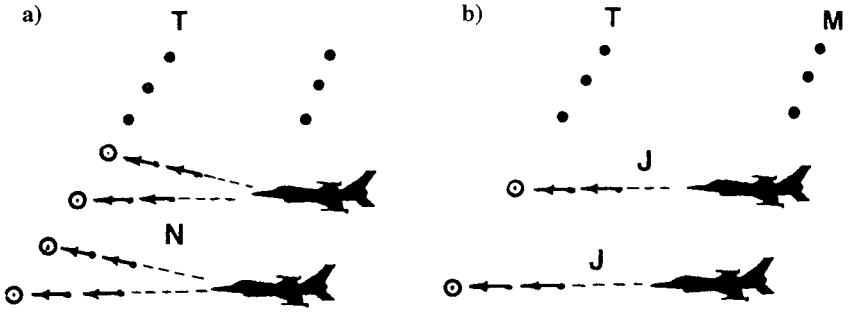


Fig. 5.19 Uniform assignment of targets: a) Uniform assignment of N shots and b) Uniform assignment of M attackers with J shots each.

where I denotes the largest integer smaller than L, and F denotes the remaining fractional part of the ratio L. In this case the optimal targeting is as follows: $I+1$ shots are allocated to FT targets and I shots to $(1-F)T$ targets. Hence *the average survival probability* S_U is given by

$$S_U(L) = FS_I(I+1) + (1-F)S_I(I) \tag{5.85}$$

where the subscript U is used to denote that this probability refers to the uniform assignment of targets.

Substituting Eq. (5.83) into Eq. (5.85),

$$S_U(L) = (1-p)^I(1-pF) \tag{5.86}$$

and the expected number of targets surviving the attack is given by

$$E_U = TS_U(L) \tag{5.87}$$

It should be noted that Eq. (5.86) is also valid for $I = 0$. In such cases $F = N/T$.

If N/T is an integer, then each target is attacked by the same number of shots (or missiles), and each target has the same probability of survival equal to $1-p$. For such cases the probability that exactly i targets out of T survive the attack is obtained from the binomial distribution [see Eq. (3.3)] as[3]

$$S_{ui} = \binom{T}{i}\left[(1-p)^{N/T}\right]^i\left[1-(1-p)^{N/T}\right]^{T-i} \tag{5.88}$$

If M attackers are assigned uniformly to T targets, i.e., $M = T$, and each attacker expends all of his J shots on the assigned target, as shown in Fig. 5.19b, then from Eq. (5.86),

$$S_{UJ}(L) = (1-p_J)^I(1-p_JF) \tag{5.89}$$

where

$$p_J = 1-(1-p)^J \tag{5.90}$$

is the probability of kill after J shots are expended on the target.

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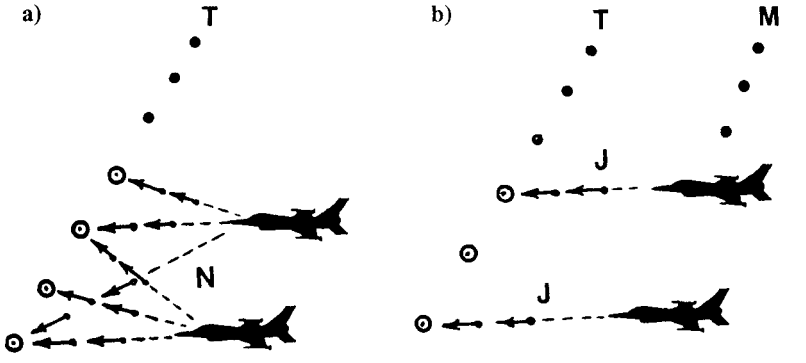


Fig. 5.20 Random assignment of targets: a) Random assignment of N shots and b) Random assignment of M attackers with J shots each.

Random Assignment of Targets

The next assignment of targets to be considered is completely random. Each shot, out of the total of N, is assigned to a target picked at random out of the total of T targets, without regard to how many other shots are assigned to that target, or whether the target had been killed by other shots (see Fig. 5.20a). For such cases the probability that a selected attacker is assigned to and destroys a particular target is p/T, which is the product of the probability of selecting a particular target $1/T$ and the probability of kill p when that particular target is actually attacked. Hence, the probability that a selected target survives an attack from the selected attacker is $1 - p/T$. Finally, to survive the battle the selected target must survive attacks from the N independent shots. Thus, the probability of survival of the arbitrarily selected target is given by

$$S_R = \left(1 - \frac{p}{T}\right)^N \tag{5.91}$$

where the subscript R is used to denote that this probability refers to the random assignment of targets. The expected number of surviving targets, E_R, is then obtained from

$$E_R = TS_R \tag{5.92}$$

Because p/T is usually small compared to unity, Eq. (5.91) can be approximated with an exponential expression. Comparing the binomial expansion of S_R,

$$S_R = 1 - \frac{N}{1!}\frac{p}{T} + \frac{N(N-1)}{2!}\left(\frac{p}{T}\right)^2 - \cdots + \left(\frac{p}{T}\right)^N \tag{5.93}$$

with

$$\exp\left(-\frac{pN}{T}\right) = 1 - \frac{Np}{T} + \frac{1}{2!}\left(\frac{Np}{T}\right)^2 - \cdots \tag{5.94}$$

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it is clear that

$$S_R \cong S_R^* = \exp\left(-\frac{pN}{T}\right) \tag{5.95}$$

can be used as an approximation, where the asterisk denotes the approximate value. The advantage of using Eq. (5.95) is that it contains only one variable N/T instead of two variables N and T appearing in Eq. (5.91). The approximate Eq. (5.95) is reasonably accurate for large values of N and T. For example, when $2 < N/T < 4$ and $T > 30$, the error in using Eq. (5.95) is less than 0.0044, diminishing even further for larger values of T. This means that for computations involving smaller values of N and T, the exact Eq. (5.91) should be used.

An analogous expression to Eq. (5.88) can also be derived for the random assignment of N shots to T targets. It was shown in Ref. 3 that the probability S_{ri} that exactly i of the T targets survive the attack from N shots is given by

$$S_{ri} = \binom{T}{i}\sum_{j=0}^{T-i}(-1)^j\binom{T-i}{j}\left[1-(i+j)\frac{p}{T}\right]^N; \qquad i = 0, 1, \ldots, T \tag{5.96}$$

The assignment case discussed thus far was for allocating the N shots randomly and independently (see Fig. 5.20a). A more restrictive random assignment is when M attackers are allocated randomly and independently for the T targets, but, once allocated, an attacker expends all of its J shots on one target (see Fig. 5.20b). Applying the results of Eq. (5.91), the probability of survival S_J is given by

$$S_J = \left(1 - \frac{p_J}{T}\right)^M \tag{5.97}$$

where

$$p_J = 1 - (1 - p)^J \tag{5.98}$$

is the probability of kill for a single attack with J shots.

Shoot-Look-Shoot Assignment of Targets

The shoot-look-shoot assignment tactics for the attackers is based on the assumption that the command and control for the battle management has the capability to assess the consequences of each shot, i.e., whether the target was missed or destroyed, and to assign successive shots only to the surviving targets. This means that the kill assessment must be perfect. A typical engagement for the shoot-look-shoot tactics is illustrated in Fig. 5.21. As before, there are T targets and N shots available, each with the kill probability p, but the number of attackers is not specified, which means that the N shots may be distributed among the attackers in an arbitrary manner.

Using again the general results of Eq. (3.3), The probability P_i of killing exactly i targets, out of the total N tries, is given by

$$P_i = \binom{N}{i}p^i(1-p)^{N-i}, \qquad N \geq T \tag{5.99}$$

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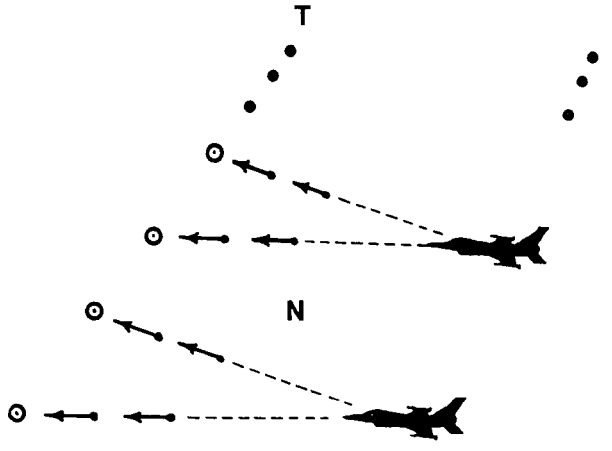


Fig. 5.21 Shoot-look-shoot assignments of attackers.

The expected number of targets destroyed is then determined from

$$T_{\mathrm{SLS}} = \sum_{i=0}^{T} i P_i + \sum_{i=T+1}^{N} T P_i, \qquad N \geq T \tag{5.100}$$

where the subscript SLS refers to the shoot-look-shoot tactics. It should also be noted that for $i \leq T$, i targets are destroyed, whereas for $i > T$, exactly T targets are destroyed. Hence, the expected number of surviving targets is

$$E_{\mathrm{SLS}} = T - T_{\mathrm{SLS}} = T S_{\mathrm{SLS}} \tag{5.101}$$

where the probability of survival is given by

$$S_{\mathrm{SLS}} = \frac{T - T_{\mathrm{SLS}}}{T} = 1 - \sum_{i=0}^{T} \frac{i}{T} P_i - \sum_{i=T+1}^{N} P_i \tag{5.102}$$

Noting that

$$1 = \sum_{i=0}^{N} P_i = \sum_{i=0}^{T} P_i + \sum_{i=T+1}^{N} P_i \tag{5.103}$$

Equation (5.102) can be simplified to

$$S_{\mathrm{SLS}} = \sum_{i=0}^{T} \left(1 - \frac{i}{T}\right) P_i \qquad \text{for} \quad N \geq T \tag{5.104}$$

When $N < T$, the corresponding equation for S_{SLS} is given by

$$S_{\mathrm{SLS}} = 1 - \frac{pN}{T} \tag{5.105}$$

The probabilities of target survival for the random, uniform, and shoot-look-shoot target assignment tactics are shown in Tables 5.5 and 5.6 for $p = 0.5$ and 0.8, respectively. The shoot-look-shoot assignment tactic is clearly superior for the

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Table 5.5 Probabilities of target survival for different target assignment tactics ($p = 0.5$)

Assignment tactics		$T^a = 1$	$T = 2$	$T = 3$	$T = 4$	$T = 5$
	Random	0.5000	0.7500	0.8333	0.8750	0.9000
$N^b = 1$	Uniform	0.5000	0.7500	0.8333	0.8750	0.9000
	Shoot-look-shoot	0.5000	0.7500	0.8333	0.8750	0.9000
	Random	0.2500	0.5625	0.6944	0.7656	0.8100
$N = 2$	Uniform	0.2500	0.5000	0.6667	0.7500	0.8000
	Shoot-look-shoot	0.2500	0.5000	0.6667	0.7500	0.8000
	Random	0.1250	0.4219	0.5787	0.6699	0.7290
$N = 3$	Uniform	0.1250	0.3750	0.5000	0.6250	0.7000
	Shoot-look-shoot	0.1250	0.3125	0.5000	0.6250	0.7000
	Random	0.0625	0.3164	0.4823	0.5862	0.6561
$N = 4$	Uniform	0.0625	0.2500	0.4167	0.5000	0.6000
	Shoot-look-shoot	0.0625	0.1875	0.3542	0.5000	0.6000
	Random	0.0313	0.2373	0.4019	0.5129	0.5905
$N = 5$	Uniform	0.0313	0.1875	0.3333	0.4375	0.5000
	Shoot-look-shoot	0.0313	0.1094	0.2396	0.3828	0.5000

[a] T = number of targets
[b] N = number of shots

Table 5.6 Probabilities of target survival for different target assignment tactics ($p = 0.8$)

Assignment tactics		$T^a = 1$	$T = 2$	$T = 3$	$T = 4$	$T = 5$
	Random	0.2000	0.6000	0.7333	0.8000	0.8400
$N^b = 1$	Uniform	0.2000	0.6000	0.7333	0.8000	0.8400
	Shoot-look-shoot	0.2000	0.6000	0.7333	0.8000	0.8400
	Random	0.0400	0.3600	0.5378	0.6400	0.7056
$N = 2$	Uniform	0.0400	0.2000	0.4667	0.6000	0.6800
	Shoot-look-shoot	0.0400	0.2000	0.4667	0.6000	0.6800
	Random	0.0080	0.2160	0.3944	0.5120	0.5927
$N = 3$	Uniform	0.0080	0.1200	0.2000	0.4000	0.5200
	Shoot-look-shoot	0.0080	0.0560	0.2000	0.4000	0.5200
	Random	0.0016	0.1296	0.2892	0.4096	0.4979
$N = 4$	Uniform	0.0016	0.0400	0.1467	0.2000	0.3600
	Shoot-look-shoot	0.0016	0.0144	0.0699	0.2000	0.3600
	Random	0.0003	0.0778	0.2121	0.3277	0.4182
$N = 5$	Uniform	0.0003	0.0240	0.0933	0.1600	0.2000
	Shoot-look-shoot	0.0003	0.0035	0.0217	0.0819	0.2000

[a] T = number of targets
[b] N = number of shots

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attacker because it leads to the lowest probability of survival of the targets under attack, whereas the random assignment is the least effective. As could be expected, for $T = 1$ (i.e., a single target), the three tactics become identical and lead to the same result.

References

[1]Taylor, J. G., *Lanchester Models of Warfare*, Vol. I, Operations Research Society of America, Arlington, VA, March 1983, pp. 397–409.

[2]Bexfield, J. N., "Lecture Notes on Many-on-Many Engagement Formulas," Air Force Inst. of Technology, Wright–Patterson AFB, OH, 1984.

[3]Thomas, C. J., "The Estimation of Bombers Surviving an Air Battle," The Assistant for Operations Analysis, Deputy Chief of Staff, Operations, Headquarters, U.S. Air Force, TM-49, Aug. 1956.

Problems

5.1. Verify that for the sequential combat duel

$$P_B(n) + P_R(n) = 1 \quad \text{and} \quad P_B^*(n) + P_R^*(n) = 1$$

when $n = \infty$.

5.2. Write a computer program for the computation of the state probabilities $P(m, n, t)$ for the directed fire one-on-one model. Check your results, at least qualitatively, with Figs. 5.8–5.11 for $\Phi_0 = 1.5$, $m = n = 20$, and $\Delta\tau = 0.01$. Compute the expected values $E[m]$ and $E[n]$ and compare them with the deterministic results from Chapter 4.

5.3. Discuss the formulation of directed fire model for two Red forces and one Blue force. Show diagrammatically the possible combat states and derive the fundamental equations for the probabilities of being in a particular state.

5.4. Show that if $P(m, n, t)$ represents the probability distribution of the states (m, n) for a directed fire combat engagement of two forces, then

$$\Sigma = \sum_{m=0}^{m_0} \sum_{n=0}^{n_0} P(m, n, t) = 1$$

Hint: It suffices to show that $d\Sigma/dt = 0$.

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6
Strategic Defense

This chapter describes the analysis of the overall effectiveness of a layered defense against ballistic missiles, as proposed originally for the Strategic Defense Initiative (SDI), including a layered defense against MIRVed attack (where MIRVs are multiple independently targeted re-entry vehicles). This concept is also used for the more recent plans for the National Missile Defense (NMD) as well as for the Theater Missile Defense (TMD), described in greater detail in Chapter 7. In addition, the terminal defense of intercontinental ballistic missile (ICBM) silos is considered for different defensive and offensive strategies through the application of game theory. The solution presented is the optimum for both the defense and offense; i.e., the defense will guarantee that at least the optimum number of silos will survive, whereas the offense cannot expect to produce a strategy that will reduce that number. That is, neither side can do any better. Next, the solution to the problem of determining optimal penetration routes for attacking bombers flying through air defenses is presented, which includes a new concept of a special threat function. Also, the surface-to-air-missile (SAM) defense against a penetrating aircraft flying along a straight path is analyzed. For a simplified analysis of the strength of nuclear strategic forces, two sufficiency models are presented: the equivalent megatonnage (EMT) model and the counter military potential (CMP) model.

6.1 Strategic Defense Initiative: Layered Defense

On 23 March 1984, President Ronald Reagan announced his famous Strategic Defense Initiative (SDI) program, dubbed in the press as the "Star Wars," and challenged American scientists to replace the concept of defense based on retaliation, popularly known as the Mutually Assured Destruction (MAD) concept, with real defense against intercontinental ballistic missiles. The underlying principle of SDI was based on multi-layered defense in which the attacking missiles could be intercepted while they passed through several defense layers (Fig. 6.1). Because of the superior effectiveness of the layered defense and the recent progress in a number of new technologies that made the layered defense plausible, the SDI concept offered a new opportunity to render offensive weapons ineffective, or at least reduce their effectiveness. This new strategic defense could also contribute to the nuclear deterrent even if only a portion of nuclear attack could be eliminated. It is interesting to note that there are some reasons to believe that President Reagan, in making his decision, was influenced by a draft circular letter issued by the American Conference of Catholic Bishops criticizing the then-current military doctrine of "Mutually Assured Destruction," which the bishops claimed to be immoral. Another reason was probably the strong evolving movement to "freeze"

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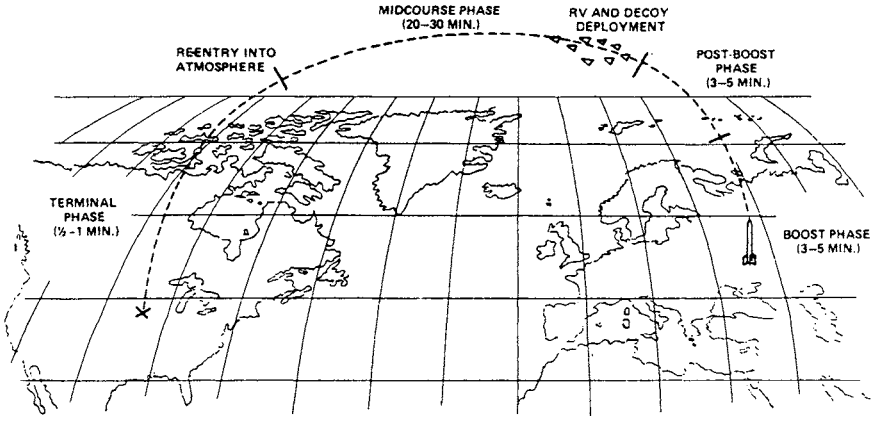


Fig. 6.1 Defense intercept phases for the Strategic Defense Initiative (SDI) proposed in 1984.

the development and testing of all nuclear weapons, and one way of counteracting it was to start some serious work on defensive systems.[1]

Prior to President Reagan's Strategic Defense Initiative, the U.S. Army was developing the Nike X-Sentinel-Safeguard* proposed in 1969. This system was a two-layer defense. The system used two anti-ballistic missiles, the Spartan and the Sprint, designed to intercept incoming warheads. The Spartan missile was a large long-range, liquid-fuel missile that would intercept warheads above the atmosphere, while the Sprint was a smaller, very high energy solid-fuel missile designed to intercept warheads at altitudes of about 100,000 feet after they had entered the atmosphere. Both of these defensive missiles carried nuclear warheads that would destroy any incoming ballistic missiles. The Spartan missile used warheads with a yield in excess of five megatons while the smaller Sprint used warheads with yields of hundreds of kilotons. The architecture of the Nike X-Sentinel-Safeguard system is shown in Fig. 6.2.

SDI research and development work was continued in the 1980s under the direction of the Space Defense Initiative Office (SDIO). Early in 1990s, the Bush Administration modified the original SDI concept to focus its direction on the Global Protection Against Limited Strikes (GPALS). The new concept reflected recognition of the vanishing strategic offensive threat from the countries of the former Soviet Union and growing concern about accidental or unauthorized launch and Third World attack by ballistic missiles, conceivably armed with warheads of mass destruction. The strategic objective of GPALS was to provide protection from such limited attacks of up to 200 reentry vehicles. Like the SDI system, the GPALS system was a layered defense containing both space- and ground-based assets. It was based on an integrated deployment of space-based interceptors ("Brilliant Pebbles"), advanced space-based sensors ("Brilliant Eyes"), ground-based battle management, and theater missile defense systems as shown in Fig. 6.3.

*Nike X was named "Sentinel" in 1967 and renamed "Safeguard" in 1969.

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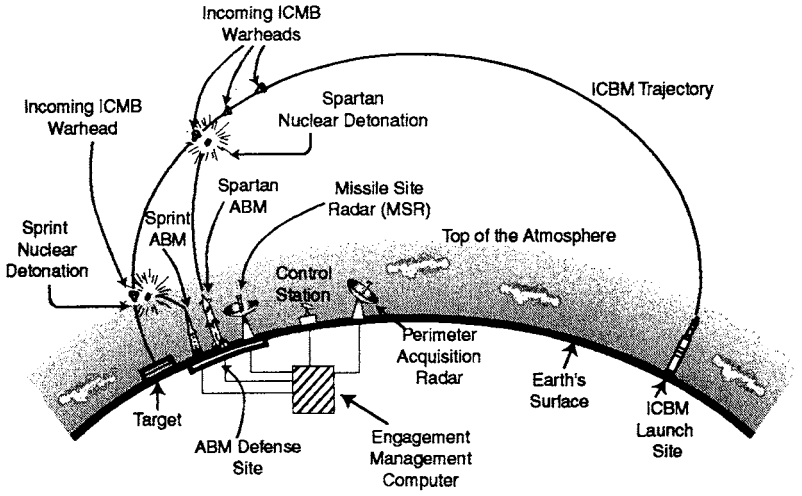


Fig. 6.2 Architecture of the Safeguard System as Proposed in 1969 (Based on Ref. 1 with permission of AIAA).

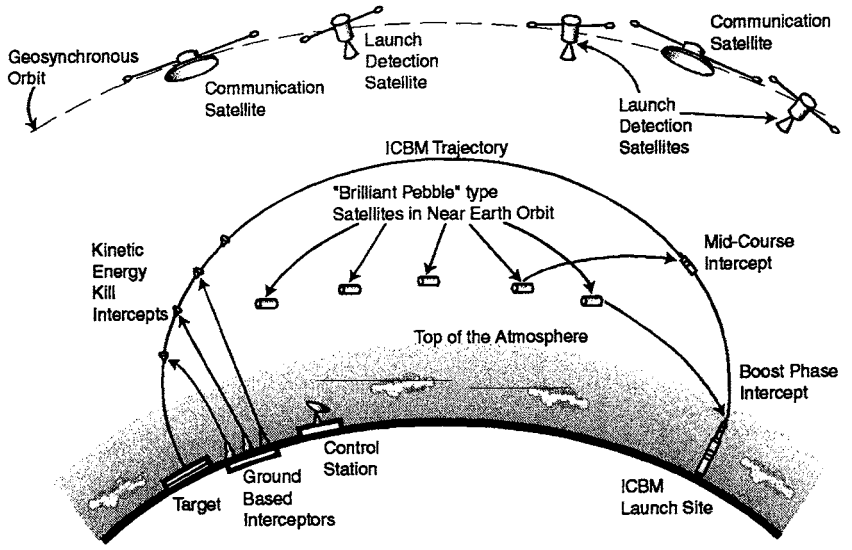


Fig. 6.3 Antiballistic Missile System as Contemplated in 1969 (Based on Ref. 1 with permission of AIAA).

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The present emphasis by the Clinton administration is primarily on the deployment of ground-based systems, such as the Patriot missile, to protect against mid-range missiles, and the deployment of space-based systems to counter intercontinental missiles is a second priority.

This section analyzes the concept of layered defense. It is assumed here that the defense has the capability of intercepting incoming missiles in an n-layered defense in which interceptors can be fired in an "unordered" manner; i.e., each deployed interceptor in a layer is fired randomly at one of the attacking missiles that has penetrated the previous layers of defense. Thus, the assumption of an unordered firing is somewhat similar to the area firing in the Lanchester combat model discussed in Sec. 4.4. Clearly the unordered firing can be improved through the use of programmed firing, but it is used here to analyze the worst case for the defense. The analysis of ordered firing for theater missile defense, such as uniform assignment of targets, is discussed in Chapter 7.

Some interceptors will be capable of multiple firings; therefore, each firing will be counted as a single intercept. It can be demonstrated that for a given number of layers there is a unique optimal way to proportion the intercept capability among the layers to minimize the number of missiles penetrating the n-layer defense.[2]

The offensive potential will be measured by the number of attacking missiles M_0; the number of surviving missiles after passing through the ith layer will be denoted by M_i. The defensive potential, on the other hand, will be measured by the number of effective intercepts obtained as the sum over all intercepts of the product of reliability and the single-shot probability of kill of an individual missile. That is, for the intercepts in the ith layer, their effective number D_i is computed from

$$D_i = \sum_r r_{ir} p_{ir} \tag{6.1}$$

where the summation is performed over all intercepts in the ith layer, r_{ir} is the reliability of the rth intercept, and p_{ir} is the single-shot kill probability of the rth intercept. The allocation of the defense intercepts is illustrated in Fig. 6.4 for

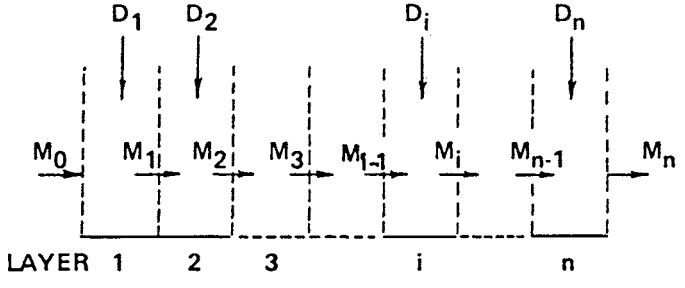


Fig. 6.4 n-Layer defense.

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n-layer defense, and the total defensive potential is obtained from

$$D_0 = \sum_{i=1}^{n} D_i \tag{6.2}$$

where n is the number of layers.

If an individual intercept has the single-shot kill probability p, then the probability of a missile surviving the fire of a given 100%-reliable intercept employing unordered firing is

$$P_0 = 1 - \frac{p}{M} \tag{6.3}$$

where M is the number of attacking missiles. (It should be noted that p/M is the product of two probabilities p and $1/M$.) Consequently, the compound probability of the same missile surviving rI intercepts is

$$P_S = \left(1 - \frac{p}{M}\right)^{rI} \tag{6.4}$$

where r is the reliability of a single intercept, and I is the actual number of intercepts employed.

Expanding Eq. (6.4),

$$\left(1 - \frac{p}{M}\right)^{rI} = 1 - \frac{rpI}{M} + O\left(\frac{p}{M}\right)^2 - \cdots$$

and comparing it with

$$\exp\left(-\frac{rpI}{M}\right) = 1 - \frac{rpI}{M} + O\left(\frac{p}{M}\right)^2 - \cdots$$

it follows that, for $p/M \ll 1$, which is the case of a large number of attacking missiles,

$$\left(1 - \frac{p}{M}\right)^{rI} = \exp\left(-\frac{rpI}{M}\right) = \exp\left(-\frac{D}{M}\right) \tag{6.5}$$

where

$$D = rpI \tag{6.6}$$

Hence, from Eqs. (6.5) and (6.4)

$$P_S = \exp\left(-\frac{D}{M}\right) \tag{6.7}$$

which, incidentally, is the same as the probability of survival for the Poisson distribution with $\lambda = D/M$ and $r = 0$ in Eq. (3.26).

Employing Eq. (6.7) the expected number of attacking missiles penetrating the first j layers of the defense is obtained from

$$M_j(D_1, D_2, \ldots, D_j) = M_{j-1} \exp\left(-\frac{D_j}{M_{j-1}}\right) \tag{6.8}$$

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in which M_0 is defined as the total number of attacking missiles entering the first layer. Ideally, a rigorous analysis would require that the complete distribution of the random variables M_j be evaluated from which an expected value of the M_j can be computed. It has been shown, however, that the stepwise expectation value model based on Eq. (6.8) is only slightly inaccurate.[2]

For an optimum defense, the problem is to minimize the function

$$M_n(D_1, D_2, \ldots, D_n)$$

subject to the constraint

$$g(D_1, D_2, \ldots, D_n) = D_1 + D_2 + \cdots + D_n - D_0 = 0 \tag{6.9}$$

The simplest method for solving this problem is to employ the Lagrange multiplier technique,[3] which requires that, at the optimum allocation of intercepts,

$$\nabla M_n = \lambda \nabla g \tag{6.10}$$

where λ is the Lagrange multiplier. Equation (6.10) implies that

$$\begin{aligned}\left[\frac{\partial M_n}{\partial D_1} \quad \frac{\partial M_n}{\partial D_2} \quad \cdots \quad \frac{\partial M_n}{\partial D_n}\right] &= \lambda \left[\frac{\partial g}{\partial D_1} \quad \frac{\partial g}{\partial D_2} \quad \cdots \quad \frac{\partial g}{\partial D_n}\right] \\ &= \lambda [1 \quad 1 \quad \cdots \quad 1]\end{aligned} \tag{6.11}$$

which in turn leads to

$$\frac{\partial M_n}{\partial D_i} = \frac{\partial M_n}{\partial D_j} \tag{6.12}$$

for all i and j less than or equal to n. Also, from the multivariable chain rule,

$$\begin{aligned}\frac{\partial M_n}{\partial D_i} &= \left(\prod_{k=i+1}^{n} \frac{\partial M_k}{\partial M_{k-1}}\right) \frac{\partial M_i}{\partial D_i} \\ &= \frac{\partial M_{i+1}}{\partial M_i} \frac{\partial M_{i+2}}{\partial M_{i+1}} \cdots \frac{\partial M_n}{\partial M_{n-1}} \frac{\partial M_i}{\partial D_i}\end{aligned} \tag{6.13}$$

and, similarly,

$$\frac{\partial M_n}{\partial D_{i-1}} = \frac{\partial M_i}{\partial M_{i-1}} \frac{\partial M_{i+1}}{\partial M_i} \cdots \frac{\partial M_n}{\partial M_{n-1}} \frac{\partial M_{i-1}}{\partial D_{i-1}} \tag{6.14}$$

Hence, substituting Eqs. (6.13) and (6.14) into Eq. (6.12),

$$\frac{\partial M_i}{\partial D_i} = \frac{\partial M_i}{\partial M_{i-1}} \frac{\partial M_{i-1}}{\partial D_{i-1}} \tag{6.15}$$

which, upon further substitution of Eq. (6.8), leads to

$$1 = \left(1 + \frac{D_i}{M_{i-1}}\right) \exp\left(-\frac{D_{i-1}}{M_{i-2}}\right) \tag{6.16}$$

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Finally, from Eqs. (6.8) and (6.16),

$$D_i = M_{i-2} - M_{i-1} \tag{6.17}$$

which states that, in order to maximize the effectiveness of the defense, the number of intercepts deployed in the ith layer should be equal the expected number of missiles destroyed by the $(i-1)$th-layer defense.

The application of this concept will now be illustrated for the case of a four-layer defense. From Eqs. (6.8), (6.17), and (6.2) the following eight nonlinear equations are obtained:

$$M_1 = M_0 \exp\left(\frac{-D_1}{M_0}\right) \qquad M_2 = M_1 \exp\left(\frac{-D_2}{M_1}\right)$$

$$M_3 = M_2 \exp\left(\frac{-D_3}{M_2}\right) \qquad M_4 = M_3 \exp\left(\frac{-D_4}{M_3}\right)$$

$$D_2 = M_0 - M_1 \qquad D_3 = M_1 - M_2$$

$$D_4 = M_2 - M_3 \qquad D_0 = D_1 + D_2 + D_3 + D_4$$

The eight independent variables are $M_1, \ldots, M_4$ and $D_1, \ldots, D_4$, which can be solved through an iterative process for a given pair of values of the number of attacking missiles M_0 and defensive intercepts D_0. Alternatively, if an admissible value of D_1 is assumed (i.e., one for which a solution exists), then $M_1, \ldots, M_4$, D_2, D_3, D_4, and D_0 can be computed directly in terms of M_0. To get an optimal allocation for a larger (or smaller) value of D_0, the process can be repeated with a larger (or smaller) value of D_1.

The allocation equations can be solved in terms of the nondimensional parameter D_0/M_0 representing a ratio of the defense/offense potential. This is illustrated in Fig. 6.5 for a two-, three-, and four-layer defense. The most important parameter, however, is the number of surviving missiles M_n, where n is the number of layers. This is shown in Fig. 6.6 as a percentage of M_n/M_0 plotted against the defense/offense ratio D_0/M_0. The dramatic decrease in the number of surviving missiles as the D_0/M_0 is increased, particularly for the cases where more than two defensive layers are employed, vividly demonstrates the superior effectiveness of the layered defense against ballistic intercontinental missiles.

The range of values of $D_0/M_0 < 1$ is of no practical interest because of the large percentage of attacking missiles penetrating the layered defense. The limiting case when $D_0 = 0$ allows, of course, 100% of missiles to penetrate the (nonexistent) defenses.

6.2 Layered Defense Against MIRVed Attack

After the boost phase the attacking missiles may deploy several independently targeted warheads (or the so-called MIRVs) and various decoys that, in the absence of reliable discrimination, may also have to be attacked by the defensive intercepts. For such cases, the analysis of the previous section must be modified. To accomplish this it will be assumed that each of the surviving missiles M_i, having passed through the first i layers, deploys α_i warheads (and decoys) as shown in

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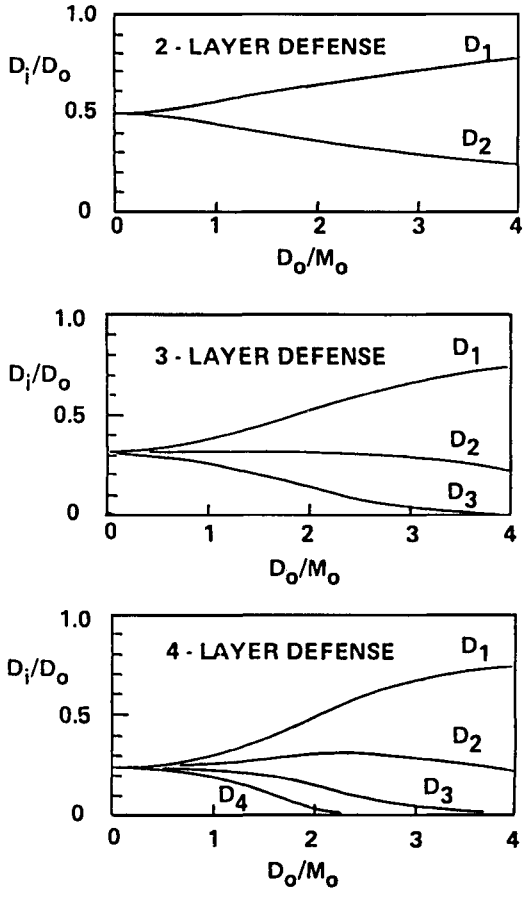


Fig. 6.5 Allocation of intercepts in two-, three-, and four-layer defense.

Fig. 6.7. When no additional warheads (and decoys) are deployed, then $\alpha_i = 1$. Therefore, Eq. (6.8) must be modified as

$$M_i(D_1, D_2, \ldots, D_i) = \alpha_{i-1} M_{i-1} \exp\left(-\frac{D_i}{\alpha_{i-1} M_{i-1}}\right) \tag{6.18}$$

The optimization process through the use of the Lagrange multiplier method is also applicable to the MIRVed deployment, and Eq. (6.15) is valid for these cases as well. Hence, the application of Eq. (6.18) in (6.15) leads to

$$1 = \alpha_{i-1}\left(1 + \frac{D_i}{\alpha_{i-1} M_{i-1}}\right)\frac{M_{i-1}}{\alpha_{i-2} M_{i-2}} \tag{6.19}$$

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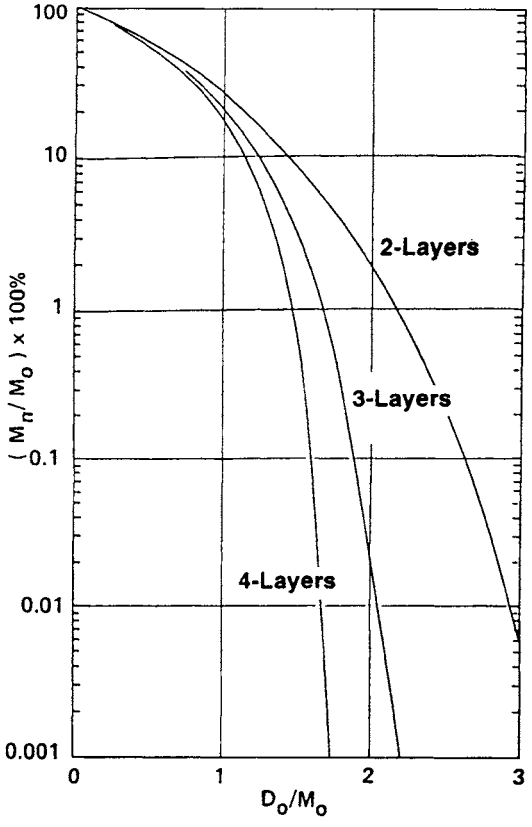


Fig. 6.6 Penetration of attacking missiles in two-, three-, and four-layer defense.

which results in a recursive relationship

$$D_i = \alpha_{i-2} M_{i-2} - \alpha_{i-1} M_{i-1} \tag{6.20}$$

This relationship states that for an optimal defense the number of intercepts deployed in the ith layer should be equal to the difference in the number of warheads (and decoys) entering the $(i-2)$th and $(i-1)$th layers. It should be noted that D_i for this case is not equal to the expected number of warheads destroyed in the $(i-1)$th layer. Also, each D_i must satisfy the condition

$$D_i \geq 0 \tag{6.21}$$

For comparison with the non-MIRVed case, the application of this analysis will be illustrated for the four-layer defense, in which the attacking missiles are MIRVed after passing through the first two defense layers. The number of warheads per missile will be denoted by α, and the resulting eight nonlinear equations are

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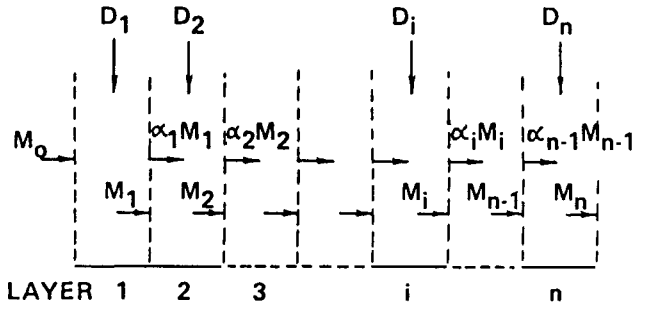


Fig. 6.7 ***n*-Layer defense against MIRVed attack.**

as follows:

$$M_1 = M_0 \exp\left(-\frac{D_1}{M_0}\right) \qquad M_2 = M_1 \exp\left(-\frac{D_2}{M_1}\right)$$

$$M_3 = \alpha M_2 \exp\left(-\frac{D_3}{\alpha M_2}\right) \qquad M_4 = M_3 \exp\left(-\frac{D_4}{M_3}\right)$$

$$D_2 = M_0 - M_1 \qquad D_3 = M_1 - \alpha M_2$$

$$D_4 = \alpha M_2 - M_3 \qquad D_0 = D_1 + D_2 + D_3 + D_4$$

The allocation of intercepts $D_1, \ldots, D_4$ for the four-layer defense for the cases of $\alpha = 1$ and 10 (i.e., no MIRVing and 10 MIRVs) is shown in Figs. 6.5 and 6.8. It is interesting to note the difference between the two cases. For $\alpha = 1$ the defensive intercepts are distributed among the four layers, whereas for $\alpha = 10$ the defense must allocate practically all intercepts to only the first two layers. This is, of course, expected in order to destroy as many attacking missiles as possible before they deploy their multiple warheads. The number of warheads penetrating the defense as a percentage of the original number of attacking missiles is shown in Fig. 6.9 for the cases of $\alpha = 1$, 3, and 10. This figure demonstrates clearly

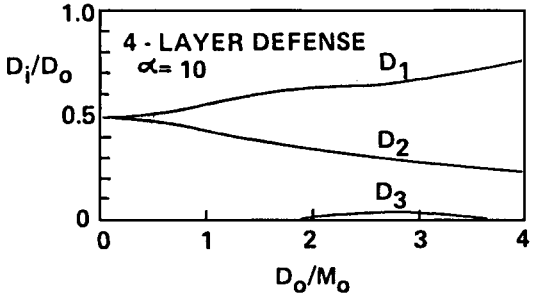


Fig. 6.8 Allocation of intercepts for a four-layer defense when α = 10: 10 RVs per missile at the end of the second layer.

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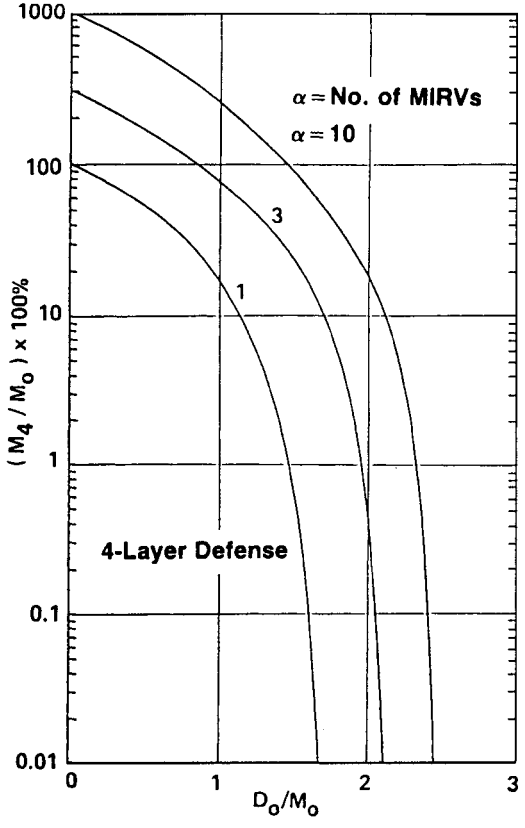


Fig. 6.9 Penetration of attacking RVs in a four-layer defense for $\alpha = 1, 3$, and 10.

that for an effective defense against a ten-tuple MIRVing the defense/offense ratio must be in the range of 2.5 to 3. Any smaller defense ratio would be unacceptable as a plausible defense.

6.3 Antiballistic Missile (ABM) Defense: Game Theory

The terminal defense of ICBM silos is of considerable interest to strategic planners, and several basing modes to enhance the survivability of such silos against a first-strike attack have been considered. These included passive defense, such as multiple shelters, mobile launchers, and defense pack configurations, and active defense, such as antiballistic missiles (ABMs). The defense can be considered successful if a relatively small fraction of the defended silos can survive, allowing for a retaliatory strike with the remaining missiles. Consequently, the potential attacker would have no clear strategic advantage from an attack and no incentive to launch a first strike against defended silos. The ABM defense can be particularly effective for hard targets, such as silos but would be of little value in area defense of population targets because the penetration of even a small number of attacking

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missiles would result in disaster for the defending side. For area defense, the previously discussed layered defense, of which terminal defense would be the last stage, is much more effective.

A defense system that could protect a substantial fraction of U.S. land-based ICBMs and the strategic command and control (C^2) installations against a first strike could be built with available technology. To be most effective, however, the direct defense with ABMs would have to be combined with the rebasing of ICBMs and with the construction of additional C^2 backup systems to ensure an adequate retaliatory capability.[4]

The ABM defense can increase the number of surviving silos by an optimal allocation of interceptors to the defended sites. This section will discuss the optimal strategies for both defense and offense, i.e., strategies of deployment of interceptors and distribution of re-entry vehicles (RVs) of the attacking side. This relatively simple analysis is performed through the application of game theory. Because few readers are familiar with the theoretical aspects of game theory, a brief introduction to the subject is included here that directly leads to the specific application of the general theory to the problem at hand.

Game Theory: Payoff Matrix[5]

Game theory is the study of outcomes in situations involving more than one decision-maker, and where the ultimate outcome (or payoff) for any one decision-maker depends not only on that decision-maker's own choices but also on the choices of others. Such studies are called "game theory" because of their similarity to common parlor games of strategy, and the decision-makers are called "players." The selection of an optimum strategy by the players is central to game theory, sometimes called "games of strategy." The analysis of strategies for the ABM defense is a game in which two players (defense and offense) have conflicting interests. The results of their decisions are measured in terms of payoff functions for a particular pair of strategies adopted by the players.

The payoff functions for the two players (the defense and offense) are the number of surviving missile silos and the number of destroyed silos, respectively. Obviously, the aim of the two players is to maximize their respective payoff functions. If the defense can employ m distinct strategies and the offense n different strategies, the corresponding payoff function can be formulated as two payoff matrices showing the payoffs of the two players given alternative possible pairs of strategies. For convenience, the two matrices are usually collapsed into a single matrix $\mathbf{\Pi}$, and each entry consists of two numbers $(\Pi^I_{ij}, \Pi^{II}_{ij})$, the first entry being the payoff to defense and the second being the payoff to offense for the ith defensive strategy and jth offensive strategy. This payoff matrix is of size $m \times n$ as given by Eq. (6.22):

$$\mathbf{\Pi} = \begin{bmatrix} (\Pi^I_{11}, \Pi^{II}_{11}) & (\Pi^I_{12}, \Pi^{II}_{12}) & \cdots & (\Pi^I_{1n}, \Pi^{II}_{1n}) \\ (\Pi^I_{21}, \Pi^{II}_{21}) & (\Pi^I_{22}, \Pi^{II}_{22}) & \cdots & (\Pi^I_{2n}, \Pi^{II}_{2n}) \\ \vdots & \vdots & \ddots & \vdots \\ (\Pi^I_{m1}, \Pi^{II}_{m1}) & (\Pi^I_{m2}, \Pi^{II}_{m2}) & \cdots & (\Pi^I_{mn}, \Pi^{II}_{mn}) \end{bmatrix} \tag{6.22}$$

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Two-Person Zero-Sum Game: Pure and Mixed Strategies

The two-person zero-sum game is defined as a game for which

$$\Pi_{ij}^{I} + \Pi_{ij}^{II} = 0 \tag{6.23}$$

where the payoff of the second player is simply the negative of the payoff of the first player, i.e.,

$$\Pi_{ij} = \Pi_{ij}^{I} = -\Pi_{ij}^{II} \tag{6.24}$$

This implies that the payoff matrix can be represented by a single entry for each pair of ij strategies, as shown by Eq. (6.25). However, it must be understood that the payoff to the first player is Π_{ij}, and the payoff to the second player is $-\Pi_{ij}$. Hence,

$$\boldsymbol{\Pi} = \begin{bmatrix} \Pi_{11}^{I} & \Pi_{12}^{I} & \dots & \Pi_{1n}^{I} \\ \Pi_{21}^{I} & \Pi_{22}^{I} & \dots & \Pi_{2n}^{I} \\ \vdots & \vdots & \ddots & \vdots \\ \Pi_{m1}^{I} & \Pi_{m2}^{I} & \dots & \Pi_{mn}^{I} \end{bmatrix} \tag{6.25}$$

The ballistic missile defense problem is in the category of constant-sum game expressed by

$$\Pi_{ij}^{I} + \Pi_{ij}^{II} = C = \text{const} \tag{6.26}$$

which can be reduced to the zero-sum game of Eq. (6.23). To demonstrate this the following definitions will be introduced:

Π_{ij}^{I} = number of silos saved

Π_{ij}^{II} = number of silos destroyed

d = total number of silos

Hence,

$$\Pi_{ij}^{I} + \Pi_{ij}^{II} = d$$

and

$$\Pi_{ij}^{II} = d - \Pi_{ij}^{I}$$

The pairs of entries in the payoff matrix will be $(\Pi_{ij}^{I}, d - \Pi_{ij}^{I})$, and subtracting $\frac{1}{2}d$ from each entry gives $(\widehat{\Pi}_{ij}^{I}, \widehat{\Pi}_{ij}^{II}) = [\Pi_{ij}^{I} - \frac{1}{2}d, -(\Pi_{ij}^{I} - \frac{1}{2}d)]$, and

$$\widehat{\Pi}_{ij}^{I} + \widehat{\Pi}_{ij}^{II} = 0 \tag{6.27}$$

The preceding equation implies that the constant-sum game can be analyzed as a zero-sum game.

The basic objective in a two-person zero-sum game is that each player seeks the maximum payoff possible regardless of what strategy the opponent chooses. This will be illustrated by an example in Fig. 6.10. It should be observed that, whatever row player 1 (defense) selects, player 2 (offense) will choose the column maximizing the return for the offense and thus minimizing the return to player 1.

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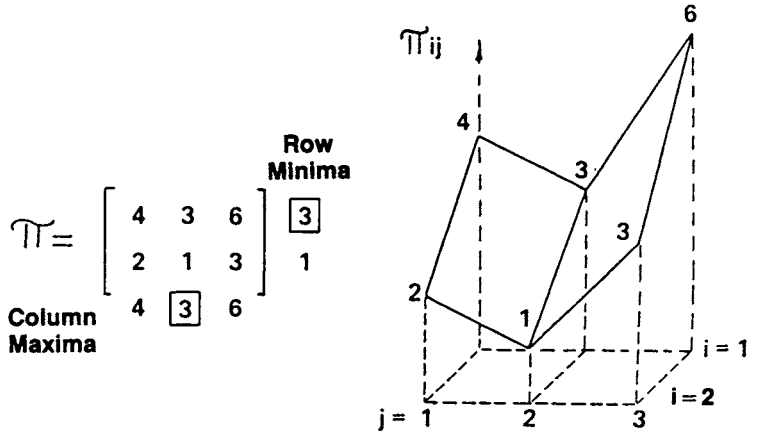


Fig. 6.10 Example of a payoff matrix in game theory.

Because the offense wants the payoff matrix value to be as low as possible (i.e., the lowest number of surviving missile silos), the strategy of player 1 (defense) is to examine row minima and select the row that gives the maximum among the minimum values. This ensures that, no matter what strategy player 2 (offense) selects, the payoff will be equal to or greater than this max-min value. Mathematically this is expressed as

$$\max_i \min_j \Pi_{ij} \tag{6.28}$$

where the maximum value selection is carried over the set of row minima.

Similarly, because the defense wants the payoff matrix value as high as possible (i.e. the highest number of surviving missile silos), the strategy for player 2 (offense) is to examine column maxima and select the row that gives the minimum among the maximum values. This ensures that, no matter what strategy player 1 (defense) selects, the payoff will be equal to or less than this min-max value. Mathematically this is expressed as

$$\min_j \max_i \Pi_{ij} \tag{6.29}$$

where the selection of the minimum value is carried over the set of column maxima. If

$$\max_i \min_j \Pi_{ij} = \min_j \max_i \Pi_{ij} = \Pi^*_{ij} \tag{6.30}$$

the payoff matrix has a saddle point at Π^*_{ij} for which the element (i, j) is both the minimum in its row and the maximum in its column. This is illustrated for the example in Fig. 6.10. Here the row minimum is equal to column maximum for the strategy $(i, j) = (1, 2)$. The payoff surface illustrates the location of the saddle point called by analogy with the surface of a saddle that curves upward in one direction and downward in the other direction. It can be seen from Fig. 6.10 that addition or subtraction of a constant to all entries in the payoff matrix does

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not change the location of the saddle point, and the optimum strategies remain unchanged.

The games with saddle point are called "strictly determined games." In general,

$$\max_i \min_j \Pi_{ij} \leq \min_j \max_i \Pi_{ij} \tag{6.31}$$

and games in which this inequality holds are called "nonstrictly determined games" without saddle point, and no solutions are available. The concept of an optimal solution developed for the strictly determined games is still valid; however, if, instead of pure strategies (i.e., single-row or single-column strategies), mixed strategies are used based on the probability distribution vectors ($\mathbf{p}^I$ and $\mathbf{p}^{II}$ for players 1 and 2) such that

$$\sum_{i=1}^{m \text{ or } n} p_i = 1 \tag{6.32}$$

where the summation is carried out from $i = 1$ to m or n, depending on whether the mixed strategy refers to player 1 or 2. The two probability vectors for players 1 and 2 are

$$\mathbf{p}^I = \begin{bmatrix} p_1^I & p_2^I & \dots & p_m^I \end{bmatrix} \tag{6.33}$$

$$\mathbf{p}^{II} = \begin{bmatrix} p_1^{II} & p_2^{II} & \dots & p_n^{II} \end{bmatrix} \tag{6.34}$$

If player 1 uses the mixed strategy $\mathbf{p}^I$ and player 2 uses the strategy $\mathbf{p}^{II}$, the payoff is

$$\Pi(\mathbf{p}^I, \mathbf{p}^{II}) = \mathbf{p}^I \boldsymbol{\Pi} (\mathbf{p}^{II})^T \tag{6.35}$$

Thus, the problem is to select a pair of probability vectors $\mathbf{p}^I$ and $\mathbf{p}^{II}$ to ensure the optimum solution. The expected minimum payoff for player 1 using the $\mathbf{p}^I$ strategy against the jth pure strategy of player 2 is

$$\Pi(\mathbf{p}^I) = \min \mathbf{p}^I \boldsymbol{\Pi}_j; \qquad j = 1, 2, \dots, n \tag{6.36}$$

where $\boldsymbol{\Pi}_j$ is the jth column of the matrix $\boldsymbol{\Pi}$. Hence, player 1 should choose $\mathbf{p}^I$ in order to maximize $\Pi(\mathbf{p}^I)$. This requires that

$$\overline{\Pi}^I = \max_{\mathbf{p}^I} \min_j \mathbf{p}^I \boldsymbol{\Pi}_j \tag{6.37}$$

where $\overline{\Pi}^I$ denotes the optimum value for $\mathbf{p}^I \boldsymbol{\Pi}$. Similarly, if player 2 chooses $\mathbf{p}^{II}$ as its strategy against the ith pure strategy of player 1,

$$\Pi(\mathbf{p}^{II}) = \max \boldsymbol{\Pi}_i (\mathbf{p}^{II})^T \tag{6.38}$$

where $\boldsymbol{\Pi}_i$ is the ith row of $\boldsymbol{\Pi}$. Here player 2 should choose $\mathbf{p}^{II}$ in order to minimize $\Pi(\mathbf{p}^{II})$. This requires that

$$\overline{\Pi}^{II} = \min_{\mathbf{p}^{II}} \max_i \boldsymbol{\Pi}_i (\mathbf{p}^{II})^T \tag{6.39}$$

where $\overline{\Pi}^{II}$ denotes the optimum value for $\Pi(\mathbf{p}^{II})$.

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The minimax theorem[5] states that there is at least a pair of mixed strategies $\mathbf{p}^I$ and $\mathbf{p}^{II}$ such that max-min equals min-max for the expected payoff, i.e.,

$$\overline{\Pi} = \overline{\Pi}^I = \overline{\Pi}^{II} \tag{6.40}$$

so that every zero-sum game with mixed strategies has a saddle point. The optimum mixed strategies $\mathbf{p}^I$ and $\mathbf{p}^{II}$ can be obtained in the general case by solving the dual linear programming problem. In the special case of strictly determined games, the optimal mixed strategies assign probabilities of 1 to the pure strategies at the saddle point. Here the optimal strategy vectors $\mathbf{p}^I$ and $\mathbf{p}^{II}$ are unit vectors.

ABM Defense Analysis

Returning now to the problem of the ABM deployment, it turns out that the payoff matrix is in the strictly determined category for which a saddle point exists. That is both defense and offense can use pure strategies for an optimal payoff to both sides.[5]

A defensive strategy denoted by $\mathcal{D}$ will be specified by a set of integers d_0, $d_1, \ldots, d_i, \ldots$ where d_i denotes the number of silos defended by i ABM interceptors. Thus, d_0 is the number of undefended silos, d_1 is the number of silos defended by one interceptor each, etc. Having defined the distribution of defensive interceptors in this manner, it follows that

$$\sum_i d_i = d \tag{6.41}$$

$$\sum_i i d_i = D \tag{6.42}$$

where d is the total number of silos, and D is the number of available interceptors. Similarly, an offensive strategy $\mathcal{O}$ will be specified by a set of integers $a_0, a_1, \ldots, a_j, \ldots$ where a_j denotes the number of silos attacked by j missiles subject to the condition

$$\sum_j a_j = d \tag{6.43}$$

$$\sum_j j a_j = A \tag{6.44}$$

where A is the number of attacking missiles.

Thus far the defensive strategies $\mathcal{D}$ and offensive strategies $\mathcal{O}$ represent all possible combinations of interceptor and attacking missile allocations. However, there is one particular set of strategies identified by Sartori[6] in which an equal number of sites are defended by one, two, etc., interceptors and in which an equal number of sites are attacked by one, two, etc., missiles, i.e.,

$$\mathcal{D} = [d_0, d_1, d_2 = d_1, d_3 = d_1, \ldots, d_p = d_1] \tag{6.45}$$

$$\mathcal{O} = [a_0, a_1, a_2 = a_1, a_3 = a_1, \ldots, a_q = a_1] \tag{6.46}$$

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Using the defensive strategy $\mathcal{D}$ and offensive strategy $\mathcal{O}$, i.e., Eqs. (6.45) and (6.46), it follows from Eqs. (6.41), (6.42), (6.43) and (6.44) that

$$d_0 = d - \frac{2D}{p+1} \tag{6.47}$$

$$d_1 = \frac{2D}{p(p+1)} \tag{6.48}$$

$$a_0 = d - \frac{2A}{q+1} \tag{6.49}$$

and

$$a_1 = \frac{2A}{q(q+1)} \tag{6.50}$$

where p and q must be chosen such that d_0, d_1, a_0, and a_1 are all positive integers.

When a silo defended by i interceptors is attacked by j RVs, it will only survive if $i \geq j$. The probability of a random silo being attacked by precisely j RVs is a_j/d; the silo will survive if, and only if, $i \geq j$. Therefore, the probability that a silo defended by i interceptors survives is

$$p_i = \sum_{j=0}^{i} \frac{a_j}{d} = \frac{1}{d} \sum_{j=0}^{i} a_j \tag{6.51}$$

which represents a ratio of the number of silos for which the offense allocated $j \leq i$ RVs to the total number of silos. Thus, the number of surviving silos (the payoff value) for a particular pair of defense and offense strategies is given by

$$\Pi(\mathcal{D}, \mathcal{O}) = \sum_i p_i d_i = \frac{1}{d} \sum_i \sum_{j=0}^{i} a_j d_i \tag{6.52}$$

in which Eq. (6.51) was used.

Substituting Eqs. (6.43)–(6.45), (6.48) and (6.49) into Eq. (6.52), it follows that

$$\Pi(\mathcal{D}, \mathcal{O}) = \frac{2D}{p} - \frac{2DA}{dp(p+1)} + a_0 \left(1 - \frac{2D}{dp}\right) + \frac{2D}{dp(p+1)}(a_{p+2} + 2a_{p+3} + \cdots) \tag{6.53}$$

Hence,

$$\min \Pi(\mathcal{D}, \mathcal{O}) = \frac{2D}{p} - \frac{2DA}{dp(p+1)}; \qquad p \geq \frac{2D}{d} \tag{6.54}$$

when

$$a_0 = a_{p+2} = a_{p+3} = \cdots = 0 \tag{6.55}$$

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Table 6.1 ICBM silos defense: defensive strategies

D/d	Defensive strategy $\mathcal{D}$	min $\Pi(\mathcal{D}, \mathcal{O})$	Range for max min $\Pi(\mathcal{D}, O)$	p
$\frac{1}{2}$	$(d/2)[1, 1]$	$(1 - A/2d)d$	$0 < A/d < \frac{3}{2}$	1
	$(d/6)[4, 1, 1]$	$(1 - A/3d)d/2$	$\frac{3}{2} < A/d < 2$	2
	$(d/12)[9, 1, 1, 1]$	$(1 - A/4d)d/3$	$2 < A/d < \frac{5}{2}$	3
	$(d/20)[16, 1, 1, 1, 1]$	$(1 - A/5d)d/4$	$\frac{5}{2} < A/d < 3$	4
	$(d/30)[25, 1, 1, 1, 1, 1]$	$(1 - A/6d)d/5$	$3 < A/d < \frac{7}{2}$	5
1	$(d/3)[1, 1, 1]$	$(1 - A/3d)d$	$0 < A/d < 2$	2
	$(d/6)[3, 1, 1, 1]$	$(1 - A/4d)2d/3$	$2 < A/d < \frac{5}{2}$	3
	$(d/10)[6, 1, 1, 1, 1]$	$(1 - A/5d)d/2$	$\frac{5}{2} < A/d < 3$	4
	$(d/15)[10, 1, 1, 1, 1, 1]$	$(1 - A/6d)2d/5$	$3 < A/d < \frac{7}{2}$	5
	$(d/21)[15, 1, 1, 1, 1, 1, 1]$	$(1 - A/7d)d/3$	$\frac{7}{2} < A/d < 4$	6
$\frac{3}{2}$	$(d/4)[1, 1, 1, 1]$	$(1 - A/4d)d$	$0 < A/d < \frac{5}{2}$	3
	$(d/20)[8, 3, 3, 3, 3]$	$(1 - A/5d)3d/4$	$\frac{5}{2} < A/d < 3$	4
	$(d/10)[5, 1, 1, 1, 1, 1]$	$(1 - A/6d)3d/5$	$3 < A/d < \frac{7}{2}$	5
	$(d/14)[8, 1, 1, 1, 1, 1, 1]$	$(1 - A/7d)d/2$	$\frac{7}{2} < A/d < 4$	6
	$(d/56)[35, 3, 3, 3, 3, 3, 3, 3]$	$(1 - A/8d)3d/7$	$4 < A/d < \frac{9}{2}$	7
2	$(d/5)[1, 1, 1, 1, 1]$	$(1 - A/5d)d$	$0 < A/d < 3$	4
	$(d/15)[5, 2, 2, 2, 2, 2]$	$(1 - A/6d)4d/5$	$3 < A/d < \frac{7}{2}$	5
	$(d/21)[9, 2, 2, 2, 2, 2, 2]$	$(1 - A/7d)2d/3$	$\frac{7}{2} < A/d < 4$	6
	$(d/14)[7, 1, 1, 1, 1, 1, 1, 1]$	$(1 - A/8d)4d/7$	$4 < A/d < \frac{9}{2}$	7
	$(d/18)[10, 1, 1, 1, 1, 1, 1, 1, 1]$	$(1 - A/9d)d/2$	$\frac{9}{2} < A/d < 5$	8
$\frac{5}{2}$	$(d/6)[1, 1, 1, 1, 1, 1]$	$(1 - A/6d)d$	$0 < A/d < \frac{7}{2}$	5
	$(d/42)[12, 5, 5, 5, 5, 5, 5]$	$(1 - A/7d)5d/6$	$\frac{7}{2} < A/d < 4$	6
	$(d/56)[21, 5, 5, 5, 5, 5, 5, 5]$	$(1 - A/8d)5d/7$	$4 < A/d < \frac{9}{2}$	7
	$(d/72)[32, 5, 5, 5, 5, 5, 5, 5, 5]$	$(1 - A/9d)5d/8$	$\frac{9}{2} < A/d < 5$	8
3	$(d/7)[1, 1, 1, 1, 1, 1, 1]$	$(1 - A/7d)d$	$0 < A/d < 4$	6
	$(d/28)[7, 3, 3, 3, 3, 3, 3, 3]$	$(1 - A/8d)6d/7$	$4 < A/d < \frac{9}{2}$	7
	$(d/12)[4, 1, 1, 1, 1, 1, 1, 1, 1]$	$(1 - A/9d)3d/4$	$\frac{9}{2} < A/d < 5$	8

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Thus, min $\Pi(\mathcal{D}, \mathcal{O})$ does not depend on any particular strategy employed by the offense. Rewriting Eq. (6.54) in normalized form for the number of surviving silos for the defensive strategies,

$$S_{\mathcal{D}} = \max \min \Pi(\mathcal{D}, \mathcal{O})/d$$
$$= \max \min \left[\frac{2D}{d} \frac{1}{p} - \frac{2D}{d} \frac{A}{d} \frac{1}{p(p+1)} \right]; \qquad p \geq \frac{2D}{d} \tag{6.56}$$

The max-min values of Eq. (6.56) are plotted in Fig. 6.11 for $D/d = 1$ as a function of A/d. Table 6.1 shows the results for the defensive strategies for a series of values of D/d. These results are also plotted in Fig. 6.12.

To determine the number of surviving silos for the offensive strategy $\mathcal{O}$, i.e., Eq. (6.46), Eqs. (6.41), (6.42), (6.46), (6.49), and (6.50) are substituted into

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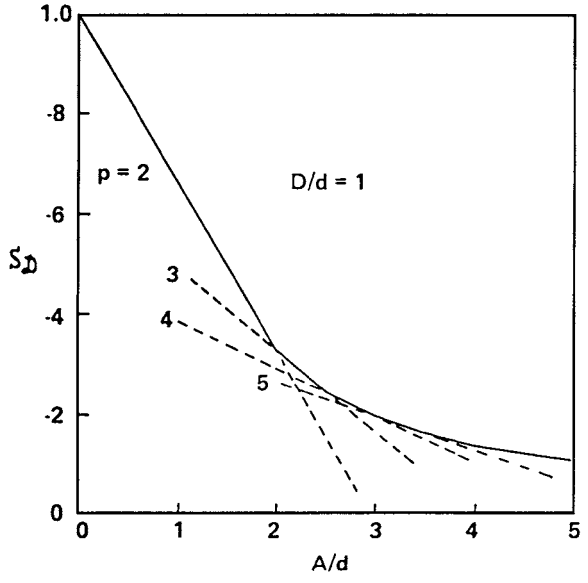


Fig. 6.11 Surviving silos for the defensive max min strategy against attacking RVs for $D/d = 1$.

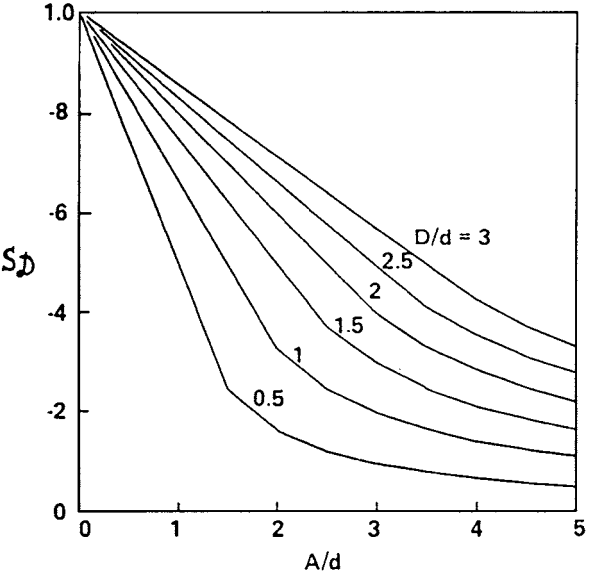


Fig. 6.12 Surviving silos for the defensive max min strategy against attacking RVs for various values of D/d.

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Eq. (6.52). This leads to

$$\Pi(\mathcal{D}, \mathcal{O}) = \left[1 - \frac{2A}{d(q+1)}\right] d + \frac{2AD}{dq(q+1)} - d_{q+1}\left[1 + \frac{2A}{dq(q+1)}\right] - d_{q+2}\left[1 + \frac{4A}{dq(q+1)}\right] - \cdots \tag{6.57}$$

whose maximum occurs when

$$d_{q+1} = d_{q+2} = \cdots = 0 \tag{6.58}$$

Hence,

$$\max\ \Pi(\mathcal{D}, \mathcal{O}) = \left[1 - \frac{2A}{d(q+1)}\right] d + \frac{2AD}{dq(q+1)} \tag{6.59}$$

As before, rewriting Eq. (6.59) in normalized form,

$$S_{\mathcal{O}} = \frac{\min \max\ \Pi(\mathcal{D}, \mathcal{O})}{d} = \min \max \left\{\left[1 - \frac{2A}{d(q+1)}\right] + \frac{2A}{d}\frac{D}{d}\frac{1}{q(q+1)}\right\} \tag{6.60}$$

The min-max values of Eq. (6.60) are plotted in Fig. 6.13 for $A/d = 1$ as a function of D/d. The set of optimal offensive strategies is displayed in Table 6.2 together

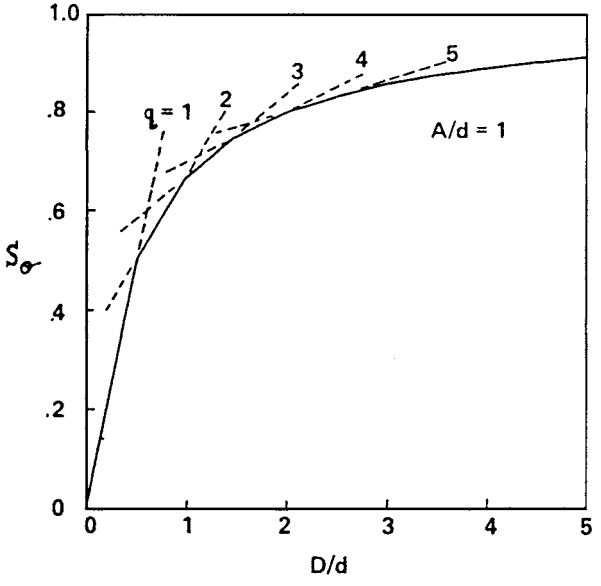


Fig. 6.13 Surviving silos for the offensive min max strategy of attacking RVs for $A/d = 1$.

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Table 6.2 ICBM silos defense: offensive strategies

A/d	Offensive strategy $\mathcal{O}$	max $\Pi(\mathcal{D}, \mathcal{O})$	Range for min max $\Pi(\mathcal{D}, O)$	q
$\frac{1}{2}$	$(d/2)[1, 1]$	$(1 + D/d)d/2$	$0 < D/d < \frac{1}{2}$	1
	$(d/6)[4, 1, 1]$	$(1 + D/4d)2d/3$	$\frac{1}{2} < D/d < 1$	2
	$(d/12)[9, 1, 1, 1]$	$(1 + D/9d)3d/4$	$1 < D/d < \frac{3}{2}$	3
	$(d/20)[16, 1, 1, 1, 1]$	$(1 + D/16d)4d/5$	$\frac{3}{2} < D/d < 2$	4
1	$d[0, 1]$	$(D/d)d$	$0 < D/d < \frac{1}{2}$	1
	$(d/3)[1, 1, 1]$	$(1 + D/d)d/3$	$\frac{1}{2} < D/d < 1$	2
	$(d/6)[3, 1, 1, 1]$	$(1 + D/3d)d/2$	$1 < D/d < \frac{3}{2}$	3
	$(d/10)[6, 1, 1, 1, 1]$	$(1 + D/6d)3d/5$	$\frac{3}{2} < D/d < 2$	4
$\frac{3}{2}$	$(d/2)[0, 1, 1]$	$(D/d)d$	$0 < D/d < 1$	2
	$(d/4)[1, 1, 1, 1]$	$(1 + D/d)d/4$	$1 < D/d < \frac{3}{2}$	3
	$(d/20)[8, 3, 3, 3, 3]$	$(1 + 3D/8d)2d/5$	$\frac{3}{2} < D/d < 2$	4
	$(d/10)[5, 1, 1, 1, 1, 1]$	$(1 + D/5d)d/2$	$2 < D/d < \frac{5}{2}$	5
2	$(d/3)[0, 1, 1, 1]$	$(D/d)d/2$	$0 < D/d < \frac{3}{2}$	3
	$(d/5)[1, 1, 1, 1, 1]$	$(1 + D/d)d/5$	$\frac{3}{2} < D/d < 2$	4
	$(d/15)[5, 2, 2, 2, 2, 2]$	$(1 + 2D/5d)d/3$	$2 < D/d < \frac{5}{2}$	5
	$(d/21)[9, 2, 2, 2, 2, 2, 2]$	$(1 + 2D/9d)3d/7$	$\frac{5}{2} < D/d < 3$	6
$\frac{5}{2}$	$(d/4)[0, 1, 1, 1, 1]$	$(D/d)d/3$	$0 < D/d < 2$	4
	$(d/6)[1, 1, 1, 1, 1, 1]$	$(1 + D/d)d/4$	$2 < D/d < \frac{5}{2}$	5
	$(d/42)[12, 5, 5, 5, 5, 5, 5]$	$(1 + 5D/12d)2d/7$	$\frac{5}{2} < D/d < 3$	6
3	$(d/5)[0, 1, 1, 1, 1, 1]$	$(D/d)d/5$	$0 < D/d < \frac{5}{2}$	5
	$(d/7)[1, 1, 1, 1, 1, 1, 1]$	$(1 + D/d)d/7$	$\frac{5}{2} < D/d < 3$	6
	$(d/28)[7, 3, 3, 3, 3, 3, 3, 3]$	$(1 + 3D/7)d/4$	$3 < D/d < \frac{7}{2}$	7
$\frac{7}{2}$	$(d/6)[0, 1, 1, 1, 1, 1, 1]$	$(D/d)d/6$	$0 < D/d$	6
	$(d/8)[1, 1, 1, 1, 1, 1, 1, 1]$	$(1 + D/d)d/8$	$3 < D/d < \frac{7}{2}$	7
	$(d/72)[16, 7, 7, 7, 7, 7, 7, 7, 7]$	$(1 + 7D/16d)2d/9$	$\frac{7}{2} < D/d < 4$	8
4	$(d/7)[0, 1, 1, 1, 1, 1, 1, 1]$	$(D/d)d/7$	$0 < D/d < \frac{7}{2}$	7
	$(d/9)[1, 1, 1, 1, 1, 1, 1, 1, 1]$	$(1 + D/d)d/9$	$\frac{7}{2} < D/d < 4$	8
	$(d/45)[9, 4, 4, 4, 4, 4, 4, 4, 4, 4]$	$(1 + 4D/9d)d/5$	$4 < D/d < \frac{9}{2}$	9

with the corresponding expressions for the min-max values for $S_{\mathcal{O}}$. Figure 6.14 also presents the same results graphically.

The combined plot of Figs. 6.12 and 6.14 shown in Fig. 6.15 demonstrates that $S_{\mathcal{D}} = S_{\mathcal{O}}$ for all pairs of D/d and A/d. This implies that the problem has a saddle point for which the relationship of Eq. (6.30) holds. The solution presented is the optimum for both the defense and offense; i.e., the defense will guarantee that at least this optimum number of silos will survive, whereas the offense cannot expect to produce a strategy that will reduce that number. That is, neither side can expect to do any better.

It should also be noted from Fig. 6.12 that a significant fraction of the silos (targets) can be saved with this preferential defense system. For example, with one

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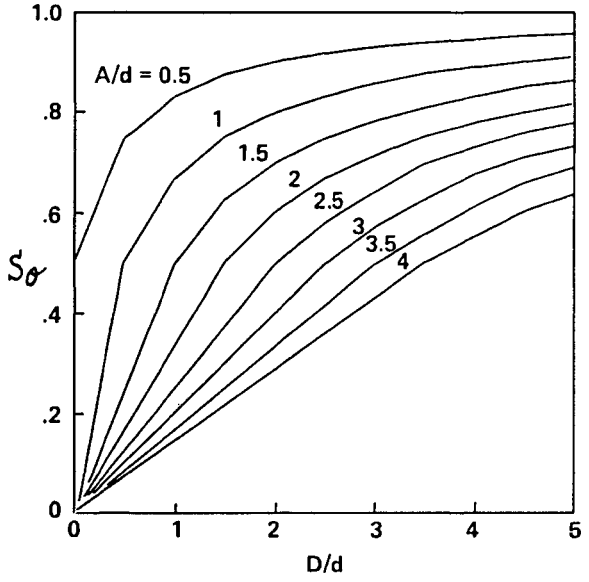


Fig. 6.14 Surviving silos for the offensive min max strategy of attacking RVs for various values of A/d.

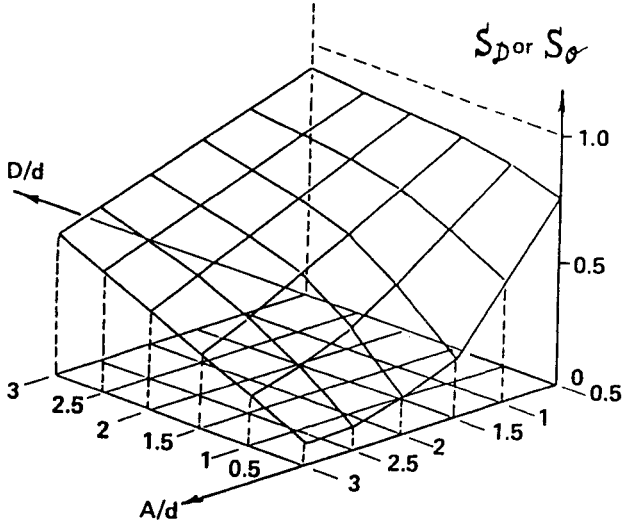


Fig. 6.15 Combined plot of the defensive max min and offensive min max strategies.

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intercept per silo, i.e., $D/d = 1$, 67% of silos can be saved for $A/d = 1$, and 20% for $A/d = 3$.

6.4 Optimal Penetration Routes Through Air Defenses: Threat Function

When a strategic bomber or a tactical aircraft is required to penetrate air defenses, it is necessary to select an optimum route that will ensure a minimum exposure to enemy fire and achieve the maximum probability of survival in accomplishing the specified mission. The mission is simply to fly from the entry point A to the target B, as shown in Fig. 6.16. The airspace between the points A and B is defended against any penetrating aircraft and the probability of kill over any *segment of flight* within that space will be assumed to be known either in terms of the actual probabilities or probability densities per unit distance or unit time.

Clearly, depending which flight route is selected, different probabilities of survival will be obtained. However, it is possible to select a route that will represent the highest probability of survival. To accomplish this a method of discrete dynamic programming is used. The overall defended area is divided into a two-dimensional grid of points through which the penetrating aircraft may pass, as illustrated in Fig. 6.16. The actual flight path will be assumed to be along the grid lines. A more accurate model would allow flight along the diagonals; however, for simplicity of explanation of the method only the simplified model shown in Fig. 6.16 will be used. The extension to the case with the diagonal flight segments is straightforward.

Along each flight segment, such as i-j, the probability of kill is specified as p_{ij} and the probability of survival as $1 - p_{ij}$. Thus, the probability of survival P_S for a set of segments leading from A to B is given by

$$P_S = \prod_{i,j}(1 - p_{ij}) \tag{6.61}$$

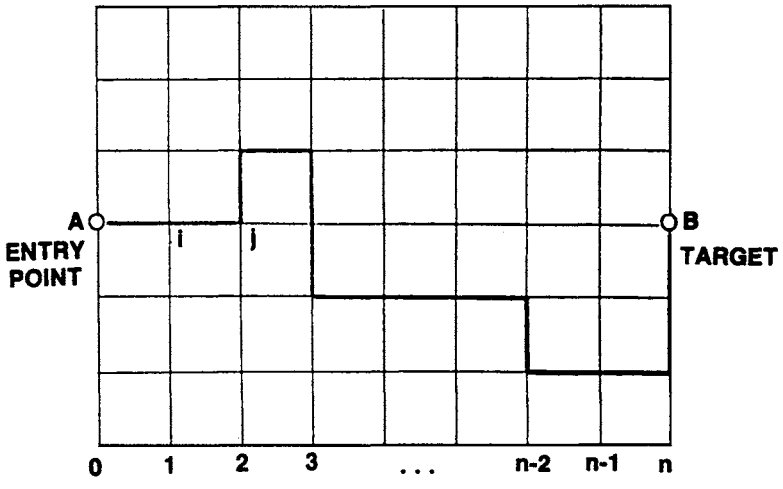


Fig. 6.16 Two-dimensional grid for discrete dynamic programming.

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The segment probability of kill can also be expressed as

$$p_{ij} = \int_0^{s_{ij}} \hat{p}_{ij}\, ds \tag{6.62a}$$

where $\hat{p}_{ij}$ is the probability density function for the i-j segment, and s_{ij} is the segment length, or alternatively as

$$q_{ij} = \int_0^{t_{ij}} \hat{q}_{ij}\, dt \tag{6.63a}$$

where $\hat{q}_{ij}$ is the probability density function, and t_{ij} is the time taken by the penetrating aircraft to traverse the i-j segment. If $\hat{p}_{ij}$ and $\hat{q}_{ij}$ are constants, then Eqs. (6.62a) and (6.63a) reduce to

$$p_{ij} = \hat{p}_{ij} s_{ij} \tag{6.62b}$$

and

$$q_{ij} = \hat{q}_{ij} t_{ij} = \frac{\hat{q}_{ij} s_{ij}}{v_{ij}} \tag{6.63b}$$

where

$$v_{ij} = \frac{s_{ij}}{t_{ij}} \tag{6.64}$$

is the constant speed of the aircraft flying over the i-j segment. It should be remembered, however, that $p_{ij} \leq 1$. When $p_{ij} = 1$, or when $P_S = 0$, this condition indicates that the aircraft has been shot down.

Taking the logarithms of both sides of Eq. (6.61), it follows that

$$\ln P_S = \ln(1 - p_{Ai}) + \ln(1 - p_{ij}) + \cdots + \ln(1 - p_{mB}) \tag{6.65}$$

Recognizing that all arguments in the preceding equation are between 0 and 1, a new variable T can be introduced such that

$$T = -\ln P_S = \sum_{i,j} T_{ij} \tag{6.66}$$

where

$$T_{ij} = -\ln(1 - p_{ij}) \tag{6.67}$$

The variable T can be described as the threat function made up of components T_{ij} for each segment over which the aircraft is flying; it can be used directly to find the probability of survival for the selected path using Eq. (6.66), i.e.,

$$P_S = e^{-T} \tag{6.68}$$

Noting that T is obtained by summing up the values of T_{ij} over all segments over which the penetrating aircraft is flying, it is clear that T_{ij} can also be obtained from

$$T_{ij} = \int_i^j \Phi\, ds \tag{6.69}$$

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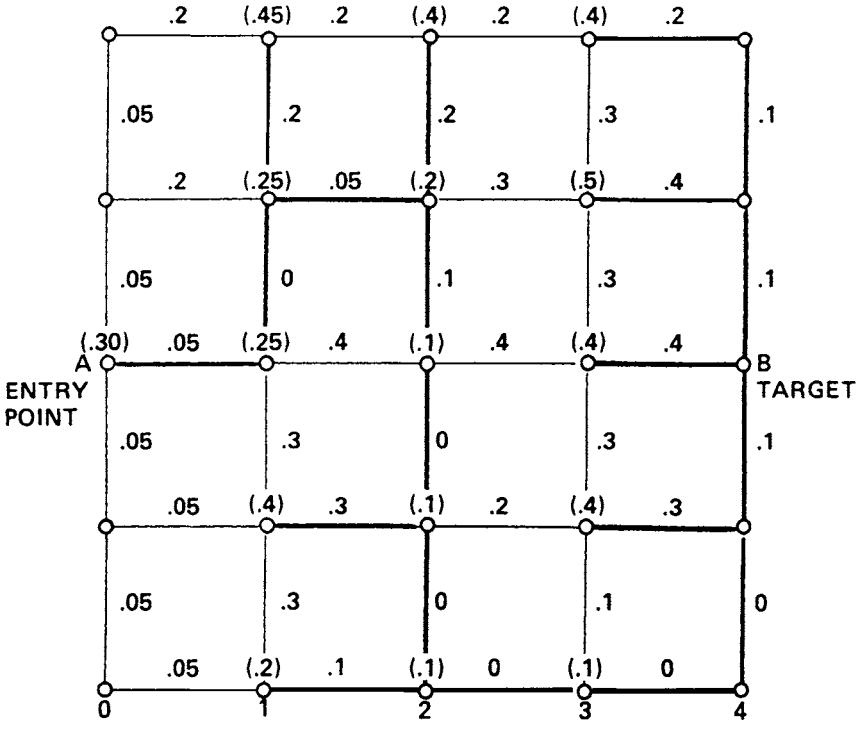


Fig. 6.17 Optimization of the flight path in a defended airspace by means of discrete dynamic programming; values of T_{ij}.

where Φ is the threat potential function, and the line integral is taken over the flight-path segment i-j. The numerical representation of such a function will be shown later for a particular combat scenario.

For the optimum penetration route the function T is a minimum. Thus, the problem is to find a set of connecting segments among all the possible ones that will render T a minimum. To accomplish this task the previously defined grid is divided into a number of stages designated by the lines 0-n, as shown in Fig. 6.17. In moving from line 0 to line 1 it is possible to chose a path for which the value of $\sum T_{ij}$ will be a minimum. But will this be a correct path? Obviously not, because if the chosen route, for example, is from A to i, this will not guarantee that a later path from the point i will not lead to some segments of very high values T_{ij} that should be avoided. It is obvious that the solution at the first stage must be chosen not on the basis of the narrow interest of this particular stage but on the basis of a much broader consideration of the operation as a whole. Thus, the solution of each stage must be adopted with the future stages taken into account. Such a solution can be obtained by considering the aircraft path in reverse.

This method will be illustrated for the example in Fig. 6.17. The values of T_{ij} for each segment in Fig. 6.17 have been selected to one-digit accuracy for

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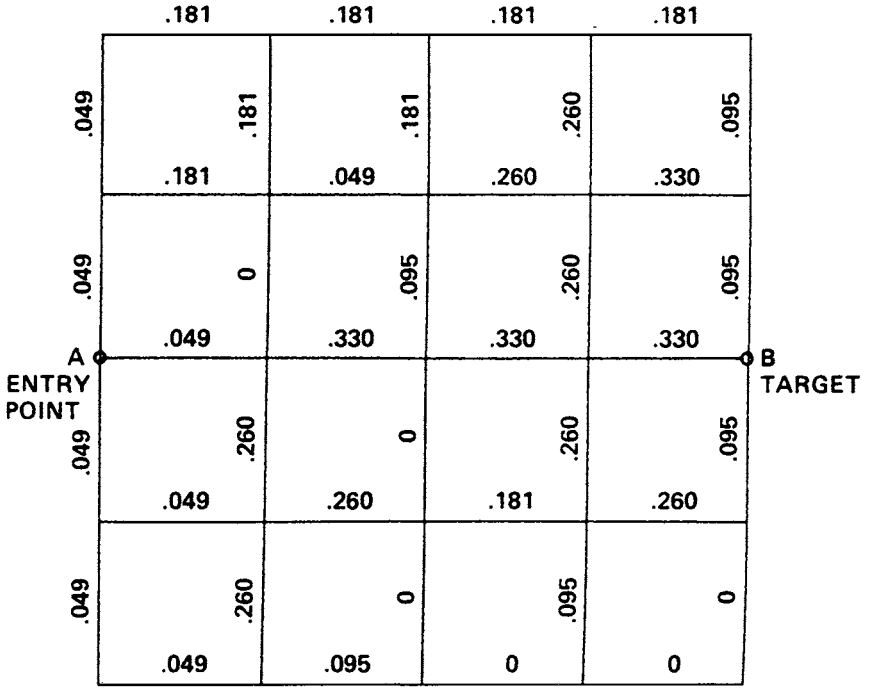


Fig. 6.18 Probabilities of kill for individual flight segments used in the numerical example: values of p_{ij}.

simplicity and correspond to values of p_{ij} shown in Fig. 6.18. The last stage is located between lines $n-1$ and n (between lines 3 and 4). On the line $n-1$ (line 3) there are five different positions (grid nodes) at the end of the next to last stage. For each of these positions there is a completely determined route to the point B (i.e., the target) that results in the lowest value of $\sum T_{ij}$. These routes are indicated by heavy solid lines, and the total minimum values for each grid point on the line $n-1$ are indicated by numbers in parentheses.

Subsequently, a solution for the next to the last stage must be chosen. Here again, there are five grid points on the line $n-2$ (line 2). In the same way as for the last stage, for each grid point the minimum T_{ij} path can be selected leading to the target as indicated by the heavy lines.

Continuing this process further for the $n-2$ line (line 2) and all of the subsequent lines up to line 0, the successive chain of solutions will produce the optimal solution for which $T = \sum T_{ij}$ is minimized. This solution is an optimal path for the aircraft, provided no "backtracking" is allowed. There may be special cases where the true optimum would require backtracking at some point. That is not allowed by the present method. Such cases may also be solved by a similar method.

It is interesting to compare the threat function for the optimal path with that for a straight line flight path from A to B. It can be seen readily from Fig. 6.17

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that

$$T_{\text{opt}} = \sum_{\substack{10 \\ \text{segments}}} T_{ij} = 0.30 \tag{6.70}$$

$$T_{\text{dir}} = \sum_{\substack{4 \\ \text{segments}}} T_{ij} = 1.25 \tag{6.71}$$

where the subscripts opt and dir indicate optimum and direct values, respectively. The corresponding probabilities of survival are

$$P_{S_{\text{opt}}} = e^{-T_{\text{opt}}} = e^{-0.30} = 0.74 \tag{6.72}$$

$$P_{S_{\text{dir}}} = e^{-T_{\text{dir}}} = e^{-1.25} = 0.29 \tag{6.73}$$

This means that approximately one aircraft out of four will be lost on this mission if the optimum path is selected. On the other hand, if the direct route is selected, then two aircraft out of three would be lost.

If the variation of Φ over the segment i-j is assumed to be linear, then it can easily be shown that

$$\frac{T_{ij}}{s_{ij}} = \Phi_{\text{midpoint of } i\text{-}j} \tag{6.74}$$

which can be used to construct the threat potential function Φ. The potential Φ for the example in Fig. 6.17 is shown in Fig. 6.19. The high threat potential is depicted there as hills, and the chosen optimal path obtained by discrete dynamic programming leads through the valleys representing the lowest threat levels.

The two-dimensional optimization analysis shown here can be extended directly to the three-dimensional cases by introducing a spatial grid with specified values of T_{ij} for each spatial segment. Here the threat dependence on altitude can be introduced through the appropriate variation of T_{ij}.

6.5 Defense Against a Penetrating Aircraft

Two similar cases of a SAM defense against a penetrating aircraft will be considered.[7] In the first case the aircraft is flying at very low altitude and is directly approaching a SAM site capable of firing a number of missiles with maximum range $R_{\max}$ and a minimum range $R_{\min}$. In the second case the aircraft is flying at a constant altitude on a straight course offset by a distance x (=CPA, closest point of approach) from the missile site. The latter case can be either at low or high altitude.

Direct Penetration of the Defended Missile Site

The directions of the aircraft and missile velocities are shown in Fig. 6.20. It will be assumed that the first intercept will occur during the inbound flight at the maximum range $R_{\max}$; therefore,

$$R_1 = R_{\max} \tag{6.75}$$

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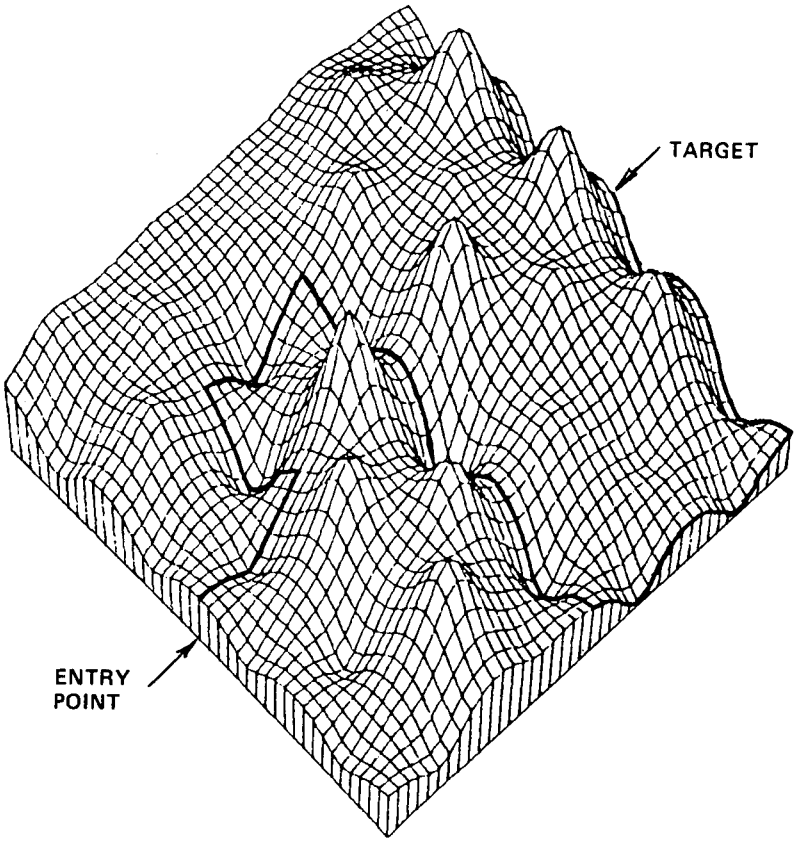


Fig. 6.19 Threat potential ϕ and the optimum flight path.

In determining the second intercept, a shoot-look-shoot firing sequence will be used with no delay time taken into account. That is, as soon as one missile intercepts the aircraft, another is fired instantaneously. Of course, if the intercept is successful and the aircraft is destroyed, no further missiles are fired. Denoting the velocities of the SAM missiles and the aircraft as V and U, respectively, it follows that the second intercept will occur at

$$R_2 = Vt = R_{\max} - Ut \tag{6.76}$$

Hence,

$$R_2 = \frac{R_{\max}}{1 + U/V} \tag{6.77}$$

Repeating this calculation for R_3, etc., it follows that, for the nth intercept,

$$R_n = R_{\max}(1 + U/V)^{1-n} \tag{6.78}$$

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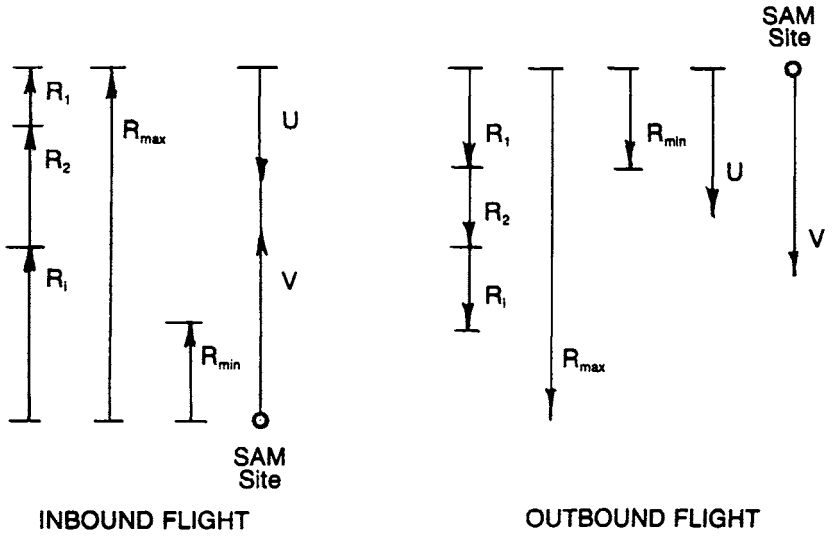


Fig. 6.20 Directions for the aircraft and missile velocities.

Clearly, $R_n \geq R_{\min}$; therefore,

$$\frac{R_{\max}}{R_{\min}} \geq \left(1 + \frac{U}{V}\right)^{n-1} \tag{6.79}$$

Because both sides are greater than 1,

$$\ln \frac{R_{\max}}{R_{\min}} \geq (n-1) \ln\left(1 + \frac{U}{V}\right)$$

and therefore,

$$n \leq -\frac{\ln(R_{\min}/R_{\max})}{\ln(1 + U/V)} + 1 \tag{6.80}$$

This means that the maximum number of SAM salvos on the inbound flight is given by

$$N_{\text{inbound}} = \text{INT}\left[-\frac{\ln(R_{\min}/R_{\max})}{\ln(1 + U/V)}\right] + 1 \tag{6.81}$$

where INT(x) denotes the largest integer that is less than or equal to x.

Similarly, for the outbound flight, it can be shown that

$$R_n = R_{\min}(1 - U/V)^{1-n} \tag{6.82}$$

In this case $R_{\max} \geq R_n$; therefore,

$$\frac{R_{\max}}{R_{\min}} \geq \left(1 - \frac{U}{V}\right)^{1-n} \tag{6.83}$$

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Hence,

$$\ell n \frac{R_{\max}}{R_{\min}} \geq (1-n)\,\ell n\left(1-\frac{U}{V}\right)$$

Therefore,

$$n \leq \frac{\ell n(R_{\min}/R_{\max})}{\ell n(1-U/V)} + 1 \tag{6.84}$$

and

$$N_{\text{outbound}} = \text{INT}\left[\frac{\ell n(R_{\min}/R_{\max})}{\ell n(1-U/V)}\right] + 1 \tag{6.85}$$

It should be noted here that, because $\ell n(1-U/V)$ is negative, division by $\ell n(1-U/V)$ changes the inequality sign.

For both the inbound and outbound flight the total number of salvos by SAM missiles is therefore given by the sum of Eqs. (6.81) and (6.85), i.e.,

$$N_{\text{total}} = \text{INT}\left[-\frac{\ell n(R_{\min}/R_{\max})}{\ell n(1+U/V)}\right] + \text{INT}\left[\frac{\ell n(R_{\min}/R_{\max})}{\ell n(1-U/V)}\right] \tag{6.86}$$

The number of salvos for the inbound flight and for the complete flight (both inbound and outbound) are shown in Figs. 6.21 and 6.22.

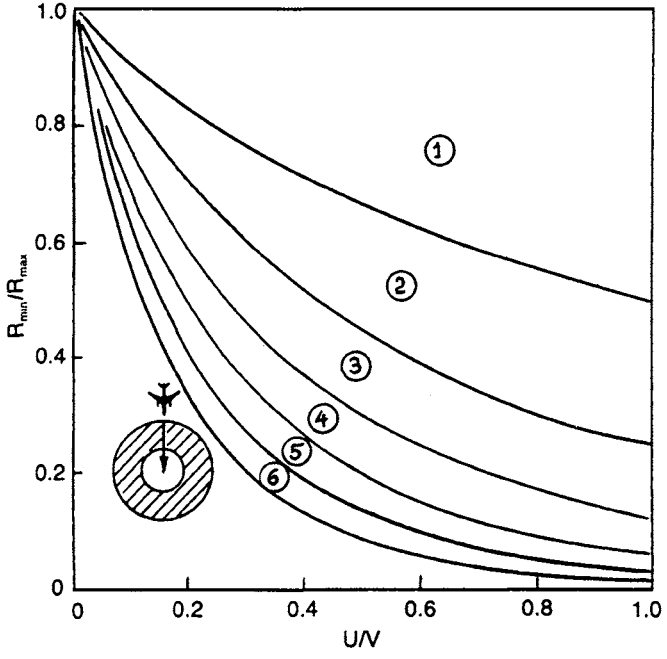


Fig. 6.21 Number of SAM salvos for an inbound flight.

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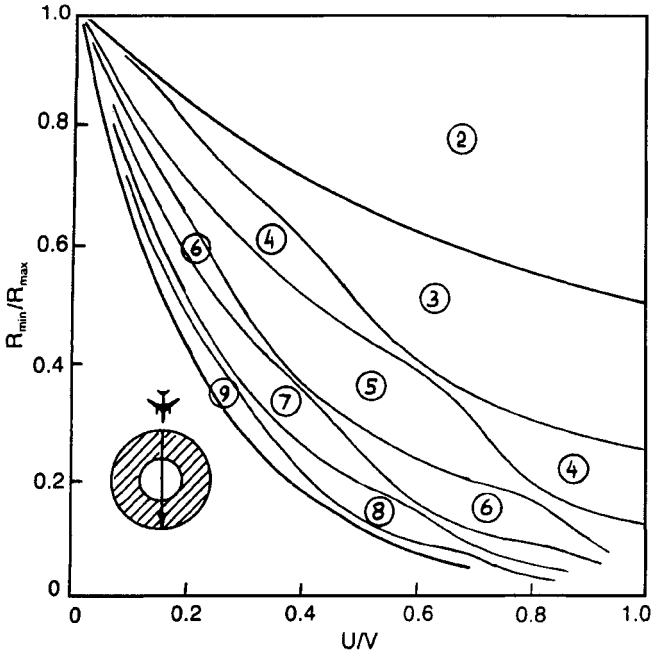


Fig. 6.22 Number of SAM salvos for the combined inbound and outbound flight.

Offset Penetration in Relation to SAM Site

If the penetrating aircraft is flying along a path that does not pass through the missile site, then two possible flight geometries can be followed: the pursuit course or intercept course, as shown in Fig. 6.23. In the pursuit course the missile is guided by constantly traveling in the direction of the current bearing to the penetrating aircraft (target). The bearing to the target is constantly changing up the point of intercept. In the intercept course, the target's speed U and its direction, as well the SAM's speed V, must be known to compute the lead angle λ (see Fig. 6.23b). The intercept point occurs at the interception of the paths of the SAM missile and the target. It can be seen from Figs. 6.23a and 6.23b that the intercept course is preferable because it leads to a shorter time of flight for the missile and reduces the time between subsequent launches if the shoot-look-shoot firing sequence is followed. In this section only the intercept course will be analyzed.

It will be assumed that the first intercept occurs at the maximum SAM range equal to R as shown in Fig. 6.24. Thus, for the first intercept

$$\beta_1 = \sin^{-1} \frac{z}{R_1} \tag{6.87}$$

For the nth intercept, the following relationship is obtained using the law of sines:

$$\frac{\sin \lambda_{n+1}}{S_{n+1}} = \frac{\sin \beta_n}{R_{n+1}} \tag{6.88}$$

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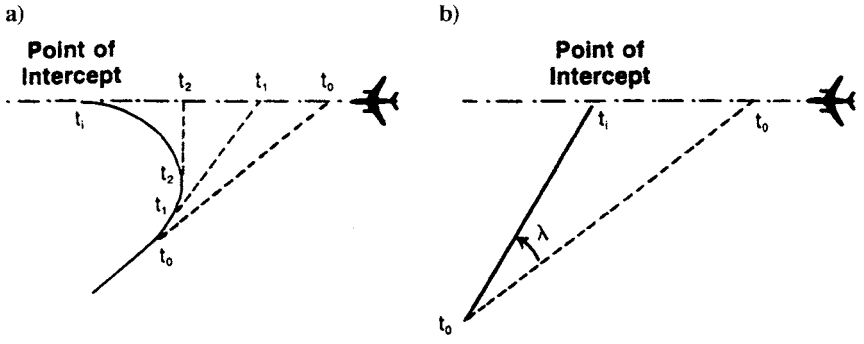


Fig. 6.23 Pursuit and intercept flight geometries: a) pursuit course and b) intercept course.

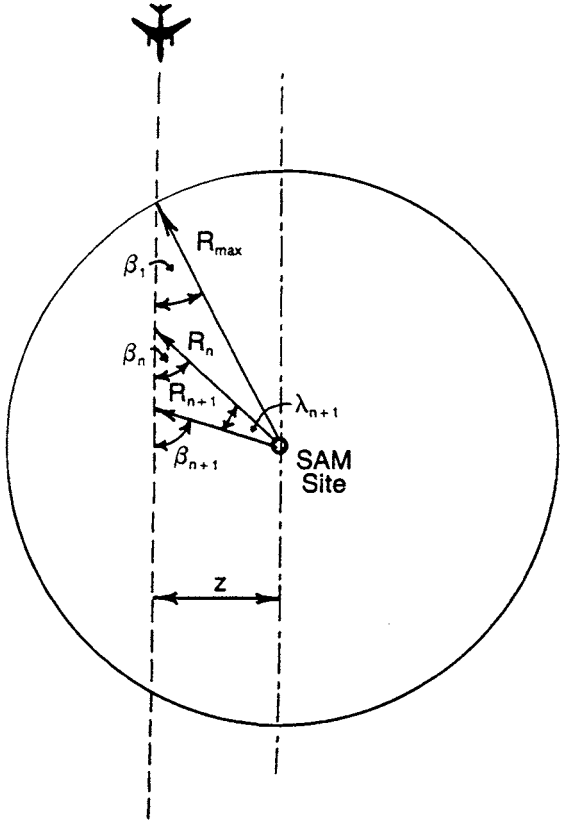


Fig. 6.24 Intercept geometries for a series of SAM launches against a penetrating aircraft.

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Noting that

$$R_{n+1} = Vt_{n+1} \tag{6.89}$$

and

$$S_{n+1} = Ut_{n+1} \tag{6.90}$$

where

t_{n+1} = time of flight for the $(n+1)$th intercept
V = SAM velocity
U = aircraft velocity

Hence, from Eq. (6.88),

$$\lambda_{n+1} = \sin^{-1}\left(\frac{U}{V}\sin\beta_n\right) \tag{6.91}$$

Also, from Fig. 6.24,

$$\beta_{n+1} = \beta_n + \lambda_{n+1} \tag{6.92}$$

Now using again the law of sines, the following two additional equations are derived:

$$R_{n+1} = R_n \frac{\sin\beta_n}{\sin\beta_{n+1}} \tag{6.93}$$

and

$$S_{n+1} = R_n \frac{\sin\lambda_{n+1}}{\sin\beta_{n+1}} \tag{6.94}$$

For given values of the offset z, the maximum SAM range R_1, and the velocities V and U, the points of intersection can be located along the path of the penetrating aircraft using Eqs. (6.92)–(6.94). Naturally, the maximum range of the SAM missiles cannot be exceeded; therefore, intercepts for which $R_n > R_1$ must be disregarded as impossible. From Fig. 6.24 it is clear that β_{n+1} cannot exceed π, i.e.,

$$\beta_{n+1} < \pi \tag{6.95}$$

Figure 6.25 shows SAM envelopes for $U/V = 0.5$ and 0.8. The envelopes show the locus of intercept points for each possible missile launch as calculated by Eqs. (6.92)–(6.95). As the penetrating aircraft traverses the SAM envelope, it is exposed to a number of missile intercepts; the greater the offset parameter z/R_1 the fewer the number of missiles that can be directed against the aircraft. Likewise, the number of possible launches decreases with an increase in the ratio U/V. For higher values of U/V, about 1 or higher, some intercepts, even when $R_n < R_1$, are not possible because of the insufficient velocity of the missile to catch up with the aircraft. This is illustrated in Fig. 6.26 for $U/V = 1$. Here the envelope is reduced, showing the region where the target is immune from an intercept within the radius R_1 from the SAM site.

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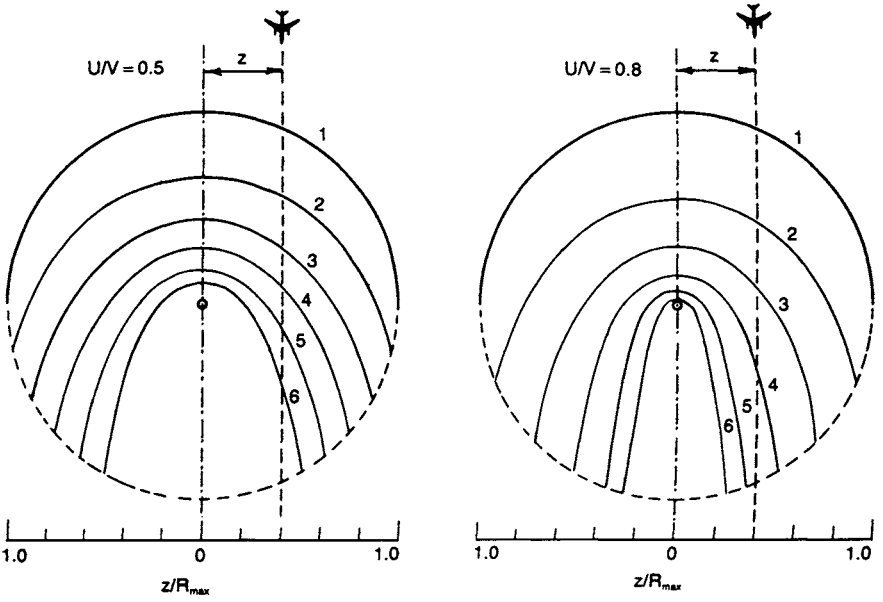


Fig. 6.25 SAM envelopes for $U/V = 0.5$ and 0.8.

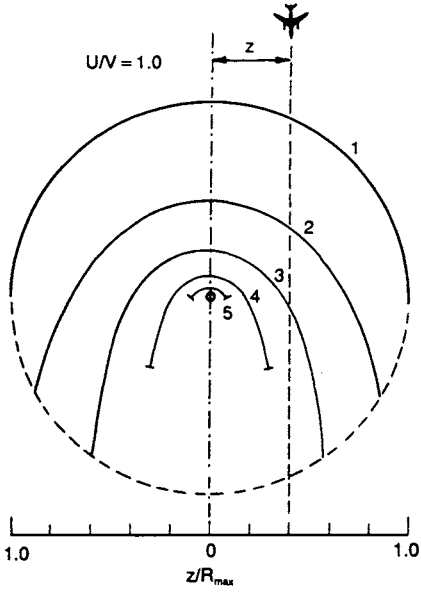


Fig. 6.26 SAM envelope for $U/V = 1$.

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The preceding analysis was based on two-dimensional consideration of the intercept; however, the equation derived for the location of the intercepts within the SAM envelope is also valid for the case where the penetrating aircraft flies at a constant altitude H. If the closest point of approach between the aircraft flight path and the missile site is D, then the two-dimensional analysis can be made applicable to the three-dimensional case by replacing z with $(D^2 + H^2)^{\frac{1}{2}}$. The SAM flight paths will then be in a plane passing through the line of flight of the penetrating aircraft and the SAM site.

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6.6 Sufficiency Models for Strategic Forces[8]

Attempts to quantify the effectiveness of strategic nuclear forces have led to a special class of models used to asses the absolute and relative sufficiency of the U.S. and foreign nuclear arsenals. These models are referred to as "sufficiency models" because they provide a useful vehicle for formulating such concepts as strategic parity, deterrence, and stability of the opposing nuclear forces. Two sufficiency models are presented in this section. The first model is the equivalent megatonnage (EMT) model. Historically this model evolved in the mid 1960s to provide an analytical assessment of the U.S. policy of assured destruction and its concomitant emphasis on the lethality of nuclear weapons in urban or industrial targeting functions. The second model is the counter military potential (CMP) model, which evolved in the 1970s as a result of the potential threat to the U.S. land-based ICBM force due to Soviet nuclear force modernization programs.

Equivalent Megatonnage (EMT)

The EMT model can be considered as a crude measure of the capability of an arsenal of strategic nuclear weapons to destroy large, geographically dispersed urban or industrial targets. The fundamental steps in generating the EMT model include the following:

1) assessment of the capability of 1 Megaton (1 MT) weapon to cover a large target with a given lethal overpressure when a nominal height of burst of the weapon is specified.

2) normalization of the region of lethal effects for weapon of arbitrary yield to that of the 1 MT weapon.

3) summation of the individual normalized yields over the entire arsenal of strategic nuclear weapons.

Assuming that the weapon radius for a 1 MT weapon is defined as $R_{1\,\mathrm{MT}}$, the area of the target thus covered by the lethal effects is given by

$$A_{1\,\mathrm{MT}} = \pi R_{1\,\mathrm{MT}}^2 \tag{6.96}$$

The weapon radius R_y for an arbitrary yield y can be obtained from the yield scaling relationship for the lethal radius

$$R_y = R_{1\,\mathrm{MT}} \left(\frac{y_{1\,\mathrm{MT}}}{y} \right)^{\alpha} \tag{6.97}$$

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where

α = scaling exponent
y = weapon yield (other than 1 MT)
$y_{1\,\mathrm{MT}}$ = 1 MT yield

Denoting the area covered by the lethal effects of a weapon of yield y by A_y,

$$A_y = \pi R_y^2 \tag{6.98}$$

it follows from the previous two equations that

$$\frac{A_{1\,\mathrm{MT}}}{A_y} = Y^{2\alpha} \tag{6.99}$$

where $Y = y/1$ MT, the ratio of the actual yield to that of 1 MT.

The scaling factor $Y^{2\alpha}$ represents an equivalence relationship to a 1 MT weapon and describes the relative size of the lethal area for yield y. The traditional scaling law uses $\alpha = \frac{1}{3}$. Because for large-yield weapons the lethal area is so large that the calculated lethal area would exceed the area of the target, it has become common practice to use $\alpha = \frac{1}{3}$ for yields below 1 MT and $\alpha = \frac{1}{4}$ for yields above 1 MT. Hence, defining EMT as

$$\mathrm{EMT} = Y^{2\alpha} \tag{6.100}$$

it follows that

$$\mathrm{EMT} = Y^{\frac{2}{3}}, \qquad y \leq 1\,\mathrm{MT} \tag{6.101}$$

$$\mathrm{EMT} = Y^{\frac{1}{2}}, \qquad y > 1\,\mathrm{MT} \tag{6.102}$$

Although the EMT model contains a number of simplifications relating to weapon effects, target vulnerability, and targeting accuracy, its simple relationship to the concept of assured destruction objective makes it one of the most convenient models for calculating the strength of the nuclear deterrence force. The overall strength is calculated by the summation of individual weapon EMTs in the available arsenal. A further refinement in the calculation may be made by introducing a fractional factor representing the expected degradation in force sufficiency that may be caused by such factors as prelaunch survivability, weapon system reliability, and defense penetrability. Hence, the EMT for the total force may be obtained from

$$\mathrm{EMT}_{\mathrm{total}} = \sum_{i=1}^{M} N_i Y_i^{2\alpha} S_i \tag{6.103}$$

where

M = number of warhead types
N_i = number of warheads of type i
$Y_i^{2\alpha}$ = EMT of warhead type i
S_i = expected degradation of a warhead of type i

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Counter Military Potential (CMP)

The CMP model, unlike the EMT model, which deals with area targets, is designed to calculate the ability of a nuclear weapon to damage a fixed target. The CMP of a weapon is a combined dimensional index expressed as a ratio of the explosive power of the warhead over the square of expected delivery accuracy. The individual indices of CMP can be added to obtain an aggregate measure of the strategic offensive force effectiveness against point targets.

The probability of destruction of a point target was shown in Chapter 3 to be given by [see Eq. (3.54)]

$$PD = 1 - \left(\tfrac{1}{2}\right)^{(R_w/\mathrm{CEP})^2} \tag{6.104}$$

where

PD = target damage probability
R_w = weapon lethal radius
CEP = circular error probable for weapon delivery

The R_w depends on target hardness and warhead yield. It is usually expressed approximately as a product of a function of nominal hardness $f(H)$ and yield y, i.e.,

$$R_w^2 \cong f(H)y^{2\alpha} \tag{6.105}$$

where α is the previously used yield scaling factor. Substituting Eq. (6.104) into Eq. (6.105), it follows that

$$PD = 1 - \left(\tfrac{1}{2}\right)^{f(H)y^{2\alpha}/\mathrm{CEP}^2} \tag{6.106}$$

The weapon-related index of CMP can therefore be defined as

$$\mathrm{CMP} = \frac{y^{2\alpha}}{\mathrm{CEP}^2} \tag{6.107}$$

As in the case of EMT, it has been found necessary to adjust the scaling index α to account for the wide range of weapon yields. The current practice has been to use

$$\alpha = \tfrac{1}{3} \quad \text{for} \quad y \geq 0.2\,\mathrm{MT} \tag{6.108}$$

and

$$\alpha = 0.4 \quad \text{for} \quad y < 0.2\,\mathrm{MT} \tag{6.109}$$

Because CEP^2 appears in the denominator, the numerical computations of CMP are very sensitive to variations in accuracy of nuclear weapons targeted for hard targets. For example, a 50% reduction in CEP produces a fourfold increase in CMP.

To determine the overall value of CMP, the individual warhead CMPs are calculated and then summed up over the total nuclear force. If necessary, the degradation factor can be applied to individual CMPs, as in the case of EMT model.

The CMP model is intended as a simplified measure of relative strengths of the counterforce capabilities defined as the strategic nuclear deterrence against hard

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Table 6.3 U.S. strategic nuclear forces

Type	Average yield	No. of launchers	N_i	Warhead EMT	$\sum$ EMT	CEP m	$\sum$ CMP $m^{-2}\ 10^{-3}$
Land-based							
Minuteman II	1×1.2 MT[a]	450	450	1.095	493	370	3.71
Minuteman III	3×170 KT[b]	211	633	0.307	194	220	3.17
Minuteman III	3×335 KT	300	900	0.482	434	220	8.97
Peacekeeper (MX)	10×350 KT	39	390	0.497	194	100	19.37
Land-based subtotals			2373		1315		35.22
Sea-based							
Poseidon C-3	10×40 KT	256	2560	0.117	300	450	0.96
Trident C-4	8×100 KT	384	3072	0.215	660	450	2.40
Sea-based subtotals			5632		960		3.37
Totals			8005		2275		38.59

[a]MT = Megaton.
[b]KT = Kiloton.

targets (point targets). This can be contrasted with the EMT model, which is used to assess the relative strengths of the countervalue force targeted against urban or industrial areas (area targets).

Comparison of Strategic Nuclear Forces[9–11]

Tables 6.3 and 6.4 show the derivation of the U.S. and Soviet EMT and CMP of land-based and sea-based strategic nuclear forces, excluding the strategic long-range bombers, as reported in Ref. 9 for 1988–89. Table 6.5 summarizes the results of Tables 6.3 and 6.4, and shows that the total number of warheads in the Soviet strategic land-based and sea-based nuclear arsenal is greater than that of the United States (9908 vs 8005). Likewise, the values of EMT and CMP are greater in the former USSR (5543 vs 2275 for EMT and 51.7×10^{-3} vs 38.6×10^{-3} for CMP). Only in one category, sea-based CMP, is the U.S. potential greater than that of the USSR (3.4×10^{-3} vs 1.6×10^{-3}). However, it should be pointed out that Tables 6.3–6.5 do not include nuclear warheads delivered by strategic bombers (the U.S. B-1B and B-52 and the Soviet Bear and Bison). If the nuclear warheads delivered are included in the total count, the strengths of the two sides in terms of numbers of warheads are comparable, with the U.S. side having a moderate advantage (14,637 vs 11,694) (see Ref. 9, p. 230).

The reported figures for 1992–93 indicated very little change in the land- and sea-based strategic nuclear forces. There was, however, some increase in the number of nuclear warheads delivered by strategic bombers of the former Soviet Union (now the Commonwealth of the Independent States) resulting essentially in parity

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Table 6.4 Soviet strategic nuclear forces

Type	Average yield	No. of launchers	N_i	Warhead EMT	$\sum$ EMT	CEP m	$\sum$ CMP $m^{-2}\ 10^{-3}$
Land-based							
SS-11 mod 2	1 × 1 MT	28[a]	28	1.000	28	1400	0.01
SS-11 mod 3	3 × 200 KT	420	1260	0.342	431	1100	0.36
SS-13 mod 2	1 × 600 KT	60	60	0.711	43	800	0.01
SS-17 mod 3	4 × 500 KT	138	552	0.630	348	400	2.17
SS-18 mod 4	10 × 500 KT	308	3080	0.630	1940	250	31.11
SS-19 mod 3	6 × 550 KT	350	2100	0.671	1409	300	15.66
SS-24	10 × 100 KT	10	100	0.215	22	200	0.40
SS-25	1 × 550 KT	100	100	0.671	67	200	0.40
Land-based subtotals			7280		4288		50.12
Sea-based							
SS-N-5	1 × 1 MT	36	36	1.000	36	2800	0.01
SS-N-6 mod 1	1 × 750 KT	256[b]	128	0.825	106	1300	0.06
SS-N-6 mod 3	2 × 500 KT		256	0.630	161	1300	0.10
SS-N-8 mod 1	1 × 750 KT	286[b]	143	0.825	118	1500	0.05
SS-N-8 mod 2	1 × 800 KT		143	0.862	123	900	0.15
SS-N-17	1 × 500 KT	12	12	0.630	7	1400	0.00
SS-N-18 mod 1	3 × 500 KT	224[b]	225	0.630	142	400	0.07
SS-N-18 mod 2	1 × 750 KT		75	0.825	62	900	0.08
SS-N-18 mod 3	5 × 500 KT		370	0.630	233	900	0.29
SS-N-20	6 × 100 KT	100	600	0.215	129	500	0.38
SS-N-23	10 × 100 KT	64	640	0.215	138	500[c]	0.41
Sea-based subtotals			2628		1255		1.60
Totals			9908		5543		51.70

[a] 1986/87 figure.
[b] Launchers are assumed to be equally distributed among different mods.
[c] CEP assumed to be the same as for SS-N-20.

Table 6.5 Comparison of U.S. & Soviet strategic nuclear forces

Country	Type	EMT	CMP	No. of warheads
U.S.	Land-based	1315	35.2×10^{-3}	2373
	Sea-based	960	3.4×10^{-3}	5632
	Totals	2275	38.6×10^{-3}	8005
USSR	Land-based	4288	50.1×10^{-3}	7280
	Sea-based	1255	1.6×10^{-3}	2628
	Totals	5543	51.7×10^{-3}	9908

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for both sides.[10] This parity was maintained in 1998 according to the number of strategic ballistic missiles (land- and sea-launched) declared by Russia and the U.S. as required by the Strategic Reduction Treaty I (START I).[11]

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Problems

6.1. Determine the percentage of RVs penetrating a three-layer defense against a MIRVed offensive system as a function of D/A and α, where the MIRVed RVs are deployed between the first and second layer. In this problem D denotes the number of intercepts, A the number of attacking missiles, and α the number of MIRVed RVs per missile.

6.2. Using the concept of strategic defense of Sec. 6.3 determine the fraction of missile silos saved when $D/d = 1$ and $A/d = 1$. What is the necessary defense strategy to achieve this?

6.3. For the defensive and offensive strategies defined in Sec. 6.3

$$\mathcal{D} = \tfrac{1}{3}d\,[0,\ 1,\ 1,\ 1]$$

$$\mathcal{O} = \tfrac{1}{2}d\,[0,\ 1,\ 1]$$

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determine the fraction of missile silos saved where d is the total number of defended silos. Is this defensive strategy the "best" possible? What are the values of A and D in terms of d, where A is the number of attacking missiles and D is the number of defensive interceptors?

6.4. Repeat Problem 6.3 but with

$$\mathcal{D} = \tfrac{1}{3}d\,[0,\ 1,\ 1,\ 1]$$

$$\mathcal{O} = \tfrac{1}{3}d\,[1,\ 1,\ 1]$$

Comment on the results.

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7
Theater Missile Defense

This chapter describes the analysis of the overall effectiveness of a layered defense against ballistic missiles proposed for the Theater Missile Defense (TMD) and the National Ballistic Missile Defense (NBMD). The analysis developed in this chapter is an extension of the strategic defense concept for intercontinental ballistic missiles discussed in Chapter 6. Three different types of targeting assignments are discussed that can be used for theater defenses against missiles and tanks: (1) random tactics, (2) uniform tactics, and (3) shoot-look-shoot tactics. In general, the first two are applicable to rocket-propelled interceptors, while the third can be used for directed energy weapon systems (e.g., airborne laser weapon). A detailed comparison is made between the first two tactics for either single-layer or two-layer defenses. In addition, the probability of zero penetrations of attacking missiles is discussed.

7.1 Concept for the Theater Missile Defense

The concept of the layered defense described in Chapter 6 for the defense against intercontinental ballistic missiles (ICBMs) can be applied for shorter range ballistic missiles or cruise missiles in a military theater of operations or even for land/air battles. The projected U.S. defenses against ballistic missiles fall into two main categories: National Missile Defense (NMD) and Theater Missile Defense (TMD). They are designed to intercept long-range or medium-range to short-range missiles, respectively. As of 1999, the United States employs a version of theater missile defense with its Patriot system but does not operate any of the more advanced TMD systems currently under development. Any future NMD will evolve through the development of the present TMD systems. With the spectacular advancements in missile technology and supporting detection, tracking, and control systems, any future upgrades to the TMD systems will be able to provide a limited defense against intercontinental missiles.

At the present time, the U.S. has five major TMD programs: (1) PATRIOT Advanced Capability-3 (PAC-3), (2) Navy Area Theater Ballistic Missile Defense (TBMD), (3) Theater High-Altitude Area Defense (THAAD), (4) Navy, Theater Wide, Medium Extended Air Defense System (MEADS), and (5) Space Based Laser (SBL), as shown in Table 7.1. The latter consists of the YAL-1A Attack Laser Aircraft using a modified 747 Boeing aircraft. This aircraft will be the world's first operational airborne high-energy laser system. It will be used to destroy Theater Ballistic Missiles (TBMs) in their boost or very earliest phase of their flight, when the TBMs display bright plumes and are under tremendous dynamic stresses, making them vulnerable to laser weapons. The airborne laser weapon can target TBMs hundreds of miles away, and thus can fly over friendly territory to kill TBMs as they are launched. The weapon is a megawatt class Chemical Oxygen-iodine laser

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Table 7.1 The U.S. Missile Defense Programs[1,2]

No.	System name	Type of warhead	Radius of defended area
	Lower Tier Point Defense		
(1)	Patriot PAC-2	Blast	10–15 km
	Patriot PAC-3	Hit-to-kill	40–50 km
(2)	Navy Area Theater Ballistic Missile Defense (TBMD)	Blast fragments	50–100 km
	Upper Tier Area Defense		
(3)	Theater High Altitude Area Defense (THAAD)	Hit-to-kill	>100 km
(4)	Navy Theater Wide Medium Extended Air Defense System (MEADS)	Hit-to-kill	>300 km
	Airborne Defense		
(5)	Airborne Laser, YAL-1A Attack Laser Aircraft	Directed energy (Laser)	>300 km

(COIL). It contains 14 COIL modules and has sufficient chemical fuel for 20–40 TBM kills.

A typical scenario for a two-layer defense would be a TMD system for South Korea with two tiers: an upper tier area defense system with THAAD types interceptors and a lower tier point defense provided by Patriot PAC-3 missiles. A conceptual arrangement for such defense is shown in Fig. 7.1. Another application of an layered defense may be considered against an armor attack where attack aircraft with air-to-ground missiles are used to destroy tanks, as shown in Fig. 7.2.

7.2 Probability of Penetration of Missiles (or Tanks)

Types of Defensive Tactics

In theater missile defense there are essentially three types of target assignment that can be considered: (1) random, (2) uniform, and (3) shoot-look-shoot. In random assignments target acquisition assigns targets [attacking missiles or armored vehicles (e.g. tanks)] in a unordered manner that implies that there is very little or no overall coordination. This means that individual targets are assigned as soon as they appear on the radar screens. In uniform assignments the overall attack assessment is carried out first before the defensive missiles (interceptors) are fired to minimize any unnecessary overlap. This tactic is, of course, much more effective than the random firing. The third type, shoot-look-shoot tactics, is the most efficient way of using defensive resources. Here after the target is acquired the interceptor is fired and an assessment of damage is made. If the target is not destroyed, then a second interceptor is fired to the same target. It should, however, be noted that there is a time lag between target acquisition and target damage assessment that may not be acceptable for the case of ballistic missiles. For stationary targets, or slow-moving targets such as tanks, the shoot-look-shoot target assignment is the

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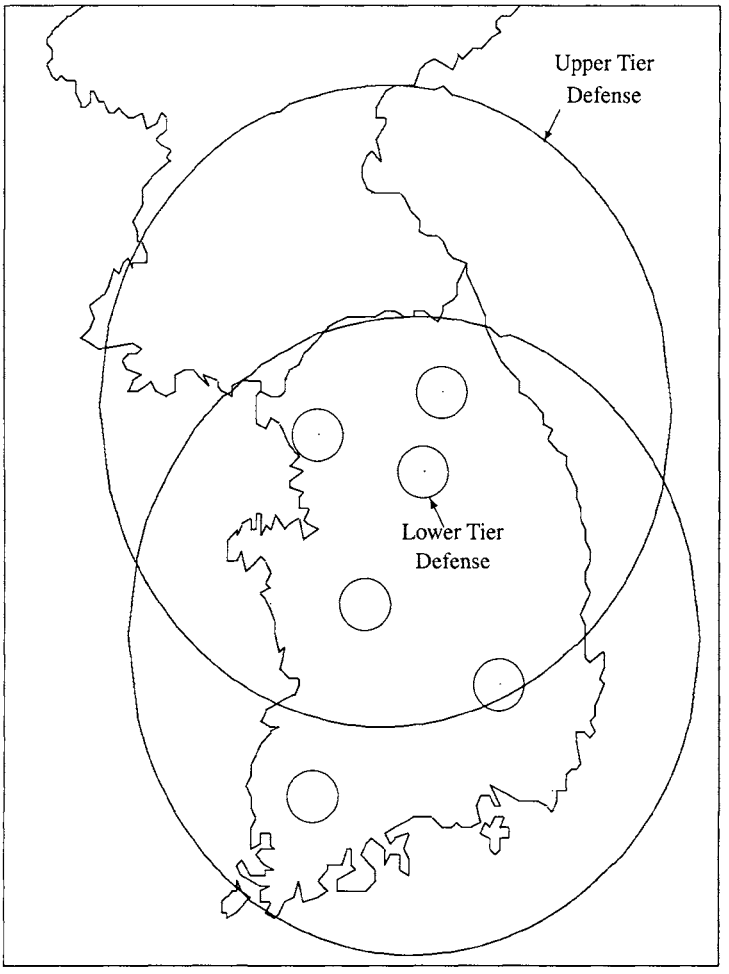


Fig. 7.1 Two-layer defense (upper and lower tier) for South Korea.

most effective method for missile interceptors. On the other hand, for direct energy weapons (e.g., laser beams), the shoot-look-shoot tactic is the best choice.

If the random or uniform assignment of targets is used, then the overall defense effectiveness can be improved through the use of layered defense. The subsequent sections provide a detailed analysis of one- and two-layer defenses, including an assessment of the zero probability of missile penetration.

Random Assignments of Targets

It will be assumed that the single-layer defense with N interceptors is attacked by W missiles (warheads) and the defense assigns interceptors against the incoming warheads in an unordered (random) manner. This, of course, is the worse possible

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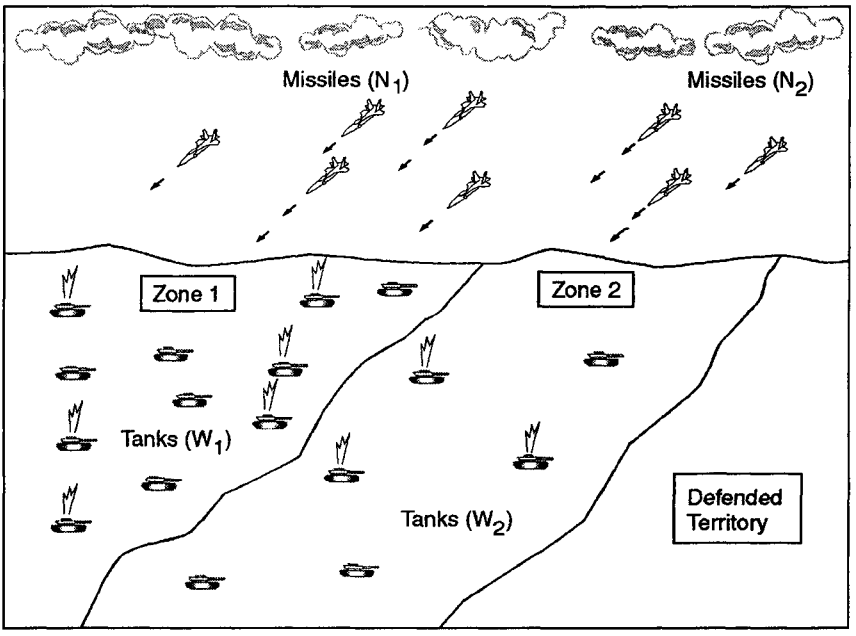


Fig. 7.2 Two-zone defense against armor attack in an air/ground battle.

tactic, but it is used here to provide a measure of effectiveness for comparison against the other two tactics: uniform and shoot-look-shoot assignments.

From Eq. (6.8) for the case of a single-layer defense for which $M_0 = W$ and $D_1 = pN$ it follows that the number of missiles penetrating the defense is given by

$$W_1 = W \exp\left(-\frac{pN}{W}\right) \tag{7.1a}$$

where p is the probability of a single interceptor killing one warhead and the subscript 1 is used to denote the first layer. Hence, the probability of penetration of W_1 warheads is given by

$$\frac{W_1}{W} = \exp\left(-\frac{pN}{W}\right) \tag{7.1b}$$

For the case of a two-layer defense with an optimum distribution of defensive intercepts N_1 and N_2, it follows from Eqs. (6.8), (6.17), and (6.2) that

$$W_1 = W \exp\left(-\frac{p_1 N_1}{W}\right) \tag{7.2}$$

$$W_2 = W_1 \exp\left(-\frac{p_2 N_2}{W_1}\right) \tag{7.3}$$

$$p_2 N_2 = W - W_1 \tag{7.4}$$

$$D_0 = p_1 N_1 + p_2 N_2 \quad \text{(Defensive potential)} \tag{7.5}$$

If the probability of destruction of warheads is the same in both layers (i.e.,

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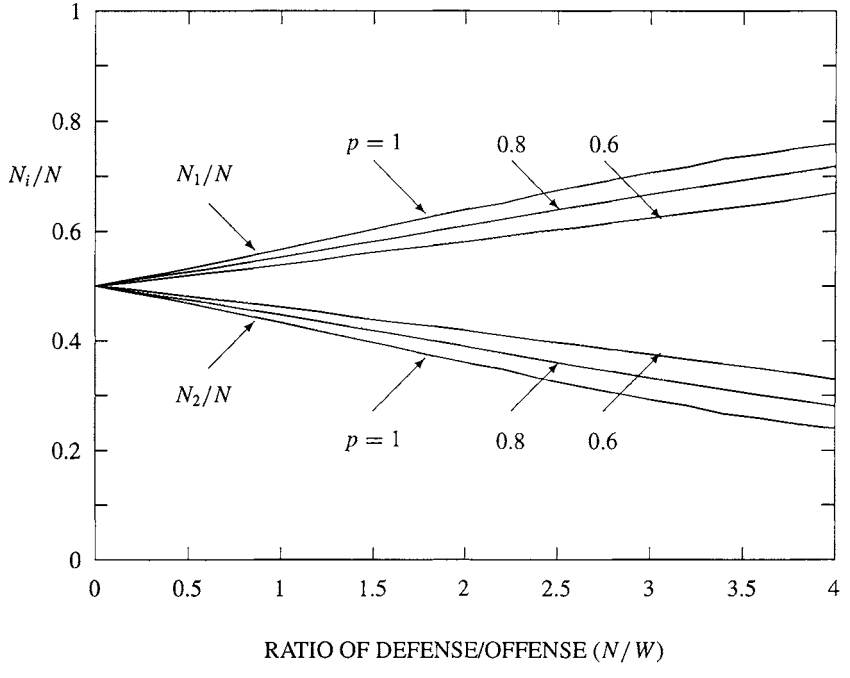


Fig. 7.3 Variation of the optimum ratio of N_1/N and N_2/N with N/W and p; random assignment of targets for a two-layer defense.

$p_1 = p_2 = p$), then

$$W_1 = W \exp\left(-\frac{pN_1}{W}\right) \tag{7.6}$$

$$W_2 = W_1 \exp\left(-\frac{pN_2}{W_1}\right) \tag{7.7}$$

$$pN_2 = W - W_1 \tag{7.8}$$

$$D_0 = pN_1 + pN_2 = pN \tag{7.9}$$

The first three nonlinear equations with three variables W_1, W_2, and N_1 or N_2 can be solved through an iterative process. The optimum values for N_1/N and N_2/N for which W_2 is a minimum are shown in Fig. 7.3 for several values of p. For the special cases when one of the layers becomes inoperative,

$$W_1 = W, \qquad W_2 = W \exp\left(-\frac{pN}{W}\right), \qquad N_1 = 0, \qquad N_2 = N \tag{7.10}$$

and

$$W_1 = W \exp\left(-\frac{pN}{W}\right), \qquad W_2 = W_1, \qquad N_1 = N, \qquad N_2 = 0 \tag{7.11}$$

Although theoretically for an optimum defense, the ratio N_1/N should be adjusted for a given value of N/W one can use a constant ratio for a wide range of N/W from

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0 to 3. For example, the results for $N_1/N = 1/2$ or $N_1/N = 2/3$ are practically indistinguishable from the optimum solution. This means that the defense tactics can be simplified because the ratio N_1/N need not be adjusted for each separate attack.

For non-optimum solutions, Eqs. (7.6) and (7.7) can be modified as

$$W_1 = W \exp\left(-\frac{p_1 N_1}{N}\frac{N}{W}\right) \tag{7.12}$$

and

$$W_2 = W_1 \exp\left(-\frac{p_2 N_2}{W_1}\right) = W \exp\left(-\frac{p_1 N_1}{N}\frac{N}{W}\right) \exp\left(-\frac{p_2 N_2}{N}\frac{N}{W_1}\right)$$

$$= W \exp\left(-\frac{p_1 N_1}{N}\frac{N}{W}\right) \exp\left[-p_2\left(1-\frac{N_1}{N}\right)\frac{N}{W} \exp\left(\frac{p_1 N_1}{N}\frac{N}{W}\right)\right] \tag{7.13}$$

where

$$N = N_1 + N_2 \tag{7.14}$$

Using Eqs. (7.1b) and (7.7) for one- and two-layer defense, the values of W_1/W and W_2/W are plotted in Fig. 7.4 as functions of N/W. This illustrates clearly the superiority of a two-layer defense over a single-layer defense. For example, when $N/W = 3$ the improvement is about three orders of magnitude. The fundamental reason for this that a two-layer defense represents an aggregated shoot-look-shoot capability in which the second layer interceptors are directed to the survivors from the first (see Chapter 5 for comparisons of random and shoot-look-shoot assignments of targets).

Uniform Assignments of Targets

Starting with the results of Section 5.5, it follows from Eq. (5.86) that for a single-layer defense

$$W_1 = W(1-p)^I(1-pF) \tag{7.15}$$

where

$$I = \text{int}\left(\frac{N}{W}\right) \tag{7.16}$$

$$F = \frac{N}{W} - \text{int}\left(\frac{N}{W}\right) \tag{7.17}$$

and I represents the integer function int(x) = largest integer that is less than or equal to x.* Hence

$$\frac{W_1}{W} = (1-p)^{\text{int}\left(\frac{N}{W}\right)}\left[1 - p\frac{N}{W} + p\,\text{int}\left(\frac{N}{W}\right)\right] \tag{7.18}$$

*The function int(x) is represented by the integer function INT(x) in BASIC and the floor function floor(x) in MATLAB.

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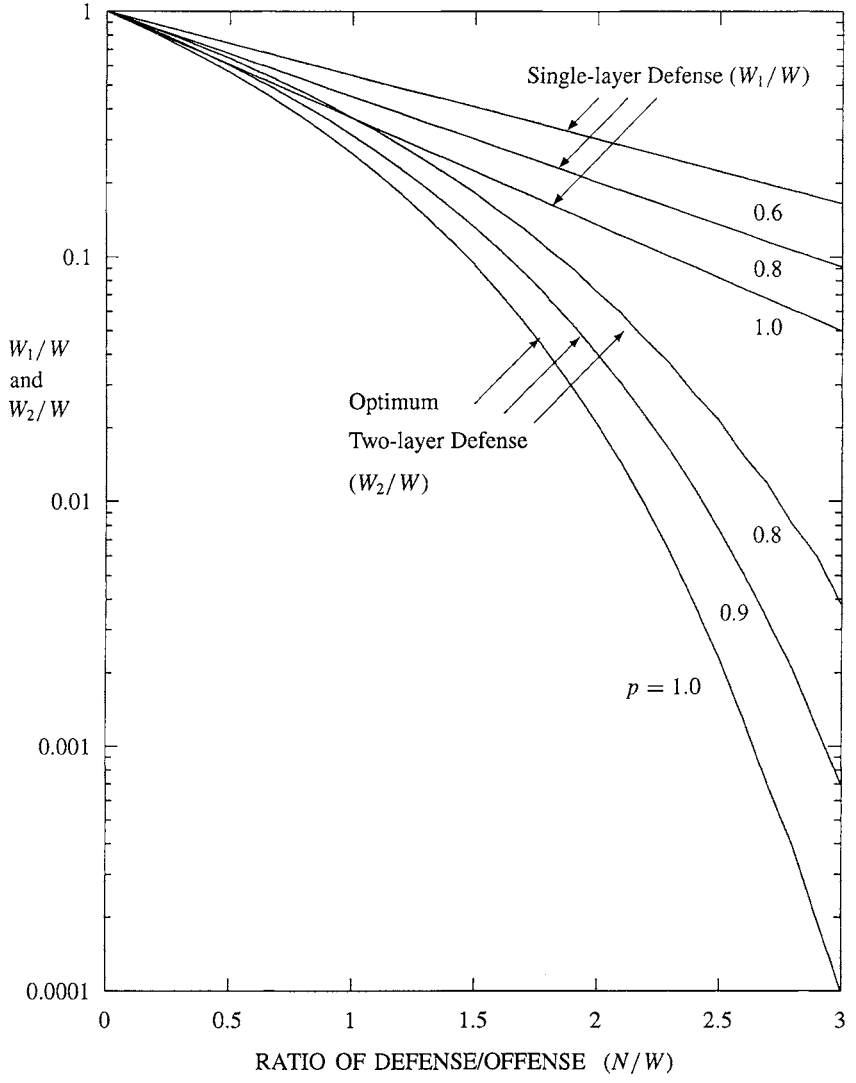


Fig. 7.4 Comparison of single- and two-layer defense effectiveness for random assignments of targets.

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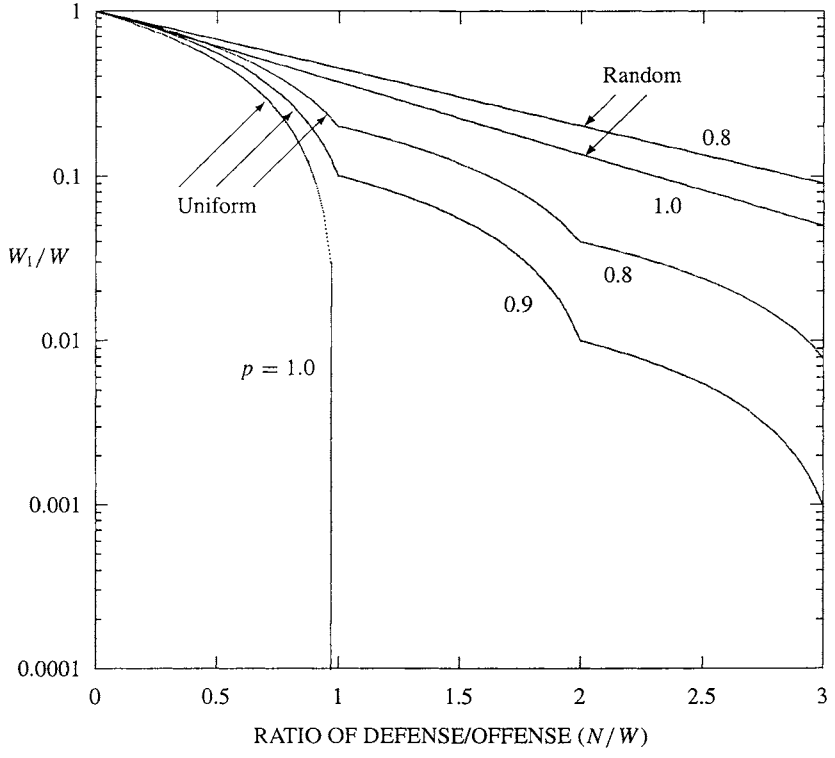


Fig. 7.5 Comparison of defense effectiveness for random and uniform target assignments; single-layer defense.

When $N/W < 1$, then $\text{int}(N/W) = 0$ and

$$\frac{W_1}{W} = 1 - p\frac{N}{W} \tag{7.19}$$

A comparison of defense effectiveness for random and uniform target assignments in a single-layer defense is shown in Fig. 7.5. This illustrates clearly the advantage of using uniform target assignments as compared with the random assignments.

For a two-layer defense the governing equations are obtained from Eq. (7.18). Hence, for the first layer

$$W_1 = W(1 - p_1)^{\text{int}\left(\frac{N_1}{W}\right)}\left[1 - p_1\frac{N_1}{W} + p_1\text{int}\left(\frac{N_1}{W}\right)\right] \tag{7.20}$$

Introducing two new expressions

$$\frac{N_1}{W} = \frac{N_1}{N}\frac{N}{W} \tag{7.21}$$

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$$\frac{N_2}{W_1} = \frac{(N - N_1)}{N}\frac{N}{W_1} = \left(1 - \frac{N_1}{N}\right)\frac{N}{W}\frac{W}{W_1} \tag{7.22}$$

$$\frac{W_1}{W} = (1 - p_1)^{\text{int}\left(\frac{N_1}{N}\frac{N}{W}\right)}\left[1 - p_1\frac{N_1}{N}\frac{N}{W} + p_1\text{int}\left(\frac{N_1}{N}\frac{N}{W}\right)\right] \tag{7.23}$$

and

$$\frac{W_2}{W} = \frac{W_1}{W}(1 - p_2)^{\text{int}\left[(1-\frac{N_1}{N})\frac{N}{W}\frac{W}{W_1}\right]} \times \left[1 - p_2\left(1 - \frac{N_1}{N}\right)\frac{N}{W}\frac{W}{W_1} + p_2\text{int}\left(\left(1 - \frac{N_1}{N}\right)\frac{N}{W}\frac{W}{W_1}\right)\right] \tag{7.24}$$

To calculate W_1/W and W_2/W values for N, N_1, W, p_1 and p_2 must be specified where p_1 and p_2 are the probabilities of single interceptor kill in layers 1 and 2 respectively.

When $N/W < 1$ then the equations for W_1 and W_2 can be simplified and

$$\frac{W_1}{W} = 1 - p_1\frac{N_1}{N}\frac{N}{W} \tag{7.25}$$

and

$$\frac{W_2}{W} = 1 - p_1\frac{N_1}{N}\frac{N}{W} - p_2\left(1 - \frac{N_1}{N}\right)\frac{N}{W} \tag{7.26}$$

For the special case of $p_1 = p_2 = p$

$$\frac{W_1}{W} = 1 - p\frac{N_1}{N}\frac{N}{W} \tag{7.27}$$

and

$$\frac{W_2}{W} = 1 - p\frac{N}{W} \tag{7.28}$$

A comparison of defense effectiveness for random and uniform target assignments in a two-layer defense is shown in Fig. 7.6. Just as in the single-layer defense, the uniform target assignments reduce the missile penetration. For example, when $N/W = 3$ the penetration W_1/W is reduced by a factor of 10, from about 0.1 to 0.001.

Shoot-Look-Shoot Assignments of Targets

The appropriate formula for computing the probability of penetration of attacking missiles in a single layer defense for the shoot-look-shoot tactics can be obtained from Eqs. (5.99), (5.104), and (5.105). Changing the symbol T to W in order to apply the previous equation to the present analysis, it follows that the probability of missile penetration is given by

$$W_1/W = \sum_{i=0}^{W}\left(1 - \frac{i}{W}\right)P_i \qquad \text{for} \quad N \geq W \tag{7.29}$$

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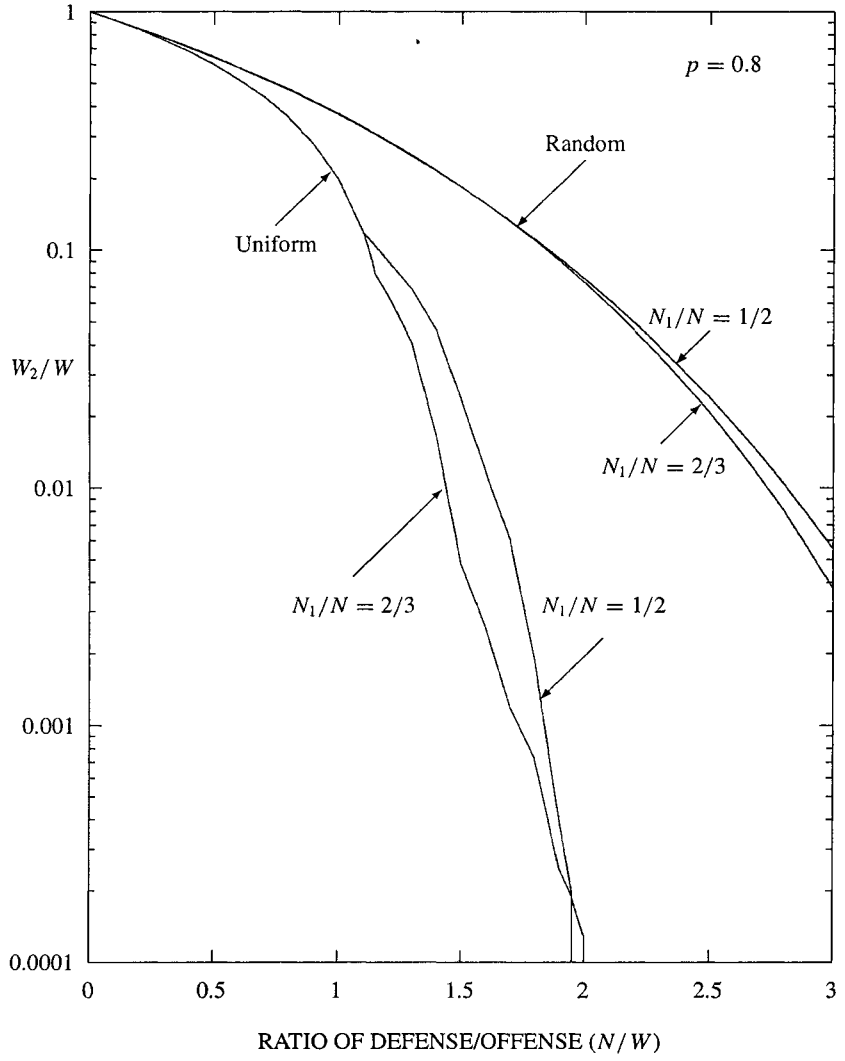


Fig. 7.6 Comparison of defense effectiveness for random and uniform target assignments for specified ratios N_1/N; two-layer defense.

where

$$P_i = \binom{N}{i} p^i (1-p)^{N-i} \qquad \text{for} \quad N \geq W \tag{7.30}$$

When $N < W$, the corresponding equation for W_1/W is given by

$$\frac{W_1}{W} = 1 - \frac{pN}{W} \tag{7.31}$$

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The corresponding equations for the case of a two-layer defense may be developed as it was shown here for the random and uniform target assignments. It should, however, be pointed out that the shoot-look-shoot tactics may not be appropriate at all unless the maximum velocity of the intercept missiles is much greater than that of the attacking missiles. The reason for this is that the attacking missile would be too close to its target for the firing of a second defensive missile. Of course the shoot-look-shoot tactics are possible with any direct energy laser weapons, such as the YAL-1A Attack Laser Aircraft.

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7.3 Probabilities of Zero Penetration

The probability of a single missile destroyed by the defense will be designated as Q. It will be also assumed for this analysis that this probability is the same for all attacking missiles (or warheads). Thus the probability of a single missile penetration is $P = 1 - Q$. This should not be confused with p, which is the probability of a single defense intercept destroying an attacking missile. It should be noted that using previous definitions

$$P = \frac{W_1}{W} \quad \text{or} \quad \frac{W_2}{W} \tag{7.32}$$

for single- and two-layer defenses, respectively.

The attacking missiles can have a single warhead (unitary missiles) or they may contain multiple warheads, in which case individual warheads are counted as separate missiles. If W unitary missiles (or warheads) are attacking, then the probability of all missiles killed by the defense is given by

$$P_W(0) = QQQ\cdots(W - \text{times}) = Q^W \tag{7.33}$$

The probability that the first missile penetrates the defense and the rest do not is

$$PQQ\cdots Q$$

where Q appears $(W-1)$ times. The probability that the second missile penetrates the defense while the other missiles do not is

$$QPQQ\cdots Q$$

where Q also appears $(W-1)$ times. And finally, the probability that the Wth missile penetrates the defense while the rest do not is

$$QQ\cdots QQP$$

Thus the probability of penetration of one missile from among W attacking missiles is the sum of all of the above probabilities, i.e.,

$$P_W(1) = WPQ^{W-1} \tag{7.34}$$

Therefore, using similar reasoning, the general case of the probability of x missiles penetrating the defense from among a total of W missiles can now be considered. The total number of events representing x penetrations of W missiles is equal to the number of ways x objects can be arranged in W spaces, which is given by the combinatorial formula

$$\binom{W}{x} = \frac{W!}{x!(W-x)!} \tag{7.35}$$

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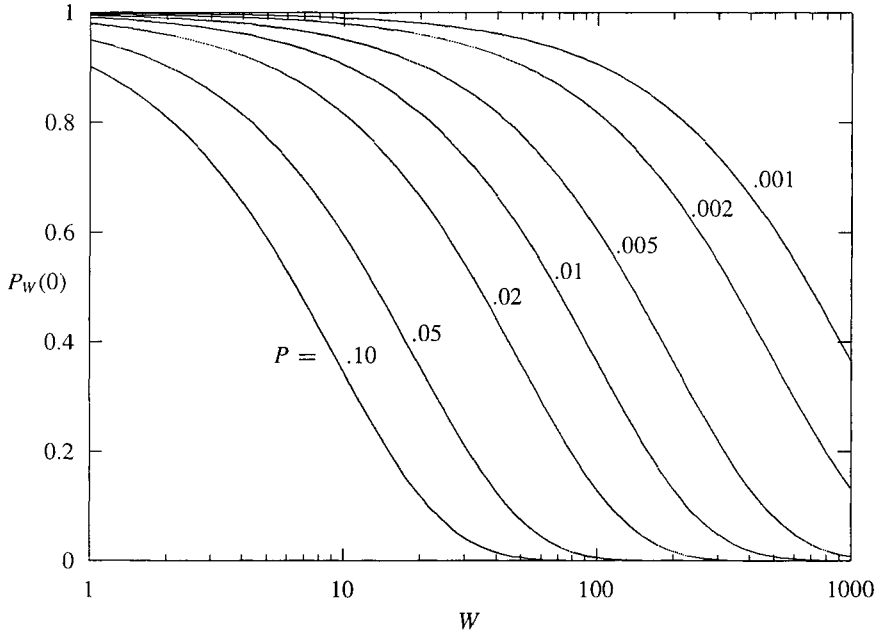


Fig. 7.7 Probability of zero penetrations as a function of the number of warheads W and the single missile penetration probability P.

Hence, the probability of x penetrations from among the W missiles is equal to

$$P_W(x) = \binom{W}{x} P^x Q^{W-x} = \binom{W}{x} (1-Q)^x Q^{W-x}$$

$$= \frac{W!}{x!(W-x)!}(1-Q)^x Q^{W-x} \tag{7.36}$$

and the probability of zero penetration from among the W missiles (i.e. $x = 0$ when all missiles are destroyed) is given by[3]

$$P_W(0) = Q^W = (1-P)^W \tag{7.37}$$

It should be noted that P is a function of the target assignment tactics and the ratio (N/W). Fig. 7.7 shows $P_W(0)$ as a function of the number of attacking missiles (W) for different values of P. For a given single missile probability of penetration p, the probability of zero penetrations $P_W(0)$ decreases with an increase in attacking warheads (W), and for a given W it increases with decreasing values of p.

From Eq. (7.45) it follows that

$$\ln(P_W(0)) = W \ln(1-P) \tag{7.38}$$

Hence

$$W = \frac{\ln(P_W(0))}{\ln(1-P)} \tag{7.39}$$

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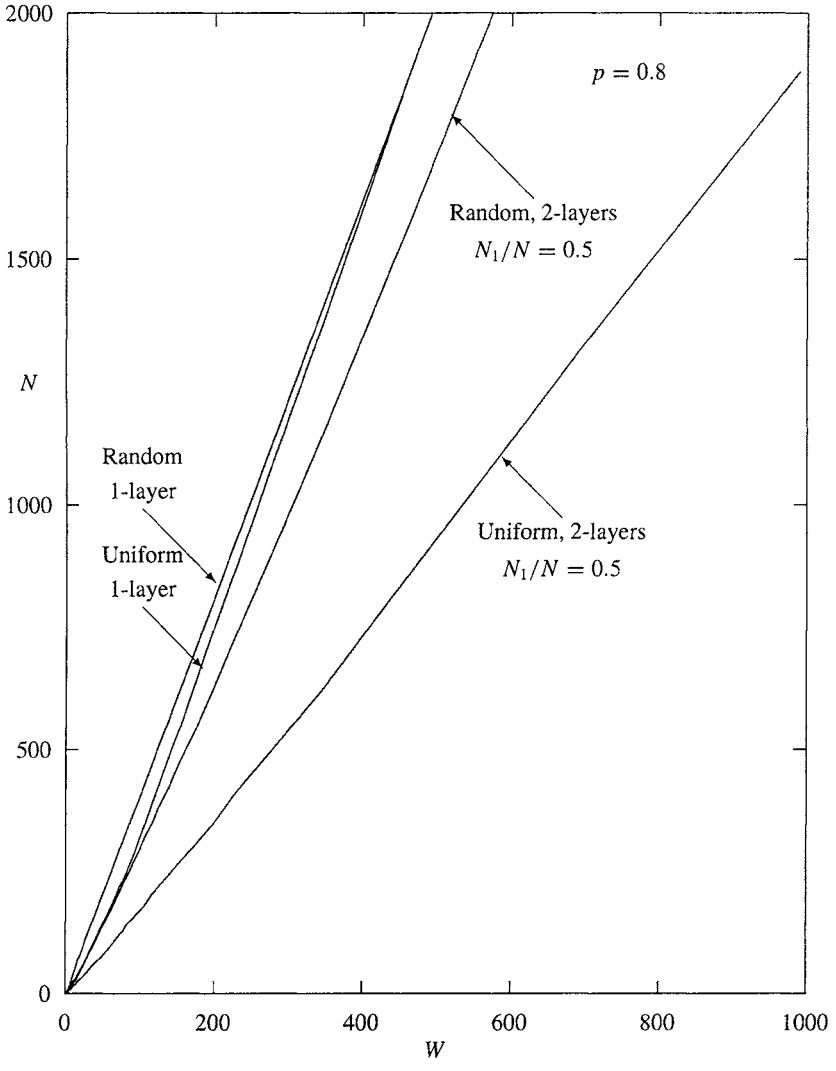


Fig. 7.8 The required number of intercepts N to achieve 50% probability of zero penetrations for a given size of attacking warheads W and different defense scenarios.

Because P is a function of the target assignment tactics and the ratio N/W, it is possible to obtain numerically a relationship between W and N for a given value of the zero probability of penetration $P_W(0)$. Fig. 7.8 shows the variation of N with W for $P_W(0) = 0.5$ for random and uniform tactics and one- and two-layer defenses. The chart was obtained through an iterative numerical solution. It demonstrates clearly the advantages of using the uniform versus the random assignments tactics. Likewise, it also shows the advantage of two-layer defense tactics.

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Problems

7.1. Verify Eqs. (7.9) and (7.10).

7.2. Prove Eqs. (7.25) to (7.28).

7.3. For a two-layer defense and random assignment of targets, calculate variation of W_2/W with N/W for $N_1/N = 1/3$ and $p = 0.8$.

7.4. For a two-layer defense and uniform assignment of targets, calculate variation of W_2/W with N/W for $N_1/N = 1/3$ and $p = 0.8$.

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8
Tactical Engagements of Heterogeneous Forces

In this chapter the mathematical modeling of combat engagements by heterogeneous tactical forces is presented and a numerical method of determining the variation of force strengths with time is derived. The modeling of combat is based on the concept of attrition coefficients, which are functions of weapon characteristics and the allocation of weapons engaged against a specified opposing tactical force. Three models are discussed in this chapter: the directed fire model, the area fire model, and the guerrilla warfare model. The directed fire model is represented by a set of first-order linear differential equations in matrix form. The details of an analytical solution to these equations for constant attrition coefficients are given in Appendix C. The analytical solution naturally leads to the concept of the superiority parameter, which represents a measure of the effectiveness of the opposing forces. This parameter is a function of both the quantitative and qualitative strengths of the opposing forces. It is defined as a product of the transpose of the left dominant eigenvector of the governing matrix of the attrition coefficients and the force strength vector. This parameter is used to establish superiority criteria for combat engagements. The application of this concept to heterogeneous forces is illustrated for the case of a "one-on-two" tactical engagement. The formulation of the second model for the area fire engagement and the third model for the guerrilla warfare results in sets of nonlinear differential equations. When these equations are rewritten in matrix notation, a special operation of changing a column matrix into a diagonal matrix must be introduced and the resulting matrix equations can then be solved numerically (See Appendix E).

8.1 Directed Fire "Many-on-Many" Engagements: Numerical Solutions

Numerical solutions to the Lanchester equations of combat for the heterogeneous forces ("many-on-many" engagements) can be obtained for specific force configurations and effective rates of fire. The governing equations for a directed fire model are given by

$$\frac{d\mathbf{M}}{dt} = -\mathbf{B}\,\mathbf{N} \tag{8.1}$$

$$\frac{d\mathbf{N}}{dt} = -\mathbf{A}\,\mathbf{M} \tag{8.2}$$

The $\mathbf{M}$ and $\mathbf{N}$ matrices are column matrices given by

$$\mathbf{M} = \{M_1 \quad M_2 \cdots M_B\} \tag{8.3}$$

$$\mathbf{N} = \{N_1 \quad N_2 \cdots N_R\} \tag{8.4}$$

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in which $M_1 \cdots M_B$ and $N_1 \cdots N_R$ represent the quantities of separate homogeneous forces of the same type. The braces are used for convenience to denote column matrices, and the subscripts B and R represent here the number of such separate forces for the Blues and Reds. The mathematical model for such engagements is represented as the state variables M_i and N_j interacting with each other through the appropriate attrition coefficients A_{ij} and B_{ij} of **A** and **B** as shown in Fig. 8.1. The attrition coefficients take into account the firing rate effectiveness and the allocation of weapons. The matrices **A** and **B** are nonnegative, a property that will be used in the subsequent analysis. Also

$$M_i,\ N_i > 0 \tag{8.5}$$

and when this condition is not satisfied, i.e. when one of the forces is annihilated (when a particular M_i or $N_i = 0$), Eqs. (8.1) and (8.2) must be reconfigured by striking out the appropriate rows and columns.

If contributions from a typical column in **A** (jth column) in Eq. (8.2) are considered, it follows then that

$$\begin{aligned}
\frac{dN_1}{dt} &= \cdots - A_{1j}M_j - \cdots \\
\frac{dN_2}{dt} &= \cdots - A_{2j}M_j - \cdots \\
&\vdots \qquad\qquad \vdots \\
\frac{dN_i}{dt} &= \cdots - A_{ij}M_j - \cdots \\
&\vdots \qquad\qquad \vdots \\
\frac{dN_R}{dt} &= \cdots - A_{Rj}M_j - \cdots
\end{aligned}$$

which indicates that M_j is engaging simultaneously the opposing forces $N_1 \cdots N_R$. Consequently, the coefficients A_{ij} must include the allocation factors, i.e. the allocation of the force M_j among all the opposing forces $N_1 \cdots N_R$. Hence the coefficient A_{ij} must be computed from

$$A_{ij} = a_{ij}\alpha_{ij} \tag{8.6}$$

where

a_{ij} = inherent weapon system kill rate when M_j is engaging only N_i

and

α_{ij} = weapon allocation factor that represents the fraction of M_j assigned to engage N_i ($0 \leq \alpha_{ij} \leq 1$).

It is clear that these fractions must satisfy the equation

$$\sum_i \alpha_{ij} \leq 1 \tag{8.7}$$

Of course, equations similar to (8.6) and (8.7) must also be true for the B_{ij} coefficients. Some more refined analyses use additional factors in Eq. (8.6) to represent the actual fraction of the allocated weapons to engage M_i or N_i.[1]

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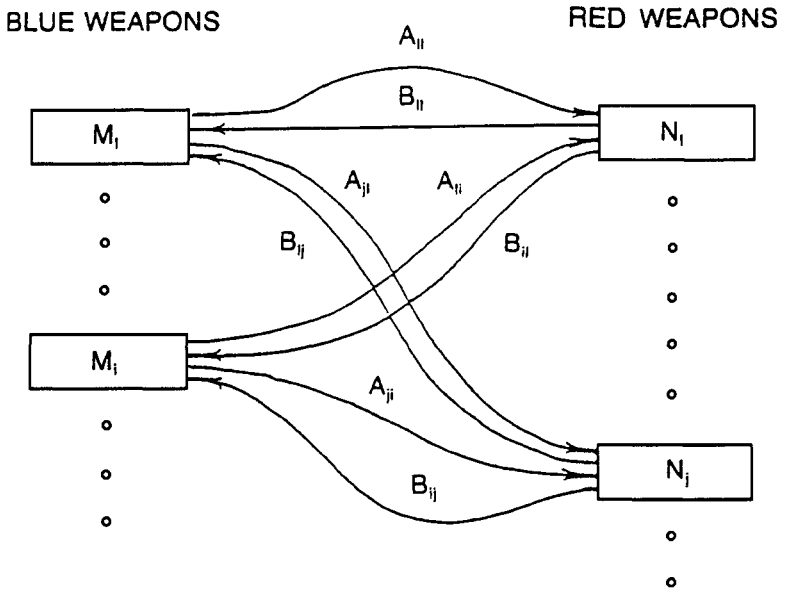


Fig. 8.1 Model of heterogeneous force engagements (many-on-many).

Equations (8.1) and (8.2) can be combined into a single matrix equation

$$\frac{d\mathbf{X}}{dt} = \dot{\mathbf{X}} = \mathbf{C}\mathbf{X} \tag{8.8}$$

where

$$\mathbf{X} = \{\mathbf{M} \quad \mathbf{N}\} \tag{8.9}$$

and

$$\mathbf{C} = \begin{bmatrix} \mathbf{O} & -\mathbf{B} \\ -\mathbf{A} & \mathbf{O} \end{bmatrix} \tag{8.10}$$

There are, of course, many different schemes for the numerical integration of Eq. (8.8), the simplest of which is the formula

$$\frac{1}{\Delta t}(\mathbf{X}_{n+1} - \mathbf{X}_n) = \mathbf{C}\mathbf{X}_n$$

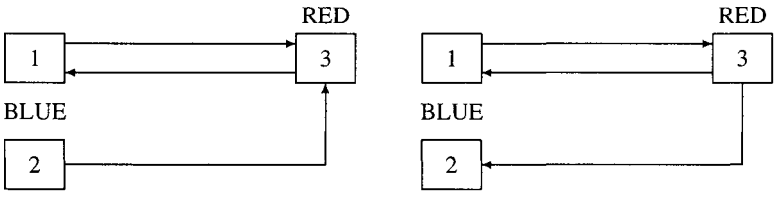
which leads to

$$\mathbf{X}_{n+1} = (\mathbf{I} + \Delta t\, \mathbf{C})\mathbf{X}_n \tag{8.11}$$

where $\mathbf{X}_n$ and $\mathbf{X}_{n+1}$ represent the values of the state vector $\mathbf{X}$ at times t_n and t_{n+1} and $\Delta t = t_{n+t} - t_n$ is the time interval used in their numerical integration. If during the time interval Δt some forces are added or withdrawn from the battlefield then

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a) Case for $\dot{X}_2 = 0$. b) Case for an inactive X_2-Force.

Fig. 8.2 Examples of Two-on-One engagements.

Eq. (8.11) can be modified as

$$\mathbf{X}_{n+1} = (\mathbf{I} + \Delta t\, \mathbf{C})\mathbf{X}_n + \mathbf{D}_n \tag{8.12}$$

where $\mathbf{D}_n$ is the column of force additions or withdrawals. Many of the complex computer models of force engagements are based on the simple representation of Eq. (8.12), except that the matrix of attrition coefficients $\mathbf{C}$ is usually a function of time.[1] A computer program for the solution of Eq. (8.12) can be found in Ref. 9 or in Appendix E.8.

If the ith row in $\mathbf{C}$ contains all zeros, then this means that the corresponding $\dot{X}_i = 0$ and the X_i-force is not being fired upon, as shown in Fig. 8.2a for the simple engagement of two Blue forces and one Red force. For such cases, the ith row in $\mathbf{C}$ must be eliminated, and the ith column multiplied by the constant force X_i must be moved to the matrix $\mathbf{D}_n$. Thus the fundamental equation for these cases is Eq. (8.12) where the $\mathbf{C}$ is matrix reduced from an $m \times m$ square matrix to an $(m-1) \times (m-1)$ matrix and $\mathbf{D}_n$ is reduced from an $m \times 1$ column matrix to an $(m-1) \times 1$ matrix.

If the jth column in $\mathbf{C}$ contains all zeros, then the X_j-force is inactive, as shown in Fig. 8.2b. For these cases, Eq. (8.11) is still applicable without any modifications.

8.2 Aggregated Forces

As discussed by Taylor,[1] several authors have introduced aggregated forces to obtain a better insight into the effects of varying force size, the weapon allocation, and the weapon effective firing rate represented by Eq. (8.8). This approach involved essentially computation of two quantities: the aggregated force strengths F_B and F_R for the Blue and Red forces, respectively, which may be defined as

$$F_B = \bar{\mathbf{v}}^T \mathbf{M} \tag{8.13}$$

$$F_R = \bar{\mathbf{w}}^T \mathbf{N} \tag{8.14}$$

where $\bar{\mathbf{v}}$ and $\bar{\mathbf{w}}$ are positive vectors representing relative values of individual weapons, while $\mathbf{M}$ and $\mathbf{N}$ are column matrices representing the force (weapon) numbers for the Blue and Red forces, respectively. Starting with the assumptions (8.13) and (8.14), a number of authors used the eigenvector method of computing $\bar{\mathbf{v}}$ and $\bar{\mathbf{w}}$ [see Eqs. (8.21) and (8.22)]. In this method, however, both vectors could only be determined up to a constant multiplier. A simple scaling relationship was

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proposed by Dare and James[2] as $\sum \bar{v}_i = \sum \bar{w}_i = 1$. Other plausible scaling methods were introduced by different authors. For example, Taylor,[1] in his book on Lanchester models of warfare, mentioned three other schemes for computing the scaling factors for the weapon value vectors $\bar{\mathbf{v}}$ and $\bar{\mathbf{w}}$.[3–6]

This chapter introduces a method of determining the eigenvectors $\bar{\mathbf{v}}$ and $\bar{\mathbf{w}}$ through a mathematical representation of the complete analytical solution and the application of the dominant left eigenvector $\mathbf{q}_1$ of the matrix of attrition coefficients.[7] This approach leads also to a measure of effectiveness $\mathbf{q}_1{}^T\mathbf{X}_0$ and its corresponding nondimensional superiority parameter S_0, both of which determine a priori the direction of the outcome of the battle engagement without the necessity of solving the differential equations representing the engagement. The application of this method of analysis is demonstrated in Sec. 8.5 for the case of a tactical engagement involving three different weapons: one Blue and two Red. In order to illustrate the nature of the analytical solution and, in particular, the presence of a dominant term in the solution that is used to derive the superiority parameter S_0, a complete discussion of the general analytical method is included in Appendix C.

8.3 Superiority Parameters[7]

The governing equations for the heterogeneous force combat engagements, as shown in Sec. 8.2, are given by

$$\frac{d\mathbf{X}}{dt} = \dot{\mathbf{X}} = \mathbf{C}\mathbf{X} \tag{8.8}$$

To derive a characteristic parameter that would indicate the trend of the solution of Eq. (8.8), that is whether or not the Blue force can expect to win the initial engagement, the left-handed eigenvector $\mathbf{q}$ of the matrix $\mathbf{C}$ and its corresponding eigenvalue λ are introduced for the subsequent analysis. The left-handed eigenvector $\mathbf{q}$ and the eigenvalue λ are determined from

$$\mathbf{C}^T\mathbf{q} = \lambda\mathbf{q} \tag{8.15}$$

The eigenvector $\mathbf{q}$ can be partitioned into rows corresponding to the Blue and Red forces, respectively, so that

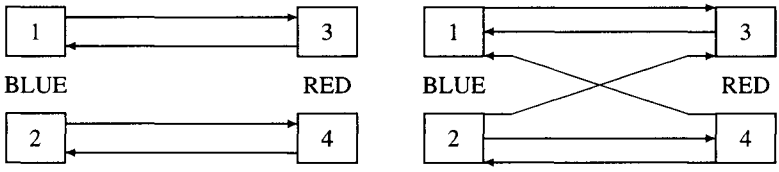
$$\mathbf{q} = \{c_1\mathbf{v} \quad c_2\mathbf{w}\} \tag{8.16}$$

where $\mathbf{v}$ and $\mathbf{w}$ are the column vectors with c_1 and c_2 as constant multipliers. The vectors $\mathbf{v}$ and $\mathbf{w}$ are determined with an arbitrary multiplier based on a convenient computational scheme (e.g., largest coefficient equal to unity).

It should be observed that Eq. (8.8) must represent a set of *coupled* equations. Uncoupled sets are not admissible in the present analysis because of the complete independence of the different sets of equations representing unrelated (uncoupled) tactical engagements. Examples of uncoupled and coupled engagements are shown in Fig. 8.3, where arrows indicate the direction of fire. For these examples, the corresponding matrices of attrition coefficients are shown by Eqs. (8.17a) and

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a) Uncoupled engagements. **b) Coupled engagements.**

Fig. 8.3 Examples of uncoupled and coupled tactical engagements.

(8.17b) where $-x$ indicates the nonzero coefficients in $\mathbf{C}$:

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -x & 0 \\ 0 & 0 & 0 & -x \\ -x & 0 & 0 & 0 \\ 0 & -x & 0 & 0 \end{bmatrix} \quad \text{uncoupled sets} \tag{8.17a}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -x & -x \\ 0 & 0 & 0 & -x \\ -x & -x & 0 & 0 \\ 0 & -x & 0 & 0 \end{bmatrix} \quad \text{coupled sets} \tag{8.17b}$$

The test of whether the matrix $\mathbf{C}$ represents uncoupled sets of equations is simply whether it can be reduced to two or more nonzero diagonal submatrices by rearranging rows and columns in $\mathbf{C}$, while all other off-diagonal submatrices are zero. Otherwise, the matrix $\mathbf{C}$ represents coupled sets of equations.

Substituting Eqs. (8.10) and (8.16) into Eq. (8.15)

$$\begin{bmatrix} \mathbf{O} & -\mathbf{A}^T \\ -\mathbf{B}^T & \mathbf{O} \end{bmatrix} \begin{bmatrix} c_1\mathbf{v} \\ c_2\mathbf{w} \end{bmatrix} = \lambda \begin{bmatrix} c_1\mathbf{v} \\ c_2\mathbf{w} \end{bmatrix} \tag{8.18}$$

from which

$$-\mathbf{A}^T c_2\mathbf{w} = \lambda c_1\mathbf{v} \tag{8.19}$$

$$-\mathbf{B}^T c_1\mathbf{v} = \lambda c_2\mathbf{w} \tag{8.20}$$

Premultiplying Eq. (8.19) by $\mathbf{B}^T$ and then substituting Eq. (8.20) and premultiplying Eq. (8.20) by $\mathbf{A}$ and then substituting Eq. (8.19), the following pair of equations is obtained

$$(\mathbf{A}^T\mathbf{B}^T - \lambda^2\mathbf{I})c_1\mathbf{v} = \mathbf{0} \tag{8.21}$$

$$(\mathbf{B}^T\mathbf{A}^T - \lambda^2\mathbf{I})c_2\mathbf{w} = \mathbf{0} \tag{8.22}$$

If the dimensions of $\mathbf{A}$ are $n \times m$ and those of $\mathbf{B}$ are $m \times n$ and if $m > n$, then there will be n common eigenvalues $\lambda^2 = \mu$ in Eqs. (8.21) and (8.22). In addition, there will be $m - n$ eigenvalues of Eq. (8.21) equal to 0 for which their corresponding eigenvectors can be obtained from

$$\mathbf{A}^T\mathbf{B}^T c_1\mathbf{v} = \mathbf{0} \tag{8.23}$$

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If those eigenvalues were not equal to 0, this would then imply that, in addition to the $2n$ eigenvalues $\lambda = \pm\sqrt{\mu}$ in $\mathbf{C}$, there would be additional $2(m-n)$ eigenvalues for a total $2n + 2(m-n) = 2m$, which would be greater than the required number $m+n$.

The matrix products $\mathbf{A}^T\mathbf{B}^T$ and $\mathbf{B}^T\mathbf{A}^T$ in Eqs. (8.21) and (8.22) are nonnegative because all A_{ij} and B_{ij} are either positive or zero. Therefore, according to the Frobenius–Perron theorem [Ref. 8, p. 193], Eqs. (8.21) and (8.22) have a real nonnegative eigenvalue λ^2 that is either equal to or exceeds the moduli of any other eigenvalue. Also, to this maximal eigenvalue there correspond eigenvectors $\mathbf{v}_1$ and $\mathbf{w}_1$ with all nonnegative elements. Because $c_1\mathbf{v}_1$ and $c_2\mathbf{w}_1$ are submatrices of the left eigenvector of $\mathbf{C}$, it follows from Eq. (8.16) that

$$\mathbf{q}_1 = \{c_1\mathbf{v}_1 \quad c_2\mathbf{w}_1\} \tag{8.24}$$

which can be determined directly from Eq. (8.15). When $\mathbf{v}_1$ and $\mathbf{w}_1$ are computed from Eqs. (8.21) and (8.22) any suitable scaling method can be used. For example the largest element can be made equal to unity, but the actual scaling between $\mathbf{v}_1$ and $\mathbf{w}_1$ is accomplished through the factors c_1 and c_2 to be determined later.

Transposing Eqs. (8.19) and (8.20) with $\mathbf{v} = \mathbf{v}_1$ and $\mathbf{w} = \mathbf{w}_1$

$$-c_2\mathbf{w}_1^T\mathbf{A} = \lambda_1 c_1\mathbf{v}_1^T \tag{8.25}$$

$$-c_1\mathbf{v}_1^T\mathbf{B} = \lambda_1 c_2\mathbf{w}_1^T \tag{8.26}$$

Premultiplying next the preceding equations by

$$\mathbf{e} = \{1 \quad 1 \quad \cdots \quad 1\} \tag{8.27}$$

and eliminating λ_1, the following relationship between c_1 and c_2 is obtained:

$$\left(\frac{c_2}{c_1}\right)^2 = \frac{\mathbf{v}_1^T\mathbf{B}\mathbf{e}}{\mathbf{w}_1^T\mathbf{A}\mathbf{e}}\,\frac{\mathbf{v}_1^T\mathbf{e}}{\mathbf{w}_1^T\mathbf{e}} \tag{8.28}$$

Hence,

$$\left(\frac{c_2}{c_1}\right) = -\left(\frac{\mathbf{v}_1^T\mathbf{B}\mathbf{e}}{\mathbf{w}_1^T\mathbf{A}\mathbf{e}}\,\frac{\mathbf{v}_1^T\mathbf{e}}{\mathbf{w}_1^T\mathbf{e}}\right)^{\frac{1}{2}} \tag{8.29}$$

The negative sign in Eq. (8.29) is selected to satisfy Eqs. (8.25) and (8.26) in which $\mathbf{A}$, $\mathbf{B}$, $\mathbf{v}_1$, and $\mathbf{w}_1$ are all nonnegative matrices and λ_1 is a positive number. Coincidently, Eq. (8.29) is of the same form as that suggested by Taylor[1] without proof for the so-called summed result in interpreting the aggregate force strengths. Equation (8.29) provides a rational relationship between $\mathbf{v}_1$ and $\mathbf{w}_1$ when these eigenvectors are calculated independently from Eqs. (8.21) and (8.22).

It should be noted that in practice the left dominant eigenvalue λ_1 and its corresponding eigenvector $\mathbf{q}_1$ can be obtained directly from Eq. (8.15) rather than from Eqs. (8.21), (8.22), and (8.29). Computer programs such as MATLAB are available that can be conveniently used to find λ_1 and $\mathbf{q}_1$. Another special computer program, DOMEIGEN, for λ_1 and $\mathbf{q}_1$ can be found in Ref. 9 and in Appendix E.

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Because of the relationship (C.28), the generalized left eigenvectors $\mathbf{q}_i$ are orthogonal to the right eigenvectors $\mathbf{p}_j$. This orthogonality property is expressed as

$$\mathbf{q}_i^T \mathbf{p}_j = \begin{cases} 1 & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases} \tag{8.30}$$

This property is used next to derive a scalar parameter from the solution of the governing equations by premultiplying $\mathbf{X}$ by the transpose of the dominant left eigenvector $\mathbf{q}_1$, corresponding to the largest eigenvalue λ_1. This parameter is a measure of merit for the combat engagement. Consequently, employing Eqs. (C.25) and (C.14b) and noting that for the notation used in Appendix C, $\lambda_1 = \lambda^{(1)}$ and $\mathbf{q}_1 = \mathbf{q}^{(1)}$

$$\begin{aligned}
\left[\mathbf{q}_1^T \mathbf{X}\right]_{t=0} &= \left[\mathbf{q}_1^T \mathbf{P} \lceil \exp(\lambda t)\mathbf{D} \rfloor \mathbf{Q}^T \mathbf{X}_0\right]_{t=0} \\
&= \left[\mathbf{q}_1^T \begin{bmatrix} \mathbf{p}^{(1)} & \mathbf{p}^{(2)} & \cdots \end{bmatrix} \lceil \exp(\lambda t)\mathbf{D} \rfloor \mathbf{Q}^T \mathbf{X}_0\right]_{t=0} \\
&= [[[1 \quad 0 \quad 0 \cdots] \quad \mathbf{0} \quad \mathbf{0} \quad \cdots] \\
&\quad \times \begin{bmatrix} \exp\left(\lambda^{(1)} t\right)\mathbf{D}^{(1)} & \mathbf{0} & \cdots \\ \mathbf{0} & \exp\left(\lambda^{(2)} t\right)\mathbf{D}^{(2)} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \ddots \end{bmatrix} \begin{bmatrix} \mathbf{q}^{(1)^T} \\ \mathbf{q}^{(2)^T} \\ \vdots \end{bmatrix} \mathbf{X}_0 \Bigg]_{t=0} \\
&= \left[\left[\exp\left(\lambda^{(1)} t\right)[1 \quad 0 \quad 0 \cdots] \quad \mathbf{0} \quad \mathbf{0} \cdots\right] \left\{\mathbf{q}^{(1)^T} \quad \mathbf{q}^{(2)^T} \cdots\right\} \mathbf{X}_0\right]_{t=0} \\
&= \left[\mathbf{q}_1^T \mathbf{X}_0 \exp\left(\lambda^{(1)} t\right)\right]_{t=0} = \mathbf{q}_1^T \mathbf{X}_0
\end{aligned} \tag{8.31}$$

Now substituting Eq. (8.24) into Eq. (8.31) and observing that $\mathbf{X}_0 = \{\mathbf{M}_0 \quad \mathbf{N}_0\}$, it follows that

$$\begin{aligned}
\mathbf{q}_1^T \mathbf{X}_0 &= \begin{bmatrix} c_1 \mathbf{v}_1^T & c_2 \mathbf{w}_1^T \end{bmatrix} \{\mathbf{M}_0 \quad \mathbf{N}_0\} \\
&= c_1 \left(\mathbf{v}_1^T \mathbf{M}_0 + \frac{c_2}{c_1} \mathbf{w}_1^T \mathbf{N}_0\right) \\
&= c_1 \left[\mathbf{v}_1^T \mathbf{M}_0 - \left(\frac{\mathbf{v}_1^T \mathbf{B}\mathbf{e}}{\mathbf{w}_1^T \mathbf{A}\mathbf{e}} \frac{\mathbf{v}_1^T \mathbf{e}}{\mathbf{w}_1^T \mathbf{e}}\right)^{\frac{1}{2}} \mathbf{w}_1^T \mathbf{N}_0\right]
\end{aligned} \tag{8.32}$$

where for convenience c_1 can be taken as one. Because $\mathbf{v}_1$ and $\mathbf{w}_1$ are both nonnegative, the terms $\mathbf{v}_1^T \mathbf{M}_0$ and $\mathbf{w}_1^T \mathbf{N}_0$ are both positive. Equation (8.32) implies that the scalar quantity $\mathbf{q}_1^T \mathbf{X}_0$ derived from the predominant term in the solution represents the difference between aggregated Blue and Red forces; therefore, it be can be taken as a measure of merit, i.e., a superiority parameter for the opposing forces.

It should be noted from Appendix C that the analytical solution for $\mathbf{X}$ is a series of terms with multiplying factors $\exp(\lambda^{(i)} t)$, which can be arranged in descending order for $\lambda^{(i)}$ as $\lambda^{(1)}, \lambda^{(2)}, \ldots, (0, \ldots 0, \text{if present}), \ldots, -\lambda^{(2)}, -\lambda^{(1)}$. Thus, the term with $+\lambda^{(1)}$ is the dominant term in the solution. If $\lambda^{(1)}$ is a repeated eigenvalue, or if other eigenvalues are close to the dominant eigenvalue, the use of $\mathbf{q}_1^T \mathbf{X}_0$ as a

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measure of merit is still possible in most cases; however, in most practical cases, $\lambda^{(1)}$ is well separated from other eigenvalues and this measure of merit is a good predictor of the battle outcome. For example, for equal coefficients in both A and B (i.e., $A_{ij} = a$ and $B_{ij} = b$ for all i and j, where a and b are constants), the eigenvalues are $\lambda^{(1)}, 0, \ldots, 0, -\lambda^{(1)}$ and $\lambda^{(1)} = \sqrt{mnab}$, where m and n refer to the number of Blue and Red forces, respectively. For these cases the dominant eigenvector is

$$\mathbf{q}_1 = \{1 \quad 1 \quad \cdots \quad m\text{-times} \quad -\sqrt{mb/na} - \sqrt{mb/na} \cdots n\text{-times}\}$$

(see Appendix C). When $\mathbf{C}$ contains multiple eigenvalues, it is usually referred to as being defective. Some examples of the multiple eigenvalue cases are given in Appendix C, where the dominant eigenvector $\mathbf{q}_1$ contains zero elements, which means that some X-forces would not be included in $\mathbf{q}_1^T\mathbf{X}_0$.

To reduce Eq. (8.32) into a nondimensional parameter, the equation can be normalized with respect to $c_1\mathbf{v}_1^T\mathbf{M}_0$ by introducing a nondimensional superiority parameter S such that

$$S = \frac{\mathbf{q}_1^T\mathbf{X}}{c_1\mathbf{v}_1^T\mathbf{M}_0} \tag{8.33}$$

and

$$S_0 = \frac{\mathbf{q}_1^T\mathbf{X}_0}{c_1\mathbf{v}_1^T\mathbf{M}_0} = 1 - \left(\frac{\mathbf{v}_1^T\mathbf{B}\mathbf{e}}{\mathbf{w}_1^T\mathbf{A}\mathbf{e}}\,\frac{\mathbf{v}_1^T\mathbf{e}}{\mathbf{w}_1^T\mathbf{e}}\right)^{\frac{1}{2}} \frac{\mathbf{w}_1^T\mathbf{N}_0}{\mathbf{v}_1^T\mathbf{M}_0}$$

$$= 1 - \Phi_0^{-1} \tag{8.34}$$

where

$$\Phi_0 = \left(\frac{\mathbf{w}_1^T\mathbf{A}\mathbf{e}}{\mathbf{v}_1^T\mathbf{B}\mathbf{e}}\,\frac{\mathbf{w}_1^T\mathbf{e}}{\mathbf{v}_1^T\mathbf{e}}\right)^{\frac{1}{2}} \frac{\mathbf{v}_1^T\mathbf{M}_0}{\mathbf{w}_1^T\mathbf{N}_0} \tag{8.35}$$

The nondimensional parameters S, S_0, and Φ_0 are all independent of the choice of the arbitrary multiplying factors with $\mathbf{v}_1$ and $\mathbf{w}_1$. The superiority parameter S is a scalar quantity measuring the contributions of the dominant term in the solution for the force levels at any given time. It should also be noted that

$$\begin{gathered} 0 < S_0 < 1.0 \quad \text{for} \quad 1.0 < \Phi_0 < \infty \\ S_0 = 0 \quad \text{for} \quad \Phi_0 = 1.0 \\ -\infty < S_0 < 0 \quad \text{for} \quad 0 < \Phi_0 < 1.0 \end{gathered} \tag{8.36}$$

The parameters $\mathbf{q}_1^T\mathbf{X}_0$ or S_0 can be used as measures of effectiveness to determine the outcome of a particular tactical engagement without actually solving the differential equations. Thus as in the case of a one-on-one engagement, the Blue forces win the engagement when $\Phi_0 > 1$ or when $S_0 > 0$, whereas the Red forces win when $\Phi_0 < 1$ or when $S_0 < 0$. When $\Phi_0 = 1$ or $S = 0$, both sides reduce their strengths proportionately. *It should be noted that the win applies here only to the first engagement and only indicates which side suffers fewer losses. When one of the forces in* M *or* N *is completely annihilated, the governing matrix equations*

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must be reconfigured and the new S_0 or S must be recalculated for the subsequent battle engagement (e.g., see Problem 8.5).

As in the case of a one-on-one engagement, the superiority parameter Φ_0 for the general case consists of a product of two components representing the qualitative and quantitative ratios. The qualitative ratio is $[(\mathbf{w}_1^T\mathbf{A}\mathbf{e})(\mathbf{w}_1^T\mathbf{e})/(\mathbf{v}_1^T\mathbf{B}\mathbf{e})(\mathbf{v}_1^T\mathbf{e})]^{\frac{1}{2}}$ and the quantitative ratio is $(\mathbf{v}_1^T\mathbf{M}_0)/(\mathbf{w}_1^T\mathbf{N}_0)$.

8.4 Aggregated Force Solution

To reduce the mathematical representation of the many-on-many engagements to a one-on-one representation, the aggregate force strengths may be introduced into Eqs. (8.1) and (8.2). Premultiplying Eq. (8.1) by $c_1\mathbf{v}_1^T$ and Eq. (8.2) by $c_2\mathbf{w}_1^T$, the following differential equations are obtained:

$$c_1\mathbf{v}_1^T\dot{\mathbf{M}} = -c_1\mathbf{v}_1^T\mathbf{B}\mathbf{N} = \beta c_2\mathbf{w}_1^T\mathbf{N} \tag{8.37}$$

$$c_2\mathbf{w}_1^T\dot{\mathbf{N}} = -c_2\mathbf{w}_1^T\mathbf{A}\mathbf{M} = \alpha c_1\mathbf{v}_1^T\mathbf{M} \tag{8.38}$$

where the overdots represent differentiation with respect to time, and the right sides of these equations, with positive constants α and β, have at this point been introduced arbitrarily. The aggregated force strengths F_B and F_R from Eqs. (8.13) and (8.14) can be written as

$$F_B = \bar{\mathbf{v}}^T\mathbf{M} = c_1\mathbf{v}_1^T\mathbf{M} \tag{8.39}$$

$$F_R = \bar{\mathbf{w}}^T\mathbf{N} = -c_2\mathbf{w}_1^T\mathbf{N} \tag{8.40}$$

where

$$\bar{\mathbf{v}} = c_1\mathbf{v}_1 \tag{8.41}$$

$$\bar{\mathbf{w}} = -c_2\mathbf{w}_1 \tag{8.42}$$

Hence, substituting Eq. (8.39) and (8.40) into Eqs. (8.37) and (8.38)

$$\dot{F}_B = -\beta F_R \tag{8.43}$$

$$\dot{F}_R = -\alpha F_B \tag{8.44}$$

which are the governing equations for a one-on-one engagement between the aggregated forces F_B and F_R; however, a word of caution is appropriate here. The aggregated solutions F_B and F_R are only approximations to the exact solutions, because they do not incorporate the condition that one of the M_i or N_j can be reduced to 0 independently of others. In the aggregated solution of Eqs. (8.43) and (8.44) all components of F_B or F_R are reduced to 0 simultaneously.

Because $\mathbf{M}$ and $\mathbf{N}$ are arbitrary, in order for the right sides of Eqs. (8.37) and (8.38) to be valid, the following relations must be true:

$$-c_1\mathbf{v}_1^T\mathbf{B} = \beta c_2\mathbf{w}_1^T \quad \text{or} \quad -\mathbf{B}^T c_1\mathbf{v}_1 = \beta c_2\mathbf{w}_1 \tag{8.45}$$

$$-c_2\mathbf{w}_1^T\mathbf{A} = \alpha c_1\mathbf{v}_1^T \quad \text{or} \quad -\mathbf{A}^T c_2\mathbf{w}_1 = \alpha c_1\mathbf{v}_1 \tag{8.46}$$

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Solving for $\mathbf{v}_1$ and $\mathbf{w}_1$,

$$\mathbf{A}^T\mathbf{B}^T c_1\mathbf{v}_1 = -\beta\mathbf{A}^T c_2\mathbf{w}_1 = \beta\alpha c_1\mathbf{v}_1$$

$$(\mathbf{A}^T\mathbf{B}^T - \alpha\beta\mathbf{I})c_1\mathbf{v}_1 = \mathbf{0} \tag{8.47}$$

$$\mathbf{B}^T\mathbf{A}^T c_2\mathbf{w}_1 = -\alpha\mathbf{B}^T c_1\mathbf{v}_1 = \alpha\beta c_2\mathbf{w}_1$$

$$(\mathbf{B}^T\mathbf{A}^T - \alpha\beta\mathbf{I})c_2\mathbf{w}_1 = \mathbf{0} \tag{8.48}$$

Hence, comparing Eqs. (8.47) and (8.48) with Eqs. (8.21) and (8.22),

$$\lambda_1^2 = \alpha\beta \tag{8.49}$$

Similarly, by comparing Eqs. (8.45) and (8.26) and Eqs. (8.46) and (8.25),

$$\alpha = \beta = \lambda_1 \tag{8.50}$$

which is the same result derived from the scaling method assumed by Holter[5] and Anderson.[6]

The preceding analysis demonstrates that a unified theory for a rational analysis of the Lanchester equations for heterogeneous forces engaged in a direct fire combat can be developed. This is accomplished by correlating the quantities $\mathbf{v}_1$, $\mathbf{w}_1$, α, and β to the left dominant eigenvector of the governing matrix of attrition coefficients and its dominant eigenvalue. This theory provides for a proper scaling of the eigenvectors $\mathbf{v}_1$ and $\mathbf{w}_1$, and leads directly to the measure of effectiveness $\mathbf{q}_1^T\mathbf{X}_0$, or the superiority parameter S_0. Although the use of the left dominant eigenvector introduced here for the combat analysis is new, a similar approach has been used in the past in other areas. For example, the use of the left dominant eigenvector in the cohort population model leads to the total reproductive value of the population as described by Luenberger (Ref. 8, p. 185).

8.5 Numerical Example: "One-on-Two" Tactical Engagement

The proposed method of analysis of heterogeneous forces combat is illustrated here for the case of a tactical engagement of one Blue force and two Red forces as shown in Fig. 8.4. For this example the equations describing the engagement have been assumed to be

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \\ \dot{X}_3 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{3}{2} & -1 \\ -\frac{4}{3} & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

It can be verified easily, that for this case,

$$\mathbf{X} = \{X_1 \quad X_2 \quad X_3\} = \{M_1 \quad M_2 \quad N_2\}$$

$$\mathbf{q}_1 = \left\{1 \quad -\frac{3}{4} \quad -\frac{1}{2}\right\}$$

$$\mathbf{v}_1^T = [\,1\,] \qquad \mathbf{w}_1^T = \begin{bmatrix} 1 & \frac{2}{3} \end{bmatrix}$$

$$\alpha = \beta = \lambda_1 = 2$$

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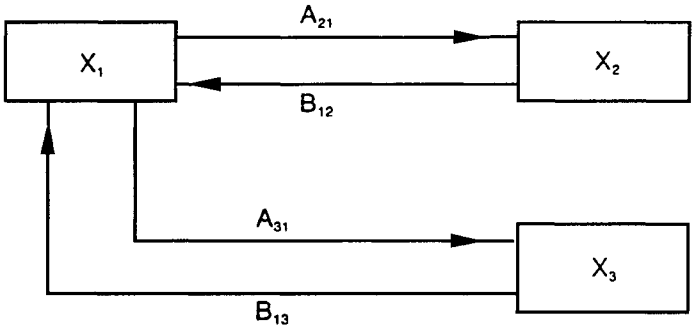


Fig. 8.4 Numerical example of one-on-two engagement.

and

$$\Phi_0 = \frac{4}{3}\frac{X_1}{X_2 + \frac{2}{3}X_3}$$

In Figs. 8.5, 8.6, and 8.7 the variation of X_1, X_2, and X_3 with the nondimensional time τ is shown for $\mathbf{X}_0 = \{200 \quad 100 \quad 100\}$, $\{125 \quad 100 \quad 100\}$, and $\{100 \quad 100 \quad 100\}$, which correspond to $\Phi_0 = 1.6$, 1.0, and 0.8, respectively. The solutions for $\mathbf{X}$ were obtained by numerical integration. For case 1 (Fig. 8.5), in which $\Phi_0 = 1.6$, $X_3 = 0$ when $\tau = 0.57$, and at that time the portion of X_1 allocated to fight X_3 is reallocated against X_2, requiring that the equations be reformulated. Subsequently, the X_2 force is reduced quickly to 0 by the X_1 force. As the battle progresses, the superiority parameter S shows a rapid increase (see Fig. 8.8). Here, the initial positive value of S is an indication of the superiority

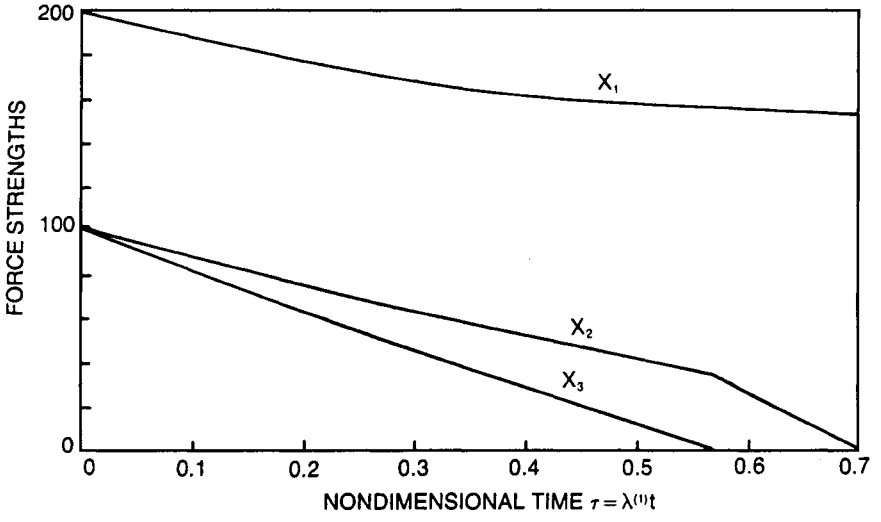


Fig. 8.5 Numerical example (case 1): Variation of force strengths X_1, X_2, and X_3 with the nondimensional time $\tau = \lambda^{(1)}t$ for $\Phi_0 = 1.6$ and $X_0 = \{200 \quad 100 \quad 100\}$.

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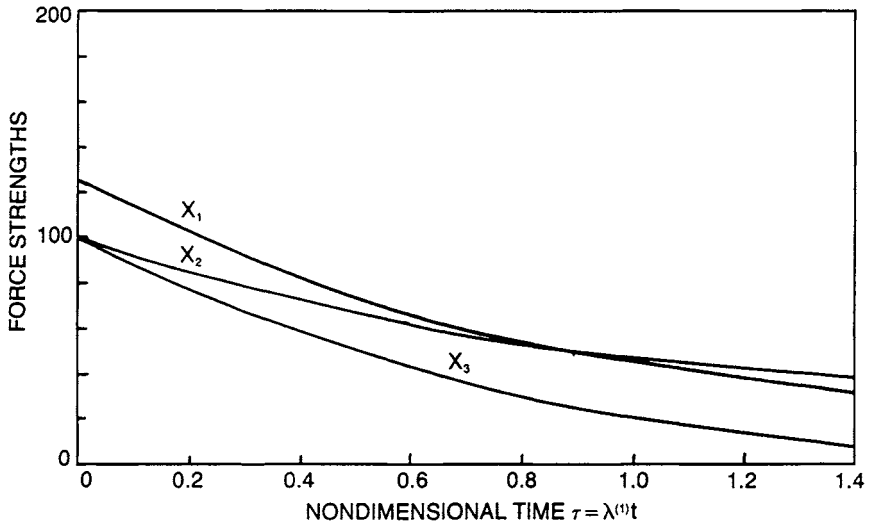


Fig. 8.6 Numerical example (case 2): Variation of force strengths X_1, X_2, and X_3 with the nondimensional time $\tau = \lambda^{(1)}t$ for $\Phi_0 = 1.0$ and $X_0 = \{125 \quad 100 \quad 100\}$.

of the Blue force. For case 2, in which $\Phi_0 = 1.0$, the opposing forces diminish their strengths gradually without any side gaining a clear advantage (see Fig. 8.6). Here $S = 0$. Case 3, for which $\Phi_0 = 0.8$, represents a rapid demise of the X_1 force (see Fig. 8.7). For this case S starts with a negative value, which diminishes as the battle progresses (see Fig. 8.8). Here, the initial negative value of S is an indication of the inferiority of the Blue force.

Because the dominant left eigenvector is the key parameter for the measure of merit S, the eigenvectors $\mathbf{q}_1$ are shown as examples in Appendix C for the cases of m-on-n engagements for which all attrition coefficients are equal.

8.6 Comments on the Directed Fire Solution

The premultiplication of the solution vector $\mathbf{X} = \{\mathbf{M} \quad \mathbf{N}\}$ for the engaging forces by the transpose of the dominant left eigenvector $\mathbf{q}_1$ of the matrix of attrition coefficients helps to clarify the meaning of the eigenvector methods of determining the relative values of heterogeneous forces in a combat engagement. Although the eigenvector methods have been around for a long time, problems existed in deciding on the best method of scaling the resulting eigenvectors. This chapter illustrates the meaning of the individual eigenvectors $\mathbf{v}_1$ and $\mathbf{w}_1$ as components of the dominant left eigenvector of the governing matrix of attrition coefficients. This, in turn, allows for a proper determination of the relative scaling of eigenvectors for the two opposing forces and leads to the nondimensional superiority parameter S_0, which can be used as a measure of merit and an indicator of the initial outcome of the battle engagement.

The practical utility of the measure of merit can be demonstrated by obtaining numerical solutions to heterogeneous force engagements while at the same time computing the parameter S introduced in this chapter. The sign of S and the extent

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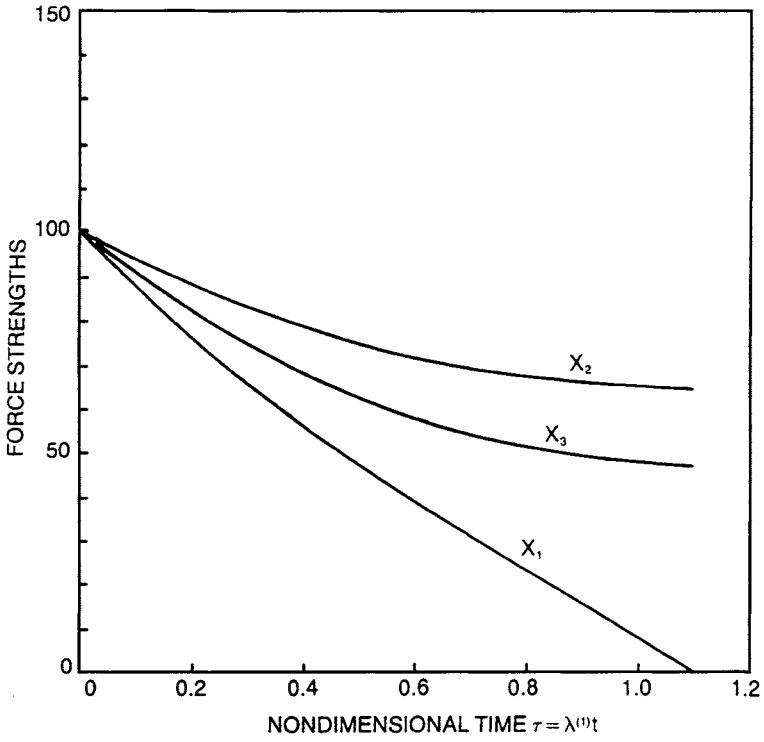


Fig. 8.7 Numerical example (case 3): Variation of force strengths X_1, X_2, and X_3 with the nondimensional time $\tau = \lambda^{(1)}t$ for $\Phi_0 = 0.8$ and $X_0 = \{100 \quad 100 \quad 100\}$.

to which it is either increasing or decreasing can be used as a measure of superiority of the Blue forces in relation to the Red forces. Although the method is valid only for constant attrition coefficients, it also can be of some value for cases with variable coefficients.

8.7 Area Fire "Many-on-Many" Engagements

Previous sections in this chapter discussed the combat model for directed fire engagements of heterogeneous forces. A similar model can be developed for area fire engagements by starting with a one-on-one model discussed in Chapter 4 and extending it to many-on-many engagements. For a one-on-one area fire engagement, the governing differential equations were derived in Section 4.4 as

$$\frac{dm_B}{dt} = -a_R m_R \frac{m_B}{M_B} \tag{4.49}$$

and

$$\frac{dm_R}{dt} = -a_B m_B \frac{m_R}{M_R} \tag{4.50}$$

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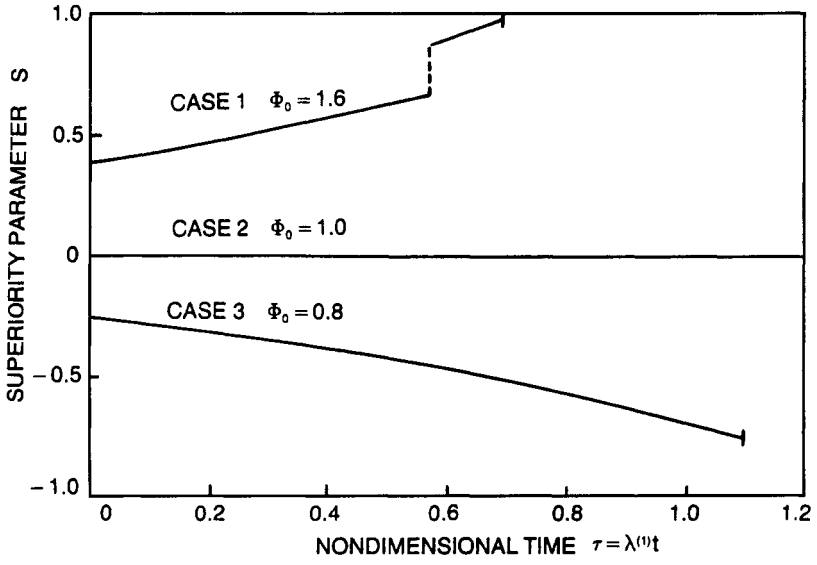


Fig. 8.8 Numerical examples (cases 1, 2, and 3): Variation of the superiority parameter S with the nondimensional time $\tau = \lambda^{(1)}t$ for $\Phi_0 = 1.6$, 1.0, and 0.6.

which imply that, by comparison with the directed fire model, the attrition coefficients for the Blue and Red forces are reduced for the area fire model by the factors m_B/M_B and m_R/M_R, respectively. Therefore, the governing equations for many-on-many engagements for the Blue forces can be written as

$$\frac{dM_1}{dt} = -B_{11}N_1\frac{M_1}{M_{10}} - B_{12}N_2\frac{M_1}{M_{10}} - \cdots - B_{1R}N_R\frac{M_1}{M_{10}}$$

$$\frac{dM_2}{dt} = -B_{21}N_1\frac{M_2}{M_{20}} - B_{22}N_2\frac{M_2}{M_{20}} - \cdots - B_{2R}N_R\frac{M_2}{M_{20}}$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$\frac{dM_B}{dt} = -B_{B1}N_1\frac{M_B}{M_{B0}} - B_{B2}N_2\frac{M_B}{M_{B0}} - \cdots - B_{BR}N_R\frac{M_B}{M_{B0}}$$

Similarly, for the Red forces

$$\frac{dN_1}{dt} = -A_{11}M_1\frac{N_1}{N_{10}} - A_{12}M_2\frac{N_1}{N_{10}} - \cdots - A_{1B}M_B\frac{N_1}{N_{10}}$$

$$\frac{dN_2}{dt} = -A_{21}M_1\frac{N_2}{N_{20}} - A_{22}M_2\frac{N_2}{N_{20}} - \cdots - A_{2B}M_B\frac{N_2}{N_{20}}$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$\frac{dN_R}{dt} = -A_{R1}M_1\frac{N_R}{N_{R0}} - A_{R2}M_2\frac{N_R}{N_{R0}} - \cdots - A_{RB}M_B\frac{N_R}{N_{R0}}$$

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where the subscripts B and R denote the number of Blue and Red forces, respectively, while M_{i0} and N_{i0} denote the values of M_i and N_i at $t = 0$. The attrition coefficients A_{ij} and B_{ij} must include the allocation factors for the distribution of fire by the jth force. Arranging the preceding equations into a matrix format, it follows that

$$\begin{bmatrix} \dot{M}_1 \\ \dot{M}_2 \\ \vdots \\ \dot{M}_B \\ \dot{N}_1 \\ \dot{N}_2 \\ \vdots \\ \dot{N}_R \end{bmatrix} = \begin{bmatrix} \frac{M_1}{M_{10}} & & & & & & & \\ & \frac{M_2}{M_{20}} & & & \mathbf{0} & & & \\ & & \ddots & & & & & \\ & & & \frac{M_B}{M_{B0}} & & & & \\ & & & & \frac{N_1}{N_{10}} & & & \\ & & & & & \frac{N_2}{N_{20}} & & \\ & & \mathbf{0} & & & & \ddots & \\ & & & & & & & \frac{N_R}{N_{R0}} \end{bmatrix}$$

$$\times \begin{bmatrix} & & & & -B_{11} & -B_{12} & \cdots & -B_{1R} \\ & & & & -B_{21} & -B_{22} & \cdots & -B_{2R} \\ & & \mathbf{0} & & \vdots & \vdots & \vdots & \vdots \\ & & & & -B_{B1} & -B_{B2} & \cdots & -B_{BR} \\ -A_{11} & -A_{12} & \cdots & -A_{1B} & & & & \\ -A_{21} & -A_{22} & \cdots & -A_{2B} & & & & \\ \vdots & \vdots & \vdots & \vdots & & \mathbf{0} & & \\ -A_{R1} & -A_{R2} & \cdots & -A_{RB} & & & & \end{bmatrix} \begin{bmatrix} M_1 \\ M_2 \\ \vdots \\ M_B \\ N_1 \\ N_2 \\ \vdots \\ N_R \end{bmatrix} \tag{8.51}$$

These equations can be rewritten in the matrix notation as

$$\dot{\mathbf{X}} = [\mathbf{D}(\mathbf{X}_0)]^{-1}\, \mathbf{D}(\mathbf{X})\mathbf{C}\mathbf{X} \tag{8.52}$$

where the matrices $\mathbf{X}$ and $\mathbf{C}$ were previously defined for the directed fire model and

$$\begin{aligned} \mathbf{X}_0 &= \text{column matrix of forces at } t = 0 \\ \mathbf{D}(\mathbf{X}) &= \text{diagonal matrix of forces} \\ \mathbf{D}(\mathbf{X}_0) &= \text{diagonal matrix of forces at } t = 0. \end{aligned}$$

For numerical solutions, Eq. (8.52) can be expressed as

$$\mathbf{X}_{n+1} - \mathbf{X}_n = \Delta t\, [\mathbf{D}(\mathbf{X}_0)]^{-1}\, \mathbf{D}(\mathbf{X}_n)\mathbf{C}\mathbf{X}_n$$

Hence,

$$\mathbf{X}_{n+1} = \left(\mathbf{I}_n + \Delta t\, [\mathbf{D}(\mathbf{X}_0)]^{-1}\, \mathbf{D}(\mathbf{X}_n)\mathbf{C}\right) \mathbf{X}_n \tag{8.53}$$

Thus, the numerical solution of the nonlinear differential equations for area fire combat requires an operation of changing a column matrix $\mathbf{X}_n$ into a diagonal matrix with the same coefficients. This operation is needed to represent the nonlinear character of Eq. (8.52).

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If during the time interval Δt some forces are added or withdrawn from the battlefield, then Eq. (8.53) must be modified as

$$\mathbf{X}_{n+1} = \left(\mathbf{I}_n + \Delta t\,[\mathbf{D}(\mathbf{X}_0)]^{-1}\,\mathbf{D}(\mathbf{X}_n)\mathbf{C}\right)\mathbf{X}_n + \mathbf{D}_n \tag{8.54}$$

where $\mathbf{D}_n$ is the column matrix of force additions or withdrawals.

Although the superiority parameter S, given by Eq. (8.33), was derived for the case of a directed fire engagement described by linear equations in X_n, this parameter also provides some useful information for the case of an area fire engagement described by the nonlinear differential equations given by Eq. (8.51). The reason for this is the strong similarity between equations

$$\dot{\mathbf{X}} = \mathbf{CX} \text{ (directed fire)} \tag{8.8}$$

and

$$\dot{\mathbf{X}} = [\mathbf{D}(\mathbf{X}_0)]^{-1}\,\mathbf{D}(\mathbf{X})\mathbf{CX} \text{ (area fire)} \tag{8.52}$$

both of which are proportional to $\mathbf{CX}$, except that in the second case the proportionality is gradually modified by $\mathbf{D}(\mathbf{X})$ as time increases. Consequently, the behavior of S for the area fire engagement is the same as for the directed fire. Thus when $S > 0$ then the Blue force has *initial* superiority over the Red force; the opposite is true when $S < 0$.

8.8 Guerrilla Warfare "Many-on-Many" Engagements

The guerrilla warfare many-on-many engagement will be defined here as a tactical engagement of the Blue forces employing the area fire defense, while the Red Forces employ the directed fire attack. The area fire by the Blue forces is simply the result of imprecise knowledge of the Red force locations. Thus the guerrilla warfare may be described as an area fire/directed fire engagement by the Blue forces ambushed by the Red forces. It was demonstrated in Section 4.5 that for a one-on-one guerrilla warfare, the governing equations are given by

$$\frac{dm_B}{dt} = -a_R m_R \tag{4.73}$$

and

$$\frac{dm_R}{dt} = -\frac{a_B m_B m_R}{M_R} \tag{4.74}$$

Therefore, extending these equations to many-on-many engagements, the governing equations for the Blue forces become

$$\frac{dM_1}{dt} = -B_{11}N_1 - B_{12}N_2 - \cdots - B_{1R}N_R$$

$$\frac{dM_2}{dt} = -B_{21}N_1 - B_{22}N_2 - \cdots - B_{2R}N_R$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$\frac{dM_B}{dt} = -B_{B1}N_1 - B_{B2}N_2 - \cdots - B_{BR}N_R$$

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Similarly, for the Red forces

$$\frac{dN_1}{dt} = -A_{11}M_1\frac{N_1}{N_{10}} - A_{12}M_2\frac{N_1}{N_{10}} - \cdots - A_{1B}M_B\frac{N_1}{N_{10}}$$

$$\frac{dN_2}{dt} = -A_{21}M_1\frac{N_2}{N_{20}} - A_{22}M_2\frac{N_2}{N_{20}} - \cdots - A_{2B}M_B\frac{N_2}{N_{20}}$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$\frac{dN_R}{dt} = -A_{R1}M_1\frac{N_R}{N_{R0}} - A_{R2}M_2\frac{N_R}{N_{R0}} - \cdots - A_{RB}M_B\frac{N_R}{N_{R0}}$$

where, as before, the subscripts $_B$ and $_R$ denote the number of Blue and Red forces, respectively, while N_{i0} denotes the values of N_i at $t = 0$. The attrition coefficients A_{ij} and B_{ij} must include the allocation factors for the distribution of fire by the jth force. Arranging the preceding equations into a matrix format, it follows that

$$\begin{bmatrix} \dot{M}_1 \\ \dot{M}_2 \\ \vdots \\ \dot{M}_B \\ \dot{N}_1 \\ \dot{N}_2 \\ \vdots \\ \dot{N}_R \end{bmatrix} = \begin{bmatrix} 1 & & & & & & & \\ & 1 & & & \mathbf{0} & & & \\ & & \ddots & & & & & \\ & & & 1 & & & & \\ & & & & \frac{N_1}{N_{10}} & & & \\ & & & & & \frac{N_2}{N_{20}} & & \\ & & \mathbf{0} & & & & \ddots & \\ & & & & & & & \frac{N_R}{N_{R0}} \end{bmatrix}$$

$$\times \begin{bmatrix} & & & & -B_{11} & -B_{12} & \cdots & -B_{1R} \\ & & & & -B_{21} & -B_{22} & \cdots & -B_{2R} \\ & & \mathbf{0} & & \vdots & \vdots & \vdots & \vdots \\ & & & & -B_{B1} & -B_{B2} & \cdots & -B_{BR} \\ -A_{11} & -A_{12} & \cdots & -A_{1B} & & & & \\ -A_{21} & -A_{22} & \cdots & -A_{2B} & & & & \\ \vdots & \vdots & \vdots & \vdots & & \mathbf{0} & & \\ -A_{R1} & -A_{R2} & \cdots & -A_{RB} & & & & \end{bmatrix} \begin{bmatrix} M_1 \\ M_2 \\ \vdots \\ M_B \\ N_1 \\ N_2 \\ \vdots \\ N_R \end{bmatrix} \tag{8.55}$$

These equations can be rewritten in the matrix notation as

$$\dot{\mathbf{X}} = [\mathbf{D}(\mathbf{Y}_0)]^{-1}\,\mathbf{D}(\mathbf{Y}_n)\mathbf{C}\mathbf{X} \tag{8.56}$$

where

$$\mathbf{Y}_0 = \{1 \quad 1 \quad \cdots \quad 1 \quad N_{10} \quad N_{20} \quad \cdots \quad N_{R0}\} \tag{8.57}$$

$$\mathbf{Y}_n = \{1 \quad 1 \quad \cdots \quad 1 \quad N_1 \quad N_2 \quad \cdots \quad N_R\} \tag{8.58}$$

$$\mathbf{D}(\mathbf{Y}_0) = \text{diagonal matrix of } \mathbf{Y}_0 \tag{8.59}$$

$$\mathbf{D}(\mathbf{Y}_n) = \text{diagonal matrix of } \mathbf{Y}_n \tag{8.60}$$

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For numerical solutions, Eq. (8.56) can be expressed as

$$\mathbf{X}_{n+1} - \mathbf{X}_n = \Delta t\,[\mathbf{D}(\mathbf{Y}_0)]^{-1}\,\mathbf{D}(\mathbf{Y}_n)\mathbf{C}\mathbf{X}_n$$

Hence,

$$\mathbf{X}_{n+1} = \left(\mathbf{I}_n + \Delta t\,[\mathbf{D}(\mathbf{Y}_0)]^{-1}\,\mathbf{D}(\mathbf{Y}_n)\mathbf{C}\right)\mathbf{X}_n \tag{8.61}$$

If during the time interval Δt some forces are added or withdrawn from the battlefield, then Eq. (8.61) must be modified as

$$\mathbf{X}_{n+1} = \left(\mathbf{I}_n + \Delta t\,[\mathbf{D}(\mathbf{Y}_0)]^{-1}\,\mathbf{D}(\mathbf{Y}_n)\mathbf{C}\right)\mathbf{X}_n + \mathbf{D}_n \tag{8.62}$$

where $\mathbf{D}_n$ is the column matrix of force additions or withdrawals.

Because the form of Eq. (8.56) is also similar to that for the directed fire model, the superiority parameter S can be used to follow the course of the battle. The significant difference, however, is that $S = 0$ does not represent the parity point. It can be observed from the solution of the one-on-one case [see Fig. 4.6 and Eqs. (8.32)–(8.35)] that when $S \geq 1 - 1/\sqrt{2}$ then the Blues cannot be defeated.

8.9 Positioning of Defense Forces

Front Line Segment Defense[10]

In defining the positioning of forces needed to protect a specified segment of the defense front, the main objectives of the aggressor must be known, or at least predicted with some degree of confidence. Some such objectives may be the following:

1) To maximize the territory control.

2) To maximize the ratio of the territory control/attack attrition, i.e., to maximize territorial gains while minimizing the attrition rate of its forces.

3) To maximize the duration of the armed conflict, i.e., to prolong the armed conflict for as long as possible to wear down the defensive forces.

4) To maximize the defensive force attrition.

5) To maximize the defender attrition/attacker attrition.

6) To maximize the territory controlled per duration.

Of course, there may be others, such as the control of specified territory. For each objective, the defense must design an appropriate strategy to minimize the attacker's objective. In this section, an effective defense proposed by Gupta[10] is discussed that minimizes territory lost to the enemy, i.e., a defense strategy that minimizes the attacker's objective of controlling as much as possible of the defender's territory.

There are essentially two possible strategies for defense: a distributed static defense or a concentrated mobile defense. In static defense, units are positioned behind the front line at specified locations in anticipation of attack, while in mobile defense a highly mobile large unit is made available to respond quickly to an attack at any point. These two approaches are shown in Fig. 8.9. As an example, consider the Red (attacker) force with 4000 units, and the Blue (defender) force also with 4000 units. The static strategy uses four separate defense areas each with 1000 units, while the mobile strategy uses all units for defense. The combat model assumed

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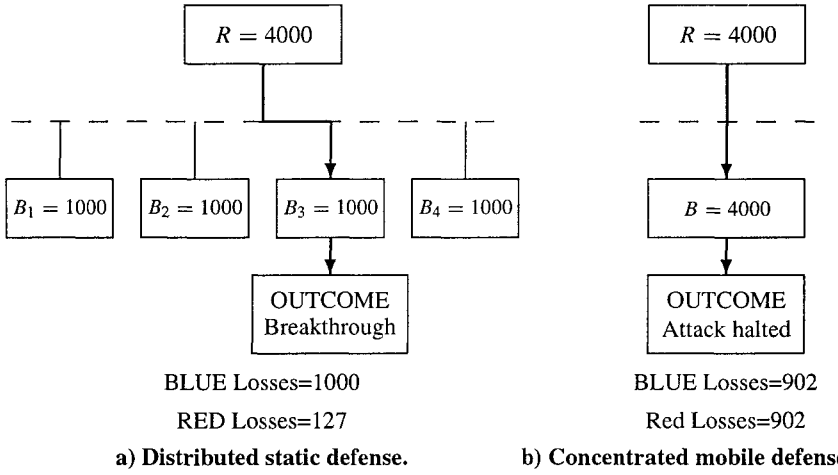


Fig. 8.9 Static and mobile defense strategies.

for this engagement is that of the direct fire exchange with the same attrition coefficients for both sides. In the case of static defense, the Red forces attack only one area (B_3), and they command clear superiority because $\Phi_0 = 0.25$ (see Section 4.3). The battle is over in time $\tau_0 = 0.255$ [Eq. (4.46c)] with 1000 casualties for the Blue force and only 127 for the Red force, and the defense line is broken. For the mobile strategy, however, $\Phi_0 = 1$, i.e., both sides are attriting their forces at the same rates. For comparison when $\tau_0 = 0.255$, both Blue and Red forces lose 902 units each, but the strategic objective of halting the attacker is fulfilled. Thus this example illustrates the advantages of the mobile strategy, which will now be discussed further.

Optimal Mobile Defense: Single Segment

In developing the mobile defensive strategy the following assumptions are made:

1) Both attacker's and defender's forces face each other across the defense line segment of length M.

2) Both are aware of each other's size, approximately equal, but the attacker's initial point of attack is not known and therefore must be treated probabilistically.

3) Since the attacker's objective is to gain as much territory as possible, the attacking force will therefore move perpendicularly to the defense line with velocity v_a.

4) Only one attacker force is expected to cross the defense line.

5) The defender's peacetime base is at the point (x, y) and its forces can move with velocity v_d towards the point (I, z) to meet the attacking force.

The tactical scenario representing the above assumptions is shown in Fig. 8.10.

The main objective of the defending force is to engage the attacker as close to the defense line as feasible. To accomplish this task the peacetime base must be placed at a location (x, y) that will minimize the infiltration of the attacking

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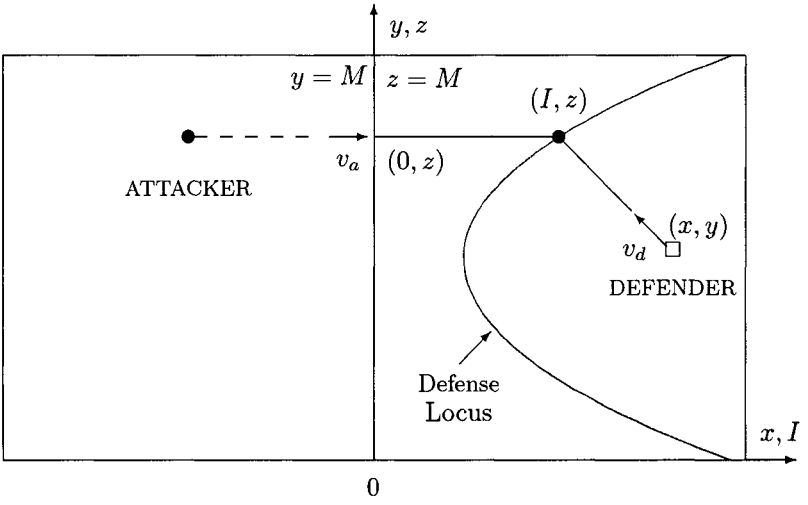


Fig. 8.10 Initial positioning of the defence units.

forces. Assuming that the infiltration takes place at the point $(0, z)$, it is required that:

> The time taken by the attacking force to traverse a distance I = The time required to transport the defending forces from their peacetime base (x, y) to the battle area location (I, z).

Hence,

$$\frac{I}{v_a} = \frac{\sqrt{(x-I)^2 + (y-z)^2}}{v_d} \tag{8.63}$$

where $\sqrt{(x-I)^2 + (y-z)^2}$ is the distance between the points (x, y) and (I, z). The exact location of the infiltration point $(0, z)$ can only be estimated by the defender. It is, however, possible to postulate a plausible probability density distribution $p(z)$ (either distributed or discrete) determined by such factors as the defender's beliefs about the strategic value attributed by the aggressor to targets in the defender's territory and the terrain topography permitting the breaching of the front with large force concentration at the point $(0, z)$. Because the attacker must breach the front at some point along the front, it follows that

$$\int_0^M p(z)\,dz = 1 \tag{8.64}$$

Several probability density distributions $p(z)$ are shown in Fig. 8.11. Depending on the relative speeds v_a and v_d, there are three different tactical scenarios: Case 1 when $v_a = v_d$, Case 2 when $v_a < v_d$, and Case 3 when $v_a > v_d$.

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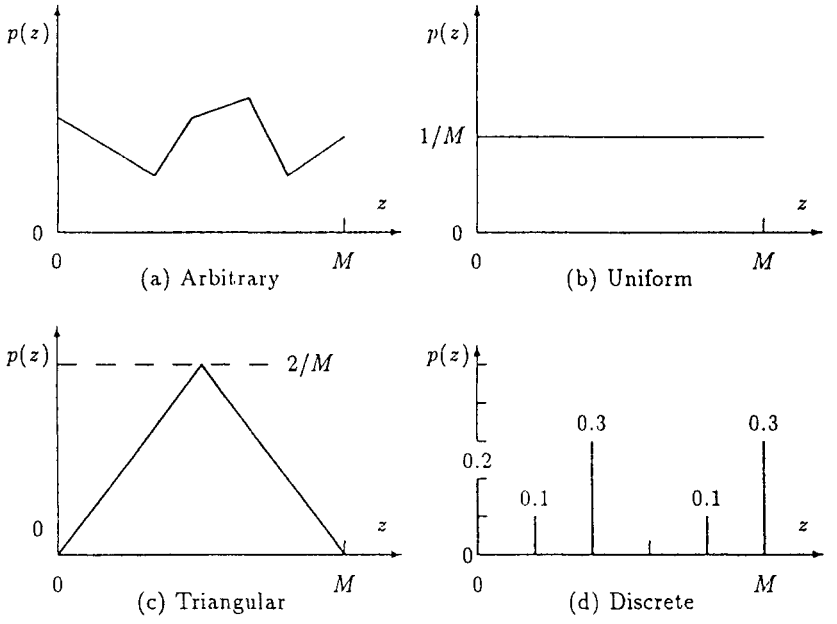


Fig. 8.11 Different types of probability density distribution $p(z)$ of an attack across the front line $z = 0$ to $z = 1.0$. These distributions apply only for $0 \leq z \leq M$; elsewhere, $p(z) = 0$.

Case 1: $v_a = v_d$

When $v_a = v_d$, it follows from Eq. (8.63) that

$$I = I(x, y; z) = \frac{x^2 + (y - z)^2}{2x} \tag{8.65}$$

The expected distance of infiltration of the invading forces into the defended territory is given by

$$\bar{I}(x, y; z) = \int_0^M p(z) I(x, y; z)\, dz \tag{8.66}$$

The defense must chose the peacetime position (x, y) to minimize the expected infiltration, i.e.,

$$\min_{x,y} \left[\int_0^M p(z) I(x, y; z)\, dz \right] \tag{8.67}$$

which requires that

$$\frac{\partial}{\partial x} \left[\int_0^M p(z) I(x, y; z)\, dz \right] = 0 \tag{8.68}$$

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and

$$\frac{\partial}{\partial y}\left[\int_0^M p(z)I(x, y; z)\,dz\right] = 0 \tag{8.69}$$

In this case, because the Leibnitz Rule for interchanging the order of differentiation and integration is applicable*, Eqs. (8.68) and (8.69) can be replaced by

$$\int_0^M \frac{\partial}{\partial x}\left[p(z)I(x, y; z)\right]\,dz = 0 \tag{8.70}$$

and

$$\int_0^M \frac{\partial}{\partial y}\left[p(z)I(x, y; z)\right]\,dz = 0 \tag{8.71}$$

Substituting the partial derivatives of I with respect to x and y into Eqs. (8.70) and (8.71), obtained from Eq. (8.65), it follows that

$$\int_0^M p(z)\frac{\partial I}{\partial x}\,dz = \int_0^M p(z)\left(\frac{1}{2} - \frac{(y-z)^2}{2x^2}\right)\,dz = 0 \tag{8.72}$$

and

$$\int_0^M p(z)\frac{\partial I}{\partial y}\,dz = \int_0^M p(z)\frac{(y-z)}{x}\,dz = 0 \tag{8.73}$$

Solving Eq. (7.73) first for y

$$y\int_0^M p(z)\,dz = \int_0^M zp(z)\,dz$$

Noting that $\int_0^M p(z)\,dz = 1$ (see Eq. (8.64))

$$y = \int_0^M zp(z)\,dz = \mu \tag{8.74}$$

where μ denotes the expected or mean value of z and it also represents the point along the front line where the aggressor is expected to attack the defender. Thus, the defense should position its forces at $y = \mu$. Replacing y with μ in the integral in Eq. (8.72) leads to

$$\int_0^M p(z)\left[1 - \frac{(\mu - z)^2}{x^2}\right]\,dz = 0$$

Hence, using again $\int_0^M p(z)\,dz = 1$

$$x^2 = \int_0^M (\mu - z)^2 p(z)\,dz = \sigma^2 \tag{8.75}$$

*The rule is applicable only under certain conditions regarding the nature of the integrand and the limits of integration (See Ref. 11 for example).

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where σ^2 denotes the variance of $p(z)$ and it represents the dispersion of the aggressor's probability density distribution about its mean value μ. For the uniform probability density distribution $p(z) = 1/M$,

$$y = \mu = \int_0^M \frac{1}{M} z\, dz = \frac{M}{2} \tag{8.76}$$

$$x^2 = \sigma^2 = \int_0^M \frac{1}{M}(\mu - z)^2\, dz = \frac{M^2}{12}$$

and

$$x = \frac{M}{\sqrt{12}} \tag{8.77}$$

Gupta[10] described the preceding results as the $\sigma - \mu$ theorem, which is stated as follows:

> The defense commander can minimize expected offensive infiltration into defense territory and simultaneously match the invader unit for unit at each focus of penetration if all the defensive units are concentrated at a distance μ along the length-M front and at a distance σ inward from the border. μ is the mean and σ is the standard deviation of the probability density function [distribution] of attack the defense expects the would-be conqueror to follow. In terms of our coordinate system established earlier, the optimal point of placement of the defense is given by (μ, σ). In deriving this result it is assumed that the aggressor and the defender possess equal velocities of transporting forces over defense territory.

Substitution of $x = M/\sqrt{12}$ and $y = M/2$ into Eq. (8.65) leads to a relationship between I and z

$$I = \frac{\frac{M^2}{12} + (\frac{M}{2} - z)^2}{\frac{2M}{\sqrt{12}}} \tag{8.78}$$

which defines the *defense locus* bounding the territory penetrated by the attacker before the defense forces are able to confront the attacker.

Case 2: $v_a < v_d$

Introducing a new parameter

$$\lambda = v_a/v_d \tag{8.79}$$

into Eq. (8.63), and then squaring both sides of this equation, it follows that

$$I^2\lambda^{-2} = (x - I)^2 + (y - z)^2$$

the positive solution of which gives

$$I = \frac{-x + \sqrt{x^2 + (\lambda^{-2} - 1)[x^2 + (y - z)^2]}}{(\lambda^{-2} - 1)} \tag{8.80}$$

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As in Case 1, Eqs. (8.70) and (8.71) are used to determine the optimal position (x, y), except that in this case, in order to obtain an analytical solution, a uniform probability density distribution $p(z) = 1/M$ will be assumed. Thus the optimality conditions become

$$\frac{1}{M}\int_0^M \frac{\partial I(x, y; z)}{\partial x}\, dz = 0 \tag{8.81}$$

$$\frac{1}{M}\int_0^M \frac{\partial I(x, y; z)}{\partial y}\, dz = 0 \tag{8.82}$$

Computing the partial derivatives with respect to x and y from Eq. (8.80)

$$\frac{\partial I}{\partial x} = \frac{1}{\lambda^{-2} - 1}\left(-1 + \frac{\lambda^{-2}x}{\sqrt{x^2 + (\lambda^{-2} - 1)[x^2 + (y - z)^2]}}\right) \tag{8.83}$$

$$\frac{\partial I}{\partial y} = \frac{\lambda(y - z)}{\sqrt{x^2 + (y - z)^2}} \tag{8.84}$$

Substitution of Eq. (8.84) into Eq. (8.82) leads to

$$\int_0^M \frac{\lambda(y - z)}{\sqrt{x^2 + (y - z)^2}}\, dz = 0 \tag{8.85}$$

which can now be integrated with the help of a new variable $u = (y - z)^2$ so that Eq. (8.85) can be rewritten as

$$-\int_{y^2}^{(y-M)^2} \frac{du}{\sqrt{x^2 + u}} = 0$$

Hence,

$$\sqrt{x^2 + u}\,\Big|_{u=y^2}^{u=(y-M)^2} = 0$$

from which it follows that

$$y = M/2 \tag{8.86}$$

Similarly, substituting Eq. (8.83) into Eq. (8.81), with $y = M/2$,

$$\int_0^M \left(-1 + \frac{\lambda^{-2}x}{\sqrt{x^2 + (\lambda^{-2} - 1)\left(\frac{M}{2} - z\right)}}\right) dz = 0$$

$$x\lambda^{-2}\int_0^M \frac{dz}{\sqrt{x^2 + (\lambda^{-2} - 1)\left(\frac{M}{2} - z\right)}} = M$$

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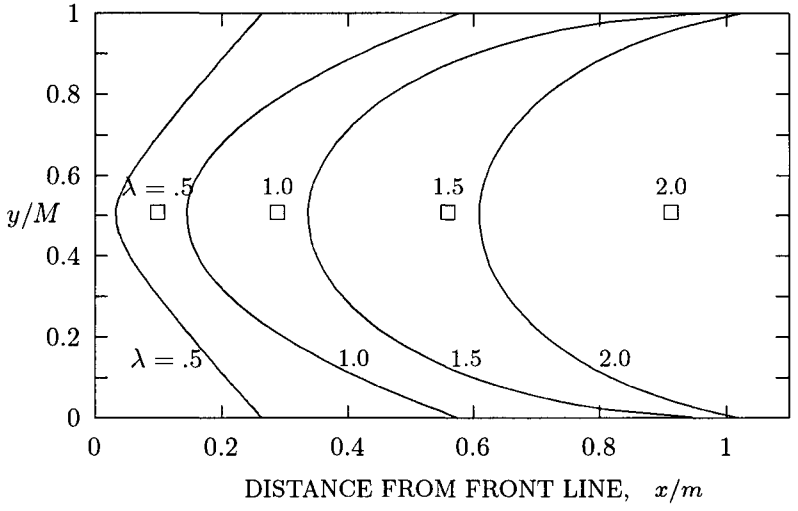


Fig. 8.12 Defense loci for $\lambda = 0.5, 1.0, 1.5$, and 2.0.

Introducing a new variable $v = \sqrt{\lambda^{-2} - 1}(\frac{M}{2} - z)$

$$x\lambda^{-2} \int_{\sqrt{\lambda^{-2}-1}\,M/2}^{-\sqrt{\lambda^{-2}-1}\,M/2} \frac{-dv}{\sqrt{\lambda^{-2}-1}\sqrt{\lambda^{-2}x^2 + v^2}} = M$$

$$\frac{x\lambda^{-2}}{\sqrt{\lambda^{-2}-1}} \ln \left| v + \sqrt{\lambda^{-2}x^2 + v^2} \right|_{v=-\sqrt{\lambda^{-2}-1}\,M/2}^{v=\sqrt{\lambda^{-2}-1}\,M/2} = M$$

$$\frac{x\lambda^{-2}}{\sqrt{\lambda^{-2}-1}} \ln \left[\frac{\sqrt{\lambda^{-2}-1}\,\frac{M}{2} + \sqrt{\lambda^{-2}x^2 + (\lambda^{-2}-1)\,\frac{M^2}{4}}}{-\sqrt{\lambda^{-2}-1}\,\frac{M}{2} + \sqrt{\lambda^{-2}x^2 + (\lambda^{-2}-1)\,\frac{M^2}{4}}} \right] = M$$

which can be further reduced to

$$\frac{x\lambda^{-2}}{\sqrt{\lambda^{-2}-1}} \ln \left(\frac{\sqrt{\frac{\lambda^{-2}x^2}{\lambda^{-2}-1} + \frac{M^2}{4}} + \frac{M}{2}}{\sqrt{\frac{\lambda^{-2}x^2}{\lambda^{-2}-1} + \frac{M^2}{4}} - \frac{M}{2}} \right) = M \tag{8.87}$$

Equation (8.87) is a transcendental equation that can only be solved numerically for x. The solution of this equation for $\lambda = 0.5$ is $x = 0.0994M$, and the corresponding optimum peacetime base location (x, y) and its defense locus are shown in Fig. 8.12.

Case 3: $v_a > v_d$

For this case $\lambda > 1$ and Eq. (8.63) can be rewritten as

$$I^2 = (\lambda - I)^2\lambda^2 + (y - z)^2\lambda^2$$

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from which

$$I = \frac{x \pm \sqrt{x^2 - (1-\lambda^{-2})[x^2 + (y-z)^2]}}{(1-\lambda^{-2})} \tag{8.88}$$

To resolve the sign ambiguity, consider the case of $z = y$ where the attack takes place diametrically opposite from (x, y). If the time that elapses before the first contact is made between the defense units and the attacker is denoted as t, then $I = tv_a$ and $x - I = tv_d$. Eliminating t between the two equations yields

$$I = \frac{x}{(1+\lambda^{-1})} \tag{8.89}$$

and therefore the negative sign must be chosen so that

$$I = \frac{x - \sqrt{x^2 - (1-\lambda^{-2})[x^2 + (y-z)^2]}}{(1-\lambda^{-2})} \tag{8.90}$$

Computing the partial derivatives of Eq. (8.90) results in

$$\frac{\partial I}{\partial x} = \frac{1}{(1-\lambda^{-2})}\left(1 - \frac{x\lambda^{-2}}{\sqrt{\lambda^{-2}x^2 - (1-\lambda^{-2})(y-z)^2}}\right) \tag{8.91}$$

$$\frac{\partial I}{\partial y} = \frac{(y-z)}{\sqrt{x^2 - (1-\lambda^{-2})[x^2 + (y-z)^2]}} \tag{8.92}$$

Substitution of Eq. (8.92) into Eq. (8.82) leads to

$$\int_0^M \frac{(y-z)\,dz}{\sqrt{x^2 - (1-\lambda^{-2})[x^2 + (y-z)^2]}} = 0 \tag{8.93}$$

which can be integrated by substituting $u = (y-z)^2$ as a new variable so that Eq. (8.93) becomes

$$\int_{z^2}^{(y-M)^2} \frac{du}{\sqrt{x^2 - (1-\lambda^{-2})(x^2+u)}} = 0$$

and therefore

$$\sqrt{\lambda^{-2}x^2 + u}\Big|_{u=y^2}^{u=(y-M)^2} = 0$$

from which it follows that

$$y = \frac{M}{2} \tag{8.94}$$

Similarly, substituting Eq. (8.91) into Eq. (8.81), and using $y = M/2$, results in

$$\frac{1}{(1-\lambda^{-2})}\int_0^M \left(1 - \frac{x\lambda^{-2}}{\sqrt{x^2\lambda^{-2} - (1-\lambda^{-2})\left(\frac{M}{2} - z\right)^2}}\right) dz = 0$$

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$$x\lambda^{-2}\int_0^M \frac{dz}{\sqrt{x^2\lambda^{-2}-(1-\lambda^{-2})\left(\frac{M}{2}-z\right)^2}} = M$$

$$\frac{x\lambda^2}{\sqrt{1-\lambda^{-2}}}\int_0^M \frac{dz}{\sqrt{\frac{x^2\lambda^{-2}}{\lambda^{-2}-1}-\left(\frac{M}{2}-z\right)^2}} = M$$

Introducing the variable $v = (\frac{M}{2} - z)$ it follows that

$$\frac{x\lambda^{-2}}{\sqrt{1-\lambda^{-2}}}\int_{-M/2}^{M/2} \frac{dv}{\sqrt{\frac{x^2\lambda^{-2}}{1-\lambda^{-2}}-v^2}} = M$$

$$\frac{x\lambda^{-2}}{\sqrt{1-\lambda^{-2}}}\sin^{-1}\left(\frac{v}{\frac{x\lambda^{-1}}{\sqrt{1-\lambda^2}}}\right)\Bigg|_{v=-M/2}^{v=M/2} = M$$

Hence,

$$\frac{2x\lambda^{-2}}{\sqrt{1-\lambda^2}}\sin^{-1}\left(\frac{\sqrt{\lambda^2-1}\,M}{2x}\right) = M$$

which after rearranging becomes

$$\sin^{-1}\left(\sqrt{\lambda^2-1}\,\frac{M}{2x}\right) = \lambda\sqrt{\lambda^2-1}\,\frac{M}{2x} \tag{8.95}$$

As in the previous case, Eq. (8.95) can only be solved numerically. The base location for $\lambda = 1.5$ and 2.0 and the corresponding defense loci are shown in Fig. 8.12. The variation of x/M with $\lambda = v_a/v_d$ is also shown in Fig. 8.13, which demonstrates that as λ is increased the location of the peacetime base for the defending forces moves away from the front line.

Optimal Mobile Defense: Multiple Segments

The previous sections addressed the case of a single force formation of N units assigned the defense of the whole length M of the front line. It may, however, be advantageous for strategic reasons to divide the defense forces into two or more segments. The simplest case is to divide the length-M into two segments, each of length $M/2$, and assign each segment to one of the two defending force formations of $N/2$. If each force formation uses the single segment defense optimum location, then it follows from previous discussion that the forces defending the lower segment must be positioned at $(M/2\sqrt{12}, M/4)$, and the upper segment forces at $(M/2\sqrt{12}, 3M/4)$. This scenario for the two-segment front is illustrated in Fig. 8.14.

The defense locus for the equal, two-segment scenarios restricts maximum offensive penetration of the attacking force to $M/\sqrt{12} = 0.2887M$, which can be contrasted with $2M/\sqrt{12} = 0.5773M$ for the single segment scenario. The penalty for having a 50% smaller penetration, however, is that the defending force of $N/2$

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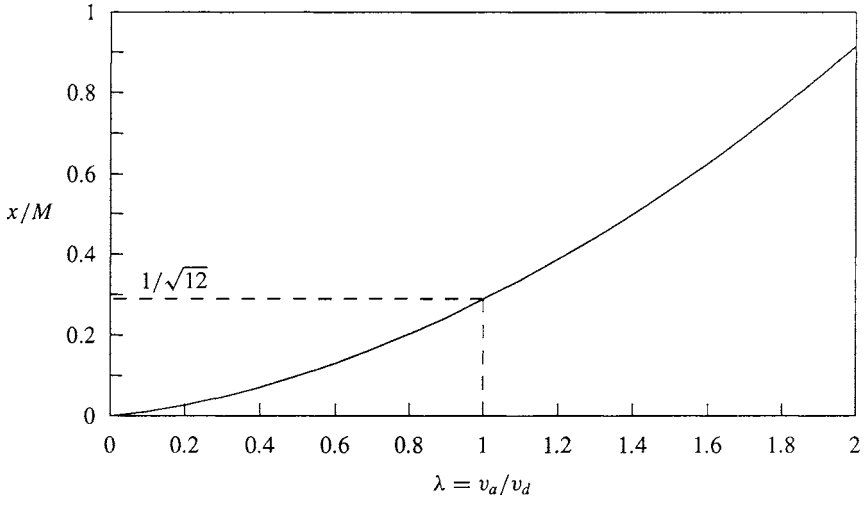


Fig. 8.13 Variation of the defensive force position x/M with $\lambda = v_a/v_d$.

units must be prepared to engage in battle with the attacking force ranging from 0 to N force units and be able to withstand the attack until reinforcing units arrive. The defending force is no longer capable of matching force unit for every unit of the attacking force at each point on the defense locus. Thus the defense has traded off its capability for perfect counterconcentration in exchange for force dispersal and reduced offensive penetration.

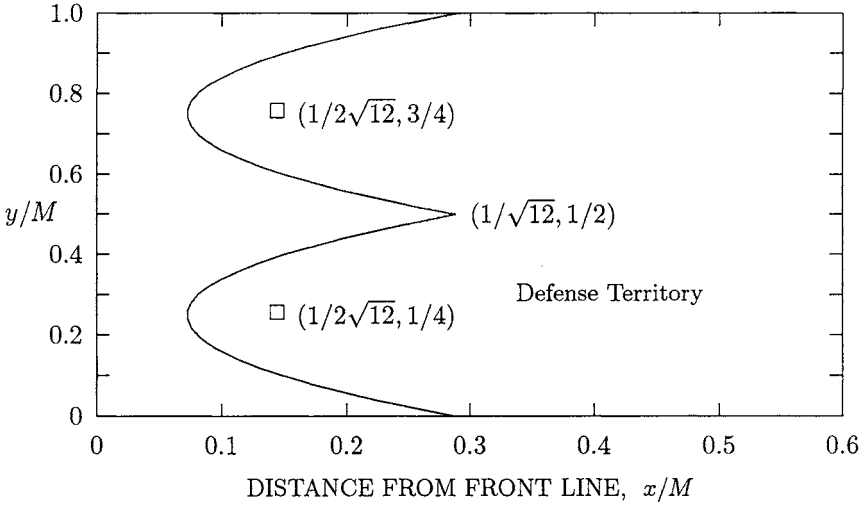


Fig. 8.14 Two-segment defense: Equal defense coverage.

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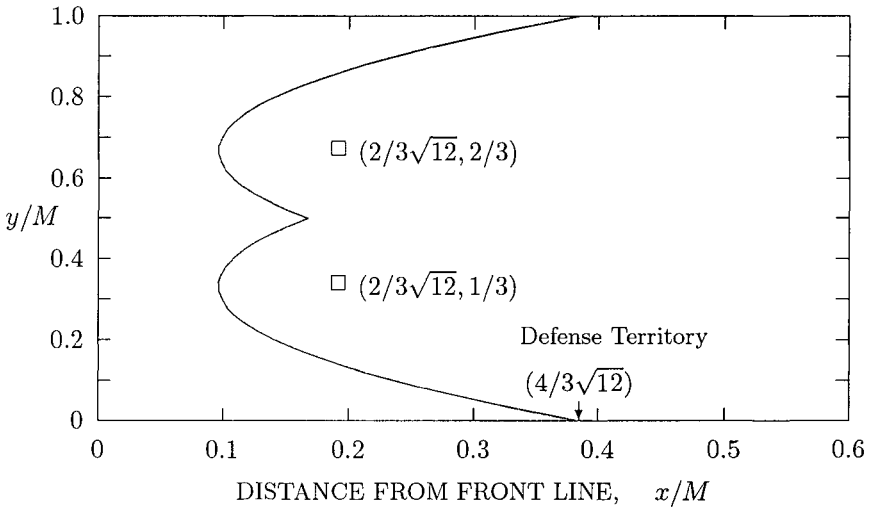


Fig. 8.15 Two-segment defense: Overlapping defense coverage.

There are also other possibilities that can be considered for a two-segment defense. For example, the y-coordinate can be taken as $y = M/3$ and $y = 2M/3$. In this case, the lower segment defense force would be responsible for the lower two-thirds of the front, while the upper segment defense force would be responsible for the upper two-thirds. The defending forces would be located at $(2M/3\sqrt{12}, M/3)$ and $(2M/3\sqrt{12}, 2M/3)$. This particular scenario with overlapping responsibilities over the middle one-third of the front is presented in Fig. 8.15. In this case, the maximum penetration of the attacking force is $4M/3\sqrt{12} = 0.3849M$.

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Problems

8.1. (a) Obtain a numerical solution* of the equations of combat represented by

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

with

$$\mathbf{X}_0 = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}_{t=0} = \begin{bmatrix} 100 \\ 125 \end{bmatrix}$$

(b) Compare the numerical solution from (a) with the exact solution obtained using the method of a "one-on-one" direct fire engagement from Chapter 4.

8.2. Show that Eq. (8.17a) represents two uncoupled sets of equations. Sketch a diagram of the engagement. Indicate the two sets of equations in terms of the original numbering of rows and columns.

8.3. Determine the superiority parameter S_0 for a tactical combat engagement represented by $\dot{\mathbf{X}} = \mathbf{C}\mathbf{X}$ where

$$\begin{matrix} \mathbf{X} = \{\mathbf{X}_B & \mathbf{X}_R\} \\ \mathbf{X}_B = \{X_1 & X_2\} \\ \mathbf{X}_R = \{X_3 & X_4\} \end{matrix} \qquad \mathbf{C} = \begin{bmatrix} 0 & 0 & -1 & -2 \\ 0 & 0 & -2 & -1 \\ -3 & -1 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{bmatrix}$$

$$\mathbf{X}_0 = \{100 \quad 100 \quad 75 \quad 50\}$$

*Use the computer program provided with this text.

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8.4. (a) Obtain a numerical solution* for a directed fire tactical engagement represented by $\dot{\mathbf{X}} = \mathbf{CX}$ where

$$\mathbf{X} = \{\mathbf{X}_B \quad \mathbf{X}_R\}$$
$$\mathbf{X}_B = \{X_1 \quad X_2\}$$
$$\mathbf{X}_R = \{X_3 \quad X_4\}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & -1 \\ -1 & -1 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{X}_0 = \{200 \quad 100 \quad 75 \quad 500\}$$

(b) Determine the variation of S with the nondimensional time $\lambda^{(1)}t$.

8.5. For cases 1, 2, and 3 in Section 8.5, obtain the aggregated force solutions for X_1, X_2, and X_3 as functions of the nondimensional time τ. Compare these approximate solutions with the numerical solutions in Figs. 8.4, 8.5, and 8.6 and comment on the results.

8.6. Obtain numerical solution* to a battle engagement represented by $\dot{\mathbf{X}} = \mathbf{CX}$ where

$$\mathbf{X} = \{\mathbf{X}_B \quad \mathbf{X}_R\}$$
$$\mathbf{X}_B = \{X_1 \quad X_2\}$$
$$\mathbf{X}_R = \{X_3 \quad X_4\}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -10 & -0.2 \\ 0 & 0 & -0.1 & -1 \\ -10 & -0.2 & 0 & 0 \\ -0.1 & -1 & 0 & 0 \end{bmatrix}$$

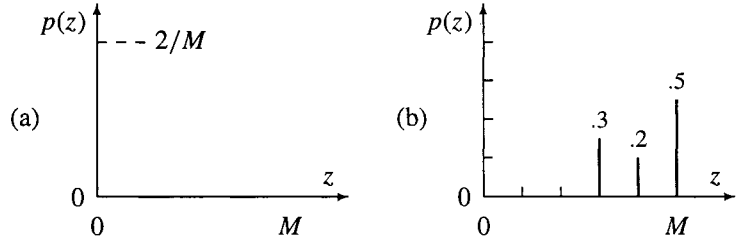
$$\mathbf{X}_0 = \{20 \quad 100 \quad 10 \quad 200\}$$

When of the the X_i-forces is reduced to zero, reconfigure the combat equation $\dot{\mathbf{X}} = \mathbf{CX}$ to determine the final solution. Comment on your results.

8.7. Obtain a numerical solution* for an area fire tactical engagement using $\mathbf{X_0}$ and $\mathbf{C}$ from Problem 8.4.

8.8. Obtain a numerical solution* for a guerrilla warfare engagement using $\mathbf{X_0}$ and $\mathbf{C}$ from Problem 8.4.

8.9. Determine the peacetime locations of the defending forces for the following probability density distributions of attack: (a) triangular and (b) discrete distribution shown below.



*Use the computer program provided with this text.

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9
Reliability of Operations and Systems

This chapter discusses the rudimentary principles of reliability in military operations and military systems that are the same as those for mechanical systems[1] in which the individual or component reliabilities are replaced by the operational reliabilities. As in mechanical systems, the operations can be treated either as series or as parallel (redundant) operations. These simplified concepts allow for a better insight into different operational scenarios to achieve a particular objective. In addition to the operational reliability concepts, weapon system reliability and reliability deterioration with time are introduced. Other topics discussed in this chapter include the maintainability function and operational availability of weapon systems.

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9.1 Reliability of Series Operations

A series system of operations, in a reliability sense, means that all components in the system must operate in order for the whole operation to be successful. The block diagram for such a system is shown in Fig. 9.1.

The reliability of a military operation is measured in terms of its probability of success. Thus, the probability of success of a series of components comprising the overall mission (operation) is the probability of success of independent events, i.e.,

$$R = R_1 R_2 \cdots R_n = \prod_{i=1}^{n} R_i \tag{9.1}$$

where R_i is the probability of success of the rth operation.

A typical example of a series system of military operations was the 1980 mission to rescue American hostages in Iran (Fig. 9.2), which consisted of six independent operations. Although no details were officially disclosed, *Newsweek* magazine postulated the following scenario[2]:

1) Desert One location: landing of three C-130 transport airplanes and six big RH-53D helicopters (Sea Stallions) at a remote desert location in Iran.

2) The RH-53D helicopters with commando troops were to have flown to a mountain hideout east of Tehran, while the C-130 airplanes would leave Iran and later return to a different area outside Tehran.

3) The following night, the troops were to drive in disguise to Tehran in prepositioned vehicles, presumably arranged by U.S. intelligence agents.

4) The rescuers were then to attack the militia guarding both the U.S. Embassy and the Iranian Foreign Ministry.

5) Once the militiamen were overpowered and the American hostages secured, the RH-53D helicopters were to land in the embassy compound and airlift all the hostages out to a nearby airstrip where they would board the C-130s.

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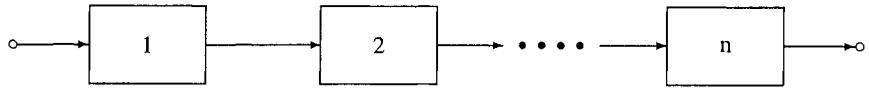


Fig. 9.1 Operations in series.

6) All of the American hostages and their rescuers were to be flown out of the country in the C-130s.

If the probability of success of each component of this mission were 0.9 (i.e., $R_i = 0.9$), then, from Eq. (9.1),

$$R = (0.9)^6 = 0.53$$

and, for $R_i = 0.8$,

$$R = (0.8)^6 = 0.26$$

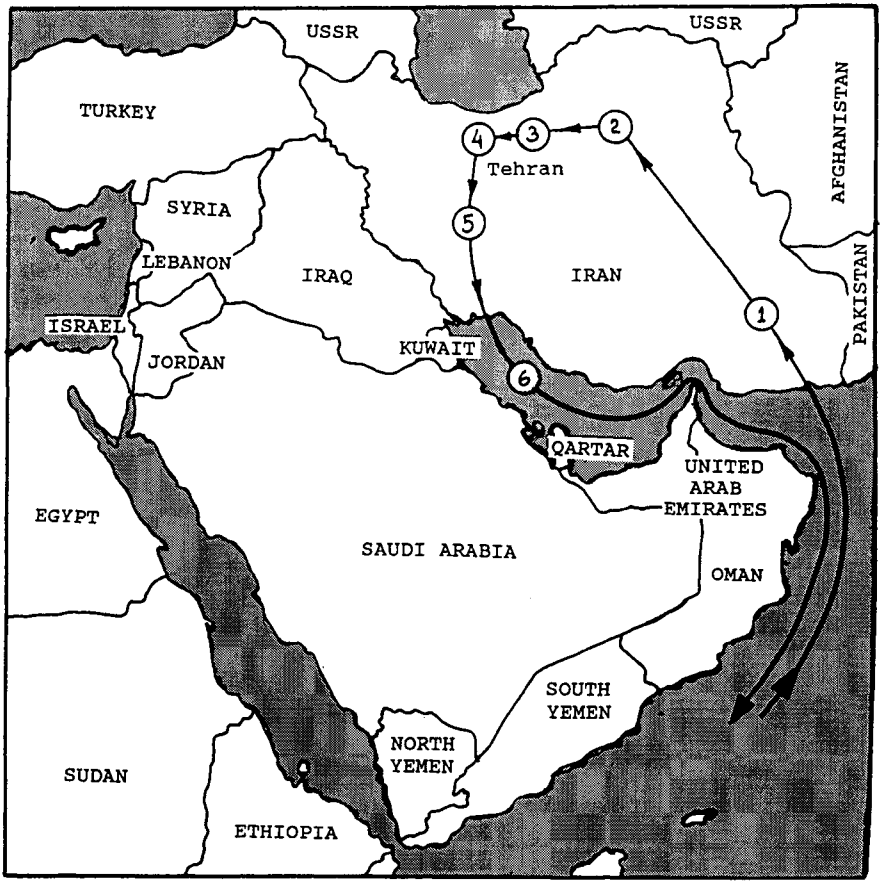


Fig. 9.2 American hostages rescue mission in Iran; six operations in series.

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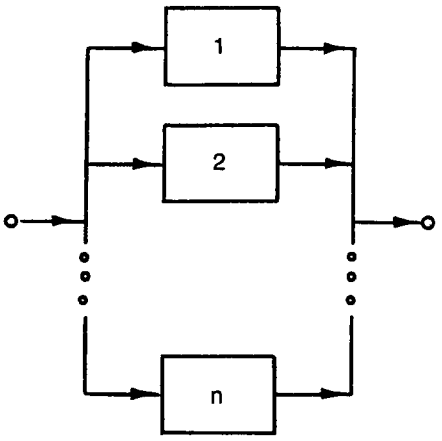


Fig. 9.3 Parallel (redundant) system of operations.

The preceding numbers clearly indicate that the lengthy series of operations could not possibly lead to an acceptable level of probability of success for this rescue mission. As it happened, the mission failed in the first component of the rescue operation.

9.2 Reliability of Parallel (Redundant) Operations

A parallel system of operations, also referred to as a redundant system, consists of two or more components. The system will fail only if all of the components will fail. The block diagram for such systems is shown in Fig. 9.3. Such systems are also described as having *active redundancy*, where all components are on-line and where the system will operate as long as at least one of the parallel components is operational. The overall operation (system) failure can be thought of as a compound event consisting of the failure of all individual components and can be expressed as

$$P = P_1 P_2 \cdots P_n = \prod_{i=1}^{n} P_i \tag{9.2}$$

where

$$P_i = 1 - R_i \tag{9.3}$$

represents the probability of failure of the ith operation (system), and R_i is the probability of success (reliability) of the ith operation. Hence, from the definition of the overall reliability,

$$R = 1 - P = 1 - (1 - R_1)(1 - R_2) \cdots (1 - R_n) \tag{9.4}$$

Another type of parallel system is one with *standby redundancy*, which means that only one component of the operation is *on-line* at a time (see Fig. 9.4). When operation 1 fails, operation 2 is implemented, etc. This scenario assumes perfect command and control for the operation in which failures are immediately detected and redundant operations introduced. The overall operation will be successful as

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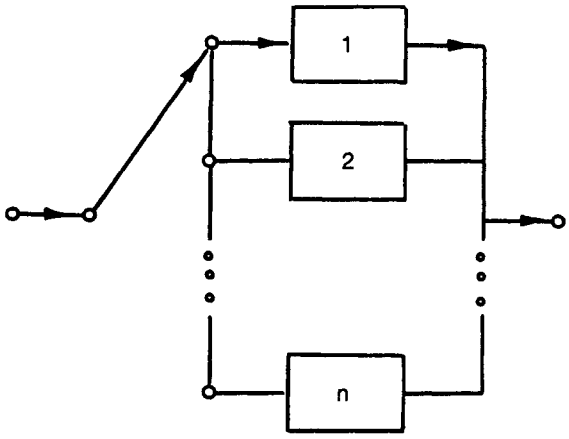


Fig. 9.4 Standby redundant system of operations.

long as the number of failures does not exceed $n-1$, where n is the total number of redundant components. The standby redundancy method is more efficient than the active redundancy, since only the active unit is being "used up" in the operation.

In military operations, the use of active redundant operations may be necessary to ensure success of the overall mission; however, if the number of redundant systems is excessive, the additional operational costs lead to diminishing returns and result in an unnecessary "overkill." Typically only one standby system is required, as illustrated by an example at the end of this section.

The standby redundancy presupposes that the reliability of the operation varies with time. It will be assumed here that the probability of success (reliability) of a standby system is determined by the Poisson distribution. The general term of a Poisson distribution from Sec. 3.3 is given by

$$P(r) = \frac{\lambda^r}{r!} e^{-\lambda} \tag{3.26}$$

where the different parameters have the following meaning assigned to a standby system as indicated in parentheses and where

$P(r)$ = the "event" probability (reliability of system for r events)
r = number of events (number of failed standby systems)
λ = rate of failure (constant)

Assuming that all standby systems are identical for which

$$\lambda = \mu t \tag{9.5}$$

the reliability $R^{(1)}$ for a single system operation becomes

$$R^{(1)} = \text{probability of success with no system failures } (r = 0)$$

$$= \frac{(\mu t)^0}{0!} e^{-\mu t} = e^{-\mu t} \tag{9.6}$$

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Note here how the probability of success (reliability of operation) diminishes with time. The operation has the highest degree of success when it is completed quickly, i.e., when the element of surprise is fully exploited. For two systems,

$$R^{(2)} = \text{probability of success with no system failure } (r = 0) + \text{probability of success with one failure } (r = 1)$$

$$= \frac{(\mu t)^0}{0!} e^{-\mu t} + \frac{(\mu t)^1}{1!} e^{-\mu t} = (1 + \mu t) e^{-\mu t} \tag{9.7}$$

and for n systems,

$$R^{(n)} = \left[1 + \mu t + \frac{(\mu t)^2}{2!} + \cdots + \frac{(\mu t)^{n-1}}{(n-1)!}\right] e^{-\mu t} \tag{9.8}$$

As an example of a three-component system (two standby systems backing up the first system) for which $\mu t = 0.5$,

$$R^{(3)} = (1 + 0.5 + 0.125) e^{-0.5} \tag{9.8}$$

$$= 0.606 + 0.303 + 0.076 = 0.985 \tag{9.9}$$

This demonstrates that the second standby system contributes only 0.076/0.985 (or 7.7%) to the overall success of the operation. Thus, in the military operation a second backup operation would not be used.

9.3 Reliability of Combined (Series and Parallel) Operations

The component operations can be combined partly in series and partly in parallel as shown in Fig. 9.5a. To obtain the overall reliability, the system must be reduced to a system of components in series as shown in Fig. 9.5b. Thus, from Eq. (9.4) the reliability of the branch 2,4 is

$$R_{2,4} = 1 - (1 - R_{2,3})(1 - R_4) \tag{9.9}$$

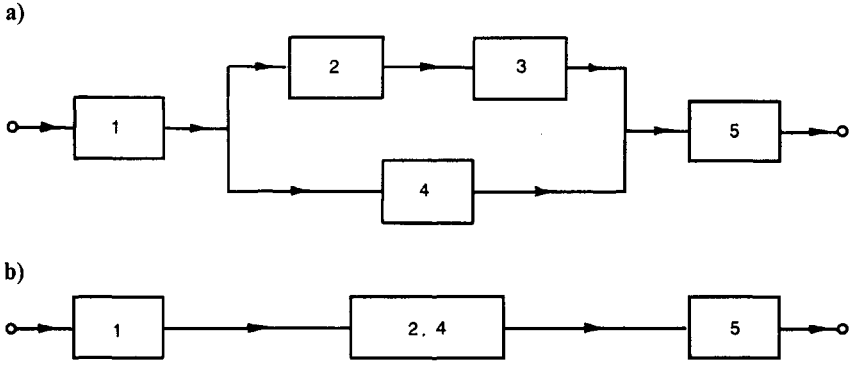


Fig. 9.5 Combined system of operations: a) Combined series and parallel operations and b) Equivalent series operations.

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Noting that, from Eq. (9.1),

$$R_{2,3} = R_2 R_3 \tag{9.10}$$

it follows that

$$\begin{aligned} R &= R_1 R_{2,4} R_5 \\ &= R_1[1 - (1 - R_2 R_3)(1 - R_4)] R_5 \end{aligned} \tag{9.11}$$

9.4 Example of an Air-to-Air Engagement

Consider the case of a fighter aircraft in an air-to-air engagement against a penetrating bomber. It will be assumed that the fighter is initially alerted and guided to the airborne target by the ground radar and then is subsequently guided by its own airborne radar. At the appropriate distance, the defending fighter aircraft releases two missiles against the penetrating bomber. The missiles are then guided by their own separate guidance systems, and the warheads are detonated by the proximity fuses. The reliabilities of different components for this air-to-air engagement are as follows:

R_g = ground radar reliability
R_a = airborne radar reliability
R_m = missile guidance system reliability
R_w = warhead fuse reliability
R_t = probability of target destruction with missile impact

The schematic representation of this engagement is shown in Fig. 9.5. Thus, the engagement scenario is represented here by a series/parallel combination of operations. Hence, using the result of Sec. 9.3, the overall probability of success (reliability) of this operation can be computed from

$$R = R_g R_a \left[1 - (1 - R_m R_w)^2\right] R_t \tag{9.12}$$

9.5 Reliability Variation with Time

The concepts of reliability developed for systems of operations or for components of weapon systems are also applicable to cases where the component reliabilities vary with time. The time variable of operations is derived from the probability density distribution of failure as a function of time. A typical probability density distribution $f(t)$ is shown in Fig. 9.7. It should be noted that the function $f(t)$ is always positive, i.e., $f(t) \geq 0$, and the area under the function between $t = 0$ and $t = \infty$ is equal to unity, i.e.,

$$\int_0^\infty f(t)\,dt = 1 \tag{9.13}$$

The function $f(t)$ satisfying Eq. (9.13) is also referred to as a single-tailed probability density distribution. In contrast, the probability density distribution $p(x)$ used in Chapter 3 was double-tailed, for which

$$\int_{-\infty}^\infty p(x)\,dx = 1 \tag{9.14}$$

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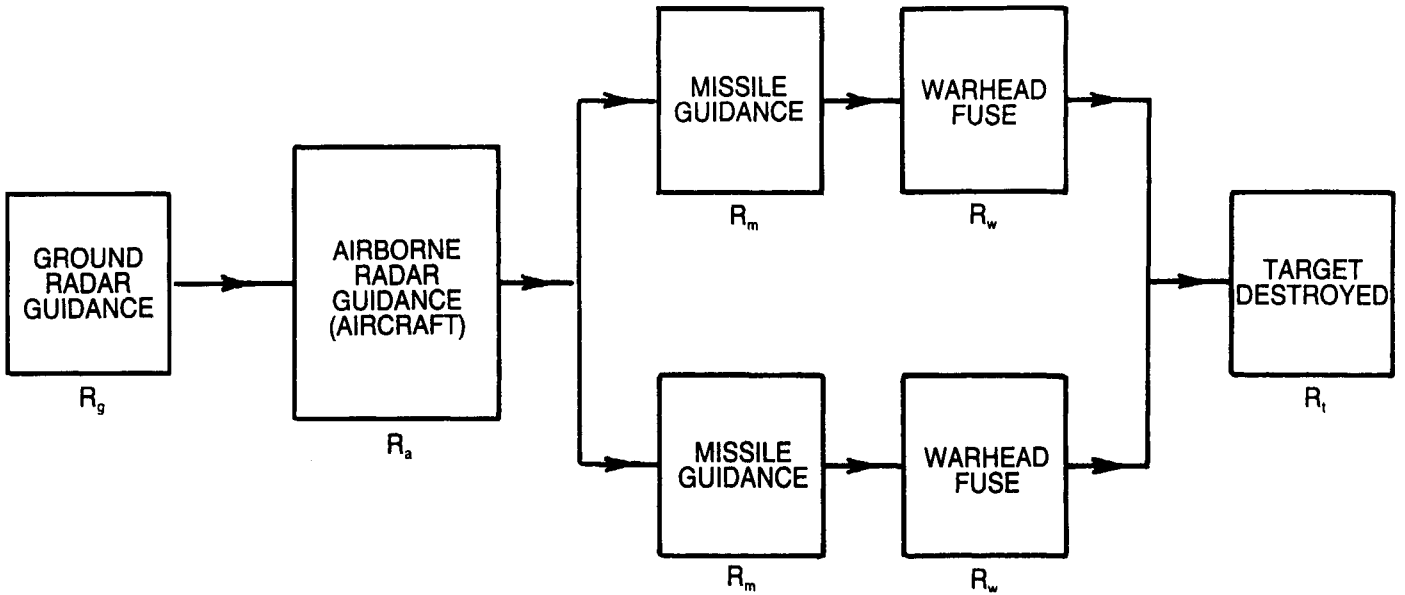


Fig. 9.6 Example of an air-to-air engagement reliability model.

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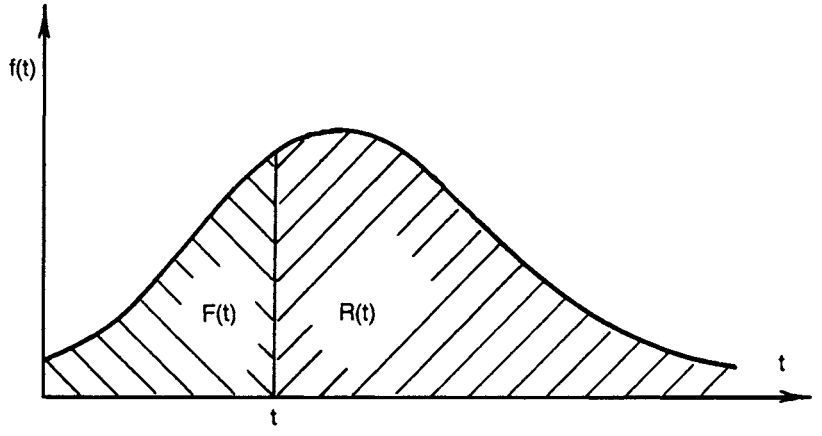


Fig. 9.7 Probability density distribution of failure.

The reliability function $R(t)$ is the probability that no failure will occur prior to time t given that the equipment is operating at time $t = 0$, and is expressed as

$$R(t) = 1 - F(t) = \int_t^\infty f(t)\,dt \tag{9.15}$$

Here the function $F(t)$ is the failure function, which is given by

$$F(t) = \int_0^t f(t)\,dt \tag{9.16}$$

which determines the probability that a failure will occur in time t, again given that the equipment is operational at time $t = 0$. The functions $F(t)$ and $R(t)$ are represented by areas under the probability density function $f(t)$ as shown in Fig. 9.7.

An important parameter in any reliability analysis is the mean time to failure (MTTF), which is defined here as the expected value of t for the probability density function $f(t)$. Hence,

$$\text{MTTF} = E[t] = \int_0^\infty tf(t)\,dt \tag{9.17}$$

The second moment of the $f(t)$ distribution about $t = \text{MTTF}$ provides a measure of the spread of the function about the its mean value. It is calculated from

$$\begin{aligned} s^2 &= \int_0^\infty (t - \text{MTTF})^2 f(t)\,dt = E[(t - \text{MTTF})^2] \\ &= \int_0^\infty t^2 f(t)\,dt - 2\text{MTTF}\int_0^\infty tf(t)\,dt + (\text{MTTF})^2 \int_0^\infty f(t)\,dt \\ &= \int_0^\infty t^2 f(t)\,dt - (\text{MTTF})^2 = E[t^2] - (\text{MTTF})^2 \end{aligned} \tag{9.18}$$

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where the parameter s^2 is described as the variance of $f(t)$. This parameter can be compared to the variance σ^2 used for the binomial and normal probability density distributions in Sec. 3.1 and 3.2.

In addition to the probability density distribution function $f(t)$, there is another important function used in reliability analysis. This function represents an instantaneous rate of failure defined as the fraction of components failing per unit time; it is usually denoted by $\gamma(t)$. Also, the function $\gamma(t)$ is referred to as the hazard function $h(t)$ [i.e., $\gamma(t) = h(t)$]. The hazard function is described in Sec. 8.7, where a simple relationship between the hazard function $h(t)$ and the probability density distribution function $f(t)$ is derived.

In the remainder of this section the four most commonly used probability density functions are discussed, and the corresponding reliability functions $R(t)$ and the MTTF formulas are derived. This discussion includes the exponential distribution, the normal distribution, the log-normal distribution, and the Weibull distribution.

Exponential Distribution

The exponential density distribution is expressed by

$$f(t) = \lambda e^{-\lambda t} \tag{9.19}$$

where λ is a constant (see Fig. 9.8a). For this distribution it can easily be shown from Eqs. (9.15) and (9.17) that

$$\begin{aligned} R(t) &= \int_t^\infty f(t)\,dt \\ &= \int_t^\infty \lambda e^{-\lambda t}\,dt = -e^{-\lambda t}\Big|_t^\infty = e^{-\lambda t} \end{aligned} \tag{9.20}$$

and

$$\begin{aligned} \text{MTTF} = E[t] &= \int_0^\infty t f(t)\,dt \\ &= \int_0^\infty t\lambda e^{-\lambda t}\,dt = -te^{-\lambda t}\Big|_0^\infty + \int_0^\infty e^{-\lambda t}\,dt \\ &= -\frac{1}{\lambda} e^{-\lambda t}\Big|_0^\infty = \frac{1}{\lambda} \end{aligned} \tag{9.21}$$

The variation of the exponential reliability function with the nondimensional time λt is shown in Fig. 9.8b.

Normal Distribution

The previously used density function $p(x)$ for the normal distribution was expressed by

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\bar{x})^2}{2\sigma^2}\right]; \qquad -\infty < x < \infty \tag{3.20}$$

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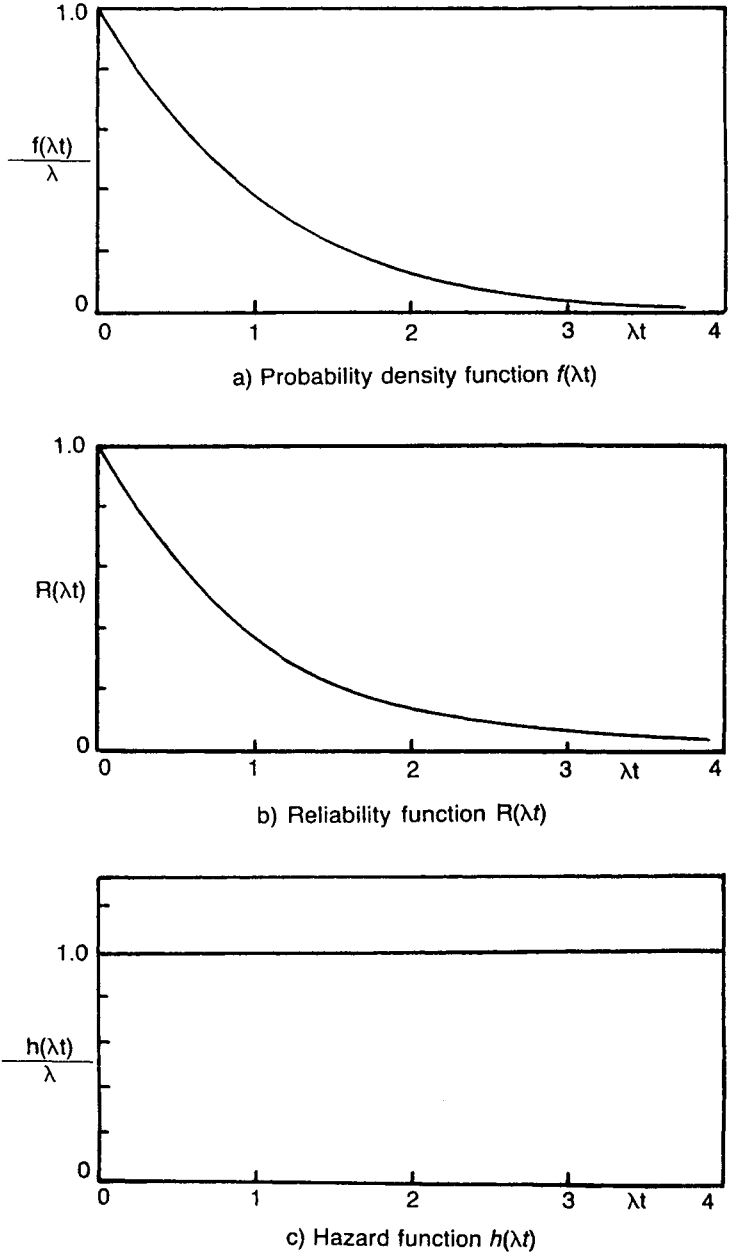


Fig. 9.8 Probability density function $f(\lambda t)$, reliability function $R(\lambda t)$, and hazard function $h(\lambda t)$ for exponential distributions.

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which satisfies the relationship

$$\int_{-\infty}^{\infty} p(x)\,dx = 1 \tag{3.31}$$

The function $p(x)$ can be modified to satisfy Eq. (9.13) by introducing a constant c and changing the spatial variable x and parameter $\bar{x}$ to the time variable t and parameter m so that

$$f(t) = \frac{c}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(t-m)^2}{2\sigma^2}\right]; \qquad 0 \le t < \infty \tag{9.22}$$

The constant c is then determined by substituting Eq. (9.22) into Eq. (9.13). Hence,

$$c = \frac{2}{1 + \text{erf}(m/\sqrt{2}\,\sigma)} \tag{9.23}$$

and

$$f(t) = \frac{\sqrt{2/\pi}}{\sigma[1 + \text{erf}(m/\sqrt{2}\,\sigma)]} \exp\left[-\frac{(t-m)^2}{2\sigma^2}\right] \tag{9.24}$$

The function $f(t)$ in Eq. (9.24) is sometimes referred to as the "truncated normal distribution." The variation of this function with the nondimensional time t/σ and the ratio m/σ is shown in Fig. 9.9a.

It can be demonstrated that for the normal distribution

$$\begin{aligned} \text{MTTF} &= \int_0^{\infty} t f(t)\,dt \\ &= \frac{\sqrt{2/\pi}}{1 + \text{erf}(m/\sqrt{2}\,\sigma)}\,\sigma \exp\left(-\frac{m^2}{2\sigma^2}\right) + m \end{aligned} \tag{9.25}$$

When $m = 0$, the preceding expression reduces to

$$\text{MTTF} = \sqrt{\frac{2}{\pi}}\,\sigma \tag{9.26}$$

Now computing the reliability function $R(t)$ from Eq. (9.15),

$$\begin{aligned} R(t) &= \int_t^{\infty} f(t)\,dt \\ &= \int_t^{\infty} \frac{\sqrt{2/\pi}}{\sigma[1 + \text{erf}(m/\sqrt{2}\sigma)]} \exp\left[-\frac{(t-m)^2}{2\sigma^2}\right] dt \end{aligned} \tag{9.27}$$

Introducing a new variable z given by

$$z = \frac{t-m}{\sigma} \tag{9.28}$$

into Eq. (9.27),

$$R(t) = \frac{2}{1 + \text{erf}(m/\sqrt{2}\sigma)} \times \frac{1}{\sqrt{2\pi}} \int_{(t-m)/\sigma}^{\infty} e^{-\frac{1}{2}z^2}\,dz \tag{9.29}$$

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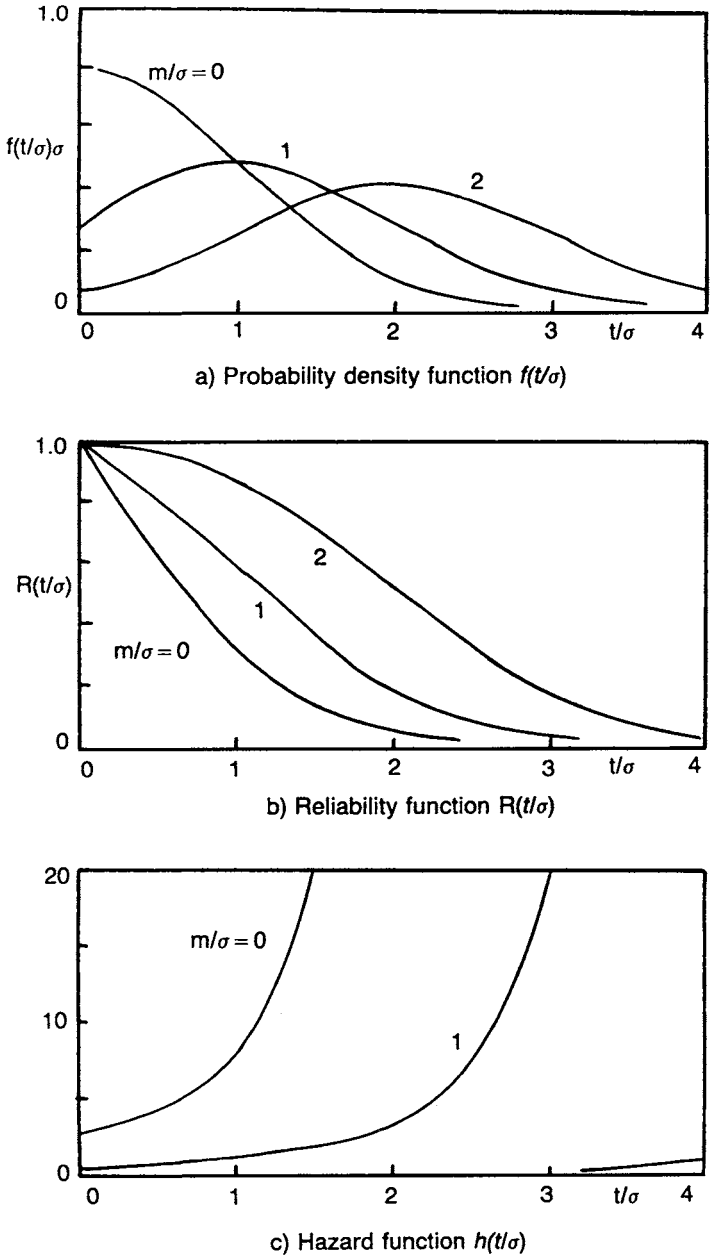


Fig. 9.9 Probability density function $f(t/\sigma)$, reliability function $R(t/\sigma)$, and hazard function $h(t/\sigma)$ for normal distributions.

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in which the expression in the factor on the right is of the same form as the normal (Gaussian) probability in Table A.3. The variation of the normal reliability function $R(t)$ with the nondimensional time t/σ for various values of the parameter m/σ is shown in Fig. 9.9b.

Log-Normal Distribution

The log-normal density distribution can be taken as

$$f(t) = \frac{1}{\sqrt{2\pi}\, t} \exp\left[-\frac{1}{2}\left(\ell_n \frac{t}{\sigma} - \frac{\mu}{\sigma}\right)^2\right] \tag{9.30}$$

where both μ and σ are constants with dimensions of time. Also, $\sigma > 0$ and $-\infty < \mu < \infty$. This function is depicted in Fig. 9.10a.

Following the integration process for the log-normal distribution of the damage function in Sec. 3.11, it can be shown that

$$R(t) = \int_t^\infty f(t)\, dt = \int_t^\infty \frac{1}{\sqrt{2\pi}\, t} \exp\left[-\frac{1}{2}\left(\ell_n \frac{t}{\sigma} - \frac{\mu}{\sigma}\right)^2\right] dt$$

$$= \frac{1}{2}\left\{1 - \text{erf}\left[\frac{1}{\sqrt{2}}\left(\ell_n \frac{t}{\sigma} - \frac{\mu}{\sigma}\right)\right]\right\} \tag{9.31}$$

and then using change of variables,

$$\text{MTTF} = \int_0^\infty t f(t)\, dt = \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\ell_n \frac{t}{\sigma} - \frac{\mu}{\sigma}\right)^2\right] dt$$

$$= \sigma \exp\left(\frac{1}{2} + \frac{\mu}{\sigma}\right) \tag{9.32}$$

The variation of the log-normal reliability function with t/σ is shown in Fig. 9.10b for $\mu/\sigma = -1, -0.5, 0, 0.5$, and 1.

Weibull Distribution

The Weibull distribution function is expressed as

$$f(t) = \begin{cases} 0 & \text{for } 0 \le t \le \gamma \\ \dfrac{\beta}{\eta}\left(\dfrac{t-\gamma}{\eta}\right)^{\beta-1} \exp\left[-\left(\dfrac{t-\gamma}{\eta}\right)^{\beta}\right] & \text{for } t > \gamma \end{cases} \tag{9.33}$$

where

t = operating time
γ = minimum guaranteed successful operating time for which $R(t) = 1$ and $0 \le t < \gamma$
β = slope (shape parameter), $\beta > 0$
η = characteristic time parameter, $\eta > 0$

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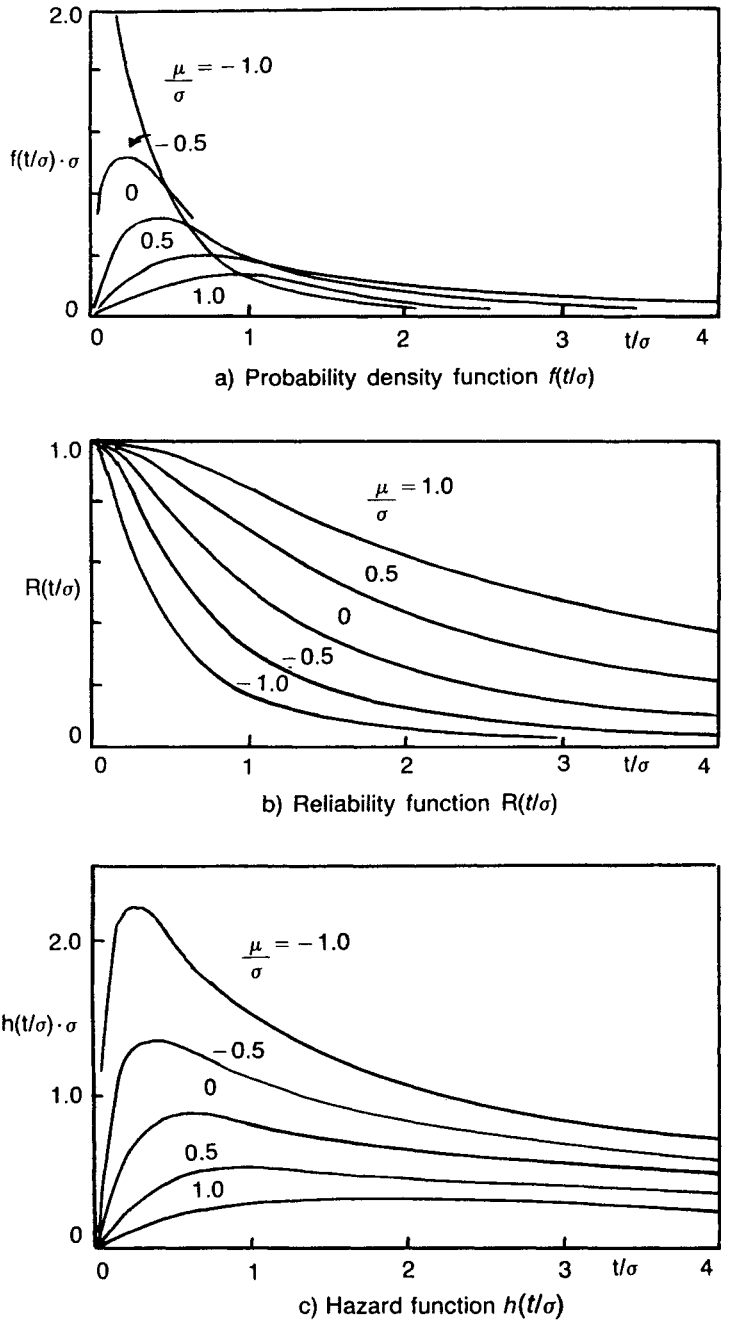


Fig. 9.10 Probability density function $f(t/\sigma)$, reliability function $R(t/\sigma)$, and hazard function $h(t/\sigma)$ for log-normal distributions.

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The Weibull density function variation with the nondimensional time t/η and the ratio γ/η is shown in Figs. 9.11a, 9.12a, 9.13a, and 9.14a for $\beta = 0.5, 1, 2$, and 3. It can be easily demonstrated that

$$R(t) = \begin{cases} 1 & \text{for } 0 \leq t \leq \gamma \\ \exp\left[-\left(\dfrac{t-\gamma}{\eta}\right)^{\beta}\right] & \text{for } t > \gamma \end{cases} \tag{9.34}$$

and

$$\text{MTTF} = \eta\Gamma\left(1 + \frac{1}{\beta}\right) + \gamma \tag{9.35}$$

where Γ is the gamma function.

When $\beta = 1$ and $\gamma = 0$, the Weibull distribution reduces to the exponential distribution. When $\gamma < 1$, the Weibull distribution takes a hyperexponential shape. For $\beta = 3.5$ the distribution is approximately symmetrical about the MTTF, whereas for $\beta > 3.5$ the distribution moves away from MTTF along the time axis and becomes skewed to the left. The variation of the Weibull reliability function with t/η, γ/η, and β is depicted in Figs. 9.11b–9.14b.

9.6 Derivation of Reliability from Probabilistic Considerations

It may now be recalled that reliability $R(t)$ is interpreted as the probability that a component will not fail before time t. This implies that $R(T)$ is equal to unity for $t = 0$ and approaches 0 as time increases to infinity. The reliabilities can be obtained from experiments for the different assumptions regarding the probability density distributions. The exponential reliability function can be easily derived from the probabilistic considerations of reliability of an assembly of components.

An experiment will be considered in which a number of components are placed in operational environment and tested over the time t. The original number of components at time $t = 0$ is N_0, and the number of components still operational in time t is N_t. To analyze the probability of failure of these components and their associated reliability, the following definitions will be introduced:

N_0 = number of components at $t = 0$
N_t = number of functioning components at t
$N_{t+\Delta t}$ = number of functioning components at time $t + \Delta t$
Δt = time increment under consideration

Using the preceding definitions, the probability

$$P(\text{failure in } \Delta t \mid \text{survival through 0 to } t) = \frac{N_t - N_{t+\Delta t}}{N_t} \tag{9.36}$$

The reliability $R(t)$ is defined as

$$R(t) = P(\text{survival through 0 to } t) \tag{9.37}$$

and the reliability $R(t + \Delta t)$ is derived from the conditional probability such that

$$R(t + \Delta t) = P(\text{survival through 0 to } t) \times P(\text{survival in } \Delta t) \tag{9.38}$$

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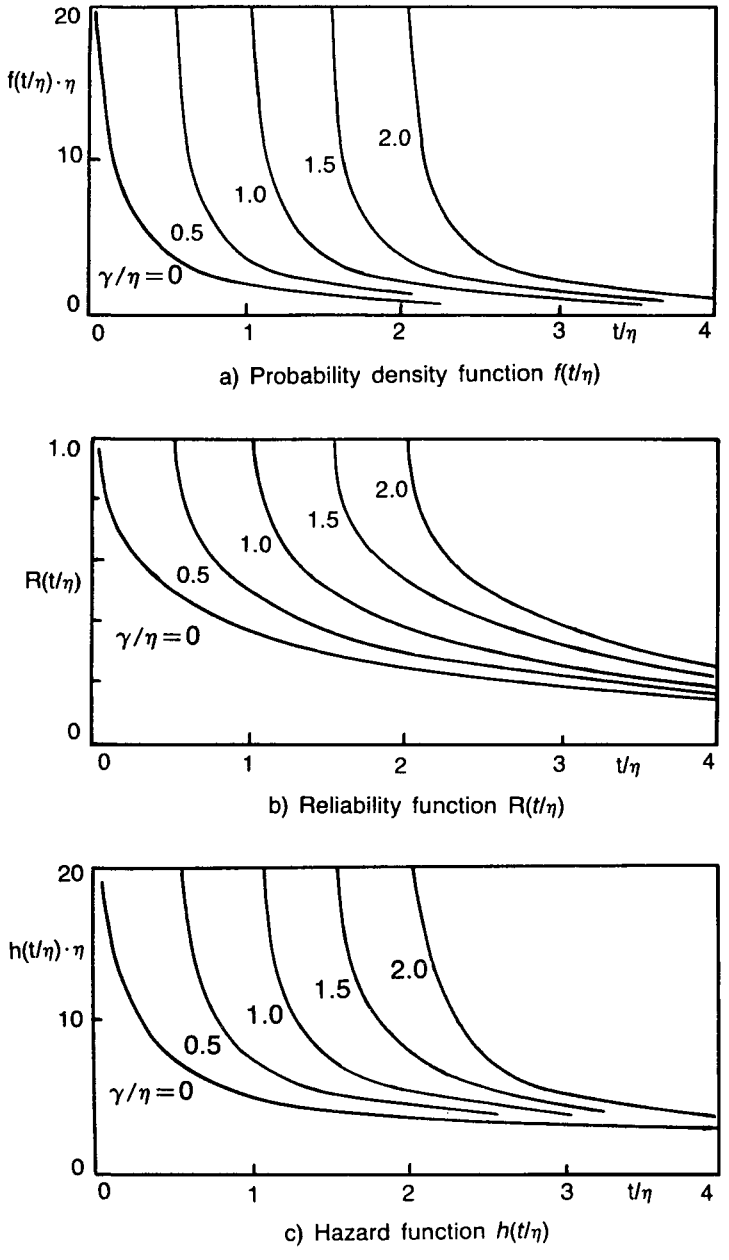


Fig. 9.11 Probability density function $f(t/\eta)$, reliability function $R(t/\eta)$, and hazard function $h(t/\eta)$ for Weibull distributions with $\beta = 0.5$.

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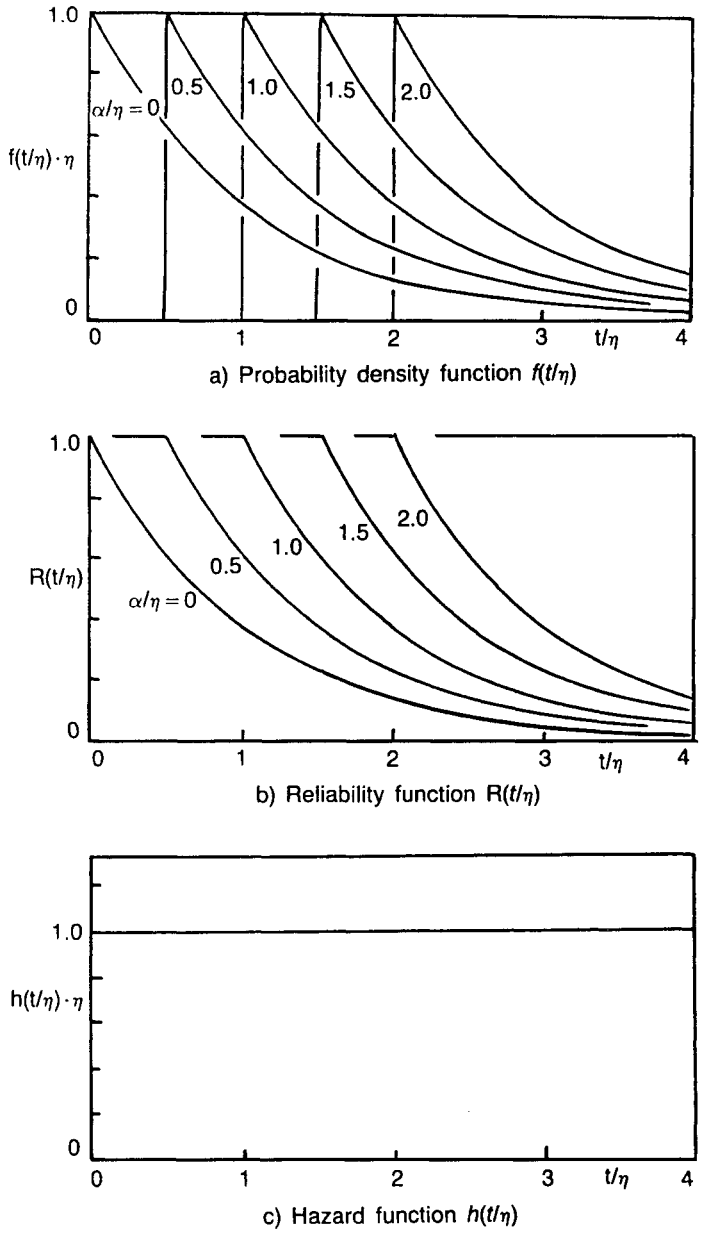


Fig. 9.12 Probability density function $f(t/\eta)$, reliability function $R(t/\eta)$, and hazard function $h(t/\eta)$ for Weibull distributions with $\beta = 1$.

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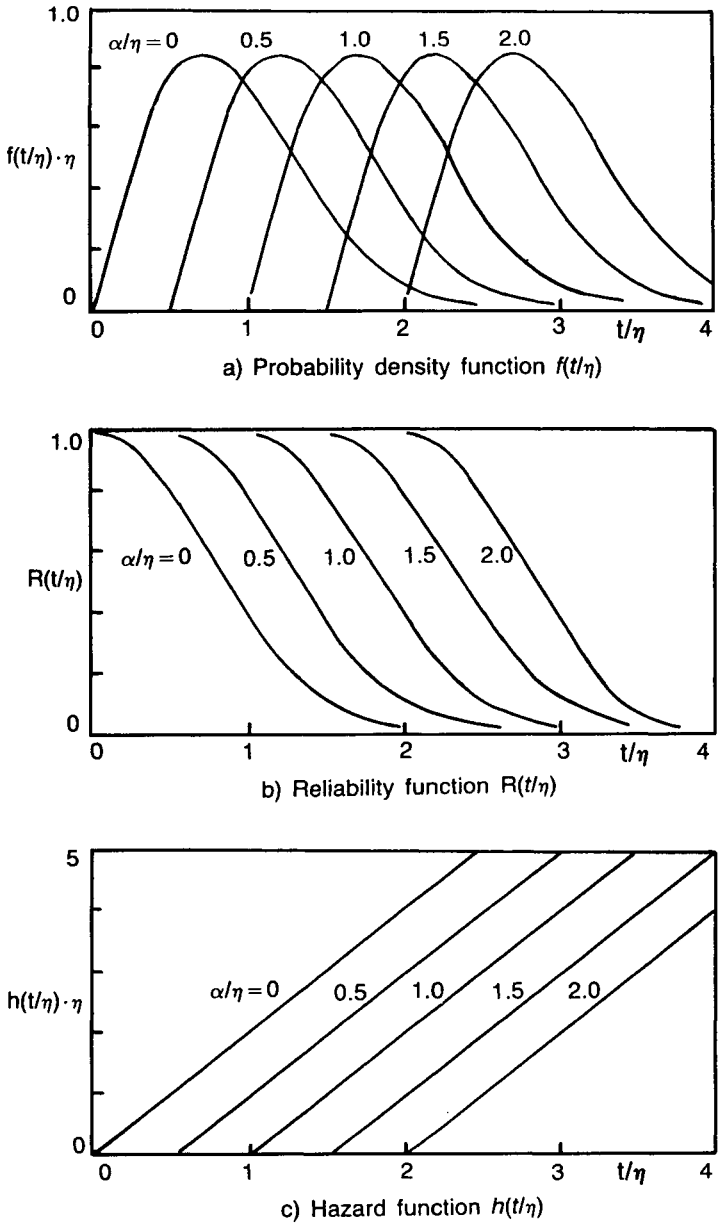


Fig. 9.13 Probability density function $f(t/\eta)$, reliability function $R(t/\eta)$, and hazard function $h(t/\eta)$ for Weibull distributions with $\beta = 2$.

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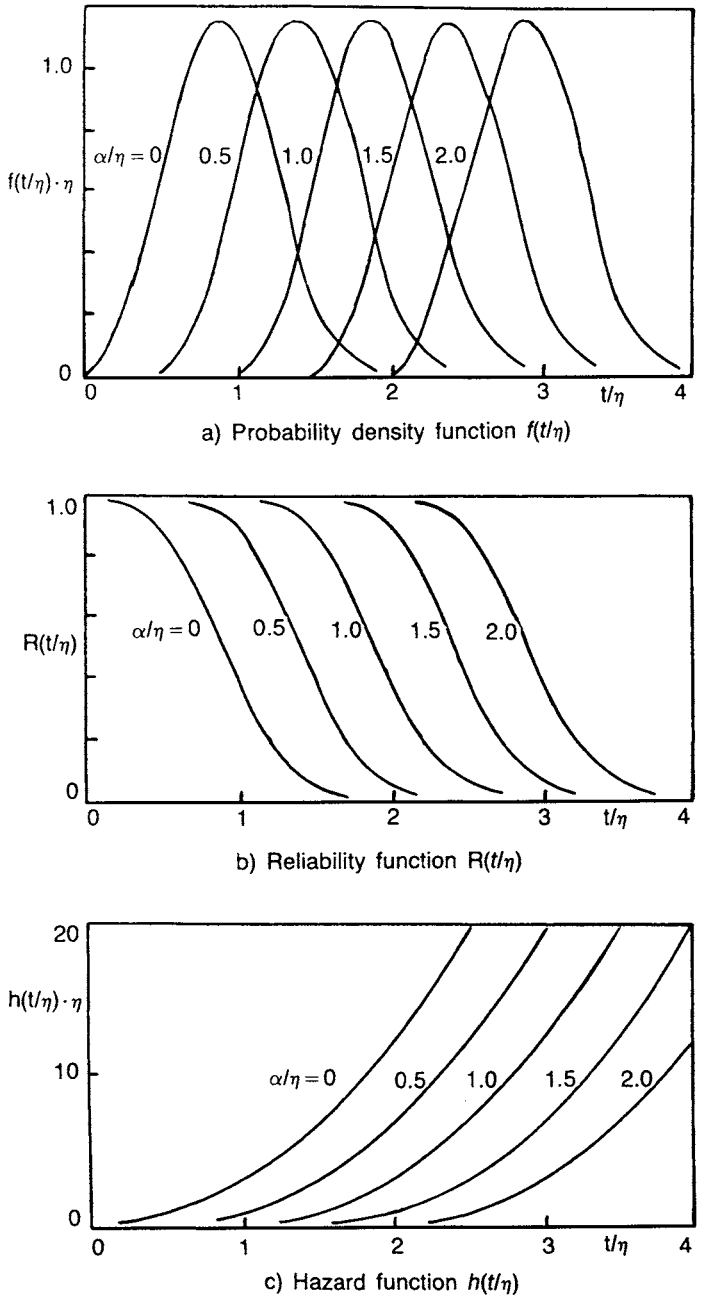


Fig. 9.14 Probability density function $f(t/\eta)$, reliability function $R(t/\eta)$, and hazard function $h(t/\eta)$ for Weibull distributions with $\beta = 3$.

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Noting that

$$P(\text{survival in } \Delta t) = 1 - \frac{N_t - N_{t+\Delta t}}{N_t} \tag{9.39}$$

it follows from Eqs. (9.37), (9.38), and (9.39) that

$$R(t + \Delta t) = R(t) \times \left(1 - \frac{N_t - N_{t+\Delta t}}{N_t}\right) \tag{9.40}$$

Rearranging Eq. (9.40) and dividing by Δt,

$$\frac{N_t - N_{t+\Delta t}}{N_t \Delta t} = -\frac{R(t + \Delta t) - R(t)}{R(t)\Delta t} \tag{9.40a}$$

and introducing the failure rate $\lambda(t)$ as the fraction of components failing per unit time during the interval Δt when Δt tends to 0 such that

$$\begin{aligned}\lambda(t) &= \lim_{\Delta t \to 0} \frac{N_t - N_{t+\Delta t}}{N_t \Delta t} \\ &= -\lim_{\Delta t \to 0} \frac{R(t + \Delta t) - R(t)}{R(t)\Delta t} \\ &= -\frac{1}{R(t)} \frac{dR(t)}{dt}\end{aligned} \tag{9.41}$$

which finally can be rearranged as

$$\frac{dR(t)}{R(t)} = -\lambda(t)\, dt \tag{9.41a}$$

Integrating Eq. (9.41) between the limits 0 and t

$$\ln R(t)\Big|_0^t = -\int_0^t \lambda(t)\, dt$$

and because $R(0) = 1$ and $\ln(1) = 0$,

$$\ln R(t) = -\int_0^t \lambda(t)\, dt \tag{9.42}$$

This equation can now be solved for $R(t)$ giving

$$R(t) = \exp\left[-\int_0^t \lambda(t)\, dt\right] \tag{9.43}$$

The function $R(t)$ represents the theoretical probability that a component will operate for t units of time without failure. This function can also be approximated by the fraction of survivors at time t, i.e.,

$$R(t) \cong \frac{N_t}{N_0} \tag{9.44}$$

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If $\lambda(t) = \text{const}$, then, from Eq. (9.43),

$$R(t) = e^{-\lambda t}; \qquad \lambda = \text{const} \tag{9.45}$$

which agrees with the previously derived result based on the exponential density distribution, Eq. (9.19). Also, using Eq. (9.44),

$$\frac{N_t}{N_0} \cong e^{-\lambda t} \tag{9.46}$$

which can be rewritten as

$$\lambda \cong -\frac{1}{t} \ln \frac{N_t}{N_0} \tag{9.47}$$

This equation gives a direct method of computing λ from recorded data for which the exponential distribution is appropriate. The exponential distribution is best suited to describe the time to failure of equipment subject to chance failure. The normal distribution, the log-normal distribution, and the Weibull distribution are best suited to describe equipment subject to wear-out types of failure. The recent popularity of the Weibull distribution, however, is mainly due to its simple mathematical representation and the three parameters that can better describe a range of failure modes. This, incidentally, includes the chance failure mode for $\beta = 1$ and $\gamma = 0$ when the Weibull distribution degenerates into an exponential distribution. The application of Eq. (8.47) to a typical reliability data for electronic equipment will be illustrated in Sec. 9.8.

9.7 Hazard Function *h(t)*

From the previous section, which discussed the derivation of the reliability function from experimental data, it should be noted that $\lambda(t)$ is an important function in reliability calculations because it specifies the instantaneous failure rate of the system or component. The function $\lambda(t)$ is sometimes referred to as the hazard function $h(t)$ so that, from Eq. (9.41),

$$h(t) = \lambda(t) = -\frac{1}{R(t)} \frac{dR(t)}{dt} \tag{9.48}$$

Using the formal definition of R(t) from Eq. (8.15),

$$\frac{dR(t)}{dt} = -f(t) \tag{9.49}$$

Hence,

$$h(t) = \frac{f(t)}{R(t)} \tag{9.50}$$

For $\lambda = \text{const}$, which represents the exponential probability density function

$$h(t) = \text{const} \tag{9.51}$$

This implies that the failure rate is the same from $t = 0$ to t. For such systems the replacement of any or all surviving units with new units will not affect the failure

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Table 9.1 Probability density distribution $f(t)$ and reliability function $R(t)$

Type	Probability density distribution $f(t)$	Reliability function $R(t)$
Exponential	$\lambda e^{-\lambda t}$	$e^{-\lambda t}$
Normal	$\dfrac{\sqrt{\frac{2}{\pi}}\exp\left[-\frac{(t-m)^2}{2\sigma^2}\right]}{\sigma\left[1+\operatorname{erf}\left(\frac{m}{\sqrt{2}\sigma}\right)\right]}$	$\dfrac{2}{1+\operatorname{erf}\left(\frac{m}{\sqrt{2}\sigma}\right)}\dfrac{1}{\sqrt{2\pi}}\displaystyle\int_{\frac{t-m}{\sigma}}^{\infty} e^{-\frac{1}{2}z^2}\,dz$
Log-normal	$\dfrac{\exp\left[-\frac{1}{2}\left(\ln\frac{t}{\sigma}-\frac{\mu}{\sigma}\right)^2\right]}{\sqrt{2\pi}\,t}$	$\dfrac{1}{2}\left\{1-\operatorname{erf}\left[\dfrac{1}{\sqrt{2}}\left(\ln\dfrac{t}{\sigma}-\dfrac{\mu}{\sigma}\right)\right]\right\}$
	0, for $0 \le t \le \gamma$	1.0, for $0 \le t \le \gamma$
Weibull	$\dfrac{\beta}{\eta}\left(\dfrac{t-\gamma}{\eta}\right)^{\beta-1}\exp\left[-\left(\dfrac{t-\gamma}{\eta}\right)^{\beta}\right]$, for $t > \gamma$	$\exp\left[-\left(\dfrac{t-\gamma}{\eta}\right)^{\beta}\right]$, for $t > \gamma$

rate of the aggregate. Thus, any preventative maintenance with replacement units is basically useless for such cases.

Equation (9.50) is the most convenient to use to determine the hazard function $h(t)$ for different probability density distributions $f(t)$. The functions $f(t)$, $R(t)$, $h(t)$, and MTTF for the four density distributions discussed in this chapter are summarized in Tables 9.1 and 9.2.

The hazard functions $h(t)$ for the four density distributions discussed in this chapter are shown in Figs. 9.8c–9.14c. One general observation is appropriate here.

Table 9.2 Hazard function $h(t)$ and MTTF

Type	Hazard function $h(t)$	Mean time to failure MTTF	Parameter range
Exponential	λ	$\dfrac{1}{\lambda}$	$\lambda > 0$
Normal	$\dfrac{\exp\left[-\frac{(t-m)^2}{2\sigma^2}\right]}{\sigma\int_{\frac{t-m}{\sigma}}^{\infty} e^{-\frac{1}{2}z^2}\,dz}$	$\sqrt{\dfrac{2}{\pi}}\dfrac{\sigma\exp\left(-\frac{m^2}{2\sigma^2}\right)}{1+\operatorname{erf}\left(\frac{m}{\sqrt{2}\sigma}\right)}+m$	$\sigma > 0$ $m \ge 0$
Log-normal	$\dfrac{\sqrt{\frac{2}{\pi}}\exp\left[-\frac{1}{2}\left(\ln\frac{t}{\sigma}-\frac{\mu}{\sigma}\right)^2\right]}{t\left\{1-\operatorname{erf}\left[\frac{1}{\sqrt{2}}\left(\ln\frac{t}{\sigma}-\frac{\mu}{\sigma}\right)\right]\right\}}$	$\sigma\exp\left(\dfrac{1}{2}+\dfrac{\mu}{\sigma}\right)$	$\sigma > 0$ $-\infty < \mu < \infty$
	0, for $0 \le t \le \gamma$		$\eta > 0$
Weibull	$\dfrac{\beta}{\eta}\left(\dfrac{t-\gamma}{\eta}\right)^{\beta-1}$, for $t > \gamma$	$\eta\Gamma\left(1+\dfrac{1}{\beta}\right)+\gamma$	$\beta > 0$ $\gamma \ge 0$

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The hazard function is *not* the probability of failure. It is the *failure rate* expressing the fraction of components failing per unit time. The Weibull hazard functions can represent decreasing rates ($\beta = 0.5$, hyperexponential type), constant rate ($\beta = 1$), and increasing rate ($\beta > 1$), which include the special cases of the exponential probability density function ($\beta = 1$) and the normal probability density function ($\beta \cong 3.5$). However, the Weibull hazard function cannot handle the cases of increasing-decreasing rates. Such cases must be represented by the log-normal distributions as shown in Fig. 9.10.

9.8 Computation of a Reliability Function from Experiment

In general, once the defective components have been weeded out through quality control and initial testing, electronic components and systems will fail randomly in time. Thus, the reliability function $R(t)$ for such systems will take the general form represented by the exponential probability density distribution. To illustrate this case, a set of data for a sample of some 120 components will be considered. The test data for a total of 3000 h are shown in Table 9.3 and Fig. 9.15.

Now computing the average value of λ over the 3000-h test period, it is found that its average value is 0.0003240 (1/h). The plot of $R(t) = N_t/N_0$ against operating time to failure from Table 8.1 is shown in Fig. 8.16. Because the data approximate to a straight line on a semilog scales, it can be concluded that the sample came from an exponential distribution. Noting that, for an exponential distribution MTTF $= 1/\lambda$, the mean time to failure can also be determined from Fig. 9.16. In this case the function $R(t)$ at time $t =$ MTTF is given by

$$R(\text{MTTF}) = e^{-\lambda t} = e^{-1} = 0.367879 \tag{9.52}$$

and therefore MTTF can be read off directly from Fig. 9.16 as approximately equal

Table 9.3 Test data for electronic components ($N_0 = 120$)

Time t, h	Number of failures	N_t	$R(t) = N_t/N_0$	$\lambda = -[\ell_n R(t)]/t$ 1/h
0–200	8	112	0.9333	0.0003450
200–400	7	105	0.8750	0.0003338
400–600	7	98	0.8167	0.0003375
600–800	6	92	0.7667	0.0003321
800–1000	6	86	0.7167	0.0003331
1000–1200	5	81	0.6750	0.0003275
1200–1400	5	76	0.6333	0.0003263
1400–1600	5	71	0.5916	0.0003280
1600–1800	4	67	0.5583	0.0003238
1800–2000	4	63	0.5250	0.0003222
2000–2200	3	60	0.5000	0.0003151
2200–2400	4	56	0.4667	0.0003176
2400–2600	3	53	0.4417	0.0003143
2600–2800	2	51	0.4250	0.0003056
2800–3000	2	49	0.4083	0.0002986

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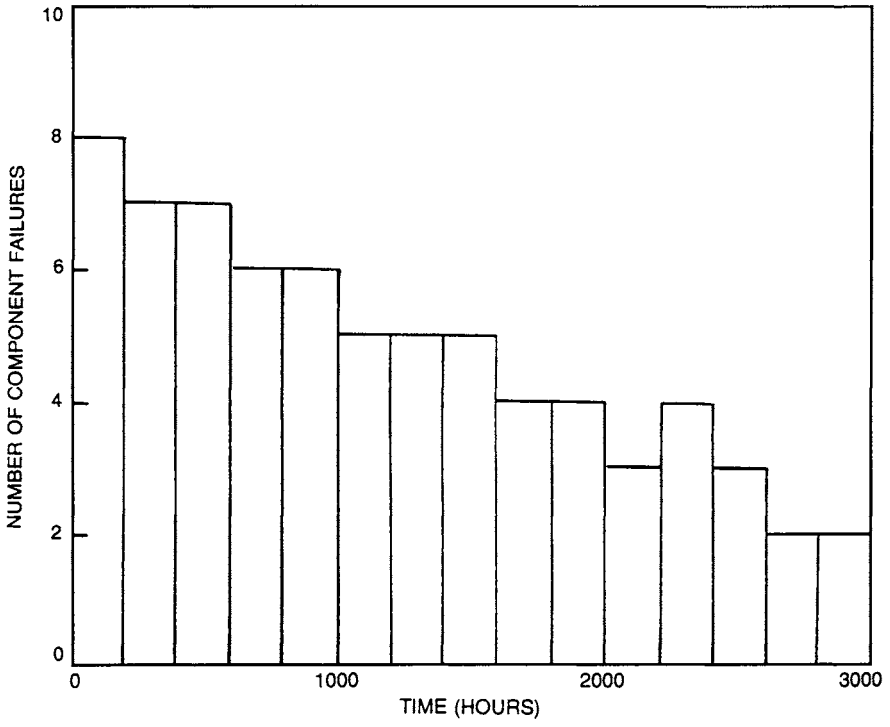


Fig. 9.15 Histogram of experimental data for electronic equipment.

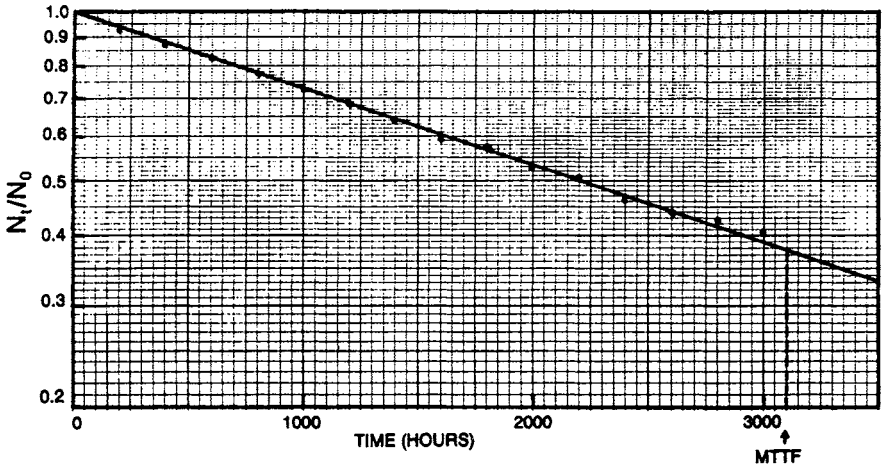


Fig. 9.16 Plot of experimental data for electronic equipment.

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to 3100 hrs. This also agrees with the MTTF computed from the average value of λ, i.e.,

$$\text{MTTF} = \frac{1}{\lambda} = \frac{1}{0.0003240} = 3086\,\text{hrs}$$

The Probability Graph Method[3]

In general, there are two basic approaches commonly used to determine which reliability model applies in a particular case: 1) the probability graph method and 2) the Kolmogorov–Smirnov (K–S) test method.

In this method the failure data on a particular type of probability graph paper are plotted to see if the data plot is approximately a straight line. This means that exponential data should plot as a straight line on semilog paper, normal data should plot as a straight line on normal probability paper, and Weibull data should plot as a straight line on Weibull log-log paper.

The Kolmogorov–Smirnov (K–S) Test Method[4,5,6]

The second method of determining the appropriate reliability model is through the application of the K–S test for goodness of fit, which is considered to be somewhat more elegant mathematically than the probability graph method. The K–S test method consists of plotting the survival curve for the sample data superimposed on a theoretical survival curve representing the reliability model being proposed. The maximum absolute difference at the order statistics of the sample curves is compared with a value from the K–S table corresponding to the sample size and to a preselected significance level. If the actual difference is greater than the value from the K–S table, then the proposed model must be rejected with the predetermined risk of being wrong. Specific details of this method can be found in Refs. 4–6. The tables in Ref. 5 are useful only when the reliability model being proposed is completely specified in advance.

If the model to be tested is based on estimating its parameters, then a different table must be used for each distribution, and the test for goodness of fit is called a modified K–S test. Details and sources for these special tables, as well as information on other goodness-of-fit tests, can be found in Ref. 6.

9.9 Maintainability of Weapon Systems

Military equipment or systems may be designed with redundancy of components in order to improve their overall reliability; however, the improved reliability is not the only factor that contributes to the effectiveness of a weapon system. If the system fails and has to be repaired, then the time to repair is also an important factor in weapon effectiveness. For example, in Fig. 9.17 two systems are compared: the system A has a higher reliability (higher MTTF) and a longer downtime to repair, whereas the system B has a lower reliability and a shorter downtime [shorter mean down time (MDT)]. Clearly, if the *operational availability* of a system, measured as the system readiness per unit time, is more important than its reliability, then system B is superior to system A, despite the fact that it has a lower reliability.

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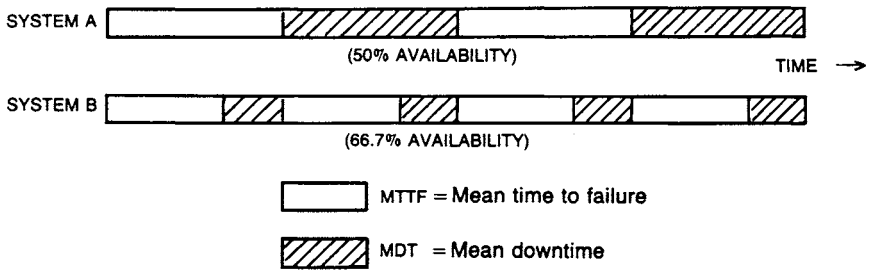


Fig. 9.17 Comparison of operational availabilities of two systems.

Thus, it is clear that the repair time of a weapon system, that is, its maintainability, is an important factor in determining the weapon operational availability. Maintainability of a weapon system is defined in terms of the total repair time, which measures the speed and ease with which malfunctions are detected and repaired and normal operations restored. In selecting competing weapon systems it is therefore necessary to analyze not only the system reliability but also its maintainability.

Maintainability can also be defined as the probability that a component in a system or a whole system itself will be restored to operational status within a specified period of time when the maintenance (including any needed repair) is performed in accordance with stated conditions and objectives. The maintainability function $M(t)$ describing this probability is defined as

$$M(t) = \int_0^t g(t)\,dt \tag{9.53}$$

where $g(t)$ is the time-to-repair probability density distribution. If the repair rate is constant, then $g(t)$ represents an exponential distribution for which

$$g(t) = \lambda_R e^{-\lambda_R t} \tag{9.54}$$

Hence, from Eqs. (9.53) and (9.54),

$$M(t) = 1 - e^{-\lambda_R t} \tag{9.55}$$

where $e^{-\lambda_R t}$ is the probability that a repair will not be completed in time t, whereas $1 - e^{-\lambda_R t}$ is the probability that it will be completed.

The mean time to repair MTTR is defined as

$$\text{MTTR} = E[t] = \int_0^t tg(t)\,dt \tag{9.56}$$

which for the exponential distribution is

$$\text{MTTR} = \frac{1}{\lambda_R} \tag{9.57}$$

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9.10 Operational Availability of Systems

From the discussion in the previous section, it follows that the *operational availability* also can be defined as the probability that a component or a system will operate satisfactorily at any given time. This probability, which will be denoted by A_0, is defined as a ratio[7]:

$$A_0 = \frac{\text{MTBM}}{\text{MTBM} + \text{MDT}} = \frac{1}{1 + \text{MDT/MTBM}} \tag{9.58}$$

where

MTBM = mean time between maintenance
MDT = mean downtime, which includes the mean active maintenance downtime and mean delay times caused by lack of logistics and/or administrative support

Equation (9.58) clearly implies that the key factor determining operational availability is the ratio MDT/MTBM of the mean down time to the mean time between maintenance, i.e., the smaller this ratio the greater the availability of an operational system.

Two additional probabilities are used to define weapon availabilities: 1) inherent availability A_I and 2) achieved availability A_A. They are expressed as ratios:[7]

$$A_I = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}} = \frac{1}{1 + \text{MTTR/MTTF}} \tag{9.59}$$

and

$$A_A = \frac{\text{MTBM}}{\text{MTBM} + \text{MAMD}} = \frac{1}{1 + \text{MAMD/MTBM}} \tag{9.60}$$

where

MTTF = mean time to failure
MTTR = mean time to repair
MAMD = mean active maintenance downtime, resulting from both preventative and corrective maintenance actions

Thus, the ratios MTTR/MTTF and MAMD/MTBM are the key ratios for A_I and A_A, respectively. When preventative maintenance downtime is zero, or is not considered, then MTBM becomes MTTF and MAMD becomes MTTR, and $A_A = A_I$.

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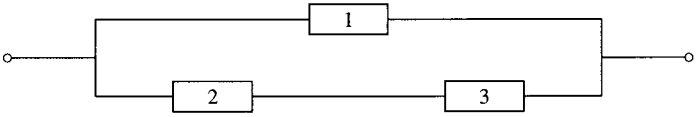
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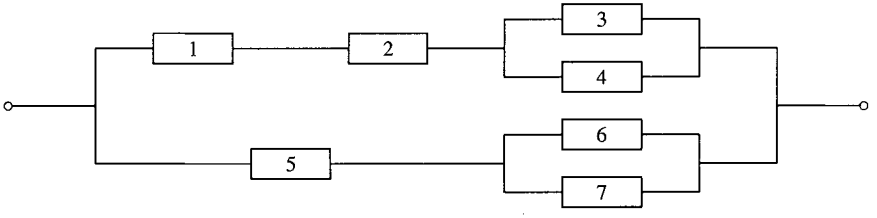
Problems

9.1. Find the overall reliability of an operation defined by the following schematic diagram:

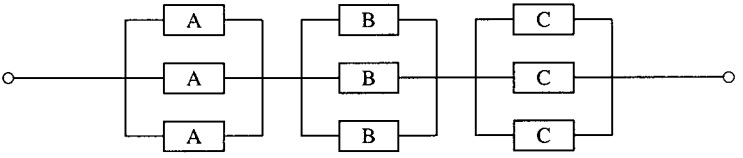


for which R_1, R_2, and R_3 are the individual component reliabilities.

9.2. Find the overall reliability for the following system:



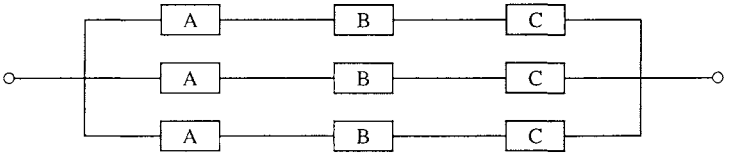
9.3. Find the overall reliability of a system made up of components A, B, and C, as shown by the following:



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9.4. Find the overall reliability of a system made up of components A, B, and C as shown by the following:



9.5. Prove that

$$\lambda(t) = -\frac{d}{dt} \ln R(t)$$

9.6. Show using the expression in Problem 8.5 and the reliability function for the Weibull distribution

$$R(t) = \exp\left[-\left(\frac{t-\gamma}{\eta}\right)^{\beta}\right]$$

that

$$\lambda(t) = \frac{\beta}{\eta}\left(\frac{t-\gamma}{\eta}\right)^{\beta-1}$$

9.7. Prove that the mean time to failure (MTTF) for the Weibull distribution is given by

$$\text{MTTF} = \eta\Gamma\left(1+\frac{1}{\beta}\right) + \gamma$$

Hint: The gamma function Γ is defined by

$$\Gamma\left(1+\frac{1}{\beta}\right) = \int_0^{\infty} e^{-u} u^{1/\beta}\, du$$

9.8. Show that the mean time to failure MTTF can also be calculated from

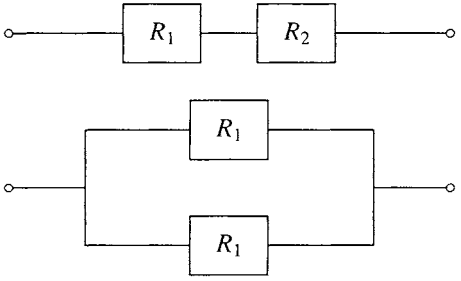
$$\text{MTTF} = \int_0^{\infty} R(t)\, dt$$

where $R(t)$ is the reliability function.

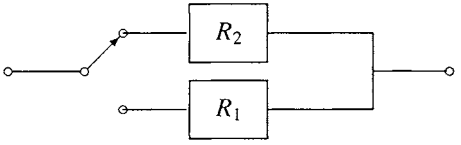
9.9. Determine MTTF for the two systems shown in the following schematic diagram in which $R_1 = R_2 = e^{-\lambda t}$ where λ is a constant and t is time.

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9.10. The reliability of the standby system shown in the following schematic diagram is described by a Poisson distribution with the rate of μt. Find the MTTF for this system.



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10
Target Detection

This chapter describes methods of computing the probability of detection of targets through a search involving either 1) intermittent search for the target or 2) continuous search for the target. The general principles involved are applicable to searches for airborne, sea, and land targets. The targets can be either stationary or moving. The variation of the detection rate with range is introduced, including two special cases of the inverse square and cubic laws of detection. To illustrate the fundamental concepts of the analysis of operational search, the probability of target detection for an airborne search encompassing a specified area is presented for three modes of search: 1) exhaustive search; 2) random search; and 3) search with an inverse cube law sensor. Two special search cases are also discussed: patrolling a line of defense and search for a target that can move with constant velocity in any direction from a point, the location of which is known initially to the searching aircraft.

10.1 Intermittent Glimpses[1]

The first type of a search to detect a particular target is the intermittent looking, in which the seeker makes a succession of brief glimpses, as shown in Fig. 10.1, where a surface-to-air missile (SAM) site is tracking a penetrating bomber. The most important parameter in this process is the glimpse probability g_i, which gives the probability of seeing the target on the ith glimpse. This probability will actually vary from glimpse to glimpse depending on changes in some physical conditions such as range, speed, illumination, weather, etc. It will be assumed here that the value of g_i is known. The methods of estimating actual values of g_i for tracking systems such as radar, infrared sensors, etc., will not be discussed here. Once these values are known, the target acquisition and tracking system effectiveness can be determined in terms of the probabilities of target detection and the expected number of glimpses required to detect the target. Of course, the overall system effectiveness will vary with range, altitude, weather conditions, etc.

For a particular search, each successive glimpse will be numbered from the beginning of the search. For the ith glimpse the probability g_i represents the probability of detecting the target provided the target has not been detected previously. It will also be assumed that the target is seen for the first time after N glimpses so that $g_N = 1$. Using these assumptions and the definition of compound probabilities,

$$\begin{aligned} Q(n) &= Q(\text{detection fails to occur on each of the first } n \text{ glimpses}) \\ &= (1-g_1)(1-g_2)(1-g_3)\cdots(1-g_n) \\ &= \prod_{i=1}^{n}(1-g_i) \end{aligned} \tag{10.1}$$

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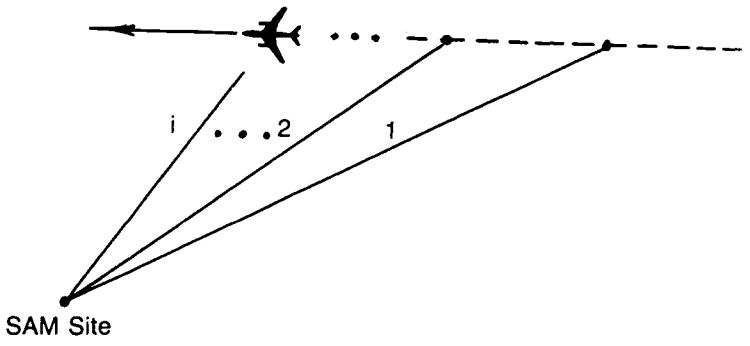


Fig. 10.1 Detection of a penetrating bomber using a succession of brief glimpses from a SAM site.

The probability of detecting the target in the first n glimpses $P(n)$ is found from

$$P(n) + Q(n) = 1 \tag{10.2}$$

which represents the basic condition for probabilities; i.e., the sum of the probability of an event occurring and the probability of the same event not occurring must be equal to certainty, which is equal to 1. Hence, from Eqs. (10.1) and (10.2) the probability of detecting the target in the first n glimpses is given by

$$P(n) = 1 - \prod_{i=1}^{n}(1 - g_i) \tag{10.3}$$

The probability of first detection of the target on the nth glimpse is obtained from

$$\begin{aligned} p(n) &= Q(\text{detection fails to occur on each of the first } n-1 \text{ glimpses}) \\ &\quad \times P(\text{detection on the } n\text{th glimpse}) \\ &= \prod_{i=1}^{n-1}(1 - g_i)g_n \\ &= \begin{cases} g_n \sum_{i=1}^{n-1}(1 - g_i) & \text{for } n > 1 \\ g_i & \text{for } n = 1 \end{cases} \end{aligned} \tag{10.4}$$

It is interesting to note here that the relationship between the probability of detecting the target in the first n glimpses and the probabilities of first detecting the target on the ith glimpse is given by

$$\begin{aligned} P(n) &= p(1) + p(2) + \cdots p(n) \\ &= \sum_{i=1}^{n} p(i) \end{aligned} \tag{10.5}$$

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Table 10.1 Probabilities of detection of a penetrating bomber

Glimpse No.	i	1	2	3	4
Probabilities	g_i	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$

Also,

$$P(N) = \sum_{n=1}^{N} p(n) = 1 \tag{10.6}$$

The expected number of glimpses required for detection may be found by computing the expected value of N from

$$E[N] = \sum_{n=1}^{N} np(n) \tag{10.7}$$

The main objective when searching for a target is to make the probability of detection as large as possible. In many cases, detection in the fewest number of glimpses may be more important in order to conceal the searching beams. Consequently, both $P(n)$ or $E[N]$ may be used as measures of effectiveness of a target acquisition system depending on the particular objective.

As an example of the computation of $p(n)$, $P(n)$, and $E[N]$, the case of an interceptor searching for a penetrating bomber will be considered. Only four separate glimpses will be considered when the range between the interceptor and bomber decreases; therefore, the probabilities of detection g_i increase. The values of g_i are assumed to be given by Table 10.1, which implies that the target is actually detected on the 4th glimpse ($g_4 = 1$).

Based on the values of g_i given in Table 10.1, the probabilities $p(n)$ of first detection of the target on the nth glimpse are computed from Eq. (10.4), and the probabilities $P(n)$ of detecting the target in the first n glimpses are computed from Eq. (10.3) or (10.5). This leads to the results given in Table 10.2.

Finally, the expected value of N is computed from Eq. (10.7) as follows:

$$\begin{aligned} E[N] &= \sum_{i=1}^{4} np(n) \\ &= 1 \times \frac{1}{4} + 2 \times \frac{3}{8} + 3 \times \frac{9}{32} + 4 \times \frac{3}{32} \\ &= \frac{71}{32} \cong 2.2 \quad \text{glimpses} \end{aligned}$$

Table 10.2 Probabilities of detection of a penetrating bomber

Number of glimpses	n	1	2	3	4
First detection on the nth glimpse	$p(n)$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{9}{32}$	$\frac{3}{32}$
Detection in the n glimpses	$P(n)$	$\frac{1}{4}$	$\frac{5}{8}$	$\frac{29}{32}$	1

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When the conditions do not change for separate glimpses, then $g_i = g$. For such cases,

$$p(n) = \begin{cases} g(1-g)^{n-1} & \text{for } n > 1 \\ g & \text{for } n = 1 \end{cases} \tag{10.8}$$

$$P(n) = 1 - (1-g)^n \tag{10.9}$$

and

$$E[N] = \frac{1}{g} \tag{10.10}$$

Here the probability of detection $P(n)$ can be made as large as desired by increasing the number of glimpses. It should also be noted that

$$P(n = \infty) = 1 .$$

10.2 Continuous Search

The previous section described searching for a target using a succession of brief glimpses. In looking for a target with a sensor (e.g., infrared detector) operating continuously, information is also being received continuously. For such cases, the probability that detection occurs at a given instant is 0. On the other hand, the probability of detection after the time t has elapsed is a meaningful measure of the effectiveness of the system. Another useful measure is the mean time to detection MTTD, which represents the expected value of time t to detect the target.

The principal characteristic of a detection system is the instantaneous detection rate $\gamma(t)$, expressed as the number of detected targets per unit of time. An equivalent concept has been used in Chapter 9 for the failure rate $\mu(t)$ for systems with time-dependent reliability. Thus, the derivation of probability of detection of targets with continuous search systems is somewhat analogous to the derivation of a failure function in the analysis of the time dependent reliability in Sec. 8.6.

To determine the probability of detection for a continuous system, the following definitions will be introduced:

T_0 = number of undetected targets at time $t = 0$
T_t = number of undetected targets at time t
$T_{t+\Delta t}$ = number of undetected targets at time $t + \Delta t$
Δt = time increment

Now introducing the probabilities P and Q,

$$Q(t) = P(\text{no detection in time } t) \tag{10.11}$$

$$P(\Delta t) = P(\text{detection in time } \Delta t) = \frac{T_t - T_{t+\Delta t}}{T_t} \tag{10.12}$$

$$Q(\Delta t) = P(\text{no detection in time } \Delta t) = 1 - P(\Delta t) \tag{10.13}$$

$$\begin{aligned} Q(t+\Delta t) &= P(\text{no detection in time } t + \Delta t) = Q(t)Q(\Delta t) \\ &= Q(t)[1 - P(\Delta t)] \\ &= Q(t)\left(1 - \frac{T_t - T_{t+\Delta t}}{T_t}\right) \end{aligned} \tag{10.14}$$

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Rewriting Eq. (10.14) and dividing by Δt,

$$\frac{T_t - T_{t+\Delta t}}{T_t \Delta t} = -\frac{Q(t+\Delta t) - Q(t)}{Q(t)\Delta t} \tag{10.15}$$

Hence, defining the instantaneous detection rate as

$$\gamma(t) = \lim_{\Delta t \to 0} \frac{T_t - T_{t+\Delta t}}{T_t \Delta t} \tag{10.16}$$

it follows from Eq. (10.15) that

$$\gamma(t) = -\lim_{\Delta t \to 0} \frac{Q(t+\Delta t) - Q(t)}{Q(t)\Delta t} = -\frac{1}{Q(t)} \frac{dQ(t)}{dt} \tag{10.17}$$

Note the complete analogy between $\gamma(t)$ and $\lambda(t) = h(t)$ defined in Chapter 8 as the instantaneous failure rate or the hazard function and between $Q(t)$ and $R(t)$ where the latter is the reliability function.

Solving Eq. (10.17) for $Q(t)$,

$$Q(t) = \exp\left[-\int_0^t \gamma(t)\, dt\right] \tag{10.18}$$

from which

$$\begin{aligned} P(t) &= P(\text{detection in time } t) \\ &= 1 - Q(t) \\ &= 1 - \exp\left[-\int_0^t \gamma(t)\, dt\right] \end{aligned} \tag{10.19}$$

If $\gamma(t) = c$ (const), then

$$P(t) = 1 - e^{-ct} \tag{10.20}$$

It should also be noted here that $P(0) = 0$ and $P(\infty) = 1$; i.e., when $t \to \infty$ the probability of detection becomes a certainty.

To find the corresponding probability density function $f(t)$, the following relationship can be used:

$$f(t) = \frac{dP(t)}{dt} \tag{10.21}$$

which generates the probability function $P(t)$ from

$$P(t) = \int_0^t f(t)\, dt \tag{10.22}$$

Hence, from Eqs. (10.19) and (10.21),

$$f(t) = \gamma(t) \exp\left[-\int_0^t \gamma(t)\, dt\right] \tag{10.23}$$

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The MTTD is then computed from

$$\text{MTTD} = E[t] = \int_0^\infty t f(t)\,dt$$

$$= \int_0^\infty t\gamma(t) \exp\left[-\int_0^t \gamma(t)\,dt\right] dt \tag{10.24}$$

If $\gamma(t) = c$, then it can be easily demonstrated that

$$\text{MTTD} = \frac{1}{c} \tag{10.25}$$

10.3 Variation of Detection Rate with Range: Inverse Square and Cube Laws of Detection

As discussed earlier in this chapter, the probabilities of detection g_i for intermittent search and the rate of detection $\gamma(t)$ for continuous search depend on the physical conditions. For example, the value of $\gamma(t)$ will usually vary as the search progresses, depending on such parameters as the range to the target; illumination, either natural or from the searching radar or laser; target contrast against the background; and weather conditions (haze, rain, snow, etc.). Regardless of the type of sensor used to detect the target, the most important parameter affecting the detection rate is the range from the sensor to the target. As an example of this dependence, a simplified analysis of the variation of the detection rate with range will be illustrated for the case of an airborne or space sensor searching for ground targets.

In the subsequent analysis the following assumptions will used[1]:

1) The sensor is at height h above the ground and distance r from the target.

2) The sensor detects the target by seeing the target footprint, which will be assumed to be a rectangle circumscribing the target.

3) The instantaneous detection rate $\gamma(t)$ is proportional to the solid angle subtended at the point of observation by the target.

4) The dimensions of the target are small in comparison with h and r.

From Fig. 10.2a by similar triangles, for small angles α,

$$\frac{h}{s} = \frac{c}{a} \tag{10.26}$$

Also, from Fig. 10.2b for small angles α and β,

$$\alpha = \frac{c}{s} \tag{10.27}$$

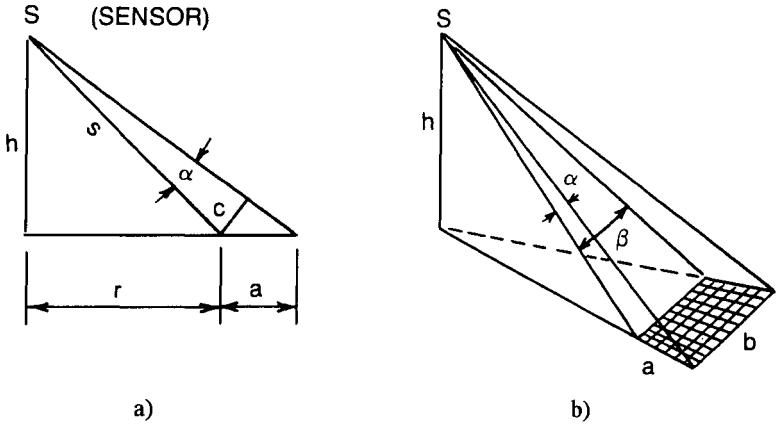
$$\beta = \frac{b}{s} \tag{10.28}$$

Noting that the solid angle φ subtended by the target is given approximately by

$$\varphi = \frac{cb}{4\pi s^2} = \frac{s\alpha\, s\beta}{4\pi s^2} = \frac{\alpha\beta}{4\pi} \tag{10.29}$$

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a) b)

Fig. 10.2 Ground target detection by an airborne sensor.

it follows from Eqs. (10.26)–(10.29) that

$$\varphi = \frac{1}{4\pi}\frac{c}{s}\frac{b}{s} = \frac{hA_T}{4\pi s^3} = \frac{hA_T}{4\pi(h^2 + r^2)^{\frac{3}{2}}} \tag{10.30}$$

where

$$A_T = ab \tag{10.31}$$

Because $\gamma(t)$ was assumed to be proportional to the solid angle φ,

$$\gamma(t) = \frac{kA_T}{4\pi}\frac{h}{(h^2 + r^2)^{\frac{3}{2}}} \tag{10.32}$$

where k is the constant of proportionality that depends on all other factors affecting the detection rate, other than the range. The dimensions of k are one over unit time.

Low Altitude Detection: $r \gg h$

If r is much larger than h, Eq. (10.32) can be approximated by

$$\gamma(t) = \frac{kA_T}{4\pi}\frac{h}{r^3} \qquad \text{for} \qquad r \gg h \tag{10.33}$$

The property of detection described by the preceding equation is called the "inverse cube law of detection." The inverse cube law represents a useful model for predicting the probability of detection, particularly if the constant k is modified to take into the account all other factors affecting the detection rate.

High Altitude Detection: $h \gg r$ (Space Surveillance)

If r is much smaller than h, Eq. (10.32) can be approximated by

$$\gamma(t) = \frac{kA_T}{4\pi}\frac{1}{h^2} \qquad \text{for} \qquad h \gg r \tag{10.34}$$

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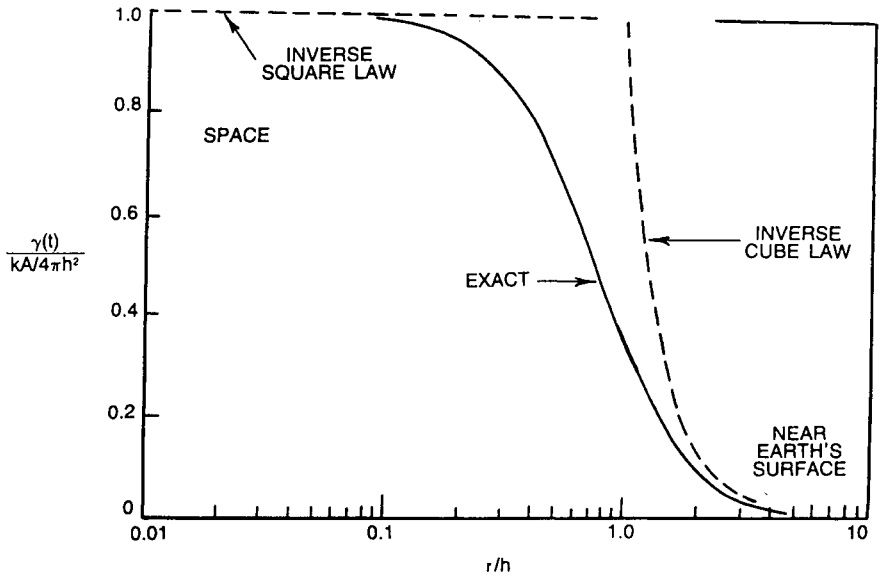


Fig. 10.3 Regions of applicability of the inverse square and cubic laws of detection.

The property of detection described by the preceding equation is called the "inverse square law of detection." The inverse square law is a useful model for predicting the probability of detection of ground targets by a space sensor.

The regions of applicability of the inverse cube and square laws of detection are illustrated in Fig. 10.3.

10.4 Probability of Detection in Search of a Given Area[2–6]

Consider a search by an airborne searcher with an observation width W in a region of area A. The search aircraft that conducts this search flies with a constant speed V continuously looking for the target, which is known to be somewhere within the area A. The target will be assumed to be stationary. Three modes of search will analyzed: 1) exhaustive search; 2) random search; and 3) inverse cube law search.

Exhaustive Search

In the exhaustive search the area A is swept systematically without any overlap (see Fig. 10.4). If the target appears within a distance $\frac{1}{2}W$ of the search aircraft, it is detected immediately. Thus, in time t the searcher will cover an area VtW, and the probability of detection will be simply the ratio of the area covered to the total area A, i.e.,

$$P_{\text{exhaustive}} = \begin{cases} z & \text{for } 0 < z \leq 1 \\ 1 & \text{for } z = 1 \end{cases} \tag{10.35}$$

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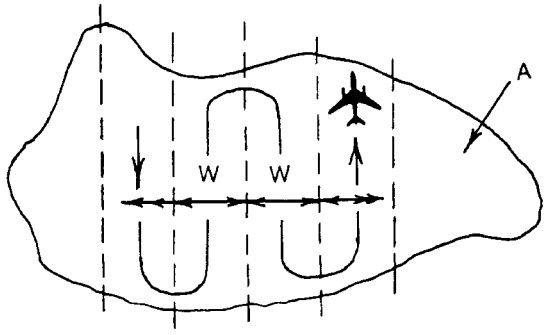


Fig. 10.4 Exhaustive search.

where

$$z = \frac{VWt}{A} \tag{10.36}$$

Random Search

If the search is carried out randomly, the aircraft does not follow any systematic pattern as shown in Fig. 10.5. The rate of detection is then computed from

$$\gamma(t) = \frac{VW}{A} \tag{10.37}$$

which represents the number of targets detected per unit time. Substituting Eq. (10.37) into Eq. (10.20),

$$P_{\text{random}} = 1 - \exp\left(-\frac{VWt}{A}\right) = 1 - e^{-z} \tag{10.38}$$

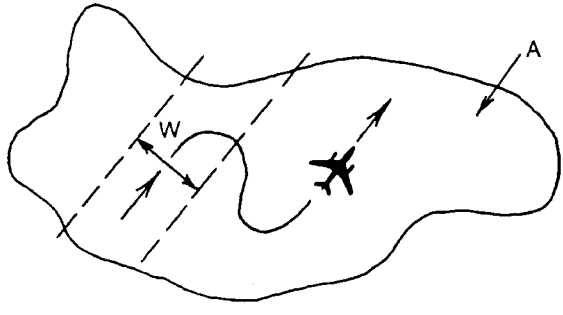


Fig. 10.5 Random search.

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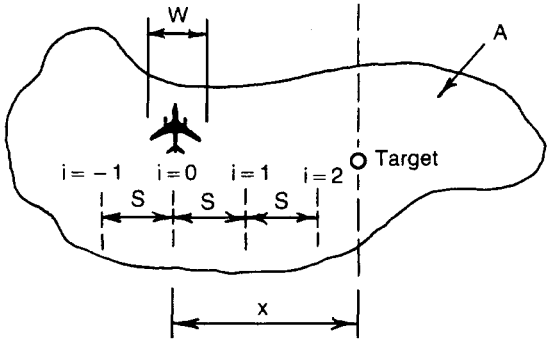


Fig. 10.6 Exhaustive search with an inverse cube law of detection.

Inverse Cube Law Search

For a low-altitude search the detection rate is approximated by

$$\gamma(t) = \frac{kA_T h}{4\pi r^3} \qquad \text{for} \qquad r \gg h \tag{10.33}$$

where, as before, A_T denotes the target footprint area. The search pattern is shown in Fig. 10.6. The aircraft traverses the search area A in a series of straight tracks denoted by $i = \ldots, -2, -1, 0, 1, 2, \ldots$, etc. A typical track for this search pattern is shown in Fig. 10.7.

Observing that

$$r^2 = x^2 + V^2t^2 \tag{10.39}$$

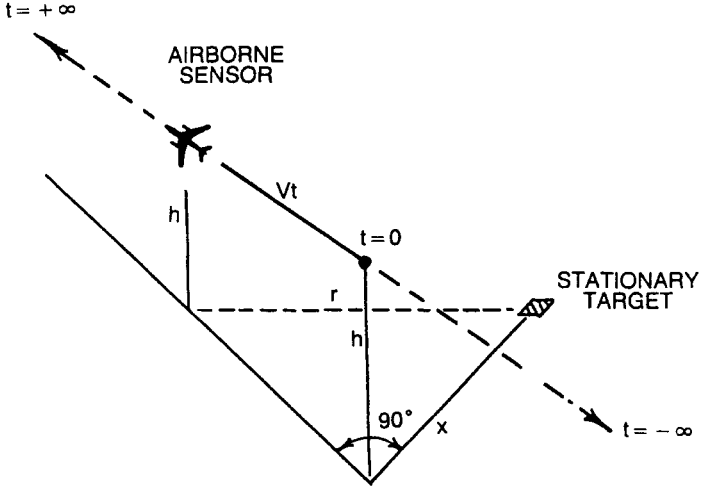


Fig. 10.7 Search along a line with an offset x from the target.

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the probability of detection on a flight at constant altitude h along the path $i = 0$ that misses the target by a distance x is[7]

$$\begin{aligned} p(x) &= 1 - \exp\left[-\int_{-\infty}^{\infty} \gamma(t)\,dt\right] \\ &= 1 - \exp\left[-\int_{-\infty}^{\infty} \frac{kA_T h}{4\pi} \frac{1}{(x^2 + V^2t^2)^{\frac{3}{2}}}\,dt\right] \\ &= 1 - \exp\left(-\frac{2m}{x^2}\right) \end{aligned} \tag{10.40}$$

where

$$m = \frac{kA_T h}{4\pi V} \tag{10.41}$$

The preceding result is obtained through a change of variables with $\cosh(y) = Vt/x$. Strictly speaking, the integration limits in Eq. (10.40) should not go beyond the boundaries of the area A, but because $\gamma(t)$ varies as $1/r^3$, the integration limits $(-\infty, +\infty)$ are used to obtain a simplified result. This is possible because the contributions to the integral in Eq. (10.40) when r is large are negligible.

The sweep width associated with the probability $p(x)$ is given by[2]

$$W = \int_{-\infty}^{\infty} p(x)\,dx = \sqrt{8\pi m} \tag{10.42}$$

The preceding result is obtained by a change in variables with $y = \sqrt{2m}/x$ and integration by parts.

Using the results of Eq. (10.40), the probability of no detection on any given track is

$$q(x, S) = \exp\left[-\frac{2m}{(x - iS)^2}\right] \tag{10.43}$$

where

$$S = \frac{A}{VT} \tag{10.44}$$

and T is the time to complete the search. Hence, the probability of no detection on tracks $i = \ldots, -2, -1, 0, 1, 2, \ldots$, etc. is

$$\begin{aligned} Q(x, S) &= \cdots \exp\left[-\frac{2m}{(x + 2S)^2}\right] \exp\left[-\frac{2m}{(x + S)^2}\right] \exp\left[-\frac{2m}{x^2}\right] \\ &\quad \times \exp\left[-\frac{2m}{(x - S)^2}\right] \exp\left[-\frac{2m}{(x - 2S)^2}\right] \\ &= e^{-N(x,S)} \end{aligned} \tag{10.45}$$

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where

$$N(x, S) = \sum_{i=-\infty}^{\infty} \frac{2m}{(x - iS)^2} \tag{10.46}$$

Using the results of Ref. 8,

$$N(x, S) = \frac{2m\pi^2}{S^2} \csc^2 \frac{\pi x}{S}$$

$$= \frac{\pi}{4} \left(\frac{W}{S}\right)^2 \csc^2 \frac{\pi x}{S} \tag{10.47}$$

Hence, the *average* probability of no detection $Q(W/S)$ over the distance $x = 0$ to $x = S$ can be found from

$$Q\left(\frac{W}{S}\right) = \frac{1}{S} \int_0^S Q(x, S)\, dx$$

$$= 1 - \text{erf} \frac{\sqrt{\pi}\, W}{2S} \tag{10.48}$$

The derivation of Eqs. (10.48) is shown in Appendix D.

Finally, the probability of detection is found from

$$\left(\frac{W}{S}\right) = 1 - Q\left(\frac{W}{S}\right) = \text{erf} \frac{\sqrt{\pi}\, W}{2S} \tag{10.49}$$

The probabilities of detection for the three modes of search are given in Fig. 10.8 as functions of the parameter $z = WVt/A$ or $z = W/S$. The exhaustive search and the inverse cube law search were carried out systematically; however, because of the inverse cube law dependence of the sensor in the second case, the probabilities are lower than for the exhaustive search. On the other hand, a completely random search gives the lowest probabilities of detection.

If a search with an inverse cube law sensor were to search randomly, the detection probability would be given by the random search curves, just as it would be for a random search with a sweep W. Thus, in a random search, the only important parameter of a sensor is its sweep width.

10.5 One-Dimensional Search[9]

In operational engagements of land forces, it is sometimes necessary to patrol a line of defense in order to detect enemy elements (troops, tanks, etc.) trying to break through the defense lines. It will be assumed that a total of n stationary observations patrols are located along the line of defense L, distance $2W$ apart, where W is the maximum radius at which the patrols can detect the enemy (see Fig. 10.9). The enemy elements are assumed to move perpendicularly to the line L (defense line). In order to detect all enemy elements, the minimum number of

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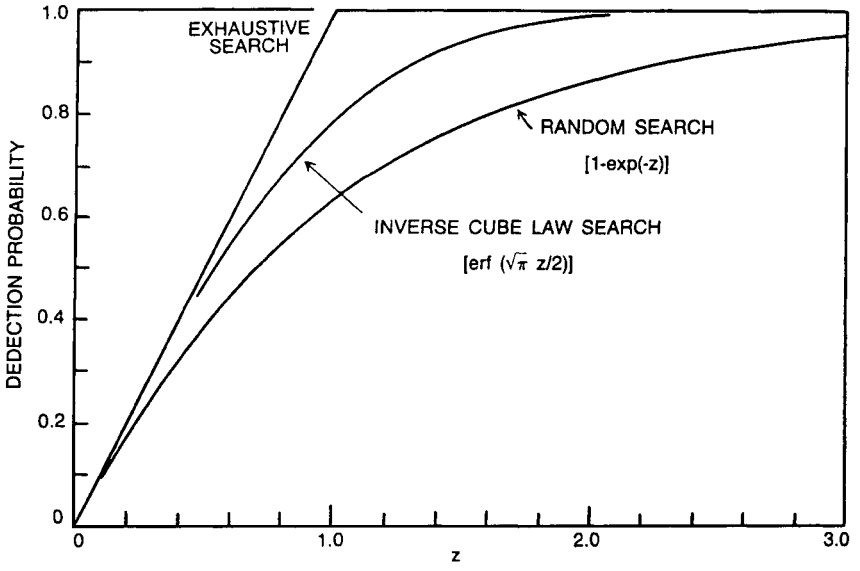


Fig. 10.8 Comparison of detection probabilities for three search modes.

patrols should be

$$n_P \geq \frac{L}{2W} \tag{10.50}$$

This number is independent of the speed of enemy movement. If there are fewer patrols such that $n < n_P$, then the probability of detecting the enemy is given by

$$P = \frac{n}{n_P} = \frac{2Wn}{L} \tag{10.51}$$

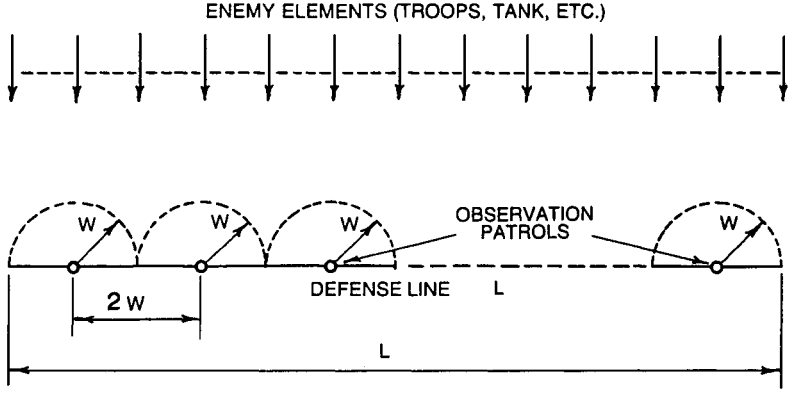


Fig. 10.9 Observation patrols along a defense line.

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If the observation patrols are mobile (i.e., moving with constant speed v_P), then it can be easily shown that

$$n_P \geq \frac{L}{W(2 + v_P/v_E)} \tag{10.52}$$

where

v_P = speed of patrol (along line L)

v_E = speed of enemy (perpendicular to L)

Equation (10.52) indicates that, if $v_P = v_E$, then the number of observation patrols required is reduced by one third. The observation patrols have to move back and forth in a synchronized fashion, covering a distance Wv_P/v_E. If there are fewer patrols employed than the number given by Eq. (10.52), then the probability of detection is obtained, as in the case of stationary patrols, from

$$P = \frac{n}{n_P} = \frac{Wn}{L}\left(2 + \frac{v_P}{v_E}\right) \tag{10.53}$$

10.6 Constant Velocity Target[9]

One interesting search problem is the problem of searching for a target that can move with a constant velocity v_T in any fixed direction from a point, the location of which is known initially to the searching aircraft. All directions for the target movement are equally probable. To intercept the target, the following search pattern is adopted: An assumption is made initially that the target is moving toward the searching aircraft; therefore, the aircraft must proceed in the direction where the target was initially located. Contact with the target is established, providing the assumption was correct, after elapsed time (see Fig. 10.10):

$$t_0 = \frac{D}{v_T + v_S} \tag{10.54}$$

where

D = distance from the aircraft to the point where the target was located at the beginning of the search
v_T = target velocity
v_S = velocity of the search aircraft; $v_S > v_T$

During this elapsed time the target would have moved a distance R_0 given by

$$R_0 = v_T t_0 = \frac{v_T D}{v_T + v_S} \tag{10.55}$$

while the search aircraft flies a distance S_0 given by

$$S_0 = v_S t_0 = \frac{v_S D}{v_T + v_S} \tag{10.56}$$

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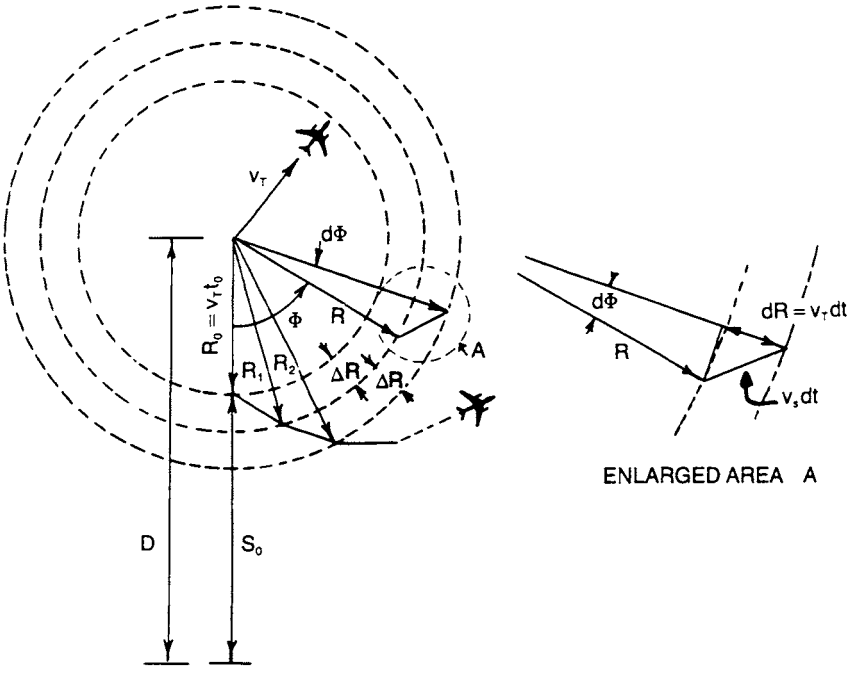


Fig. 10.10 Logarithmic spiral search for a constant-velocity target.

If contact is not established, the searching aircraft should leave this hypothetical point of contact reached after elapsed time t_0 and move toward the next most probable point at which the target would be found. During the time increment Δt the target should be found somewhere on the circle of radius $R_1 = v_T(t + \Delta t)$, and the aircraft would move a distance $v_S \Delta t$. If the direction of aircraft flight in Δt is toward the intersection of arcs R_1 and $v_S \Delta t$, the latter struck from the hypothetical contact point, then the target would be intercepted between the radii R_0 and R_1 if the target was moving within the segment $R_1\, d\phi$. Subsequently, if no contact is made, a new direction for the aircraft is determined from the intersection of arcs of $R_2 = v_T(t_0 + 2\Delta t)$ and $v_S \Delta t$. Thus, proceeding in time increments of Δt, a curve comprised of straight line segments is obtained. This curve is the logarithmic spiral. If the aircraft follows this curve, interception of the target is assured, regardless of the direction selected by the target. The exact shape of this spiral will be determined next. This problem was formulated in Ref. 9 without providing an analytical solution for the search spiral.

From Fig. 10.10 it is clear, using the Pythagorean principle, that

$$(R \sin d\phi)^2 + \left[dR + R \sin d\phi \tan\left(\tfrac{1}{2} d\phi\right)\right]^2 = (v_S\, dt)^2$$

which, when second-order quantities are neglected, reduces to

$$R^2 (d\phi)^2 + (dR)^2 = v_S^2 (dt)^2 \tag{10.57}$$

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Also,

$$v_T\,dt = dR \tag{10.58a}$$

Using Eqs. (10.57) and (10.58a),

$$\frac{d\phi}{dt} = \frac{v_T}{R}\left(\frac{v_S^2}{v_T^2} - 1\right)^{\frac{1}{2}} = \frac{v_T}{R}\frac{1}{k} \tag{10.59}$$

where

$$k = \left(\frac{v_S^2}{v_T^2} - 1\right)^{-\frac{1}{2}} \tag{10.60}$$

Integrating Eq. (10.59),

$$k\int_0^{\phi} d\phi = \int_0^t \frac{v_T}{R}\,dt = \int_{R_0}^{R} \frac{dR}{R}$$

Hence,

$$k\phi = \ell_n \frac{R}{R_0}$$

and

$$R = R_0 e^{k\phi} \tag{10.61}$$

Starting with Eq. (10.58a),

$$\frac{dR}{dt} = v_T \tag{10.58b}$$

it follows that

$$R = R_0 + v_T t \tag{10.62}$$

Next, substituting Eq. (10.61) into Eq. (10.62),

$$R_0 e^{k\phi} = R_0 + v_T t$$

which yields

$$k\phi = \ell_n\left(1 + \frac{v_T t}{R_0}\right) \tag{10.63}$$

and

$$R = R_0 \exp\left[\ell_n\left(1 + \frac{v_T t}{R_0}\right)\right] \tag{10.64}$$

Thus, Eqs. (10.61) and (10.64) describe completely the path of the searching aircraft. This path ensures that the intercept is somewhere between $\phi = 0$ and 2π.

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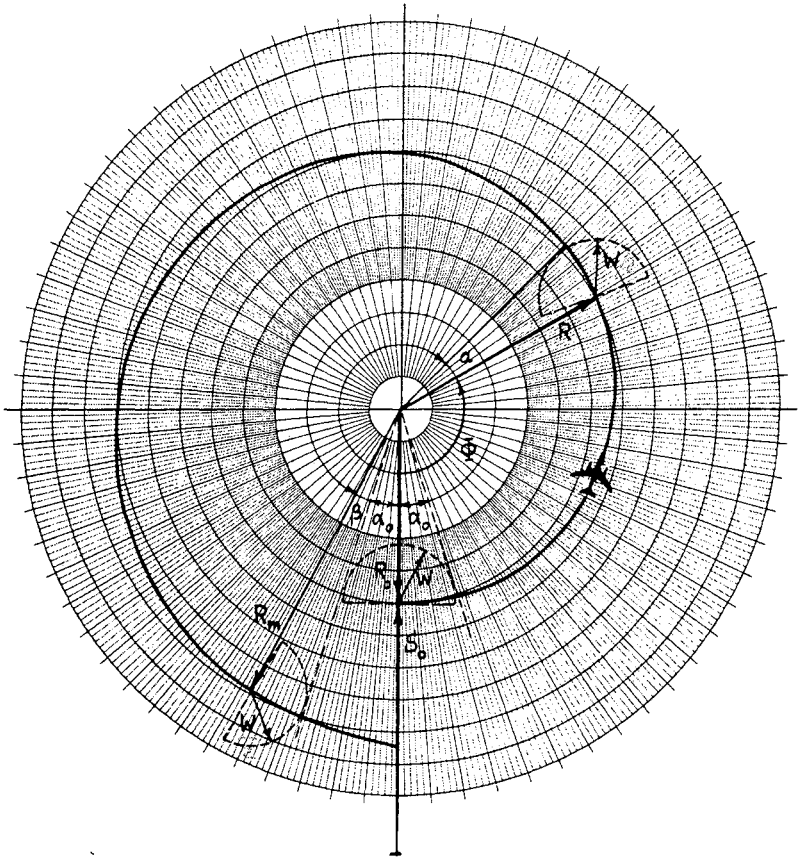


Fig. 10.11 Logarithmic spiral search with a 180-deg radar sweep of radius W.

In the logarithmic spiral search one is naturally interested in the estimates of the probability of detection of the target. In reality the searching aircraft does not have to intercept the target. A long-range radar may be used, for example, to detect the target. It will be assumed here that the search is conducted by sweeping the forward 180-deg arc with an effective radius W (see Fig. 10.11).

When the aircraft arrives at the initial contact point the probability of an encounter with the target is

$$P_0 = \frac{2R_0\alpha_0}{2\pi R_0} = \frac{\alpha_0}{\pi} = \frac{1}{\pi}\sin^{-1}\frac{W}{R_0} \tag{10.65}$$

where

$$\alpha_0 = \sin^{-1}\frac{W}{R_0} \tag{10.66}$$

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It should be noted here that the probability density distribution for this case is constant (equal to $1/2\pi$). If the aircraft does not find its target, it then turns to starboard and follows the logarithmic spiral path with its search radar turned on. Along this spiral path, the probability of detection at radius R is obtained simply from

$$\begin{aligned} P &= \frac{R(\alpha_0 + \phi + \alpha)}{2\pi R} \\ &= \frac{\alpha_0}{2\pi} + \frac{\phi}{\pi} + \frac{\alpha}{2\pi} \\ &= \frac{1}{2\pi}\sin^{-1}\frac{W}{R_0} + \frac{\phi}{2\pi} + \frac{1}{2\pi}\sin^{-1}\frac{W}{R} \end{aligned} \tag{9.67a}$$

When $\phi = 0$, then $R = R_0$ and $P = P_0$. The probability P increases from P_0 at $\phi = 0$ to 1 at $\phi = 2\pi - \alpha_0 - \beta$, where $\beta = \sin^{-1}(W/R_0)$, and R_m is the maximum spiral radius needed for 100% detection (see Fig. 10.11). The reader can easily verify that indeed Eq. (9.67a) is equal to 1.0 when $\phi = 2\pi - \alpha_0 - \beta$. Substituting Eq. (10.61) or (10.63) into (10.67a), two alternative expressions for the probability P can be obtained:

$$P = \frac{1}{2\pi}\sin^{-1}\frac{W}{R_0} + \frac{\phi}{2\pi} + \frac{1}{2\pi}\sin^{-1}\left(\frac{W}{R_0}e^{-k\phi}\right) \tag{10.67b}$$

or

$$\begin{aligned} P = {} & \frac{1}{2\pi}\sin^{-1}\frac{W}{R_0} + \frac{1}{2k\pi}\ln\left(1 + \frac{v_T t}{R_0}\right) \\ & + \frac{1}{2\pi}\sin^{-1}\left\{\frac{W}{R_0}\exp\left[-\ln\left(1 + \frac{v_T t}{R_0}\right)\right]\right\} \end{aligned} \tag{10.67c}$$

When $W = 0$ the probability P reduces to a very simple form:

$$P = \frac{\phi}{2\pi} = \frac{1}{2k\pi}\ln\left(1 + \frac{v_T t}{R_0}\right) \tag{10.67d}$$

10.7 Detection of Electromagnetic Radiation from a Target[10]

A radiating target is known to be located in a specified area that can be overflown by an aircraft with a search radar. To find this target, a search of the whole area will be conducted without overlaps. It will be assumed that the radiation source may be at any point with equal probability, and during the search of the area the target is switched on for time t_1 and switched off for time t_2. Thus, the total search time is $t_1 + t_2$. The search aircraft is equipped with a sensor for detecting electromagnetic radiation with a field of view ϕ, both in the direction of flight and in the transverse direction. The width of sweep is calculated from (see Fig. 10.12)

$$W = 2h\tan\tfrac{1}{2}\phi \tag{10.68}$$

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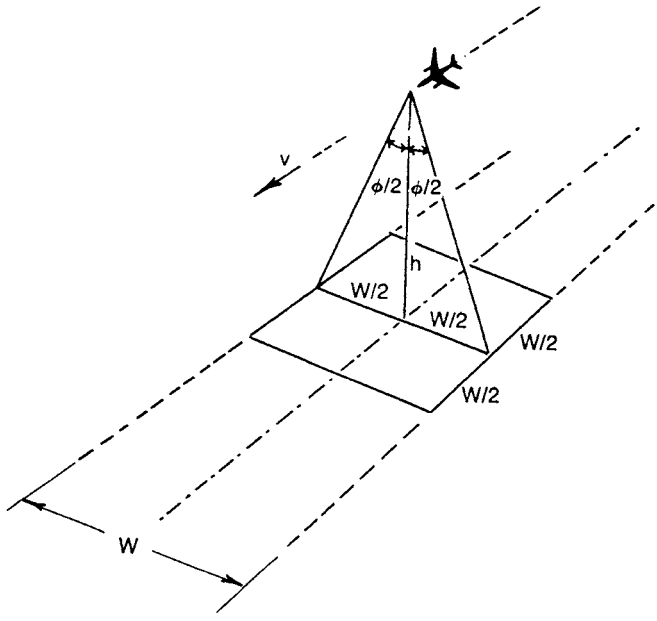


Fig. 10.12 Airborne search for a radiating target.

where h is the flight altitude. Because the sweep is carried out in both directions, any point on the surface (i.e., potential target) will be observed for the time t_s given by

$$t_s = \frac{2h \tan \frac{1}{2}\phi}{v} \tag{10.69}$$

where v is the velocity of the aircraft.

The probability of signal detection from the radiating target, P_D, is obtained from the product

$$P_D = P_A P_T P_S \tag{10.70}$$

where

P_A = probability that the target is in the swept area
P_T = probability that the target is emitting radiation
P_S = probability of target detection by the sensor

The probability P_A is simply the ratio of the swept area over the total area A, i.e.

$$P_A = \frac{2h \tan \frac{1}{2}\phi \times vt}{A} \tag{10.71}$$

where t is the elapsed time of search, so that when the search is completed

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$t = t_1 + t_2$. The probability P_T is given by

$$P_T = \frac{t_1}{t_1 + t_2} \tag{10.72}$$

and the probability of target detection, P_S, can be obtained from Eq. (9.20) for an instantaneous detection rate $\gamma(t) = \gamma = \text{const}$ and $t = t_S$, i.e.,

$$P_S = 1 - e^{-\gamma t_S}$$

$$= 1 - \exp\left(-\frac{2\gamma h \tan \frac{1}{2}\phi}{v}\right) \tag{10.73}$$

Hence, from Eqs. (9.70)–(9.73),

$$P_D(t) = \frac{2h \tan \frac{1}{2}\phi \times vt}{A} \times \frac{t_1}{t_1 + t_2}\left[1 - \exp\left(-\frac{2\gamma h \tan \frac{1}{2}\phi}{v}\right)\right] \tag{10.74}$$

At the end of the search, when $t = t_1 + t_2$,

$$P_D = \frac{2h \tan \frac{1}{2}\phi \times vt_1}{A}\left[1 - \exp\left(-\frac{2\gamma h \tan \frac{1}{2}\phi}{v}\right)\right] \tag{10.75}$$

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Problems

10.1. Show that the mean time to detection MTTD for a continuous search may be computed from

$$\text{MTTD} = E[T] = \int_0^\infty Q(t)\,dt$$

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10.2. The probability of detection g_i of an attacking ballistic missile for the first five glimpses ($i = 1, 2, \ldots, 5$) by radar is given by

$i =$	1	2	3	4	5
$g_i =$	0.1	0.2	0.4	0.8	1.0

Determine:
a) the probability of detecting the missile on the ith glimpse,
b) the probability of detecting the missile in the first i glimpses, and
c) the expected number of glimpses to detect the missile.

10.3. If $g_i = 0.2$ and $N = \infty$, what will be the expected number of glimpses i needed to detect the attacking missile?

10.4. For a constant velocity target in Section 9.6, determine the time to complete the logarithmic search spiral for the following two cases:
a) Search without radar detection ($W = 0$), and
b) Search with radar sweep of radius W.

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11
Optimization Methods

This chapter is intended as an introduction to the most common optimization techniques and their applications to some problems in defense analyses. The mathematical level has been kept as elementary as possible, with emphasis on presenting the mathematical tools rather than on rigorous proofs and derivations of the fundamentals.

11.1 Mathematical Optimization

The various mathematical techniques can be classified according to the nature of constraints, number of decision makers, and treatment of time, i.e., for use in static or dynamic problems. This classification is illustrated in Table 11.1.[1] Most of the defense applications of these techniques are for the static domain and therefore discussion will be limited to the various types of *static* programming. The reader has already been introduced to the rudimentary concepts of game theory in Chapter 6, where the game theory was applied to ballistic missile defense allocations. In this section, only a general overview of the fundamental concepts of classical, nonlinear, and linear programming is presented, emphasizing the differences in constraints and in types of objective functions (functions to be optimized either as maxima or minima). For further details of the dynamic optimization techniques, such as calculus of variations, dynamic programming, Maximum Principle, and differential games, the reader may consult Ref. 1.

Classical Programming

The classical programming problem is to maximize or minimize a given function $F(\mathbf{x})$, usually referred to as the objective function, subject to a given set of *equality* constraints, which can be expressed for the classical programming maximum problem as

$$\max_{\mathbf{x}} F(\mathbf{x}), \quad \text{subject to } \mathbf{g}(\mathbf{x}) = \mathbf{b} \tag{11.1}$$

where the independent variables (or parameters) $\mathbf{x}$ are given by

$$\mathbf{x} = \{x_1, x_2, \ldots, x_n\} \tag{11.2}$$

The function $F(\mathbf{x})$ is the objective function, $\mathbf{g}(\mathbf{x})$ is the the constraint function, and $\mathbf{b}$ is a column matrix of the constraint constants. Equation (11.1) can be written in full as

$$\max_{x_1, x_2, \ldots, x_n} F(x_1, x_2, \ldots, x_n) \tag{11.3}$$

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Table 11.1 Classification of optimization techniques[1]

Nature of Constraints; Number of decision-makers.	Treatment of time	
	Static	Dynamic
No constraints or equality constraints; One decision-maker.	Classical programming	Calculus of Variations
Inequality constraints; One decision-maker.	Nonlinear programming Linear programming	Dynamic programming Maximum Principle
Two or more decision-makers.	Game theory	Differential games

subject to

$$\begin{aligned} g_1(x_1, x_2, \ldots, x_n) &= b_1 \\ g_2(x_1, x_2, \ldots, x_n) &= b_2 \\ \vdots \qquad\qquad &\vdots \\ g_m(x_1, x_2, \ldots, x_n) &= b_m \end{aligned} \tag{11.4}$$

It is assumed here that $n > m$ where the difference $n - m$ is the number of *degrees of freedom* in the problem. Furthermore, it is also assumed that both $F(\mathbf{x})$ and $\mathbf{g}(\mathbf{x})$ are continuously differentiable with respect to $\mathbf{x}$.

Maximizing $F(\mathbf{x})$ is equivalent to maximizing $a + bF(\mathbf{x})$ with $b > 0$, and to minimizing $a + bF(\mathbf{x})$ with $b < 0$, where in both cases a is an additive constant. Thus, multiplication of $F(\mathbf{x})$ by -1 can be used to convert maximization problems to minimization problems.

A special case of classical programming is the case of the objective function $F(\mathbf{x})$ represented by a quadratic function of $\mathbf{x}$. In this case

$$\min_{\mathbf{x}} F(\mathbf{x}) = a + \mathbf{cx} + \tfrac{1}{2}\mathbf{x}^{\mathrm{T}}\mathbf{Qx} \tag{11.5}$$

subject to

$$\mathbf{Ax} = \mathbf{b} \tag{11.6}$$

and

$$\mathbf{x} \geq \mathbf{0} \tag{11.7}$$

where a is a constant, $\mathbf{c}$ is a row matrix of constant coefficients, and $\mathbf{A}$ is a $(m \times n)$ matrix of constant coefficients. The matrix $\mathbf{Q}$ is a square matrix and is positive semi-definite, which is what makes the quadratic program convex, and is equivalent to the requirement that

$$\mathbf{z}^{\mathrm{T}}\mathbf{Qz} \geq \mathbf{0} \text{ for all } \mathbf{z} \tag{11.8}$$

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An algorithm for the solution to convex quadratic programs was proposed by Jensen and King.[2]

An efficient method for the solution of classical programming problems can be obtained through the application of Lagrange multipliers. In this method a column matrix $\mathbf{\Lambda}$ is defined as

$$\mathbf{\Lambda} = \{\lambda_1, \lambda_2, \ldots, \lambda_m\} \tag{11.9}$$

where $\lambda_1, \lambda_2, \ldots, \lambda_m$ represent m Lagrange multipliers, and the Lagrangian function $L(\mathbf{x}, \mathbf{\Lambda})$ is given by

$$L(\mathbf{x}, \mathbf{\Lambda}) = F(\mathbf{x}) + \mathbf{\Lambda}^T(\mathbf{b} - \mathbf{g}(\mathbf{x})) \tag{11.11}$$

The solution to the problem is the point $(\mathbf{x}^*, \mathbf{\Lambda}^*)$, which may be obtained from the requirement that all first order partial derivatives of the Lagrangian vanish. Thus, the required differentiation with respect to $\mathbf{x}$ and $\mathbf{\Lambda}$ leads to a pair of sets of equations

$$\frac{\partial L(\mathbf{x}^*, \mathbf{\Lambda}^*)}{\partial \mathbf{x}} = \frac{\partial F(\mathbf{x}^*)}{\partial \mathbf{x}} - (\mathbf{\Lambda}^*)^T \frac{\partial \mathbf{g}(\mathbf{x}^*)}{\partial \mathbf{x})} = \mathbf{0} \tag{11.12}$$

and

$$\frac{\partial L(\mathbf{x}^*, \mathbf{\Lambda}^*)}{\partial \mathbf{\Lambda}} = \mathbf{b} - \mathbf{g}(\mathbf{x}^*) = \mathbf{0} \tag{11.13}$$

The first set of n equations can be written as

$$\nabla F(\mathbf{x}^*) = (\mathbf{\Lambda}^*)^T \frac{\partial \mathbf{g}(\mathbf{x}^*)}{\partial \mathbf{x}} \tag{11.14}$$

where, written in full,

$$\nabla F(\mathbf{x}^*) = \left[\frac{\partial F(\mathbf{x}^*)}{\partial x_1} \quad \frac{\partial F(\mathbf{x}^*)}{\partial x_2} \cdots \frac{\partial F(\mathbf{x}^*)}{\partial x_n}\right] \tag{11.15}$$

and

$$(\mathbf{\Lambda}^*)^T \frac{\mathbf{g}(\mathbf{x}^*)}{\partial \mathbf{x}} = \left[\lambda_1^*, \lambda_2^*, \ldots, \lambda_m^*\right] \begin{bmatrix} \frac{\partial g_1(\mathbf{x}^*)}{\partial x_1} & \frac{\partial g_1(\mathbf{x}^*)}{\partial x_2} & \cdots & \frac{\partial g_1(\mathbf{x}^*)}{\partial x_n} \\ \frac{\partial g_2(\mathbf{x}^*)}{\partial x_1} & \frac{\partial g_2(\mathbf{x}^*)}{\partial x_2} & \cdots & \frac{\partial g_2(\mathbf{x}^*)}{\partial x_n} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial g_m(\mathbf{x}^*)}{\partial x_1} & \frac{\partial g_m(\mathbf{x}^*)}{\partial x_2} & \cdots & \frac{\partial g_m(\mathbf{x}^*)}{\partial x_n} \end{bmatrix}_{m \times n} \tag{11.16}$$

The second set of equations is the constraint

$$\mathbf{g}(\mathbf{x}^*) = \mathbf{b} \tag{11.17}$$

It should be noted that the $(n + m)$ equations in (11.14) and (11.17) are sufficient to solve for $\mathbf{x}^*$ and $\mathbf{\Lambda}^*$ so that at $(\mathbf{x}^*, \mathbf{\Lambda}^*)$

$$L(\mathbf{x}^*, \mathbf{\Lambda}^*) = F(\mathbf{x}^*) \tag{11.18}$$

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A typical application of the Langrange multiplier method was already discussed in Section 6.1 for the analysis of layered defense against intercontinental ballistic missiles. Another application is shown in Section 10.2.

Classical Programming: Unconstrained Optimization

A special case of classical programming is a class of problems with *unconstrained* optimization, where the objective function $F(\mathbf{x})$ is *not* subject to any constraints. For these cases, the objective function at the optimum solution (either maximum or minimum) at $\mathbf{x} = \mathbf{x}^*$ must satisfy the equation

$$\left.\frac{\partial F(\mathbf{x})}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}^*} = \nabla F(\mathbf{x}^*) = \left[\frac{\partial F(\mathbf{x}^*)}{\partial x_1}\ \frac{\partial F(\mathbf{x}^*)}{\partial x_2} \cdots \frac{\partial F(\mathbf{x}^*)}{\partial x_n}\right] = \mathbf{0} \tag{11.19}$$

which represents the condition that gradients in the direction of x_i, where $i = 1, 2, \ldots n$, are all equal to zero, i.e.,

$$\frac{\partial F(\mathbf{x})}{\partial x_i} = 0; \quad i = 1, 2, \ldots n, \quad \text{at } \mathbf{x} = \mathbf{x}^* \tag{11.20}$$

Obtaining the value of $\mathbf{x}^*$ that causes $F(\mathbf{x})$ to be optimized can be difficult if the dimension of $\mathbf{x}$ is high or if the shape of $F(\mathbf{x})$ is intricate. In such cases, an iterative numerical technique can be used, as derived below.

If the first two derivatives are continuous, then $\nabla F(\mathbf{x})$ can be expanded as a Taylor series about $\mathbf{x} = \mathbf{x}_0$ so that

$$\nabla F(\mathbf{x}) = \frac{\partial F(\mathbf{x})}{\partial \mathbf{x}} = \left.\frac{\partial F(\mathbf{x})}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0} + (\mathbf{x} - \mathbf{x}_0)^T \left.\frac{\partial^2 F(\mathbf{x})}{\partial \mathbf{x}^2}\right|_{\mathbf{x}=\mathbf{x}_0} + \text{higher order terms in } (\mathbf{x} - \mathbf{x}_0) \tag{11.21}$$

For $\mathbf{x} = \mathbf{x}^*$, i.e. at the optimum point, and $\mathbf{x}_0$ near $\mathbf{x}^*$, it follows from Eqs. (11.21) that when higher order terms are neglected.

$$\frac{\partial F(\mathbf{x}^*)}{\partial \mathbf{x}} \approx \left.\frac{\partial F(\mathbf{x})}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0} + (\mathbf{x}^* - \mathbf{x}_0)^T \left.\frac{\partial^2 F(\mathbf{x})}{\partial \mathbf{x}^2}\right|_{\mathbf{x}=\mathbf{x}_0} \tag{11.22}$$

The expression $\partial^2 F(\mathbf{x})/\partial \mathbf{x}^2$ is a symmetric matrix, as long as the order of differentiation is immaterial. Therefore, from Eqs. (11.19) and (11.22) after matrix transposition,

$$\left.\frac{\partial^2 F(\mathbf{x})}{\partial \mathbf{x}^2}\right|_{\mathbf{x}=\mathbf{x}_0} (\mathbf{x}^* - \mathbf{x}_0) = -\left(\left.\frac{F(\mathbf{x})}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0}\right)^T$$

or

$$\mathbf{x}^* = \mathbf{x}_0 - \left(\left.\frac{\partial^2 F(\mathbf{x})}{\partial \mathbf{x}^2}\right|_{\mathbf{x}=\mathbf{x}_0}\right)^{-1} \left(\left.\frac{F(\mathbf{x})}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0}\right)^T$$

$$= \mathbf{x}_0 - (\mathbf{H}(\mathbf{x}_0))^{-1} (\nabla F(\mathbf{x}_0))^T \tag{11.23}$$

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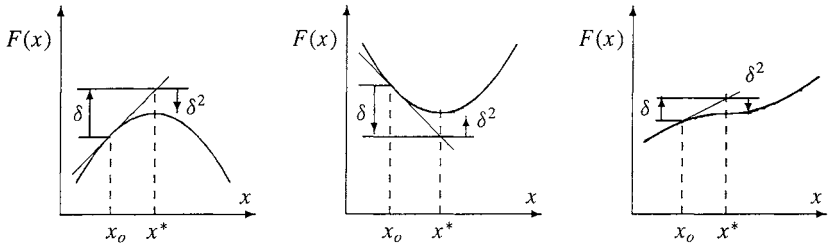


Fig. 11.1 The first and second variations for $x^* > x_0$.

where

$$\mathbf{H}(\mathbf{x}_0) = \begin{bmatrix} \frac{\partial^2 F(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 F(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 F(\mathbf{x})}{\partial x_1 \partial x_n} \\ \frac{\partial^2 F(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 F(\mathbf{x})}{\partial x_2^2} & \cdots & \frac{\partial^2 F(\mathbf{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 F(\mathbf{x})}{\partial x_n \partial x_1} & \frac{\partial^2 F(\mathbf{x})}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 F(\mathbf{x})}{\partial x_n^2} \end{bmatrix}_{\mathbf{x}=\mathbf{x}_0} \tag{11.24}$$

and

$$\nabla F(\mathbf{x}_0) = \left[\frac{\partial F(\mathbf{x}_0)}{\partial x_1} \; \frac{\partial F(\mathbf{x}_0)}{\partial x_2} \cdots \frac{\partial F(\mathbf{x}_0)}{\partial x_n} \right] \tag{11.25}$$

The matrix $\mathbf{H}$ is referred to as the Hessian of the function $F(\mathbf{x})$.

To understand the behavior of the function $F(\mathbf{x})$ at the optimum point $\mathbf{x}^*$, the function can be expanded as a Taylor series so that

$$F(\mathbf{x}^*) = F(bfx_0) + \left.\frac{\partial F(\mathbf{x})}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0} (\mathbf{x}^* - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \frac{\partial^2 F(\mathbf{x})}{\partial \mathbf{x}^2} + \cdots$$

$$= F(\mathbf{x}_0) + \delta + \delta^2 + \cdots \tag{11.26}$$

where

$$\delta = \left.\frac{\partial F(\mathbf{x})}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0} (\mathbf{x}^* - \mathbf{x}_0) = \text{First variation} \tag{11.27}$$

and

$$\delta^2 = \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \frac{\partial^2 F(\mathbf{x})}{\partial \mathbf{x}^2} = \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \mathbf{H}_0 (\mathbf{x}^* - \mathbf{x}_0)^T = \text{Second variation} \tag{11.28}$$

There are three possibilities for the function $F(\mathbf{x})$ at $\mathbf{x} = \mathbf{x}^*$: (a) $F(\mathbf{x}^*)$ is a local maximum, (b) $F(\mathbf{x}^*)$ is a local minimum, and (c) $F(\mathbf{x}^*)$ is a saddle point (i.e., an inflection point), as shown in Figs. 11.1 and 11.2. Examining Figs. 11.1 and 11.2 it is clear that the following conditions must be satisfied:

When $x^* > x_0$ or $x^* < x_0$ and $F(\mathbf{x}^*)$ is a maximum, then $\delta > 0$ and $\delta^2 < 0$.

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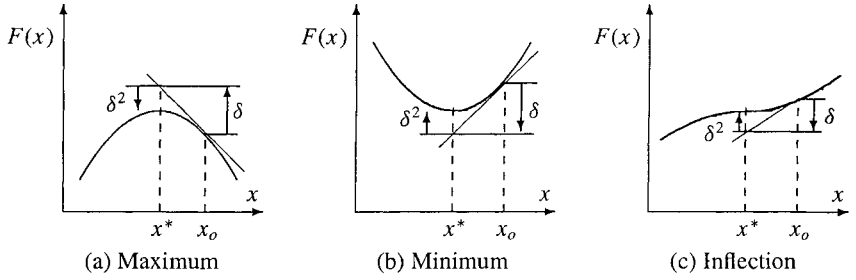


Fig. 11.2 The first and second variations for $x^* < x_0$.

When $x^* > x_0$ or $x^* < x_0$ and $F(\mathbf{x}^*)$ is a minimum, then $\delta < 0$ and $\delta^2 > 0$.
When $x^* > x_0$ and $F(\mathbf{x}^*)$ is a saddle point, then $\delta > 0$ and $\delta^2 < 0$, but when $x^* < x_0$, then $\delta < 0$ and $\delta^2 > 0$.
The scalar product $(\mathbf{x}^* - \mathbf{x}_0)^T \mathbf{H}_0 (\mathbf{x}^* - \mathbf{x}_0)^T$ for $\mathbf{x}_0$ close to $\mathbf{x}^*$ is:

$$\text{Positive definite when } F(\mathbf{x}^*) = \min_{\mathbf{x}}$$

$$\text{Negative definite when } F(\mathbf{x}^*) = \max_{\mathbf{x}}$$

Neither positive nor negative definite when $F(\mathbf{x}^*)$ is at an inflection point.

Equation (11.23) will not yield $\mathbf{x}^*$ on the first try of an assumed column matrix $\mathbf{x}_0$. It is therefore necessary to use an iteration, which should converge to the optimum solution as presented below:

$$\mathbf{x}_k = \mathbf{x}_{k-1} - \left(\frac{\partial^2 F(\mathbf{x})}{\partial \mathbf{x}^2} \bigg|_{\mathbf{x}=\mathbf{x}_{k-1}} \right)^{-1} \left(\frac{F(\mathbf{x})}{\partial \mathbf{x}} \bigg|_{\mathbf{x}=\mathbf{x}_{k-1}} \right)^{T} \tag{11.31}$$

The preceding formula is the so-called Newton–Raphson algorithm. It should, however, be noted that for multidimensional cases $\nabla F(\mathbf{x}^*) = \mathbf{0}$ may have several solutions for $\mathbf{x}^*$. For these cases, for a given initial value of $\mathbf{x}_k$, the algorithm will converge to the nearest optimum set $\mathbf{x}^*$.

Nonlinear Programming

The nonlinear programming problem is defined as choosing nonnegative values of the independent variables $\mathbf{x}$ so as to maximize or minimize a given function $F(\mathbf{x})$, subject to a given set of inequality constraints. This can be expressed for a maximization problem as

$$\max_{x} F(\mathbf{x}) \text{ subject to } g(\mathbf{x}) \leq \mathbf{b} \text{ and } \mathbf{x} \geq \mathbf{0} \tag{11.32}$$

which can be written in full as

$$\max_{x_1, x_2, \ldots, x_n} F(x_1, x_2, \ldots, x_n) \tag{11.33}$$

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subject to

$$\begin{aligned} g_1(x_1, x_2, \ldots, x_n) &\leq b_1 \\ g_2(x_1, x_2, \ldots, x_n) &\leq b_2 \\ \vdots \quad & \quad \vdots \quad \vdots \\ g_m(x_1, x_2, \ldots, x_n) &\leq b_m \end{aligned} \tag{11.34}$$

and

$$x_1 \geq 0, x_2 \geq 0, \ldots, x_n \geq 0 \tag{11.35}$$

Linear Programming

The linear programming problem is defined as choosing nonnegative values of $\mathbf{x}$ so as to maximize or minimize a given linear function $F(\mathbf{x}) = \mathbf{cx}$, subject to a given set of linear inequality constraints. This can be expressed for a maximization problem as

$$\max_{\mathbf{x}} F(\mathbf{x}) = \mathbf{cx}, \text{ subject to } \mathbf{Ax} \leq \mathbf{b} \text{ and } \mathbf{x} \geq \mathbf{0} \tag{11.36}$$

where $\mathbf{c}$ is a row matrix and $\mathbf{A}$ is a rectangular matrix, both of which contain only constant coefficients. Equation (11.36) can be written fully as

$$\max_{x_1, x_2, \ldots, x_n} F(x_1, x_2, \ldots, x_n) = c_1 x_1 + c_2 +, \ldots, + c_n x_n \tag{11.37}$$

subject to

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 +, \ldots, + a_{1n}x_n &\leq b_1 \\ a_{21}x_1 + a_{22}x_2 +, \ldots, + a_{2n}x_n &\leq b_2 \\ \vdots \quad & \quad \vdots \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 +, \ldots, + a_{mn}x_n &\leq b_m \end{aligned} \tag{11.38}$$

and

$$x_1 \geq 0, \; x_2 \geq 0, \ldots, x_n \geq 0 \tag{11.39}$$

A more general case of linear programming is the case of constraints involving both the inequality and equality constraints. This is expressed as

$$\max_{\mathbf{x}} F(\mathbf{x}) = \mathbf{cx}, \text{ subject to } \mathbf{Ax} \leq \mathbf{b}, \mathbf{Gx} = \mathbf{a} \text{ and } \mathbf{x} \geq \mathbf{0} \tag{11.40}$$

where $\mathbf{G}$ is a rectangular matrix and $\mathbf{a}$ a column matrix, both of which contain only constant coefficients. The additional equality constraints $\mathbf{Gx=a}$ can be written as

$$\begin{aligned} g_{11}x_1 + g_{12}x_2 +, \ldots, + g_{1n}x_n &= a_1 \\ g_{21}x_1 + g_{22}x_2 +, \ldots, + g_{2n}x_n &= a_2 \\ \vdots \quad & \quad \vdots \quad \vdots \\ g_{p1}x_1 + g_{p2}x_2 +, \ldots, + g_{pn}x_n &= a_p \end{aligned} \tag{11.41}$$

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The constraints imposed in a particular problem can lead to (1) unique solution; (2) nonunique solution (multiple solutions); or (3) no solution, representing the three alternative possibilities for a linear programming problem. Until recently, the algorithms used to obtain numerical solutions to linear programming were based on the simplex method developed by Dantzig[3,4] in the late 1940s, and numerous commercial linear programming codes have been available for routine computations. A list of these codes may be found in the IBM Engineering and Scientific Subroutine Library Guide and Reference.[5] Applications of linear programming have become more diverse and now include distribution, transportation, scheduling, finance, and defense problems. Recently, another group of codes has been developed based on the interior point methods,[6] which are computationally more efficient for very large problems. Beyond the linear programming, the main new area for computer codes and applications is nonlinear programming. Only very limited progress has been made so far in this area. The best known code is MINOS, which linearizes the nonlinear parts of the objective function.[7]

11.2 Application of the Lagrange Multiplier Method: A Cluster Bomb[8]

A cluster bomb delivery will be examined to determine the optimum configuration of bomblets for a maximum probability of destroying the target. The pattern of a cluster bomb carrying a number of bomblets delivered from an aircraft can be assumed as an approximation to be a circle. The radius of this pattern W can be altered by adjusting a mechanism on a bomb that controls its angular velocity in free fall. For small angular velocity, the bomblets will be grouped tightly together, while for large angular velocity, the bomblets will spread over a large area. By selecting an appropriate value of W, the probability of a bomblet destroying the target can be maximized for a given size of the target and the bomb characteristics.

Once the bomblets are free from the bomb casing, they may be assumed to disperse according to a uniform probability distribution within the circular pattern. The probability that one of the bomblets destroys a fixed single-point target on the ground is the product of two probabilities: the probability of impact within radius W from the target, P_1, and the probability that the target is destroyed, given that it is within the cluster bomb pattern, P_2. The first probability is given by

$$P_1 = 1 - \exp\left(-\frac{W^2}{2\sigma^2}\right) \tag{11.42}$$

which is obtained from Eq. (3.49) by substituting $r = W$. Observing that from Eq. (3.51)

$$2\sigma^2 = (\text{CEP})^2 / \ell n 2 \tag{11.43}$$

where the circular error probable $\text{CEP} = R_0$, and substituting Eq. (11.43) into Eq. (11.42)

$$P_1 = 1 - \exp\left[-\left(\frac{W}{\text{CEP}}\right)^2 \ell n\, 2\right] \tag{11.44}$$

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The second probability P_2 is obtained as follows. If the effective target area (vulnerable area) is A, then the conditional probability of survival of the target attacked by N bomblets is given by

$$S_2 = \left(1 - \frac{A}{\pi W^2}\right)^N \tag{11.45}$$

Typically, the ratio $A/\pi W^2$ is small so that

$$\left(1 - \frac{A}{\pi W^2}\right)^N \approx \exp\left(-\frac{AN}{\pi W^2}\right) \tag{11.46}$$

which can be demonstrated by expanding both sides of this equation [Compare this with a similar expression used in Eq. (6.5)]. Thus, the conditional probability of destroying the target is given by

$$P_2 = 1 - S_2 = 1 - \exp\left(-\frac{AN}{\pi W^2}\right) \tag{11.47}$$

and the probability of destroying the target is then given by

$$P_K = P_1 \times P_2 = \left(1 - \exp\left[-\left(\frac{W}{\text{CEP}}\right)^2 \ell_n 2\right]\right)\left(1 - \exp\left(-\frac{AN}{\pi W^2}\right)\right) \tag{11.48}$$

The optimum W is the one that maximizes P_K. The direct approach of solving $dP_K/dW = 0$ from Eq. (11.48) leads to a lengthy equation that is not amenable to solution. The Lagrange multiplier method, on the other hand, leads to a surprisingly simple solution. Introducing

$$x_1 = \left(\frac{W}{\text{CEP}}\right)^2 \ell_n 2 \tag{11.49}$$

and

$$x_2 = \frac{AN}{\pi W^2} \tag{11.50}$$

the objective function $F(\mathbf{x})$ becomes

$$F(\mathbf{x}) = (1 - e^{-x_1})(1 - e^{-x_2}) \tag{11.51}$$

Noting from Eqs. (11.49) and (11.50) that for any value of W $x_1 x_2 = constant$, the constraints on x_1 and x_2 at the optimal value of W can be expressed as

$$g(\mathbf{x}^*) = x_1^* x_2^* = b \tag{11.52}$$

Applying, therefore, Eq. (11.14) to this problem, it follows that

$$\left[e^{-x_1^*}\left(1 - e^{-x_2^*}\right) \quad e^{-x_2^*}\left(1 - e^{-x_1^*}\right)\right] = \lambda\left[x_2^* \quad x_1^*\right] \tag{11.53}$$

which leads to

$$e^{-x_1^*}(1 - e^{-x_2^*}) = \lambda x_2^*$$

$$e^{-x_2^*}(1 - e^{-x_1^*}) = \lambda x_1^*$$

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The solution to the above equations is

$$x_1^* = x_2^* \tag{11.54}$$

Substituting now Eqs. (11.49) and (11.50) into Eq. (11.54)

$$\left(\frac{W}{\text{CEP}}\right)^2 \ell_n 2 = \frac{AN}{\pi W^2}$$

and

$$W = \left(\frac{AN(\text{CEP})^2}{\pi \ell_n 2}\right)^{\frac{1}{4}} \tag{11.55}$$

Subsequent substitution of Eq. (11.55) into Eq. (11.48) produces an optimum $P_K(\text{opt})$ given by

$$P_K(\text{opt}) = \left(1 - \exp\left(-\sqrt{\frac{\ell_n 2}{\pi}}\sqrt{\frac{AN}{(\text{CEP})^2}}\right)\right)^2 \tag{11.56}$$

11.3 Examples of Linear Programming

Formulation of linear programming problems will be illustrated for four examples involving allocation of targets or resources in typical military applications. If the number of variables is two, then the problem can be solved graphically. Such solutions are shown for the first three problems. When the number of variables is greater than two, then the solution must be obtained using one of the standard computer programs available for PCs or mainframe computers, such as SIMPLEX method. One example involving three variables is shown in this section.

Example 11.1

A wing of F-16 aircraft, consisting of 72 aircraft, is tasked to accomplish a mission against tanks and Scud missile launchers with the objective of inflicting maximum damage to targets, determined by the highest value of the objective function. Targets have different merit values; 10 units are assigned for destroying a Scud launcher, but only one unit for destroying a tank. The kill rates (target attrition) are:

$a_S = 0.2$ per F-16 sortie against Scud launchers, and
$a_T = 1.0$ per F-16 sortie against tanks

Introducing the two variables

X_S = allocation of F-16 sorties to Scud launchers, and
X_T = allocation of F-16 sorties to tanks,

the objective function F is represented by

$$F_{\max} = 10 \times 0.2X_S + 1.0 \times 1.0X_T = 2X_S + X_T \tag{11.57}$$

There are three operational constraints: (1) at least five Scud launchers must be destroyed; (2) at least 25 tanks must be destroyed; and (3) only one sortie per

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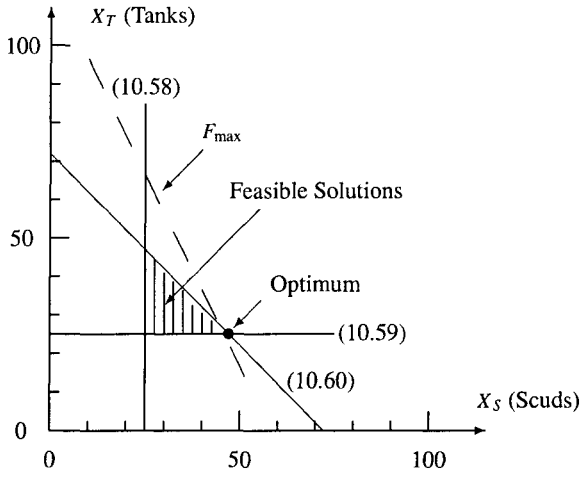


Fig. 11.3 F-16 mission allocations of targets.

aircraft can be used, for a total of 72 aircraft. Mathematically these constraints are expressed as*

$$0.2X_S \geq 5 \tag{11.58}$$

$$1.0X_T \geq 25 \tag{11.59}$$

$$X_S + X_T \leq 72 \tag{11.60}$$

A graphical solution of this problem is represented in Fig. 11.3, where the optimal allocation of the F-16s for this mission is as follows: $X_S = 47$, $X_T = 25$, and the value of the objective function $F_{max} = 119$.

Example 11.2

Example 10.1 will now be modified by introducing an additional constraint that at least as many tank sorties X_T as Scud launcher sorties X_S will be flown, i.e., $X_T \geq X_S$ or $-X_S + X_T \geq 0$. Hence, the mathematical representation for this new problem is

$$F_{max} = 2X_S + X_T \tag{11.61}$$

$$0.2X_S \geq 5 \tag{11.62}$$

$$1.0X_T \geq 25 \tag{11.63}$$

$$X_S + X_T \leq 72 \tag{11.64}$$

$$-X_S + X_T \geq 0 \tag{11.65}$$

*Equations (11.58) and (11.59) can be modified into the standard form of Eq. (11.38) by multiplying by -1 to produce $-0.2X_S \leq -5$ and $-1.0 \leq -25$.

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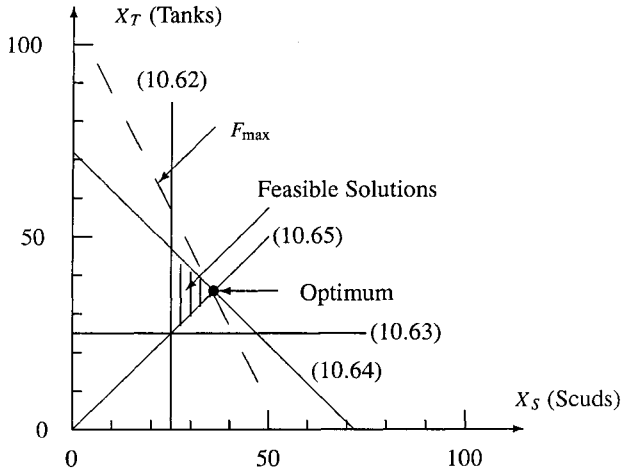


Fig. 11.4 F-16 modified mission allocations of targets.

A graphical solution of this new problem is represented in Fig. 11.4, where the optimal allocation of the F-16s for this modified mission is as follows: $X_S = 36$, $X_T = 36$, and the value of the objective function $F_{max} = 108$.

Example 11.3

It is required to fly as large a tonnage of supplies as possible to a regional conflict area using two types of aircraft:

20 C-141 aircraft, which can carry 30 tons, and
12 C-5A/B aircraft, which can carry 120 tons.

To load a C-141 it takes a crew of four for two hours; to load a C-5A/B it takes a crew of six for two hours. The total loading effort cannot exceed 240 man-hours. It is also assumed that each aircraft can only make one trip and the loading crews are interchangeable. Assigning the variables X_1 and X_2 to the numbers of C-141 and C-5A/B aircraft, respectively, the problem may be stated as follows:

$$F_{max} = 30X_1 + 120X_2 \tag{11.66}$$

$$X_1 \leq 20 \tag{11.67}$$

$$X_2 \leq 12 \tag{11.68}$$

$$8X_1 + 12X_2 \leq 240 \tag{11.69}$$

A graphical solution for this problem is presented in Fig. 11.5, where the optimal allocation of the transport aircraft is as follows: $X_1 = 12$, $X_2 = 12$, and the value of the objective function $F_{max} = 1800$.

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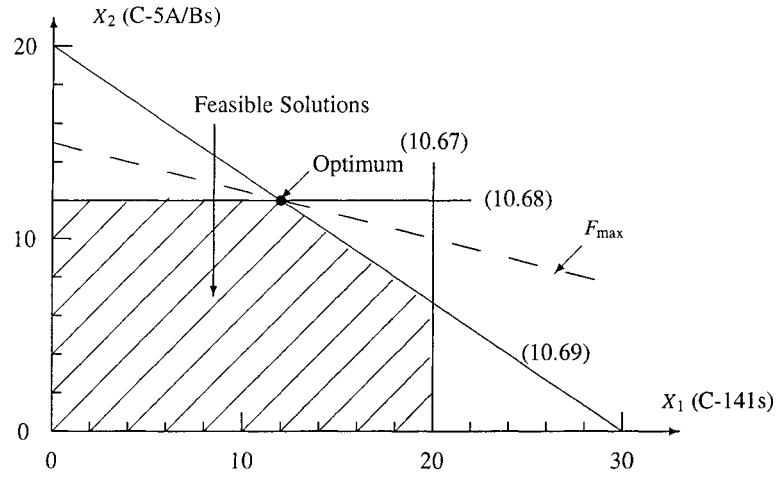


Fig. 11.5 Transportation of supplies by C-141 and C-5A/B aircraft.

Example 11.4

The next example is the transhipment problem. There are two depots D_1 and D_2 in which supplies of the same kind are stored. These supplies can be shipped by trucks from D_1 to the staging points P_1 and P_2, and from D_2 to the staging point P_2, as shown in Fig. 11.6. Thus, the supplies in D_2, because of the terrain difficulties, can only be moved to P_2. The staging points are connected to the depots by a network of roads and the shortest distances d_{ij} between the ith depot and the jth staging point ($i = 1, 2$ and $j = 1, 2$) are:

$$d_{11} = 50 \text{ miles}$$
$$d_{12} = 75 \text{ miles}$$
$$d_{22} = 100 \text{ miles}$$

The demands l_1 and l_2 for the supplies in the staging points are:

$$l_1 = 2000 \text{ units}$$
$$l_2 = 1500 \text{ units}$$

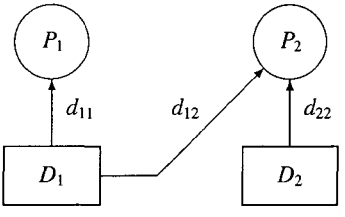


Fig. 11.6 Transhipment of supplies.

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The two depots hold the following quantities of supplies:

$$s_1 = 3000 \text{ units}$$
$$s_2 = 1000 \text{ units}$$

Denoting by X_{ij} the quantity of supply units taken from the ith depot to the jth staging point, the following mathematical inequalities must be satisfied: (1) The total quantity taken from each depot must not be greater than the supply stored in a given depot, i.e.,

$$X_{11} + X_{12} \leq s_1$$
$$X_{22} \leq s_2$$

and (2) The total quantity of supplies delivered to each staging point must not be less than the quantity requested at each point, i.e.,

$$X_{11} \geq l_1$$
$$X_{12} + X_{22} \geq l_2$$

It is required that the total logistics transhipment must be minimal in terms of track-miles expended to accomplish the given task. Assuming that each truck is capable of moving c units of supplies, it follows that X_{ij}/c trucks are needed to transfer X_{ij} supplies from the ith depot the jth staging point. This in turn consumes $d_{ij}X_{ij}/c$ truck-miles. Hence, the total truck-miles (objective function F) are determined from:

$$F_{\min} = \frac{d_{11}}{c}X_{11} + \frac{d_{12}}{c}X_{12} + \frac{d_{22}}{c}X_{22}$$

Assuming that $c = 25$, this linear programming problem is represented by the following set of equations and inequalities:

$$F_{\min} = 2X_{11} + 3X_{12} + 4X_{22} \tag{11.70}$$
$$X_{11} + X_{12} \leq 3000 \tag{11.71}$$
$$X_{22} \leq 1000 \tag{11.72}$$
$$X_{11} \geq 2000 \tag{11.73}$$
$$X_{12} + X_{22} \geq 1500 \tag{11.74}$$

The solution to this problem was obtained using one of the standard linear programming software packages. The results for the optimum solution are as follows: $X_{11} = 2000$, $X_{12} = 1000$, $X_{22} = 500$, and $F_{\min} = 11000$.

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Problems

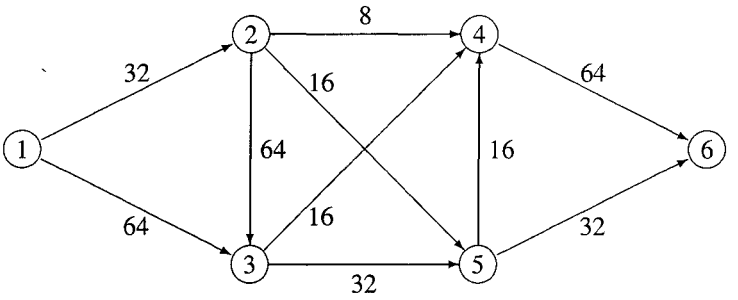
11.1. If in Example 11.1 the first constraint (Eq. (11.45)) is replaced with $X_S = 25$, describe on Fig. 11.1 what will be:

(a) the new domain of feasible solutions, and

(b) the new optimum solution.

11.2. Solve the problem in Example 11.1 by *minimizing* the objective function $F(\mathbf{x})$.

11.3. Messages between nodes 1 and 6 of a communications system can be rerouted through the network shown below:



The nodes 1 and 6 can be interconnected through intermediary nodes 2, 3, 4, and 5, but each communications link between any two nodes can only handle simultaneously a maximum number of messages indicated on the diagram. For example, the number of messages from nodes 1 to 2 cannot exceed 32 at any one time. All messages can be routed only in the directions indicated by the arrows, and the intermediary nodes cannot generate their own messages. Using a standard computer program for linear programming, determine the maximum number of messages transmitted from nodes 1 to 6 at any particular time.

Hint: Use X_{ij} as the number of messages from node i to node j; the objective function will then be $F = X_{12} + X_{13}$.

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11.4. Using the graphical method of solution, shown in Examples 11.1–11.3, solve the linear programming problem for which the objective function $F = -X_1 + X_2$ and the constraints are

$$X_1 - X_2 \leq 2$$
$$X_1 + X_2 \leq 8$$
$$X_1 \geq 0, \quad X_2 \geq 0$$

11.5. Using the graphical method, comment on the results for the case of $F = X_1 + X_2$, subject to the following constraints:

$$-X_1 + X_2 \leq 1$$
$$X_1 + X_2 \leq 2$$
$$X_1 \geq 0, \quad X_2 \geq 0$$

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12
Modeling

This chapter provides a cursory overview of the modeling techniques applied to military operations. For further study of military modeling, Ref. 7 is recommended, where more than 60 models used in the U.S. Department of Defense are discussed. The fundamental concept of the state variable used in modeling is described, and the applications of this concept are presented for an air combat mission and for servicing military equipment. A numerical method of solving the differential equations representing the state variables is derived. This method forms the basis of solutions in many cases of military modeling.

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12.1 Models

A model may be defined as an "analog of reality" that provides an insight into the real world.[1] Models can be classified into one of three descriptive types: iconic models, analog models, or symbolic models. Typical examples of iconic models are photographs, paintings, airplane models, etc. In analog models a physical relationship found in nature is represented by relationships between other physical quantities. Thus, in an electrical analog, voltages may represent other physical quantities; in a slide rule, distances are made proportional to their logarithms; and in a hydraulic analog of the national economy, fluid represents money in the system. In symbolic models, abstract symbols or quantities are used to describe the real world. These models in turn are categorized into descriptive and mathematical models. The descriptive models are words or diagrams, such as those found in a typical organizational chart. An example of such a model is given in Fig. 12.1, which represents model classifications. Mathematical models, on the other hand, represent the "reality" by logical or quantitative relationships. These in turn can be either static or dynamic, and this overall category can be further subdivided into analytical or simulation models.[2] The analytical model determines exact quantitative numbers representing the reality, whereas the simulation model is, in general, a series of "models that may be used to converge on solutions to very complex problems involving uncertainty (probabilistic occurrence) and risk."[3] Simulation models can be further subdivided into deterministic and stochastic (probabilistic) models. Examples of the deterministic models are Lanchester equations for combat analysis, whereas stochastic models are found in simulations of theater operations where random number generators (representing random events) are used to determine the outcomes.

In simulating reality, the military models cover the whole spectrum of degree of abstraction, from the complete mathematical description of the process and events to the complete involvement of human operators and decision-makers found in war games or military exercises. The examples representing this spectrum consist of

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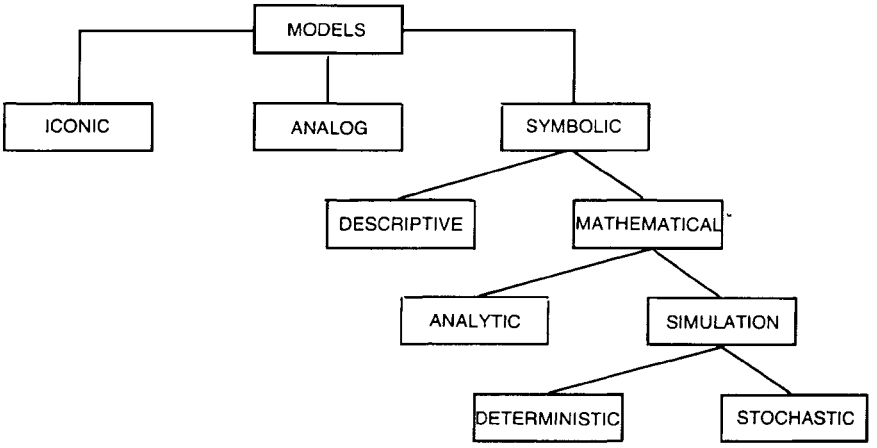


Fig. 12.1 Model classification.

the following: mathematical analyses, computer simulation, interactive computer games, computer-assisted war games, manual war games, and military exercises and experiments. These models represent different degrees of operational realism, abstraction, level of outcome reproducibility, and convenience and flexibility[4] as indicated in Fig. 12.2.

To be realistic, combat models must consider the physical process involved in combat, which is determined by weapon effects, human behavior, and decision-making (command and control). Weapon effects are typically described by the

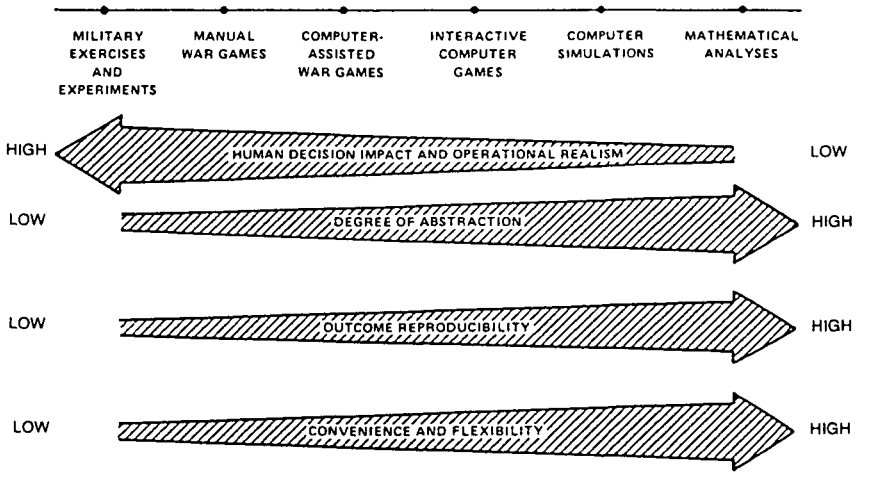


Fig. 12.2 Model spectrum and characteristic trends.[4] (Reproduced with the permission of the Military Operations Research Society, Alexandria, VA.)

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attrition rates that profoundly affect the results. Realistic estimates of these rates must conform with the specified real world to be of any value in defense decisions. Attrition rates are better understood, and received greater attention in combat modeling in the past. Human behavior and command decision, the other two equally important aspects of combat modeling, are much more difficult to model and only recently are receiving more attention. In particular, modeling of human behavior makes it very difficult to assess the outcome results. This difficulty was perhaps best described by Bonder[5] in his assessment of the value of war-gaming:

> A war game is a model which is, in a sense, a step removed from the reality of a field exercise wherein only teams of players representing the commanding officers and their staffs are included. Since decisions are made by humans, it is not unreasonable to expect a high output variance if different decision makers were used; however, the long operation time usually precludes more than one realization of the process. It is my personal view that this type of model is not a feasible mechanism for analyzing a broad spectrum of system alternatives in a responsive manner to meet planning cycle requirements. Experience has shown, however, that they are diagnostic in the sense that they reveal problems that need to be solved with future systems, and are viable mechanisms for training decision makers.

As already pointed out in Chapter 1, simulated warfare provides a means of gaining experience, exploring the consequences of alternative strategies and tactics, identifying weaknesses, and improving skills without the necessity of acting out the real situation in the real world. Gaming exercises stimulate the research for novel and more effective strategies and tactics and encourage innovation. But, more importantly, motivations aroused in war-gaming in peacetime may have a carryover value that will pay off handsomely in the ultimate test of actual war.

In exercising war games, as pointed out by Weiner,[6] the golden rule must be that "the validity of a war game does not ordinarily lie in the accuracy with which the mathematical computations or arithmetic are done, but rather in the extent to which the sides can be faithfully represented and the rules be designed to bear some relationship to real operations."

Because the main purpose here is to present the various mathematical methods used in defense analyses, any detailed description of the mechanics of modeling, as well as its state of the art, would be beyond the scope of this text. For further information the reader is referred to an excellent text edited by Battilega and Grange.[7] Only the basic mathematical concepts used in modeling of military operations will be introduced in this chapter.

12.2 Modeling of Military Operations

Simulation models are used to provide a detailed mathematical representation of the actual sequence of physical events that occur during a military operation. Thus, they are appropriate for a time-varying process in which the various interrelationships between the variables in the process can be established. Many of these

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relationships are known with certainty, whereas others are probabilistic in nature. In the latter case Monte Carlo techniques are used to obtain a probability distribution on the outcomes. The models may contain many variables (in fact some of the currently used models contain thousands of variables); therefore, computer times per scenario are relatively long, particularly if a large number of Monte Carlo replications is needed. If all relationships are deterministic, then a single set of time-varying outcomes is obtained; otherwise, the outcomes are obtained as probability distributions or probabilities of occurrences.

The fundamental concept in modeling is the state variable, which describes a particular state in the process, such as a battle engagement or logistics support of the front line. In general, these variables describe the level of resources at time t. Such a concept was already introduced in Chapter 5 for the continuous combat probability $P(m, n, t)$, which specified the probability of the battle state with the Blue force equal to m and the Red force equal to n at time t. The state variables are linked together by specified flow rates, which will be illustrated for two simple examples.

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Combat Mission

The state variable system in Fig. 12.3 represents six possible states for an aircraft or group of aircraft on a combat mission. The six states are 1) aircraft on air base, undamaged; 2) aircraft undamaged on combat mission; 3) aircraft shot down; 4) aircraft damaged, must return to base; 5) aircraft on base, damaged; and 6) aircraft being repaired. The arrows in Fig. 12.3 represent the direction of transition (flow) from one state into another. With the exception of states 1 and 2, where the transition may occur in both directions, all transitions occur only in one direction. For example, an aircraft on a combat mission can be shot down (it would then transition from state 2 to state 3); however, it is not possible for a downed aircraft to transition into the combat mission (from state 3 into state 2).

Denoting the ith state variable as N_i and the flow rates between i and j as $\dot{X}_{ij}$, the equations connecting the six state variables in the preceding example consist

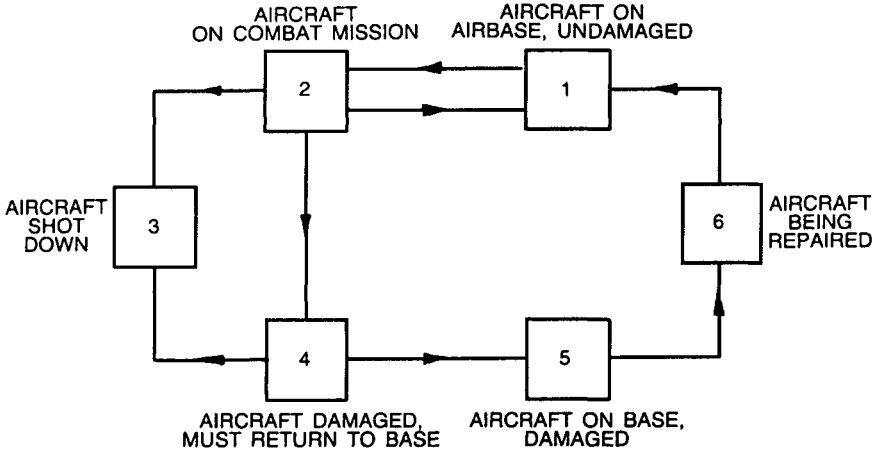


Fig. 12.3 Combat mission model.

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of the following:

$$\begin{aligned}
\dot{N}_1 &= \dot{X}_{21} - \dot{X}_{12} + \dot{X}_{61} \\
\dot{N}_2 &= \dot{X}_{12} - \dot{X}_{21} - \dot{X}_{23} - \dot{X}_{24} \\
\dot{N}_3 &= \dot{X}_{23} + \dot{X}_{43} \\
\dot{N}_4 &= \dot{X}_{24} - \dot{X}_{43} - \dot{X}_{45} \\
\dot{N}_5 &= \dot{X}_{45} - \dot{X}_{56} \\
\dot{N}_6 &= \dot{X}_{56} - \dot{X}_{61}
\end{aligned} \tag{12.1}$$

where overdots over N_i represent differentiation with respect to time. All state variables naturally must be nonnegative; i.e., they must satisfy the relationship

$$N_i \geq 0, \qquad i = 1, 2, \ldots, 6 \tag{12.2}$$

If the initial values of N_i at $t = 0$ and the flow rates $\dot{X}_{ij}$ are known, then the preceding equations for N_i can be solved numerically. This gives the values of N_i as functions of time.

Equipment

Another simple example is shown in Fig. 12.4. Here the flow between states 1 and 3 can occur both ways because there is a possibility of reinstating a piece of equipment, i.e., returning it from the state of being repaired to the state of being serviceable. The appropriate equations for these states are

$$\begin{aligned}
\dot{N}_1 &= \dot{X}_{31} - \dot{X}_{13} - \dot{X}_{12} \\
\dot{N}_2 &= \dot{X}_{12} \\
\dot{N}_3 &= \dot{X}_{13} - \dot{X}_{31} \\
N_i &\geq 0, \qquad i = 1, 2, 3
\end{aligned} \tag{12.3}$$

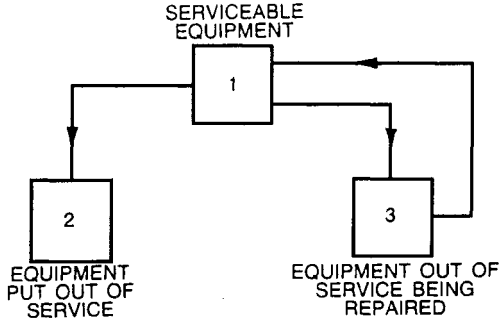


Fig. 12.4 Model of equipment servicing process.

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General Models

The preceding two simple examples illustrate the overall concept of modeling by means of state variables and their interrelationships. It should be noted that a pair of states may be joined by a directional flow path, which indicates the flow of resources from one state to another. In general the resources in each state may have different dimensions, and not all states are linked together by the flow paths. For example, some states may refer to F-16 aircraft, whereas others may be tanks or trucks. Some directional flows are irreversible; e.g., an aircraft being shot down is an irreversible event. The flow of resources between the states is controlled by flow rates across the flow paths depicting the physical situation. Thus, the key to success in obtaining meaningful solutions is the modeling of the flow rates. The simplest form of modeling of the flow rate is represented by the equation

$$\dot{X}(t) = -\frac{N(t)}{T} \tag{12.4}$$

where

$\dot{X}(t)$ = instantaneous flow rate of resources out of a state at time t
$N(t)$ = level of resources in the state at time t
T = mean time a resource element spends in the state

Equation (12.4) states that the flow rate is proportional to the current resource level in the state. If the transition occurs between the ith and jth states, then Eq. (12.4) can be written as

$$\dot{X}_{ij}(t) = -\frac{N_i(t)}{T_i} \tag{12.5}$$

For any individual state N_i there will be a flow of resources out of and into the state as depicted in Fig. 12.5. Thus, for the general case,

$$\dot{N}_i(t) = \sum_k \dot{X}_{ki}(t) - \sum_j \dot{X}_{ij}(t) \tag{12.6}$$

where

$\dot{N}_i(t)$ = rate at which state i is changing at time t
$\sum_k \dot{X}_{ki}(t)$ = flow rates into state i at time t
$\sum_j \dot{X}_{ij}(t)$ = flow rates out from state i at time t

The first summation over k is carried out over all of the states from which resources flow into state i, and the second summation over j is over all of the states into which resources flow from state i. Noting that, for the time increment Δt,

$$\dot{N}_i(t) \cong \frac{N_i(t + \Delta t) - N_i(t)}{\Delta t}$$

Hence,

$$\begin{aligned} N_i(t + \Delta t) &\cong N_i(t) + \dot{N}_i(t)\,\Delta t \\ &= N_i(t) + \left(\sum_k \dot{X}_{ki}(t) - \sum_j \dot{X}_{ij}(t)\right)\Delta t \end{aligned} \tag{12.7}$$

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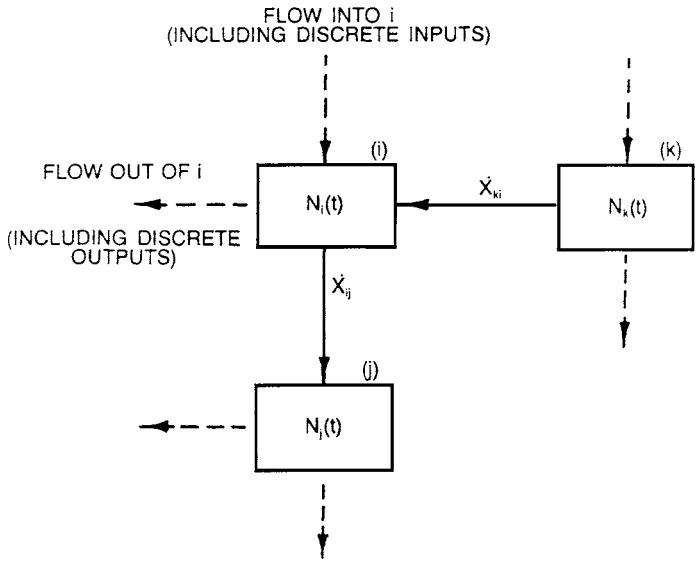


Fig. 12.5 Flow of resources into and out of state i.

In a more general case there will also be a discrete input into N_i from other states or a discrete output of some of the N_i resources into other states in time Δt as indicated in Fig. 12.5. For such cases Eq. (12.7) is modified as

$$N_i(t+\Delta t) = N_i(t) + \left(\sum_k \dot{X}_{ki}(t) - \sum_j \dot{X}_{ij}(t)\right)\Delta t + \sum_k \Delta N_{ki}(t) - \sum_j \Delta N_{ij}(t) \tag{12.8}$$

where

$\Delta N_{ki}(t)$ = discrete allocation of some of the N_k resources into N_i during time Δt

$\Delta N_{ij}(t)$ = discrete allocation of some of the N_i resources into N_j during time Δt

As a more complex example of military modeling, Fig. 12.6 represents an air base flow diagram in which some 15 states are used. The two additional states, "departing aircraft" (state 0) and "returning aircraft" (state 16), can be linked to another model representing the actual combat mission similar to the model diagram in Fig. 12.3. Here each state is connected to several other states. For example, the inflow of resources into state 5 occurs from states 4, 6, 7, 8, 9, 10, and 11, whereas the outflow occurs into states 2 and 3.

The mean time T used in Eq. 12.5 may, in some cases, be a function of time. The unscheduled maintenance flow rate on an air base in Fig. 12.6 may be

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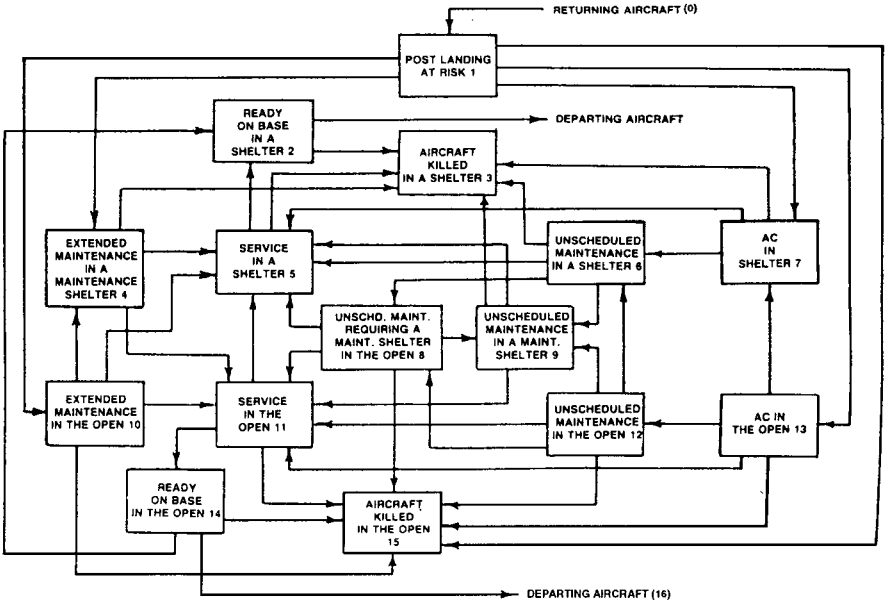


Fig. 12.6 Air base flow model.

represented by[8]

$$T = KR^{c_1}t^{c_2}S^{c_3} \tag{12.9}$$

in which

K, c_1, c_2, and c_3 = constants
R = ratio of spare parts to aircraft remaining
t = time elapsed since start of flying aircraft
S = the sorties per day per primary assigned aircraft

Reference 7 describes some 64 different models used by the U.S. Department of Defense. Some of these models require a very large number of state variables to describe the physical situation. In fact, models using thousands of variables are not uncommon for which the solutions are obtained typically in one-minute time intervals. However, a detailed discussion of these models would be beyond the scope of the present text.

The state variables in military models are treated as continuous, which is a reasonable approximation when there are large numbers of elements in most of the states. For such cases fractional aircraft or SAMs are not important, and in most modeling programs there is a subroutine included that periodically rounds off the results to integer values. Thus, the modeling concept presented here is indeed appropriate for use in a dense environment with a large number of elements. For cases where the numbers of elements are small, or if only certain vital elements are few in number, this approach may not be appropriate. For example, in the case of defense using AWAC aircraft, where there will be only a very

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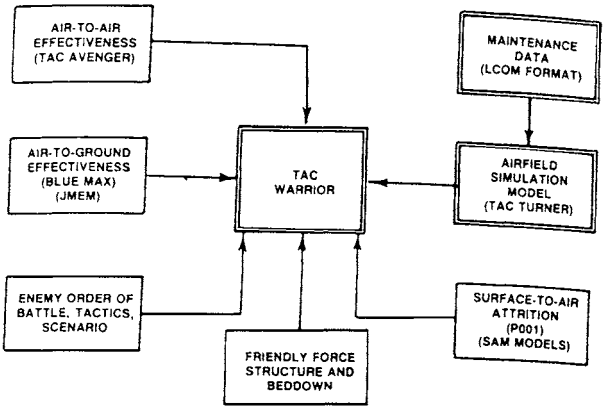


Fig. 12.7 Hierarchical model (TAC WARRIOR).

few such aircraft in the air on each side at any given point in time, a loss of one AWAC may have a tremendous, or maybe even decisive, effect on the outcome of the air battle, unlike the killing of a single combat aircraft or a SAM launcher when their numbers are large. Thus, caution must always be exercised to ensure that the modeling is appropriately formulated for the particular military operation. The analyst must always be aware of the assumptions and validity of the ranges for which the model employed may be expected to provide meaningful results.

Figure 12.7 illustrates a typical hierarchical model, TAC WARRIOR, which depends on a set of lower-level models for many of its inputs. This model is designed to evaluate tactical air weapon systems in an air battle and to examine the variations in effectiveness resulting from different force structures or different tactics and scenarios. The model is essentially a deterministic, differential equation model that treats uncertainties by setting the parameters of probability density functions and then computing the expected values of the required random variables. The TAC WARRIOR model relies on the input from the following submodels: air-to-air effectiveness (TAC AVENGER); air-to-ground effectiveness (BLUE MAX, JMEM); enemy order of battle, tactics, and scenarios; maintenance data (LCOM FORMAT); airfield simulation model (TAC TURNER, Fig. 12.6); surface-to-air attrition (POO1, GREAT SAM, BIGSAM, BOLD SAM); and friendly force structure and beddown.

The TAC WARRIOR model itself consists of a series of loops and states. A typical loop is shown in Fig. 12.8 representing a closed-loop state variable system in which a particular resource can exist (in this case an aircraft) throughout the course of the battle. The loop becomes broken when an aircraft is engaged and shot down. The model accommodates 34 aircraft loops and has more than 4000 differential equations describing the interaction between the state variables within the model as well as the supporting models (e.g., the airfield simulation model).

It should be noted that the TAC WARRIOR model, including its supporting models, is a three-level model, because it requires inputs from a level 3 model

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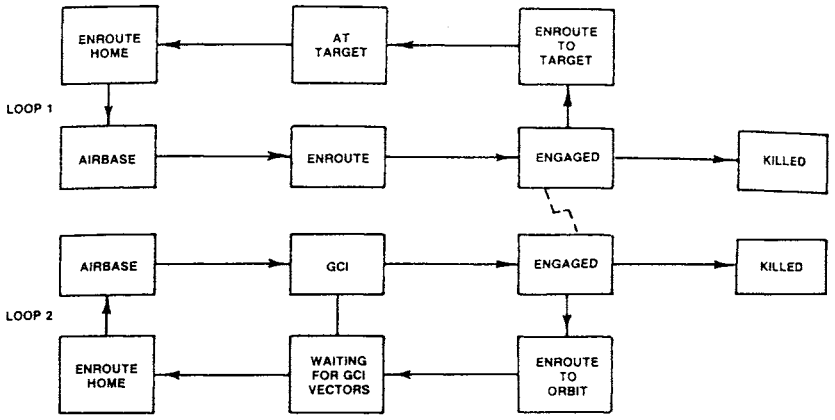


Fig. 12.8 TAC WARRIOR loops.

(LCOM FORMAT) feeding information into a level 2 model (TAC TURNER) and then finally into the level 1 model (TAC WARRIOR) (see Fig.12.8). All other models are level 2 models.

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Appendix A
Probability Tables

A.1 Cumulative Binomial Probabilities

Table A.1 gives the probability P_n of obtaining *at least* $r = r_{\min}$ or more hits on a target for n independent shots when the single-hit probability is p. This probability is obtained from the formula

$$P_n(r \geq r_{\min}) = \sum_{r=r_{\min}}^{n} \binom{n}{r} p^r (1-p)^{n-r} \tag{3.5}$$

Where there is no entry for a particular pair of values of r and p, the corresponding probability is less than 0.00005. Similarly, a tabulated value of 1.0000 represents a probability greater than 0.99995, except for the case of $r \geq 0$ where the entry is exact. As a reasonable approximation to cumulative binomial probabilities, 1) use the Poisson approximation if $p < 0.1$ by substituting $\lambda = np$, and 2) use the Gaussian approximation* if $0.1 \leq p \leq 0.9$, $np > 5$, and $n(1-p) > 5$ by substituting $\bar{x} = np$, $\sigma = [np(1-p)]^{\frac{1}{2}}$ and $x = r$.

A.2 Cumulative Poisson Probabilities

Table A.2 gives the probability that at least a specified number of hits on a target $r = r_{\min}$ will occur in the time interval when the average number of such hits per interval is λ. This probability is obtained from

$$P(r \geq r_{\min}) = \sum_{r=r_{\min}}^{\infty} \frac{\lambda^r}{r!} e^{-\lambda} = 1 - \sum_{r=0}^{r_{\min}-1} \frac{\lambda^r}{r!} e^{-\lambda} \tag{3.28}$$

Where there is no entry for a particular pair of values of r and λ, the corresponding probability is less than 0.00005. Similarly, a tabulated value of 1.0000 represents a probability greater than 0.99995, except for the case of $r = 0$, where the entry is exact. When $\lambda > 15$ the normal (Gaussian) probability may be used by substituting $\bar{x} = \lambda$ and $\sigma = \sqrt{\lambda}$.

*Accuracy of this approximation can be improved by using a correction for continuity (see, for example, Snedecor, G.W., and Cochran, W.G., *Statistical Methods*, 7th ed., Iowa State Univ. Press, Ames, IA, 1980, pp. 117–119. The correction for continuity requires simply a subtraction of 0.5 from r, i.e., the corrected r is equal to $r - 0.5$.)

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Table A.1 Cumulative binomial probabilities

$p =$		0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
$n = 2$											
$r \geq$	0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	1	0.0975	0.1900	0.2775	0.3600	0.4375	0.5100	0.5775	0.6400	0.6975	0.7500
	2	0.0025	0.0100	0.0225	0.0400	0.0625	0.0900	0.1225	0.1600	0.2025	0.2500
$n = 4$											
$r \geq$	0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	1	0.1855	0.3439	0.4780	0.5904	0.6836	0.7599	0.8215	0.8704	0.9085	0.9375
	2	0.0140	0.0523	0.1095	0.1808	0.2617	0.3483	0.4370	0.5248	0.6090	0.6875
	3	0.0005	0.0037	0.0120	0.0272	0.0508	0.0837	0.1265	0.1792	0.2415	0.3125
	4		0.0001	0.0005	0.0016	0.0039	0.0081	0.0150	0.0256	0.0410	0.0625
$n = 6$											
$r \geq$	0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	1	0.2649	0.4686	0.6229	0.7379	0.8220	0.8824	0.9246	0.9533	0.9723	0.9844
	2	0.0328	0.1143	0.2235	0.3446	0.4661	0.5798	0.6809	0.7667	0.8364	0.8906
	3	0.0022	0.0158	0.0473	0.0989	0.1694	0.2557	0.3529	0.4557	0.5585	0.6563
	4	0.0001	0.0013	0.0059	0.0170	0.0376	0.0705	0.1174	0.1792	0.2553	0.3438
	5		0.0001	0.0004	0.0016	0.0046	0.0109	0.0223	0.0410	0.0692	0.1094
	6				0.0001	0.0002	0.0007	0.0018	0.0041	0.0083	0.0156
$n = 8$											
$r \geq$	0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	1	0.3366	0.5695	0.7275	0.8322	0.8999	0.9424	0.9681	0.9832	0.9916	0.9961
	2	0.0572	0.1869	0.3428	0.4967	0.6329	0.7447	0.8309	0.8936	0.9368	0.9648
	3	0.0058	0.0381	0.1052	0.2031	0.3215	0.4482	0.5722	0.6846	0.7799	0.8555
	4	0.0004	0.0050	0.0214	0.0563	0.1138	0.1941	0.2936	0.4059	0.5230	0.6367
	5		0.0004	0.0029	0.0104	0.0273	0.0580	0.1061	0.1737	0.2604	0.3633
	6			0.0002	0.0012	0.0042	0.0113	0.0253	0.0498	0.0885	0.1445
	7				0.0001	0.0004	0.0013	0.0036	0.0085	0.0181	0.0352
	8						0.0001	0.0002	0.0007	0.0017	0.0039
$n = 10$											
$r \geq$	0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	1	0.4013	0.6513	0.8031	0.8926	0.9437	0.9718	0.9865	0.9940	0.9975	0.9990
	2	0.0861	0.2639	0.4557	0.6242	0.7560	0.8507	0.9140	0.9536	0.9767	0.9893
	3	0.0115	0.0702	0.1798	0.3222	0.4744	0.6172	0.7384	0.8327	0.9004	0.9453
	4	0.0010	0.0128	0.0500	0.1209	0.2241	0.3504	0.4862	0.6177	0.7340	0.8281
	5	0.0001	0.0016	0.0099	0.0328	0.0781	0.1503	0.2485	0.3669	0.4956	0.6230
	6		0.0001	0.0014	0.0064	0.0197	0.0473	0.0949	0.1662	0.2616	0.3770
	7			0.0001	0.0009	0.0035	0.0106	0.0260	0.0548	0.1020	0.1719
	8				0.0001	0.0004	0.0016	0.0048	0.0123	0.0274	0.0547
	9						0.0001	0.0005	0.0017	0.0045	0.0107
	10								0.0001	0.0003	0.0010

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A.3 The Normal (Gaussian) Probabilities

Table A.3 gives the area in the tail end of the normal (Gaussian) probability distribution calculated from

$$P(u_1 < u < \infty) = \frac{1}{\sqrt{2\pi}} \int_{u_1}^{\infty} e^{-\frac{1}{2}u^2} du \tag{3.30b}$$

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Table A.2 Cumulative Poisson probabilities

$\lambda =$	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.00
$r =$										
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	0.0952	0.1813	0.2592	0.3297	0.3935	0.4512	0.5034	0.5507	0.5934	0.6321
2	0.0047	0.0175	0.0369	0.0616	0.0902	0.1219	0.1558	0.1912	0.2275	0.2642
3	0.0002	0.0011	0.0036	0.0079	0.0144	0.0231	0.0341	0.0474	0.0629	0.0803
4		0.0001	0.0003	0.0008	0.0018	0.0034	0.0058	0.0091	0.0135	0.0190
5				0.0001	0.0002	0.0004	0.0008	0.0014	0.0023	0.0037
6							0.0001	0.0002	0.0003	0.0006
7										0.0001
$\lambda =$	1.10	1.20	1.30	1.40	1.50	1.60	1.70	1.80	1.90	2.00
$r =$										
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	0.6671	0.6988	0.7275	0.7534	0.7769	0.7981	0.8173	0.8347	0.8504	0.8647
2	0.3010	0.3374	0.3732	0.4082	0.4422	0.4751	0.5068	0.5372	0.5663	0.5940
3	0.0996	0.1205	0.1429	0.1665	0.1912	0.2166	0.2428	0.2694	0.2963	0.3233
4	0.0257	0.0338	0.0431	0.0537	0.0656	0.0788	0.0932	0.1087	0.1253	0.1429
5	0.0054	0.0077	0.0107	0.0143	0.0186	0.0237	0.0296	0.0364	0.0441	0.0527
6	0.0010	0.0015	0.0022	0.0032	0.0045	0.0060	0.0080	0.0104	0.0132	0.0166
7	0.0001	0.0003	0.0004	0.0006	0.0009	0.0013	0.0019	0.0026	0.0034	0.0045
8			0.0001	0.0001	0.0002	0.0003	0.0004	0.0006	0.0008	0.0011
9							0.0001	0.0001	0.0002	0.0002
$\lambda =$	2.10	2.20	2.30	2.40	2.50	2.60	2.70	2.80	2.90	3.00
$r =$										
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	0.8775	0.8892	0.8997	0.9093	0.9179	0.9257	0.9328	0.9392	0.9450	0.9502
2	0.6204	0.6454	0.6691	0.6916	0.7127	0.7326	0.7513	0.7689	0.7854	0.8009
3	0.3504	0.3773	0.4040	0.4303	0.4562	0.4816	0.5064	0.5305	0.5540	0.5768
4	0.1614	0.1806	0.2007	0.2213	0.2424	0.2640	0.2859	0.3081	0.3304	0.3528
5	0.0621	0.0725	0.0838	0.0959	0.1088	0.1226	0.1371	0.1523	0.1682	0.1847
6	0.0204	0.0249	0.0300	0.0357	0.0420	0.0490	0.0567	0.0651	0.0742	0.0839
7	0.0059	0.0075	0.0094	0.0116	0.0142	0.0172	0.0206	0.0244	0.0287	0.0335
8	0.0015	0.0020	0.0026	0.0033	0.0042	0.0053	0.0066	0.0081	0.0099	0.0119
9	0.0003	0.0005	0.0006	0.0009	0.0011	0.0015	0.0019	0.0024	0.0031	0.0038
10	0.0001	0.0001	0.0001	0.0002	0.0003	0.0004	0.0005	0.0007	0.0009	0.0011
11					0.0001	0.0001	0.0001	0.0002	0.0002	0.0003
12									0.0001	0.0001
$\lambda =$	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90	4.00
$r =$										
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	0.9550	0.9592	0.9631	0.9666	0.9698	0.9727	0.9753	0.9776	0.9798	0.9817
2	0.8153	0.8288	0.8414	0.8532	0.8641	0.8743	0.8838	0.8926	0.9008	0.9084
3	0.5988	0.6201	0.6406	0.6603	0.6792	0.6973	0.7146	0.7311	0.7469	0.7619
4	0.3752	0.3975	0.4197	0.4416	0.4634	0.4848	0.5058	0.5265	0.5467	0.5665
5	0.2018	0.2194	0.2374	0.2558	0.2746	0.2936	0.3128	0.3322	0.3516	0.3712
6	0.0943	0.1054	0.1171	0.1295	0.1424	0.1559	0.1699	0.1844	0.1994	0.2149
7	0.0388	0.0446	0.0510	0.0579	0.0653	0.0733	0.0818	0.0909	0.1005	0.1107
8	0.0142	0.0168	0.0198	0.0231	0.0267	0.0308	0.0352	0.0401	0.0454	0.0511
9	0.0047	0.0057	0.0069	0.0083	0.0099	0.0117	0.0137	0.0160	0.0185	0.0214
10	0.0014	0.0018	0.0022	0.0027	0.0033	0.0040	0.0048	0.0058	0.0069	0.0081
11	0.0004	0.0005	0.0006	0.0008	0.0010	0.0013	0.0016	0.0019	0.0023	0.0028
12	0.0001	0.0001	0.0002	0.0002	0.0003	0.0004	0.0005	0.0006	0.0007	0.0009
13				0.0001	0.0001	0.0001	0.0001	0.0002	0.0002	0.0003
14									0.0001	0.0001

(*Cont.*)

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Table A.2 (Continued)

$\lambda =$	4.10	4.20	4.30	4.40	4.50	4.60	4.70	4.80	4.90	5.00
$r =$										
0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
1	0.9834	0.9850	0.9864	0.9877	0.9889	0.9899	0.9909	0.9917	0.9925	0.9932
2	0.9155	0.9220	0.9281	0.9337	0.9389	0.9437	0.9481	0.9522	0.9560	0.9595
3	0.7762	0.7898	0.8026	0.8148	0.8264	0.8373	0.8477	0.8574	0.8666	0.8753
4	0.5858	0.6046	0.6228	0.6405	0.6577	0.6743	0.6903	0.7057	0.7206	0.7349
5	0.3907	0.4102	0.4296	0.4488	0.4679	0.4867	0.5054	0.5237	0.5417	0.5594
6	0.2307	0.2468	0.2633	0.2801	0.2970	0.3142	0.3315	0.3490	0.3664	0.3840
7	0.1213	0.1325	0.1442	0.1563	0.1689	0.1819	0.1954	0.2092	0.2233	0.2377
8	0.0573	0.0639	0.0710	0.0786	0.0866	0.0950	0.1039	0.1133	0.1231	0.1333
9	0.0245	0.0279	0.0317	0.0358	0.0402	0.0450	0.0502	0.0558	0.0617	0.0680
10	0.0095	0.0111	0.0129	0.0149	0.0171	0.0195	0.0222	0.0251	0.0283	0.0318
11	0.0034	0.0041	0.0048	0.0057	0.0066	0.0078	0.0090	0.0104	0.0119	0.0136
12	0.0011	0.0014	0.0017	0.0020	0.0024	0.0028	0.0034	0.0039	0.0046	0.0054
13	0.0003	0.0004	0.0005	0.0006	0.0008	0.0010	0.0011	0.0014	0.0016	0.0019
14	0.0001	0.0001	0.0001	0.0002	0.0002	0.0003	0.0004	0.0004	0.0005	0.0006
15				0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002

where

$$u = \frac{x - \bar{x}}{\sigma} \tag{3.32a}$$

For negative values of u_1 the probability is calculated from

$$P(-u_1 < u < \infty) = 1 - P(u_1 < u < \infty) \tag{A.1}$$

A.4 Error Function erf(z)

In calculating probabilities it is often necessary to compute integrals of the form

$$P = \frac{1}{\sqrt{2\pi}} \int_{-u}^{u} e^{-\frac{1}{2}u^2}\, du \tag{A.2}$$

where $u = x/\sigma$. Substituting

$$z = \frac{u}{\sqrt{2}} = \frac{x}{\sqrt{2}\,\sigma} \tag{A.3}$$

the preceding integral becomes

$$P = \frac{2}{\sqrt{\pi}} \int_{0}^{z} e^{-z^2}\, dz = \text{erf(z)} = \text{erf}\frac{\text{x}}{\sqrt{2}\,\sigma} \tag{3.92}$$

Table A.4 shows the error function erf(z) computed for a range of values of z from 0 to 3.0 at 0.01 intervals.

A.5 Single-Shot Probability of Hit on a Rectangular Target

Table A.5 can be used to determine the single-shot probability of hit on a rectangular target by missiles having a normal distribution with specified range error

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Table A.3 The normal (Gaussian) probabilities

u	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.10	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.20	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.30	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.40	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.50	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.60	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.70	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.80	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.90	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.00	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.10	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.20	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.30	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.40	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.50	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.60	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.70	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.80	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.90	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.00	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.10	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.20	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.30	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.40	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.50	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.60	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.70	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.80	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.90	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.00	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010

probable REP $= 0.6745\sigma_x$ and cross-range error probable CREP $= 0.6745\sigma_y$, where σ_x and σ_y are the standard deviations in the range and cross-range directions. The probability is obtained from

$$P = P_{\text{range}} \times P_{\text{cross range}} \tag{3.38}$$

where P_{range} and $P_{\text{cross range}}$ are obtained from Table A.5 in which

$$z = \tfrac{1}{2} \times \text{target range dimension (or target cross-range dimension)/REP (or CREP)} \tag{A.4}$$

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Table A.4 Error function erf(z)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.0000	0.0113	0.0226	0.0338	0.0451	0.0564	0.0676	0.0789	0.0901	0.1013
0.10	0.1125	0.1236	0.1348	0.1459	0.1569	0.1680	0.1790	0.1900	0.2009	0.2118
0.20	0.2227	0.2335	0.2443	0.2550	0.2657	0.2763	0.2869	0.2974	0.3079	0.3183
0.30	0.3286	0.3389	0.3491	0.3593	0.3694	0.3794	0.3893	0.3992	0.4090	0.4187
0.40	0.4284	0.4380	0.4475	0.4569	0.4662	0.4755	0.4847	0.4937	0.5027	0.5117
0.50	0.5205	0.5292	0.5379	0.5465	0.5549	0.5633	0.5716	0.5798	0.5879	0.5959
0.60	0.6039	0.6117	0.6194	0.6270	0.6346	0.6420	0.6494	0.6566	0.6638	0.6708
0.70	0.6778	0.6847	0.6914	0.6981	0.7047	0.7112	0.7175	0.7238	0.7300	0.7361
0.80	0.7421	0.7480	0.7538	0.7595	0.7651	0.7707	0.7761	0.7814	0.7867	0.7918
0.90	0.7969	0.8019	0.8068	0.8116	0.8163	0.8209	0.8254	0.8299	0.8342	0.8385
1.00	0.8427	0.8468	0.8508	0.8548	0.8586	0.8624	0.8661	0.8698	0.8733	0.8768
1.10	0.8802	0.8835	0.8868	0.8900	0.8931	0.8961	0.8991	0.9020	0.9048	0.9076
1.20	0.9103	0.9130	0.9155	0.9181	0.9205	0.9229	0.9252	0.9275	0.9297	0.9319
1.30	0.9340	0.9361	0.9381	0.9400	0.9419	0.9438	0.9456	0.9473	0.9490	0.9507
1.40	0.9523	0.9539	0.9554	0.9569	0.9583	0.9597	0.9611	0.9624	0.9637	0.9649
1.50	0.9661	0.9673	0.9684	0.9695	0.9706	0.9716	0.9726	0.9736	0.9745	0.9755
1.60	0.9763	0.9772	0.9780	0.9788	0.9796	0.9804	0.9811	0.9818	0.9825	0.9832
1.70	0.9838	0.9844	0.9850	0.9856	0.9861	0.9867	0.9872	0.9877	0.9882	0.9886
1.80	0.9891	0.9895	0.9899	0.9903	0.9907	0.9911	0.9915	0.9918	0.9922	0.9925
1.90	0.9928	0.9931	0.9934	0.9937	0.9939	0.9942	0.9944	0.9947	0.9949	0.9951
2.00	0.9953	0.9955	0.9957	0.9959	0.9961	0.9963	0.9964	0.9966	0.9967	0.9969
2.10	0.9970	0.9972	0.9973	0.9974	0.9975	0.9976	0.9977	0.9979	0.9980	0.9980
2.20	0.9981	0.9982	0.9983	0.9984	0.9985	0.9985	0.9986	0.9987	0.9987	0.9988
2.30	0.9989	0.9989	0.9990	0.9990	0.9991	0.9991	0.9992	0.9992	0.9992	0.9993
2.40	0.9993	0.9993	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995	0.9995	0.9996
2.50	0.9996	0.9996	0.9996	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
2.60	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9999	0.9999
2.70	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
2.80	0.9999	0.9999	0.9999	0.9999	1.0000	0.9999	1.0000	1.0000	1.0000	1.0000
2.90	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

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Table A.5 Single-shot component probability P_{range} or $P_{\text{cross-range}}$ of hit on a rectangular target

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.00	0.0000	0.0054	0.0108	0.0161	0.0215	0.0269	0.0323	0.0377	0.0430	0.0484
0.10	0.0538	0.0591	0.0645	0.0699	0.0752	0.0806	0.0859	0.0913	0.0966	0.1020
0.20	0.1073	0.1126	0.1180	0.1233	0.1286	0.1339	0.1392	0.1445	0.1498	0.1551
0.30	0.1604	0.1656	0.1709	0.1761	0.1814	0.1866	0.1919	0.1971	0.2023	0.2075
0.40	0.2127	0.2179	0.2230	0.2282	0.2334	0.2385	0.2436	0.2488	0.2539	0.2590
0.50	0.2641	0.2691	0.2742	0.2793	0.2843	0.2893	0.2944	0.2994	0.3044	0.3093
0.60	0.3143	0.3193	0.3242	0.3291	0.3340	0.3389	0.3438	0.3487	0.3535	0.3584
0.70	0.3632	0.3680	0.3728	0.3776	0.3823	0.3871	0.3918	0.3965	0.4012	0.4059
0.80	0.4105	0.4152	0.4198	0.4244	0.4290	0.4336	0.4381	0.4427	0.4472	0.4517
0.90	0.4562	0.4606	0.4651	0.4695	0.4739	0.4783	0.4827	0.4871	0.4914	0.4957
1.00	0.5000	0.5043	0.5085	0.5128	0.5170	0.5212	0.5254	0.5295	0.5337	0.5378
1.10	0.5419	0.5460	0.5500	0.5541	0.5581	0.5621	0.5660	0.5700	0.5739	0.5778
1.20	0.5817	0.5856	0.5894	0.5933	0.5971	0.6008	0.6046	0.6083	0.6121	0.6158
1.30	0.6194	0.6231	0.6267	0.6303	0.6339	0.6375	0.6410	0.6445	0.6480	0.6515
1.40	0.6550	0.6584	0.6618	0.6652	0.6686	0.6719	0.6753	0.6786	0.6818	0.6851
1.50	0.6883	0.6916	0.6947	0.6979	0.7011	0.7042	0.7073	0.7104	0.7134	0.7165
1.60	0.7195	0.7225	0.7255	0.7284	0.7314	0.7343	0.7371	0.7400	0.7429	0.7457
1.70	0.7485	0.7513	0.7540	0.7567	0.7595	0.7621	0.7648	0.7675	0.7701	0.7727
1.80	0.7753	0.7779	0.7804	0.7829	0.7854	0.7879	0.7904	0.7928	0.7952	0.7976
1.90	0.8000	0.8024	0.8047	0.8070	0.8093	0.8116	0.8138	0.8161	0.8183	0.8205
2.00	0.8227	0.8248	0.8270	0.8291	0.8312	0.8333	0.8353	0.8374	0.8394	0.8414
2.10	0.8434	0.8453	0.8473	0.8492	0.8511	0.8530	0.8549	0.8567	0.8585	0.8604
2.20	0.8622	0.8639	0.8657	0.8675	0.8692	0.8709	0.8726	0.8743	0.8759	0.8776
2.30	0.8792	0.8808	0.8824	0.8840	0.8855	0.8871	0.8886	0.8901	0.8916	0.8930
2.40	0.8945	0.8960	0.8974	0.8988	0.9002	0.9016	0.9029	0.9043	0.9056	0.9069
2.50	0.9083	0.9095	0.9108	0.9121	0.9133	0.9146	0.9158	0.9170	0.9182	0.9194
2.60	0.9205	0.9217	0.9228	0.9239	0.9250	0.9261	0.9272	0.9283	0.9293	0.9304
2.70	0.9314	0.9324	0.9334	0.9344	0.9354	0.9364	0.9373	0.9383	0.9392	0.9401
2.80	0.9411	0.9420	0.9428	0.9437	0.9446	0.9454	0.9463	0.9471	0.9479	0.9487
2.90	0.9495	0.9503	0.9511	0.9519	0.9526	0.9534	0.9541	0.9549	0.9556	0.9563
3.00	0.9570	0.9577	0.9583	0.9590	0.9597	0.9603	0.9610	0.9616	0.9622	0.9629

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Appendix B
Derivation of the Characteristic Function $\Phi_N(s)$

Equation (4.21) is given by

$$\Phi_N(s) = \sum_{N=0}^{\infty} e^{isN} f_z(N) \tag{B.1}$$

where by definition $N \geq z$; i.e., the total number of shots is greater than or equal to the number of hits of the target. The expression for $f_z(N)$ is given by Eq. (4.20) as

$$f_z(N) = P_1 u^{z-1} \qquad \text{for } N = z \tag{B.2a}$$

$$f_z(N) = P_1 \sum_{k=2}^{z} \binom{z-1}{k-1} \binom{N-z-1}{k-2} u^{z-k}(1-u)^{k-1} v^{k-1}(1-v)^{N-z-k+1}$$

$$+ (1-P_1) \sum_{k=1}^{z} \binom{z-1}{k-1} \binom{N-z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k}(1-v)^{N-z-k}$$

$$\text{for } N > z \tag{B.2b}$$

To calculate the expansion in Eq. (B.1), the following combinatorial identities will be used in which α, k, and n are real positive integers:[1]

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k \tag{B.3}$$

for all real n and all single-valued x with $|x| < 1$,

$$\binom{-\alpha}{k} = (-1)^k \binom{\alpha+k-1}{k} \tag{B.4}$$

$$\binom{n}{k} = \binom{n}{n-k} \tag{B.5}$$

$$\binom{n}{k} = 0 \tag{B.6}$$

if $k > n$ and n is a positive integer, and

$$\binom{n}{-k} = 0 \tag{B.7}$$

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Substituting Eq. (B.2) into Eq. (B.1),

$$\Phi_N(s) = P_1 u^{z-1} e^{isz} + P_1 \sum_{N=z+1}^{\infty} e^{isN} \sum_{k=2}^{z} \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k-1} \binom{N-z-1}{k-2} q^{N-z-k+1} + (1-P_1) \sum_{N=z+1}^{\infty} e^{isN} \sum_{k=1}^{z} \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k} \binom{N-z-1}{k-1} q^{N-z-k} \tag{B.8}$$

where

$$q = 1 - v \tag{B.9}$$

Introducing a new variable,

$$M = N - (z+1) - k + 2 \tag{B.10}$$

so that

$$N = M + (z+1) + k - 2 \tag{B.11}$$

and rearranging the order of summation,

$$\Phi_N(s) = P_1 u^{z-1} e^{isz} + P_1 \sum_{k=2}^{z} \sum_{M=-k+2}^{\infty} \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} \times v^{k-1} \binom{M+k-2}{k-2} q^{M} e^{isM} e^{is(z+k-1)} + (1-P_1) \sum_{k=1}^{z} \sum_{M=-k+2}^{\infty} \binom{z-1}{k-1} \times u^{z-k}(1-u)^{k-1} v^{k} \binom{M+k-2}{k-1} q^{M-1} e^{is(M-1)} e^{is(z+k)} \tag{B.12}$$

Noting that, by Eqs. (B.5) and (B.4),

$$\binom{M+k-2}{k-2} = \binom{M+k-2}{M} = (-1)^M \binom{-(k-1)}{M} \tag{B.13}$$

and

$$\binom{M+k-2}{k-1} = \binom{M+k-2}{M-1} = (-1)^{M-1} \binom{-k}{M} \tag{B.14}$$

it follows with the use of Eq. (B.7) that

$$\Phi_N(s) = P_1 u^{z-1} e^{isz} + P_1 \sum_{k=2}^{\infty} \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k-1} e^{is(z+k-1)} \sum_{M=0}^{\infty} \binom{-(k-1)}{M} (-q^{is})^M + (1-P_1) \sum_{k=1}^{z} \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k} e^{is(z+k)} \sum_{M=0}^{\infty} \binom{-k}{M-1} (-q^{is})^{M-1} \tag{B.15}$$

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Hence, using Eqs. (B.3) and (B.6),

$$\Phi_N(s) = P_1 u^{z-1} e^{isz} + P_1 \sum_{k=2}^{z} \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k-1} e^{is(z+k-1)} \frac{1}{(1-qe^{is})^{k-1}} + (1-P_1) \sum_{k=1}^{z} \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k} e^{is(z+k)} \frac{1}{(1-qe^{is})^{k}} \tag{B.16}$$

Now noting that, for $k = 1$,

$$P_1 \binom{z-1}{k-1} u^{z-k}(1-u)^{k-1} v^{k-1} e^{is(z+k-1)} \frac{1}{(1-qe^{is})^{k-1}} = P_1 u^{z-1} e^{isz} \tag{B.17}$$

$$\Phi_N(s) = P_1 \sum_{k=1}^{z} \binom{z-1}{k-1} u^{z-k} e^{isz} \left[\frac{(1-u)ve^{is}}{1-qe^{is}}\right]^{k-1} + (1-P_1) \sum_{k=1}^{z} \binom{z-1}{k-1} \times u^{z-k} e^{isz} \left[\frac{(1-u)ve^{is}}{1-qe^{is}}\right]^{k-1} \frac{ve^{is}}{1-qe^{is}} = P_1 e^{isz} \left[u + \frac{(1-u)ve^{is}}{1-qe^{is}}\right]^{z-1} + (1-P_1) e^{isz} \frac{ve^{is}}{1-qe^{is}} \left[u + \frac{(1-u)ve^{is}}{1-qe^{is}}\right]^{z-1} = e^{isz} \left[P_1 + (1-P_1)\frac{ve^{is}}{1-qe^{is}}\right] \left[u + \frac{(1-u)ve^{is}}{1-qe^{is}}\right]^{z-1} \tag{B.18}$$

Reference

[1]Riordan, J., *Combinatorial Identities*, Wiley, New York, 1968, pp. 1–2.

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Appendix C
Analytical Solution of Equations of Combat

C.1 General Solution of $\dot{\mathbf{X}} = \mathbf{CX}$

The general solution of the fundamental equation for the direct fire model

$$\dot{\mathbf{X}} = \mathbf{CX} \tag{C.1}$$

can be obtained as a product solution of the form

$$\mathbf{X} = \mathbf{P}\varphi \tag{C.2}$$

where $\mathbf{P}$ is the matrix of *generalized* right eigenvectors and φ is the column matrix whose elements are function of time. Substituting Eq. (C.2) into Eq. (C.1),

$$\dot{\mathbf{X}} = \mathbf{P}\dot{\varphi} = \mathbf{CP}\varphi$$

Hence,

$$\dot{\varphi} = \mathbf{P}^{-1}\mathbf{CP}\varphi = \mathbf{J}\varphi \tag{C.3}$$

where

$$\varphi = \left\{ \varphi^{(1)} \quad \varphi^{(2)} \quad \ldots \quad \varphi^{(i)} \quad \ldots \right\} \tag{C.4}$$

$$\dot{\varphi} = \left\{ \dot{\varphi}^{(1)} \quad \dot{\varphi}^{(2)} \quad \ldots \quad \dot{\varphi}^{(i)} \quad \ldots \right\} \tag{C.5}$$

and

$$\mathbf{J} = \mathbf{P}^{-1}\mathbf{CP} = \begin{bmatrix} \mathbf{J}^{(1)} & & & \\ & \mathbf{J}^{(2)} & & \mathbf{O} \\ & & \ddots & \\ & \mathbf{O} & & \mathbf{J}^{(i)} \\ & & & & \ddots \end{bmatrix} \tag{C.6}$$

is the Jordan canonical form of the matrix $\mathbf{C}$, modified to take into the account the dimensionality of the matrix elements. The governing equation matrix $\mathbf{C}$, its eigenvalues, and the matrix $\mathbf{J}$ are of the dimension 1/time. Also, for reasons explained later, $\mathbf{J}$ is taken here as the lower triangular matrix instead of the conventional upper triangular matrix.

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A typical term in Eq. (C.3) is of the form

$$\dot{\varphi}^{(i)} = \mathbf{J}^{(i)}\varphi^{(i)} \tag{C.7}$$

$$\dot{\varphi}^{(i)} = \left\{\dot{\varphi}_1^{(i)} \quad \dot{\varphi}_2^{(i)} \quad \cdots \quad \dot{\varphi}_k^{(i)}\right\}_{k\times 1} \tag{C.8}$$

$$\varphi^{(i)} = \left\{\varphi_1^{(i)} \quad \varphi_2^{(i)} \quad \cdots \quad \varphi_k^{(i)}\right\}_{k\times 1} \tag{C.9}$$

where $\mathbf{J}^{(i)}$ is essentially one of the three types, or combinations thereof, as discussed below.[1]

Type 1:

$$\mathbf{J}^{(i)} = \lambda^{(i)} \tag{C.10}$$

when the algebraic and geometric multiplicity of the eigenvalue $\lambda^{(i)}$ is equal to 1.

Type 2:

$$\mathbf{J}^{(i)} = \lambda^{(i)}\mathbf{I} \tag{C.11}$$

when the algebraic and geometric multiplicity of the eigenvalue $\lambda^{(i)}$ is equal to k with $k \geq 2$, and the corresponding eigenvectors are linearly independent.

Type 3:

$$\mathbf{J}^{(i)} = \lambda^{(i)}\begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & 1 \end{bmatrix}_{k\times k} \tag{C.12}$$

when the algebraic multiplicity of the eigenvalue $\lambda^{(i)}$ exceeds its geometric multiplicity. The expression for $\mathbf{J}^{(i)}$ has been modified here, from its standard textbook form in which only the diagonal terms are equal to $\lambda^{(i)}$, to ensure proper dimensionality of all of its elements. The eigenvalues $\lambda^{(i)}$ in two different $\mathbf{J}^{(i)}$ can be the same. Also, Type 3 may appear in combination with Types 1 and 2. When $k = 1$, Type 3 reduces to Type 1.

It is easy to verify that the solution to Eq. (C.7) with $\mathbf{J}^{(i)}$ given by Eq. (C.12) (i.e., Type 3) can be obtained by solving first for $\varphi_1^{(i)}$ and then substituting it into the equation for $\dot{\varphi}_2^{(i)}$. This process is repeated until the solution for $\varphi_k^{(i)}$ is found. These solutions can be written concisely as

$$\varphi^{(i)} = \exp\left(\lambda^{(i)}t\right)\left[\mathbf{I} + \frac{\lambda^{(i)}t}{1!}\mathbf{H}_1 + \frac{\left(\lambda^{(i)}t\right)^2}{2!}\mathbf{H}_2 + \cdots\right]\mathbf{A}^{(i)} \tag{C.13a}$$

$$= \exp\left(\lambda^{(i)}t\right)\left[\mathbf{I} + \frac{\lambda^{(i)}t}{1!}\mathbf{H} + \frac{\left(\lambda^{(i)}t\right)^2}{2!}\mathbf{H}^2 + \cdots\right]\mathbf{A}^{(i)} \tag{C.13b}$$

$$= \exp\left(\lambda^{(i)}t\right)\mathbf{D}^{(i)}\mathbf{A}^{(i)} \tag{C.13c}$$

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where

$$\mathbf{D}^{(i)} = \mathbf{I} + \frac{\lambda^{(i)}t}{1!}\mathbf{H} + \frac{\left(\lambda^{(i)}t\right)^2}{2!}\mathbf{H}^2 + \cdots + \frac{\left(\lambda^{(i)}t\right)^{k-1}}{(k-1)!}\mathbf{H}^{k-1} \tag{C.14a}$$

$$= \begin{bmatrix} 1 & & & & \\ \lambda^{(i)}t & 1 & & \mathbf{O} & \\ \frac{1}{2}\lambda^{(i)}t)^2 & \lambda^{(i)}t & 1 & & \\ \frac{1}{6}\lambda^{(i)}t)^3 & \frac{1}{2}\lambda^{(i)}t)^2 & \lambda^{(i)}t & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{C.14b}$$

and

$$\mathbf{A}^{(i)} = \begin{Bmatrix} A_1^{(i)} & A_2^{(i)} & \cdots & A_k^{(i)} \end{Bmatrix} \tag{C.15}$$

is the column matrix of constants of integration. The $\mathbf{H}$ matrices are given by

$$\mathbf{H}_1 = \mathbf{H} = \begin{bmatrix} 0 & & & & \\ 1 & 0 & & \mathbf{O} & \\ 0 & 1 & 0 & & \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}_{k\times k} \tag{C.16}$$

$$\mathbf{H}_2 = \mathbf{H}^2 = \begin{bmatrix} 0 & & & & & \\ 0 & 0 & & & \mathbf{O} & \\ 1 & 0 & 0 & & & \\ 0 & 1 & 0 & 0 & & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}_{k\times k} \tag{C.17}$$

etc., which finally leads to

$$\mathbf{H}^k = \mathbf{O} \tag{C.18}$$

Substituting Eq. (C.13c) into Eqs. (C.4) and (C.2), it follows that the general solution of $\dot{\mathbf{X}} = \mathbf{C}\mathbf{X}$ can be written as

$$\mathbf{X} = \mathbf{P}\,\varphi = \mathbf{P}\lceil\mathbf{exp}(\lambda t)\mathbf{D}\rfloor\mathbf{A} \tag{C.19}$$

where

$$\lceil\mathbf{exp}(\lambda t)\mathbf{D}\rfloor = \begin{bmatrix} \exp\left(\lambda^{(1)}t\right)\mathbf{D}^{(1)} & \exp\left(\lambda^{(2)}t\right)\mathbf{D}^{(2)} & \cdots \end{bmatrix} \tag{C.20}$$

$$\mathbf{A} = \begin{Bmatrix} \mathbf{A}^{(1)} & \mathbf{A}^{(2)} & \cdots & \mathbf{A}^{(i)} & \cdots \end{Bmatrix} \tag{C.21}$$

At time $t = 0$, $\mathbf{X} = \mathbf{X}_0$; therefore, from Eq. (C.19),

$$\mathbf{X}_0 = \mathbf{P}\mathbf{A} \tag{C.22}$$

since $\exp(0) = 1$ and $\mathbf{D} = \mathbf{I}$ for $t = 0$. Hence,

$$\mathbf{A} = \mathbf{P}^{-1}\mathbf{X}_0 = \mathbf{Q}^T\mathbf{X}_0 \tag{C.23}$$

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where the matrix $\mathbf{Q}$ is defined as

$$\mathbf{Q} = (\mathbf{P}^{-1})^T \tag{C.24}$$

Hence, from Eqs. (C.21) and (C.24),

$$\mathbf{X} = \mathbf{P}\lceil\exp(\lambda t)\mathbf{D}\rfloor\mathbf{Q}^T\mathbf{X}_0 \tag{C.25}$$

If the matrix $\mathbf{C}$ is not defective, i.e., its geometric multiplicity is equal to the algebraic multiplicity for all eigenvalues $\lambda^{(i)}$, then $\mathbf{D}^{(i)} = 1$, and the solution for $\mathbf{X}$ simplifies to

$$\mathbf{X} = \mathbf{P}\lceil\exp(\lambda t)\rfloor\mathbf{Q}^T\mathbf{X}_0 \tag{C.26}$$

C.2 Right and Left Generalized Eigenvectors

From Eqs. (C.6a), (C.6b), and (C.24) it follows that

$$\mathbf{CP} = \mathbf{PJ} \tag{C.27}$$

$$\mathbf{Q}^T\mathbf{P} = \mathbf{I} \tag{C.28}$$

$$\mathbf{PQ}^T = \mathbf{I} \tag{C.29}$$

and

$$\begin{aligned}\mathbf{Q}^T\mathbf{C} &= \mathbf{P}^{-1}\mathbf{C}\\ &= \mathbf{P}^{-1}\mathbf{CPP}^{-1}\\ &= \mathbf{JQ}^T\end{aligned} \tag{C.30}$$

Equation (C.30) implies that $\mathbf{Q}$ is the matrix of the *generalized* left eigenvectors, whereas Eqs. (C.28) and (C.29) indicate orthogonality between the right and left *generalized* eigenvectors.

Introducing the individual generalized eigenvectors into $\mathbf{P}$ and $\mathbf{Q}$ such that

$$\mathbf{P} = \begin{bmatrix}\mathbf{p}^{(1)} & \mathbf{p}^{(2)} & \dots & \mathbf{p}^{(i)} & \dots\end{bmatrix} \tag{C.31}$$

and

$$\mathbf{Q} = \begin{bmatrix}\mathbf{q}^{(1)} & \mathbf{q}^{(2)} & \dots & \mathbf{q}^{(i)} & \dots\end{bmatrix} \tag{C.32}$$

it can be demonstrated that the application of Eqs. (C.27) and (C.30) for $\mathbf{J}^{(i)}$ of Type 3 and order $k \times k$ leads to the following relations:

$$\begin{aligned}(\mathbf{C}-\lambda\mathbf{I})\mathbf{p}_1 &= \lambda\mathbf{p}_2 \quad \text{or} \quad (\mathbf{C}-\lambda\mathbf{I})^k\mathbf{p}_1 = \mathbf{0}\\ (\mathbf{C}-\lambda\mathbf{I})\mathbf{p}_2 &= \lambda\mathbf{p}_3 \quad \text{or} \quad (\mathbf{C}-\lambda\mathbf{I})^{k-1}\mathbf{p}_2 = \mathbf{0}\\ &\vdots \qquad\quad \vdots \qquad\quad \vdots\\ (\mathbf{C}-\lambda\mathbf{I})\mathbf{p}_{k-1} &= \lambda\mathbf{p}_k \quad \text{or} \quad (\mathbf{C}-\lambda\mathbf{I})^2\mathbf{p}_{k-1} = \mathbf{0}\\ (\mathbf{C}-\lambda\mathbf{I})\mathbf{p}_k &= \mathbf{0} \quad \text{or} \quad (\mathbf{C}-\lambda\mathbf{I})\mathbf{p}_k = \mathbf{0}\end{aligned} \tag{C.33}$$

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and

$$\begin{array}{rcl} \mathbf{q}_1^T(\mathbf{C}-\lambda\mathbf{I}) = \mathbf{0} & \text{or} & \mathbf{q}_1^T(\mathbf{C}-\lambda\mathbf{I}) = \mathbf{0} \\ \mathbf{q}_2^T(\mathbf{C}-\lambda\mathbf{I}) = \lambda\mathbf{q}_1^T & \text{or} & \mathbf{q}_2^T(\mathbf{C}-\lambda\mathbf{I})^2 = \mathbf{0} \\ \vdots \quad\quad \vdots & \vdots & \vdots \quad\quad \vdots \\ \mathbf{q}_{k-1}^T(\mathbf{C}-\lambda\mathbf{I}) = \lambda\mathbf{q}_{k-2}^T & \text{or} & \mathbf{q}_{k-1}^T(\mathbf{C}-\lambda\mathbf{I})^{k-1} = \mathbf{0} \\ \mathbf{q}_k^T(\mathbf{C}-\lambda\mathbf{I}) = \lambda\mathbf{q}_{k-1}^T & \text{or} & \mathbf{q}_k^T(\mathbf{C}-\lambda\mathbf{I})^k = \mathbf{0} \end{array} \tag{C.34}$$

where for simplicity all superscripts (i) have been omitted, and the subscripts $1-k$ refer to the eigenvectors comprising the columns in $\mathbf{p}^{(i)}$ and $\mathbf{q}^{(i)}$. Because the analysis of the relative strengths in combat engagements involves the calculation of the dominant left eigenvector $\mathbf{q}_1$, it is clear that the selection of $\mathbf{J}$ as the lower triangular matrix was more convenient. In the case of defective matrices $\mathbf{C}$ it resulted in a simple expression:

$$\mathbf{q}_1^T(\mathbf{C}-\lambda\mathbf{I}) = \mathbf{0} \tag{C.35a}$$

or

$$(\mathbf{C}^T-\lambda\mathbf{I})\mathbf{q}_1 = \mathbf{0} \tag{C.35b}$$

for the calculation of $\mathbf{q}_1$.

C.3 Examples of the Dominant Left Eigenvectors

The dominant left eigenvectors for the matrix $\mathbf{C}$ are illustrated here for several (m-on-n) engagements for equal attrition coefficients in both $\mathbf{A}$ and $\mathbf{B}$ (i.e., $A_{ij} = a$ and $B_{ij} = b$ for all i and j where a and b are constants). The number of engaging types of weapons is indicated by the subscripts (m, n) with each matrix $\mathbf{C}$. All eigenvalues of $\mathbf{C}$ are shown, with the dominant eigenvalue(s) listed first. The dominant eigenvector $\mathbf{q}_1$ has been normalized on its largest element for the Blue forces.

$$\mathbf{C} = \begin{bmatrix} 0 & -b \\ -a & 0 \end{bmatrix}_{(1,1)}; \quad \lambda = \sqrt{ab}, -\sqrt{ab}; \quad \mathbf{q}_1 = \left\{ 1 \quad -\sqrt{b/a} \right\}_{2\times 1}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -b \\ 0 & 0 & -b \\ a & -a & 0 \end{bmatrix}_{(2,1)}; \quad \begin{array}{l} \lambda = \sqrt{2ab}, 0, -\sqrt{2ab}; \\ \mathbf{q}_1 = \left\{ 1 \quad 1 \quad -\sqrt{2b/a} \right\}_{3\times 1} \end{array}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & \dots & -b \\ 0 & 0 & \dots & -b \\ \vdots & \vdots & \ddots & \vdots \\ -a & -a & \dots & 0 \end{bmatrix}_{(m,1)};$$

$$\lambda = \sqrt{mab}, 0, \dots (m-1) \text{ zeros } \dots, 0, -\sqrt{mab};$$

$$\mathbf{q}_1 = \left\{ 1 \quad 1 \dots 1 \quad -\sqrt{mb/a} \right\}_{(m+1)\times 1}$$

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$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -b & -b \\ 0 & 0 & -b & -b \\ -a & -a & 0 & 0 \\ -a & -a & 0 & 0 \end{bmatrix}_{(2,2)}; \qquad \lambda = 2\sqrt{ab}, 0, 0, -2\sqrt{ab};$$

$$\mathbf{q}_1 = \left\{1 \quad 1 \quad -\sqrt{b/a} \quad -\sqrt{b/a}\right\}_{4\times 1};$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 & -b & -b \\ 0 & 0 & 0 & -b & -b \\ 0 & 0 & 0 & -b & -b \\ -a & -a & -a & 0 & 0 \\ -a & -a & -a & 0 & 0 \end{bmatrix}_{(3,2)};$$

$$\lambda = \sqrt{6ab}, 0, 0, 0, -\sqrt{6ab};$$

$$\mathbf{q}_1 = \left\{1 \quad 1 \quad 1 \quad -\sqrt{3b/2a} \quad -\sqrt{3b/2a}\right\}_{5\times 1}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 & -b & -b & -b \\ 0 & 0 & 0 & -b & -b & -b \\ 0 & 0 & 0 & -b & -b & -b \\ -a & -a & -a & 0 & 0 & 0 \\ -a & -a & -a & 0 & 0 & 0 \\ -a & -a & -a & 0 & 0 & 0 \end{bmatrix}_{(3,3)};$$

$$\lambda = 3\sqrt{ab}, 0, 0, 0, 0, -3\sqrt{ab};$$

$$\mathbf{q}_1 = \left\{1 \quad 1 \quad 1 \quad -\sqrt{b/a} \quad -\sqrt{b/a} \quad -\sqrt{b/a}\right\}_{6\times 1}$$

For the general case of m Blue forces on n Red forces (m-on-n) with equal coefficients in **A** and **B**

$$\lambda = \lambda^{(1)}, 0, 0, \ldots, 0, 0, -\lambda^{(1)};$$

$$\lambda^{(1)} = \sqrt{mnab}$$

$$\mathbf{q}_1 = \left\{1 \quad 1 \ldots (m \text{ times}) - \sqrt{mb/na} - \sqrt{mb/na} \ldots (n \text{ times})\right\}_{(m+n)\times 1}$$

The following two examples with some zero attrition coefficients represent cases of defective matrices **C** with repeated eigenvalues.

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & -b & -b \\ 0 & 0 & 0 & -b \\ -a & 0 & 0 & 0 \\ 0 & -a & 0 & 0 \end{bmatrix}_{(2,2)}; \qquad \lambda = \sqrt{ab}, \sqrt{ab}, -\sqrt{ab}, -\sqrt{ab};$$

$$\mathbf{q}_1 = \{0 \quad 1 \quad 0 \quad -\sqrt{b/a}\}_{4\times 1}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 & -b & -b & -b \\ 0 & 0 & 0 & 0 & 0 & -b \\ 0 & 0 & 0 & 0 & 0 & -b \\ -a & 0 & 0 & 0 & 0 & 0 \\ 0 & -a & 0 & 0 & 0 & 0 \\ 0 & 0 & -a & 0 & 0 & 0 \end{bmatrix}_{(3,3)};$$

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$$\lambda = \sqrt{ab}, \sqrt{ab}, 0, 0, -\sqrt{ab}, -\sqrt{ab};$$

$$\mathbf{q}_1 = \left\{0 \quad 0 \quad 1 \quad 0 \quad 0 \quad -\sqrt{b/a}\right\}_{6\times 1}$$

C.4 Computation of the Dominant Left Eigenvectors and Their Eigenvalues

In addition to the dominant eigenvalue λ_1, the matrix $\mathbf{C}^T$ also contains an eigenvalue equal to $-\lambda_1$. Consequently, because there are two eigenvalues with equal absolute values, the standard power method of computing the largest eigenvalue and its corresponding eigenvector cannot be applied; however, the power method with a shift of origin can be used to find both λ_1 and $\mathbf{q}_1$.[2]

Starting with the basic equation (C.35) with $\lambda = \lambda_1$,

$$(\mathbf{C}^T - \lambda_1 \mathbf{I})\mathbf{q}_1 = \mathbf{0} \tag{C.35a}$$

and adding $\mu \mathbf{I}\mathbf{q}_1$ to both sides of Eq. (C.35a) where μ is a constant (e.g., $\mu = 1$),

$$(\mathbf{C}^T + \mu \mathbf{I})\mathbf{q}_1 = (\lambda_1 + \mu)\mathbf{q}_1 \tag{C.36}$$

Hence,

$$\bar{\mathbf{C}}^T \mathbf{q}_1 = \bar{\lambda}\mathbf{q}_1 \tag{C.37}$$

where

$$\bar{\mathbf{C}}^T = \mathbf{C} + \mu \mathbf{I} \tag{C.38}$$

$$\lambda_1 = \bar{\lambda} - \mu \tag{C.39}$$

In this way the shift of the origin for the eigenvalues λ_1 and $-\lambda_1$ has resulted in $|\lambda_1 + \mu|$ being greater than $|-\lambda_1 + \mu|$; therefore, the standard power method can be applied to Eq. (C.37) to find $\mathbf{q}_1$ and $\bar{\lambda}$ and then λ_1 from Eq. (C.39). A computer program for the calculation of the dominant eigenvalue of $\mathbf{C}$ and its corresponding left eigenvector, based on the power method, is provided with this text (See Appendix E, Section E.10).

References

[1]Young, D. M., and Gregory, R. T., *A Survey of Numerical Mathematics*, Vol. II, Addison–Wesley, Reading, MA, 1973, pp. 745–752.

[2]Young, D. M., and Gregory, R. T., *A Survey of Numerical Mathematics*, Vol. II, Addison–Wesley, Reading, MA, 1973, pp. 915–918.

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Appendix D
Calculation of the Average Probability of No Detection

Introducing the new variable θ and the parameter λ given by

$$\theta = \frac{\pi x}{S} \tag{D.1}$$

$$\lambda = \frac{\pi}{4}\left(\frac{W}{S}\right)^2 \tag{D.2}$$

and using Eqs. (9.45) and (9.47), Eq. (9.48) for the average probability of no detection can be rewritten as

$$Q\left(\frac{W}{S}\right) = \frac{1}{S}\int_0^S Q(x, S)\,dx = \frac{1}{S}\int_0^S \exp\left[-\frac{\pi}{4}\left(\frac{W}{S}\right)^2 \csc^2\frac{\pi x}{S}\right] dx$$

$$= \frac{1}{\pi}\int_0^{\pi} \exp(-\lambda \csc^2\theta)\,d\theta = \frac{2}{\pi}\int_0^{\frac{1}{2}\pi} \exp(-\lambda \csc^2\theta)\,d\theta \tag{D.3}$$

To evaluate the preceding integral, the following procedure can be adopted. A new variable

$$\Phi(\lambda) = Q\left(\frac{W}{S}\right) = \frac{2}{\pi}\int_0^{\frac{1}{2}\pi} \exp(-\lambda \csc^2\theta)\,d\theta \tag{D.4}$$

is introduced and then differentiated with respect to λ, i.e.,

$$\Phi'(\lambda) = -\frac{2}{\pi}\int_0^{\frac{1}{2}\pi} \csc^2\theta \exp(-\lambda \csc^2\theta)\,d\theta \tag{D.5}$$

which can be integrated by the substitution of

$$y = \sqrt{\lambda}\cot\theta \tag{D.6}$$

so that

$$\Phi'(\lambda) = -\frac{2}{\pi}\left(-e^{-\lambda}\int_{\infty}^0 e^{-y^2}\frac{dy}{\sqrt{\lambda}}\right) = -\frac{2e^{-\lambda}}{\pi\sqrt{\lambda}}\int_0^{\infty} e^{-y^2}\,dy$$

$$= -\frac{1}{\sqrt{\pi\lambda}}e^{-\lambda} \tag{D.7}$$

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where the formula

$$\int_0^\infty e^{-y^2}\,dy = \frac{\sqrt{\pi}}{2} \tag{D.8}$$

was used. Now integrating $\Phi'(\lambda)$ with respect to λ and noting from Eq. (D.4) that $\Phi(0) = 1$,

$$\Phi(\lambda) = 1 - \frac{1}{\sqrt{\pi}}\int_0^\lambda e^{-\lambda}\frac{d\lambda}{\sqrt{\lambda}} = 1 - \frac{2}{\sqrt{\pi}}\int_0^{\sqrt{\lambda}} e^{-z^2}\,dz = 1 - \text{erf}\sqrt{\lambda} \tag{D.9}$$

where

$$z = \sqrt{\lambda} \tag{D.10}$$

Hence, from Eq. (D.2), (D.4), and (D.9),

$$Q\left(\frac{W}{S}\right) = \Phi(\lambda) = 1 - \text{erf}\sqrt{\lambda} = 1 - \text{erf}\left(\frac{\sqrt{\pi}\,W}{2S}\right) \tag{10.48}$$

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Appendix E
Defense Analyses Software

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E.1 Introduction

COMPUTER REQUIREMENTS: IBM PC, PC XT, PC AT, PS/2, or a compatible computer with 512K minimum memory and one double-sided disk drive. MS DOS version 2.0 or later with accompanying BASICA interpreter.

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E.2 Subroutine Instructions

The *"Defense Analyses Software"* compendium of subroutines has been developed specifically for this textbook. These subroutines were used to compute numerical values or tables developed in this text. Many of these subroutines can also be used in other applications and, if necessary, can be linked together to generate more extensive programs. The subroutines are organized into sections, the titles of which correspond to chapter titles in this textbook. Each subroutine is provided with a detailed reference to the equation(s) from which it was derived to ensure that its intended application corresponds to the assumptions used in the derivation. It should be noted that, in some cases, the variable symbols used in the programs may be different from those used in the textbook (e.g., Greek letters have been changed into their phonetic equivalents).

The defense analyses software has been written in BASIC to allow for the widest possible use on personal computers. Each subroutine has been tested in IBM PC BASIC Version 2.0 under an IBM PC DOS 3.3 operating system, using an IBM PS/2 PC with 1MB memory, and also in QuickBASIC.

The "*Defense Analyses Software*" subroutines are provided on a 3.5 in disk for use on IBM-compatible personal computers (PCs). Each subroutine consists of

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two separate files designated as:

Filename.DAT
and Filename.SUB

In some cases, an additional file Filename.TAB is included. Files with an extension .DAT provide the input data to generate a single numerical value of the required variable from the subroutine Filename.SUB. Files with an extension .TAB are used when tables of variables are generated; they provide input data for a range of parameters and their increments. All subroutine files Filename.SUB have their own line numbers starting with 10,000. This allows different subroutine files to be merged in order to generate programs for other applications. Only the Filename.DAT and Filename.TAB files are reproduced in this appendix; however, all Filename.SUB files can be printed directly from the program disk if required.

To operate the program in the BASICA Interpreter environment, place the DOS disk in the default drive and enter BASICA and then press the return key. If a hard disk is used, simply enter BASICA and then press the return key. Next load the program disk into the drive A, enter Filename.DAT or Filename.TAB, depending whether single value is needed or tables are to be generated, and again press the return key. Using the instructions in the program, modify the input data and then enter MERGE "A:Filename.SUB" to merge Filename.DAT or Filename.TAB with Filename.SUB, and then press the return key. The program can now be RUN. Assuming that the Defense Analyses Subroutines program disk is placed in the disk drive A, the following steps are needed:

Step 1. Enter BASICA [CR]
Step 2. Place the program disk in drive A
Step 3. Enter LOAD"A:Filename.DAT" (or LOAD"A:Filename.TAB") [CR]
Step 4. Edit input data in Filename.DAT (or Filename.TAB), i.e., replace the test data with the required data for the computation
Step 5. Enter MERGE "A:Filename.SUB" [CR]
Step 6. Enter RUN [CR] (or F2)

where [CR] denotes the return key (carriage return).

To operate the program in the QuickBasic (QBASIC) environment, the following steps are needed:

Step 1. Place the program disk in drive A
Step 2. Merge Filename.DAT (or Filename.TAB) with Filename.SUB by using DOS command COPY Filename.DAT (or Filename.TAB) + Filename.SUB C:Filename.QBA [CR]. The extension QBA is used here to indicate that the new file, which is placed in drive C, is intended for the QBasic environment.
Step 3. Use any suitable editor to modify the required input data in Filename.QBA and then save the file.
Step 4. Enter QBASIC [CR]
Step 5. Enter LOAD"C:Filename.QBA" [CR]
Step 6. Enter F5

CAUTION: Practically all Filename.DAT and Filename.TAB files and over one half of the Filename.SUB files in this compendium contain both PRINT (display on

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the monitor screen) and LPRINT (print to the system printer) commands, which means an attached printer is required to use them as is. If there is no printer connected to the computer, an apostrophe (') must be placed directly in front of LPRINT command. This can accomplished as a global change from LPRINT to 'LPRINT.

Each subroutine can be tested with the input from the corresponding Filename.DAT or Filename.TAB. The resulting output data should agree with the listed output provided under the Test Example subsections of this compendium.

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E.3 Passive Targets

E.3.1 Binomial Probabilities

Subroutines:
BINOMIAL.DAT
BINOMIAL.TAB
BINOMIAL.SUB

Purpose: To compute individual value or table of values of the binomial probability (BP) that at least R hits will occur, when N shots are expended against a target with single probability of hit P.

Source: Eq. (3.5).

Inputs:
BINOMIAL.DAT
The following parameters must be specified:
N = Number of independent shots (an integer).
R = Minimum number of hits (an integer).
P = Single shot probability of hit.
BINOMIAL.TAB
The ranges of N, R, and P must be specified to create the required tables.

Program Files: BINOMIAL.DAT, and BINOMIAL.TAB

```
100 '*****************   BINOMIAL.DAT   ********************
110 'BINOMIAL PROBABILITIES
120 'Input data for the subroutine BINOMIAL.SUB which
     calculates the probability (BP) of at least R hits
     when N shots are expended against a target with
     a single shot probability of hit P.
130 'Values of N, R, and P must be specified here for a
     single value of BP (Lines 150, 160, and 170).
140 'N=number of independent shots, R=minimum number of
     hits, P=single shot probability of hit.
150 N=5
160 R=3
170 P=.3
180 GOSUB 10000  'BINOMIAL.SUB
190 LPRINT "Binomial probability (BP) of at least R hits
    when N shots are expended against a target with single
    shot probability of hit P."
```

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```
200  PRINT "Binomial probability (BP) of at least R hits
     when N shots are expended against a target with single
     shot probability of hit P."
210 LPRINT "N =";N;
220  PRINT "N =";N;
230 LPRINT "  R =";R;
240  PRINT "  R =";R;
250 LPRINT "  P =";P
260  PRINT "  P =";P
270 BP=E
280 LPRINT "BP =";
290  PRINT "BP =";
300 IF BP>.00005 THEN LPRINT USING " #.####";BP; ELSE
    LPRINT " 0.0000"
310 IF BP>.00005 THEN  PRINT USING " #.####";BP; ELSE
     PRINT " 0.0000"
320 END

100 '****************   BINOMIAL.TAB   ********************
110 'BINOMIAL PROBABILITIES
120 'Input ranges of variables into BINOMIAL.SUB subroutine
     to create tables of binomial probabilities (BP) for at
     least R hits when N shots are expended against a target
     with single shot probability of hit P.
130 'Ranges of values for R, N, and P and their increments
     must be specified here to create the tables (Lines 190,
     200, and 240).
140 'N=Number of independent shots, R=Minimum number of
     hits, and P= single shot probability of hit.
150 LPRINT "Binomial Probabilities"
160  PRINT "Binomial Probabilities"
170 LPRINT " P =      ";
180  PRINT " P =      ";
190 DP=.05
200 FOR P=.05 TO .55 STEP DP
210 LPRINT USING " #.##  ";P;
220  PRINT USING " #.##  ";P;
230 NEXT P
240 FOR N=2 TO 4 STEP 2
250 LPRINT "-------------------------------------------------
---------------------------------";"N=";N;" R>=";
260  PRINT "-------------------------------------------------
---------------------------------";"N=";N;" R>=";
270 LPRINT
280  PRINT
290 DR=1
300 FOR R=0 TO N STEP DR
310 LPRINT "       ";
320  PRINT "       ";
330 LPRINT USING "## ";R;
340  PRINT USING "## ";R;
350 FOR P=.05 TO .55 STEP DP
```

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```
360 GOSUB 10000 'BINOMIAL.SUB
370 BP=E
380 IF BP>.00005 THEN LPRINT USING " #.####";BP; ELSE LPRINT
    " ";
390 IF BP>.00005 THEN  PRINT USING " #.####";BP; ELSE  PRINT
    " ";
400 NEXT P
410 NEXT R
420 NEXT N
430 END
```

Test Example:

```
Binomial probability (BP) of at least R hits when N shots
are expended against a target with single shot probability
of hit P.
N = 5   R = 3   P = .3
BP = 0.1631
```

E.3.2 Poisson Probabilities

Subroutines:
POISSON.DAT
POISSON.TAB
POISSON.SUB

Purpose: To compute individual value or table of values of the Poisson probability (PP) of at least R hits against a target when the shots are expended at an average rate L.

Source: Eq. (3.28).

Inputs:
POISSON.DAT
The following parameters must be specified:
R = Minimum number of hits (an integer).
L = Rate of fire.
POISSON.TAB
The ranges of R and L must be specified to create the required tables.

Program Files: POISSON.DAT and POISSON.TAB

```
100 '*****************   POISSON.DAT   ********************
110 'POISSON PROBABILITIES
120 'Input data for the subroutine POISSON.SUB which
     calculates the Poisson probability (PP) of at least R
     hits against a target when the shots are expended at an
     average rate L.
130 'Values of R and L must be specified here to determine a
     single value of PP (Lines 160 and 170).
140 'R=Minimum number of hits against a target
150 'L=Average rate of hits against a target
160 R=2
```

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```
170 L=1.2
180 GOSUB 11000
190 LPRINT "Poisson probability (PP) of at least R hits when the
    shots are expended at an average rate of L against the target"
200 PRINT "Poisson probability (PP) of at least R hits when the
    shots are expended at an average rate of L against the target"
210 LPRINT "R =";R;
220 PRINT "R =";R;
230 LPRINT " L =";L
240 PRINT " L =";L
250 PP=E
260 PRINT "PP =";
270 IF PP>.00005 THEN LPRINT USING " #.####";PP; ELSE LPRINT
    "0.0000"
280 IF PP>.00005 THEN PRINT USING " #.####";PP; ELSE PRINT "0.0000"
290 END

100 '******************   POISSON.TAB   ********************
110 'POISSON PROBABILITIES
120 'Input ranges of variables into POISSON.SUB to create tables
     of Poisson probabilities (PP) of at least R hits when the
     shots are expended at an average rate L against the target.
130 'Range of values for R and L and their increments must
     be specified here to create the tables (Line numbers
     180, 190, 290, and 310).
140 LPRINT "-----------------------------------------------
---------------------------------";
150 PRINT "-----------------------------------------------
---------------------------------";
160 LPRINT "L =        ";
170 PRINT "L =        ";
180 DL=.1
190 FOR L=.1 TO 1.1 STEP DL
200 LPRINT USING " #.##  ";L;
210 PRINT USING " #.##  ";L;
220 NEXT L
230 LPRINT "-----------------------------------------------
---------------------------------";
240 PRINT "-----------------------------------------------
---------------------------------";
250 LPRINT " R=       ";
260  PRINT " R=       ";
270 LPRINT
280  PRINT
290 DR=1
300 W=1
310 FOR R=0 TO 7 STEP DR
320 LPRINT USING "   ##      ";R;
330  PRINT USING "   ##      ";R;
340 FOR L=.1 TO 1.1 STEP DL
350 GOSUB 11000
360 PP=E
```

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```
370 IF PP>.00005 THEN LPRINT USING " #.####";PP; ELSE LPRINT
    "      ";
380 IF PP>.00005 THEN  PRINT USING " #.####";PP; ELSE PRINT
    "      ";
390 NEXT L
400 IF W/5=INT(W/5) THEN LPRINT
410 IF W/5=INT(W/5) THEN  PRINT
420 W=W+1
430 NEXT R
440 END
```

Test Example:

```
Poisson probability (PP) of at least R hits when the shots
are expended at an average rate of L against the target
R = 2  L = 1.2
PP = 0.3374
```

E.3.3 Normal (Gaussian) Probabilities

Subroutines:
NORMAL.DAT
NORMAL.TAB
NORMAL.SUB

Purpose: To compute individual value or table of values of the normal (Gaussian) probability (NP).

Source: Eq. (3.30b).

Inputs:
NORMAL.DAT
The following parameters must be specified:
X = Value of the independent variable.
MEANX = Mean value of the independent variable.
SIGMA = Standard deviation.
NORMAL.TAB
The ranges of X and SIGMA must be specified to create the required tables.

Program Files: NORMAL.DAT and NORMAL.TAB

```
100 '******************   NORMAL.DAT   **********************
110 'NORMAL (GAUSSIAN) PROBABILITIES
120 'Input data for the subroutine NORMAL.SUB which
     calculates the normal (Gaussian) probability (NP).
130 'Values of X, MEANX, AND SIGMA must be specified on
     lines 150, 160, and 170
140 'U=(X-MEANX)/SIGMA, where SIGMA is the standard deviation.
150 X=2.2
160 MEANX=1
170 SIGMA=1
180 U=(X-MEANX)/SIGMA
190 LPRINT "Normal (Gaussian) Probability"
```

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```
200 PRINT "Normal (Gaussian) Probability"
210 LPRINT "X =";X; "MEANX ="; MEANX; "SIGMA ="; SIGMA; "U
    =(X-MEANX)/SIGMA =";U
220 PRINT "X =";X; "MEANX ="; MEANX; "SIGMA ="; SIGMA; "U
    =(X-MEANX)/SIGMA =";U
230 T=U/SQR(2)
240 Z=T
250 GOSUB 12000
260 LPRINT "NP="; NP
270 PRINT "NP="; NP
280 END

100 '******************   NORMAL.TAB   **********************
110 'NORMAL (GAUSSIAN) PROBABILITIES
120 'Input data for the subroutine NORMAL.SUB to generate
     tables of the normal probabilities NP.
130 'Values of DX and DY must be specified on lines 160 and
    170, and ranges of values for X and Y on lines 290 and 320.
140 LPRINT "Normal (Gaussian) Probabilities"
150 PRINT "Normal (Gaussian) Probabilities"
160 DX=.1
170 DY=.01
180 LPRINT "   U   ";
190 PRINT "   U   ";
200 FOR Y=0 TO 9.000001E-02 STEP DY
210 LPRINT USING " #.##  ";Y;
220 PRINT USING " #.##  ";Y;
230 NEXT Y
240 LPRINT"-------------------------------------------------
----------------------------"
250  PRINT"-------------------------------------------------
----------------------------"
260 LPRINT
270 PRINT
280 W=1
290 FOR X=0 TO 3 STEP DX
300 LPRINT USING " #.##  ";X;
310 PRINT USING " #.##  ";X;
320 FOR Y=0 TO 9.000001E-02 STEP DY
330 T=(X+Y)/SQR(2)
340 Z=T
350 GOSUB 12000
360 LPRINT USING " #.####";NP;
370 PRINT USING " #.####";NP;
380 NEXT Y
390 LPRINT
400 PRINT
410 IF W/5=INT(W/5) THEN LPRINT
420 IF W/5=INT(W/5) THEN PRINT
430 W=W+1
440 NEXT X
450 END
```

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Test Example:

```
Normal (Gaussian) Probability
X = 2.2 MEANX = 1 SIGMA = 1 U =(X-MEANX)/SIGMA = 1.2
NP= .1150697
```

E.3.4 Error Function

Subroutines:
ERF(Z).DAT
ERF(Z).TAB
ERF(Z).SUB

Purpose: To compute individual value or table of values of the error function erf(z).

Source: Eq. (3.92).

Inputs:
ERF(Z).DAT
The value of the argument Z must be specified.
ERF(Z).TAB
To generate tables of the error function the range of Z and its increments must be specified.

Program Files: ERF(Z).DAT and ERF(Z).TAB

```
100 '****************   ERF(Z).DAT   **********************
110 'ERROR FUNCTION ERF(Z)
120 'Input data for the subroutine ERF(Z).SUB to calculate
     the error function ERF(Z).
130 'Value of Z must be specified on line number 140
140 Z=2
150 LPRINT "Z= ", Z
160  PRINT "Z= "; Z
170 GOSUB 13000
180 ERFZ=E
190 LPRINT "ERF(Z)="; ERFZ
200  PRINT "ERF(Z)="; ERFZ
210 END
```

```
100 '****************   ERF(Z).TAB   **********************
110 'ERROR FUNCTION ERF(Z)
120 'Input ranges of variables into ERF(Z).SUB subroutine to
     compute ERF(Z) tables.
130 'Range of values of Z=X+Y and their increments must be
     specified in Lines 170, 180, 210, 280, and 310).
140 LPRINT "                    TABLE OF ERROR FUNCTION
    ERF(Z)"
150  PRINT "                    TABLE OF ERROR FUNCTION
     ERF(Z)"
160 LPRINT
170  PRINT
180 DX=.1
190 DY=.01
```

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```
200 LPRINT "   Z   ";
210  PRINT "   Z   ";
220 FOR Y=0 TO 9.000001E-02 STEP DY
230 LPRINT USING " #.##  ";Y;
240  PRINT USING " #.##  ";Y;
250 NEXT Y
260 LPRINT"-----------------------------------------------
----------------------------"
270  PRINT"-----------------------------------------------
----------------------------"
280 I=1
290 FOR X=0 TO 1 STEP DX
300 LPRINT USING " #.##  ";X;
310  PRINT USING " #.##  ";X;
320 FOR Y=0 TO 9.000001E-02 STEP DY
330 T=X+Y
340 Z=T
350 GOSUB 13000
360 ERFZ=E
370 LPRINT USING " #.####";ERFZ;
380  PRINT USING " #.####";ERFZ;
390 NEXT Y
400 LPRINT
410  PRINT
420 IF I/5=INT(I/5) THEN LPRINT
430 IF I/5=INT(I/5) THEN  PRINT
440 I=I+1
450 NEXT X
460 END
```

Test Example:

```
Z=  2
ERF(Z)= .9953224
```

E.3.5 Probability of Hit of a Rectangular Target

Subroutines:
PRECTANG.DAT
PRECTANG.TAB
PRECTANG.SUB

Purpose: To compute individual value or table of values of the component probability (range or cross-range) of hit of a rectangular target.

Source: Eq. (3.38).

Inputs:
PRECTANG.DAT
The following parameters must be specified:
Z = (1/2)x(Target range or cross-range dimension/REP or CREP).
REP=Range error probable.

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CREP=Cross-range error probable.
PRECTANG.TAB
The range of Z and its increment must be specified to create the required tables.

Program Files: PRECTANG.DAT and PRECTANG.TAB

```
100 '***************   PRECTANG.DAT   *****************
110 'PROBABILITY OF HIT OF A RECTANGULAR TARGET
120 'Input data for the subroutine PRECTANG.SUB to calculate the
    range (or cross-range) probability of hit of a
    rectangular target.
130 'Value of Z must be specified on line 150.
140 'Z=(1/2)x(Target range or cross-range dimension/REP
     (or CREP)), where REP = Range error probable, and CREP=
     Cross-range error probable.
150 Z=1.2
160 T=Z*.6745/SQR(2)
170 GOSUB 14000
180 ERFT=E
190 LPRINT "Z= "; Z
200  PRINT "Z= "; Z
210 LPRINT "Range or cross-range probability of hit P= ";
     ERFT
220  PRINT "Range or cross-range probability of hit P= ";
     ERFT
230 END

100 '***************   PRECTANG.TAB   *****************
110 'PROBABILITY OF HIT OF A RECTANGULAR TARGET
120 'Input ranges of variables into the subroutine
     PRECTANG.SUB to compute tables of the range or cross-
     range probabilities of hit of a rectangular target.
130 'Range of values of Z=X+Y and their increments must be
     specified on lines 140, 150, 270, and 300.
140 DX=.1
150 DY=.01
160 LPRINT "   Z   ";
170  PRINT "   Z   ";
180 FOR Y=0 TO 9.000001E-02 STEP DY
190 LPRINT USING " #.##  ";Y;
200  PRINT USING " #.##  ";Y;
210 NEXT Y
220 LPRINT"-------------------------------------------------
----------------------------"
230  PRINT"-------------------------------------------------
----------------------------"
240 LPRINT
250  PRINT
260 I=1
270 FOR X=0 TO 3 STEP DX
280 LPRINT USING " #.##  ";X;
290  PRINT USING " #.##  ";X;
300 FOR Y=0 TO 9.000001E-02 STEP DY
```

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```
310 Z=X+Y
320 T=Z*.6745/SQR(2)
330 GOSUB 14000
340 ERFT=E
350 LPRINT USING " #.####";ERFT;
360  PRINT USING " #.####";ERFT;
370 NEXT Y
380 LPRINT
390  PRINT
400 IF I/5=INT(I/5) THEN PRINT
410 I=I+1
420 NEXT X
430 END
```

Test Example:

```
Z=  1.2
Range or cross-range probability of hit P=  .5817148
```

E.3.6 Probability of Destruction of a Point Target with Offset Distribution

Subroutines:
PD2.DAT
PD2.SUB

Purpose: To compute the probability of destruction of a point target by warhead with a lethal weapon radius R, aiming offset RO, and standard deviation SIGMA.

Source: Eqs. (3.69), (3.70), and (3.71).

Inputs:
PD2.DAT
The following ratios must be specified:
R/SIGMA and RO/SIGMA.

Program Files: PD2.DAT and PD2.TAB

```
100 '***************   PD2.DAT   **********************
110 'PROBABILITY OF DESTRUCTION OF A SURFACE POINT TARGET
120 'Input data for the subroutine PD2.SUB which calculates
     the probability of destruction of point target by warhead
     with a lethal weapon radius R and RO as the offset from
     the mean point of impact and the standard deviation SIGMA.
130 'Values of X = R/SIGMA and Y = RO/SIGMA must be
     specified on lines 140 and 150.
140 X=1
150 Y=1
160 GOSUB 15000
170 LPRINT "X=R/SIGMA= "; X
180  PRINT "X=R/SIGMA= "; X
190 LPRINT "Y=RO/SIGMA= "; Y
200  PRINT "Y=RO/SIGMA= "; Y
```

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```
210 LPRINT "Probability of Destruction PD2= "; PD2
220  PRINT "Probability of Destruction PD2= "; PD2
230 END
```

Test Example:

```
X=R/SIGMA=  1
Y=RO/SIGMA=  1
Probability of Destruction PD2=  .2671202
```

E.3.7 Probability of Destruction of a Space Point Target with Offset Distribution

Subroutines:
PD3.DAT
PD3.SUB

Purpose: To compute the probability of destruction of a point target in space by a warhead with the lethal radius R, the offset from the mean point of detonation RO, and the standard deviation SIGMA.

Source: Eq. (3.91) and Ref. 1

Inputs:
PD3.DAT
The following ratios must be specified:
R/SIGMA and RO/SIGMA.

Program Files: PD3.DAT and PD3.SUB

```
100 '***************   PD3.DAT   **********************
110 'PROBABILITY OF DESTRUCTION OF A POINT TARGET IN SPACE
120 'Input data for the subroutine PD3.SUB which calculates
     the probability of destruction of a point target in space
     by a warhead with the lethal radius R, the offset from
     the mean point of impact RO, and the standard deviation SIGMA.
130 'Values of X=R/SIGMA and Y=RO/SIGMA must be specified on
     lines 140 and 150.
140 X=2
150 Y=1
160 GOSUB 16000
170 LPRINT "X=R/SIGMA= "; X
180  PRINT "X=R/SIGMA= "; X
190 LPRINT "Y=RO/SIGMA= "; Y
200  PRINT "Y=RO/SIGMA= "; Y
210 LPRINT "Probability of Destruction PD3= "; PD3
220  PRINT "Probability of Destruction PD3= "; PD3
230 END
```

Test Example:

```
X=R/SIGMA=  2
Y=RO/SIGMA=  1
Probability of Destruction PD3=  .6024558
```

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E.3.8 Expected Fraction of Damage of a Circular Target

Subroutines:
ED.DAT
ED.SUB

Purpose: To compute the expected fraction of damage for a circular surface target of radius K when the lethal radius of the warhead is R and its standard deviation is SIGMA.

Source: Eq. (3.114)

Inputs:
ED.DAT
The following ratios must be specified:
R/SIGMA and K/SIGMA.

Program File: ED.DAT

```
100 '***************   ED.DAT   **********************
110 'EXPECTED FRACTION OF DAMAGE OF A CIRCULAR TARGET
120 'Input data for the subroutine ED.SUB which calculates
     the expected fraction of damage (EDRK) for a circular
     surface target of radius K when the lethal radius of the
     warhead is R.  The standard deviation for the warhead is
     SIGMA.
130 'The values of RS = R/SIGMA and KS = K/SIGMA must be
     specified on lines 140 and 150.
140 RS=2
150 KS=3
160 LPRINT "RS= "; RS
170  PRINT "RS= "; RS
180 LPRINT "KS= "; KS
190  PRINT "KS= "; KS
200 GOSUB 17000
210 LPRINT "Expected Fraction of Damage EDRK= "; EDRK
220  PRINT "Expected Fraction of Damage EDRK= "; EDRK
230 END
```

Test Example:

```
RS=  2
KS=  3
Expected Fraction of Damage EDRK=  .4009691
```

E.3.9 Probability of Destruction of a Point Target with the Exponential Damage Characteristics

Subroutines:
PDEXPDF.DAT
PDEXPDF.SUB

Purpose: To compute the probability of destruction of a point surface target with the exponential damage characteristics (Parameter B) and the warhead standard deviation SIGMA.

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Source: Ref. 1: Eq. (3.118)

Inputs:
PDEXPDF.DAT
The following parameters must be specified:
B = constant in the exponential damage function.
SIGMA = Standard deviation.

Program File: PDEXPDF.DAT

```
100 '***************   PDEXPDF.DAT   ******************
110 'PROBABILITY OF TARGET DESTRUCTION (EXPONENTIAL DAMAGE FUNCTION)
120 'Input data for the subroutine PDEXPDF.SUB which calculates
     the probability of destruction of a point surface
     target with the exponential damage characteristics.
130 'The value of SIGMA and B must be specified on lines 120
     and 130, where SIGMA= standard deviation for the
     warhead and B=parameter in the damage function D(R/B).
140 'D(R/B)=EXP(-R/B) and R=radial distance from the point target.
150 SIGMA=2
160 B=1
170 LPRINT "SIGMA= "; SIGMA
180  PRINT "SIGMA= "; SIGMA
190 LPRINT "B= "; B
200  PRINT "B= "; B
210 GOSUB 18000
220 LPRINT "PROBABILITY OF DESTRUCTION (Exponential damage)=
    ";PD
230  PRINT "PROBABILITY OF DESTRUCTION (Exponential damage)=
    ";PD
240 END
```

Test Example:

```
SIGMA=  2
B=  1
PROBABILITY OF DESTRUCTION (Exponential damage)=  .1572587
```

E.3.10 Probability of Destruction of a Surface Point Target with the Inclined Step Damage Function

Subroutines:
PDINSDF.DAT
PDINSDF.SUB

Purpose: To compute the probability of destruction of a point surface target with the inclined step damage characteristics and the warhead standard deviation SIGMA.

Source: Eq. (3.123)

Inputs:
PDINSDF.DAT
The following parameters must be specified:
RSK = Sure kill radius.

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RSS = Sure save radius.
SIGMA = Standard deviation for the warhead.

Program Files: PDINSDF.DAT and PDINSDF.SUB

```
100 '***************   PDINSDF.DAT   ******************
110 'PROBABILITY OF TARGET DESTRUCTION (INCLINED STEP DAMAGE
    FUNCTION)
120 'Input data for the subroutine PDEXPDF.SUB which
    calculates the probability of destruction of a point
    surface target with the inclined step damage  characteristics.
130 'The value of SIGMA, RSK, and RSS must be specified on lines
    140, 150, and 160 where SIGMA = standard deviation for the
    warhead, RSK = sure kill radius, and RSS = sure save
    radius [NOTE: Use the same units for SIGMA, RSK, and SIGMA.
140 SIGMA=2
150 RSK=1
160 RSS=3
170 LPRINT "SIGMA= "; SIGMA
180  PRINT "SIGMA= "; SIGMA
190 LPRINT "RSK= "; RSK
200  PRINT "RSK= "; RSK
210 LPRINT "RSS= "; RSS
220  PRINT "RSS= "; RSS
230 GOSUB 19000
240 LPRINT "PROBABILITY OF DESTRUCTION (Inclined step
   damage)= ";PD
250  PRINT "PROBABILITY OF DESTRUCTION (Inclined step
    damage)= ";PD
260 END
```

Test Example:

```
SIGMA=  2
RSK=  1
RSS=  3
PROBABILITY OF DESTRUCTION (Inclined step damage)=  .3940716
```

E.3.11 Probability of Destruction of a Surface Point Target with the Log-Normal Damage Characteristics

Subroutines:
PDLOGNDF.DAT
PDLOGNDF.SUB

Purpose: To compute the probability of destruction of a point surface target with the log-normal damage characteristics and warhead standard deviation SIGMA.

Source: Eq. (3.137)

Inputs:
PDLOGNDF.DAT
The following parameters must be specified:

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RSK = Sure kill radius.
RSS = Sure save radius.
SIGMA = standard deviation.

Program File: PDLOGNDF.DAT

```
100 '***************   PDLOGNDF.DAT   *****************
110 'PROBABILITY OF TARGET DESTRUCTION (LOG-NORMAL DAMAGE FUNCTION)
120 'Input data for the subroutine PDLOGNDF.SUB which
     calculates the probability of destruction (PD) of a
     point target with the log-normal damage
     characteristics.
130 'The values of SIGMA, RSK, and RSS must be specified on lines
     140, 150, and 160, where SIGMA = Standard deviation,
     RSK = Sure kill radius, and RSS = Sure save radius
     [NOTE: Use the same units for SIGMA, RSK, and RSS].
140 SIGMA=2
150 RSK=.2
160 RSS=2
170 QR=RSK/RSS
180 X=RSS/SIGMA
190 LPRINT "SIGMA= "; SIGMA
200  PRINT "SIGMA= "; SIGMA
210 LPRINT "RSK= "; RSK
220  PRINT "RSK= "; RSK
230 LPRINT "RSS= "; RSS
240  PRINT "RSS= "; RSS
250 GOSUB 20000
260 LPRINT "Probability of Destruction (Log-normal damage)=
    "; PD
270  PRINT "Probability of Destruction (Log-normal damage)=
    "; PD
280 END
```

Test Example:

```
SIGMA=  2
RSK=  .2
RSS=  2
Probability of Destruction (Log-normal damage)=  .2907641
```

E.3.12 Weapon Effective Radius for Normal (Gaussian) Damage Function

Subroutines:
WRADGAU.DAT
WRADGAU.SUB

Purpose: To compute the weapon effective radius for a point target with Gaussian damage characteristics.

Source: Eq. (3.153)

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Inputs:
WRADGAU.DAT
The following parameters must be specified:
B = Constant in the Gaussian function.
SIGMA = Standard deviation.

Program File: WRADGAU.DAT

```
100 '***************   WRADGAU.DAT   ******************
110 'WEAPON EFFECTIVE RADIUS (GAUSSIAN DAMAGE FUNCTION)
120 'Input data for the subroutine WRADGAU.SUB which
     calculates the weapon effective radius (WR) for a point
     target with Gaussian damage characteristics.
130 'The values of SIGMA and B must be specified on
     lines 150 and 160, where SIGMA=standard deviation
     for the warhead and B=parameter in the damage  function D(R/B).
140 'D(R/B)=EXP((-R^2/2*B^2) and R= radial distance from the
     point target.
150 SIGMA=2
160 B=2
170 X=B/SIGMA
180 'WR=effective weapon radius/SIGMA
190 LPRINT "SIGMA= "; SIGMA
200  PRINT "SIGMA= "; SIGMA
210 LPRINT "B= "; B
220  PRINT "B= "; B
230 GOSUB 21000
240 LPRINT "Weapon effective radius (Gaussian damage)/SIGMA=
    "; WR
250  PRINT "Weapon effective radius (Gaussian damage)/SIGMA=
    "; WR
260 END
```

Test Example:

```
SIGMA=  2
B=  2
Weapon effective radius (Gaussian damage)/SIGMA=  1.17741
```

E.3.13 Weapon Effective Radius for Exponential Damage Function

Subroutines:
WRADEXP.DAT
WRADEXP.SUB

Purpose: To compute the weapon effective radius for a point target with exponential damage characteristics.

Source: Eq. (3.154)

Inputs:
WRADEXP.DAT
The following parameters must be specified:

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B = Constant in the exponential damage function.
SIGMA = standard deviation.

Program File: WRADEXP.DAT

```
100 '***************   WRADEXP.DAT   ******************
110 'WEAPON EFFECTIVE RADIUS (EXPONENTIAL DAMAGE FUNCTION)
120 'Input data for the subroutine WRADEXP.SUB which
     calculates the weapon effective radius (WR) for a point
     target with the exponential damage.
130 'The values of SIGMA, B and R must be specified on lines
     150, 160, and 170, where SIGMA=standard deviation for
     the warhead, B=parameter in the damage function D(R/B),
     and R= radial distance from the point target.
140 'D(R/B)=EXP(-R/B).
150 SIGMA=2
160 B=2
170 R=2
180 X=B/SIGMA
190 'WR=effective weapon radius/SIGMA
200 LPRINT "SIGMA= "; SIGMA
210  PRINT "SIGMA= "; SIGMA
220 LPRINT "B= "; B
230  PRINT "B= "; B
240 GOSUB 22000
250 LPRINT "Weapon effective radius (Exponential damage)/
     SIGMA="; WR
260  PRINT "Weapon effective radius (Exponential damage)/
     SIGMA="; WR
270 END
```

Test Example:

```
SIGMA=  2
B=  2
Weapon effective radius (Exponential damage)/SIGMA= .9187848
```

E.3.14 Weapon Effective Radius for Inclined Step Damage Function

Subroutines:
WRADINS.DAT
WRADINS.SUB

Purpose: To compute the weapon effective radius for a point target with inclined step damage characteristics.

Source: Eq. (3.154)

Inputs:
POISSON.DAT
The following parameters must be specified:
RSK = Sure kill radius.
RSS = Sure save radius.
SIGMA = Standard deviation.

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Program Files: WRADINS.DAT

```
100 '***************   WRADINS.DAT   ******************
110 'WEAPON EFFECTIVE RADIUS (INCLINED STEP DAMAGE FUNCTION)
120 'Input data for the subroutine WRADINS.SUB which
     calculates the weapon effective radius (WR)
     for a point surface target with inclined step damage
     characteristics.
130 'The value of SIGMA, RSK, and RSS must be specified on lines
    140, 150, and 160 where SIGMA = standard deviation for the
    warhead, RSK = sure kill radius, and RSS = sure save radius
    [NOTE: Use the same units for SIGMA, RSK, and RSS].
140 SIGMA=2
150 RSK=1
160 RSS=3
170 LPRINT "SIGMA= "; SIGMA
180  PRINT "SIGMA= "; SIGMA
190 LPRINT "RSK= "; RSK
200  PRINT "RSK= "; RSK
210 LPRINT "RSS= "; RSS
220  PRINT "RSS= "; RSS
230 GOSUB 23000
240 LPRINT "Weapon effective radius (Inclined step
    damage)/SIGMA= ";WR
250  PRINT "Weapon effective radius (Inclined step
     damage)/SIGMA= ";WR
260 END
```

Test Example:

```
SIGMA=  2
RSK=  1
RSS=  3
Weapon effective radius (Inclined step damage)/SIGMA=1.000993
```

E.3.15 Weapon Effective Radius for Log-Normal Damage Function

Subroutines:
WRADLOGN.DAT
WRADLOGN.SUB

Purpose: To compute the weapon effective radius for a point target with the log-normal damage characteristics.

Source: Eq. (3.155)

Inputs:
WRADLOGN.DAT
The following parameters must be specified:
RSK = Sure kill radius.
RSS = Sure save radius.
SIGMA = Standard deviation.

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Program Files: WRADLOGN.DAT

```
100 '****************   WRADLOGN.DAT   *****************
110 'WEAPON EFFECTIVE RADIUS (LOG-NORMAL DAMAGE FUNCTION)
120 'Input data for the subroutine WRADLOGN.SUB which
     calculates the weapon effective radius (WR) for a point
     target with log-normal damage characteristics.
130 'The values of SIGMA, RSK, and RSS must be specified on
     lines 140, 150, and 160, where SIGMA = Standard deviation,
     RSK = Sure kill radius, and RSS = Sure save radius
     [NOTE: Use the same units for SIGMA, RSK, and RSS].
140 SIGMA=2
150 RSK=.2
160 RSS=2
170 QR=RSK/RSS
180 X=RSS/SIGMA
190 LPRINT "SIGMA= "; SIGMA
200  PRINT "SIGMA= "; SIGMA
210 LPRINT "RSK= "; RSK
220  PRINT "RSK= "; RSK
230 LPRINT "RSS= "; RSS
240  PRINT "RSS= "; RSS
250 GOSUB 24000
260 LPRINT "Weapon effective radius (Log-normal
    damage)/SIGMA= "; WR
270  PRINT "Weapon effective radius (Log-normal
     damage)/SIGMA= "; WR
280 END
```

Test Example:

```
SIGMA=  2
RSK=  .2
RSS=  2
Weapon effective radius (Log-normal damage)/SIGMA=  .4124173
```

E.3.16 Weapon Effective Radius for Shrapnel Damage Function

Subroutines:
WRADSHRA.DAT
WRADSHRA.SUB

Purpose: To compute the weapon effective radius for a point target with the shrapnel damage characteristics.

Source: Eq. (3.158).

Inputs:
WRADSHRA.DAT
The following parameters must be specified:
SIGMA = Standard deviation.
RASIGMA = Standard deviation.

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Program File: WRADSHRA.DAT

```
100 '***************   WRADSHRA.DAT   ***************
110 'WEAPON EFFECTIVE RADIUS (SHRAPNEL DAMAGE FUNCTION FOR
    SURFACE TARGETS)
120 'Input data for the subroutine WRADSHRA.SUB which
    calculates the weapon effective radius (WR) for a point
    surface target with shrapnel damage characteristics.
130 'The values of SIGMA and RA must be specified on lines 140
    and 150 where SIGMA = standard deviation for the warhead
    and RA = radius for the single shrapnel to hit
    the target.
140 SIGMA=2
150 RA=1
160 LPRINT "SIGMA= "; SIGMA
170  PRINT "SIGMA= "; SIGMA
180 LPRINT "RA= "; RA
190  PRINT "RA= "; RA
200 GOSUB 46000
210 LPRINT "Weapon effective radius (shrapnel damage function
   for surface targets= "; WR
220  PRINT "Weapon effective radius (shrapnel damage function
   for surface targets= "; WR
230 END
```

Test Example:

```
SIGMA=  2
RA= 1
Weapon effective radius (shrapnel damage function for surface
targets) = 2.369097
```

E.4 Deterministic Combat Models

E.4.1 Directed Fire Lanchester Deterministic Model

Subroutines:
DFLANMOD.DAT
DFLANMOD.SUB

Purpose: To compute the nondimensional force strengths MUB and MUR of the Blue and Red forces engaged in combat with a directed fire exchange, where MUB=Blue force level/The initial Blue force level and MUR=Red force level/The initial Red force level.

Source: Eqs. (4.38) and (4.39).

Inputs:
BINOMIAL.DAT
The following parameters must be specified:
PHI = Superiority parameter.
DT = Nondimensional time increment.
Z = Maximum number of time steps in the computation.

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Program File: DFLANMOD.DAT

```
100 '***************   DFLANMOD.DAT   *****************
110 'DIRECTED FIRE COMBAT DETERMINISTIC MODEL
120 'Input data for the calculation of the force strengths MUB
     and MUR engaged in combat with the directed fire exchange.
130 'Values of PHI, DT, and Z must be specified on lines
     140, 150 and 160, where PHI=Superiority parameter,
     DT=Nondimensional time increment, and Z=Maximum number
     of time steps, specifying the range of computations.
140 PHI=.5
150 DT=.1
160 Z=10
170 LPRINT "Directed fire combat model"
180  PRINT "Directed fire combat model"
190 LPRINT
200  PRINT
210 LPRINT "Superiority parameter PHI= "; PHI
220  PRINT "Superiority parameter PHI= "; PHI
230 LPRINT "Nondimensional time increment DT= "; DT
240  PRINT "Nondimensional time increment DT= "; DT
250 LPRINT "Required number of time steps Z= "; Z
260  PRINT "Required number of time steps Z= "; Z
270 LPRINT
280  PRINT
290 LPRINT "TAU=Nondimensional time"
300  PRINT "TAU=Nondimensional time"
310 LPRINT "MUB=Blue force strength/Initial strength (MB)"
320  PRINT "MUB=Blue force strength/Initial strength (MB)"
330 LPRINT "MUR=Red force strength/Initial strength (MR)"
340  PRINT "MUR=Red force strength/Initial strength (MR)"
350 LPRINT
360  PRINT
370 LPRINT "  TAU    "; "    MUB    "; "    MUR    "
380  PRINT "  TAU    "; "    MUB    "; "    MUR    "
390 LPRINT USING "#.####    "; T; MUB; MUR
400  PRINT USING "#.####    "; T; MUB; MUR
410 GOSUB 25000
420 END
```

Test Example:

```
Directed fire combat model
Superiority parameter PHI=  .5
Nondimensional time increment DT=  .1
Required number of time steps Z=  10
TAU=Nondimensional time
MUB=Blue force strength/Initial strength (MB)
MUR=Red force strength/Initial strength (MR)

  TAU        MUB        MUR
0.0000     0.0000     0.0000
```

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```
0.0000    1.0000    1.0000
0.1000    0.8047    0.9549
0.2000    0.6174    0.9194
0.3000    0.4363    0.8931
0.4000    0.2596    0.8757
0.5000    0.0854    0.8671
```

E.4.2 Area Fire Lanchester Deterministic Model

Subroutines:
AFLANMOD.DAT
AFLANMOD.SUB

Purpose: To compute the nondimensional force strengths MUB and MUR of the Blue and Red forces engaged in combat with an area fire exchange, MUB=Blue force level/The initial Blue force level and MUR=Red force level/The initial Red force level.

Source: Eqs. (4.67), (4.68), and (4.71).

Inputs:
AFLANMOD.DAT
The following parameters must be specified:
PHI = Superiority parameter.
DT = Nondimensional time increment.
Z = Maximum number of time steps in the computation.

Program File: AFLANMOD.DAT

```
100 '***************   AFLANMOD.DAT   *****************
110 'AREA FIRE COMBAT DETERMINISTIC MODEL
120 'Input data for the calculation of the force strengths
     MUB and MUR engaged in combat with the area fire
     exchange.
130 'Values of PHI, DT, and Z must be specified on lines
     140, 150, and 160, where PHI=Superiority parameter,
     DT=Nondimensional time increment, and Z=Maximum number
     of time steps, specifying the range of computations.
140 PHI=1
150 DT=.1
160 Z=10
170 LPRINT "Area fire combat model"
180  PRINT "Area fire combat model"
190 LPRINT
200  PRINT
210 LPRINT "Superiority parameter PHI= "; PHI
220  PRINT "Superiority parameter PHI= "; PHI
230 LPRINT "Nondimensional time increment DT= "; DT
240  PRINT "Nondimensional time increment DT= "; DT
250 LPRINT "Required number of time steps Z= "; Z
260  PRINT "Required number of time steps Z= "; Z
270 LPRINT
280  PRINT
290 LPRINT "TAU=Nondimensional time"
```

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```
300  PRINT "TAU=Nondimensional time"
310 LPRINT "MUB=Blue force strength/Initial force strength
(MB)"
320  PRINT "MUB=Blue force strength/Initial force strength
(MB)"
330 LPRINT "MUR=Red force strength/Initial force strength
(MR)"
340  PRINT "MUR=Red force strength/Initial force strength
(MR)"
350 LPRINT
360  PRINT
370 LPRINT "  TAU    "; "    MUB    "; "    MUR    "
380  PRINT "  TAU    "; "    MUB    "; "    MUR    "
390 GOSUB 26000
400 END
```

Test Example:

```
Area fire combat model
Superiority parameter PHI=  1
Nondimensional time increment DT=  .1
REQUIRED NUMBER OF TIME STEPS Z=  10
TAU=Nondimensional time
MUB=BLUE FORCE STRENGTH/INITIAL FORCE STRENGTH (MB)
MUR=Red force strength/Initial force strength (MR)

  TAU         MUB         MUR
0.0000      1.0000      1.0000
0.1000      0.9091      0.9091
0.2000      0.8333      0.8333
0.3000      0.7692      0.7692
0.4000      0.7143      0.7143
0.5000      0.6667      0.6667
0.6000      0.6250      0.6250
0.7000      0.5882      0.5882
0.8000      0.5556      0.5556
0.9000      0.5263      0.5263
1.0000      0.5000      0.5000
```

E.4.3 Guerrilla Warfare Deterministic Model

Subroutines:
GUERRMOD.DAT
GUERRMOD.SUB

Purpose: To compute the force level strengths MUB and MUR of the Blue and Red forces engaged in guerrilla warfare combat (Blue forces with an area fire and Red forces with a directed fire), where MUB=Blue force level/The initial Blue force level and MUR=Red force level/The initial Red force level.

Source: Eqs. (4.84), (4.85), (4.89), (4.90), (4.93) and (4.94).

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Inputs:
GUERRMOD.DAT
The following parameters must be specified:
PHI = Superiority parameter.
DT = Nondimensional time increment.
Z = Maximum number of time steps in the computation.

Program File: GUERRMOD.DAT

```
100 '***************   GUERRMOD.DAT   *****************
110 'GUERRILLA WARFARE DETERMINISTIC MODEL
120 'Input data for the calculation of the force strength ratios
     MUB and MUR engaged in combat with the guerrilla fire
     exchange.
130 'Values of PHI, DT, and Z must be specified on lines 140,
      150, and 160, where PHI=Superiority parameter,
      DT=Nondimensional time increment, and Z=Maximum number of
      time steps, specifying the range of computations.
140 PHI=1
150 DT=.1
160 Z=10
170 LPRINT "Guerrilla fire combat model"
180  PRINT "Guerrilla fire combat model"
190 LPRINT
200  PRINT
210 LPRINT "Superiority parameter PHI= "; PHI
220  PRINT "Superiority parameter PHI= "; PHI
230 LPRINT "Nondimensional time increment DT= "; DT
240  PRINT "Nondimensional time increment DT= "; DT
250 LPRINT "Required number of time steps Z= "; Z
260  PRINT "Required number of time steps Z= "; Z
270 LPRINT
280  PRINT
290 LPRINT "TAU=Nondimensional time"
300  PRINT "TAU=Nondimensional time"
310 LPRINT "MUB=Blue force strength/Initial strength (MB)"
320  PRINT "MUB=Blue force strength/Initial strength (MB)"
330 LPRINT "MUR=Red force strength/Initial strength (MR)"
340  PRINT "MUR=Red force strength/Initial strength (MR)"
350 LPRINT
360  PRINT
370 LPRINT "  TAU    "; "    MUB    "; "    MUR    "
380  PRINT "  TAU    "; "    MUB    "; "    MUR    "
390 GOSUB 27000
400 END
```

Test Example:

```
Guerrilla fire combat model
Superiority parameter PHI=  1
Nondimensional time increment DT=  .1
Required number of time steps Z=  10
```

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```
TAU=Nondimensional time
MUB=Blue force strength/Initial strength (MB)
MUR=Red force strength/Initial strength (MR)

  TAU        MUB        MUR
0.0000     1.0000     1.0000
0.1000     0.9047     0.9092
0.2000     0.8176     0.8343
0.3000     0.7374     0.7719
0.4000     0.6629     0.7197
0.5000     0.5932     0.6759
0.6000     0.5275     0.6391
0.7000     0.4652     0.6082
0.8000     0.4057     0.5823
0.9000     0.3486     0.5607
1.0000     0.2934     0.5430
```

E.5 Probabilistic Combat Models

E.5.1 Probabilistic Directed Fire Model

Subroutines:
PDFMODEL.DAT
PDFMODEL.SUB

Purpose: To compute the probability P(I,J) of Blue and Red forces being in state (I,J) when these forces are engaged in the directed fire exchange.

Source: Eqs. (5.47) and (5.51)–(5.56).

Inputs:
PDFMODEL.DAT
The following parameters must be specified:
PHI = Superiority parameter.
M = Number of Blue troops (or weapons).
N = Number of Red troops (or weapons).
T = Nondimensional time increment.
Z = Number of specified time steps for the solution.

Program File: PDFMODEL.DAT

```
100 '***************   PDFMODEL.DAT   *****************
110 'DIRECTED FIRE COMBAT (PROBABILISTIC MODEL)
120 'Input data for the calculation of the probability
    P(I,J) of being in combat states (I,J) when engaged in
    the directed fire exchange.
130 'Values of PHI, M, N, T, and Z must be specified on line
    160, where PHI= Superiority parameter, M=Number of Blue
    troops (or weapons), N=Number of Red troops (or
    weapons), T=Nondimensional time increment, and Z=Number
    of specified time steps for the required solution.\\
140 'Nondimensional time for computed values of P2(I,J)=T*Z.
```

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```
150 'Output P(I,J)=P2(I,J)=Probability of being in state
    (I,J).  This output is also recorded on disk drive "A"
    as A:PDFPLOT.DAT in the following format I, J, P(I,J)
    for subsequent plotting.
160 PHI=1.5: M=10: N=10: T=.01: Z=5
170 TAU=T*Z
180 LPRINT "Probabilistic directed fire model"
190  PRINT "Probabilistic directed fire model"
200 LPRINT
210  PRINT
220 LPRINT "Probability table of combat states P(I,J); ";
    "TAU="; TAU; "PHI="; PHI
230  PRINT "Probability table of combat states P(I,J); ";
     "TAU="; TAU; "PHI="; PHI
240 LPRINT
250  PRINT
260 GOSUB 28000
270 END
```

Test Example:

```
The results are recorded as the A:PDFPLOT.DAT file;
however, they are too voluminous to be reproduced here.
```

E.5.2 Probabilistic Area Fire Model

Subroutines:
PAFMODEL.DAT
PAFMODEL.SUB

Purpose: To compute the probability P(I,J) of Blue and Red forces being in state (I,J) when these forces are engaged in the area fire exchange.

Source: Eq. (5.69) and Eqs. (5.72)–(5.77).

Inputs:
PAFMODEL.DAT
The following parameters must be specified:
PHI = Superiority parameter.
M = Number of Blue troops (or weapons).
N = Number of Red troops (or weapons).
T = Nondimensional time increment.
Z = Number of specified time steps for the solution.

Program Files: PAFMODEL.DAT

```
100 '***************   PAFMODEL.DAT   *****************
110 'AREA FIRE COMBAT (PROBABILISTIC MODEL)
120 'Input data for the calculation of the probability
    P(I,J) of being in combat states (I,J) when engaged in
    the area fire exchange.
130 'Values of PHI, M, N, T, and Z must be specified on line
    160, where PHI=Superiority parameter, M=Number of Blue
```

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```
     troops (or weapons), N=Number of Red troops (or
     weapons), T=Nondimensional time increment, and Z=Number
     of specified time steps for the required solution.\\
140 'Nondimensional time for computed values of P2(I,J)=T*Z.
150 'Output P(I,J)=P2(I,J)=Probability of being in state (I,J).
     This output is also recorded on disk drive "A" as A:PDFPLOT.DAT
     in the following format I, J, P(I,J) for subsequent plotting.
160 PHI=1.5: M=10: N=10: T=.01: Z=5
170 TAU=T*Z
180 LPRINT "Probabilistic area fire model"
190  PRINT "Probabilistic area fire model"
200 LPRINT
210  PRINT
220 LPRINT "Probability table of combat states P(I,J); ";
    "TAU="; TAU; "PHI="; PHI
230  PRINT "Probability table of combat states P(I,J); ";
     "TAU="; TAU; "PHI="; PHI
240 LPRINT
250  PRINT
260 GOSUB 29000
270 END
```

Test Example:

```
The results are recorded as the A:PAFPLOT.DAT file;
however, they are too voluminous to be reproduced here.
```

E.6 Strategic Defense

E.6.1 SAM Defense for Direct Penetration

Subroutines:
SAMDEFDP.TAB
SAMDEFDP.SUB

Purpose: To compute a table of the number of possible SAM launches against the penetrating aircraft heading toward the SAM site.

Source: Eqs. (6.81) and (6.85).

Inputs:
SAMDEFPD.DAT
The following parameters must be specified:
M = A digit specifying the range for X=0.1xM.
N = A digit specifying the range for Y=0.1xN.
M,N < 10.
X = Aircraft velocity/SAM velocity.
Y = SAM minimum range/SAM maximum range.

Program Files: SAMDEFDP.DAT and SAMDEFDP.SUB

```
100 '***************    SAMDEFDP.TAB    *****************
110 'SAM DEFENSE FOR DIRECT PENETRATION
```

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```
120 'Input ranges of variables into SAMDEFDP.SUB to create a
    table of the number of possible launches against the
    penetrating aircraft.
130 'SAM missiles are launched against the penetrating
    aircraft for both the inward and outward flight.
140 'U=Penetrating aircraft velocity.
150 'V=SAM missile velocity.
160 'X=U/V
170 'RMAX=Maximum SAM range; RMIN=Minimum SAM range.
180 'R=RMIN/RMAX.
190 'N(X,Y)=Number of possible SAM launches.
200 'Values of M and N must be specified on lines 210 and 220;
    this produces ranges of X=0.1*I with I=1,2,...,M, and
    Y=0.1*J with J=1,2,...,N, where M,N<10.
210 M=9
220 N=9
230 GOSUB 30000
240 END
```

Test Example:

```
TABLE OF THE NUMBER OF POSSIBLE SAM LAUNCHES
          [For INWARD flight only]
X=U/V=Aircraft velocity/SAM velocity
Y=RMIN/RMAX=SAM minimum range/SAM maximum range
```

X\Y	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.1	25	17	13	10	8	6	4	3	2
0.2	13	9	7	6	4	3	2	2	1
0.3	9	7	5	4	3	2	2	1	1
0.4	7	5	4	3	3	2	2	1	1
0.5	6	4	3	3	2	2	1	1	1
0.6	5	4	3	2	2	2	1	1	1
0.7	5	4	3	2	2	1	1	1	1
0.8	4	3	3	2	2	1	1	1	1
0.9	4	3	2	2	2	1	1	1	1

```
TABLE OF THE NUMBER OF POSSIBLE SAM LAUNCHES
       [For INWARD and OUTWARD flight]
X=U/V=Aircraft velocity/SAM velocity
Y=RMIN/RMAX=SAM minimum range/SAM maximum range
```

X\Y	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.1	47	33	25	19	15	11	8	6	3
0.2	24	17	13	11	8	6	4	4	2
0.3	16	12	9	7	5	4	4	2	2
0.4	12	9	7	5	5	4	3	2	2

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0.5	10	7	5	5	4	3	2	2	2
0.6	8	6	5	3	3	3	2	2	2
0.7	7	6	5	3	3	2	2	2	2
0.8	6	4	4	3	3	2	2	2	2
0.9	5	4	3	3	3	2	2	2	2

E.6.2 SAM Defense for Offset Penetration

Subroutines:
SAMDEFOP.TAB
SAMDEFOP.SUB

Purpose: To compute a table of the number of possible SAM launches against the penetrating aircraft heading toward the SAM site along an offset path.

Source: Eqs. (6.87)–(6.94).

Inputs:

SAMDEFOP.TABT
The following parameters must be specified:
C = U/V = Penetrating aircraft velocity/SAM velocity.
$0 < C \leq 1$.
K = Z/R = Distance to the offset path of penetration.
$0 < K < 1$.
M = A digit specifying the range for CX=0.1xM.
N = A digit specifying the range for KY=0.1xN.
M,N < 10.

Program File: SAMDEFOP.TAB

```
100 '***************   SAMDEFOP.TAB   *****************
110 'SAM DEFENSE FOR OFFSET PENETRATION
120 'Input ranges of variables into SAMDEFOP.SUB to create a
     table of the number of possible launches against the
     penetrating aircraft on an offset path.
130 'SAM missiles are launched against the penetrating
     aircraft following a path offset in relation to the SAM
     site.
140 'U=Penetrating aircraft velocity
150 'V=SAM missile velocity
160 'R=SAM maximum range.
170 'Z=Distance of the SAM site from the offset path of
     penetration.
180 'C=U/V; 0<C<1 or =1
190 'K=Z/R; 0<K<1
200 'N(C,K)=Number of possible launches.
210 'Values of M and N must be specified on lines 220 and
     230; this produces ranges of C=0.1*I with I=1,2,...,M
     and K=0.1*J with J=1,2,...,N, where 0<M<11 and
     0<N<10.
220 M=10
```

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```
230 N=9
240 GOSUB 31000
250 END

TABLE OF THE NUMBER OF POSSIBLE SAM LAUNCHES
           [Offset penetration]
C=U/V=Aircraft velocity/SAM velocity
K=Z/R=Offset distance/Sam maximum range
```

C\K	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.1	60	46	38	32	27	22	18	14	10
0.2	30	23	19	16	14	11	9	7	5
0.3	20	16	13	11	9	8	6	5	4
0.4	15	12	10	8	7	6	5	4	3
0.5	12	9	8	7	6	5	4	3	2
0.6	10	8	7	6	5	4	3	3	2
0.7	9	7	6	5	4	3	3	2	2
0.8	7	6	5	4	4	3	3	2	2
0.9	6	5	4	4	3	3	2	2	1
1.0	5	4	4	3	3	2	2	2	1

E.7 Theater Missile Defense

E.7.1 Missile Penetration for Random Assignments of Targets; One-Layer Defense

Subroutines:
MPENRAN1.DAT
MPENRAN1.SUB

Purpose: To compute probabilities of missile penetrations for random assignment of targets (attacking missiles) in a single-layer penetration when the individual interceptor probability of kill is given (P). The computations are carried out for a range of values of the ratio X=N/W, where N is the number of intercepts and W is the number of attacking missiles.

Source: Eq. (7.6)

Inputs:
MPENRAN1.DAT
The following values must be specified:
P=probability of interceptor kill.
DX=steps for X
MX=maximum value for X

Program Files: MPENRAN1.DAT and MPENRAN1.SUB

```
100  '**********************   MPENRAN1.DAT   **********************
110 'MISSILE PENETRATION FOR RANDOM ASSIGNMENT OF TARGETS;
111 'ONE-LAYER DEFENSE.
120 'Input data for the subroutine MPENRAN1.SUB
130 'The following values must be specified:
```

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```
131 'P=Interceptor probability of kill,
132 'DX=Steps for X,
133 'MX=Maximum value for X, where X=N/W, N=number of interceptors,
134 'and W=Number of warheads.
140 P=.8
150 DX=.2
160 MX=3!
170 GOSUB 1000
180 END
```

Test Example:

```
P=          0.8
X=          Y=
0           1.0
0.2         0.8521438
0.4         0.726149
0.6         0.6187835
0.8         0.5272925
1.0         0.449329
1.2         0.3828929
1.4         0.3262798
1.6         0.2780373
1.8         0.2369277
2.0         0.2018965
2.2         0.1720448
2.4         0.1466069
2.6         0.1249302
2.8         0.1064585
```

E.7.2 Missile Penetration for Random Assignments of Targets; Two-Layer Defense

Subroutines:
MPENRAN2.DAT
MPENRAN2.SUB

Purpose: To compute probabilities of missile penetrations for random assignment of targets (attacking missiles) in a two-layer penetration when the individual interceptor probability of kill is given (P). The computations are carried out for a range of values of the ratio X=N/W, where N is the number of intercepts and W is the number of attacking missiles.

Source: Eq. (7.13)

Inputs:
MPENRAN2.DAT
The following values must be specified:
P=probability of interceptor kill.
DX=steps for X
MX=maximum value for X

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Program Files: MPENRAN2.DAT and MPENRAN2.SUB

```
100  '**********************   MPENRAN2.DAT   **********************
110 'MISSILE PENETRATION FOR RANDOM ASSIGNMENT OF TARGETS;
111 'TWO-LAYER DEFENSE.
120 'Input data for the subroutine MPENRAN2.SUB
130 'The following values must be specified:
131 'P=interceptor probability of kill,
132 'DX=steps for X,
133 'MX=maximum value for X, where X=N/W, N=number of interceptors,
134 'and W=number of warheads.
135 'Z=N1/N where N1 is the number of interceptors in the 1st layer.
140 P=.8
150 DX=.2
160 MX=3.1
170 Z=.5
180 GOSUB 2000
190 END
```

Test Example:

```
P=          0.8
Z=N1/N=     0.5
X=          Y=
0           1.0
0.2         0.8464849
0.4         0.7062671
0.6         0.579784
0.8         0.4673481
1.0         0.3690845
1.2         0.2848719
1.4         0.2143004
1.6         0.156649
1.8         0.1108927
2.0         0.0757412
2.2         4.970787E-02
2.4         3.120336E-02
2.6         1.864109E-02
2.8         1.053937E-02
3.0         5.604746E-03
```

E.7.3 Missile Penetration for Uniform Assignments of Targets; One-Layer Defense

Subroutines:
MPENUNI1.DAT
MPENUNI1.SUB

Purpose: To compute probabilities of missile penetrations for uniform assignment of targets (attacking missiles) in a single-layer penetration when the individual interceptor probability of kill is given (P). The computations are carried out for a range of values of

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the ratio X=N/W, where N is the number of intercepts and W is the number of attacking missiles.

Source: Eq. (7.18)

Inputs:
MPENRUNI.DAT
The following values must be specified:
P=probability of interceptor kill.
DX=steps for X
MX=maximum value for X

Program Files: MPENUNI1.DAT and MPENUNI1.SUB

```
100  '*********************   MPENUNI1.DAT   ***********************
110 'MISSILE PENETRATION FOR UNIFORM ASSIGNMENT OF TARGETS;
111 'ONE-LAYER DEFENSE.
120 'Input data for the subroutine MPENUNI1.SUB
130 'The following values must be specified:
131 'P=interceptor probability of kill,
132 'DX=steps for X,
133 'MX=maximum value for X, where X=N/W, N=number of interceptors,
134 'and W=number of warheads.
140 P=.8
150 DX=.2
160 MX=3.1
170 GOSUB 3000
180 END
```

Test Example:

```
P=          0.8
X=          Y=
0           1.0
0.2         0.8400001
0.4         0.68
0.6         0.52
0.8         0.36
1.0         0.2
1.2         0.168
1.4         0.136
1.6         0.104
1.8         7.199996E-02
2.0         3.999999E-02
2.2         3.359999E-02
2.4         2.719999E-02
2.6         2.079998E-02
2.8         1.439999E-02
3.0         7.999997E-03
```

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E.7.4 Missile Penetration for Uniform Assignments of Targets; Two-Layer Defense

Subroutines:
MPENUNI2.DAT
MPENUNI2.SUB

Purpose: To compute probabilities of missile penetrations for random assignment of targets (attacking missiles) in a two-layer penetration when the individual interceptor probability of kill is given (P). The computations are carried out for a range of values of the ratio X=N/W, where N is the number of intercepts and W is the number of attacking missiles.

Source: Eq. (7.24)

Inputs:
MPENUNI2.DAT
The following values must be specified:
P=probability of interceptor kill.
DX=steps for X
MX=maximum value for X

Program Files: MPENUNI2.DAT and MPENUNI2.SUB

```
100  '**********************   MPENUNI2.DAT   **********************
110 'MISSILE PENETRATION FOR UNIFORM ASSIGNMENT OF TARGETS;
111 'TWO-LAYER DEFENSE.
120 'Input data for the subroutine MPENUNI2.SUB
130 'The following values must be specified:
131 'P=interceptor probability of kill,
132 'DX=steps for X,
133 'MX=maximum value for X, where X=N/W, N=number of interceptors,
134 'and W=number of warheads.
135 'Z=N1/N where N1 is number of intercepts in the 1st layer.
140 P=.8
150 DX=.2
160 MX=3.1
170 Z=.5
180 GOSUB 4000
190 END
```

Test Example:

```
P=          0.8
Z=          0.5
X=          Y=
0           1.0
0.2         0.8400001
0.4         0.68
0.6         0.52
0.8         0.36
1.0         0.2
1.2         9.119998E-02
1.4         4.639996E-02
```

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```
1.6          1.183999@-02
1.8          1.855997E-03
2.0          6.399991E-05
2.2          1.279992E-05
2.4          1.904633E-06
2.6          2.170865E-07
2.8          1.064955E-08
3.0          2.949096E-10
```

E.7.5 Probability of Zero Missile Penetrations for Specified Number of Warheads W

Subroutines:
PROBZERO.DAT
PROBZERO.SUB

Purpose: To compute probabilities of zero missile penetrations for a specified number of warheads W and the probability P of penetration of a single warhead (This should not be confused with p, which is the probability of kill by a single interceptor, as used in Sections 7.1–7.4).

Source: Eq. (7.45)

Inputs:
PROBZERO.DAT
The following values must be specified:
P=probability of warhead penetration.
DW=steps for W
MW=maximum value for W

Program Files: PROBZERO.DAT and PROBZERO.SUB

```
100  '**********************   PROBZERO.DAT   **********************
110 'PROBABILITY OF ZERO MISSILE PENETRATION
120 'Input data for the subroutine PROBZERO.SUB
130 'The following values must be specified:
131 'DW=steps for W,
132 'MW=maximum value of W, and
133 'P=probability of single missile penetration.
140 DW=10
150 MW=200
160 P=.01
170 GOSUB 5000
180 END
```

Test Example:

```
P=            .01
W=          PW(0)=
0           1.0
10          0.9043822
20          0.8179071
```

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```
30          0.7397008
40          0.6689721
50          0.6050065
60          0.5471571
70          0.4948392
80          0.4475237
90          0.4047325
100         0.3660329
110         0.3310336
120         0.2993809
130         0.2707547
140         0.2448658
150         0.2214522
160         0.2002775
170         0.1811274
180         0.1638084
190         0.1481454
200         0.13398
```

E.8 Tactical Engagements of Heterogeneous Forces

E.8.1 Heterogeneous Force Levels in Tactical Engagements (Directed Fire Model)

Subroutines:
HFDFMOD.DAT
HFDFMOD.SUB

Purpose: To compute the force levels XN(I) and the superiority parameter S in a tactical engagement with directed fire for a specified matrix of attrition coefficients C(I,J) and initial force levels XN(I).

Source: Eqs. (8.11) and (8.15), and Ref. 2.

Inputs:
HFMODEL.DAT
The following input data must be specified:
B = Number of BLUE forces.
R = Number of RED forces.
Z = Number of time steps in the integration.
DT = Nondimensional time step.
T = Number of iterations to compute the dominant eigenvalue.
C(I,J) = Matrix of attrition coefficients.
XN(I) = Column matrix of force levels at time T=0.

Program File: HFDFMOD ¿DAT

```
100 '***************   HFDFMOD.DAT   ***************
110 'HETEROGENEOUS FORCE COMBAT: DIRECTED FIRE; Lanchester
     model for tactical directed fire engagements.
```

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```
120 'Input data for the subroutine HFDFMOD.SUB for the
    calculation of the force levels in a tactical directed
    fire engagement of heterogeneous forces.
130 'Size of the square matrix C(I,J) is specified on line
    260, i.e. N=B+R.
140 'Number of BLUE forces B must be specified on line 240.
150 'Number of RED forces R must be specified on line 250.
160 'Z=number of time steps must be specified on line 280.
170 'DT=nondimensional time step must be specified on line 290.
180 'T=number of iterations used in HFDFMOD.SUB to calculate
    the dominant eigenvalue LAMBDA1 for the transpose of
    matrix C(I,J) must be specified on line 300.
190 'C(I,J)=matrix of attrition coefficients, specified as
    number of forces or weapons attritted per unit time,
    e.g. seconds, minutes, hours, or days.
200 'Matrix coefficients in C(I,J), and XN(J) must be
    specified on lines 400..., and 490...; additional lines
    may be added for larger size matrices.
210 'XN(I)=matrix of force levels.
220 'S=superiority parameter.
230 'TAU=nondimensional time (time x dominant eigenvalue
    LAMBDA1 of the transpose of C(I,J)).
240 B=2
250 R=2
260 N=B+R
270 DIM C(N,N), XN(N), I(N,N), K(N,N), Q1(N), D(N,N), XP(N)
280 Z=20
290 DT=.05
300 T=100
310 FOR I=1 TO N
320 FOR J=1 TO N
330 READ C(I,J)
340 NEXT J
350 NEXT I
360 FOR I=1 TO N
370 READ XN(I)
380 NEXT I
390 'DATA below specify the matrix coefficients in C(I,J)
    by rows.
400 DATA 0., 0., -1., -1.
410 DATA 0., 0., 0., -1.
420 DATA -1., -1., 0., 0.
430 DATA -1., -1., 0., 0.
440 'Here additional data for C(I,J) should be inserted when N>4.
450 '
460 '
470 '
480 'DATA below specify the initial values of XN(J) by rows, i.e.
    values of XN(J) at t=0.
490 DATA 200
500 DATA 150
510 DATA 75
```

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```
520 DATA 200
530 'Here additional data for XN(J) should be inserted when N>4.
540 '
550 '
560 '
570 FOR I=1 TO N
580 FOR J=1 TO N
590 IF I=J THEN I(I,J)=1 ELSE I(I,J)=0
600 NEXT J
610 NEXT I
620 'Assumed values of Q1(I) for the first iteration cycle in
     HFDFMOD.SUB
630 FOR I=1 TO B
640 Q1(I)=1
650 NEXT I
660 FOR I=B+1 TO N
670 Q1(N)=-1
680 NEXT I
690 GOSUB 32000
700 END
```

Test Example:

```
TABLE OF HETEROGENEOUS FORCE LEVELS: DIRECTED FIRE
[ X1, X2, X3, X4, etc. = force levels of the Blue and Red
forces ]
[ TAU=nondimensional time (time x dominant eigenvalue
LAMBDA1) ]
```

TAU	TIME	S	X1	X2	X3	X4
0.000	0.000	0.22	200	150	75	200
0.050	0.029	0.23	192	144	65	190
0.100	0.058	0.24	185	139	55	180
0.150	0.087	0.25	178	134	46	171
0.200	0.115	0.26	172	129	37	162
0.250	0.144	0.28	166	124	28	153
0.300	0.173	0.29	161	120	20	145
0.350	0.202	0.30	156	115	12	137
0.400	0.231	0.32	152	111	4	129

```
Dominant eigenvalue LAMBDA1 =  1.732051
Attrition coefficients in C(I,J)
 0            0           -1           -1
 0            0            0           -1
-1           -1            0            0
-1           -1            0            0
```

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E.8.2 Heterogeneous Force Levels in Tactical Engagements (Area Fire Model)

Subroutines:
HFAFMOD.DAT
HFAFMOD.SUB

Purpose: To compute the force levels XN(I) and the superiority parameter S in a tactical engagement with area fire for a specified matrix of attrition coefficients C(I,J) and initial force levels XN(I).

Source: Eqs. (8.53) and (8.15), and Ref. 2.

Inputs:
HFMODEL.DAT
The following input data must be specified:
B = Number of BLUE forces.
R = Number of RED forces.
Z = Number of time steps in the integration.
DT = Nondimensional time step.
T = Number of iterations to compute the dominant eigenvalue.
C(I,J) = Matrix of attrition coefficients.
XN(I) = Column matrix of force levels at time T=0.

Program Files: HFAFMOD.DAT

```
100 '***************   HFAFMOD.DAT   ***************
110 'HETEROGENEOUS FORCE COMBAT: AREA FIRE; Lanchester model
    for tactical area fire engagements.
120 'Input data for the subroutine HFAFMOD.SUB for the
    calculation of the force levels in a tactical area fire
    engagement of heterogeneous forces.
130 'Values of B, R, Z, and DT must be specified on lines 200,
    210, 240, and 250 where B=number of Blue forces, R=number
    of Red forces, Z=number of time steps in the integration,
    and DT=nondimensional time step. N=number of rows or
    columns.
140 'T=number of iterations used in HFAFMOD.SUB to calculate the
    dominant eigenvalue LAMBDA1 for the matrix C(I,J) must be
    specified on line 260.
150 'C(I,J)=matrix of attrition coefficients, specified as number
    of forces or weapons attrited per unit time, e.g., seconds,
    minutes, hours, or days.
160 'Matrix coefficients in C(I,J) and XN(J) must be specified on
    lines 360... and 450...; additional lines may be added for
    larger size matrices.
170 'XN(I)=column matrix of force levels.
180 'S=superiority parameter.
190 'TAU=nondimensional time (TIME x dominant eigenvalue LAMBDA1
    of the transpose of C(I,J)).
200 B=2
210 R=2
220 N=B+R
```

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```
230 DIM C(N,N), XN(N), I(N,N), K(N,N), Q1(N), D(N,N), XP(N)
240 Z=10
250 DT=.05
260 T=100
270 FOR I=1 TO N
280 FOR J=1 TO N
290 READ C(I,J)
300 NEXT J
310 NEXT I
320 FOR I=1 TO N
330 READ XN(I)
340 NEXT I
350 'DATA below specify the matrix coefficients in C(I,J) by rows.
360 DATA 0., 0., -1., -1.
370 DATA 0., 0., 0., -1.
380 DATA -1., -1., 0., 0.
390 DATA -1., -1., 0., 0.
400 ' Here additional data for C(I,J) should be inserted when N>4.
410 '
420 '
430 '
440 'DATA below specify the initial values of XN(J) by rows, i.e.,
     values of XN(J) at T=0.
450 DATA 200
460 DATA 150
470 DATA 75
480 DATA 150
490 'Here additional data for XN(J) should be inserted when N>4.
500 '
510 '
520 '
530 FOR I=1 TO N
540 FOR J=1 TO N
550 IF I=J THEN I(I,J)=1 ELSE I(I,J)=0
560 NEXT J
570 NEXT I
580 'Assumed values of Q1(I) for the first iteration cycle in
     HFAFMOD.SUB
590 FOR I=1 TO B
600 Q1(I)=1
610 NEXT I
620 FOR I=B+1 TO N
630 Q1(N)=-1
640 NEXT I
650 GOSUB 50000
660 END
```

Test Example:

```
TABLE OF HETEROGENEOUS FORCE LEVELS: AREA FIRE
[ X1, X2, X3, X4, etc. = force levels of the Blue and
Red forces ]
```

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```
[ TAU=nondimensional time (time x dominant eigenvalue
LAMBDA1) ]

============================================================
    TAU       TIME        S       X1       X2       X3      X4
------------------------------------------------------------
   0.000     0.000     0.38      200      150       75     150
   0.050     0.029     0.40      194      146       65     140
   0.100     0.058     0.42      188      142       56     131
   0.150     0.087     0.43      183      138       49     122
   0.200     0.115     0.44      178      135       43     115
   0.250     0.144     0.46      174      132       38     108
   0.300     0.173     0.47      170      129       34     102
   0.350     0.202     0.47      167      127       30      96
   0.400     0.231     0.48      164      124       26      90
   0.450     0.260     0.49      161      122       23      85
   0.500     0.289     0.50      159      120       21      81
============================================================

Dominant eigenvalue LAMBDA1 =  1.732051
Attrition coefficients in C(I,J)
 0            0           -1           -1
 0            0            0           -1
-1           -1            0            0
-1           -1            0            0
```

E.8.3 *Heterogeneous Force Levels in Tactical Engagements (Guerrilla Warfare Model)*

Subroutines:
HFGWMOD.DAT
HFGWMOD.SUB

Purpose: To compute the force levels XN(I) and the superiority parameter S in a tactical engagement with guerrilla warfare fire for a specified matrix of attrition coefficients C(I,J) and initial force levels XN(I).

Source: Eqs. (8.61) and (8.15), and Ref. 2.

Inputs:
HFMODEL.DAT
The following input data must be specified:
B = Number of BLUE forces.
R = Number of RED forces.
Z = Number of time steps in the integration.
DT = Nondimensional time step.
T = Number of iterations to compute the dominant eigenvalue. C(I,J) = Matrix of attrition coefficients.
XN(I) = Column matrix of force levels at time T=0.

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Program Files: HFGWMOD.DAT

```
100 '***************   HFGWMOD.DAT   ***************
110 'HETEROGENEOUS FORCE COMBAT: GUERRILLA WARFARE; Lanchester
     model for tactical guerrilla warfare engagements.
120 'Input data for the subroutine HFGWMOD.SUB for the
     calculation of the force levels in a tactical guerrilla
     warfare engagement of heterogeneous forces.
130 'Values of B, R, Z, and DT must be specified on lines 200,
     210, 240, and 260 where B=number of Blue forces, R=number
     of Red forces, Z=number of time steps in the integration,
     and DT=nondimensional time step. N=number of rows or columns.
140 'T=number of iterations used in HFGWMOD.SUB to calculate the
     dominant eigenvalue LAMBDA1 for the matrix C(I,J) must be
     specified on line 260.
150 'C(I,J)=matrix of attrition coefficients, specified as number
     of forces or weapons attrited per unit time, e.g., seconds,
     minutes, hours, or days.
160 'Matrix coefficients in C(I,J) and XN(J) must be specified on
     lines 360... and 450...; additional lines may be added for
     larger size matrices.
170 'XN(I)=column matrix of force levels.
180 'S=superiority parameter.
190 'TAU=nondimensional time (TIME x dominant eigenvalue LAMBDA1
     of the transpose of C(I,J)).
200 B=2
210 R=2
220 N=B+R
230 DIM C(N,N), XN(N), I(N,N), K(N,N), Q1(N), D(N,N), YO(N), XP(N)
240 Z=40
250 DT=.05
260 T=100
270 FOR I=1 TO N
280 FOR J=1 TO N
290 READ C(I,J)
300 NEXT J
310 NEXT I
320 FOR I=1 TO N
330 READ XN(I)
340 NEXT I
350 'DATA below specify the matrix coefficients in C(I,J) by rows.
360 DATA 0., 0., -1., -1.
370 DATA 0., 0., -1., -1.
380 DATA -1., -1., 0., 0.
390 DATA -1., -1., 0., 0.
400 ' Here additional data for C(I,J) should be inserted when N>4.
410 '
420 '
430 '
440 'DATA below specify the initial values of XN(J) by rows, i.e.,
     values of XN(J) at T=0.
450 DATA 150
460 DATA 150
```

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```
470 DATA 150
480 DATA 150
490 'Here additional data for XN(J) should be inserted when N>4.
500 '
510 '
520 '
530 FOR I=1 TO N
540 FOR J=1 TO N
550 IF I=J THEN I(I,J)=1 ELSE I(I,J)=0
560 NEXT J
570 NEXT I
580 'Assumed values of Q1(I) for the first iteration cycle in
     HFGWMOD.SUB
590 FOR I=1 TO B
600 Q1(I)=1
610 NEXT I
620 FOR I=B+1 TO N
630 Q1(N)=-1
640 NEXT I
650 GOSUB 52000
660 END
```

Test Example:

```
TABLE OF HETEROGENEOUS FORCE LEVELS: GUERRILLA WARFARE
[ X1, X2, X3, X4, etc. = force levels of the Blue and
Red forces ]
[ TAU=nondimensional time (time x dominant eigenvalue
LAMBDA1) ]
```

TAU	TIME	S	X1	X2	X3	X4
0.000	0.000	0.00	150	150	150	150
0.050	0.025	0.00	143	143	143	143
0.100	0.050	-0.00	135	135	136	136
0.150	0.075	-0.01	129	129	130	130
0.200	0.100	-0.01	122	122	124	124
0.250	0.125	-0.02	116	116	119	119
0.300	0.150	-0.03	110	110	114	114
0.350	0.175	-0.04	104	104	110	110
0.400	0.200	-0.05	99	99	106	106
0.450	0.225	-0.06	93	93	103	103
0.500	0.250	-0.08	88	88	100	100
0.550	0.275	-0.09	83	83	97	97
0.600	0.300	-0.10	78	78	94	94
0.650	0.325	-0.12	74	74	92	92
0.700	0.350	-0.13	69	69	89	89
0.750	0.375	-0.15	65	65	87	87
0.800	0.400	-0.17	60	60	85	85

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```
   0.850     0.425     -0.18      56        56        84        84
   0.900     0.450     -0.20      52        52        82        82
   0.950     0.475     -0.22      48        48        81        81
   1.000     0.500     -0.24      44        44        79        79
   1.050     0.525     -0.26      40        40        78        78
   1.100     0.550     -0.28      36        36        77        77
   1.150     0.575     -0.30      32        32        76        76
   1.200     0.600     -0.32      28        28        75        75
   1.250     0.625     -0.34      24        24        75        75
   1.300     0.650     -0.36      21        21        74        74
   1.350     0.675     -0.38      17        17        74        74
   1.400     0.700     -0.40      13        13        73        73
   1.450     0.725     -0.42      10        10        73        73
   1.500     0.750     -0.44       6         6        73        73
   1.550     0.775     -0.47       2         2        73        73
================================================================
Dominant eigenvalue LAMBDA1 =  2
Attrition coefficients in C(I,J)
 0            0           -1           -1
 0            0            0           -1
-1           -1            0            0
-1           -1            0            0
```

E.9 Reliability of Operations and Systems

E.9.1 Normal Probability Density Function and MTTF

Subroutines:
NORDFUNC.TAB
NORDFUNC.SUB

Purpose: To compute tables of nondimensional normal density function.

Source: Eqs. (9.24) and (9.25).

Inputs:
BINOMIAL.TAB
The following parameters must be specified:
DX = Time increment/SIGMA.
DY = Increment in MU/SIGMA.
M = Number of increments in DX.
N = Number of increments in DY.
SIGMA = Standard deviation.

Program File: NORDFUNC.TAB

```
100 '***************   NORDFUNC.TAB   ********************
110 'NORMAL PROBABILITY DENSITY FUNCTION AND MTTF
120 'Input ranges of variables into NORDFUNC.SUB
```

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```
     subroutine to create tables of the nondimensional
     normal density function (probability density function
     x SIGMA)  where SIGMA=parameter in the density function.
130 'Values of DX=TIME INCREMENT/SIGMA and DY=MU INCREMENT/
     SIGMA must be specified on lines 140 and 150, as well
     as their ranges M and N.
140 DX=.1
150 DY=1!
160 M=10
170 N=4
180 GOSUB 34000
190 END
```

Test Example:

TABLE OF THE NORMAL DENSITY FUNCTION x SIGMA
X=Time/Sigma
Y=MU/Sigma

X\Y	0.0	1.0	2.0	3.0
0.0	0.7979	0.2876	0.0552	0.0044
0.1	0.7939	0.3163	0.0671	0.0060
0.2	0.7821	0.3443	0.0808	0.0079
0.3	0.7628	0.3711	0.0962	0.0104
0.4	0.7365	0.3961	0.1135	0.0136
0.5	0.7041	0.4185	0.1325	0.0176
0.6	0.6664	0.4377	0.1532	0.0224
0.7	0.6245	0.4533	0.1754	0.0284
0.8	0.5794	0.4648	0.1987	0.0355
0.9	0.5322	0.4718	0.2229	0.0440
1.0	0.4839	0.4742	0.2476	0.0541

TABLE OF MTTF/SIGMA (NORMAL DISTRIBUTION)
Y=MU/SIGMA

Y	0.0	1.0	2.0	3.0
MTTFNOR	0.7979	1.1744	2.0075	3.0000

E.9.2 Normal Reliability Function

Subroutines:
NORRFUNC.TAB
NORRFUNC.SUB

Purpose: To compute tables of normal reliability function.

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Source: Eq. (9.27)

Inputs:
NORRFUNC.TAB
The following parameters must be specified:
DX = Time increment/SIGMA.
DY = Increment in MU/SIGMA.
M = Number of increments in DX.
N = Number of increments in DY.
SIGMA = Standard deviation.

Program File: NORRFUNC.TAB

```
100 '***************   NORRFUNC.TAB   ********************
110 'NORMAL RELIABILITY FUNCTION
120 'Input ranges of variables into NORRFUNC.SUB subroutine
     to create tables of the normal reliability function.
130 'Values of DX=TIME INCREMENT/SIGMA and DY=INCREMENT IN
     M/SIGMA must be specified on lines 140 and 150, as well
     as their ranges M and N on lines 160 and 170.
140 DX=.1
150 DY=1!
160 M=10
170 N=4
180 GOSUB 35000
190 END
```

Test Example:

TABLE OF THE NORMAL RELIABILITY FUNCTION
X=Time/Sigma
Y=M/Sigma

X\Y	0.0	1.0	2.0	3.0
0.0	1.0000	1.0000	1.0000	1.0000
0.1	0.9203	0.9698	0.9939	0.9995
0.2	0.8415	0.9368	0.9865	0.9988
0.3	0.7642	0.9010	0.9777	0.9979
0.4	0.6892	0.8626	0.9672	0.9967
0.5	0.6171	0.8219	0.9549	0.9951
0.6	0.5485	0.7790	0.9406	0.9931
0.7	0.4839	0.7344	0.9242	0.9906
0.8	0.4237	0.6885	0.9055	0.9874
0.9	0.3681	0.6416	0.8845	0.9835
1.0	0.3173	0.5943	0.8609	0.9786

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E.9.3 Normal Hazard Function

Subroutines:
NORHFUNC.TAB
NORHFUNC.SUB

Purpose: To compute tables of the nondimensional normal hazard function.

Source: Table 9.2

Inputs:
NORHFUNC.TAB
The following parameters must be specified:
DX = Time increment/SIGMA.
DY = Increment in MU/SIGMA.
M = Number of increments in DX.
N = Number of increments in DY.
SIGMA = Standard deviation.

Program File: NORHFUNC.TAB

```
100 '***************   NORHFUNC.TAB   ********************
110 'NORMAL HAZARD FUNCTION
120 'Input ranges of variables into NORHFUNC.SUB subroutine
     to create tables of the normal hazard function x SIGMA,
     where SIGMA is a parameter in the normal density function.
130 'Values of DX=TIME INCREMENT/SIGMA and DY=INCREMENT IN
     M/SIGMA must be specified on lines 140 and 150, as well
     as their ranges M and N on lines 160 and 170.
140 DX=.1
150 DY=1!
160 M=10
170 N=4
180 GOSUB 36000
190 END
```

```
TABLE OF THE NORMAL HAZARD FUNCTION x SIGMA
X=Time/Sigma
Y=M/Sigma
-------------------------------------------------------
 X\Y        0.0        1.0        2.0        3.0
-------------------------------------------------------
 0.0      2.5066     0.5480     0.0235     0.0002
 0.1      2.7236     0.5651     0.0236     0.0002
 0.2      2.9788     0.5850     0.0238     0.0002
 0.3      3.2802     0.6082     0.0240     0.0002
 0.4      3.6372     0.6353     0.0243     0.0002
 0.5      4.0621     0.6668     0.0246     0.0002
 0.6      4.5699     0.7035     0.0250     0.0002
 0.7      5.1798     0.7462     0.0254     0.0002
 0.8      5.9159     0.7960     0.0259     0.0002
 0.9      6.8093     0.8541     0.0266     0.0002
 1.0      7.8996     0.9221     0.0273     0.0002
-------------------------------------------------------
```

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E.9.4 Log-Normal Density Function and MTTF

Subroutines:
LOGNDFUN.TAB
LOGNDFUN.SUB

Purpose: To compute tables of the nondimensional log-normal density function.

Source: Eqs. (9.30) and (9.32).

Inputs:
LOGNDFUN.TAB
The following parameters must be specified:
DX = Time increment/SIGMA.
DY = Increment in MU/SIGMA.
M = Number of increments in DX.
N = Number of increments in DY.
SIGMA = Parameter in the log-normal density function.

Program Files: LOGNDFUN.TAB

```
100 '***************   LOGNDFUN.TAB   ********************
110 'LOG-NORMAL PROBABILITY DENSITY FUNCTION AND MTTF
120 'Input ranges of variables into LOGNDFUN.SUB subroutine
     to create tables of the nondimensional log-normal
     probability density function (probability density
     function x SIGMA), where SIGMA=parameter in the density
     function.
130 'Values of DX=TIME INCREMENT/SIGMA and DY=INCREMENT IN
     MU/SIGMA must specified on lines 140 and 150, as well
     as their ranges M and N on lines 160 and 170.
140 DX=.1
150 DY=.5
160 M=10
170 N=5
180 GOSUB 37000
190 END
```

Test Example:

```
TABLE OF THE LOG-NORMAL DENSITY FUNCTION x SIGMA
X=Time/Sigma
Y=MU/Sigma
```

X\Y	-1.0	-.5	0.0	0.5	1.0
0.1	1.7079	0.7858	0.2816	0.0786	0.0171
0.2	1.6566	1.0780	0.5463	0.2156	0.0663
0.3	1.3024	1.0379	0.6442	0.3114	0.1172
0.4	0.9939	0.9146	0.6554	0.3658	0.1590
0.5	0.7612	0.7831	0.6275	0.3916	0.1903

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```
 0.6       0.5899      0.6649      0.5836      0.3989      0.2124
 0.7       0.4634      0.5641      0.5348      0.3949      0.2271
 0.8       0.3688      0.4799      0.4864      0.3839      0.2360
 0.9       0.2971      0.4101      0.4408      0.3691      0.2406
 1.0       0.2420      0.3521      0.3989      0.3521      0.2420
-------------------------------------------------------------------

TABLE OF MTTF/SIGMA (LOG-NORMAL DISTRIBUTION)
Y=MU/SIGMA
-------------------------------------------------------------------
Y              -1.0       -.5        0.0        0.5        1.0
-------------------------------------------------------------------
MTTF/SIGMA    0.6065     1.0000     1.6487     2.7183     4.4817
-------------------------------------------------------------------
```

E.9.5 Log-Normal Reliability Function

Subroutines:
LOGNRFUN.TAB
LOGNRFUN.SUB

Purpose: To compute tables of the log-normal reliability function.

Source: Eq. (9.31).

Inputs:
LOGNRFUN.TAB
The following parameters must be specified:
DX = Time increment/SIGMA.
DY = Increment in MU/SIGMA.
M = Number of increments in DX.
N = Number of increments in DY.
SIGMA = Parameter in the log-normal density function.

Program File: LOGNRFUN.TAB

```
100 '***************   LOGNRFUN.TAB   ******************
110 'LOG-NORMAL RELIABILITY FUNCTION
120 'Input ranges of variables into LOGNRFUN.SUB subroutine
     to create tables of the log-normal reliability function.
130 'Values of DX=TIME INCREMENT/SIGMA and DY=INCREMENT IN
     MU/SIGMA must be specified on lines 140 and 150, as
     well as their ranges M and N on lines 160 and 170.
140 DX=.1
150 DY=.5
160 M=10
170 N=5
180 GOSUB 38000
190 END
```

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Test Example:

TABLE OF THE NORMAL LOG-RELIABILITY FUNCTION
X=Time/Sigma
Y=MU/Sigma

X\Y	-1.0	-.5	0.0	0.5	1.0
0.1	0.9036	0.9643	0.9893	0.9975	0.9995
0.2	0.7289	0.8664	0.9462	0.9825	0.9955
0.3	0.5808	0.7593	0.8857	0.9558	0.9862
0.4	0.4666	0.6614	0.8202	0.9217	0.9723
0.5	0.3795	0.5766	0.7559	0.8836	0.9548
0.6	0.3124	0.5043	0.6953	0.8440	0.9346
0.7	0.2600	0.4430	0.6393	0.8042	0.9126
0.8	0.2186	0.3909	0.5883	0.7652	0.8894
0.9	0.1855	0.3466	0.5420	0.7275	0.8655
1.0	0.1587	0.3085	0.5000	0.6915	0.8413

E.9.6 Log-Normal Hazard Function

Subroutines:
LOGNHFUN.TAB
LOGNHFUN.SUB

Purpose: To compute tables of the nondimensional log-normal hazard function.

Source: Table 9.2

Inputs:
LOGNHFUN.TAB
The following parameters must be specified:
DX = Time increment/SIGMA.
DY = Increment in MU/SIGMA.
M = Number of increments in DX.
N = Number of increments in DY.
SIGMA = Parameter in the log-normal density function.

Program File: LOGNHFUN.TAB

```
100 '***************   LOGNHFUN.TAB   ******************
110 'LOG-NORMAL HAZARD FUNCTION
120 'Input ranges of variables into LOGNHFUN.SUB subroutine
     to create tables of the log-normal hazard function x
     SIGMA, where SIGMA is a parameter in the normal density
     function.
130 'Values of DX=TIME INCREMENT/SIGMA and DY=INCREMENT IN
     MU/SIGMA must be specified on lines 140 and 150, as
     well as their ranges M and N on lines 160 and 170.
140 DX=.1
```

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```
150 DY=.5
160 M=10
170 N=5
180 GOSUB 39000
190 END
```

Test Example:

```
TABLE OF THE LOG-NORMAL HAZARD FUNCTION x SIGMA
X=Time/Sigma
Y=MU/Sigma
```

X\Y	-1.0	-.5	0.0	0.5	1.0
0.1	1.8901	0.8149	0.2846	0.0788	0.0171
0.2	2.2728	1.2442	0.5773	0.2194	0.0666
0.3	2.2424	1.3670	0.7273	0.3258	0.1189
0.4	2.1298	1.3828	0.7991	0.3969	0.1635
0.5	2.0059	1.3583	0.8301	0.4432	0.1993
0.6	1.8886	1.3183	0.8394	0.4727	0.2272
0.7	1.7822	1.2733	0.8365	0.4910	0.2488
0.8	1.6869	1.2276	0.8268	0.5018	0.2654
0.9	1.6016	1.1833	0.8134	0.5073	0.2780
1.0	1.5251	1.1411	0.7979	0.5092	0.2876

E.9.7 Weibull Probability Density Function and MTTF

Subroutines:
WEIDFUNC.TAB
WEIDFUNC.SUB

Purpose: To compute tables of the nondimensional Weibull density function.

Source: Eqs. (9.33) and (9.35) and Ref. 1.

Inputs:
WEIDFUNC.TAB
The following parameters must be specified:
DX = Time increment/ETA.
DY = Increment in GAMMA/ETA.
M = Number of increments in DX.
N = Number of increments in DY.
ETA = Parameter in the Weibull density function.

Program File: WEIDFUNC.TAB

```
100 '***************   WEIDFUNC.TAB   ********************
110 'WEIBULL PROBABILITY DENSITY FUNCTION AND MTTF
120 'Input ranges of variables into WEIDFUNC.SUB subroutine
     to create tables of nondimensional Weibull probability
```

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```
     density function (probability density function x ETA),
     where ETA=parameter in the density function.
130 'Values of DX=TIME INCREMENT/ETA and DY=INCREMENT IN
     GAMMA/ETA must be specified on lines 140 and 150,
     as well as their ranges M and N on lines 160 and
     170. Also value of BETA must be specified on line 180.
140 DX=.1
150 DY=.5
160 M=10
170 N=5
180 BETA=2
190 GOSUB 40000
200 END
```

Test Example:

```
TABLE OF THE WEIBULL DENSITY FUNCTION x ETA
X=Time/Eta
Y=Gamma/Eta
BETA=  2
```

X\Y	0.0	0.5	1.0	1.5	2.0
0.0	0.0000	0.0000	0.0000	0.0000	0.0000
0.1	0.1980	0.0000	0.0000	0.0000	0.0000
0.2	0.3843	0.0000	0.0000	0.0000	0.0000
0.3	0.5484	0.0000	0.0000	0.0000	0.0000
0.4	0.6817	0.0000	0.0000	0.0000	0.0000
0.5	0.7788	0.0000	0.0000	0.0000	0.0000
0.6	0.8372	0.1980	0.0000	0.0000	0.0000
0.7	0.8577	0.3843	0.0000	0.0000	0.0000
0.8	0.8437	0.5484	0.0000	0.0000	0.0000
0.9	0.8007	0.6817	0.0000	0.0000	0.0000
1.0	0.7358	0.7788	0.0000	0.0000	0.0000

```
TABLE OF MTTF/ETA (WEIBULL DISTRIBUTION)
Y=Gamm a/Eta
BETA= 2
```

Y	0.0	0.5	1.0	1.5	2.0
MTTF/ETA	0.8862	1.3862	1.8862	2.3862	2.8862

E.9.8 Weibull Reliability Function

Subroutines:
WEIRFUNC.TAB
WEIRFUNC.SUB

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Purpose: To compute tables of the Weibull reliability function.

Source: Eq. (9.34).

Inputs:
WEIRFUNC.TAB
The following parameters must be specified:
DX = Time increment/ETA.
DY = Increment in GAMMA/ETA.
M = Number of increments in DX.
N = Number of increments in DY.
ETA = Parameter in the Weibull density function.

Program File: WEIRFUNC.TAB

```
100 '***************   WEIRFUNC.TAB   ******************
110 'WEIBULL RELIABILITY FUNCTION
120 'Input ranges of variables into WEIRFUNC.SUB subroutine
     to create tables of the Weibull reliability function.
130 'Values of DX=TIME INCREMENT/SIGMA and DY=INCREMENT IN
     MU/SIGMA must be specified on lines 140 and 150, as
     well as their ranges M and N on lines 160 and 170.
140 DX=.1
150 DY=.5
160 M=10
170 N=5
180 BETA=1
190 GOSUB 42000
200 END
```

Test Example:

TABLE OF THE WEIBULL RELIABILITY FUNCTION
X=Time/Eta, Y=Gamma/Eta, BETA=1

X\Y	0.0	0.5	1.0	1.5	2.0
0.0	1.0000	1.0000	1.0000	1.0000	1.0000
0.1	0.9048	1.0000	1.0000	1.0000	1.0000
0.2	0.8187	1.0000	1.0000	1.0000	1.0000
0.3	0.7408	1.0000	1.0000	1.0000	1.0000
0.4	0.6703	1.0000	1.0000	1.0000	1.0000
0.5	0.6065	1.0000	1.0000	1.0000	1.0000
0.6	0.5488	0.9048	1.0000	1.0000	1.0000
0.7	0.4966	0.8187	1.0000	1.0000	1.0000
0.8	0.4493	0.7408	1.0000	1.0000	1.0000
0.9	0.4066	0.6703	1.0000	1.0000	1.0000
1.0	0.3679	0.6065	1.0000	1.0000	1.0000

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E.9.9 Weibull Hazard Function

Subroutines:
WEIHFUNC.TAB
WEIHFUNC.SUB

Purpose: To compute tables of the nondimensional Weibull hazard function.

Source: Table 9.2.

Inputs:
WEIHFUNC.TAB
The following parameters must be specified:
DX = Time increment/ETA.
DY = Increment in GAMMA/ETA.
M = Number of increments in DX.
N = Number of increments in DY.
ETA = Parameter in the Weibull density function.

Program File: WEIHFUNC.TAB

```
100 '***************   WEIHFUNC.TAB   ********************
110 'WEIBULL HAZARD FUNCTION
120 'Input ranges of variables into WEIHFUNC.SUB subroutine
     to create tables of the Weibull hazard function x ETA,
     where ETA is a parameter in the Weibull density function.
130 'Values of DX=TIME INCREMENT/ETA and DY=INCREMENT IN
     GAMMA/ETA must be specified on lines 140 and 150, as
     well as their ranges M and N on lines 160 and 170. The
     required value of BETA must be specified on line 180.
140 DX=.1
150 DY=.5
160 M=10
170 N=5
180 BETA=1
190 GOSUB 43000
200 END
```

Test Example:

```
TABLE OF THE WEIBULL HAZARD FUNCTION x ETA
X=Time/Eta
Y=Gamma/Eta
BETA=  1
```

X\Y	0.0	0.5	1.0	1.5	2.0
0.0	1.0000	0.0000	0.0000	0.0000	0.0000
0.1	1.0000	0.0000	0.0000	0.0000	0.0000
0.2	1.0000	0.0000	0.0000	0.0000	0.0000
0.3	1.0000	0.0000	0.0000	0.0000	0.0000
0.4	1.0000	0.0000	0.0000	0.0000	0.0000

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0.5	1.0000	1.0000	0.0000	0.0000	0.0000
0.6	1.0000	1.0000	0.0000	0.0000	0.0000
0.7	1.0000	1.0000	0.0000	0.0000	0.0000
0.8	1.0000	1.0000	0.0000	0.0000	0.0000
0.9	1.0000	1.0000	0.0000	0.0000	0.0000
1.0	1.0000	1.0000	1.0000	0.0000	0.0000

E.10 Target Detection and Search

E.10.1 Search for Constant Velocity Target

Subroutines:
SPSEARCH.DAT
SPSEARCH.SUB

Purpose: To compute the position of an aircraft searching for a constant velocity target. The search path is a spiral, ensuring that the target will be located prior to or at the completion of the spiral flight.

Source: Eqs. (10.67a) and (10.67b).

Inputs:
SPSEARCH.DAT
The following parameters must be specified:
C = VS/VT, where VS = velocity of the search aircraft and VT = velocity of the target.
RADW = W/D, where W = radar sweep radius and D = initial distance from the target.

Program File: SPSEARCH.DAT

```
100 '***************   SPSEARCH.DAT   ******************
110 'SEARCH FOR CONSTANT VELOCITY TARGET
120 'Determine the position of an aircraft searching for a
     constant velocity target.  The search path is a spiral
     ensuring that the target is going to be located prior
     to or at the completion of the spiral flight.
130 'Input data for the subroutine SPSEARCH.SUB : values of
     C=VS/VT, and RADW = W/D must be specified on lines 170
     and 180.
140 'VS = velocity of search aircraft, VT = velocity of
     target aircraft, W= radar sweep radius, and D = initial
     distance from the target.
150 'The following output variables are used: X=x/D and
     Y=y/D, where x,y are the coordinates for the search aircraft
     measured from the starting point of the spiral search.
160 'Additional output variables are: P=probability of finding
     the target with the radar turned off PR=probability
     of finding the target with the radar turned on.
170 C=3
180 RADW=.3
190 GOSUB 44000
200 END
```

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Test Example:

```
POSITION OF SEARCH AIRCRAFT AND PROBABILITY OF TARGET
DETECTION
[PHI=Location angle, TAU=VS*T/D, X=x/D, Y=y/D,
R=Radius]
[P=Probability of finding the target without radar]
[PR=Probability of finding the target with radar]
```

PHI	TAU	X	Y	R	P	PR
0.000	0.000	0.000	-0.250	0.250	0.000	0.307
0.628	0.704	0.183	-0.253	0.312	0.100	0.362
1.257	1.582	0.371	-0.120	0.390	0.200	0.428
1.885	2.679	0.463	0.150	0.487	0.300	0.502
2.513	4.049	0.357	0.492	0.608	0.400	0.582
3.142	5.760	-0.000	0.759	0.759	0.500	0.666
3.770	7.897	-0.557	0.767	0.948	0.600	0.753
4.398	10.565	-1.126	0.366	1.184	0.700	0.843
5.027	13.896	-1.406	-0.457	1.478	0.800	0.935
5.655	18.056	-1.085	-1.493	1.846	0.900	1.000
6.283	23.251	0.000	-2.305	2.305	1.000	1.000

E.11 Miscellaneous

E.11.1 Dominant Eigenvalue and Left Eigenvector

Subroutines:
DOMEIGEN.DAT
DOMEIGEN.SUB

Purpose: To compute the dominant eigenvalue and the corresponding left eigenvector of the matrix C(I,J) (LAMBDA1 and Q1(I)).

Source: Ref. 2.

Inputs:
DOMEIGEN.DAT
The following input data must be specified:
B = Number of BLUE forces.
R = Number of RED forces.
T = Number of iterations.
C(I,J) = Coefficients in the C matrix.

Program Files: DOMEIGEN.DAT and DOMEIGEN.SUB

```
100 '***************   DOMEIGEN.DAT   ********************
110 'DOMINANT EIGENVALUE AND LEFT EIGENVECTOR
```

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```
120 'Input data for the determination of the dominant
    eigenvalue LAMBDA1 and the corresponding left
    eigenvector Q1(I) for the TRANSPOSE of the matrix of
    attrition coefficients C(I,J).
130 'Size of the square matrix C(I,J) is specified on line
    200; i.e., N=B+R.
140 'Number of BLUE forces B must be specified on line 180.
150 'Number of RED Forces R must be specified on line 190.
160 'CAUTION: The number of iterations T on line 210, used
    in DOMEIGEN.SUB, may have to be increased from 100 to
    500 to improve the accuracy for cases of defective
    matrices C(I,J); i.e., matrices with multiple
    eigenvalues.
170 'Matrix coefficients in C(I,J) must be specified on
    lines 290...; additional lines may be added for larger
    size matrices (i.e., lines 340 ...).
180 B=2
190 R=2
200 N=B+R
210 T=100
220 DIM C(N,N), I(N,N), Q1(N)
230 FOR I=1 TO N
240 FOR J=1 TO N
250 READ C(I,J)
260 NEXT J
270 NEXT I
280 'DATA below specify the matrix coefficients in C(I,J)
    by rows.
290 DATA 0., 0., -1., -1.
300 DATA 0., 0., 0., -1.
310 DATA -1., -1., 0., 0.
320 DATA -1., -1., 0., 0.
330 'Here additional data for C(I,J) should be inserted
    when N>4.
340 '
350 '
360 '
370 FOR I=1 TO N
380 FOR J=1 TO N
390 IF I=J THEN I(I,J)=1 ELSE I(I,J)=0
400 NEXT J
410 NEXT I
420 'Assumed values of Q1(I) for the first iteration
    cycle.
430 FOR I=1 TO B
440 Q1(I)=1
450 NEXT I
460 FOR I=B+1 TO N
470 Q1(I)=-1
480 NEXT I
490 GOSUB 45000
500 END
```

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Test Example:

```
Matrix C(I,J)=
 0  0 -1 -1
 0  0  0 -1
-1 -1  0  0
-1 -1  0  0
Dominant eigenvalue LAMBDA1=  1.732051
Dominant left eigenvector Q1(I)=
 .8660254
 .8660254
-.5
-1
```

References

[1]Hastings, C., Jr., "Approximations for Digital Computers," Princeton Univ. Press, Princeton, NJ, 1955, p. 169.

[2]Young, D. M., and Gregory, R. T., "A Survey of Numerical Mathematics," Vol. 2, Addison–Wesley Publishing Co., Reading, MA, 1973, pp. 915–918.

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